

**TO**

Michelle Unger, B.Sc., Manager, Environmental
Performance, Legacy Properties

DATE

July 10, 2024

FROM

Craig Pinnell, P.Eng., Project Manager

REF

697296

SUBJECT

Post-Liner Installation Interim Water Quality Monitoring Results, Sunro Mine

Introduction

The following factual memo summarizes the current results for the ongoing water sampling program at Sunro Mine near Jordan River, BC (the Site) completed following the North Waste Rock Pile (NWRP) temporary liner installation. The work was completed as per the document titled *2023 to 2024 Water Sampling and Risk Opinion Cost Estimate at Sunro Mine, Jordan River, BC* dated June 1, 2023, and in accordance with the master services agreement #TR00013 between Teck and AtkinsRéalis Canada Inc. (AtkinsRéalis) and Teck Metals Ltd. (Teck). It is assumed that readers of this memo are familiar with the Site and as such a detailed background is not provided herein. This factual memo is intended to present the results to date (January 2022 to November 2023) with a more comprehensive report to be completed following the planned 2024 summer and winter sampling events.

Objectives and Scope of Work

The objectives of the water sampling program at Sunro Mine are to:

- measure water quality in the Jordan River downstream of the cover and 5100 Portal to document changes;
- build upon the existing dataset to assess trends and support future remedial efforts at the Site; and
- assess the impact the cover has had on overall metal concentrations in the surface water of the Jordan River, downstream of the 5100 Portal. Specifically, the sample locations were selected from the baseline dataset (pre-liner locations) to determine if the seasonal increases in metals concentrations observed in porewater have been mitigated.

To support these objectives five water monitoring and sampling events were completed between January 2022 and November 2023 (February 2022, October 2022, February 2023, July 2023, November 2023).

This is a factual memo presenting the water quality monitoring results following the installation of the temporary liner in the fall of 2021. As such temporal and spatial trends are not discussed in this report but will be addressed in the comprehensive report to be prepared after the 2024 summer and winter sampling events.

The sample locations and analytical program are described in Table A. Sample locations are shown on the attached Drawing 697296-001. All samples were collected following AtkinsRéalisis' preferred operating procedures which are in alignment with the British Columbia Field Sampling Manual¹ and industry standards.

Table A: Sample Identifications and Analytical Program

Media	Sample IDs	N	Parameters
Surface Water	SW15-01a to SW15-04 and SW15-07	5	Total and dissolved metals, total dissolved solids (TDS), hardness, major anions ² , total and speciated alkalinity and acidity.
Porewater	PW15-08 to PW15-11, PW17-17, PW17-19, PW17-21, PW18-22, PW23-29 ¹	9	Dissolved metals, TDS, hardness, major anions, total and speciated alkalinity and acidity.
Jordan River	SW15-08 to SW15-11, SW17-17, SW17-19, SW17-21, SW18-22, SW22-28 ² , SW23-29 ¹	10	Same as surface water, with sulphide, total suspended solids, and dissolved organic carbon.
Groundwater	MW18-01 to MW18-07 and MW18-09	8	Dissolved metals, TDS, hardness, major anions, total and speciated alkalinity and acidity.

¹ SW/PW23-29 was added during the October 2022 sampling event.

² SW23-28 was added during the July 2023 sampling event.

Not all sample locations could be sampled during each field program. A summary is provided below by event indicating when and where samples could not be collected, and the reason.

- February 2022: MW18-01 was dry. SW15-03 and SW15-04 were dry.
- October 2022: MW18-06 and MW18-05 were dry. SW15-03 and SW15-04 were dry.
- February 2023: MW18-01 was dry. PW15-08 was not sampled due to elevated water levels causing unsafe access conditions. SW15-03 and SW15-04 were dry.
- July 2023: MW18-01 was dry. SW15-03 and SW15-04 were dry.
- November 2023: MW18-01 was dry. Locations PW15-08 and PW23-29 were not sampled due to elevated water levels causing unsafe access conditions. The downstream location SW23-29 was not sampled due to elevated water levels creating unsafe access conditions. SW15-03 and SW15-04 were dry.

¹ British Columbia Ministry of Environment and Climate Change Strategy. 2013. *British Columbia Field Sampling Manual: For Continuous Monitoring Plus the Collection of Air, Air Emission, Water, Wastewater, Soil, Sediment and Biological Samples*, 2013 Edition (Permittee). Water, Air and Climate Branch, Ministry of Water, Land and Air Protection: Victoria, BC.

² Includes sulphate, chloride, bromide, fluoride, ammonia, nitrate and nitrite.

Groundwater wells were monitored for depth to water and depth to bottom prior to sampling using a water level probe which is cleaned between locations using a mild phosphate free soap and distilled water. All wells were sampled by conventional purge methods using an inertial lift pump which consists of a foot valve connected to new or dedicated high density polyethylene (HDPE) tubing. During purging field parameters were collected using a YSI Multiparameter Probe which is calibrated daily. The wells were purged at a rate of less than 4 litres/minute until a minimum purge volume³ was achieved and field parameters stabilized⁴. Once the wells were purged the samples were collected in new pre-labelled laboratory supplied bottles which are pre-charged with preservatives where applicable. Dissolved parameters were field filtered using new 0.45 µm inline filter that is purged a minimum of three filter volumes prior to sample collection.

Porewater samples are collected by advancing a 20 mm stainless steel probe with a 150 mm screened tip to a depth of approximately 0.5 m. The probes are purged using low flow sampling techniques which utilize new HDPE tubing inserted into the probe and connected to a peristaltic pump using silicon tubing. Field parameters are monitored using a YSI Multiparameter Probe connected to a flow through cell. The probes are purged at a flow rate of less than 500 mL/second until field parameters stabilize. Once the wells were purged the samples were collected in new pre-labelled laboratory supplied bottles which are pre-charged with preservatives where applicable. Dissolved parameters were field filtered using new 0.45 µm inline filter.

All surface water samples were collected by submerging, where possible, laboratory supplied bottles directly into the water feature. Sample bottles with pre-charged preservative are plunged beneath the surface with the opening facing directly down. Once submerged the opening is rotated into the current with care taken not to introduce suspended sediment or materials into the sample. Once the bottles are full, they are removed from the flow by bringing them into the current and upwards and the cap is replaced immediately. For dissolved parameters, bottles pre-charged with preservatives or when flows are too low to submerge the sample bottles, a sterile syringe is used to collect the samples. For filtered samples a 0.45 µm disc filter connected to a syringe is used.

All samples were packed on ice and shipped to Caro Analytical Services (Caro) in Burnaby, BC for the analysis indicated in Table A. For quality assurance and quality control (QA/QC) purposes blind field duplicates were collected and analyzed. Additionally, blanks (trip blank, field blank and equipment blank) were analyzed to examine any potential sample contamination sources from collection or shipping.

Regulatory Framework

The results of the sampling program were compared to the aquatic life (AW) standards and/or guidelines included in the BC *Contaminated Sites Regulation*⁵ (CSR) under the British Columbia *Environmental Management Act* (EMA), as well as the *British Columbia Water Quality Guidelines*⁶ (BCWQG). Results were also compared to the CSR Protocol 11⁷ Upper Cap Concentrations (UCCs) to assess the presence of “high risk” conditions.

³ The minimum purge volume is considered to be one well volume (i.e., one well water volume is equal to the volume of standing water within the well scree, saturated sand pack and riser pipe).

⁴ Parameters are considered stabilized when three successive readings are within these limits: pH $\pm$ 0.1 units, specific conductivity $\pm$ 3%, redox $\pm$ 10 mV, and temperature $\pm$ 0.2 °C.

⁵ *Contaminated Sites Regulation* (CSR), B.C. Reg. 375/96. Last amended March 1, 2023 by B.C. Reg. 133/2022.

⁶ *British Columbia Approved Water Quality Guidelines: Aquatic Life, Wildlife & Agriculture* (BCWQG). British Columbia Ministry of Environment & Climate Change Strategy, updated November 2023.

⁷ Protocol 11; Protocol for Contaminated Sites – *Upper Cap Concentrations for Substances Listed in the Contaminated Sites Regulation*, BC Ministry of Environment & Climate Change Strategy, Version 5, March 20, 2023.

Results

Results are presented as box and whisker plots below for groundwater, porewater and surface water (upstream to downstream, where applicable). The box plots include all available data (between 2015 to present) from each location. Sample concentrations collected following the liner installation are presented for reference. The analytical laboratory results are included in Attachment A and are compared to the applicable guidelines and standards in attached Tables 1 to 7. The presentation of results focuses primarily on dissolved copper concentrations as this is the primary contaminant of concern in water at the Site.

Groundwater

Groundwater monitoring results are presented in Table 1 and analytical results compared to the applicable standards are presented in Table 2, attached. Dissolved copper concentrations in groundwater are illustrated in **Figure 1** below.

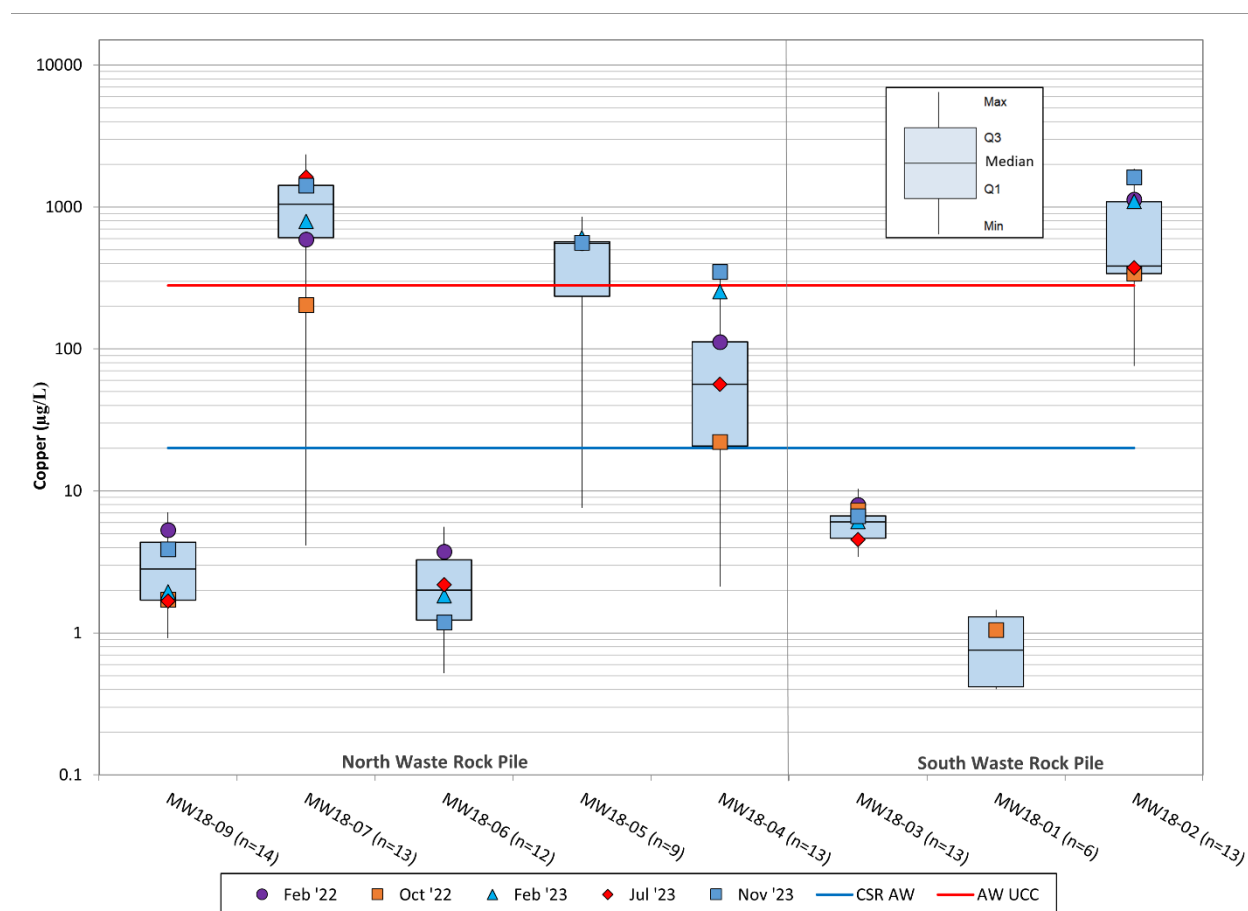


Figure 1: Dissolved Copper Concentrations in Groundwater

The figure and tabulated results show the following:

- AW UCC exceedances were observed for dissolved copper in groundwater samples collected from MW18-02, MW18-04, MW18-05, and MW18-07.
- Dissolved copper concentrations at the remaining four locations were less than the AW UCC.
- Dissolved cobalt in all samples and zinc in select samples collected from MW18-07 exceeded the CSR AW.
- Dissolved cobalt and zinc in select samples collected from MW18-02 exceeded the CSR AW.
- Overall results were generally within historical limits. One exception was for dissolved copper in the November 2023 sample collected from MW18-04 which was the new maximum concentration for this location.

Porewater

Porewater field parameters are presented in Table 3 and analytical results compared to the applicable standards and guidelines are presented in Table 4, attached. Dissolved copper concentrations are shown in **Figure 2** below:

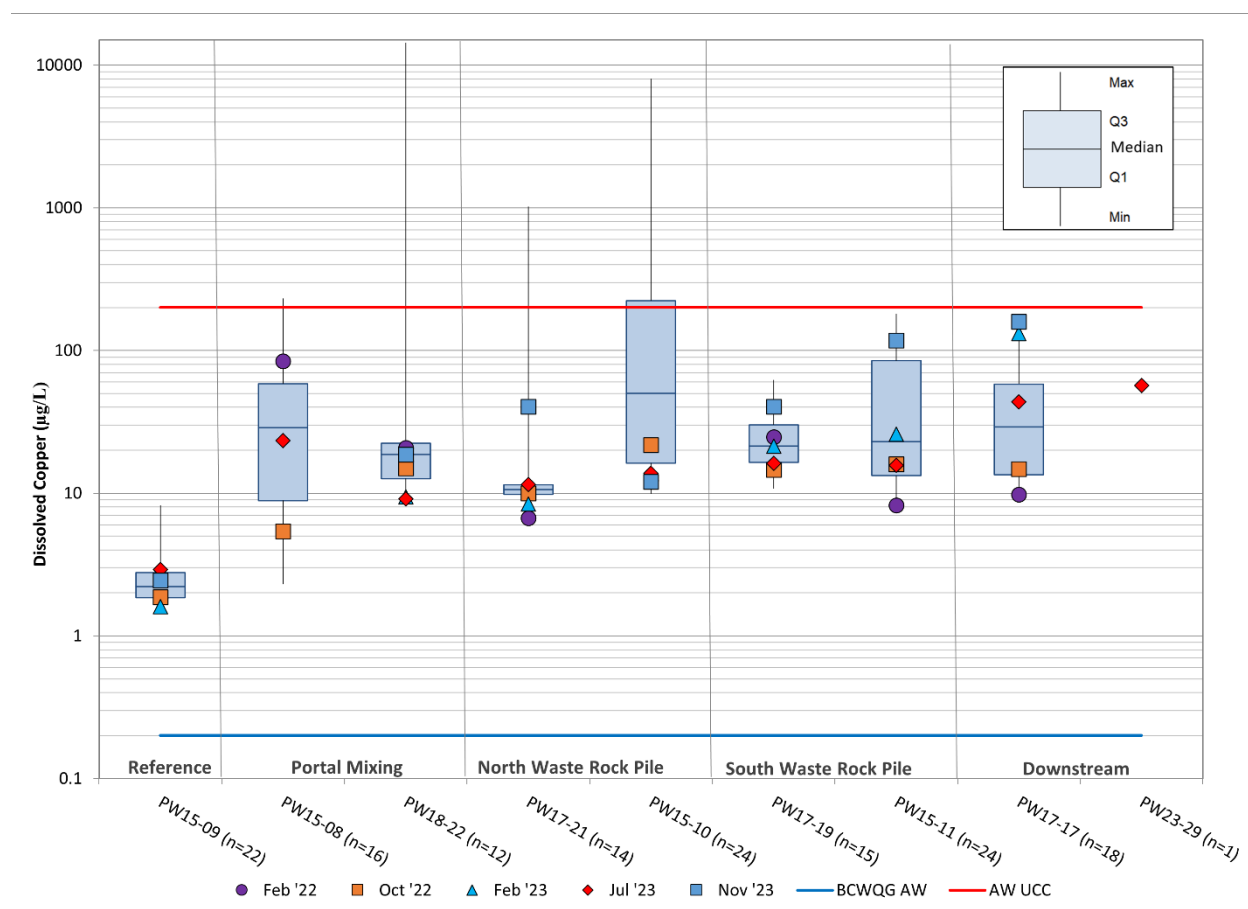


Figure 2: Dissolved Copper Concentrations in Porewater

The figure and tabulated porewater results show the following:

- There were no AW UCC exceedances in porewater samples.
- All results were generally within historical limits with the exception of dissolved copper in the sample collected from PW17-17 which was the highest concentration observed in samples collected from this location.
- Dissolved copper exceeded the BCWQG AW in all porewater samples, including the reference location PW15-09.
- Dissolved aluminum and zinc exceeded the BCWQG AW in select samples collected from PW15-08.
- Dissolved aluminum exceeded the BCWQG AW in select samples collected from PW15-08, PW15-10, PW17-19, PW17-21 and PW18-22. The pH of select samples was also slightly less than the BCWQG AW range.
- Dissolved aluminum and cadmium exceeded the BCWQG AW in select samples collected from PW15-11. The pH of one sample was slightly less than the BCWQG AW range.
- Dissolved aluminum, cadmium and zinc exceeded the BCWQG AW in select samples collected from PW17-17. The pH of one sample was slightly less than the BCWQG AW range.
- The pH of the sample collected from PW18-22 was slightly less than the BCWQG AW range.

Surface Water

Surface water monitoring results are presented in Table 5 and analytical results compared to the applicable guidelines and standards are presented in Tables 6 and 7, attached. Surface water results for dissolved copper are illustrated separately with Portal results presented in **Figure 3** and Jordan River results in **Figure 4**.

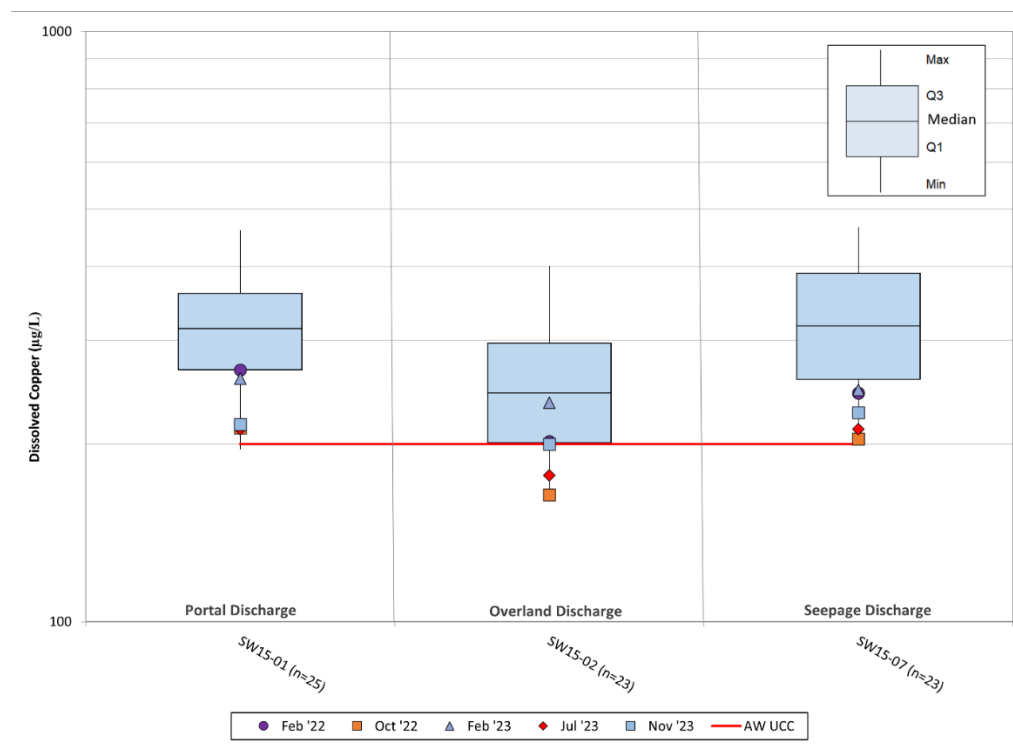


Figure 3: Dissolved Copper Concentrations in Surface Water from the 5100 Portal

The figure and tabulated results show the following:

- Consistent with previous results, dissolved copper exceeded the AW UCC in all samples collected from SW15-01 and SW15-07.
- Dissolved copper exceeded the AW UCC in select samples collected from SW15-02. The October 2022 and August 2023 samples had the lowest and second lowest dissolved copper concentrations of samples collected from SW15-02.
- All samples had dissolved cadmium, copper and zinc, as well as total aluminum and cobalt exceeding the BCWQG AW. One sample collected from SW15-02 also had concentrations of total chromium, nickel and silver exceeding the BCWQG AW.
- Generally, all results were within or near historical limits. Of note the dissolved copper concentrations were generally less than or near the lower quartile.

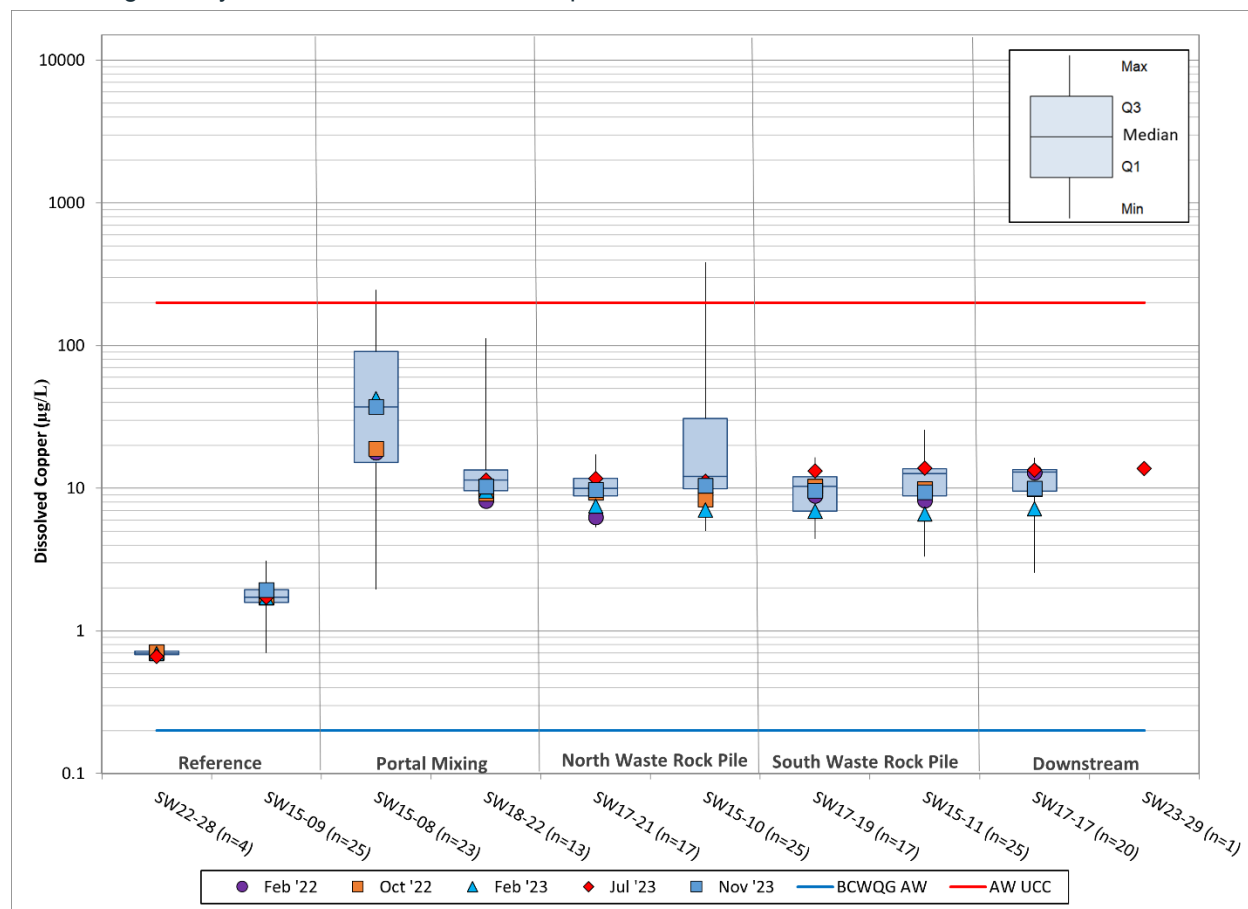


Figure 4: Dissolved Copper Concentrations in Surface Water in the Jordan River

The figure and tabulated results show the following:

- All samples collected from the Jordan River had concentrations of parameters less than the AW UCC. However, dissolved copper exceeded the BCWQG AW in all samples collected from the Jordan River.
- All dissolved copper concentrations were within historical limits with the exception of the far upstream reference location (SW22-28) which had the highest concentration to date. It is of note that this location has limited samples (n = 4) and the overall variability is low (0.66 µg/L to 0.75 µg/L).

- The historical seasonal spikes in copper concentrations in water samples collected from the Jordan River below the North Waste Rock Pile were not observed. The seeps from the NWRP (SW15-03 and SW15-04) have not been observed flowing since installation of the temporary liner over the NWRP. These seeps have historically discharged high concentrations of dissolved copper ($> 29,000 \mu\text{g/L}$) from the NWRP to the Jordan River.
- Total aluminium was the only other parameter to exceed the BCWQG AW at all water sample locations, including the reference location SW22-28. Total chromium exceeded the BCWQG AW in select samples collected from SW15-08 and SW18-22.
- Dissolved zinc exceeded BCWQG AW in a sample collected from SW15-08.
- All other parameters in samples collected from the Jordan River were less than the BCWQG AW.

Quality Assurance/Quality Control

For the five sampling events discussed in this memo 125 samples (35 groundwater, 29 porewater, 61 surface water) and 18 field duplicates (5 groundwater, 5 porewater, 8 surface water) were submitted for analysis. With a minimum duplicate submission ratio of 10%, this event exceeded at 14%. RPD (Relative Percent Difference) values ranged between 0% and 41% for groundwater, between 0% and 28% for porewater and between 0% and 15% for surface water. All duplicate RPDs were less than the RPD alert limit of 50%. The trip blank, equipment blank and field blank produced concentrations less than the reportable detection limit for all parameters, except for total alkalinity in a trip blank and dissolved silver in a field blank. For the trip blank the concentration was 2mg/L which is equal to the alert limit. The dissolved silver concentration was less than two times the reporting detection limit and is therefore considered acceptable. Alkalinity is not a contaminant of concern at the Site and given the overall blank performance the results are considered acceptable. The results are considered acceptable and with good reproducibility and therefore do not affect the interpretation of this memo.

The laboratory QA/QC results are included with the laboratory analytical results provided in the Appendix A. Overall, the laboratories considered the results acceptable based on the quality control performance. All quality control flags from the laboratory were associated with nitrite/nitrate and respective holding times, as well as one for bismuth. These parameters are not contaminants of concern at the Site and therefore do not affect the interpretation of this report.

Summary

The groundwater, porewater and surface water results were generally within historical results. Two exceptions are new maximum dissolved copper concentrations in the samples collected from MW18-04 (located within the tailings area) and PW17-17 (the downstream location). The concentration of dissolved copper in surface water samples generally stayed consistent in the Jordan River between the 5100 Portal Mixing Zone (SW18-22) and downstream location (SW17-17). This indicates there is no significant loading of dissolved copper to the Jordan River as it flows past the North and South Waste Rock Piles, after the 5100 Portal Discharge. An additional sampling event is planned for the summer and winter of 2024 and a more comprehensive report will be prepared following those events.

Closure

We trust this meets your needs at this time. If you have any questions please reach out to the undersigned.



Craig Pinnell, P.Eng.
Project Engineer

Environment Practice
Engineering Services Canada

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TABLES

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- 3: Porewater Log
- 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)
- 5: Surface Water Log
- 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)
- 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

TABLE 1: Summary of Monitoring Well Locations, Elevations, Well Completion Details, Elevation of Bedrock and Potentiometric Elevations

MW ID	Northing	Easting	Reference Elevation (masl)	Ground Surface Elevation (masl)	Depth to Top of Screen (mbgs)	Depth to Bottom of Screen (mbgs)	Depth to Bottom (mbtoc)	Depth to Bedrock (mbgs)	Bedrock Elevation (masl)	Date Monitored (yyyy/mm/dd)	Depth to Water (mbtoc)	Potentiometric Elevation (masl)
MW18-01	5364759	422259	24.651	23.643	16.31	19.35	19.164	-	-	2018-05-17	18.415	6.236
										2018-07-23	17.802	6.849
										2018-10-29	18.270	6.381
										2019-01-21	18.145	6.506
										2019-09-24	18.234	6.417
										2020-07-28	Dry	-
										2021-01-11	Dry	-
										2021-08-11	Dry	-
										2022-02-09	Dry	-
										2022-10-24	18.297	6.354
										2023-02-21	Dry	-
										2023-07-31	Dry	-
										2023-11-13	Dry	-
MW18-02	5364728	422277	23.511	22.681	18.29	21.34	22.196	-	-	2018-05-17	20.786	2.725
										2018-07-23	20.980	2.531
										2018-10-29	20.481	3.030
										2019-01-21	19.984	3.527
										2019-09-23	20.946	2.565
										2019-12-04	20.140	3.371
										2020-07-28	21.143	2.368
										2021-01-11	19.587	3.924
										2021-08-10	21.222	2.289
										2022-02-09	20.401	3.110
										2022-10-24	21.248	2.263
										2023-02-21	20.130	3.381
										2023-07-31	21.225	2.286
										2023-11-13	19.850	3.661
MW18-03	5364759	422295	32.509	31.517	4.27	7.32	8.479	-	-	2018-05-17	6.894	25.615
										2018-07-23	6.999	25.510
										2018-10-29	7.051	25.458
										2019-01-21	7.019	25.490
										2019-09-23	7.169	25.340
										2019-12-04	7.117	25.392
										2020-07-28	7.208	25.301
										2021-01-11	7.014	25.495
										2021-08-10	7.116	25.393
										2022-02-09	7.100	25.409
										2022-10-24	7.150	25.359
										2023-02-21	7.085	25.424
										2023-07-31	7.245	25.264
										2023-11-13	7.134	25.375
MW18-04	5364795	422256	25.518	24.583	4.88	7.92	8.984	7.6	16.98	2018-05-17	7.442	18.076
										2018-07-23	7.716	17.802
										2018-10-29	7.270	18.248
										2019-01-21	6.878	18.640
										2019-09-23	7.425	18.093
										2019-12-04	7.900	17.618
										2020-07-28	7.751	17.767
										2021-01-11	6.885	18.633

- not recorded

m btoc - Metres Below Top of Casing

m bgs - Metres Below Ground Surface

m asl - Meters Above Sea Level

TABLE 1: Summary of Monitoring Well Locations, Elevations, Well Completion Details, Elevation of Bedrock and Potentiometric Elevations

MW ID	Northing	Easting	Reference Elevation (masl)	Ground Surface Elevation (masl)	Depth to Top of Screen (mbgs)	Depth to Bottom of Screen (mbgs)	Depth to Bottom (mbtoc)	Depth to Bedrock (mbgs)	Bedrock Elevation (masl)	Date Monitored (yyyy/mm/dd)	Depth to Water (mbtoc)	Potentiometric Elevation (masl)
MW18-04	5364795	422256	25.518	24.583	4.88	7.92	8.984	7.6	16.98	2021-08-10	7.848	17.670
										2022-02-09	6.885	18.633
										2022-10-24	7.857	17.661
										2023-02-21	7.960	17.558
										2023-07-31	7.740	17.778
										2023-11-13	6.556	18.962
MW18-05	5364823	422260	39.931	39.992	4.57	7.62	7.679	-	-	2018-05-17	7.589	32.342
										2018-07-23	7.610	32.321
										2018-10-29	7.289	32.642
										2019-01-21	5.987	33.944
										2019-09-23	7.033	32.898
										2019-12-04	6.220	33.711
										2020-07-28	Dry	-
										2021-01-11	5.885	34.046
										2022-02-09	6.110	33.821
										2022-10-24	Dry	-
										2023-02-21	5.395	34.536
MW18-06	5364824	422238	28.085	27.274	9.14	12.19	13.263	11.9	15.37	2018-05-17	12.642	15.443
										2018-07-23	12.904	15.181
										2018-10-29	12.331	15.754
										2019-01-21	11.810	16.275
										2019-09-23	11.951	16.134
										2019-12-04	11.375	16.710
										2020-07-29	12.639	15.446
										2021-01-11	11.150	16.935
										2021-08-10	12.805	15.280
										2022-02-10	11.200	16.885
										2022-10-24	12.925	15.160
MW18-07	5364854	422229	29.411	28.698	20.73	23.77	24.785	-	-	2023-02-21	11.185	16.900
										2023-07-31	12.808	15.277
										2023-11-13	10.819	17.266
										2018-05-17	21.171	8.240
										2018-07-23	21.281	8.130
										2018-10-29	21.311	8.100
										2019-01-21	21.124	8.287
										2019-09-23	21.434	7.977
										2019-12-04	21.141	8.270
										2020-07-28	21.225	8.186
										2021-01-11	21.087	8.324

- not recorded

m btoc - Metres Below Top of Casing

m bgs - Metres Below Ground Surface

m asl - Meters Above Sea Level

TABLE 1: Summary of Monitoring Well Locations, Elevations, Well Completion Details, Elevation of Bedrock and Potentiometric Elevations

MW ID	Northing	Easting	Reference Elevation (masl)	Ground Surface Elevation (masl)	Depth to Top of Screen (mbgs)	Depth to Bottom of Screen (mbgs)	Depth to Bottom (mbtoc)	Depth to Bedrock (mbgs)	Bedrock Elevation (masl)	Date Monitored (yyyy/mm/dd)	Depth to Water (mbtoc)	Potentiometric Elevation (masl)
MW18-07	5364854	422229	29.411	28.698	20.73	23.77	24.785	-	-	2021-08-10	21.315	8.096
										2022-02-09	21.143	8.268
										2022-10-24	21.355	8.056
										2023-02-22	21.072	8.339
										2023-07-31	22.709	6.702
										2023-11-13	21.093	8.318
MW18-08	5364885	422234	31.823	30.824	14.18	17.23	20.726	19.5	11.324	2018-05-17	18.112	13.711
										2018-07-23	18.177	13.646
										2018-10-29	18.186	13.637
										2019-01-21	17.981	13.842
MW18-09	5364897	422236	31.411	30.442	16.77	19.82	20.725	19.2	11.2	2018-05-17	18.471	12.940
										2018-07-23	19.765	11.646
										2018-10-29	18.246	13.165
										2019-01-21	17.867	13.544
										2019-09-23	19.946	11.465
										2019-12-04	19.674	11.737
										2021-01-11	19.502	11.909
										2021-08-10	19.950	11.461
										2022-02-09	19.510	11.901
										2022-10-24	20.002	11.409
										2023-02-21	19.549	11.862
										2023-07-31	19.923	11.488
										2023-11-13	19.700	11.711

- not recorded

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m asl - Meters Above Sea Level

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters			Field Parameters					Geochemical Indicators																						
			pH	Hardness mg/L	Total Hardness mg/L	pH (field)	Temperature °C	Conductivity µS/cm	ORP mV	Dissolved Oxygen mg/L	Total Dissolved Solids mg/L	Ammonia, Total (as N) µg/L	Nitrate (as N) µg/L	Nitrite (as N) µg/L	Chloride mg/L	Fluoride µg/L	Sulfate mg/L	Total Alkalinity mg/L	Alkalinity, Bicarbonate (as CaCO3) mg/L	Alkalinity, Carbonate (as CaCO3) mg/L	Alkalinity, Hydroxide (as CaCO3) mg/L	Alkalinity, Phenolphthalein (as CaCO3) mg/L	Bromide mg/L	Total Acidity mg/L	Acidity (as CaCO3) mg/L	Dissolved Aluminum µg/L	Dissolved Calcium mg/L	Dissolved Iron µg/L	Dissolved Magnesium mg/L	Dissolved Manganese µg/L	Dissolved Potassium mg/L	Dissolved Sodium mg/L	
BC Standard																																	
CSR Aquatic Life (AW) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	18,400-18,500 ^b	400,000	400-2,000 ^c	1,500	2,000-3,000 ^d	2,180-4,290 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^b	4,000,000	4,000-20,000 ^c	15,000	20,000-30,000 ^d	21,800-42,900 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
MW18-01	MW18-01-180310	2018 03 10	6.53	-	37.1	5.79	8.57	83	-118.9	7.73	-	-	< 15.0	< 5.0	4.70	< 100	< 1.0	48.0	48.0	< 1.0	< 1.0	< 1.0	-	13.4	-	31.9	11.2	7,200	2.19	219	0.67	6.05	
	MW18-A-180310	Duplicate	6.53	-	37.0	5.79	-	-	-	-	-	-	22.2	< 5.0	4.69	< 100	< 1.0	47.7	47.7	< 1.0	< 1.0	< 1.0	-	8.4	-	27.4	11.2	7,200	2.20	219	0.67	6.27	
	QA/QC RPD%		< 1	-	< 1	0	*	*	*	*	*	-	-	*	*	< 1	*	*	1	1	*	*	*	-	*	-	15	0	0	< 1	0	0	4
	MW18-01-180725	2018 07 25	6.32	-	34.1	6.38	10.08	90	23.6	3.87	69	-	< 10	< 10	5.07	< 100	1.3	46.3	46.3	< 1.0	< 1.0	< 1.0	-	25.7	-	35.9	9.93	7,940	2.25	151	0.52	4.92	
	MW18-01-181030	2018 10 31	6.88	-	38.1	5.94	9.7	94.4	71	-	86	-	< 10	< 10	4.50	< 100	1.4	47.2	47.2	< 1.0	< 1.0	< 1.0	< 0.10	13.9	-	50.4	11.4	9,110	2.33	143	0.65	5.63	
	MW18-01-190122	2019 01 22	6.02	-	28.9	6.44	9.1	114	66	-	-	-	< 10	< 10	4.13	< 100	< 1.0	39.6	39.6	< 1.0	< 1.0	< 1.0	< 0.10	8.5	-	33.4	8.25	7,040	2.00	127	0.49	4.54	
	MW18-01-190924	2019 09 24	6.98	-	35.8	5.68	10.7	85	-	-	47	-	< 10	< 10	4.14	< 100	1.2	40.7	40.7	< 1.0	< 1.0	< 1.0	< 0.10	7.0	-	26.4	10.8	5,530	2.15	171	0.66	5.06	
	MW18-01-221024	2022 10 24	-	-	36.6	6.42	10	1,205	-	-	46	204	< 10	< 10	4.46	< 100	2.2	41.5	41.5	< 1.0	< 1.0	< 1.0	< 0.10	17.4	-	36.3	11.1	7,090	2.18	123	0.63	5.20	
MW18-02	MW18-02-180310	2018 03 10	6.65	-	58.5	6.34	10.63	1,160	-138.6	12.63	-	-	496	< 5.0	7.95	< 100	32.3	38.9	38.9	< 1.0	< 1.0	< 1.0	-	5.1	-	5.1	18.4	< 10	3.03	95.1	0.66	6.91	
	MW18-02-181030	2018 10 30	7.11	-	51.1	6.22	9.8	105.3	208	-	124	-	413	< 10	6.16	< 100	23.1	45.1	45.1	< 1.0	< 1.0	< 1.0	< 0.10	11.3	-	46.5	16.1	52	2.66	23.1	0.88	7.24	
	MW18-02-190122	2019 01 22	6.78	-	44.3	6.36	9.4	149.3	165.7	-	-	-	1,030	< 10	4.43	< 100	36.3	24.2	24.2	< 1.0	< 1.0	< 1.0	< 0.10	7.7	-	9.5	13.0	< 10	2.84	9.75	0.41	6.90	
	MW18-02-190923	2019 09 23	6.92	-	46.4	6.55	10.3	111	-	-	-	-	699	< 10	4.79	< 100	29.9	32.3	32.3	< 1.0	< 1.0	< 1.0	< 0.10	9.4	-	67.4	14.0	75	2.75	6.96	0.55	6.49	
	MW18-02-191205	2019 12 05	6.76	-	48.7	6.1	9.9	153	210	-	125	-	1,780	< 5.0	4.97	< 100	39.7	21.4	21.4	< 1.0	< 1.0	< 1.0	-	7.5	-	1,890	13.9	2,120	3.43	163	0.55	7.38	
	MW18-02-200728	2020 07 28	7.00	51.4	-	5.5	10.5	103.6	208	9.15	111	-	793	< 1.0	4.68	42	33.2	29.0	29.0	< 1.0	< 1.0	< 1.0	< 0.050	-	6.1	10.1	15.6	< 10	3.03	2.12	0.600	6.85	
	MW18-02-210111	2021 01 11	6.76	43.8	-	5.29	10.2	106.2	249	-	118	< 5.0	1,110	< 1.0	4.49	37	39.4	15.9	15.9	< 1.0	< 1.0	< 1.0	< 0.050	-	9.7	8.2	13.1	< 10	2.69	5.51	0.510	7.08	
	MW18-02-210811	2021 08 11	-	-	42.4	5.72	10.8	148.9	287	7.69	108	-	622	< 10	4.51	< 100	31.2	49.2	49.2	< 1.0	< 1.0	< 1.0	< 0.10	8.2	-	< 5.0	12.5	< 10	2.70	0.82	0.56	6.76	
	MW-B-210811	Duplicate	-	-	41.8	5.72	-	-	-	-	105	-	617	< 10	4.56	< 100	31.4	51.3	51.3	< 1.0	< 1.0	< 1.0	< 0.10	8.0	-	< 5.0	12.3	< 10	2.70	0.79	0.54	6.68	
	QA/QC RPD%		-	-	1	0	*	*	*	*	*	3	-	1	*	1	*	1	4	4	*	*	*	*	*	-	*	2	*	0	*	4	1
	MW18-02-220210	2022 02 10	-	-	42.2	5.88	9.9	133.1	258	9.62	105	154	377	< 5.0	3.67	< 100	41.9	15.7	15.7	< 1.0	< 1.0	< 1.0	< 0.10	6.8	-	17.4	12.8	< 10	2.50	2.22	0.48	6.74	
	MW18-02-221024	2022 10 24	-	-	44.8	6.25	9.9	131.6	-	-	77	< 50	467	< 10	4.82	< 100	27.2	27.3	27.3	< 1.0	< 1.0	< 1.0	< 0.10	13.2	-	5.4	13.9	< 10	2.45	0.83	0.56	6.30	
MW18-02-230221	2023 02 21	-	-	41.4	6.09	10.1	0.141	203.1	9.98	101	58	616	< 10	5.07	< 100	33.4	14.1	14.1	< 1.0	< 1.0	< 1.0	< 0.10	8.3	-	10.4	12.5	< 10	2.48	4.46	0.51	7.32		
MW18-02-230801	2023 08 01	-	42.3	-	6.22	11.8	124.4	70.3	9.13	66	60	604	< 10	4.71	< 100	25.3	22.7	22.7	< 1.0	< 1.0	< 1.0	< 0.10	11.8	-	5.2	13.1	< 2.0	2.31	0.570	0.581	6.57		
MW18-02-231113	2023 11 13	-	51.1	-	5.85	10.4	145.6	116.4	9.91	92	< 50	961	< 10	5.61	< 100	45.3	12.1	12.1	< 1.0	< 1.0	< 1.0	< 0.1	11.2	-	13.9	15.7	4.8	2.87	7.71	0.563	8.11		
MW18-03	MW18-03-180306	2018 03 06	6.87	-	35.9	7.27	8.9	103.4	118.4	5.17	-	-	294	< 5.0	5.16	< 100	5.3	38.6	38.6	< 1.0	< 1.0	< 1.0	-	3.3	-	< 5.0	10.1	21	2.61	58.4	0.64	5.55	
	MW18-03-181030	2018 10 30	7.22	-	39.9	6.77	10.1	79.8	187	-	82	-	295	< 10	5.39	< 100	3.8	45.5	45.5	< 1.0	< 1.0	< 1.0	< 0.10	7.7	-	7.3	11.2	20	2.91	0.69	0.62	6.36	
	MW18-03-190122	2019 01 22	7.05	-	31.2	6.74	9.1	98.6	155.8	-	-	-	265	< 10	5.17	< 100	5.2	38.7	38.7	< 1.0	< 1.0	< 1.0	< 0.10	5.6	-	5.0	8.29	< 10	2.53	0.36	0.48	5.05	
	MW18-03-190923	2019 09 23	7.20	-	36.2	6.61	10.1</																										

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																							
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Strontium µg/L	Thallium µg/L	Tin µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^f µg/L
BC Standard			90	50	10,000	1.5	12,000	0.5-4 ^d	10 ^e	40	20-90 ^d	40-160 ^d	n/a	0.25	10,000	250-1,500 ^d	20	0.5-15 ^d	n/a	3	n/a	1,000	n/a	85	n/a	75-2,400
CSR Aquatic Life (AW) ^a			900	500	100,000	15	120,000	5-40 ^d	100 ^e	400	200-900 ^d	400-1,600 ^d	n/a	2.5	100,000	2,500-15,000 ^d	200	5-150 ^d	n/a	30	n/a	10,000	n/a	850	n/a	750-24,000
MW18-01	MW18-01-180310	2018 03 10	< 0.20	< 0.50	6.2	< 0.10	20.9	0.012	0.68	1.01	1.38	< 0.20	0.58	< 0.010	0.36	0.67	< 0.50	0.101	34.0	< 0.020	< 0.20	< 5.0	< 1.0	0.027	< 1.0	5.6
	MW18-A-180310	Duplicate	< 0.20	< 0.50	6.2	< 0.10	18.4	< 0.010	0.67	1.00	0.54	< 0.20	0.60	< 0.010	0.34	0.66	< 0.50	< 0.050	33.9	< 0.020	< 0.20	< 5.0	< 1.0	0.027	< 1.0	4.9
	QA/QC RPD%		*	*	*	*	*	*	*	1	*	*	3	*	*	*	*	*	< 1	*	*	*	*	*	*	*
	MW18-01-180725	2018 07 25	< 0.20	< 0.50	5.5	< 0.10	16.7	< 0.010	0.81	0.13	< 0.40	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	30.9	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	MW18-01-181030	2018 10 31	< 0.20	< 0.50	6.6	< 0.10	11.9	< 0.010	0.85	0.22	0.47	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	31.1	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	1.5	< 4.0
	MW18-01-190122	2019 01 22	< 0.20	< 0.50	5.3	< 0.10	13.5	< 0.010	0.67	0.13	< 0.40	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	27.5	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	1.0	< 4.0
	MW18-01-190924	2019 09 24	< 0.20	< 0.50	5.3	< 0.10	22.7	< 0.010	0.74	0.41	1.45	< 0.20	0.15	< 0.010	< 0.10	0.60	< 0.50	< 0.050	30.2	< 0.020	0.32	< 5.0	< 1.0	< 0.020	< 1.0	7.6
MW18-02	MW18-01-221024	2022 10 24	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.060	1.73	0.15	1.05	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	28.4	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 5.0	5.8
	MW18-02-180310	2018 03 10	< 0.20	< 0.50	14.6	< 0.10	29.7	0.148	< 0.50	28.4	75.8	< 0.20	1.33	< 0.010	0.25	29.1	< 0.50	0.163	46.9	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	47.6
	MW18-02-181030	2018 10 30	< 0.20	< 0.50	15.3	< 0.10	24.4	0.147	< 0.50	23.1	116	< 0.20	0.34	< 0.010	0.25	26.5	< 0.50	0.052	36.2	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	50.0
	MW18-02-190122	2019 01 22	< 0.20	< 0.50	15.3	< 0.10	21.0	0.263	< 0.50	31.2	644	< 0.20	0.33	< 0.010	0.12	36.8	< 0.50	< 0.050	38.6	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	71.8
	MW18-02-190923	2019 09 23	< 0.20	< 0.50	15.1	< 0.10	22.0	0.198	< 0.50	27.3	295	< 0.20	0.32	< 0.010	< 0.10	31.6	0.67	< 0.050	39.2	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	64.7
	MW18-02-191205	2019 12 05	< 0.20	0.64	33.3	< 0.10	31.9	0.381	3.22	48.1	1,860	0.66	1.28	< 0.010	0.12	53.3	0.65	0.100	44.6	< 0.020	< 0.20	41.0	< 1.0	0.089	5.3	121
	MW18-02-200728	2020 07 28	< 0.10	< 0.10	14.1	< 0.020	20	0.215	0.11	30.9	369	< 0.050	< 1.0	< 0.0050	0.113	36.1	0.596	0.012	40.5	< 0.010	< 0.10	< 10.0	0.34	< 0.010	< 0.50	65.8
	MW18-02-210111	2021 01 11	< 0.10	< 0.10	12.6	< 0.020	20	0.276	0.14	29.5	856	< 0.050	< 1.0	< 0.0050	0.082	36.0	0.501	0.020	47.5	< 0.010	< 0.10	< 10.0	< 0.10	0.011	< 0.50	76.5
	MW18-02-210811	2021 08 11	< 0.20	< 0.50	13.2	< 0.10	< 50.0	0.215	< 0.50	29.0	384	< 0.20	0.22	< 0.010	< 0.10	35.2	0.63	< 0.050	36.4	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	65.4
	MW-B-210811	Duplicate	< 0.20	< 0.50	13.5	< 0.10	< 50.0	0.223	< 0.50	29.1	384	< 0.20	0.19	< 0.010	< 0.10	35.0	0.69	< 0.050	35.6	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	65.0
	QA/QC RPD%		*	*	*	*	*	4	*	< 1	0	*	*	*	*	1	*	*	2	*	*	*	*	*	*	1
	MW18-02-220210	2022 02 10	< 0.20	< 0.50	17.9	< 0.10	< 50.0	0.387	< 0.50	44.3	1,130	< 0.20	0.45	< 0.010	< 0.10	53.0	0.71	< 0.050	36.5	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	106
	MW18-02-221024	2022 10 24	< 0.20	< 0.50	12.0	< 0.10	< 50.0	0.202	0.71	24.5	340	< 0.20	0.27	< 0.010	< 0.10	29.2	0.63	< 0.050	34.4	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 5.0	57.1
	MW18-02-230221	2023 02 21	< 0.20	< 0.50	14.6	< 0.10	< 50.0	0.307	< 0.50	35.4	1,090	< 0.20	0.35	< 0.010	< 0.10	43.4	0.54	< 0.050	36.8	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 5.0	88.5
	MW18-02-230801	2023 08 01	< 0.050	< 0.050	11.8	< 0.010	21.5	0.191	< 0.50	27.2	373	< 0.050	0.234	< 0.0050	0.060	31.2	0.59	0.011	34.9	< 0.0040	< 0.050	< 0.20	0.20	0.0028	< 1.00	61.4
MW18-02-231113	2023 11 13	< 0.050	< 0.050	21.9	0.018	23.0	0.471	< 0.50	50.2	1,620	< 0.050	0.437	< 0.0050	0.058	61.6	0.80	0.017	45.9	0.0047	< 0.050	< 0.20	< 0.20	0.0018	< 1.00	122	
MW18-03	MW18-03-180306	2018 03 06	< 0.20	< 0.50	< 5.0	< 0.10	20.0	< 0.010	< 0.50	0.63	4.63	< 0.20	0.41	< 0.010	0.53	0.70	< 0.50	< 0.050	23.9	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	6.9
	MW18-03-181030	2018 10 30	< 0.20	< 0.50	< 5.0	< 0.10	20.5	0.010	< 0.50	0.17	3.41	< 0.20	< 0.10	< 0.010	0.32	1.91	< 0.50	< 0.050	24.2	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	MW18-03-190122	2019 01 22	< 0.20	< 0.50	< 5.0	< 0.10	19.1	< 0.010	< 0.50	0.18	3.98	< 0.20	0.11	< 0.010	0.32	0.40	< 0.50	< 0.050	21.5	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	MW18-03-190923	2019 09 23	< 0.20	< 0.50	< 5.0	< 0.10	24.5	0.014	< 0.50	0.28	4.69	< 0.20	0.12	< 0.010	0.37	0.64	< 0.50	< 0.050	26.6	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	MW-A-190923	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	24.3	< 0.010	< 0.50	0.27	4.59	< 0.20	0.12	< 0.010												

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters			Field Parameters					Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Temperature	Conductivity	ORP	Dissolved Oxygen	Total Dissolved Solids (Laboratory)	Ammonia, Total (as N)	Nitrate (as N)	Nitrite (as N)	Chloride	Fluoride	Sulfate	Total Alkalinity	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (as CaCO3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Standard																																	
CSR Aquatic Life (AW) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	18,400-18,500 ^b	400,000	400-2,000 ^c	1,500	2,000-3,000 ^d	2,180-4,290 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^b	4,000,000	4,000-20,000 ^c	15,000	20,000-30,000 ^d	21,800-42,900 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
MW18-04	MW18-04-180306	2018 03 06	7.25	-	64.8	7.77	9.6	226.3	125.5	7.17	-	-	456	< 5.0	5.62	120	30.7	72.1	72.1	< 1.0	< 1.0	< 1.0	-	5.0	-	< 5.0	19.8	< 10	3.71	38.0	0.87	15.3	
	MW18-04-181029	2018 10 29	7.85	-	107	6.76	10	187.2	181.7	-	177	-	1,280	< 10	3.00	< 100	27.0	102	102	< 1.0	< 1.0	< 1.0	< 0.10	2.3	-	< 5.0	34.7	< 10	4.91	1.43	0.99	7.03	
	MW-A-181029	Duplicate	7.89	-	103	6.76	-	-	-	-	169	-	1,230	< 10	3.07	< 100	27.3	105	105	< 1.0	< 1.0	< 1.0	< 0.10	2.2	-	< 5.0	33.2	< 10	4.83	1.38	0.98	6.94	
	QA/QC RPD%		1	-	4	0	*	*	*	-	5	-	4	*	2	*	1	3	3	*	*	*	*	*	-	*	4	*	2	4	1	1	
	MW18-04-190122	2019 01 22	7.44	-	74.0	7	9.5	196.9	178.2	-	-	-	378	< 10	3.82	< 100	24.7	75.5	75.5	< 1.0	< 1.0	< 1.0	< 0.10	4.2	-	6.1	23.7	< 10	3.60	9.38	0.43	3.97	
	MW18-04-190923	2019 09 23	7.82	-	102	7.53	10.1	172	-	-	-	-	1,260	< 10	3.18	< 100	25.3	85.2	85.2	< 1.0	< 1.0	< 1.0	< 0.10	2.3	-	21.8	33.4	< 10	4.56	1.07	0.66	4.50	
	MW18-04-191205	2019 12 05	7.65	-	91.5	6.96	9.8	240	202	-	176	-	847	< 5.0	4.38	< 100	25.9	89.8	89.8	< 1.0	< 1.0	< 1.0	-	4.2	-	10.7	29.5	< 10	4.31	5.13	0.49	4.58	
	MW18-04-200728	2020 07 28	7.84	92.0	-	7.28	10.7	153	167	7.6	154	-	557	< 1.0	3.17	28	23.6	73.4	73.4	< 1.0	< 1.0	< 1.0	< 0.050	-	2.6	4.3	30.2	< 10	4.03	0.36	0.700	4.86	
	MW18-04-210111	2021 01 11	7.15	67.0	-	6.68	9.7	116.8	186	-	131	< 5.0	386	< 1.0	4.38	< 20	23.7	48.1	48.1	< 1.0	< 1.0	< 1.0	< 0.050	-	7.1	5.2	20.5	< 10	3.84	10.7	0.479	4.15	
	MW18-04-210810	2021 08 10	-	-	62.8	7.17	13.6	182.2	204.6	6.20	135	80	352	< 10	3.24	< 100	22.0	123	123	< 1.0	< 1.0	< 1.0	< 0.10	4.6	-	< 5.0	19.1	< 10	3.67	< 0.20	0.54	5.56	
	MW18-04-220209	2022 02 09	-	-	46.9	6.65	9.5	137.8	288	8.18	119	107	258	< 10	2.87	< 100	22.3	64.2	64.2	< 1.0	< 1.0	< 1.0	< 0.10	2.8	-	14.4	13.5	31	3.20	5.35	0.34	3.40	
	MW18-04-221024	2022 10 24	-	-	73.3	7.26	10	190.3	-	-	93	< 50	313	< 10	3.37	< 100	21.4	79.9	79.9	< 1.0	< 1.0	< 1.0	< 0.10	3.4	-	< 5.0	22.6	< 10	4.07	0.35	0.54	5.19	
	MW-A-221024	Duplicate	-	-	72.8	7.26	10	190.3	-	-	89	53	321	< 10	3.37	< 100	21.7	70.8	70.8	< 1.0	< 1.0	< 1.0	< 0.10	2.7	-	< 5.0	22.2	< 10	4.21	0.33	0.53	5.20	
	QA/QC RPD%		-	-	1	*	*	*	-	-	4	*	3	*	0	*	1	12	12	*	*	*	*	*	-	*	2	*	3	*	2	0	
	MW18-04-230221	2023 02 21	-	-	44.6	6.47	9.32	0.116	318.8	10.16	95	59	707	< 10	4.35	< 100	23.8	28.5	28.5	< 1.0	< 1.0	< 1.0	< 0.10	5.2	-	19.5	12.5	16	3.23	9.12	0.43	4.29	
MW18-04-230731	2023 07 31	-	67.6	-	7.13	12.9	148.4	95.1	7.43	113	60	518	< 10	3.65	< 100	21.5	61.3	61.3	< 1.0	< 1.0	< 1.0	< 0.10	3.2	-	8.3	19.3	5.5	4.7	0.672	0.467	4.82		
MW-B-230731	Duplicate	-	67.4	-	7.13	12.9	148.4	95.1	7.43	92	< 50	518	< 10	3.64	< 100	21.6	60.4	60.4	< 1.0	< 1.0	< 1.0	< 0.10	2.2	-	5.5	19.2	4.0	4.71	0.772	0.464	4.88		
QA/QC RPD%		-	0	-	*	*	*	*	*	20	*	0	*	0	*	0	1	1	*	*	*	*	*	-	41	1	*	0	14	1	1		
MW18-04-231113	2023 11 13	-	61.8	-	6.17	9.7	136.9	189.4	10.26	135	< 50	1,200	< 10	4.61	< 100	25.6	43.3	43.3	< 1.0	< 1.0	< 1.0	< 0.1	6.7	-	11.2	17.3	13.9	4.48	9.74	0.519	5.17		
MW18-05	MW18-05-180310	2018 03 10	7.23	-	153	6.30	9.89	204	-134.6	9.35	-	-	25.4	7.7	5.83	100	38.0	142	142	< 1.0	< 1.0	< 1.0	-	5.9	-	16.3	45.2	220	9.61	828	0.96	5.53	
	MW18-05-181029	2018 10 29	7.54	-	179	6.71	10.5	279.2	149.8	-	231	-	1,010	20	4.67	< 100	34.4	163	163	< 1.0	< 1.0	< 1.0	< 0.10	7.9	-	< 5.0	61.4	28	6.30	768	1.11	4.86	
	MW18-05-190121	2019 01 21	6.99	-	117	6.63	10	296	167	-	-	-	208	< 10	6.00	< 100	41.3	97.2	97.2	< 1.0	< 1.0	< 1.0	< 0.10	15.1	-	< 5.0	35.9	< 10	6.50	767	0.42	4.47	
	MW18-05-190924	2019 09 24	7.42	-	263	6.72	10.3	422	-	-	357	-	1,040	< 10	5.38	< 100	43.4	247	247	< 1.0	< 1.0	< 1.0	< 0.10	16.7	-	< 5.0	91.0	< 10	8.55	982	1.29	4.78	
	MW18-05-191204	2019 12 04	7.09	-	121	6.49	10.4	298.3	165	-	201	-	189	< 10	6.66	< 100	37.2	113	113	< 1.0	< 1.0	< 1.0	< 0.10	23.8	-	< 5.0	35.5	< 10	7.79	675	0.50	4.47	
	MW18-05-210111	2021 01 11	6.82	103	-	6.01	10.6	185.4	189	-	188	< 5.0	272	1.0	6.58	24	50.3	65.6	65.6	< 1.0	< 1.0	< 1.0	< 0.050	-	16.9	< 3.0	28.8	14	7.63	729	0.472	5.25	
	MW18-05-220209	2022 02 09	-	-	76.3	6.10	10	518.2	443.9	4.31	139	< 50	152	< 5.0	5.06	< 100	40.1	47.5	47.5	< 1.0	< 1.0	< 1.0	< 0.10	9.0	-	13.4	20.3	< 10	6.23	527	0.37	4.65	
	MW18-05-230221	2023 02 21	-	-	167	6.33	9.72	281	177	1.17	235	< 50	273	< 10	7.57	< 100	58.2	108	108	< 1.0	< 1.0	< 1.0	< 0.10	10.0	-	< 5.0	51.0	17	9.52	451	0.57	6.28	
	MW18-05-231114	2023 11 14	-	269	-	5.97	10.0	437.7	214.6	1.80	358	< 50	534																				

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																							
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Strontium µg/L	Thallium µg/L	Tin µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^f µg/L
BC Standard			90	50	10,000	1.5	12,000	0.5-4 ^d	10 ^e	40	20-90 ^d	40-160 ^d	n/a	0.25	10,000	250-1,500 ^d	20	0.5-15 ^d	n/a	3	n/a	1,000	n/a	85	n/a	75-2,400
CSR Aquatic Life (AW) ^a			900	500	100,000	15	120,000	5-40 ^d	100 ^e	400	200-900 ^d	400-1,600 ^d	n/a	2.5	100,000	2,500-15,000 ^d	200	5-150 ^d	n/a	30	n/a	10,000	n/a	850	n/a	750-24,000
MW18-04	MW18-04-180306	2018 03 06	< 0.20	0.54	6.4	< 0.10	24.1	0.011	< 0.50	0.45	2.11	< 0.20	2.00	< 0.010	3.70	1.12	0.58	< 0.050	53.0	< 0.020	< 0.20	< 5.0	< 1.0	0.376	< 1.0	< 4.0
	MW18-04-181029	2018 10 29	< 0.20	< 0.50	22.5	< 0.10	13.6	< 0.010	< 0.50	0.21	4.51	< 0.20	1.00	< 0.010	0.67	1.67	< 0.50	< 0.050	68.4	< 0.020	< 0.20	< 5.0	1.1	0.338	< 1.0	< 4.0
	MW-A-181029	Duplicate	< 0.20	< 0.50	21.8	< 0.10	13.0	< 0.010	< 0.50	0.19	4.41	< 0.20	0.96	< 0.010	0.66	1.51	< 0.50	< 0.050	65.3	< 0.020	< 0.20	< 5.0	1.1	0.322	< 1.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	2	*	4	*	2	*	*	*	5	*	*	*	*	5	*	*
	MW18-04-190122	2019 01 22	< 0.20	< 0.50	15.7	< 0.10	8.4	0.109	< 0.50	6.88	110	< 0.20	0.45	< 0.010	< 0.10	13.3	< 0.50	< 0.050	50.1	< 0.020	< 0.20	< 5.0	< 1.0	0.071	< 1.0	32.9
	MW18-04-190923	2019 09 23	< 0.20	< 0.50	23.2	< 0.10	10.1	0.021	< 0.50	0.17	15.8	< 0.20	0.64	0.011	0.38	3.43	< 0.50	< 0.050	63.7	< 0.020	< 0.20	< 5.0	4.5	0.194	< 1.0	6.2
	MW18-04-191205	2019 12 05	< 0.20	< 0.50	17.4	< 0.10	16.3	0.083	< 0.50	3.16	74.7	< 0.20	0.43	< 0.010	0.13	9.59	< 0.50	< 0.050	58.4	< 0.020	0.21	< 5.0	1.8	0.139	< 1.0	25.2
	MW18-04-200728	2020 07 28	< 0.10	0.49	18.3	< 0.020	11	0.0240	< 0.10	0.17	20.7	< 0.050	< 1.0	< 0.0050	0.740	2.38	0.179	< 0.010	47.8	< 0.010	< 0.10	< 10.0	10.9	0.200	< 0.50	3.9
	MW18-04-210111	2021 01 11	< 0.10	0.14	14.3	< 0.020	< 10	0.194	< 0.10	13.2	250	< 0.050	< 1.0	< 0.0050	< 0.050	19.3	0.175	< 0.010	52.1	< 0.010	< 0.10	< 10.0	1.43	0.030	< 0.50	59.5
	MW18-04-210810	2021 08 10	< 0.20	< 0.50	13.3	< 0.10	< 50.0	0.039	1.08	0.52	29.1	< 0.20	0.59	< 0.010	0.39	3.46	< 0.50	< 0.050	30.4	< 0.020	< 0.20	< 5.0	4.9	0.080	< 1.0	7.1
	MW18-04-220209	2022 02 09	< 0.20	< 0.50	10.4	< 0.10	< 50.0	0.118	< 0.50	7.54	112	< 0.20	0.28	< 0.010	< 0.10	12.9	< 0.50	< 0.050	31.4	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	41.7
	MW18-04-221024	2022 10 24	< 0.20	< 0.50	16.0	< 0.10	< 50.0	0.039	< 0.50	0.63	22.2	< 0.20	0.58	< 0.010	0.24	3.51	< 0.50	< 0.050	33.8	< 0.020	< 0.20	< 5.0	4.5	0.083	< 5.0	9.4
	MW-A-221024	Duplicate	< 0.20	< 0.50	15.6	< 0.10	< 50.0	0.042	< 0.50	0.62	22.1	< 0.20	0.56	< 0.010	0.22	3.50	< 0.50	< 0.050	34.4	< 0.020	< 0.20	< 5.0	4.4	0.080	< 5.0	9.3
	QA/QC RPD%		*	*	*	*	*	*	*	2	0	*	4	*	*	0	*	*	2	*	*	*	*	*	*	*
	MW18-04-230221	2023 02 21	< 0.20	< 0.50	9.9	< 0.10	< 50.0	0.164	< 0.50	13.9	253	< 0.20	0.34	< 0.010	< 0.10	17.9	< 0.50	< 0.050	36.9	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 5.0	55.6
	MW18-04-230731	2023 07 31	< 0.050	0.192	9.10	< 0.010	15.9	0.0511	< 0.50	1.48	56.4	< 0.050	0.392	< 0.0050	0.153	5.13	0.21	< 0.010	36.2	0.0050	< 0.050	< 0.20	3.75	0.0297	< 1.00	10.8
	MW-B-230731	Duplicate	< 0.050	0.204	9.50	< 0.010	15.4	0.0516	< 0.50	1.46	56.3	< 0.050	0.396	< 0.0050	0.175	5.14	0.22	< 0.010	36.3	0.0048	< 0.050	< 0.20	4.35	0.0307	< 1.00	11.1
	QA/QC RPD%		*	*	4	*	3	1	*	1	0	*	1	*	13	0	*	*	0	*	*	*	15	3	*	3
	MW18-05	MW18-04-231113	2023 11 13	< 0.050	0.126	18.7	< 0.010	11.2	0.210	< 0.50	17.3	349	< 0.050	0.414	< 0.0050	0.028	24.6	0.14	< 0.010	49.1	0.0103	< 0.050	< 0.20	0.43	0.0115	< 1.00
MW18-05-180310		2018 03 10	< 0.20	0.52	< 5.0	< 0.10	20.6	0.054	< 0.50	6.05	7.56	< 0.20	2.25	< 0.010	6.33	2.98	< 0.50	0.146	119	< 0.020	0.81	< 5.0	< 1.0	0.200	< 1.0	4.6
MW18-05-181029		2018 10 29	< 0.20	< 0.50	19.3	< 0.10	8.8	0.144	< 0.50	18.1	16.6	< 0.20	1.33	< 0.010	0.36	3.27	< 0.50	< 0.050	105	0.027	< 0.20	< 5.0	< 1.0	0.332	< 1.0	8.1
MW18-05-190121		2019 01 21	< 0.20	< 0.50	13.9	< 0.10	6.1	0.261	< 0.50	21.4	447	< 0.20	0.75	< 0.010	< 0.10	7.02	< 0.50	< 0.050	91.8	< 0.020	< 0.20	< 5.0	< 1.0	0.080	< 1.0	40.3
MW18-05-190924		2019 09 24	< 0.20	< 0.50	34.7	< 0.10	16.6	0.233	< 0.50	17.8	235	< 0.20	2.24	< 0.010	0.38	7.65	< 0.50	< 0.050	142	0.032	0.35	< 5.0	1.9	0.543	< 1.0	28.8
MW18-05-191204		2019 12 04	< 0.20	< 0.50	13.0	< 0.10	9.8	0.318	< 0.50	28.3	856	< 0.20	0.81	< 0.010	0.14	11.3	< 0.50	< 0.050	103	< 0.020	< 0.20	< 5.0	< 1.0	0.091	< 1.0	63.9
MW18-05-210111		2021 01 11	< 0.10	0.23	8.49	< 0.020	< 10	0.286	< 0.10	22.4	571	< 0.050	< 1.0	< 0.0050	0.074	8.78	0.076	< 0.010	106	0.011	< 0.10	< 10.0	0.65	0.037	< 0.50	52.3
MW18-05-220209		2022 02 09	< 0.20	< 0.50	5.3	< 0.10	< 50.0	0.267	< 0.50	20.5	556	< 0.20	0.65	< 0.010	< 0.10	9.52	< 0.50	< 0.050	77.2	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 1.0	51.0
MW18-05-230221		2023 02 21	< 0.20	< 0.50	11.2	< 0.10	< 50.0	0.265	< 0.50	14.5	607	< 0.20	0.88	< 0.010	< 0.10	10.0	< 0.50	< 0.050	135	< 0.020	< 0.20	< 5.0	< 1.0	0.092	< 5.0	53.7
MW18-05-231114	2023 11 14	< 0.050	0.198	21.6	0.013	8.7	0.241	< 0.50	10.6	561	< 0.050	1.23	< 0.0050	0.119	7.93	0.12	< 0.010	185	0.0190	< 0.050	< 0.20	2.03	0.225	< 1.00	48.0	
MW18-06	MW18-06-180310	2018 03 10	0.26	0.58	20.9	< 0.10	56.5	0.019	< 0.50	1.26	2.66	< 0.20														

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters			Field Parameters					Geochemical Indicators																								
			pH	Hardness mg/L	Total Hardness mg/L	pH (field)	Temperature °C	Conductivity µS/cm	ORP mV	Dissolved Oxygen mg/L	Total Dissolved Solids (Laboratory) mg/L	Ammonia, Total (as N) µg/L	Nitrate (as N) µg/L	Nitrite (as N) µg/L	Chloride mg/L	Fluoride µg/L	Sulfate mg/L	Total Alkalinity mg/L	Alkalinity, Bicarbonate (as CaCO3) mg/L	Alkalinity, Carbonate (as CaCO3) mg/L	Alkalinity, Hydroxide (as CaCO3) mg/L	Alkalinity, Phenolphthalein (as CaCO3) mg/L	Bromide mg/L	Total Acidity mg/L	Acidity (as CaCO3) mg/L	Dissolved Aluminum µg/L	Dissolved Calcium mg/L	Dissolved Iron µg/L	Dissolved Magnesium mg/L	Dissolved Manganese µg/L	Dissolved Potassium mg/L	Dissolved Sodium mg/L			
BC Standard																																			
CSR Aquatic Life (AW) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	18,400-18,500 ^b	400,000	400-2,000 ^c	1,500	2,000-3,000 ^d	2,180-4,290 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^b	4,000,000	4,000-20,000 ^c	15,000	20,000-30,000 ^d	21,800-42,900 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
MW18-07	MW18-07-180306	2018 03 06	7.20	-	108	7.15	10.9	307.5	129.6	7.15	-	-	226	< 5.0	8.63	200	60.2	78.2	78.2	< 1.0	< 1.0	< 1.0	-	6.8	-	24.3	33.3	< 10	6.10	69.1	1.26	18.5			
	MW18-07-181029	2018 10 29	7.07	-	132	6.11	11	266.1	211.1	-	242	-	550	< 10	33.3	110	87.6	32.4	32.4	< 1.0	< 1.0	< 1.0	< 0.10	7.0	-	6.1	38.0	< 10	9.02	28.0	0.54	17.0			
	MW18-07-190122	2019 01 22	6.52	-	68.4	6.24	10	222.5	192.4	-	-	-	316	< 10	10.2	< 100	66.0	16.6	16.6	< 1.0	< 1.0	< 1.0	< 0.10	10.0	-	25.5	18.8	10	5.18	51.5	0.35	8.38			
	MW-A-190122	Duplicate	6.54	-	68.7	6.24	-	-	-	-	-	-	306	< 10	10.2	100	66.1	16.6	16.6	< 1.0	< 1.0	< 1.0	< 0.10	8.2	-	32.4	18.9	18	5.23	50.6	0.37	8.49			
	QA/QC RPD%		< 1	-	< 1	0	*	*	*	*	-	-	-	3	*	0	*	< 1	0	0	*	*	*	*	*	-	24	1	*	1	2	*	1		
	MW18-07-190923	2019 09 23	6.80	-	138	6.05	11.4	260	-	-	-	-	556	< 10	29.7	110	96.4	24.7	24.7	< 1.0	< 1.0	< 1.0	< 0.10	9.2	-	7,780	38.7	3,710	9.97	167	0.67	15.1			
	MW18-07-191205	2019 12 05	6.65	-	94.6	5.92	10.2	284	301	-	215	-	634	< 5.0	10.5	180	83.9	21.3	21.3	< 1.0	< 1.0	< 1.0	< 1.0	8.7	-	664	26.2	259	7.03	71.9	0.47	10.4			
	MW-A-191205	Duplicate	6.70	-	96.3	5.92	10.2	284	301	-	204	-	551	< 5.0	10.5	180	80.2	20.2	20.2	< 1.0	< 1.0	< 1.0	-	7.7	-	743	26.6	290	7.24	75.5	0.48	10.7			
	QA/QC RPD%		1	-	2	0	*	*	*	*	-	5	-	14	*	0	*	5	5	5	*	*	*	-	*	-	11	2	11	3	5	*	3		
	MW18-07-200728	2020 07 28	6.70	106	-	5.85	11.8	220	196	8.93	236	-	668	< 1.0	16.0	130	94.5	14.9	14.9	< 1.0	< 1.0	< 1.0	0.071	-	6.8	31.3	29.9	23	7.47	16.4	0.527	11.4			
	MW18-07-210111	2021 01 11	6.77	58.5	-	5.99	10.7	130.9	212	-	137	< 5.0	272	< 1.0	9.85	90	46.7	23.0	23.0	< 1.0	< 1.0	< 1.0	< 0.050	-	8.6	15.9	16.6	10	4.14	15.0	0.424	8.73			
	MW18-A-210111	Duplicate	6.59	56.9	-	5.99	-	-	-	-	147	< 5.0	272	< 1.0	9.88	89	46.5	22.9	22.9	< 1.0	< 1.0	< 1.0	< 0.050	-	9.9	14.9	15.8	10	4.23	14.9	0.434	8.68			
	QA/QC RPD%		3	3	-	0	*	*	*	*	-	7	*	0	*	< 1	*	< 1	< 1	< 1	*	*	*	*	-	*	-	*	*	5	*	2	1	2	1
	MW18-07-210810	2021 08 10	-	-	95.4	6.03	15.2	292.3	237.4	7.20	202	159	440	< 10	20.9	< 100	78.3	34.5	34.5	< 1.0	< 1.0	< 1.0	< 0.10	10.8	-	98.2	26.5	33	7.08	8.12	0.54	11.9			
	MW18-07-220209	2022 02 09	-	-	55.0	6.01	10.9	195.5	330	8.65	132	95	151	7.7	9.89	140	46.3	25.1	25.1	< 1.0	< 1.0	< 1.0	< 0.10	4.3	-	21.2	15.6	21	3.90	5.66	0.36	7.87			
MW18-07-221024	2022 10 24	-	-	105	5.92	11.3	300	-	-	154	< 50	507	< 10	29.8	< 100	77.1	10.1	10.1	< 1.0	< 1.0	< 1.0	< 0.10	12.9	-	41.4	30.0	< 10	7.20	7.87	0.51	12.6				
MW18-07-230222	2023 02 22	-	-	81.7	6.19	9.34	211	388	-	148	55	619	< 10	11.0	110	65.1	23.8	23.8	< 1.0	< 1.0	< 1.0	< 0.10	5.8	-	34.8	23.2	56	5.78	10.4	0.47	11.2				
MW18-07-230731	2023 07 31	-	109	-	6.02	14.6	264.6	274.8	7.1	172	< 50	653	< 10	20.7	150	79.9	15.3	15.3	< 1.0	< 1.0	< 1.0	< 0.10	12.7	-	98.5	31.6	35.3	7.37	12.2	0.541	12				
MW18-07-231113	2023 11 13	-	149	-	5.98	12.1	348.1	225.1	10.6	242	< 50	1,570	< 10	14.4	< 100	138	18.0	18.0	< 1.0	< 1.0	< 1.0	< 0.1	8.3	-	25.5	40.8	5.9	11.5	15.1	0.618	16.5				
MW18-09	MW18-09-180306	2018 03 06	7.33	-	59.0	-	-	-	-	-	-	-	154	< 5.0	10.0	160	16.9	63.1	63.1	< 1.0	< 1.0	< 1.0	-	4.3	-	12.6	18.3	< 10	3.23	10.8	0.98	11.3			
	MW18-09-180725	2018 07 25	6.92	-	112	7.75	14.91	246	146.8	9.13	203	-	< 10	< 10	14.2	260	28.1	117	117	< 1.0	< 1.0	< 1.0	< 1.0	-	6.0	-	22.9	35.8	21	5.54	136	1.59	11.0		
	MW18-09-181029	2018 10 29	7.68	-	133	7.17	10.9	236	156.9	-	209	-	940	< 10	8.04	< 100	80.2	62.4	62.4	< 1.0	< 1.0	< 1.0	< 0.10	2.6	-	93.6	40.7	83	7.64	17.6	1.64	13.0			
	MW18-09-190121	2019 01 21	7.38	-	49.0	7.11	10.9	155.6	138.3	-	-	-	44	< 10	14.4	< 100	8.6	47.7	47.7	< 1.0	< 1.0	< 1.0	< 0.10	2.8	-	5.7	14.6	< 10	3.02	1.70	0.58	7.47			
	MW18-09-190924	2019 09 24	7.60	-	111	7.22	14.6	233	-	-	205	-	735	< 10	7.86	< 100	80.2	48.8	48.8	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	10.3	35.0	< 10	5.68	7.96	1.14	10.4			
	MW18-09-191205	2019 12 04/05	7.57	-	62.1	7.09	10.5	175.7	203.4	-	125	-	139	< 5.0	14.0	< 100	15.7	59.6	59.6	< 1.0	< 1.0	< 1.0	-	3.0	-	10.3	18.0	13	4.12	6.35	0.66	8.97			
	MW18-09-200729	2020 07 29	7.82	104	-	7.51	11.5	195	134	9.28	170	-	387	< 1.0	10.4	60	45.2	54.5	54.5	< 1.0	< 1.0	< 1.0	< 0.050	-	< 2.0	16.1	32.2	17	5.64	2.35	1.14	10.7			
	MW18-09-210111	2021 01 11	7.48	50.5	-	6.82	9.9	109.9	182	-	124	< 5.0	41.9	< 1.0	15.0	31	10.2	45.1	45.1	< 1.0	< 1.0	< 1.0	< 0.050	-	3.1	91.5	14.8	85	3.29	6.25	0.645	8.34			
	MW18-09-210811	2021 08 11	-	-	101	7.06	13	307.9	266.4	9.08	186	-	409	< 10	11.3	< 100	44.8	107	107	< 1.0	< 1.0	< 1.0	< 0.10	3.0	-	14.6	29.8	11	6.42</						

Associated ALS file(s): VA20B1567, VA20B1740, VA20B

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																							
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Strontium µg/L	Thallium µg/L	Tin µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^f µg/L
BC Standard			90	50	10,000	1.5	12,000	0.5-4 ^d	10 ^e	40	20-90 ^d	40-160 ^d	n/a	0.25	10,000	250-1,500 ^d	20	0.5-15 ^d	n/a	3	n/a	1,000	n/a	85	n/a	75-2,400
CSR Aquatic Life (AW) ^a			900	500	100,000	15	120,000	5-40 ^d	100 ^e	400	200-900 ^d	400-1,600 ^d	n/a	2.5	100,000	2,500-15,000 ^d	200	5-150 ^d	n/a	30	n/a	10,000	n/a	850	n/a	750-24,000
MW18-07	MW18-07-180306	2018 03 06	0.30	0.67	< 5.0	< 0.10	51.6	0.011	< 0.50	1.52	4.13	< 0.20	8.51	< 0.010	12.5	2.93	< 0.50	< 0.050	63.9	< 0.020	< 0.20	< 5.0	4.8	0.469	1.0	< 4.0
	MW18-07-181029	2018 10 29	< 0.20	< 0.50	< 5.0	< 0.10	112	0.700	< 0.50	116	783	< 0.20	0.42	< 0.010	0.20	171	< 0.50	< 0.050	105	< 0.020	< 0.20	< 5.0	3.9	< 0.020	< 1.0	222
	MW18-07-190122	2019 01 22	< 0.20	< 0.50	< 5.0	< 0.10	36.9	0.501	< 0.50	74.7	1,050	< 0.20	0.47	< 0.010	0.17	112	< 0.50	< 0.050	62.2	< 0.020	< 0.20	< 5.0	1.4	0.051	< 1.0	146
	MW-A-190122	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	36.4	0.484	< 0.50	73.5	1,040	< 0.20	0.46	< 0.010	0.18	110	< 0.50	< 0.050	61.6	< 0.020	< 0.20	< 5.0	1.3	0.050	< 1.0	166
	QA/QC RPD%		*	*	*	*	1	3	*	2	1	*	*	*	*	2	*	*	1	*	*	*	*	*	*	13
	MW18-07-190923	2019 09 23	< 0.20	0.60	20.6	0.14	106	0.934	7.53	141	2,340	0.62	1.14	< 0.010	0.20	222	< 0.50	< 0.050	136	< 0.020	< 0.20	138	5.0	0.075	10.1	319
	MW18-07-191205	2019 12 05	< 0.20	< 0.50	6.0	< 0.10	52.2	0.764	0.76	110	1,610	< 0.20	0.55	< 0.010	0.11	167	< 0.50	< 0.050	94.6	< 0.020	0.35	11.1	6.7	0.039	< 1.0	256
	SU-GW-A-191205	Duplicate	< 0.20	< 0.50	6.2	< 0.10	52.2	0.766	0.78	111	1,630	< 0.20	0.57	< 0.010	0.14	167	< 0.50	< 0.050	96.0	< 0.020	0.35	11.8	6.7	0.038	< 1.0	268
	QA/QC RPD%		*	*	*	*	0	< 1	*	1	1	*	4	*	*	0	*	*	1	*	*	*	0	*	*	5
	MW18-07-200728	2020 07 28	< 0.10	< 0.10	6.06	0.047	64	0.727	0.12	110	1,380	< 0.050	< 1.0	< 0.0050	0.073	171	0.216	< 0.010	93.0	< 0.010	< 0.10	< 10.0	2.94	0.017	< 0.50	221
	MW18-07-210111	2021 01 11	< 0.10	< 0.10	3.97	0.021	35	0.338	0.18	43.9	608	< 0.050	< 1.0	< 0.0050	< 0.050	69.8	0.089	< 0.010	54.0	< 0.010	< 0.10	< 10.0	1.78	0.012	< 0.50	100
	MW18-A-210111	Duplicate	< 0.10	< 0.10	4.11	0.020	33	0.344	0.16	44.5	616	< 0.050	< 1.0	< 0.0050	< 0.050	71.7	0.152	< 0.010	55.5	< 0.010	< 0.10	< 10.0	1.85	0.012	< 0.50	104
	QA/QC RPD%		*	*	3	*	*	2	*	1	1	*	*	*	*	3	*	*	3	*	*	*	4	*	*	4
	MW18-07-210810	2021 08 10	< 0.20	< 0.50	5.3	< 0.10	80.6	0.690	1.08	89.2	1,280	< 0.20	0.54	< 0.010	< 0.10	150	< 0.50	< 0.050	79.8	< 0.020	0.20	< 5.0	3.5	0.021	< 1.0	199
	MW18-07-220209	2022 02 09	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.344	< 0.50	46.3	590	< 0.20	0.29	< 0.010	< 0.10	78.9	< 0.50	< 0.050	43.9	< 0.020	< 0.20	< 5.0	1.8	< 0.020	< 1.0	102
	MW18-07-221024	2022 10 24	< 0.20	< 0.50	< 5.0	< 0.10	104	0.818	< 0.50	101	1,540	< 0.20	0.39	< 0.010	< 0.10	166	< 0.50	< 0.050	89.9	< 0.020	< 0.20	< 5.0	1.9	< 0.020	< 5.0	219
	MW18-07-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.451	< 0.50	56.4	791	< 0.20	0.26	< 0.010	< 0.10	95.8	< 0.50	< 0.050	64.7	< 0.020	< 0.20	< 5.0	2.4	< 0.020	< 5.0	125
	MW18-07-230731	2023 07 31	< 0.050	0.076	7.38	0.070	77.8	0.755	< 0.50	91.0	1,620	< 0.050	0.344	< 0.0050	0.037	153	0.20	< 0.010	87.8	< 0.0040	< 0.050	0.29	2.23	0.0120	< 1.00	207
	MW18-07-231113	2023 11 13	< 0.050	< 0.050	2.83	0.045	56.6	0.814	< 0.50	97.8	1,420	< 0.050	0.254	< 0.0050	0.048	172	0.40	0.015	126	0.0045	< 0.050	< 0.20	2.23	0.0061	< 1.00	214
MW18-09	MW18-09-180306	2018 03 06	< 0.20	< 0.50	< 5.0	< 0.10	47.3	< 0.010	< 0.50	0.12	1.57	< 0.20	2.79	< 0.010	2.78	< 0.40	< 0.50	< 0.050	21.9	< 0.020	< 0.20	< 5.0	4.2	0.092	1.4	< 4.0
	MW18-09-180725	2018 07 25	0.31	3.57	14.9	< 0.10	46.4	< 0.010	< 0.50	0.48	0.92	< 0.20	4.40	< 0.010	19.6	1.36	< 0.50	< 0.050	53.0	< 0.020	0.85	< 5.0	4.5	1.20	2.1	< 4.0
	MW18-09-181029	2018 10 29	< 0.20	< 0.50	13.5	< 0.10	32.1	< 0.010	< 0.50	0.42	5.09	< 0.20	1.08	< 0.010	2.37	0.59	3.14	< 0.050	42.0	< 0.020	< 0.20	< 5.0	6.1	0.126	1.2	< 4.0
	MW18-09-190121	2019 01 21	< 0.20	< 0.50	< 5.0	< 0.10	22.6	< 0.010	< 0.50	0.15	3.05	< 0.20	0.23	< 0.010	0.29	< 0.40	< 0.50	< 0.050	17.9	< 0.020	< 0.20	< 5.0	2.9	< 0.020	< 1.0	< 4.0
	MW18-09-190924	2019 09 24	< 0.20	0.61	11.2	< 0.10	35.4	< 0.010	< 0.50	< 0.10	2.59	< 0.20	0.63	< 0.010	2.26	< 0.40	3.04	< 0.050	35.4	< 0.020	2.17	< 5.0	22.9	0.077	1.2	< 4.0
	MW18-09-191205	2019 12 05	< 0.20	< 0.50	6.3	< 0.10	36.0	0.011	< 0.50	0.17	4.51	< 0.20	0.27	< 0.010	0.29	0.43	< 0.50	< 0.050	23.3	< 0.020	< 0.20	< 5.0	3.3	< 0.020	1.0	< 4.0
	MW18-09-200729	2020 07 29	0.19	0.72	14.7	< 0.020	34	0.0146	0.17	< 0.10	3.79	0.549	< 1.0	< 0.0050	2.23	0.52	1.50	< 0.010	33.0	< 0.010	0.75	< 10.0	13.8	0.259	1.28	2.7
	MW18-09-210111	2021 01 11	< 0.10	0.17	4.65	< 0.020	25	0.0156	0.25	0.20	7.06	< 0.050	< 1.0	< 0.0050	0.212	0.59	0.219	< 0.010	20.4	< 0.010	< 0.10	< 10.0	2.55	0.011	1.27	1.6
	MW18-09-210811	2021 08 11	< 0.20	0.61	16.1	< 0.10	< 50.0	0.012	< 0.50	0.24	1.70	< 0.20	0.57	< 0.010	3.89	0.52	1.71	< 0.050	30.8	< 0.020	< 0.20	< 5.0	8.6	0.170	< 1.0	< 4.0
	MW18-09-220209	2022 02 09	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	5.29	< 0.20	0.20	< 0.010	0.20	0.47	< 0.50	< 0.050	15.5	< 0.020	< 0.20	< 5.0	3.0	< 0.020	1.7	< 4.0
	MW-A-220209	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.011	< 0.50	< 0.10	5.24	< 0.20	0.21	< 0.010	0.23	0.47	< 0.50	< 0.050	14.8	< 0.020	< 0.20	< 5.0	2.9	< 0.020	1.6	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	5	*	*	*	3	*	*	*
	MW18-09-221025	2022 10 25	< 0.20	< 0.50	11.8	< 0.10	< 50.0	0.668																		

Associated ALS file(s): VA20B1567, VA20B1740, VA20B1766, VA21A0691, VA21A0719.

Associated Caro file(s): 8030934, 8072648, 8102765, 8102781, 8110206, 9011666, 9011884, 9092510, 9092670, 9120554, 9120637, 21H1466, 21H1583, 22B1773, 22J3877, 23B2664,23H1041, 23K2213, 23K2233.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times R

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters			Field Parameters					Geochemical Indicators																						
			pH	Hardness mg/L	Total Hardness mg/L	pH (field)	Temperature °C	Conductivity µS/cm	ORP mV	Dissolved Oxygen mg/L	Total Dissolved Solids (Laboratory) mg/L	Ammonia, Total (as N) µg/L	Nitrate (as N) µg/L	Nitrite (as N) µg/L	Chloride mg/L	Fluoride µg/L	Sulfate mg/L	Total Alkalinity mg/L	Alkalinity, Bicarbonate (as CaCO3) mg/L	Alkalinity, Carbonate (as CaCO3) mg/L	Alkalinity, Hydroxide (as CaCO3) mg/L	Alkalinity, Phenolphthalein (as CaCO3) mg/L	Bromide mg/L	Total Acidity mg/L	Acidity (as CaCO3) mg/L	Dissolved Aluminum µg/L	Dissolved Calcium mg/L	Dissolved Iron µg/L	Dissolved Magnesium mg/L	Dissolved Manganese µg/L	Dissolved Potassium mg/L	Dissolved Sodium mg/L	
BC Standard																																	
CSR Aquatic Life (AW) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	18,400-18,500 ^b	400,000	400-2,000 ^c	1,500	2,000-3,000 ^d	2,180-4,290 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^b	4,000,000	4,000-20,000 ^c	15,000	20,000-30,000 ^d	21,800-42,900 ^d	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Field Blank	Field Blank-221026	2022 10 26	-	-	< 0.500	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	-	-	-	-	-	< 0.10	-	-	< 5.0	< 0.20	< 10	< 0.010	< 0.20	< 0.10	< 0.10	
	Field Blank-230222	2023 02 22	-	-	< 0.500	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	< 5.0	< 0.20	< 10	< 0.010	< 0.20	< 0.10	< 0.10		
	Field Blank-230801	2023 08 01	-	-	< 0.125	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	-	-	-	-	-	-	-		
	Field Blank-231114	2023 11 14	-	-	< 0.125	-	-	-	-	-	< 10	< 50	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.1	< 2.0	-	-	-	-	-	-	-	-	
Trip Blank	Trip Blank-221027	2022 10 27	-	-	< 0.500	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	-	-	-	-	-	< 0.10	-	-	-	-	-	-	-	-	-	-
	Trip Blank-230223	2023 02 23	-	-	< 0.500	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	-	-	-	-	-	-	-		
	Trip Blank-230727	2023 07 27	-	-	< 0.125	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	-	-	-	-	-	-	-		
	Trip Blank-231116	2023 11 16	-	-	< 0.125	-	-	-	-	-	< 10	< 50	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.1	< 2.0	-	-	-	-	-	-	-	-		
Equipment Blank	Equipment Blank-230801	2023 08 01	-	< 0.125	< 0.125	-	-	-	-	-	< 15	< 50	< 10	< 10	< 0.10	< 100	< 1.0	2.0	2.0	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	< 1.0	< 0.05	< 2.0	< 0.0050	< 0.050	< 0.02	< 0.02	
	Equipment Blank-231114	2023 11 14	-	< 0.125	< 0.125	-	-	-	-	-	< 10	-	< 10	< 10	< 0.10	< 100	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.1	< 2.0	-	< 1.0	< 0.05	< 2.0	< 0.0050	< 0.050	< 0.02	< 0.02		

Associated ALS file(s): VA20B1567, VA20B1740, VA20B1766, VA21A0691, VA21A0719.

Associated Caro file(s): 8030934, 8072648, 8102765, 8102781, 8110206, 9011666, 9011884, 9092510, 9092670, 9120554, 9120637, 21H1466, 21H1583, 22B1773, 22J3877, 23B2664,23H1041, 23K2213, 23K2233.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

^a Standard to protect freshwater aquatic life.

^b Standard varies with pH and Temperature. Temperature assumed 10C.

^c Standard varies with Chloride.

^d Standard varies with Hardness.

^e Individual standards exist for Cr +3 and Cr +6. Reported value represents more stringent standard.

^f There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

BOLD	Concentration greater than CSR Aquatic Life (AW) standard
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

Table 2: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Groundwater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																							
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Strontium µg/L	Thallium µg/L	Tin µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^f µg/L
BC Standard																										
CSR Aquatic Life (AW) ^a			90	50	10,000	1.5	12,000	0.5-4 ^d	10 ^e	40	20-90 ^d	40-160 ^d	n/a	0.25	10,000	250-1,500 ^d	20	0.5-15 ^d	n/a	3	n/a	1,000	n/a	85	n/a	75-2,400
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^a			900	500	100,000	15	120,000	5-40 ^d	100 ^e	400	200-900 ^d	400-1,600 ^d	n/a	2.5	100,000	2,500-15,000 ^d	200	5-150 ^d	n/a	30	n/a	10,000	n/a	850	n/a	750-24,000
Field Blank	Field Blank-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	< 0.40	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 1.0	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	Field Blank-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	< 0.40	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	0.063	< 1.0	< 0.020	< 0.20	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	Field Blank-230801	2023 08 01	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Field Blank-231114	2023 11 14	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Trip Blank	Trip Blank-221027	2022 10 27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Trip Blank-230223	2023 02 23	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Trip Blank-230727	2023 07 27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Trip Blank-231116	2023 11 16	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Equipment Blank	Equipment Blank-230801	2023 08 01	< 0.050	< 0.050	< 0.10	< 0.010	< 2.0	< 0.0020	< 0.50	< 0.0050	< 0.10	< 0.050	< 0.050	< 0.0050	< 0.010	< 0.040	< 0.10	< 0.010	< 0.10	< 0.0040	< 0.050	< 0.20	< 0.20	< 0.0010	< 1.00	< 1.0
	Equipment Blank-231114	2023 11 14	< 0.050	< 0.050	< 0.10	< 0.010	< 2.0	< 0.0020	< 0.50	< 0.0050	< 0.10	< 0.050	< 0.050	< 0.0050	< 0.010	< 0.040	< 0.10	< 0.010	< 0.10	< 0.0040	< 0.050	< 0.20	< 0.20	< 0.0010	< 1.00	< 1.0

Associated ALS file(s): VA20B1567, VA20B1740, VA20B1766, VA21A0691, VA21A0719.
Associated Caro file(s): 8030934, 8072648, 8102765, 8102781, 8110206, 9011666, 9011884, 9092510, 9092670, 9120554, 9120637, 21H1466, 21H1583, 22B1773, 22J3877, 23B2664,23H1041, 23K2213, 23K2233.
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^a Standard to protect freshwater aquatic life.
^b Standard varies with pH and Temperature. Temperature assumed 10C.
^c Standard varies with Chloride.
^d Standard varies with Hardness.
^e Individual standards exist for Cr +3 and Cr +6. Reported value represents more stringent standard.
^f There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

BOLD	Concentration greater than CSR Aquatic Life (AW) standard
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

Table 3: Porewater Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)*	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Location Description
PW15-08	5364926	422197	PW15-08-170731	2017-07-31	6.26	0.028	-	154.7	17.3	0.0208	0.02	Jordan River where flow from 5100' portal enters
			PW15-08-170821	2017-08-21	7.09	0.034	8.36	168.6	14.9	0.0286	0.02	
			PW15-08-170926	2017-09-26	7.23	0.060	-	145.5	14.7	0.0384	0.03	
			PW15-08-171031	2017-10-31	6.10	0.031	1.88	189.6	10.2	0.0202	0.01	
			PW15-08-180306	2018-03-06	7.25	0.022	12.48	162.0	4.0	0.0143	-	
			PW15-08-180725	2018-07-25	6.27	0.093	8.74	160.3	16.4	0.0760	0.06	
			PW15-08-190124	2019-01-24	6.48	0.022	9.93	163.4	6.7	0.0143	0.01	
			PW15-08-191203	2019-12-03	5.35	0.040	10.75	214.1	6.7	0.0270	0.02	
			PW15-08-210811	2021-08-11	7.05	0.043	9.33	37.1	17.1	-	-	
			PW15-08-220211	2022-02-11	6.50	0.037	10.10	209.6	5.9	-	-	
			PW15-08-221025	2022-10-25	7.03	0.033	9.05	136.0	10.6	-	-	
			PW15-08-230802	2023-08-02	6.85	0.025	9.48	-146.5	15.1	0.0240	-	
PW15-09	5364951	422200	PW15-09-151123	2015-11-23	7.55	0.043	13.07	129.2	6.2	0.0430	0.03	Jordan River upstream from 5,100' portal (reference sample)
			PW15-09-160629	2016-06-29	7.29	0.019	8.91	171.8	16.1	0.0150	0.01	
			PW15-09-160906	2016-09-06	-	0.023	9.23	31.7	14.6	0.0182	0.01	
			PW15-09-170801	2017-07-31	6.85	0.018	-	164.2	16.5	0.0152	0.01	
			PW15-09-170821	2017-08-21	7.26	0.016	7.37	186.6	-	0.0130	0.01	
			PW15-09-170926	2017-09-26	7.39	0.026	-	141.5	15.1	0.0169	0.01	
			PW15-09-171031	2017-10-31	6.56	0.023	5.19	226.4	8.7	0.0150	0.01	
			PW15-09-171122	2017-11-22	6.36	0.033	10.62	261.7	5.8	0.0221	0.02	
			PW15-09-171212	2017-12-12	7.47	0.024	13.87	162.4	5.2	-	0.01	
			PW15-09-180306	2018-03-06	7.74	0.021	12.66	151.7	3.7	0.0137	-	
			PW15-09-180725	2018-07-25	6.79	0.031	7.99	115.3	17.5	0.0230	0.02	
			PW15-09-181030	2018-10-30	5.71	0.019	13.30	208.0	9.6	0.0176	0.01	
			PW15-09-190123	2019-01-23	6.54	0.018	10.97	145.0	6.7	0.0117	0.01	
			PW15-09-190926	2019-09-26	6.38	0.020	10.86	170	13.6	0.0160	0.01	
			PW15-09-191204	2019-12-04	6.37	0.018	12.63	172.2	6.5	0.0110	0.01	
			PW15-09-200729	2020-07-29	6.33	0.021	10.02	93.9	15.8	0	0.01	
			PW15-09-210811	2021-08-11	7.04	0.027	9.22	20.7	17.0	-	-	
			PW15-09-220211	2022-02-11	6.41	0.014	11.60	214.7	5.4	-	-	
			PW15-09-221025	2022-10-25	7.21	0.032	9.77	108.9	10.2	-	-	
			PW15-09-230223	2023-02-23	6.69	0.018	13.42	209.6	2.44	-	-	
			PW15-09-230802	2023-08-02	6.99	0.021	8.52	151.4	15.5	0.1390	-	
			PW15-09-231115	2023 11 15	6.30	0.0313	18.77	28.1	6.4	-	-	
PW15-10	5364824	422197	PW15-10-151123	2015-11-23	5.98	0.175	10.13	168.6	7.3	0.1710	0.13	Jordan River at the base of North Waste Rock Pile
			PW15-10-160629	2016-06-29	7.02	0.024	7.60	185.2	16.2	0.0190	0.01	
			PW15-10-160907	2016-09-07	-	0.082	6.12	128.2	15.0	0.0663	0.05	
			PW15-10-161018	2016-10-18	6.83	0.042	8.84	192.5	10.8	0.0370	0.03	
			PW15-10-161213	2016-12-13	5.05	0.360	8.35	198.7	8.0	0.2340	0.17	
			PW15-10-170801	2017-08-01	6.88	0.022	-	155.1	16.0	0.0175	0.01	
			PW15-10-170926	2017-09-26	6.21	0.033	-	194.5	16.9	0.0215	0.01	
			PW15-10-171031	2017-10-31	5.51	0.044	-	236.8	10.3	0.0286	0.02	
			PW15-10-171123	2017-11-23	6.48	0.038	10.27	218.9	6.1	0.0240	0.02	
			PW15-10-171213	2017-12-13	5.78	0.281	11.33	211.2	5.6	0.1826	0.14	
			PW15-10-180309	2018-03-09	6.83	0.081	13.64	-153.7	5.2	0.0510	0.04	
			PW15-10-180724	2018-07-24	5.96	0.039	5.78	197.1	18.4	0.0280	0.02	
			PW15-10-181031	2018-10-31	5.22	0.030	12.70	263.0	10.3	0.0273	0.02	
			PW15-10-190124	2019-01-24	6.49	0.020	11.14	154.0	6.6	0.0130	0.01	
			PW15-10-190925	2019-09-25	6.59	0.020	11.02	147	13.8	0.0170	0.01	
			PW15-10-191203	2019-12-03	5.86	0.058	10.7	183.9	5.9	0.0380	0.03	
			PW15-10-191204	2019-12-04	6.21	0.019	12.55	168.7	6.5	0.0120	0.01	
			PW15-10-200730	2020-07-30	6.62	0.022	9.45	151.8	17.2	0.017	0.01	
			PW15-10-210113	2021-01-13	5.63	0.013	13.49	220.4	5.5	0.009	0	
			PW15-10-210811	2021-08-11	7.45	0.032	7.77	60.7	18.3	-	-	
			PW15-10-220210	2022-02-10	6.53	0.015	12.02	190.2	6	-	-	
			PW15-10-221027	2022-10-27	6.87	0.042	9.67	115	10.4	-	-	
			PW15-10-230222	2023-02-22	6.67	0.020	12.76	266	3.98	-	-	
			PW15-10-230802	2023-08-02	6.95	0.023	8.31	-119.2	16.5	-	-	
			PW15-10-231114	2023 11 14	6.27	0.0275	14.84	159.2	7.0	-	-	

* pH meter malfunction for September sampling event.
- not recorded
n/a not applicable

Table 3: Porewater Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)*	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Location Description
PW15-11	5364682	422228	PW15-11-151123	2015-11-23	6.99	0.102	11.10	161.7	7.1	0.1010	-	Jordan River downgradient of the South Waste Rock Pile
			PW15-11-160630	2016-06-30	7.09	0.022	8.91	195.8	15.3	0.0170	0.01	
			PW15-11-160907	2016-09-07	-	0.027	8.39	60.7	15.0	0.0215	0.01	
			PW15-11-161018	2016-10-18	6.09	0.063	7.48	216.9	11.4	0.0553	0.04	
			PW15-11-161213	2016-12-13	6.80	0.060	10.75	145.6	7.2	0.0390	0.03	
			PW15-11-170801	2017-08-01	6.48	0.020	-	159.3	17.8	0.0156	0.01	
			PW15-11-170822	2017-08-22	6.20	0.020	8.46	174.6	15.2	0.0163	0.01	
			PW15-11-170927	2017-09-27	7.53	0.029	-	158.5	13.8	0.0189	0.01	
			PW15-11-171101	2017-11-01	6.23	0.028	-	183.8	8.7	0.0182	0.01	
PW15-11	5364682	422228	PW15-11-171123	2017-11-23	6.11	0.062	9.62	263.3	8.0	0.0397	0.03	Jordan River downgradient of the South Waste Rock Pile
			PW15-11-171213	2017-12-13	7.12	0.031	14.84	136.1	5.8	0.0201	0.01	
			PW15-11-180309	2018-03-09	6.23	0.034	11.82	-134.4	5.3	0.0220	0.02	
			PW15-11-180724	2018-07-24	7.13	0.021	9.25	140.6	17.3	0.0170	0.01	
			PW15-11-181031	2018-10-31	5.56	0.023	11.80	239.0	10.4	0.0202	0.01	
			PW15-11-190124	2019-01-24	6.44	0.027	9.68	160.0	6.9	0.0176	0.01	
			PW15-11-190924	2019-09-24	6.16	0.032	6.2	184	15.2	0.0260	0.02	
			PW15-11-191203	2019-12-03	6.5	0.046	11.29	158	6.5	0.0300	0.02	
			PW15-11-200730	2020-07-30	6.38	0.023	9.9	136.2	17.3	0.017	0.01	
			PW15-11-210812	2021-08-12	6.88	0.022	9.60	194.5	18.3	-	-	
			PW15-11-220210	2022-02-10	6.72	0.017	11.52	163.9	6.1	-	-	
			PW15-11-221026	2022-10-26	7.09	0.038	10.23	93	10.3	-	-	
			PW15-11-230222	2023-02-22	6.72	0.021	15.02	151.3	4.5	-	-	
			PW15-11-230802	2023-08-02	6.74	0.026	6.73	161.9	16.1	-	-	
PW15-12	5364421	422208	PW15-11-231114	2023 11 14	5.97	0.0329	10.34	163.4	8.7	-	-	Jordan River further downgradient near a former industrial site
			PW15-12-151123	2015-11-23	6.97	0.041	15.01	113.1	6.2	0.0420	0.03	
			PW15-12-160630	2016-06-30	6.91	0.192	9.23	166.7	15.6	0.1400	0.19	
			PW15-12-160906	2016-09-06	-	0.028	8.80	-51.8	15.6	0.0221	0.01	
			PW15-12-161017	2016-10-17	5.80	0.016	10.63	228.3	10.6	0.0143	0.01	
			PW15-12-161213	2016-12-13	6.30	5.623	13.01	227.2	4.0	3.5484	2.81	
			PW15-12-170801	2017-08-01	6.49	0.020	-	154.4	18.3	0.0156	0.01	
			PW15-12-170821	2017-08-21	7.25	0.007	7.35	124.4	17.3	0.0052	0.00	
			PW15-12-170927	2017-09-27	6.99	0.032	-	155.3	14.7	0.0208	0.01	
			PW15-12-171101	2017-11-01	6.58	-	-	184.4	8.9	-	-	
			PW15-12-171122	2017-11-22	7.40	0.015	12.66	248.1	5.5	0.0097	0.01	
PW17-14	5364955.5	422192.3	PW15-12-171212	2017-12-12	6.74	0.603	7.11	174.1	5.8	-	0.29	Opposite side of river from SW-09
			PW17-14-170731	2017-07-31	6.63	0.018	-	156.6	17.5	0.0130	0.01	
			PW17-14-170821	2017-08-21	7.36	0.019	7.79	185.2	11.1	0.0156	0.01	
			PW17-14-170926	2017-09-26	7.36	0.027	-	135.7	16.2	0.0176	0.01	
			PW17-14-171031	2017-10-31	6.31	0.023	3.68	207.9	9.0	0.0150	0.01	
PW17-16	5364776.7	422197.4	PW17-16-170801	2017-08-01	5.67	0.019	-	199.7	16.3	0.0143	0.01	10m downstream of SW15-10
			PW17-16-170926	2017-09-26	6.91	0.028	-	168.9	14.1	0.0182	0.01	
			PW17-16-171031	2017-10-31	5.90	0.027	5.70	214.6	9.0	0.0176	0.01	
			PW17-16-171213	2017-12-13	5.45	0.135	12.65	211.4	5.7	0.0917	0.07	
PW17-17	5364613.5	422239.2	PW17-17-170801	2017-08-01	6.50	0.020	-	145.1	18.5	0.0150	0.01	10m downstream of SW15-11 near metal plate sticking out of water
			PW17-17-170822	2017-08-22	6.58	0.020	8.56	137.3	15.2	0.0156	0.01	
			PW17-17-170927	2017-09-27	7.22	0.029	-	152.1	14.0	0.0189	0.01	
			PW17-17-171101	2017-11-01	6.82	0.027	-	170.0	8.6	0.0176	0.01	
			PW17-17-171213	2017-12-13	6.93	0.035	14.46	153.7	6.5	0.0228	0.02	
			PW17-17-180724	2018-07-24	7.80	0.060	12.31	149.4	16.2	0.0470	0.03	
			PW17-17-181101	2018-11-01	6.74	0.028	14.80	158.0	10.8	0.0247	0.02	
			PW17-17-190124	2019-01-24	6.23	0.037	8.03	169.5	7.4	0.0241	0.02	
			PW17-17-190924	2019-09-24	6.36	0.024	10.4	161	14.5	0.0200	0.01	
			PW17-17-191203	2019-12-03	6.51	0.058	10.42	167.9	7.2	0.0380	0.03	
			PW17-17-200730	2020-07-30	5.7	0.022	9.97	153.5	16.6	-	0.01	
			PW17-17-210112	2021-01-12	5.59	0.016	12.98	191.3	6.3	0.0100	0.01	
			PW17-17-210812	2021-08-12	6.82	0.034	8.61	176.3	18.1	-	-	
			PW17-17-220210	2022-02-10	6.68	0.017	11.39	170.1	6.0	-	-	
			PW17-17-221026	2022-10-26	7.21	0.038	9.31	87.5	10.2	-	-	
			PW17-17-230222	2023-02-22	6.66	0.031	11.31	283	5.88	-	-	
			PW17-17-230802	2023-08-02	6.75	0.025	8.38	-4.0	16.1	0.0150	-	
			PW17-17-231114	2023 11 14	5.92	0.0825	9.46	50.9	9.2	-	-	
PW17-18	5364362.2	422173.1	PW17-18-170801	2017-08-01	6.49	0.024	-	144.3	19.4	0.0175	0.01	25m downstream of SW15-12 in middle of river

* pH meter malfunction for September sampling event.

- not recorded

n/a not applicable

Table 3: Porewater Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)*	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Location Description
PW17-19	5364719.7	422219.1	PW17-19-170822	2017-08-22	6.95	0.020	7.07	134.0	15.5	0.0163	0.01	10m downstream of SW15-05 as it exits pool
			PW17-19-170927	2017-09-27	7.59	0.030	-	164.7	13.7	0.0195	0.01	
			PW17-19-171101	2017-11-01	5.89	0.024	-	185.6	9.0	0.0189	0.01	
			PW17-19-180309	2018-03-09	5.92	0.020	12.18	-154.3	4.5	0.0130	0.01	
			PW17-19-180724	2018-07-24	6.30	0.025	5.32	153.2	19.4	0.0180	0.01	
			PW17-19-181031	2018-10-31	5.11	0.020	6.10	271.0	10.1	0.0182	0.01	
			PW17-19-190124	2019-01-24	6.49	0.021	10.18	159.0	6.7	0.0137	0.01	
			PW17-19-190925	2019-09-25	6.39	0.021	9.8	197	13.4	0.0180	0.01	
			PW17-19-191204	2019-12-04	6.03	0.024	12.21	207.8	6.8	0.0160	0.01	
			PW17-19-210812	2021-08-12	6.78	0.006	10.18	194.9	20.0	-	-	
			PW17-19-220210	2022-02-10	6.53	0.016	13.33	173.5	6.1	-	-	
			PW17-19-221026	2022-10-26	6.5	0.035	7.81	117	10.5	-	-	
			PW17-19-230223	2023-02-23	7.01	0.012	12.25	127.6	4.27	-	-	
			PW17-19-230802	2023-08-02	6.49	0.024	7.0	164.3	16.6	-	-	
			PW17-19-231114	2023 11 14	6.25	0.0336	12.22	35.8	7.6	-	-	
PW17-20	5364804.6	422184.9	PW17-20-170822	2017-08-22	6.99	0.020	8.54	129.3	16.5	0.0156	0.01	10m upstream of SW15-10 above big rock
			PW17-20-180309	2018-03-09	5.64	0.018	13.91	-128.7	4.5	0.0120	0.01	
PW17-21	5364849.8	422185.1	PW17-21-170926	2017-09-26	6.75	0.029	-	182.0	15.5	0.0189	0.01	Jordan River below the North Waste Rock Pile (south region)
			PW17-21-171031	2017-10-31	6.05	0.026	-	232.0	8.6	0.0169	0.01	
			PW17-21-171123	2017-11-23	5.34	0.300	11.24	257.2	6.1	0.0188	0.01	
			PW17-21-180724	2018-07-24	6.92	0.022	9.92	24.8	17.9	0.0160	0.01	
			PW17-21-181031	2018-10-31	5.74	0.018	14.49	233.3	9.8	0.0169	0.01	
			PW17-21-190124	2019-01-24	6.48	0.020	10.38	155.0	6.7	0.0130	0.01	
			PW17-21-190925	2019-09-25	6.7	0.021	11.6	138	14.2	0.017	0.01	
			PW17-21-191203	2019-12-03	6.26	0.025	12.79	160.8	5.5	0.016	0.01	
			PW17-21-210811	2021-08-11	7.29	0.032	8.57	59.6	18.3	-	-	
			PW17-21-220210	2022-02-10	6.59	0.015	11.63	175.8	6.1	-	-	
			PW17-21-221026	2022-10-26	7.15	0.038	10.33	136	10.3	-	-	
			PW17-21-230222	2023-02-22	6.69	0.019	12.52	140.2	3.87	-	-	
			PW17-21-230802	2023-08-02	6.89	0.023	10.40	133.3	15.3	-	-	
			PW17-21-231114	2023 11 14	6.30	0.0317	13.55	40.5	7.0	-	-	
PW18-22	5364865	422190	PW18-22-180726	2018-07-26	7.08	0.055	9.23	176.0	17.5	0.0420	0.03	Jordan River near inferred bedrock at North Waste Rock Pile
			PW18-22-181031	2018-10-31	5.77	0.019	13.84	231.6	10.0	0.0169	0.01	
			PW18-22-190124	2019-01-24	4.61	0.331	9.13	228.0	6.7	0.2150	0.16	
			PW18-22-190925	2019-09-25	6.61	0.021	10.36	150	14.1	0.018	0.01	
			PW18-22-191203	2019-12-03	6.37	0.026	12.21	158.8	6.7	0.017	0.01	
			PW18-22-200729	2020-07-29	7.01	0.022	9.84	132.3	16.9	-	0.01	
			PW18-22-210811	2021-08-11	7.44	0.033	8.37	56.5	18.5	-	-	
			PW18-22-220210	2022-02-10	6.39	0.015	11.95	183.4	6.1	-	-	
			PW18-22-221026	2022-10-26	7.05	0.038	10.53	92.9	10.3	-	-	
			PW18-22-230223	2023-02-23	6.68	0.021	12.61	113	2.47	-	-	
			PW18-22-230802	2023-08-02	7.01	0.001	9.35	113.3	15.2	-	-	
			PW18-22-231115	2023 11 15	6.16	0.0336	17.00	-1.1	6.6	-	-	
PW23-29	-	-	PW23-29-230802	2023 08 02	6.41	0.049	9.20	-190.8	15.8	0.0310	-	

* pH meter malfunction for September sampling event.
- not recorded
n/a not applicable

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																							
			pH	Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Acidity (as CaCO3)	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^g	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	767-2,338 ^g	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
PW15-08	PW15-08-151123	2015 11 23	6.76	34.5	7.52	52	20	-	< 20	144	< 10	-	13.4	< 100	13.2	16	16	< 1	< 1	< 1	< 0.1	-	-	< 2	36	11.6	32	1.32	4.7	0.22	7.98	
	PW15-DUPB-151123	Duplicate	6.65	32.9	7.52	53	7	-	< 20	133	< 10	-	12.9	< 100	14.4	16	16	< 1	< 1	< 1	< 0.1	-	-	< 2	36	10.9	33	1.36	4.9	0.23	8.19	
	QA/QC RPD%		2	5	0	2	*	-	*	8	*	-	4	*	9	0	0	*	*	*	*	-	-	*	0	6	*	3	4	4	3	
	PW15-08-160629	2016 06 29	8.15	14.9	7.13	-	-	-	-	< 10	< 10	-	9.41	100	2.7	5	5	< 1	< 1	< 1	< 0.10	-	-	< 2	65	4.8	117	0.69	4.0	0.25	3.66	
	PW15-08-160906	2016 09 06	6.92	12.5	-	-	-	-	-	161	< 10	-	7.21	< 100	2.1	7	7	< 1	< 1	< 1	< 0.10	-	-	3	73	3.9	589	0.69	5.5	0.34	3.03	
	PW15-08-161213	2016 12 13	6.98	26.9	7	-	-	-	-	114	< 10	-	7.95	< 100	9.4	17	17	< 1	< 1	< 1	-	-	-	5	25	8.9	32	1.12	3.0	0.32	5.56	
	PW15-08-170731	2017 07 31	6.71	7.07	6.26	-	-	-	-	83	< 10	-	3.81	140	1.4	5.1	5.1	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	97.3	1.97	156	0.523	5.36	0.24	1.95	
	PW15-08-170821	2017 08 21	6.78	11.4	7.09	-	-	-	-	94	< 10	-	6.15	< 100	2.2	5.9	5.9	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	45.9	3.61	123	0.573	2.23	-	2.79	
	PW15-08-170926	2017 09 26	6.35	13.3	7.23	-	-	-	-	88	< 10	-	7.68	< 100	2.9	6.7	6.7	< 1.0	< 1.0	< 1.0	-	-	-	4.2	39.3	4.23	155	0.653	3.62	0.26	3.26	
	PW15-08-171031	2017 10 31	6.04	7.18	6.1	-	-	-	-	92	< 10	-	3.25	< 100	1.8	4.6	4.6	< 1.0	< 1.0	< 1.0	-	-	-	4.1	112	1.95	98	0.559	4.80	0.21	2.11	
	PW15-08-180306	2018 03 06	6.41	4.92	7.25	-	-	-	-	58.5	< 5.0	58.5	2.77	< 100	1.1	3.4	3.4	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	95.8	1.26	55	0.428	3.78	0.11	1.73	
	PW15-08-180725	2018 07 25	6.03	29.9	6.27	84	169	-	-	59	< 10	-	24.4	< 100	5.2	3.4	3.4	< 1.0	< 1.0	< 1.0	-	-	-	3.1	41.5	10.4	43	0.937	7.14	0.23	6.36	
	PW15-08-190124	2019 01 24	6.43	5.64	6.48	51	294	-	-	17	< 10	-	3.08	< 100	1.6	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.5	137	1.51	88	0.454	7.44	0.13	2.02	
	PW15-08-191203	2019 12 03	6.89	13.3	5.35	71	8.4	-	-	31	< 10	-	5.53	< 100	3.9	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.10	-	-	375	89.8	3.93	54	0.843	3.86	0.20	3.32	
	PW15-08-210811	2021 08 11	-	-	7.05	26	-	-	-	142	< 10	-	5.27	< 100	1.7	7.9	7.9	< 1.0	< 1.0	< 1.0	< 0.10	-	3.1	-	34.6	3.71	195	0.699	1.23	0.30	3.06	
	PW15-08-220211	2022 02 11	-	-	6.50	39	-	-	< 50	26	< 10	-	5.21	< 100	4.2	5.0	5.0	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	-	91.5	3.36	31	0.541	1.62	0.13	2.83	
PW15-08-221025	2022 10 25	-	8.88	7.03	29	-	-	< 50	150	< 10	-	3.85	< 100	1.2	6.4	6.4	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	42.2	2.39	266	0.702	1.99	0.33	2.53		
PW15-08-230802	2023 08 02	-	8.26	6.85	24	-	-	< 50	103	< 10	-	2.91	< 100	1.9	4.9	4.9	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	38.7	2.48	77.8	0.5	1.33	0.242	2.25		
PW15-09	PW15-09-151123	2015 11 23	5.73	4.58	7.55	< 10	237	-	< 20	56	< 10	-	2.99	< 100	< 1	4	4	< 1	< 1	< 1	< 0.1	-	-	2	128	1.3	300	0.35	4.6	0.06	1.66	
	PW15-09-160629	2016 06 29	6.77	6.80	7.29	-	-	-	-	79	< 10	-	2.62	< 100	< 1.0	5	5	< 1	< 1	< 1	< 0.10	-	-	< 2	73	1.9	197	0.50	3.0	0.25	1.94	
	PW16-A-160629	Duplicate	6.79	6.66	7.29	-	-	-	-	73	< 10	-	3.21	< 100	1.1	5	5	< 1	< 1	< 1	< 0.10	-	-	< 2	73	1.9	200	0.49	2.7	0.24	1.94	
	QA/QC RPD%		< 1	2	0	-	-	-	-	8	*	-	20	*	*	0	0	*	*	*	*	-	-	*	0	0	2	2	11	4	0	
	PW15-09-160906	2016 09 06	6.90	7.96	-	-	-	-	-	168	< 10	-	2.91	< 100	1.2	7	7	< 1	< 1	< 1	< 0.10	-	-	3	53	2.3	717	0.55	2.8	0.35	2.04	
	PW15-09-170801	2017 07 31	6.72	6.34	6.85	-	-	-	-	78	< 10	-	1.86	< 100	1.0	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	64.2	1.76	132	0.470	2.62	0.26	1.89	
	PW15-09-170821	2017 08 21	6.70	6.40	7.26	-	-	-	-	96	< 10	-	1.40	< 100	< 1.0	5.1	5.1	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	66.1	1.84	218	0.441	2.54	-	1.53	
	PW15-09-170926	2017 09 26	6.37	7.05	7.39	-	-	-	-	77	< 10	-	1.94	< 100	< 1.0	7.2	7.2	< 1.0	< 1.0	< 1.0	-	-	-	3.0	50.3	1.97	253	0.518	2.95	0.27	1.74	
	PW15-09-171031	2017 10 31	6.02	6.12	6.56	-	-	-	-	68	< 10	-	4.38	260	< 1.0	4.1	4.1	< 1.0	< 1.0	< 1.0	-	-	-	3.7	87.4	1.64	60	0.493	1.31	0.18	1.79	
	PW15-09-171122	2017 11 22	6.43	10.5	6.36	-	-	-	-	19	< 10	-	9.02	160	1.4	6.5	6.5	< 1.0	< 1.0	< 1.0	< 1.0	-	-	-	3.5	307	2.71	114	0.894	5.33	0.48	3.51
	PW15-09-171212	2017 12 12	6.36	5.61	7.47	-	-	-	-	27	< 10	-	1.92	< 100	1.0	4.7	4.7	< 1.0	< 1.0	< 1.0	-	-	-	3.3	84.8	1.49	26	0.457	0.96	0.13	1.74	
	PW15-09-180306	2018 03 06	6.33	4.80	7.74	-	-	-	-	27.9	< 5.0	27.9	2.50	< 100	3.3	3.0	3.0	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	68.2	1.22	16	0.424	0.54	0.10	1.73	
	PW15-09-180725	2018 07 25	6.11	5.83	6.79	< 15	39.2	-	-	60	< 10	-	2.89	< 100	< 1.0	2.6	2.6	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	71.5	1.64	172	0.418	2.77	0.16	1.48	
	PW15-09-181030	2018 10 30	6.33	6.35	5.71	124	36.5	-	-	36	< 10	-	4.11	< 100	< 1.0	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.1	158	1.57	48	0.589	1.38	0.13	2.55	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration)

of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.

^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^m Guideline is hardness and DOC dependent.

ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,i}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g
PW15-08	PW15-08-151123	2015 11 23	< 0.1	< 0.5	< 5	< 0.1	42	0.09	< 0.5	7.24	232	< 0.1	0.3	< 0.02	0.2	10.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	21
	PW15-DUPB-151123	Duplicate	< 0.1	< 0.5	< 5	< 0.1	39	0.1	< 0.5	7.28	236	< 0.1	0.2	< 0.02	0.2	10.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	22
	QA/QC RPD%		*	*	*	*	7	11	*	1	2	*	*	*	*	0	*	*	*	*	-	*	*	5
	PW15-08-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	15	0.03	< 0.5	2.44	49.3	< 0.1	0.1	< 0.02	< 0.1	3.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	9
	PW15-08-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	14	< 0.01	< 0.5	0.85	13.1	< 0.1	0.1	< 0.02	< 0.1	1.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW15-08-161213	2016 12 13	< 0.1	< 0.5	< 5	< 0.1	34	0.09	0.6	5.39	133	< 0.1	0.1	< 0.02	0.1	7.9	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	18
	PW15-08-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	7.5	< 0.010	0.51	0.43	8.66	< 0.10	0.12	< 0.010	0.10	1.34	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	6.2
	PW15-08-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	15.0	0.027	< 0.50	1.69	35.8	< 0.20	0.13	< 0.010	< 0.10	2.82	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	6.2
	PW15-08-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	13.4	0.027	< 0.50	1.96	34.0	< 0.20	0.11	0.034	< 0.10	3.12	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	7.5
	PW15-08-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	5.5	< 0.010	< 0.50	0.37	8.95	< 0.20	< 0.10	< 0.010	< 0.10	1.05	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-08-180306	2018 03 06	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.15	2.29	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-08-180725	2018 07 25	< 0.20	< 0.50	< 5.0	< 0.10	34.3	0.086	< 0.50	5.96	70.2	< 0.20	0.25	< 0.010	0.11	10.3	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	18.9
	PW15-08-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.38	4.02	0.21	0.11	< 0.010	< 0.10	1.30	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	10.1
	PW15-08-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	24.7	0.014	0.60	1.55	54.6	1.09	0.11	< 0.010	0.14	2.80	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	8.7
	PW15-08-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.017	< 0.50	0.89	16.9	< 0.20	< 0.10	< 0.010	< 0.10	2.17	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
PW15-08-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.031	< 0.50	2.50	84.4	< 0.20	< 0.10	< 0.010	< 0.10	3.73	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	9.2	
PW15-08-221025	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.17	5.40	< 0.20	< 0.10	< 0.010	< 0.10	0.76	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
PW15-08-230802	2023 08 02	< 0.050	0.085	2.28	< 0.010	7.0	0.0098	< 0.50	0.754	23.4	< 0.050	0.102	< 0.0050	0.034	1.24	< 0.10	< 0.010	< 0.0040	0.34	< 0.20	< 0.0010	< 1.00	3.0	
PW15-09	PW15-09-151123	2015 11 23	< 0.1	< 0.5	< 5	< 0.1	6	< 0.01	< 0.5	0.16	2.8	< 0.1	0.1	< 0.02	< 0.1	0.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW15-09-160629	2016 06 29	< 0.1	< 0.5	6	< 0.1	16	< 0.01	0.9	0.06	2.2	< 0.1	< 0.1	< 0.02	< 0.1	0.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW16-A-160629	Duplicate	< 0.1	< 0.5	< 5	< 0.1	10	< 0.01	0.7	0.05	1.8	< 0.1	< 0.1	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	QA/QC RPD%		*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*
	PW15-09-160906	2016 09 06	< 0.1	< 0.5	7	< 0.1	10	< 0.01	< 0.5	0.05	1.6	< 0.1	0.1	< 0.02	< 0.1	0.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW15-09-170801	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	7.0	< 0.010	0.70	< 0.10	1.91	< 0.10	0.13	< 0.010	< 0.10	0.53	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW15-09-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	6.8	< 0.010	0.62	< 0.10	2.20	< 0.20	0.13	< 0.010	< 0.10	0.47	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW15-09-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	8.1	< 0.010	< 0.50	< 0.10	2.11	< 0.20	< 0.10	0.026	< 0.10	0.55	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW15-09-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	2.35	< 0.20	< 0.10	< 0.010	< 0.10	0.45	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-09-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	9.0	< 0.010	< 0.50	0.22	8.23	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-09-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.78	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-09-180306	2018 03 06	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.47	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-09-180725	2018 07 25	< 0.20	< 0.50	< 5.0	< 0.10	10.4	< 0.010	< 0.50	< 0.10	2.73	< 0.20	0.14	< 0.010	< 0.10	0.71	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
PW15-09-181030	2018 10 30	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	2.93	< 0.20	< 0.10	< 0.010	< 0.10	0.41	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.
^d Standard to protect freshwater aquatic life.
^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^f Guideline/standard varies with Chloride.
^g Guideline/standard varies with Hardness.
^h Guideline/standard varies with pH.
ⁱ Guideline is temperature, pH, DOC and hardness dependent.
^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.
^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.
^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^m Guideline is hardness and DOC dependent.
ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																							
			pH pH	Hardness mg/L	pH (field) pH	Total Dissolved Solids mg/L	Total Suspended Solids mg/L	Dissolved Organic Carbon mg/L	Ammonia, total (as N) µg/L	Nitrate (as N) µg/L	Nitrite (as N) µg/L	Nitrate+Nitrite Nitrogen µg/L	Chloride mg/L	Fluoride µg/L	Sulfate mg/L	Total Alkalinity (as CaCO3) mg/L	Alkalinity, Bicarbonate (as CaCO3) mg/L	Alkalinity, Carbonate (as CaCO3) mg/L	Alkalinity, Hydroxide (as CaCO3) mg/L	Alkalinity, Phenolphthalein (as CaCO3) mg/L	Bromide mg/L	Acidity (as CaCO3) mg/L	Total Acidity mg/L	Acidity (pH 8.3) mg/L	Dissolved Aluminum µg/L	Dissolved Calcium mg/L	Dissolved Iron µg/L	Dissolved Magnesium mg/L	Dissolved Manganese µg/L	Dissolved Potassium mg/L	Dissolved Sodium mg/L	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^g	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	767-2,338 ^g	n/a	n/a	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
PW15-09	PW15-09-190123	2019 01 23	-	4.24	6.54	19	< 4.0	-	-	< 10	< 10	-	2.64	< 100	< 1.0	1.9	1.9	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.9	130	1.07	30	0.378	0.77	< 0.10	1.63	
	PW15-09-190926	2019 09 26	6.08	6.72	6.38	< 15	19.0	-	-	49	< 10	-	2.98	< 100	< 1.0	2.0	2.0	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.5	182	1.88	153	0.489	3.86	0.38	2.09	
	PW15-09-191204	2019 12 04	6.54	4.45	6.37	36	2.2	-	-	23	< 10	-	2.26	< 100	1.0	8.1	8.1	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.9	194	1.14	68	0.389	0.90	< 0.10	1.70	
	PW15-09-200729	2020 07 29	6.82	7.00	6.33	24	-	-	-	91.6	< 1.0	-	2.35	< 20	0.77	5.9	5.9	< 1.0	< 1.0	< 1.0	< 0.050	2.1	-	-	66.1	1.90	194	0.547	2.06	0.182	1.95	
	PW15-09-210811	2021 08 11	-	-	7.04	25	-	-	-	149	< 10	-	1.93	< 100	< 1.0	7.2	7.2	< 1.0	< 1.0	< 1.0	< 0.10	-	3.0	-	56.7	2.12	476	0.642	1.86	0.30	2.18	
	PW15-09-220211	2022 02 11	-	-	6.41	22	-	-	< 50	14	< 10	-	1.81	< 100	< 1.0	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	-	89.8	0.84	18	0.353	< 0.20	< 0.10	1.50	
	PW15-09-221015	2022 10 25	-	8.57	7.21	41	-	-	< 50	143	< 10	-	3.45	< 100	1.1	6.6	6.6	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	34.1	2.35	284	0.653	1.50	0.33	2.35	
	PW15-09-230223	2023 02 23	-	4.72	6.69	30	-	-	< 50	18	< 10	-	3.05	< 100	< 1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	90.9	1.16	16	0.445	0.34	< 0.10	2.06	
PW15-10	PW15-09-230802	2023 08 02	-	84.5	6.99	139	-	-	< 50	487	< 10	-	8.99	< 100	33.5	68.9	68.9	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	13.4	25.8	8.7	4.86	12.5	0.931	9.54	
	PW15-09-231115	2023 11 15	-	5.50	6.30	38	-	-	< 50	30	< 10	-	3.34	< 100	1.0	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.1	-	3.1	-	113	1.36	31.1	0.511	0.944	0.123	2.34	
	PW15-10-151123	2015 11 23	5.55	76.5	5.98	123	693	-	68	389	< 10	-	3.51	< 100	75.7	6	6	< 1	< 1	< 1	< 0.1	-	-	9	50	24.3	43	3.87	31.4	0.67	7.35	
	PW15-10-160629	2016 06 29	6.63	7.78	7.02	-	-	-	-	89	< 10	-	5.01	110	1.3	6	6	< 1	< 1	< 1	< 0.10	-	-	2	109	2.2	180	0.57	5.4	0.23	2.20	
	PW15-10-160907	2016 09 07	6.19	36.7	-	-	-	-	-	227	< 10	-	5.80	< 100	30.1	5	5	< 1	< 1	< 1	< 0.10	-	-	5	61	12.0	49	1.65	55.1	0.51	4.54	
	PW15-10-161018	2016 10 18	6.80	20.5	6.83	-	-	-	-	112	< 10	-	5.15	110	11.6	7	7	< 1	< 1	< 1	-	-	-	< 2	59	7.5	69	0.43	3.1	0.39	2.30	
	PW15-10-161213	2016 12 13	4.56	122	5.05	-	-	-	-	376	< 5	376	5.05	< 100	164	1	1	< 1	< 1	< 1	-	-	-	21	933	40.4	33	5.07	189	0.90	9.58	
	PW15-10-170801	2017 08 01	6.40	8.32	6.88	-	-	-	-	99	< 10	-	3.24	140	2.5	5.1	5.1	< 1.0	< 1.0	< 1.0	-	-	-	3.5	584	2.36	638	0.592	43.5	0.25	2.15	
	PW15-10-170926	2017 09 26	6.12	9.03	6.21	-	-	-	-	126	< 10	-	2.79	< 100	3.9	4.2	4.2	< 1.0	< 1.0	< 1.0	-	-	-	4.9	52.1	2.86	68	0.456	12.0	0.22	2.08	
	PW15-10-171031	2017 10 31	5.94	13.7	5.51	-	-	-	-	96	< 10	-	5.67	290	8.0	3.9	3.9	< 1.0	< 1.0	< 1.0	-	-	-	5.5	81.8	3.93	55	0.953	16.3	0.29	2.95	
	PW15-10-171123	2017 11 23	6.49	13.6	6.48	-	-	-	-	27	< 10	-	3.29	140	6.0	6.3	6.3	< 1.0	< 1.0	< 1.0	-	-	-	3.7	48.8	4.98	53	0.286	2.62	0.19	1.46	
	PW15-10-171213	2017 12 13	5.33	112	5.78	-	-	-	-	240	< 10	-	3.68	110	170	1.0	1.0	< 1.0	< 1.0	< 1.0	-	-	-	16.9	304	36.7	19	4.89	150	0.68	8.56	
	PW15-10-180309	2018 03 09	6.41	5.37	6.83	-	-	-	-	30.4	< 5.0	30.4	2.90	< 100	5.2	3.2	3.2	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	106	1.46	80	0.416	1.45	0.25	1.85	
	PW15-10-180724	2018 07 24	6.09	12.3	5.96	-	-	-	-	100	< 10	-	8.09	460	7.7	2.9	2.9	< 1.0	< 1.0	< 1.0	-	-	-	4.1	59.2	3.77	53	0.693	11.0	0.27	2.36	
	PW15-10-181031	2018 10 31	6.06	9.23	5.22	35	17.0	-	-	56	< 10	-	4.10	< 100	6.1	1.8	1.8	< 1.0	< 1.0	< 1.0	< 0.10	-	-	3.2	137	2.79	47	0.545	8.26	0.12	2.33	
	PW15-10-190124	2019 01 24	6.38	5.04	6.49	61	5.2	-	-	< 10	< 10	-	2.91	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.4	105	1.35	26	0.403	1.11	< 0.10	1.82	
	PW-B-190124	Duplicate	6.45	5.05	6.49	66	4.4	-	-	< 10	< 10	-	2.91	< 100	1.1	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.4	105	1.36	26	0.403	1.10	< 0.10	1.82	
	QA/QC RPD%			1	< 1	0	*	*	-	-	*	*	-	0	*	*	*	*	*	*	*	-	-	*	0	1	*	0	1	*	0	
	PW15-10-190925	2019 09 25	6.23	7.72	6.59	29	3.2	-	-	44	< 10	-	2.70	< 100	1.1	2.0	2.0	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.1	162	2.19	261	0.546	5.36	0.20	2.11	
	PW15-10-191203	2019 12 03	6.62	19.1	5.86	57	16.6	-	-	38	< 10	-	3.49	< 100	14.4	4.5	4.5	< 1.0	< 1.0	< 1.0	< 0.10	-	-	3.1	86.1	5.42	34	1.35	7.55	0.21	2.95	
	PW15-10-191204	2019 12 04	6.54	4.43	6.21	33	< 2.0	-	-	10	< 10	-	2.29	< 100	1.2	9.4	9.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.6	186	1.14	65	0.383	1.16	< 0.10	1.63	
	PW15-10-200730	2020 07 30	7.00	7.37	6.62	28	-	-	-	95.7	< 1.0	-	2.89	< 20	1.15	6.1	6.1	< 1.0	< 1.0	< 1.0	< 0.050	2.5	-	-	99.5	2.08	229	0.527	3.63	0.222	2.07	
PW-B-200730	Duplicate	6.90	7.48	6.62	27	-	-	-	96.5	< 1.0	-	2.90	< 20	1.15	5.7	5.7	< 1.0	< 1.0	< 1.0	< 0.050	2.2	-	-	102	2.05	228	0.573	3.97	0.213	2.27		
QA/QC RPD%			1	1	0	*	-	-	-	1	*	-	< 1	*	*	7	7	*	*	*	*	*	-	-	2	1	< 1	8	9	*	9	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5%

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																						
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L	
BC Guideline/Standard																									
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m	
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,i}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g	
PW15-09	PW15-09-190123	2019 01 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.65	< 0.10	<u>2.22</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-09-190926	2019 09 26	< 0.20	< 0.50	< 5.0	< 0.10	19.6	< 0.010	0.56	0.11	<u>4.25</u>	< 0.20	0.18	0.150	< 0.10	0.58	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-09-191204	2019 12 04	< 0.20	< 0.50	< 5.0	< 0.10	10.2	< 0.010	< 0.50	< 0.10	<u>2.98</u>	0.59	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-09-200729	2020 07 29	< 0.10	0.12	1.89	< 0.020	< 10	< 0.0050	0.15	< 0.10	<u>2.31</u>	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0	
	PW15-09-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	<u>1.65</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.96	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-09-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	<u>1.85</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-09-221015	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	<u>1.87</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	PW15-09-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	<u>1.60</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
PW15-10	PW15-09-230802	2023 08 02	0.167	0.256	7.20	< 0.010	31.2	0.0152	< 0.50	0.193	<u>2.91</u>	< 0.050	0.429	< 0.0050	0.961	0.628	1.37	< 0.010	0.0041	< 0.20	9.13	0.0300	1.16	4.4	
	PW15-09-231115	2023 11 15	< 0.050	0.054	1.89	< 0.010	2.8	0.0039	< 0.50	0.0504	<u>2.45</u>	< 0.050	0.078	< 0.0050	0.020	0.388	< 0.10	< 0.010	< 0.0040	0.71	< 0.20	0.0016	< 1.00	1.8	
	PW15-10-151123	2015 11 23	< 0.1	< 0.5	27	< 0.1	15	0.25	< 0.5	38.3	<u>376</u>	< 0.1	0.6	< 0.02	0.1	53.9	2	< 0.05	< 0.02	< 5	-	< 0.02	< 1	<u>61</u>	
	PW15-10-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	6	0.01	0.6	0.64	<u>33.5</u>	0.2	0.1	< 0.02	< 0.1	1.0	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	PW15-10-160907	2016 09 07	< 0.1	< 0.5	9	< 0.1	17	<u>0.21</u>	< 0.5	21.2	<u>970</u>	< 0.1	0.6	< 0.02	< 0.1	34.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	<u>52</u>	
	PW15-10-161018	2016 10 18	0.2	< 0.5	< 5	< 0.1	6	0.09	0.6	3.36	<u>108</u>	1.1	0.1	< 0.02	0.3	3.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	<u>7</u>	
	PW15-10-161213	2016 12 13	0.1	< 0.5	15	0.1	27	<u>1.22</u>	< 0.5	<u>141</u>	<u>8,010</u>	1.0	1.2	< 0.02	< 0.1	224	1.1	0.08	< 0.02	< 5	-	< 0.02	< 1	<u>259</u>	
	PW15-10-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	7.7	0.041	1.45	<u>4.56</u>	<u>244</u>	1.11	0.31	< 0.010	< 0.10	6.25	< 0.50	< 0.050	< 0.020	17.4	-	< 0.020	1.3	<u>11.5</u>	
	PW15-10-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	10.7	<u>0.054</u>	< 0.50	<u>4.75</u>	<u>217</u>	< 0.20	0.25	0.016	< 0.10	7.68	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	<u>16.0</u>	
	PW15-10-171031	2017 10 31	< 0.20	< 0.50	5.1	< 0.10	6.2	0.067	1.00	5.81	<u>184</u>	< 0.20	0.20	< 0.010	< 0.10	9.73	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>16.4</u>	
	PW15-10-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	5.2	0.058	< 0.50	2.78	<u>94.9</u>	< 0.20	< 0.10	< 0.010	0.26	3.13	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-10-171213	2017 12 13	< 0.20	< 0.50	33.5	< 0.10	20.5	<u>0.805</u>	< 0.50	102	<u>5,150</u>	< 0.20	0.64	< 0.010	< 0.10	164	0.80	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>200</u>	
	PW15-10-180309	2018 03 09	< 0.20	< 0.50	< 5.0	< 0.10	6.0	< 0.010	< 0.50	0.28	<u>15.6</u>	0.22	0.13	< 0.010	0.10	0.74	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	6.4	
	PW15-10-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	6.9	0.056	< 0.50	<u>4.94</u>	<u>206</u>	< 0.20	0.22	< 0.010	< 0.10	8.22	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>11.6</u>	
	PW15-10-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	5.0	<u>0.057</u>	< 0.50	4.78	<u>397</u>	0.22	0.25	< 0.010	< 0.10	7.71	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>10.8</u>	
	PW15-10-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.30	<u>17.8</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.82	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW-B-190124	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.31	<u>18.0</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.83	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	QA/QC RPD%			*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	
	PW15-10-190925		2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	11.3	< 0.010	< 0.50	0.29	<u>18.8</u>	< 0.20	0.17	0.086	< 0.10	1.09	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-10-191203		2019 12 03	< 0.20	< 0.50	5.4	< 0.10	14.2	0.052	< 0.50	4.71	<u>67.0</u>	0.26	0.15	< 0.010	< 0.10	7.16	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>12.0</u>
	PW15-10-191204		2019 12 04	< 0.20	< 0.50	< 5.0	< 0.10	8.4	< 0.010	< 0.50	0.28	<u>14.4</u>	0.45	0.10	< 0.010	< 0.10	0.50	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-10-200730		2020 07 30	< 0.10	0.12	1.84	< 0.020	< 10	0.0052	0.27	0.30	<u>16.5</u>	0.162	< 1.0	< 0.0050	< 0.050	0.76	0.064	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	2.0
PW-B-200730		Duplicate	< 0.10	0.20	1.87	< 0.020	< 10	0.0080	0.26	0.35	<u>17.5</u>	0.162	< 1.0	< 0.0050	< 0.050	0.78	0.069	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.4	
QA/QC RPD%			*	*	2	*	*	*	*	*	6	*	*	*	*	*	*	*	*	*	*	*	*		

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

- ^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.
^d Standard to protect freshwater aquatic life.
^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^f Guideline/standard varies with Chloride.
^g Guideline/standard varies with Hardness.
^h Guideline/standard varies with pH.
ⁱ Guideline is temperature, pH, DOC and hardness dependent.
^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.
^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.
^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^m Guideline is hardness and DOC dependent.
ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																							
			pH	Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Acidity (as CaCO3)	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^g	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	767-2,338 ^g	n/a	n/a	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
PW15-10	PW15-10-210113	2021 01 13	-	3.32	5.63	22	9,000	-	< 5.0	26.8	< 1.0	-	1.76	< 20	0.69	3.2	3.2	< 1.0	< 1.0	< 1.0	< 0.050	2.1	-	-	133	0.875	39	0.277	2.09	0.131	1.21	
	PW15-10-210811	2021 08 11	-	-	7.45	17	-	-	-	152	< 10	-	3.07	< 100	1.3	6.9	6.9	< 1.0	< 1.0	< 1.0	< 0.10	-	2.8	-	34.7	2.59	220	0.641	1.40	0.29	2.49	
	PW15-10-220210	2022 02 10	-	-	6.53	< 15	-	-	< 50	22.1	< 5.0	22.1	1.99	< 100	1.3	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	-	89.6	1.27	21	0.356	0.57	0.11	1.56	
	PW15-10-221027	2022 10 27	-	10.6	6.87	26	-	-	< 50	439	< 10	-	5.09	< 100	1.4	5.5	5.5	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	44.8	2.94	177	0.796	2.71	0.31	2.88	
	PW15-10-230222	2023 02 22	-	4.99	6.67	28	-	-	< 50	21	< 10	-	3.24	< 100	1.3	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	106	1.30	25	0.426	0.86	0.11	2.11	
	PW15-10-230801	2023 08 01	-	7.31	6.95	< 15	-	-	< 50	111	< 10	-	2.31	< 100	1.4	5.2	5.2	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	34.4	2.11	74.4	0.494	0.958	0.248	2.02	
PW15-10-231114	2023 11 14	-	5.98	6.27	32	-	-	< 50	30	< 10	-	3.59	< 100	1.3	3.3	3.3	< 1.0	< 1.0	< 1.0	< 0.1	-	2.1	-	118	1.55	30.6	0.511	0.929	0.112	2.41		
PW15-11	PW15-11-151123	2015 11 23	6.48	14.8	6.99	23	478	-	45	87	< 10	-	3.58	< 100	1.6	17	17	< 1	< 1	< 1	< 0.1	-	-	< 2	24	4	25	1.16	1	0.24	4.13	
	PW15-11-160630	2016 06 29	6.80	7.60	7.09	-	-	-	-	70	< 10	-	3.41	< 100	1.3	5	5	< 1	< 1	< 1	< 0.10	-	-	< 2	65	2.2	160	0.52	1.8	0.23	2.15	
	PW15-11-160907	2016 09 07	6.91	9.38	-	-	-	-	-	147	< 10	-	4.11	< 100	1.4	7	7	< 1	< 1	< 1	< 0.10	-	-	< 2	44	2.8	511	0.59	2.0	0.32	2.39	
	PW15-11-161018	2016 10 18	6.45	28.9	6.09	-	-	-	-	784	< 10	-	5.37	< 100	11.1	16	16	< 1	< 1	< 1	-	-	-	4	154	7.9	154	2.21	45.0	1.42	4.36	
	PW15-11-161213	2016 12 13	6.95	17.7	6.8	-	-	-	-	263	< 5	263	3.91	< 100	5.5	15	15	< 1	< 1	< 1	-	-	-	3	18	5.0	15	1.26	0.7	0.32	3.97	
	PW-A-161213	Duplicate	6.43	18.0	6.8	-	-	-	-	244	< 5	244	4.22	< 100	5.2	14	14	< 1	< 1	< 1	-	-	-	< 2	19	5.1	16	1.28	0.6	0.33	4.01	
	QA/QC RPD%		8	2	0	-	-	-	-	7	*	7	8	*	6	7	7	*	*	*	-	-	-	*	*	*	2	*	2	*	3	1
	PW15-11-170801	2017 08 01	6.76	8.62	6.48	-	-	-	-	86	< 10	-	3.52	160	1.2	5.2	5.2	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	70.2	2.52	150	0.563	2.19	0.25	2.20	
	PW15-11-170822	2017 08 22	6.80	7.17	6.2	-	-	-	-	127	< 10	-	2.28	< 100	1.2	6.5	6.5	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	53.6	2.04	174	0.501	1.98	-	1.84	
	PW15-11-170927	2017 09 27	6.35	8.13	7.53	-	-	-	-	81	< 10	-	2.81	< 100	1.1	6.2	6.2	< 1.0	< 1.0	< 1.0	-	-	-	3.0	42.4	2.33	240	0.560	2.22	0.29	2.07	
	PW15-11-171101	2017 11 01	6.09	7.82	6.23	-	-	-	-	60	< 10	-	3.58	140	1.8	5.3	5.3	< 1.0	< 1.0	< 1.0	-	-	-	4.1	89.8	2.20	78	0.563	3.24	0.23	2.10	
	PW15-11-171123	2017 11 23	6.77	21.4	6.11	-	-	-	-	340	< 10	-	5.01	< 100	3.4	20.2	20.2	< 1.0	< 1.0	< 1.0	-	-	-	6.0	38.0	5.91	32	1.61	1.21	0.61	4.00	
	PW-A-171123	Duplicate	6.71	20.9	6.11	-	-	-	-	319	< 10	-	5.50	< 100	3.4	17.6	17.6	< 1.0	< 1.0	< 1.0	-	-	-	10.8	36.7	5.80	30	1.56	1.12	0.59	3.89	
	QA/QC RPD%		1	2	0	-	-	-	-	6	*	-	9	*	*	14	14	*	*	*	-	-	-	*	3	2	*	3	8	3	3	
	PW15-11-171213	2017 12 13	6.44	8.97	7.12	-	-	-	-	52	< 10	-	2.48	< 100	1.8	8.2	8.2	< 1.0	< 1.0	< 1.0	-	-	-	3.0	86.0	2.45	67	0.691	2.82	0.19	2.47	
	PW15-11-180309	2018 03 09	6.39	10.9	6.23	-	-	-	-	69.3	< 5.0	69.3	3.13	< 100	2.5	8.5	8.5	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	38.1	3.23	42	0.677	1.85	0.22	3.07	
	PW15-11-180724	2018 07 24	6.35	6.82	7.13	-	-	-	-	60	< 10	-	6.71	380	1.6	2.4	2.4	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	68.1	1.98	128	0.457	3.03	0.26	1.80	
	PW15-11-181031	2018 10 31	6.30	8.78	5.56	95	8,250	-	-	< 10	< 10	-	4.71	< 100	1.8	14.8	14.8	< 1.0	< 1.0	< 1.0	< 0.10	-	-	17.6	259	2.39	180	0.682	26.4	0.48	2.29	
	PW15-11-190124	2019 01 24	6.53	7.44	6.44	55	81.2	-	-	23	< 10	-	3.01	< 100	1.9	4.8	4.8	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.8	98.3	2.12	62	0.519	3.89	0.13	2.10	
	PW15-11-190924	2019 09 24	6.60	13.2	6.16	33	9.2	-	-	64	< 10	-	3.05	< 100	1.9	7.5	7.5	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.2	83.3	4.05	118	0.752	2.83	0.29	3.28	
	PW15-11-191203	2019 12 03	7.03	14.4	6.5	45	28.3	-	-	44	< 10	-	3.34	< 100	3.2	12.1	12.1	< 1.0	< 1.0	< 1.0	< 0.10	-	-	3.2	54.2	4.18	74	0.961	1.89	0.26	3.19	
	PW15-11-200730	2020 07 30	6.98	7.45	6.38	28	-	-	-	89.8	< 1.0	-	2.91	< 20	1.23	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	2.2	-	-	77.5	2.12	207	0.527	3.23	0.213	2.05	
	PW15-11-210812	2021 08 12	-	-	6.88	< 15	-	-	-	154	< 10	-	3.08	< 100	1.2	7.5	7.5	< 1.0	< 1.0	< 1.0	< 0.10	-	2.8	-	63.2	2.69	483	0.642	2.23	0.30	2.51	
	PW15-11-220210	2022 02 10	-	-	6.72	19	-	-	< 50	24.2	< 5.0	24.2	2.06	< 100	1.2	4.1	4.1	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	98.1	1.54	25	0.406	0.49	0.12	1.67	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.

^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^m Guideline is hardness and DOC dependent.

ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																						
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L	
BC Guideline/Standard																									
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m	
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,j}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g	
PW15-10	PW15-10-210113	2021 01 13	< 0.10	< 0.10	2.00	< 0.020	< 10	< 0.0050	0.17	0.14	<u>9.85</u>	< 0.050	< 1.0	< 0.0050	< 0.050	1.15	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.4	
	PW15-10-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.016	< 0.50	0.48	<u>24.1</u>	0.73	< 0.10	< 0.010	< 0.10	1.57	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-10-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.24	<u>12.3</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.76	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-10-221027	2022 10 27	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.010	< 0.50	0.53	<u>21.8</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.29	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	PW15-10-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.29	<u>12.6</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.72	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	PW15-10-230801	2023 08 01	< 0.050	0.073	2.01	< 0.010	5.5	0.0078	< 0.50	0.303	<u>13.8</u>	< 0.050	0.097	< 0.0050	0.031	0.671	< 0.10	< 0.010	< 0.0040	0.22	< 0.20	< 0.0010	< 1.00	1.6	
	PW15-10-231114	2023 11 14	< 0.050	0.054	1.91	< 0.010	3.3	0.0050	< 0.50	0.281	<u>12.0</u>	0.059	0.074	0.0058	0.035	0.710	< 0.10	< 0.010	< 0.0040	0.63	< 0.20	< 0.0010	< 1.00	2.0	
PW15-11	PW15-11-151123	2015 11 23	< 0.1	< 0.5	< 5	< 0.1	7	< 0.01	< 0.5	0.08	<u>10.3</u>	< 0.1	< 0.1	< 0.02	< 0.1	0.3	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	PW15-11-160630	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	< 0.5	0.39	<u>13.2</u>	< 0.1	< 0.1	< 0.02	< 0.1	0.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	PW15-11-160907	2016 09 07	< 0.1	< 0.5	< 5	< 0.1	8	< 0.01	< 0.5	0.41	<u>11.4</u>	< 0.1	0.1	< 0.02	< 0.1	0.9	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	PW15-11-161018	2016 10 18	< 0.1	< 0.5	7	< 0.1	10	0.05	1.0	<u>8.66</u>	<u>108</u>	< 0.1	0.2	< 0.02	0.2	9.4	0.8	< 0.05	< 0.02	5	-	0.03	< 1	<u>12</u>	
	PW15-11-161213	2016 12 13	< 0.1	< 0.5	< 5	< 0.1	8	<u>0.10</u>	< 0.5	2.42	<u>92.8</u>	< 0.1	< 0.1	< 0.02	< 0.1	5.6	< 0.5	<u>0.06</u>	< 0.02	< 5	-	< 0.02	< 1	<u>12</u>	
	PW-A-161213	Duplicate	< 0.1	< 0.5	< 5	< 0.1	8	<u>0.10</u>	< 0.5	2.49	<u>92.0</u>	< 0.1	< 0.1	< 0.02	0.1	5.8	< 0.5	<u>0.07</u>	< 0.02	< 5	-	< 0.02	< 1	<u>11</u>	
	QA/QC RPD%		*	*	*	*	*	0	*	3	1	*	*	*	*	4	*	*	*	*	-	*	*	*	
	PW15-11-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	6.7	< 0.010	0.79	0.43	<u>14.7</u>	< 0.10	0.12	< 0.010	< 0.10	0.87	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	PW15-11-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	19.6	< 0.010	< 0.50	0.34	<u>13.3</u>	< 0.20	0.17	< 0.010	< 0.10	0.95	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	PW15-11-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	8.6	< 0.010	< 0.50	0.39	<u>12.6</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.87	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	PW15-11-171101	2017 11 01	0.81	< 0.50	< 5.0	< 0.10	5.8	0.013	< 0.50	0.60	<u>33.0</u>	<u>6.20</u>	< 0.10	< 0.010	< 0.10	1.37	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.4	
	PW15-11-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	7.6	0.011	0.73	0.60	<u>63.2</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW-A-171123	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	6.7	0.012	0.50	0.58	<u>61.6</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	QA/QC RPD%		*	*	*	*	*	*	*	3	3	*	*	*	*	*	*	*	*	*	*	*	*	*	
	PW15-11-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.013	0.62	0.60	<u>27.1</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.34	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-11-180309	2018 03 09	< 0.20	< 0.50	< 5.0	< 0.10	5.9	<u>0.065</u>	< 0.50	1.51	<u>94.2</u>	< 0.20	0.10	< 0.010	0.12	3.88	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>11.6</u>	
	PW15-11-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	5.0	0.012	< 0.50	0.55	<u>21.7</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.18	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-11-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.054</u>	< 0.50	2.41	<u>180</u>	1.42	0.12	< 0.010	< 0.10	4.54	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.2	
	PW15-11-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.012	< 0.50	0.71	<u>24.1</u>	0.95	0.11	< 0.010	< 0.10	2.67	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW15-11-190924	2019 09 24	< 0.20	< 0.50	< 5.0	< 0.10	8.8	<u>0.104</u>	< 0.50	1.65	<u>167</u>	0.25	0.14	< 0.010	< 0.10	5.35	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>9.9</u>	
	PW15-11-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	12.0	<u>0.072</u>	< 0.50	1.90	<u>82.0</u>	0.63	< 0.10	< 0.010	< 0.10	4.81	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>9.8</u>	
	PW15-11-200730	2020 07 30	0.19	0.12	1.63	< 0.020	< 10	0.0081	0.27	0.45	<u>14.3</u>	0.367	< 1.0	< 0.0050	0.086	0.90	0.065	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.4	
	PW15-11-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.36	<u>11.1</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.86	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
		PW15-11-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.20	<u>8.24</u>	< 0.20	0.12	< 0.010	< 0.10	1.07	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD
SHADOW
SHADED

Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.
^d Standard to protect freshwater aquatic life.
^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^f Guideline/standard varies with Chloride.
^g Guideline/standard varies with Hardness.
^h Guideline/standard varies with pH.
ⁱ Guideline is temperature, pH, DOC and hardness dependent.
^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.
^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.
^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^m Guideline is hardness and DOC dependent.
ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																							
			pH	Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Acidity (as CaCO3)	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^g	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	767-2,338 ^g	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
PW15-11	PW15-11-221026	2022 10 26	-	9.73	7.09	44	-	-	< 50	358	< 10	-	4.64	< 100	1.5	5.7	5.7	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	48.4	2.66	251	0.750	2.04	0.34	2.83	
	PW-A-221026	Duplicate	-	10.3	-	48	-	-	< 50	370	< 10	-	4.64	< 100	1.5	5.7	5.7	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	45.8	2.90	242	0.747	1.92	0.34	2.81	
	QA/QC RPD%		-	6	-	*	-	-	*	3	*	-	0	*	*	0	0	*	*	*	*	-	*	-	6	9	4	0	6	*	1	
	PW15-11-230222	2023 02 22	-	5.57	6.72	38	-	-	66	28	< 10	-	3.23	< 100	1.1	2.7	2.7	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	101	1.47	36	0.460	1.01	0.13	2.15	
	PW15-11-230801	2023 08 01	-	8.03	6.74	< 15	-	-	92	123	< 10	-	2.34	< 100	1.6	6.1	6.1	< 1.0	< 1.0	< 1.0	< 0.10	-	2.0	-	53.0	2.29	92.1	0.558	3.20	0.272	2.09	
	PW-B-230801	Duplicate	-	7.95	6.74	< 15	-	-	< 50	118	< 10	-	2.60	< 100	1.6	5.9	5.9	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	52.1	2.29	91.9	0.54	3.18	0.274	2.03	
	QA/QC RPD%		-	1	*	*	-	-	*	4	*	-	11	*	*	3	3	*	*	*	*	-	*	-	2	0	0	3	1	1	3	
PW15-12	PW15-11-231114	2023 11 14	-	8.94	5.97	52	-	-	< 50	32	< 10	-	3.14	< 100	1.9	6.5	6.5	< 1.0	< 1.0	< 1.0	< 0.1	-	3.5	-	141	2.56	103	0.614	5.35	0.228	2.27	
	PW15-12-151123	2015 11 23	5.6	4.01	6.97	< 10	152	-	< 20	25	< 10	-	1.45	< 100	< 1	3	3	< 1	< 1	< 1	< 0.1	-	-	2	153	1	100	0.34	6.8	0.13	1.41	
	PW15-12-160630	2016 06 29	6.87	31.5	6.91	-	-	-	-	65	< 10	-	79.9	< 100	11.3	5	5	< 1	< 1	< 1	0.26	-	-	< 2	53	3.4	75	5.60	1.8	1.89	48.1	
	PW15-12-160906	2016 09 06	5.69	8.74	-	-	-	-	-	146	< 10	-	4.14	< 100	1.5	10	10	< 1	< 1	< 1	< 0.10	-	-	18	55	1.9	498	0.96	6.0	0.45	3.01	
	PW15-12-161017	2016 10 17	6.26	5.86	5.8	-	-	-	-	< 10	< 10	-	4.06	< 100	1.3	3	3	< 1	< 1	< 1	-	-	-	3	264	1.4	228	0.56	17.4	0.41	1.80	
	PW15-12-161213	2016 12 13	5.94	394	6.3	-	-	-	-	54	< 5	54	770	< 100	78.6	8	8	< 1	< 1	< 1	-	-	-	4	107	41.6	33	70.3	13.3	14.6	462	
	PW15-12-170801	2017 08 01	6.75	7.12	6.49	-	-	-	-	82	< 10	-	3.99	< 100	1.4	5.1	5.1	< 1.0	< 1.0	< 1.0	-	-	-	2.2	78.1	1.72	155	0.684	6.02	0.30	2.38	
	PW15-12-170821	2017 08 21	6.78	36.3	7.25	-	-	-	-	132	< 10	-	96.8	160	14.1	6.4	6.4	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	36.9	3.54	66	6.66	2.43	-	57.3	
	PW15-12-170927	2017 09 27	6.37	7.92	6.99	-	-	-	-	80	< 10	-	3.10	< 100	1.2	6.9	6.9	< 1.0	< 1.0	< 1.0	-	-	-	3.1	46.6	2.16	242	0.615	3.59	0.30	2.24	
	PW15-12-171101	2017 11 01	6.02	164	6.58	-	-	-	-	89	< 10	-	489	< 100	70.3	5.8	5.8	< 1.0	< 1.0	< 1.0	-	-	-	5.6	67.1	13.0	26	31.8	2.58	9.86	261	
	PW15-12-171122	2017 11 22	6.17	3.81	7.4	-	-	-	-	17	< 10	-	4.27	140	< 1.0	1.7	1.7	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	146	1.01	44	0.313	2.98	0.18	1.22	
	PW15-12-171212	2017 12 12	6.34	39.2	6.74	-	-	-	-	42	< 10	-	165	480	26.0	8.6	8.6	< 1.0	< 1.0	< 1.0	-	-	-	4.5	62.0	3.69	35	7.29	109	5.19	108	
PW17-14	PW17-14-170731	2017 07 31	6.76	6.74	6.63	-	-	-	-	138	< 10	-	2.37	< 100	1.0	15.9	15.9	< 1.0	< 1.0	< 1.0	-	-	-	2.0	68.7	1.91	140	0.474	2.74	0.24	1.81	
	PW17-14-170821	2017 08 21	6.68	6.58	7.36	-	-	-	-	94	< 10	-	1.41	< 100	< 1.0	4.9	4.9	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	62.2	1.90	206	0.446	2.27	-	1.57	
	PW17-14-170926	2017 09 26	6.42	7.27	7.36	-	-	-	-	74	< 10	-	1.94	< 100	< 1.0	7.9	7.9	< 1.0	< 1.0	< 1.0	-	-	-	3.0	44.0	2.06	243	0.515	2.27	0.28	1.79	
	PW17-14-171031	2017 10 31	6.19	6.42	6.31	-	-	-	-	81	< 10	-	5.76	320	1.0	5.6	5.6	< 1.0	< 1.0	< 1.0	-	-	-	3.2	93.3	1.72	63	0.515	1.49	0.19	1.89	
	PW-B-171031	Duplicate	6.01	6.36	6.31	-	-	-	-	104	< 10	-	3.31	< 100	1.1	4.0	4.0	< 1.0	< 1.0	< 1.0	-	-	-	3.9	92.7	1.71	62	0.508	1.44	0.19	1.84	
	QA/QC RPD%		3	1	0	-	-	-	-	25	*	-	54	*	*	*	*	*	*	*	-	-	-	*	1	1	2	1	3	0	3	
PW17-16	PW17-16-170801	2017 08 01	6.67	6.98	5.67	-	-	-	-	122	< 10	-	2.97	< 100	1.3	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	72.1	2.00	144	0.481	2.30	0.23	1.94	
	PW17-16-170926	2017 09 26	6.35	8.09	6.91	-	-	-	-	143	< 10	-	8.51	840	1.8	6.5	6.5	< 1.0	< 1.0	< 1.0	-	-	-	3.4	44.5	2.33	209	0.554	2.78	0.29	2.06	
	PW17-16-171031	2017 10 31	6.03	7.01	5.9	-	-	-	-	81	< 10	-	6.90	430	1.6	4.6	4.6	< 1.0	< 1.0	< 1.0	-	-	-	4.7	95.1	1.93	80	0.533	3.45	0.19	2.08	
	PW17-16-171213	2017 12 13	5.29	76.5	5.45	-	-	-	-	310	< 10	-	3.04	< 100	143	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	-	-	-	17.1	1,160	25.1	1,180	3.36	148	0.77	7.25	
	PW-A-171213	Duplicate	5.29	78.1	5.45	-	-	-	-	260	< 10	-	3.04	< 100	128	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	-	-	-	14.8	1,110	25.6	988	3.43	150	0.80	7.51	
QA/QC RPD%			0	2	0	-	-	-	-	18	*	-	0	*	11	*	*	*	*	*	-	-	-	14	4	2	18	2	1	4	4	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.

^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^m Guideline is hardness and DOC dependent.

ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,j}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g
PW15-11	PW15-11-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.36	16.0	< 0.20	< 0.10	< 0.010	< 0.10	0.90	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	PW-A-221026	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.36	15.6	< 0.20	< 0.10	< 0.010	< 0.10	0.85	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	3	*	*	*	*	*	*	*	*	*	*	*	*	*
	PW15-11-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.011	< 0.50	0.29	25.9	0.23	< 0.10	< 0.010	< 0.10	0.73	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	PW15-11-230801	2023 08 01	< 0.050	0.067	2.22	< 0.010	5.2	0.0125	< 0.50	0.436	15.7	0.075	0.109	< 0.0050	0.041	0.872	< 0.10	< 0.010	< 0.0040	1.12	< 0.20	0.0010	< 1.00	3.4
	PW-B-230801	Duplicate	< 0.050	0.078	2.19	< 0.010	4.6	0.0103	< 0.50	0.438	15.7	0.077	0.101	< 0.0050	0.042	0.850	< 0.10	< 0.010	< 0.0040	1.48	< 0.20	< 0.0010	< 1.00	1.9
	QA/QC RPD%		*	*	1	*	*	19	*	0	0	*	*	*	*	3	*	*	*	28	*	*	*	*
	PW15-11-231114	2023 11 14	< 0.050	0.054	1.24	< 0.010	3.4	0.0537	< 0.50	1.13	118	0.174	0.087	< 0.0050	0.082	2.77	< 0.10	0.017	< 0.0040	3.93	< 0.20	0.0022	< 1.00	6.3
PW15-12	PW15-12-151123	2015 11 23	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	< 0.5	0.21	5.6	< 0.1	0.2	< 0.02	< 0.1	0.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW15-12-160630	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	22	< 0.01	0.6	0.35	12.6	< 0.1	0.9	< 0.02	< 0.1	0.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW15-12-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	9	< 0.01	< 0.5	0.28	10.8	< 0.1	0.6	< 0.02	0.1	0.8	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	PW15-12-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	8	< 0.01	0.5	0.40	6.1	< 0.1	0.3	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	9	-	< 0.02	< 1	6
	PW15-12-161213	2016 12 13	< 0.1	< 0.5	32	< 0.1	112	0.05	< 0.5	0.51	5.8	< 0.1	4.6	< 0.02	< 0.1	5.0	< 0.5	0.07	< 0.02	< 5	-	< 0.02	< 1	9
	PW15-12-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	12.5	< 0.010	0.58	0.41	13.4	< 0.10	0.53	< 0.010	< 0.10	0.70	< 0.50	< 0.050	< 0.020	< 5.0	-	0.025	< 1.0	< 4.0
	PW15-12-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	39.7	< 0.010	< 0.50	0.29	9.75	< 0.20	1.29	< 0.010	0.12	0.75	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW15-12-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	8.0	< 0.010	< 0.50	0.35	11.7	< 0.20	0.23	< 0.010	< 0.10	0.70	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW15-12-171101	2017 11 01	< 0.20	< 0.50	8.0	< 0.10	106	0.012	< 0.50	0.36	10.5	< 0.20	4.28	< 0.010	0.27	1.16	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	0.066	< 1.0	< 4.0
	PW15-12-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	2.47	< 0.20	0.15	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW15-12-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	56.4	0.012	< 0.50	1.20	6.05	< 0.20	2.83	< 0.010	0.11	0.87	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	0.030	< 1.0	5.9
	PW17-14-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	7.7	< 0.010	< 0.50	< 0.10	2.06	< 0.10	0.13	< 0.010	< 0.10	0.69	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	4.9
PW17-14	PW17-14-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	7.1	< 0.010	< 0.50	< 0.10	1.82	< 0.20	0.13	< 0.010	< 0.10	0.48	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-14-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	8.2	< 0.010	0.59	< 0.10	1.54	< 0.20	< 0.10	< 0.010	< 0.10	0.46	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-14-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	5.5	< 0.010	< 0.50	< 0.10	2.20	< 0.20	< 0.10	< 0.010	< 0.10	0.54	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW-B-171031	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	5.1	< 0.010	0.51	< 0.10	2.18	< 0.20	< 0.10	< 0.010	< 0.10	0.84	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	*
	PW17-16	PW17-16-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	6.8	< 0.010	0.61	0.39	18.2	< 0.10	0.12	< 0.010	< 0.10	0.86	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0
PW17-16	PW17-16-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	6.8	< 0.010	< 0.50	0.39	16.9	< 0.20	< 0.10	< 0.010	< 0.10	0.98	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-16-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	6.9	< 0.010	< 0.50	0.48	34.2	1.03	< 0.10	< 0.010	< 0.10	1.17	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-16-171213	2017 12 13	< 0.20	< 0.50	16.2	< 0.10	16.1	0.641	1.81	85.9	4,380	6.26	0.67	< 0.010	< 0.10	138	1.38	0.095	0.025	34.0	< 1.0	< 0.020	1.7	159
	PW-A-171213	Duplicate	< 0.20	< 0.50	16.1	< 0.10	17.5	0.675	2.30	88.5	4,510	6.27	0.63	< 0.010	< 0.10	142	0.91	0.083	< 0.020	25.0	< 1.0	< 0.020	1.7	164
	QA/QC RPD%		*	*	*	*	*	5	*	3	3	< 1	6	*	*	3	*	*	*	31	*	*	*	3

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.

^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^m Guideline is hardness and DOC dependent.

ⁿ Guideline is pH, DOC and hardness dependent.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																								
			pH	Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Acidity (as CaCO3)	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissoved Potassium	Dissoved Sodium		
BC Guideline/Standard																																	
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^h	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	n/a	767-2,338 ^h	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
PW17-17	PW17-17-170801	2017 08 01	6.69	7.08	6.5	-	-	-	-	64	< 10	-	4.36	110	1.2	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	68.5	2.04	126	0.482	2.61	0.24	1.94		
	PW17-17-170822	2017 08 22	6.71	7.10	6.58	-	-	-	-	94	< 10	-	2.20	< 100	1.2	5.6	5.6	< 1.0	< 1.0	< 1.0	-	-	-	12.6	55.4	2.03	188	0.491	1.93	-	1.89		
	PW17-17-170927	2017 09 27	6.39	7.92	7.22	-	-	-	-	112	< 10	-	7.06	170	1.3	6.7	6.7	< 1.0	< 1.0	< 1.0	-	-	-	3.0	44.4	2.28	240	0.539	2.62	0.28	2.02		
	PW17-17-171101	2017 11 01	6.12	7.24	6.82	-	-	-	-	77	< 10	-	4.88	280	1.7	6.1	6.1	< 1.0	< 1.0	< 1.0	-	-	-	3.7	88.6	2.01	60	0.538	1.66	0.19	2.05		
	PW17-17-171213	2017 12 13	6.38	10.5	6.93	-	-	-	-	78	< 10	-	2.66	< 100	2.5	8.8	8.8	< 1.0	< 1.0	< 1.0	-	-	-	3.4	53.7	2.91	26	0.780	1.19	0.20	2.74		
	PW17-17-180724	2018 07 24	6.56	9.42	7.8	-	-	-	-	75	< 10	-	4.93	220	1.9	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	-	2.2	39.7	2.74	65	0.623	1.18	0.25	2.17		
	PW-A-180724	Duplicate	6.52	9.63	7.8	-	-	-	-	79	< 10	-	3.06	< 100	1.6	5.4	5.4	< 1.0	< 1.0	< 1.0	-	-	-	2.0	26.9	2.83	62	0.618	1.12	0.24	2.16		
	QA/QC RPD%		1	2	*	-	-	-	-	5	*	-	47	*	*	11	11	*	*	*	*	-	-	-	*	38	3	5	1	5	*	< 1	
	PW17-17-181101	2018 11 01	6.29	5.81	6.74	38	8.0	-	-	50	< 10	-	3.50	< 100	< 1.0	2.0	2.0	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.4	182	1.48	121	0.512	10.3	0.34	2.37		
	PW17-17-190124	2019 01 24	6.55	10.1	6.23	80	66.7	-	-	127	< 10	-	5.56	< 100	2.3	7.8	7.8	< 1.0	< 1.0	< 1.0	< 0.10	-	-	4.0	63.9	2.88	88	0.698	4.93	0.21	3.09		
	PW17-17-190924	2019 09 24	6.48	9.02	6.36	25	8.0	-	-	55	< 10	-	3.08	< 100	1.3	4.1	4.1	< 1.0	< 1.0	< 1.0	< 0.10	-	-	< 2.0	115	2.57	192	0.630	4.73	0.26	3.05		
	PW17-17-191203	2019 12 03	7.17	17.8	6.51	56	2.6	-	-	72	< 10	-	3.53	< 100	3.3	17.6	17.6	< 1.0	< 1.0	< 1.0	< 0.10	-	-	3.2	14.6	5.11	13	1.22	0.72	0.29	3.70		
	PW17-17-200730	2020 07 30	7.02	7.37	5.7	24	-	3.18	-	94.3	< 1.0	-	2.93	< 20	1.21	5.9	5.9	< 1.0	< 1.0	< 1.0	< 0.050	2.2	-	-	62.5	2.09	171	0.524	2.04	0.204	2.06		
	PW17-17-210112	2021 01 12	-	3.90	5.59	22	10,000	-	< 5.0	40.4	< 1.0	-	2.18	< 20	0.81	3.2	3.2	< 1.0	< 1.0	< 1.0	< 0.050	3.5	-	-	135	1.01	53	0.336	3.80	0.166	1.43		
	PW17-B-210112	Duplicate	-	3.99	5.59	26	-	-	< 5.0	37.9	< 1.0	-	2.19	< 20	0.82	3.2	3.2	< 1.0	< 1.0	< 1.0	< 0.050	2.2	-	-	139	1.04	52	0.339	3.82	0.180	1.48		
	QA/QC RPD%		-	2	0	*	-	-	*	6	*	-	*	*	*	*	*	*	*	*	*	*	-	-	*	3	3	2	*	1	*	3	
	PW17-17-210812	2021 08 12	-	-	6.82	< 15	-	-	-	152	< 10	-	2.99	< 100	1.3	8.1	8.1	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.8	41.1	2.66	308	0.624	1.34	0.29	2.42		
	PW-A-210812	Duplicate	-	-	6.82	< 15	-	-	-	152	< 10	-	3.02	< 100	1.2	7.8	7.8	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.7	47.2	2.69	365	0.674	1.49	0.30	2.51		
	QA/QC RPD%		-	-	0	*	*	-	-	0	*	-	1	*	*	4	4	*	*	*	*	*	-	-	*	14	1	17	8	11	*	4	
	PW17-17-220210	2022 02 10	-	-	6.68	< 15	-	-	< 50	24.5	< 5.0	24.5	2.07	< 100	1.2	3.9	3.9	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.2	89.2	1.39	21	0.388	0.51	0.11	1.61		
PW17-17-221026	2022 10 26	-	10.3	7.21	55	-	-	< 50	339	< 10	-	4.62	< 100	1.5	5.7	5.7	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	52.8	2.85	254	0.778	2.96	0.35	2.85			
PW17-17-220222	2023 02 22	-	10.1	6.66	38	-	-	< 50	53	< 10	-	3.39	< 100	2.3	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	41.5	2.98	29	0.641	0.69	0.18	2.42			
PW17-17-230801	2023 08 01	-	7.55	6.75	15	-	-	< 50	105	< 10	-	2.33	< 100	1.4	6.0	6.0	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	-	24.3	2.2	52.5	0.497	1.36	0.241	2.07			
PW17-17-231114	2023 11 14	-	27.9	5.92	12	-	-	< 50	297	< 10	-	3.93	< 100	11.6	17.3	17.3	< 1.0	< 1.0	< 1.0	< 0.1	-	8.1	-	21.2	8.26	16.7	1.75	0.712	0.344	4.11			
PW17-18	PW17-18-170801	2017 08 01	6.81	6.49	6.49	-	-	-	-	67	< 10	-	5.14	< 100	1.3	5.4	5.4	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	74.1	1.74	130	0.517	3.09	0.34	3.00		
PW17-19	PW17-19-170822	2017 08 22	6.74	6.80	6.95	-	-	-	-	106	< 10	-	2.21	< 100	1.2	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	-	2.4	45.2	1.88	114	0.513	2.00	-	1.79		
	PW17-19-170927	2017 09 27	6.37	7.93	7.59	-	-	-	-	93	< 10	-	5.17	570	1.3	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	-	3.3	44.3	2.27	228	0.545	3.56	0.30	2.05		
	PW17-19-171101	2017 11 01	6.05	7.43	5.89	-	-	-	-	86	< 10	-	5.78	300	1.8	5.1	5.1	< 1.0	< 1.0	< 1.0	-	-	-	4.6	76.0	2.09	55	0.535	3.02	0.20	2.11		
	PW17-19-180309	2018 03 09	6.17	5.25	5.92	-	-	-	-	33.0	< 5.0	33.0	3.20	< 100	4.1	3.2	3.2	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	100	1.46	92	0.387	6.38	0.16	1.76		
	PW17-19-180724	2018 07 24	6.31	7.36	6.3	-	-	-	-	62	< 10	-	4.56	230	2.3	3.3	3.3	< 1.0	< 1.0	< 1.0	-	-	-	4.5	74.1	2.04	103	0.547	3.69	0.20	1.85		
	PW17-19-181031	2018 10 31	5.63	4.77	5.11	69	696	-	-	< 10	< 10	-	4.70	< 100	1.2	3.1	3.1	< 1.0	< 1.0	< 1.0	< 0.10	-	-	15.2	256	1.23	185	0.414	12.2	0.34	2.24		
	PW17-19-190124	2019 01 24	6.44	5.43	6.49	63	194	-	-	12	< 10	-	2.97	< 100	1.1	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.7	84.3	1.56	44	0.375	2.42	0.13	1.92		
	PW17-19-190925	2019 09 25	6.39	7.66	6.39	28	94.0	-	-	42	< 10	-	2.70	< 100	1.0	3.2	3.2	< 1.0	< 1.0	< 1.0	< 0.10	-	-	< 2.0	141	2.24	197	0.501	9.44	0.18	2.11		
	PW17-19-191204	2019 12 04	6.53	4.45	6.03	42	57.1	-	-	< 10	< 10	-	2.33	< 100	1.1	9.1	9.1	< 1.0	< 1.0	< 1.0	< 0.10	-	-	3.2	185	1.16	80	0.377	2.58	0.12	1.62		

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 810202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline

RPD Denotes relative percent difference

* RPDs are not calculated where one or more concentrations are less than five times RDL

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic")

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH

ⁱ Guideline is temperature, pH, DOC

^j Total Mercury guideline is based on the % of MethylMercury.

of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg

^k There is no Zinc standard specified for $H \geq 400$; therefore, the standard for $H=30$

¹ Guideline is for Antimony (III and V) Guideline came into affect November 17, 2023

^m Guideline is hardness and DOC dependent

ⁿ Guideline is pH, DOC and hardness dependent

...and the following expression:

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,i}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g
PW17-17	PW17-17-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	6.8	< 0.010	< 0.50	0.38	<u>13.8</u>	< 0.10	0.12	< 0.010	< 0.10	1.05	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-17-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	7.5	< 0.010	< 0.50	0.37	<u>13.4</u>	< 0.20	0.13	< 0.010	< 0.10	0.92	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-17-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	6.4	< 0.010	< 0.50	0.37	<u>12.7</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.08	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-17-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	7.1	< 0.010	0.99	0.39	<u>14.4</u>	< 0.20	< 0.10	< 0.010	< 0.10	2.15	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-17-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.042</u>	0.67	0.94	<u>45.7</u>	< 0.20	< 0.10	< 0.010	< 0.10	2.39	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.3
	PW17-17-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	5.1	<u>0.043</u>	< 0.50	1.13	<u>59.7</u>	< 0.20	< 0.10	< 0.010	< 0.10	2.85	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>6.2</u>
	PW-A-180724	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	9.2	<u>0.039</u>	< 0.50	1.15	<u>60.4</u>	< 0.20	< 0.10	< 0.010	< 0.10	2.67	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	5.5
	QA/QC RPD%		*	*	*	*	*	*	*	2	1	*	*	*	*	7	*	*	*	*	*	*	*	*
	PW17-17-181101	2018 11 01	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.045</u>	0.55	0.35	<u>45.3</u>	< 0.20	0.12	< 0.010	< 0.10	1.23	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-17-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.047</u>	< 0.50	1.10	<u>75.7</u>	< 0.20	0.12	< 0.010	< 0.10	6.89	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>7.4</u>
	PW17-17-190924	2019 09 24	< 0.20	< 0.50	< 5.0	< 0.10	7.3	0.030	< 0.50	0.63	<u>53.5</u>	< 0.20	0.12	< 0.010	< 0.10	3.53	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-17-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	11.7	<u>0.075</u>	< 0.50	2.38	<u>110</u>	0.41	< 0.10	< 0.010	< 0.10	5.58	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>10.5</u>
	PW17-17-200730	2020 07 30	< 0.10	0.11	1.68	< 0.020	< 10	< 0.0050	0.20	0.29	<u>13.7</u>	< 0.050	< 1.0	< 0.0050	< 0.050	0.69	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	2.0
	PW17-17-210112	2021 01 12	< 0.10	< 0.10	1.66	< 0.020	< 10	0.0050	0.38	0.32	<u>13.0</u>	< 0.050	< 1.0	< 0.0050	< 0.050	5.00	< 0.0500	< 0.010	< 0.010	< 10.0	0.18	< 0.010	< 0.50	1.5
	PW17-B-210112	Duplicate	< 0.10	< 0.10	1.75	< 0.020	< 10	< 0.0050	0.33	0.34	<u>13.4</u>	< 0.050	< 1.0	< 0.0050	< 0.050	5.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.2
	QA/QC RPD%		*	*	5	*	*	*	*	*	3	*	*	*	*	10	*	*	*	*	*	*	*	*
	PW17-17-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.33	<u>10.0</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.95	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW-A-210812	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.34	<u>10.1</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.06	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	*
	PW17-17-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.23	<u>9.76</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.70	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
PW17-17-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.012	< 0.50	0.50	<u>14.8</u>	< 0.20	0.11	< 0.010	< 0.10	1.03	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
PW17-17-220222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	<u>0.063</u>	< 0.50	1.21	<u>131</u>	< 0.20	< 0.10	< 0.010	< 0.10	3.18	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	<u>6.1</u>	
PW17-17-230801	2023 08 01	< 0.050	0.059	0.43	< 0.010	5.3	0.0292	< 0.50	0.576	<u>43.6</u>	0.063	0.081	< 0.0050	0.053	1.77	< 0.10	< 0.010	< 0.0040	0.26	< 0.20	< 0.0010	< 1.00	3.9	
PW17-17-231114	2023 11 14	< 0.050	< 0.050	1.35	< 0.010	6.7	<u>0.159</u>	< 0.50	3.94	<u>159</u>	< 0.050	0.131	< 0.0050	0.073	9.35	0.82	0.011	< 0.0040	0.22	< 0.20	0.0012	< 1.00	<u>18.1</u>	
PW17-18	PW17-18-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	7.5	< 0.010	< 0.50	0.34	<u>13.7</u>	< 0.10	0.52	< 0.010	< 0.10	0.88	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
PW17-19	PW17-19-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	7.2	0.016	< 0.50	0.22	<u>20.3</u>	< 0.20	0.10	< 0.010	< 0.10	0.92	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-19-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	6.5	< 0.010	< 0.50	0.39	<u>11.7</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.94	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	PW17-19-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	7.4	0.012	< 0.50	0.32	<u>16.6</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.11	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-19-180309	2018 03 09	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.021	< 0.50	0.72	<u>28.9</u>	1.46	< 0.10	< 0.010	< 0.10	0.82	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-19-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	5.6	0.018	< 0.50	0.45	<u>24.2</u>	0.78	< 0.10	< 0.010	< 0.10	0.96	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-19-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.029</u>	< 0.50	2.30	<u>62.3</u>	<u>5.45</u>	0.23	< 0.010	< 0.10	2.14	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-19-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.027</u>	< 0.50	0.43	<u>31.2</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.50	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.9
	PW17-19-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	7.4	0.022	< 0.50	0.95	<u>57.6</u>	0.93	0.15	<u>0.092</u>	< 0.10	2.59	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW17-19-191204	2019 12 04	< 0.20	< 0.50	< 5.0	< 0.10	7.1	< 0.010	< 0.50	0.36	<u>17.3</u>	0.43	0.10	< 0.010	< 0.10	0.51	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD

Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)

SHADOW

Concentration greater than BCWQG Aquatic Life Long-term Average (AW)

SHADED

Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.

^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.

^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^m Guideline is hardness and DOC dependent.

ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																							
			pH pH	Hardness mg/L	pH (field) pH	Total Dissolved Solids mg/L	Total Suspended Solids mg/L	Dissolved Organic Carbon mg/L	Ammonia, total (as N) µg/L	Nitrate (as N) µg/L	Nitrite (as N) µg/L	Nitrate+Nitrite Nitrogen µg/L	Chloride mg/L	Fluoride µg/L	Sulfate mg/L	Total Alkalinity (as CaCO3) mg/L	Alkalinity, Bicarbonate (as CaCO3) mg/L	Alkalinity, Carbonate (as CaCO3) mg/L	Alkalinity, Hydroxide (as CaCO3) mg/L	Alkalinity, Phenolphthalein (as CaCO3) mg/L	Bromide mg/L	Acidity (as CaCO3) mg/L	Total Acidity mg/L	Acidity (pH 8.3) mg/L	Dissolved Aluminum µg/L	Dissolved Calcium mg/L	Dissolved Iron µg/L	Dissolved Magnesium mg/L	Dissolved Manganese µg/L	Dissolved Potassium mg/L	Dissolved Sodium mg/L	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^g	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	767-2,338 ^g	n/a	n/a	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
PW17-19	PW17-19-210812	2021 08 12	-	-	6.78	< 15	-	-	-	156	< 10	-	2.98	< 100	1.2	6.9	6.9	< 1.0	< 1.0	< 1.0	< 0.10	-	2.7	-	62.4	2.66	503	0.631	2.07	0.33	2.50	
	PW17-19-220210	2022 02 10	-	-	6.53	< 15	-	-	< 50	< 10.0	19.1	23.1	2.05	< 100	1.7	3.7	3.7	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	-	118	1.31	55	0.355	1.56	0.12	1.55	
	PW17-19-221026	2022 10 26	-	9.76	6.5	35	-	-	< 50	276	< 10	-	4.29	< 100	1.6	5.4	5.4	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	35.9	2.66	145	0.754	2.86	0.26	2.61	
	PW17-19-230222	2023 02 22	-	4.91	7.01	33	-	-	199	17	< 10	-	3.23	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	107	1.30	43	0.402	1.53	0.12	2.08	
	PW17-19-230801	2023 08 01	-	7.48	6.49	< 15	-	-	< 50	118	< 10	-	2.35	< 100	1.5	5.5	5.5	< 1.0	< 1.0	< 1.0	< 0.10	-	2.0	-	59.4	2.1	94.5	0.542	2.64	0.251	2.03	
	PW17-19-231114	2023 11 14	-	6.67	6.25	55	-	-	< 50	30	< 10	-	3.48	< 100	1.3	4.0	4.0	< 1.0	< 1.0	< 1.0	< 0.1	-	3.6	-	150	1.85	120	0.494	3.41	0.162	2.26	
PW17-20	PW17-20-170822	2017 08 22	6.79	6.84	6.99	-	-	-	-	96	< 10	-	2.22	< 100	1.1	5.8	5.8	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	57.1	1.97	199	0.464	1.91	-	1.79	
	PW-D-170822	Duplicate	6.69	6.63	6.99	-	-	-	-	97	< 10	-	2.23	< 100	1.1	5.4	5.4	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	56.2	1.91	192	0.452	1.87	-	1.75	
	QA/QC RPD%		1	3	0	-	-	-	-	1	*	-	< 1	*	*	7	7	*	*	*	-	-	-	*	2	3	4	3	2	-	2	
PW17-21	PW17-20-180309	2018 03 09	6.09	4.89	5.64	-	-	-	-	30.2	< 5.0	30.2	3.56	< 100	< 1.0	2.2	2.2	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	97.8	1.28	27	0.412	0.79	0.14	2.00	
	PW17-21-170926	2017 09 26	6.36	7.91	6.75	-	-	-	-	75	< 10	-	2.75	< 100	1.0	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	-	2.6	42.9	2.27	251	0.539	1.83	0.28	2.04	
	PW17-21-171031	2017 10 31	6.04	7.03	6.05	-	-	-	-	461	< 10	-	6.46	370	1.5	4.7	4.7	< 1.0	< 1.0	< 1.0	-	-	-	3.8	98.6	1.92	78	0.541	3.33	0.19	2.08	
	PW17-21-171123	2017 11 23	4.96	5.91	5.34	-	-	-	-	28	< 10	-	2.16	< 100	6.3	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	-	-	-	5.3	258	1.83	54	0.326	13.9	0.15	1.35	
	PW17-21-180724	2018 07 24	6.34	6.72	6.92	-	-	-	-	61	< 10	-	2.96	< 100	1.1	2.2	2.2	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	62.8	1.97	126	0.437	1.63	0.23	1.75	
	PW17-21-181031	2018 10 31	6.35	5.68	5.74	48	21.5	-	-	28	< 10	-	4.24	< 100	1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.4	138	1.41	60	0.522	2.11	< 0.10	2.19	
	PW17-21-190124	2019 01 24	6.42	5.14	6.48	56	80.0	-	-	< 10	< 10	-	2.93	< 100	1.1	3.0	3.0	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.4	118	1.36	50	0.422	2.90	0.11	1.81	
	PW17-21-190925	2019 09 25	6.28	7.82	6.7	20	8.4	-	-	46	< 10	-	3.05	< 100	< 1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.1	146	2.13	197	0.608	2.82	0.26	2.37	
	PW17-21-191203	2019 12 03	6.64	6.77	6.26	24	7.6	-	-	18	< 10	-	3.03	< 100	1.5	4.4	4.4	< 1.0	< 1.0	< 1.0	< 0.10	-	-	< 2.0	115	1.83	40	0.535	1.11	0.14	2.12	
	PW17-21-210811	2021 08 11	-	-	7.29	44	-	-	-	141	< 10	-	3.10	< 100	1.2	8.0	8.0	< 1.0	< 1.0	< 1.0	< 0.10	-	3.9	-	60.3	2.69	487	0.633	1.94	0.29	2.48	
	PW17-21-220210	2022 02 10	-	-	6.59	15	-	-	< 50	21.3	< 5.0	21.3	2.00	< 100	1.2	3.3	3.3	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	90.3	1.22	18	0.353	0.37	0.10	1.54	
	PW-A-220210	Duplicate	-	-	6.59	< 15	-	-	< 50	20.2	< 5.0	20.2	1.98	< 100	1.1	3.7	3.7	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	89.1	1.20	18	0.358	0.31	< 0.10	1.54	
	QA/QC RPD%		-	-	0	*	-	-	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	-	1	2	*	*	*	*	0	
	PW17-21-221026	2022 10 26	-	10.3	7.15	45	-	-	< 50	372	< 10	-	4.79	< 100	1.5	5.5	5.5	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	41.0	2.87	232	0.762	1.37	0.34	2.84	
	PW17-21-230222	2023 02 22	-	4.72	6.69	29	-	-	< 50	15	< 10	-	3.22	< 100	1.1	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	104	1.20	18	0.417	0.51	< 0.10	2.08	
	PW-A-230222	Duplicate	-	4.77	6.69	33	-	-	53	16	< 10	-	3.23	< 100	1.1	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	104	1.21	17	0.424	0.50	< 0.10	2.06	
	QA/QC RPD%		-	1	*	*	-	-	*	*	*	-	0	*	*	*	*	*	*	*	*	-	*	-	0	1	*	2	*	*	1	
	PW17-21-230802	2023 08 02	-	7.44	6.89	< 15	-	-	< 50	102	< 10	-	2.34	< 100	1.3	5.3	5.3	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	40.8	2.17	92.6	0.489	1.28	0.25	1.97	
	PW17-21-231114	2023 11 14	-	6.20	6.30	60	-	-	< 50	30	< 10	-	3.59	< 100	1.4	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.1	-	3.2	-	128	1.62	51.0	0.522	2.15	0.127	2.42	
	PW-A-231114	Duplicate	-	6.06	-	40	-	-	< 50	32	< 10	-	3.59	< 100	1.4	4.1	4.1	< 1.0	< 1.0	< 1.0	< 0.1	-	5.8	-	126	1.58	51.7	0.509	2.16	0.122	2.37	
	QA/QC RPD%		-	2	-	*	-	-	*	*	*	-	0	*	*	*	*	*	*	*	*	-	*	-	2	3	1	3	0	4	2	
PW18-22	PW18-22-180726	2018 07 26	6.11	6.97	7.08	-	-	-	-	31	< 10	-	1.44	< 100	< 1.0	3.4	3.4	< 1.0	< 1.0	< 1.0	-	-	-	< 2.0	68.4	2.06	150	0.446	1.72	0.17	1.86	
	PW18-22-181031	2018 10 31	6.35	5.76	5.77	40	< 5.0	-	-	26	< 10	-	4.25	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.5	135	1.44	43	0.523	1.36	< 0.10	2.19	
	PW18-22-190124	2019 01 24	4.62	99.3	4.61	276	25.0	-	-	70	< 10	-	3.68	< 100	173	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.10	-	-	47.6	2,600	34.8	4,270	2.99	320	0.44	5.22	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.

Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.

^d Standard to protect freshwater aquatic life.

^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^f Guideline/standard varies with Chloride.

^g Guideline/standard varies with Hardness.

^h Guideline/standard varies with pH.

ⁱ Guideline is temperature, pH, DOC and hardness dependent.

^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.

^k There is no Zinc standard specified for H > 400; therefore

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																						
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L	
BC Guideline/Standard																									
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m	
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,i}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g	
PW17-19	PW17-19-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.36	<u>10.8</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.78	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-19-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.33	<u>24.7</u>	< 0.20	< 0.10	< 0.010	0.23	1.03	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-19-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.018	< 0.50	0.37	<u>14.6</u>	0.32	< 0.10	< 0.010	< 0.10	1.04	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	PW17-19-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	0.50	0.26	<u>21.4</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.81	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	5.9	
	PW17-19-230801	2023 08 01	0.053	0.068	1.21	< 0.010	5.3	0.0119	< 0.50	0.280	<u>16.2</u>	0.173	0.086	< 0.0050	0.050	0.847	< 0.10	< 0.010	< 0.0040	1.43	< 0.20	0.0014	< 1.00	2.6	
	PW17-19-231114	2023 11 14	< 0.050	0.073	1.24	< 0.010	3.5	0.0242	< 0.50	0.509	<u>40.4</u>	24.7	0.069	< 0.0050	0.041	1.22	< 0.10	0.014	< 0.0040	3.45	< 0.20	0.0026	< 1.00	4.4	
PW17-20	PW17-20-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	9.3	< 0.010	< 0.50	0.31	<u>10.3</u>	< 0.20	0.13	< 0.010	< 0.10	0.82	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	PW-D-170822	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	8.7	< 0.010	< 0.50	0.30	<u>10.1</u>	< 0.20	0.12	< 0.010	< 0.10	0.82	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	QA/QC RPD%		*	*	*	*	*	*	*	3	2	*	*	*	*	*	*	*	*	*	-	*	*	*	
PW17-21	PW17-20-180309	2018 03 09	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.12	<u>5.45</u>	< 0.20	0.10	< 0.010	< 0.10	0.44	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>9.0</u>	
	PW17-21-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	5.6	< 0.010	< 0.50	0.34	<u>9.74</u>	< 0.20	< 0.10	0.023	< 0.10	0.71	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	PW17-21-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	6.7	< 0.010	0.76	0.44	<u>11.4</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.10	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	<u>0.060</u>	< 0.50	8.85	<u>1.020</u>	0.78	0.29	< 0.010	< 0.10	14.0	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>11.3</u>	
	PW17-21-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.020	< 0.50	0.39	<u>11.1</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.63	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.69	0.22	<u>9.95</u>	1.09	< 0.10	< 0.010	< 0.10	0.55	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.37	<u>11.2</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.59	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	6.9	< 0.010	< 0.50	0.20	<u>6.79</u>	< 0.20	0.16	0.067	< 0.10	0.79	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	9.6	< 0.010	< 0.50	0.31	<u>11.5</u>	0.44	< 0.10	< 0.010	< 0.10	0.77	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.35	<u>10.1</u>	< 0.20	< 0.10	< 0.010	< 0.10	1.04	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW17-21-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.15	<u>6.69</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.49	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW-A-220210	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.17	<u>6.74</u>	< 0.20	0.11	< 0.010	< 0.10	0.50	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	QA/QC RPD%		*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	*	
	PW17-21-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.34	<u>10.0</u>	< 0.20	0.10	< 0.010	< 0.10	0.69	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	PW17-21-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.18	<u>8.36</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.53	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	PW-A-230222	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.18	<u>8.41</u>	< 0.20	< 0.10	< 0.010	< 0.10	0.51	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
	QA/QC RPD%		*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	*	
PW17-21-230802	2023 08 02	< 0.050	0.087	2.28	< 0.010	5.4	0.0053	0.91	0.346	<u>11.5</u>	< 0.050	0.105	< 0.0050	0.069	0.981	< 0.10	< 0.010	< 0.0040	0.29	< 0.20	0.0011	< 1.00	4.4		
PW17-21-231114	2023 11 14	< 0.050	0.052	1.76	< 0.010	3.4	0.0085	< 0.50	0.346	<u>22.0</u>	0.111	0.075	< 0.0050	0.029	0.776	< 0.10	< 0.010	< 0.0040	1.44	< 0.20	0.0017	< 1.00	1.7		
PW-A-231114	Duplicate	< 0.050	0.059	1.91	< 0.010	3.5	0.0072	< 0.50	0.338	<u>22.0</u>	0.099	0.077	< 0.0050	0.033	0.723	< 0.10	< 0.010	< 0.0040	1.44	< 0.20	0.0017	< 1.00	1.5		
QA/QC RPD%		*	*	8	*	*	*	*	2	0	*	*	*	*	7	*	*	*	0	*	*	*	*		
PW18-22	PW18-22-180726	2018 07 26	< 0.20	< 0.50	< 5.0	< 0.10	9.4	< 0.010	0.50	0.48	<u>24.6</u>	< 0.20	0.15	< 0.010	< 0.10	1.09	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW18-22-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.24	<u>18.7</u>	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	PW18-22-190124	2019 01 24	0.23	1.78	15.9	0.15	11.4	<u>1.03</u>	< 0.50	<u>140</u>	<u>14,300</u>	13.6	1.82	< 0.010	0.56	209	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	<u>247</u>	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.
^d Standard to protect freshwater aquatic life.
^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^f Guideline/standard varies with Chloride.
^g Guideline/standard varies with Hardness.
^h Guideline/standard varies with pH.
ⁱ Guideline is temperature, pH, DOC and hardness dependent.
^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.
^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.
^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^m Guideline is hardness and DOC dependent.
ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters						Geochemical Indicators																							
			pH pH	Hardness mg/L	pH (field) pH	Total Dissolved Solids mg/L	Total Suspended Solids mg/L	Dissolved Organic Carbon mg/L	Ammonia, total (as N) µg/L	Nitrate (as N) µg/L	Nitrite (as N) µg/L	Nitrate+Nitrite Nitrogen µg/L	Chloride mg/L	Fluoride µg/L	Sulfate mg/L	Total Alkalinity (as CaCO3) mg/L	Alkalinity, Bicarbonate (as CaCO3) mg/L	Alkalinity, Carbonate (as CaCO3) mg/L	Alkalinity, Hydroxide (as CaCO3) mg/L	Alkalinity, Phenolphthalein (as CaCO3) mg/L	Bromide mg/L	Acidity (as CaCO3) mg/L	Total Acidity mg/L	Acidity (pH 8.3) mg/L	Dissolved Aluminum µg/L	Dissolved Calcium mg/L	Dissolved Iron µg/L	Dissolved Magnesium mg/L	Dissolved Manganese µg/L	Dissolved Potassium mg/L	Dissolved Sodium mg/L	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^e	32,800	60-600 ^f	n/a	600	400-1,870 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	815-3,390 ^g	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			6.5 - 9	n/a	6.5 - 9	n/a	n/a	1,770 ^e	3,000	20-200 ^f	n/a	150	n/a	128-429 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	20-300 ^{c,n}	n/a	n/a	n/a	767-2,338 ^g	n/a	n/a	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	n/a	n/a	184,000-185,000 ^e	4,000,000	2,000-20,000 ^f	4,000,000	15,000	20,000-30,000 ^g	12,800-42,900 ^g	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
PW18-22	PW18-22-190925	2019 09 25	6.18	7.69	6.61	26	15.2	-	-	45	< 10	-	3.15	< 100	1.1	2.7	2.7	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.3	142	2.25	194	0.499	6.33	0.23	2.14	
	PW-D-190925	Duplicate	6.18	7.60	6.61	< 15	10.4	-	-	47	< 10	-	3.17	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.3	142	2.22	186	0.496	5.86	0.24	2.12	
	QA/QC RPD%		0	1	0	*	37	-	-	*	*	-	1	*	*	*	*	*	*	*	*	-	-	*	0	1	4	1	8	*	1	
	PW18-22-191203	2019 12 03	6.62	6.93	6.37	33	20.8	-	-	20	< 10	-	3.18	< 100	1.7	4.6	4.6	< 1.0	< 1.0	< 1.0	< 0.10	-	-	2.2	126	1.89	63	0.537	2.47	0.13	2.17	
	PW18-22-200729	2020 07 29	6.82	7.77	7.01	28	-	-	-	88.4	< 1.0	-	2.86	< 20	1.11	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	2.3	-	-	76.6	2.18	209	0.561	2.39	0.190	2.08	
	PW18-22-210811	2021 08 11	-	-	7.44	< 15	-	-	-	144	< 10	-	3.30	< 100	1.3	7.4	7.4	< 1.0	< 1.0	< 1.0	< 0.10	-	3.3	-	51.7	2.76	349	0.661	2.03	0.30	2.51	
	PW18-22-220210	2022 02 10	-	-	6.39	< 15	-	-	< 50	26.3	< 5.0	26.3	1.96	< 100	1.3	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	-	91.5	1.27	25	0.370	0.73	0.12	1.58	
	PW18-22-221026	2022 10 26	-	9.76	7.05	21	-	-	< 50	388	< 10	-	4.82	< 100	1.5	5.6	5.6	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	42.5	2.65	248	0.762	1.78	0.34	2.88	
	PW18-22-230223	2023 02 23	-	5.14	6.68	30	-	-	< 50	18	< 10	-	3.22	< 100	1.2	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	96.5	1.34	17	0.436	0.65	0.10	2.12	
	PW18-22-230802	2023 08 02	-	7.53	7.01	< 15	-	-	< 50	114	< 10	-	2.31	< 100	1.4	5.4	5.4	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	-	46.8	2.15	85.5	0.524	1.96	0.259	1.98	
PW18-22-231115	2023 11 15	-	6.34	6.16	28	-	-	< 50	31	< 10	-	3.64	< 100	1.5	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.1	-	3.6	-	110	1.66	36.1	0.532	0.851	0.125	2.45		
PW23-29	PW18-29-230802	2023 08 02	-	4.50	6.41	31	-	-	< 50	132	< 10	-	7.93	< 100	3.4	9.8	9.8	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	-	25.0	1.07	42.0	0.443	0.949	0.508	9.59	

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD

SHADOW

SHADED

Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.
^d Standard to protect freshwater aquatic life.
^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^f Guideline/standard varies with Chloride.
^g Guideline/standard varies with Hardness.
^h Guideline/standard varies with pH.
ⁱ Guideline is temperature, pH, DOC and hardness dependent.
^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.
^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.
^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^m Guideline is hardness and DOC dependent.
ⁿ Guideline is pH, DOC and hardness dependent.

Table 4: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Porewater (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc ^k µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^{a,c}			250 ^{c,i}	5	n/a	n/a	n/a	0.038-2.41 ^g	n/a	110	0.2-2.4 ⁱ	3-417 ^e	n/a	n/a	2,000	n/a	n/a	0.1-3.0 ^f	n/a	n/a	n/a	n/a	n/a	5.6-63.1 ^m
BCWQG Aquatic Life Long-term Average (AW) ^{b,c}			74 ^{c,i}	n/a	1,000	0.13	1,200	0.019-0.457 ^g	1 (Cr(+6))	4	0.2-0.4 ⁱ	3-19.6 ^e	n/a	0.02 ^j	1,000	25-150 ^g	2	0.05-1.5 ^f	0.8	n/a	n/a	8.5	n/a	2.1-45.4 ⁿ
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			900	500	100,000	15	120,000	5-40 ^g	100	400	200-900 ^g	400-1,600 ^e	n/a	2.5	100,000	2,500-15,000 ^g	200	5-150 ^f	30	10,000	n/a	850	n/a	750-24,000 ^g
PW18-22	PW18-22-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	6.0	< 0.010	< 0.50	0.32	21.7	< 0.20	0.13	0.103	< 0.10	1.06	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW-D-190925	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	5.6	< 0.010	< 0.50	0.32	21.2	< 0.20	0.14	0.154	< 0.10	1.50	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	2	*	*	40	*	*	*	*	*	*	*	*	*	*
	PW18-22-191203	2019 12 03	0.27	< 0.50	< 5.0	< 0.10	9.4	0.018	< 0.50	0.44	51.4	2.46	< 0.10	< 0.010	< 0.10	1.06	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW18-22-200729	2020 07 29	< 0.10	0.12	1.88	< 0.020	< 10	< 0.0050	0.16	0.30	9.83	< 0.050	< 1.0	< 0.0050	< 0.050	0.64	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.8
	PW18-22-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.28	13.6	< 0.20	< 0.10	< 0.010	< 0.10	1.05	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW18-22-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.19	20.8	< 0.20	< 0.10	< 0.010	< 0.10	0.63	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	PW18-22-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.44	15.0	< 0.20	< 0.10	< 0.010	< 0.10	1.06	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	PW18-22-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.24	9.41	< 0.20	< 0.10	< 0.010	< 0.10	0.51	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	PW18-22-230802	2023 08 02	< 0.050	0.078	2.36	< 0.010	4.9	0.0051	< 0.50	0.209	9.11	< 0.050	0.092	< 0.0050	0.034	0.600	< 0.10	< 0.010	< 0.0040	1.01	< 0.20	0.0011	< 1.00	< 1.0
PW18-22-231115	2023 11 15	0.064	< 0.050	1.81	< 0.010	3.5	0.0059	< 0.50	0.249	18.6	0.613	0.070	< 0.0050	0.022	0.686	< 0.10	< 0.010	< 0.0040	0.94	< 0.20	0.0014	< 1.00	1.9	
PW23-29	PW18-29-230802	2023 08 02	< 0.050	0.065	0.26	< 0.010	9.1	0.0156	< 0.50	0.385	56.7	< 0.050	0.256	< 0.0050	0.096	1.08	0.13	0.012	< 0.0040	0.27	< 0.20	0.0027	< 1.00	2.2

Associated ALS file(s): VA20B1740, VA20B1766, VA21A0691, VA21A0716.
Associated Caro file(s): 5111661, 6070220, 6090497, 6101221, 6121082, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8030934, 8072387, 8072648, 8110202, 8110213, 9011882, 9011884, 9011882, 9011884, 9092682, 9092818, 9092827, 9120466, 9120554, 21H1548, 22B1772, 22B2327, 22J3877, 23B2664, 23H1041, 23K2229..
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW)
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW)
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

- ^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Guidelines for total metals have been applied to dissolved values as a conservative comparison.
^d Standard to protect freshwater aquatic life.
^e Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^f Guideline/standard varies with Chloride.
^g Guideline/standard varies with Hardness.
^h Guideline/standard varies with pH.
ⁱ Guideline is temperature, pH, DOC and hardness dependent.
^j Total Mercury guideline is based on the % of MethylMercury present. WQG = 0.0001 / (MeHg/total Hg), where MeHg is mass (or concentration) of methyl mercury and THg. Guideline shown assumes MeHg<0.5% of Total Hg.
^k There is no Zinc standard specified for H > 400; therefore, the standard for H=300-<400 is applied as a conservative comparison.
^l Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^m Guideline is hardness and DOC dependent.
ⁿ Guideline is pH, DOC and hardness dependent.

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW15-01	5364917	422242	SW15-01-151117	2015-11-17	6.39	0.104	10.41	144.7	10.2	0.0930	0.07	25	5,100' Portal discharge at pipe
			SW15-01A-160629	2016-06-29	7.01	0.125	11.04	234.7	9.6	0.1130	0.08	0.38	Drainage from rock/large fallen tree below 5100' portal
			SW15-01A-160906	2016-09-06	-	0.194	11.13	75.1	10.8	0.1710	0.13	0.03	Weir attached to 5100' portal
			SW15-01-161017	2016-10-17	7.16	0.111	9.64	272.9	9.9	0.1014	0.07	14.5	
			SW15-01-161212	2016-12-12	5.80	0.082	10.11	177.3	9.9	0.0533	0.04	6.3	
			SW15-01-170525	2017-05-25	6.84	0.064	11.09	218.4	7.6	-	-	0.30	
			SW15-01A-170731	2017-07-31	7.78	0.120	-	162.0	9.8	0.1099	0.08	1.00	Drainage from rock/large fallen tree below 5100' portal
			SW15-01A-170821	2017-08-21	7.41	0.116	12.52	283.4	9.5	0.1060	0.08	0.38	Weir attached to 5100' portal
			SW15-01A-170926	2017-09-26	7.97	0.156	-	191.9	10.4	0.1014	0.07	0.30	
			SW15-01-171031	2017-10-31	6.43	0.117	5.20	244.7	10.1	0.0761	0.05	0.30	
			SW15-01-171122	2017-11-22	6.80	0.117	10.99	281.0	8.0	0.0701	0.06	20	
			SW15-01-171212	2017-12-12	7.73	0.072	12.08	194.8	9.2	-	0.03	10	
			SW15-01-180305	2018-03-05	7.13	0.073	10.18	202.8	7.7	0.0475	-	1.13	
			SW15-01A-180723	2018-07-23	8.13	0.196	13.55	242.0	12.4	0.1670	0.12	0.2	
			SW15-01a-181029	2018-10-29	6.56	0.104	144.30	230.0	10.3	0.0943	0.07	0.167	
			SW15-01-190123	2019-01-23	6.44	0.075	11.31	174.7	8.4	0.0488	0.03	25	Combined flow with observations as SW15-01A (5 L/sec from weir).
			SW15-01a-190123	2019-01-23	6.97	0.076	13.54	163.5	8.5	0.0494	0.04	20	
			SW15-01A-190925	2019-09-25	6.27	0.096	11.67	222	10	0.087	0.06	2	
			SW15-01-191205	2019-12-05	6.16	0.091	10.07	212	9.7	0.06	0.04	5	
			SW15-01A-191205	2019-12-05	6.64	0.091	11.26	168	9.7	0.059	0.04	20	
			SW15-01-200729	2020-07-29	6.33	0.068	11.45	175.1	9.3	0	0.05	2	
			SW15-01-210112	2021-01-12	6.45	0.044	11.71	256.7	8	0.03	0.02	500	
			SW15-01-210811	2021-08-11	6.25	183.6	11.9	260.2	10	-	-	1.5	
			SW15-01-220211	2022-02-11	6.14	0.061	11.73	250.1	7.1	-	-	4.2	
			SW15-01A-220211	2022-02-11	6.14	0.055	11.85	256.8	7.1	-	-	-	
			SW15-01A-221025	2022-10-25	6.84	0.081	10.2	97	11.7	-	-	-	
			SW15-01A-230223	2023-02-23	7.28	0.061	12.01	245.8	7.28	-	-	-	
			SW15-01A-230802	2023-08-02	6.73	0.0565	11.45	109.6	10.3	0.035	-	10	
			SW15-01A-231113	2023-11-13	6.50	0.0668	13.75	157.3	10.6	-	-	25	
SW15-02	5364924	422203	SW15-02-151117	2015-11-17	7.38	0.125	11.85	134.5	10.5	0.1110	0.08	25	5,100' Portal discharge before confluence with Jordan River
			SW15-02-160629	2016-06-29	7.56	0.123	11.52	186.7	9.9	0.1120	0.08	50	
			SW15-02-160906	2016-09-06	-	0.188	10.37	82.6	10.5	0.1690	0.12	0.17	
			SW15-02-161017	2016-10-17	6.94	0.112	11.14	237.3	10.1	0.1014	0.07	-	
			SW15-02-161212	2016-12-12	6.94	0.081	11.58	155.2	8.8	0.0527	0.04	16.67	
			SW15-02-170731	2017-07-31	6.56	0.114	-	197.6	9.8	0.1040	0.07	1.33	
			SW15-02-170821	2017-08-21	6.97	0.107	10.07	256.6	9.7	0.1027	0.07	0.38	
			SW15-02-170926	2017-09-26	7.66	0.149	-	212.0	10.2	0.0969	0.07	2.50	
			SW15-02-171031	2017-10-31	6.55	0.118	10.02	245.3	9.8	0.0767	0.06	11	
			SW15-02-171122	2017-11-22	6.88	0.110	11.07	259.8	8.2	0.0722	0.05	27	
			SW15-02-171212	2017-12-12	7.19	0.076	11.77	183.0	9.0	-	0.04	20	
			SW15-02-180305	2018-03-05	7.01	0.072	11.00	191.4	7.6	0.0468	-	6.67	
			SW15-02-180724	2018-07-24	7.17	0.148	12.01	78.8	11.1	0.1310	0.10	1	
			SW15-02-181029	2018-10-29	6.69	0.103	15.09	219.0	10.2	0.0936	0.07	1.667	
			SW15-02-190123	2019-01-23	7.00	0.071	10.24	165.0	8.5	0.0455	0.03	20	
			SW15-02-190925	2019-09-25	6.61	0.098	11.65	202	10.7	0.088	0.06	6	
			SW15-02-191205	2019-12-05	7.07	0.091	11.45	169.4	9.5	0.059	0.04	20	
			SW15-02-210812	2021-08-12	6.45	0.205	11.62	274.3	10.3	-	-	-	
			SW15-02-220211	2022-02-11	6.29	0.060	11.96	256.0	7.0	-	-	-	
			SW15-02-221025	2022-10-25	7.05	0.078	9.21	58	11.2	-	-	-	
			SW15-02-230223	2023-02-23	7.32	0.043	12.89	243.4	6.47	-	-	-	
			SW15-02-230802	2023-08-02	6.69	0.0563	11.03	-126.0	11.8	0.037	-	30	
			SW15-02-231115	2023-11-15	6.42	0.0853	15.79	70.5	10.2	-	-	-	

* Average (rounded) discharge rates between 8-5 pm on November 18th as measured at the Elliot Dam located approx. 7.6 km upstream on Jordan River.

** pH meter malfunction for September sampling event.

*** possible notation errors.

- not recorded

n/a not applicable

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW15-03	5364846	422200	SW15-03-151117	2015-11-17	4.34	0.437	9.68	216.4	11.3	0.3850	0.29	0.17	Seepage #1 from the North Waste Rock Pile (south region)
			-	2016-06-29	-	-	-	-	-	-	-	DRY	
			-	2016-09-06	-	-	-	-	-	-	-	DRY	
			-	2016-10-17	-	-	-	-	-	-	-	DRY	
			-	2016-12-12	-	-	-	-	-	-	-	DRY	
			SW15-03-171123	2017-11-23	3.98	0.500	9.81	403.3	7.9	0.3250	0.24	0.33	
			SW15-03-181101	2018-11-01	3.88	0.729	13.09	325.0	11.3	0.6435	0.49	0.017	
SW15-04	5364831	422215	-	2019-01-23	-	-	-	-	-	-	-	DRY	Seepage #2 from the North Waste Rock Pile (south region)
			SW15-04-151117	2015-11-17	3.88	0.499	10.60	280.7	12.1	0.4310	0.32	0.1	
			-	2016-06-29	-	-	-	-	-	-	-	DRY	
			-	2016-09-06	-	-	-	-	-	-	-	DRY	
			-	2016-10-17	-	-	-	-	-	-	-	DRY	
			-	2016-12-12	-	-	-	-	-	-	-	DRY	
			SW15-04-181101	2018-11-01	3.78	0.711	12.52	375.0	11.9	0.4615	0.35	0.017	
SW15-05	5364765	422217	-	2019-01-23	-	-	-	-	-	-	-	DRY	Unnamed Creek #1 between the waste rock piles
			SW15-05-151117	2015-11-17	7.06	0.071	11.71	134.0	9.9	0.0660	0.05	20	
			SW15-05-160629	2016-06-29	7.05	0.051	10.92	175.6	12.5	0.0430	0.03	30	
			SW15-05-160907	2016-09-07	-	0.058	10.36	83.6	12.0	0.0500	0.04	0.5	
			SW15-05-161018	2016-10-18	7.03	0.050	11.10	186.1	10.4	0.0448	0.03	2.5	
			SW15-05-161212	2016-12-12	6.99	0.048	13.38	146.6	6.6	0.0312	0.02	6.67	
			SW15-05-170801	2017-08-01	6.12	0.048	-	169.8	13.1	0.0410	0.03	0.67	
			SW15-05-170822	2017-08-22	7.13	0.051	8.69	120.5	12.6	0.0436	0.03	0.40	
			SW15-05-170927	2017-09-27	7.84	0.075	-	176.9	11.5	0.0488	0.03	0.67	
			SW15-05-171101	2017-11-01	6.88	0.071	-	150.8	9.0	0.0462	0.03	1.83	
			SW15-05-171123	2017-11-23	6.85	0.059	10.37	236.2	7.8	0.0383	0.03	20	
			SW15-05-171213	2017-12-13	6.81	0.046	14.20	167.6	7.6	0.0299	0.02	15	
			SW15-05-180305	2018-03-05	7.08	0.051	11.68	173.7	7.4	0.0332	-	13	
SW15-06	5364720	422245	SW15-06-151117	2015-11-17	7.28	0.069	11.65	138.6	9.7	0.0650	0.05	50	Unnamed Creek #2 down river from South Waste Rock Pile
			SW15-06-160630	2016-06-29	8.31	0.059	10.25	186.4	11.4	0.0520	0.04	20	
			SW15-06-160907	2106-09-07	-	0.063	10.24	76.1	11.9	0.0546	0.04	0.25	
			SW15-06-161018	2016-10-18	6.54	0.050	10.83	195.3	10.5	0.0533	0.04	0.25	
			SW15-06-161212	2016-12-12	6.87	0.050	12.80	146.1	6.7	0.0319	0.02	3.33	
			SW15-06-170801	2017-08-01	6.24	0.049	-	167.8	12.2	0.0422	0.03	1.67	
			SW15-06-170822	2017-08-22	6.57	0.052	9.79	147.7	12.0	0.0448	0.03	0.40	
			SW15-06-170927	2017-09-27	7.51	0.075	-	173.9	11.2	0.0488	0.03	1.33	
			SW15-06-171101	2017-11-01	6.85	0.074	-	152.0	9.1	0.0484	0.03	2.67	
			SW15-06-171123	2017-11-23	6.89	0.054	10.98	231.6	7.7	0.3570	0.03	10	
			SW15-06-171213	2017-12-13	7.48	0.046	14.25	137.1	7.8	0.0299	0.02	15	
			SW15-06-180305	2018-03-05	7.07	0.051	11.36	172.9	7.4	0.0332	-	6.67	
SW15-07	5364901	422212	SW15-07-151117	2015-11-17	6.69	0.123	11.43	176.3	10.3	0.1110	0.08	8	Seepage #3 from the North Waste Rock Pile (central region)
			SW15-07-160629	2016-06-29	6.78	0.120	12.75	223.8	8.9	0.1120	0.08	0.58	
			SW15-07-160906	2016-09-06	-	0.189	12.01	133.7	9.8	0.1729	0.13	0.5	
			SW15-07-161017	2016-10-17	6.72	0.113	11.00	240.2	10.0	0.1027	0.08	1.34	
			SW15-07-161212	2016-12-12	6.96	0.085	11.42	157.2	9.9	0.0553	0.04	3.67	
			SW15-07-170731	2017-07-31	5.99	0.111	-	196.7	9.2	0.1040	0.08	4.17	
			SW15-07-170821	2017-08-21	6.98	0.153	10.56	200.9	-	0.0988	0.07	0.38	
			SW15-07-170927	2017-09-27	6.73	0.149	-	222.3	9.9	0.0969	0.07	1.00	
			SW15-07-171031	2017-10-31	6.26	0.118	7.94	257.7	10.2	0.0767	0.06	5.83	
			SW15-07-171122	2017-11-22	6.62	0.120	10.91	260.8	8.1	0.0780	0.06	6.67	
			SW15-07-171212	2017-12-12	7.07	0.077	13.41	190.4	9.2	-	0.04	5.00	
			SW15-07-180305	2018-03-05	6.97	0.074	11.42	202.8	7.8	0.0481	-	6.67	
			SW15-07-180725	2018-07-25	6.30	0.321	11.88	186.9	9.0	0.3000	0.22	0.5	
			SW15-07-181030	2018-10-30	6.13	0.109	15.76	226.0	10.1	0.0988	0.07	0.667	
			SW15-07-190123	2019-01-23	6.62	0.078	9.98	167.0	8.5	0.0507	0.04	10	
			SW15-07-190926	2019-09-26	6.43	0.081	11.63	222	10.2	0.0850	0.06	5	

* Average (rounded) discharge rates between 8-5 pm on November 18th as measured at the Elliot Dam located approx. 7.6 km upstream on Jordan River.

** pH meter malfunction for September sampling event.

*** possible notation errors.

- not recorded

n/a not applicable

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW15-07	5364901	422212	SW15-07-191205	2019-12-05	6.95	0.092	11.33	153.1	9.7	0.0600	0.04	10	Seepage #3 from the North Waste Rock Pile (central region)
			SW15-07-210812	2021-08-12	6.86	0.205	11.43	286.4	9.4	-	-	-	
			SW15-07-220211	2022-02-11	6.44	0.061	11.91	263.5	7.1	-	-	-	
			SW15-07-221025	2022-10-25	6.92	0.081	9.61	120	11.8	-	-	-	
			SW15-07-230223	2023-02-23	7.32	0.041	12.83	297	7.36	-	-	-	
			SW15-07-230802	2023-08-02	6.44	0.0566	11.55	-77.2	10.4	0.0410	-	27.69	
			SW15-07-231115	2023-11-15	6.42	0.0871	15.46	96.9	10.40	-	-	10	
SW15-08	5364926	422197	SW15-08-151118	2015-11-18	7.06	0.098	13.02	87.5	8.9	0.0730	0.05	50,000*	Jordan River at confluence with 5,100' Portal discharge
			SW15-08-160629	2016-06-29	7.39	0.080	18.54	190.8	13.2	0.0670	0.05	500	
			SW15-08-160906	2016-09-06	-	0.089	7.94	73.9	13.3	0.0748	0.05	400	
			SW15-08-161017	2016-10-17	6.72	0.109	11.20	244.9	10.0	0.0897	0.07	24,000	
			SW15-08-161212	2016-12-12	6.98	0.086	10.85	153.3	8.6	0.0559	0.04	550	
			SW15-08-170731	2017-07-31	6.31	0.049	-	154.5	15.3	0.0390	0.02	450	
			SW15-08-170821	2017-08-21	6.95	0.061	9.06	184.5	13.3	0.0507	0.04	430	
			SW15-08-170926	2017-09-26	7.24	0.030	-	149.6	14.1	0.0195	0.01	350	
			SW15-08-171031	2017-10-31	6.02	0.022	7.17	197.3	8.2	0.0143	0.01	440	
			SW15-08-171122	2017-11-22	6.32	0.016	12.74	267.8	5.2	0.0104	0.01	80,000	
			SW15-08-171212	2017-12-12	7.40	0.062	13.97	180.0	7.8	-	0.03	450	
			SW15-08-180305	2018-03-05	7.30	0.033	12.90	182.6	4.6	0.0215	-	450	
			SW15-08-180725	2018-07-25	6.61	0.102	9.35	147.7	14.4	-	0.06	450	
			SW15-08-181030	2018-10-30	6.01	0.019	16.00	194.0	9.5	0.0169	0.01	27,400	
			SW15-08-190123	2019-01-23	6.12	0.018	11.25	170.0	6.5	0.0117	0.01	6,600	
			SW15-08-190926	2019-09-26	6.34	0.019	11.5	170	13.4	0.0160	0.01	6,750****	
			SW15-08-191203	2019-12-03	5.78	0.070	11.9	211.1	7.6	0.0460	0.03	450****	
			SW15-08-210811	2021-08-11	7.67	0.029	8.96	34.4	17.1	-	-	-	
			SW15-08-220211	2022-02-11	6.63	0.028	12.03	224.1	5.7	-	-	-	
			SW15-08-221025	2022-10-25	7.18	0.040	10.45	134	10.5	-	-	-	
			SW15-08-230223	2023-02-23	7.11	0.048	13.08	288.7	5.09	-	-	-	
			SW15-08-230802	2023-08-02	6.86	0.0193	9.90	-134.2	15.3	0.0210	-	-	
			SW15-08-231115	2023-11-15	6.57	0.0506	16.88	68.1	7.7	-	-	-	
SW15-09	5364953	422198	SW15-09-151118	2015-11-18	6.74	0.040	13.35	72.5	7.3	0.0390	0.03	50,000*	Jordan River upstream from 5,100' portal (reference sample)
			SW15-09-160629	2016-06-29	7.41	0.018	9.09	175.9	16.0	0.0140	0.01	500	
			SW15-09-160906	2016-09-06	-	0.016	9.38	26.9	14.6	0.0169	0.01	400	
			SW15-09-161017	2016-10-17	6.79	0.014	11.59	224.5	10.1	0.0130	0.01	24,000	
			SW15-09-161212	2016-12-12	7.82	0.022	14.40	135.7	3.2	0.0143	0.01	550	
			SW15-09-170731	2017-07-31	6.76	0.017	-	165.9	15.8	0.0136	0.01	450	
			SW15-09-170821	2017-08-21	7.38	0.160	8.09	184.5	14.9	0.0123	0.01	430	
			SW15-09-170926	2017-09-26	7.19	0.024	-	150.6	14.1	0.0156	0.01	350	
			SW15-09-171031	2017-10-31	6.24	0.022	8.01	220.3	8.0	0.0143	0.01	440	
			SW15-09-171122	2017-11-22	6.49	0.014	12.90	269.7	5.1	0.0091	0.01	80,000	
			SW15-09-171212	2017-12-12	6.83	0.019	14.50	161.3	4.6	-	0.01	450	
			SW15-09-180305	2018-03-05	7.25	0.020	12.71	193.8	3.5	0.0130	-	450	
			SW15-09-180725	2018-07-25	7.03	0.017	9.46	113.6	16.6	0.0130	0.01	450	
			SW15-09-181030	2018-10-30	5.56	0.018	16.52	209.7	9.5	0.0163	0.01	27,400	
			SW15-09-190123	2019-01-23	6.04	0.018	11.58	172.8	6.4	0.0117	0.01	6,600	
			SW15-09-190926	2019-09-26	6.42	0.018	11.29	174	13.4	0.0150	0.01	6,750****	
			SW15-09-191204	2019-12-04	6.32	0.017	12.71	138	6.2	0.0110	0.01	4,500****	
			SW15-09-200729	2020-07-29	6.43	0.020	10.03	95.8	16	-	0.01	4,000	
			SW15-09-210112	2021-01-12	5.08	0.014	13.48	273.3	5.7	0.009	0.01	60,000	
			SW15-09-210811	2021-08-11	7.13	0.028	6.36	23.3	16.8	-	-	250	
			SW15-09-220211	2022-02-11	6.62	0.015	12.03	229.9	5.3	-	-	4,900	
			SW15-09-221025	2022-10-25	7.17	0.033	7.95	126	10.4	-	-	-	
			SW15-09-230223	2023-02-23	6.52	0.018	15.02	203.5	2.3	-	-	-	
			SW15-09-230802	2023-08-02	6.93	0.0214	8.39	149.1	15.1	-	-	-	
			SW15-09-231115	2023-11-15	6.31	0.0313	17.88	31.4	6.3	-	-	-	

* Average (rounded) discharge rates between 8-5 pm on November 18th as measured at the Elliot Dam located approx. 7.6 km upstream on Jordan River.

** pH meter malfunction for September sampling event.

*** possible notation errors.

**** Flowrate estimate made by qualitative comparison to base discharge from Elliot Dam.

- not recorded

n/a not applicable

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW15-10	5364827	422199	SW15-10-151118	2015-11-18	6.60	0.041	12.90	97.9	7.4	0.0410	0.03	50,000*	Jordan River at the base of North Waste Rock Pile
			SW15-10-160629	2016-06-29	7.36	0.022	9.75	177.4	16.1	0.0170	0.01	500	
			SW15-10-160907	2016-09-07	-	0.027	8.04	32.7	14.7	0.0215	0.01	400	
			SW15-10-161018	2016-10-18	6.93	0.018	11.28	199.0	10.0	0.0169	0.01	24,000	
			SW15-10-161212	2016-12-12	7.03	0.035	13.47	139.7	3.6	0.0228	0.02	550	
			SW15-10-170801	2017-08-01	5.74	0.020	-	200.8	16.1	0.0156	0.01	450	
			SW15-10-170822	2017-08-22	7.04	0.019	8.72	117.9	15.7	0.0150	0.01	430	
			SW15-10-170926	2017-09-26	6.88	0.028	-	195.2	14.0	0.0182	0.01	350	
			SW15-10-171031	2017-10-31	5.68	0.028	1.71	227.2	8.5	0.0182	0.01	440	
			SW15-10-171123	2017-11-23	6.07	0.016	12.01	234.9	5.6	0.0091	0.01	80,000	
			SW15-10-171213	2017-12-13	7.24	0.024	13.73	120.0	5.0	0.0156	0.01	450	
			SW15-10-180305	2018-03-05	6.92	0.022	12.17	178.7	4.0	0.0137	-	450	
			SW15-10-180724	2018-07-24	7.16	0.020	10.05	151.0	17.0	0.0160	0.01	450	
			SW15-10-181031	2018-10-31	5.62	0.018	15.24	248.9	9.5	0.0169	0.01	27,400	
			SW15-10-190124	2019-01-24	6.51	0.020	12.09	122.0	6.5	0.0130	0.01	6,600	
			SW15-10-190925	2019-09-25	6.69	0.020	11.06	150	13.3	0.0170	0.01	6,750****	
			SW15-10-191203	2019-12-03	6.2	0.026	12.88	174	5	0.0170	0.01	450****	
			SW15-10-191204	2019-12-04	6.32	0.018	12.59	179.8	6.2	0.0120	0.01	4,500****	
			SW15-10-200730	2020-07-30	6.68	0.022	9.92	167.5	16.7	17	0.01	3,000	
			SW15-10-210113	2021-01-13	5.47	0.008	13.72	216.8	5.3	0.008	0	60,000	
			SW15-10-210810	2021-08-10	7.3	0.033	7.6	39.5	18.6	-	-	250	
			SW15-10-220210	2022-02-10	6.50	0.015	13.53	207.6	5.6	-	-	2,700	
			SW15-10-221027	2022-10-27	6.95	0.041	10.8	118	10.2	-	-	5.33	
SW15-11	5364682	422232	SW15-10-230222	2023-02-22	6.71	0.019	13.91	250	3.98	-	-	-	Jordan River downgradient of the South Waste Rock Pile
			SW15-10-230801	2023-08-02	6.86	0.0232	9.57	160.5	16.3	-	-	1,726	
			SW15-10-231114	2023-08-02	6.14	0.0271	16.45	160.1	6.9	-	-	-	
			SW15-11-151118	2015-11-18	6.63	0.040	13.05	79.3	7.4	0.0390	0.03	50,000*	
			SW15-11-160630	2016-06-30	6.99	0.022	9.68	183.0	15.3	0.0180	0.01	500	
			SW15-11-160907	2016-09-07	-	0.026	10.14	72.3	14.7	0.0215	0.01	400	
			SW15-11-161018	2016-10-18	6.05	0.015	11.33	219.5	10.0	0.0143	0.01	24,000	
			SW15-11-161212	2016-12-12	6.81	0.036	13.74	152.8	4.7	0.0234	0.02	550	
			SW15-11-170801	2017-08-01	6.35	0.020	-	159.7	17.6	0.0150	0.01	450	
			SW15-11-170822	2017-08-22	6.67	0.020	9.03	164.5	15.1	0.0560	0.01	430	
			SW15-11-170927	2017-09-27	7.36	0.028	-	151.3	13.8	0.0182	0.01	350	
			SW15-11-171101	2017-11-01	6.05	0.026	-	183.4	8.2	0.0169	0.01	440	
			SW15-11-171123	2017-11-23	6.12	0.026	11.84	263.1	5.9	0.0136	0.01	80,000	
			SW15-11-171213	2017-12-13	7.12	0.024	15.44	135.4	5.2	0.0156	0.01	450	
			SW15-11-180305	2018-03-05	7.29	0.027	12.85	170.8	4.7	0.0169	-	450	
			SW15-11-180724	2018-07-24	7.20	0.021	10.81	131.6	16.54	0.0160	0.01	450	
			SW15-11-181031	2018-10-31	5.70	0.018	16.19	235.6	9.6	0.0163	0.01	27,400	
			SW15-11-190124	2019-01-24	6.10	0.021	10.92	178.0	6.6	0.0137	0.01	6,600	
			SW15-11-190924	2019-09-24	6.52	0.019	10.94	131	14.3	0.0160	0.01	6,750****	
			SW15-11-191203	2019-12-03	6.85	0.026	12.83	129	5.1	0.0170	0.01	450****	
			SW15-11-200730	2020-07-30	6.57	0.022	10.05	136.2	16.9	0.0170	0.01	2,500	
			SW15-11-210112	2021-01-12	5.73	0.014	13.21	196.8	5.8	0.009	0.01	30,000	
			SW15-11-210812	2021-08-12	6.92	0.034	9.47	199.1	18	-	-	3,800	
			SW15-11-220210	2022-02-11	6.54	0.016	13.85	179.3	5.7	-	-	5,800	
			SW15-11-221026	2022-10-26	7.24	0.038	9.87	115	10.2	-	-	8.78	
			SW15-11-230222	2023-02-22	5.98	0.018	13.31	174.9	3.95	-	-	-	
			SW15-11-230802	2023-08-02	7.59	0.0252	9.58	182.6	16.2	-	-	2,368	
			SW15-11-231114	2023-11-01	6.40	0.0277	16.21	138.1	6.9	-	-	-	

* Average (rounded) discharge rates between 8-5 pm on November 18th as measured at the Elliot Dam located approx. 7.6 km upstream on Jordan River.

** pH meter malfunction for September sampling event.

*** possible notation errors.

**** Flowrate estimate made by qualitative comparison to base discharge from Elliot Dam.

- not recorded

n/a not applicable

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW15-12	5364421	422206	SW15-12-151118	2015-11-18	6.65	0.042	13.06	82.1	7.4	0.0410	0.03	50,000*	Jordan River further downgradient near a former industrial site
			SW15-12-160630	2016-06-30	6.98	0.233	9.30	158.9	15.6	0.1840	0.14	500	
			SW15-12-160906	2016-09-06	-	0.028	9.66	-84***	15.2	0.0221	0.01	400	
			SW15-12-161017	2016-10-17	6.50	0.015	10.83	223.6	10.2	0.0136	0.01	24,000	
			SW15-12-161212	2016-12-12	5.89	16.789	13.02	253.0	5.2	10.4545	8.75	50,000	Field measurement and flow data collected on Dec 13.
			SW15-12-170801	2017-08-01	6.32	0.021	-	155.1	18.3	0.0156	0.01	450	Jordan River further downgradient near a former industrial site
			SW15-12-170821	2017-08-21	7.50	0.314	8.97	114.3	17.0	0.2405	0.18	430	
			SW15-12-170927	2017-09-27	7.10	0.029	-	155.5	14.3	0.0189	0.01	350	
			SW15-12-171101	2017-11-01	6.47	-	-	194.4	8.4	-	0.93	440	
			SW15-12-171122	2017-11-22	6.23	0.013	12.40	290.7	5.0	0.0084	0.01	80,000	
SW15-12-171212	2017-12-12	6.86	0.136	11.20	144.2	5.5	-	6.95	450				
SW16-13	5364823.8	422350.4	SW16-13-160629	2016-06-29	7.18	0.049	10.52	171.5	11.1	0.0430	0.03	30	Approximately 25 m upstream from road on Unnamed creek #2
			SW16-13-160906	2016-09-06	-	0.066	10.11	-122.1***	11.5	0.0572	0.04	0.5	
			SW16-13-161018	2016-10-18	6.87	0.055	11.04	179.8	10.1	0.5000	0.04	0.84	
			SW16-13-161212	2016-12-12	6.34	0.051	12.56	165.7	7.2	0.0332	0.02	2	
			SW16-13-170801	2017-08-01	6.48	0.047	-	140.0	12.1	0.0410	0.03	1.95	
			SW16-13-170821	2017-08-21	-	0.050	10.60	144.7	11.6	0.0429	0.03	0.40	
			SW16-13-170927	2017-09-27	7.32	0.073	-	200.3	10.8	0.0475	0.03	1.50	
			SW16-13-171101	2017-11-01	6.84	0.072	-	199.2	8.9	0.0468	0.03	3.24	
			SW16-13-171122	2017-11-22	6.91	0.073	10.88	271.1	7.4	0.0422	0.03	20	
			SW16-13-171212	2017-12-12	7.33	0.045	12.42	146.6	8.5	-	0.02	20	
SW17-14	5364955	422195	SW16-13-180306	2018-03-06	7.53	0.052	11.76	157.9	6.8	0.0332	-	-	Opposite side of river from SW-09
			SW17-14-170731	2017-07-31	6.26	0.015	-	161.9	16.1	0.0117	0.01	450	
			SW17-14-170821	2017-08-21	7.24	0.016	7.78	166.0	-	0.0130	0.01	430	
			SW17-14-170926	2017-09-26	7.31	0.024	-	160.2	14.1	0.0156	0.01	350	
			SW17-14-171031	2017-10-31	6.26	0.022	4.43	200.5	8.1	0.0143	0.01	440	
			SW17-14-171212	2017-12-12	6.86	0.019	15.33	141.3	4.6	-	0.01	450	
SW17-15	5364900	422182	SW17-14-180305	2018-03-05	7.00	0.020	13.18	164.1	3.7	0.0130	-	450	10m South of SW15-08 in middle of river flow
			SW17-15-170731	2017-07-31	6.53	0.029	-	145.8	15.9	0.0221	0.01	450	
			SW17-15-170821	2017-08-21	7.02	0.024	8.94	154.3	14.9	0.0196	0.01	430	
			SW17-15-170926	2017-09-26	7.17	0.037	-	169.0	13.8	0.0241	0.02	350	
			SW17-15-171031	2017-10-31	6.06	0.033	9.98	234.7	8.3	0.0215	0.01	440	
			SW17-15-171212	2017-12-12	7.39	0.030	15.32	164.6	5.5	-	0.01	450	
SW17-16	5364776	422195	SW17-15-180305	2018-03-05	7.22	0.021	13.32	205.3	3.7	0.0137	-	450	10m downstream of SW15-10
			SW17-16-170801	2017-08-01	5.93	0.019	-	179.9	16.5	0.0143	0.01	450	
			SW17-16-170926	2017-09-26	6.90	0.030	-	182.8	14.1	0.0195	0.01	350	
			SW17-16-171031	2017-10-31	5.90	0.027	2.47	223.6	8.2	0.0182	0.01	440	
			SW17-16-171123	2017-11-23	5.38	0.015	12.21	316.1	5.5	0.0091	0.01	80,000	
			SW17-16-171213	2017-12-13	6.60	0.024	14.82	173.2	5.0	0.0156	0.01	450	
			SW17-16-180305	2018-03-05	6.86	0.022	12.72	174.2	4.0	0.0143	-	450	

* Average (rounded) discharge rates between 8-5 pm on November 18th as measured at the Elliot Dam located approx. 7.6 km upstream on Jordan River.

** pH meter malfunction for September sampling event.

*** possible notation errors.

**** Flowrate estimate made by qualitative comparison to base discharge from Elliot Dam.

- not recorded

n/a not applicable

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW17-17	5364617	422239	SW17-17-170801	2017-08-01	6.35	0.019	-	148.0	17.7	0.0143	0.01	450	10m downstream of SW15-11 near metal plate sticking out of water
			SW17-17-170822	2017-08-22	6.41	0.019	9.14	147.9	15.1	0.0156	0.01	430	
			SW17-17-170927	2017-09-27	7.29	0.028	-	159.7	13.8	0.0182	0.01	350	
			SW17-17-171101	2017-11-01	6.39	0.021	-	178.3	8.2	0.0169	0.01	440	
			SW17-17-171123	2017-11-23	5.55	0.007	12.20	286.0	5.6	0.0084	0.01	80,000	
			SW17-17-171213	2017-12-13	7.01	0.023	15.59	138.0	5.2	0.0150	0.01	450	
			SW17-17-180724	2018-07-24	7.24	0.021	10.38	144.3	16.2	0.0160	0.01	450	
			SW17-17-181101	2018-11-01	5.23	0.017	16.26	250.0	10.2	0.0150	0.01	27,400	
			SW17-17-190124	2019-01-24	5.97	0.023	10.71	181.0	6.7	0.0150	0.01	6,600	
			SW17-17-190924	2019-09-24	6.33	0.019	10.92	140	14.2	0.0160	0.01	6,750****	
			SW17-17-191203	2019-12-03	6.81	0.030	12.63	149.9	5.6	0.0190	0.01	450****	
			SW17-17-200730	2020-07-30	6.06	0.022	10.02	148.7	16.5	0	0.01	2,500	
			SW17-17-210112	2021-01-12	5.56	0.009	13.16	232.3	5.9	0.01	0.01	-	
			SW17-17-210812	2021-08-12	5.81	0.034	9.72	201	17.9	-	-	9,600	
			SW17-17-220210	2022-02-10	6.43	0.016	15.41	205.2	5.7	-	-	8,800	
			SW17-17-221026	2022-10-26	7.44	0.038	10.93	140.8	10.1	-	-	-	
			SW17-17-230222	2023-02-22	6.79	0.020	14.3	317	4.15	-	-	-	
			SW17-17-230802	2023-08-02	7.04	0.0237	10.53	-	16.4	-	-	418	
			SW17-17-231114	2023-11-14	6.32	0.0347	14.22	15.4	6.9	-	-	-	
SW17-18	5364366	422170	SW17-18-170801	2017-08-01	6.04	0.021	-	166.6	18.4	0.0156	0.01	450	25m downstream of SW15-12 in middle of river (not sampled)
SW17-19	5364722	422216	SW17-19-170822	2017-08-22	7.26	0.019	6.50	114.5	15.0	0.0150	0.01	430	10m downstream of SW15-05 as it exits pool
			SW17-19-170927	2017-09-27	7.27	0.028	-	164.0	13.6	0.0182	0.01	350	
			SW17-19-171101	2017-11-01	6.08	0.027	-	174.9	8.3	0.0176	0.01	440	
			SW17-19-171123	2017-11-23	6.51	0.014	12.04	248.0	5.6	0.0091	0.01	80,000	
			SW17-19-171213	2017-12-13	7.47	0.025	15.09	146.7	5.3	0.0163	0.01	450	
			SW17-19-180305	2018-03-05	7.22	0.023	12.11	164.7	4.1	0.0150	-	450	
			SW17-19-180724	2018-07-24	7.17	0.021	9.84	142.6	16.9	0.0160	0.01	450	
			SW17-19-181031	2018-10-31	5.62	0.018	15.61	236.0	9.7	0.0163	0.01	27,400	
			SW17-19-190124	2019-01-24	6.36	0.021	10.56	170.0	6.6	0.0137	0.01	6,600	
			SW17-19-190925	2019-09-25	7.5	0.022	11.19	165	13	0.0190	0.01	6,750****	
			SW17-19-191204	2019-12-04	6.24	0.018	12.65	173.8	6.1	0.0120	0.01	4,500*****	
			SW17-19-210812	2021-08-12	6.03	0.034	9.49	251	18.1	-	-	2,050	
			SW17-19-220210	2022-02-10	6.72	0.015	13.23	173.4	5.7	-	-	2,400	
			SW17-19-221026	2022-10-26	6.92	0.038	10.24	118	10.3	-	-	-	
			SW17-19-230222	2023-02-22	6.01	0.019	13.43	179.5	4.03	-	-	-	
			SW17-19-230802	2023-08-02	6.88	0.0235	8.66	15.8	16.7	-	-	-	
			SW17-19-231114	2023-11-14	6.37	0.0326	13.6	40.9	6.7	-	-	-	
SW17-20	5364804	422188	SW17-20-170822	2017-08-22	7.26	0.020	8.72	121.1	15.9	0.0156	0.01	430	10m upstream of SW15-10 above big rock
SW17-21	5364850	422187	SW17-21-170926	2017-09-26	7.03	0.028	-	170.0	14.1	0.0182	0.01	350	Jordan River below the North Waste Rock Pile (south region)
			SW17-21-171031	2017-10-31	5.89	0.026	9.38	234.6	8.2	0.0169	0.01	440	
			SW17-21-171123	2017-11-23	7.33	0.014	12.45	199.6	5.5	0.0091	0.01	80,000	
			SW17-21-171213	2017-12-13	7.17	0.022	14.60	117.9	4.9	0.0143	0.01	450	
			SW17-21-180305	2018-03-05	6.72	0.022	12.98	201.2	3.9	0.0143	-	450	
			SW17-21-180724	2018-07-24	6.68	0.020	10.15	36.3	16.9	0.0150	0.01	450	
			SW17-21-181031	2018-10-31	5.63	0.018	16.45	242.9	9.5	0.0163	0.01	27,400	
			SW17-21-190124	2019-01-24	6.21	0.020	10.74	170.0	6.5	0.0130	0.01	6,600	
			SW17-21-190925	2019-09-25	6.49	0.020	11.1	163	13.3	0.016	0.01	6,750****	
			SW17-21-191203	2019-12-03	6.41	0.025	13	148.8	4.9	0.016	0.01	450****	
			SW17-21-210811	2021-08-11	7.93	0.032	8.19	63	17.9	-	-	1,050	
			SW17-21-220210	2022-02-10	6.47	0.015	13.60	193.9	5.7	-	-	4,400	
			SW17-21-221026	2022-10-26	7.19	0.037	11.16	27	10.3	-	-	-	
			SW17-21-230222	2023-02-22	5.98	0.018	13.31	174.9	3.95	-	-	11	
			SW17-21-230802	2023-08-02	6.57	0.0238	9.27	-55.9	16.1	-	-	1,839	
			SW17-21-231114	2023-11-14	6.30	0.0318	13.40	63.5	6.7	-	-	-	

* Average (rounded) discharge rates between 8-5 pm on November 18th as measured at the Elliot Dam located approx. 7.6 km upstream on Jordan River.

** pH meter malfunction for September sampling event.

*** possible notation errors.

**** Flowrate estimate made by qualitative comparison to base discharge from Elliot Dam.

- not recorded

n/a not applicable

Table 5: Surface Water Log

Location	Northing (m)	Easting (m)	Sample ID	Sample Date (yyyy/mm/dd)	pH (pH)**	Conductivity (mS/cm)	DO (mg/L)	ORP (mV)	Temp. (°C)	TDS (g/L)	Salinity (ppt)	Approximate Flow Rate L/sec	Location Description
SW18-22	5364865	422190	SW18-22-180726	2018-07-26	7.05	0.082	10.62	186.4	16.3	0.0640	0.05	450	Jordan River near inferred bedrock at North Waste Rock Pile
			SW18-22-181031	2018-10-31	5.80	0.018	15.98	299.1	9.6	0.0169	0.01	27,400	
			SW18-22-190124	2019-01-24	6.17	0.024	10.52	178.0	6.5	0.0156	0.01	6,600	
			SW18-22-190925	2019-09-25	6.82	0.02	11.17	144	13.3	0.017	0.01	6,750*****	
			SW18-22-191203	2019-12-03	6.56	0.026	12.98	154.2	5.1	0.017	0.01	450*****	
			SW18-22-200729	2020-07-29	6.97	0.022	9.93	131.1	16.7	0	1.70	4,000	
			SW18-22-210113	2021-01-13	6.00	0.014	13.65	203	5.6	0.009	0.01	50,000	
			SW18-22-210811	2021-08-11	7.35	0.032	8.12	60.2	17.8	-	-	1,100	
			SW18-22-220210	2022-02-10	6.69	0.015	13.78	189.1	5.9	-	-	2,400	
			SW18-22-221026	2022-10-26	7.16	0.038	11.13	36	10.3	-	-	-	
			SW18-22-230223	2023-02-23	5.7	0.019	13.28	174	2.45	-	-	-	
			SW18-22-230802	2023-08-02	7.01	0.0237	9.15	195.3	15.8	-	-	1,578	
			SW18-22-231115	2023-11-15	6.96	0.0339	18.01	-77.1	6.5	-	-	-	
SW22-28	5368445	425461	SW22-28-221027	2022-10-27	7.16	0.032	10.30	129	10.8	-	-	-	Jordan River upstream of mine break through.
			SW22-28-230223	2023-02-23	6.43	0.014	13.89	203	2.4	-	-	-	
			SW22-28-230802	2023-08-02	7.36	0.0197	9.96	70.8	13.9	-	-	-	
			SW22-28-231116	2023-11-16	6.65	0.0225	11.38	147.3	5.8	-	-	-	
SW23-29	-	-	SW22-29-230802	2023-08-02	6.55	0.0274	10.48	-122.6	17.2	0.021	-	1,173	

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																					
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium
BC Guideline/Standard																															
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-01	SW15-01-151117	2015 11 17	6.83	43.3	-	6.39	90	2	-	< 20	169	< 10	-	18.2	< 100	16.8	16	16	< 1	< 1	< 1	< 0.1	-	5	38	14.4	31	1.78	6.4	0.39	8.63
	SW15-01A-160629	2016 06 29	6.95	47.1	-	7.01	-	-	-	-	79	< 10	-	35.2	< 100	12.5	7	7	< 1	< 1	< 1	0.11	-	2	40	17.0	32	1.16	11.5	0.29	10.5
	SW15-01A-160906	2016 09 06	7.03	71.7	-	-	-	-	-	-	119	< 10	-	59.1	< 100	12.5	7	7	< 1	< 1	< 1	0.19	-	< 2	22	26.1	27	1.59	14.1	0.36	15.9
	SW16-A-160906	Duplicate	7.01	73.8	-	-	-	-	-	-	119	< 10	-	59.1	< 100	12.5	8	8	< 1	< 1	< 1	0.19	-	< 2	22	26.9	27	1.63	14.4	0.38	16.4
	QA/QC RPD%		< 1	3	-	-	-	-	-	-	0	*	-	0	*	0	13	13	*	*	*	*	-	*	*	3	*	2	2	*	3
	SW15-01-161017	2016 10 17	6.49	46.3	-	7.16	-	-	-	-	164	< 10	-	25.8	< 100	22.1	6	6	< 1	< 1	< 1	-	-	3	26	16.1	46	1.52	12.8	0.34	9.69
	SW15-01-161212	2016 12 12	6.89	23.5	-	5.8	-	-	-	-	154	< 5	154	6.44	< 100	7.7	18	18	< 1	< 1	< 1	-	-	3	43	7.4	37	1.20	6.7	0.30	5.18
	SW15-01-170525	2017 05 25	6.66	16.5	-	6.84	-	-	-	24	47	< 10	-	6.48	< 100	11.4	7.2	7.2	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	35.4	5.53	22	0.660	5.63	0.15	4.25
	SW15-01A-170731	2017 07 31	6.69	43.5	-	7.78	-	-	-	-	64	< 10	-	34.5	< 100	11.4	6.3	6.3	< 1.0	< 1.0	< 1.0	-	-	2.6	23.8	15.4	18	1.21	9.59	0.26	10.4
	SW15-01A-170821	2017 08 21	6.86	42.5	-	7.41	-	-	-	-	80	< 10	-	32.9	< 100	10.6	6.4	6.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	27.7	15.3	26	1.06	8.44	-	9.57
	SW15-01A-170926	2017 09 26	6.43	37.7	-	7.97	-	-	-	-	121	30	-	31.2	< 100	11.0	7.4	7.4	< 1.0	< 1.0	< 1.0	-	-	3.6	28.8	13.4	38	1.00	8.14	0.23	8.95
	SW15-01-171031	2017 10 31	6.11	31.7	-	6.43	-	-	-	-	180	< 10	-	19.6	150	16.7	6.4	6.4	< 1.0	< 1.0	< 1.0	-	-	7.8	38.0	11.1	39	0.947	10.7	0.22	7.37
	SW15-01-171122	2017 11 22	6.51	31.8	-	6.8	-	-	-	-	117	< 10	-	19.5	< 100	12.4	9.4	9.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	34.1	10.8	27	1.15	8.82	0.20	6.78
	SW15-01-171212	2017 12 12	6.33	22.2	-	7.73	-	-	-	-	111	< 10	-	6.65	< 100	10.6	12.6	12.6	< 1.0	< 1.0	< 1.0	-	-	5.4	43.1	7.26	27	0.988	5.85	0.24	4.92
	SW15-01-180305	2018 03 05	6.56	19.4	-	7.13	-	-	-	-	81.1	< 5.0	81.1	7.02	< 100	10.3	10.5	10.5	< 1.0	< 1.0	< 1.0	-	-	4.4	49.7	6.33	26	0.874	6.22	0.24	4.60
	SW15-01A-180723	2018 07 23	6.05	55.4	-	8.13	-	-	-	-	66	< 10	-	47.6	< 100	9.6	5.4	5.4	< 1.0	< 1.0	< 1.0	-	-	6.6	26.9	20.3	19	1.15	10.3	0.23	10.5
	SW15-01a-181029	2018 10 29	6.90	39.8	-	6.56	106	< 4.0	-	-	66	< 10	-	28.1	< 100	12.7	6.6	6.6	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	38.9	14.2	38	1.03	9.95	0.23	9.08
	SW-B-181029	Duplicate	6.88	39.5	-	6.56	98	< 4.0	-	-	66	< 10	-	27.7	< 100	12.7	6.7	6.7	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	37.3	14.1	38	1.02	9.87	0.22	9.08
	QA/QC RPD%		< 1	1	-	0	8	*	-	-	0	*	-	1	*	0	2	2	*	*	*	*	-	*	4	1	*	1	1	*	0
	SW15-01-190123	2019 01 23	-	21.1	-	6.44	50	< 4.0	-	-	55	< 10	-	7.85	< 100	11.2	9.4	9.4	< 1.0	< 1.0	< 1.0	< 0.10	-	4.0	53.2	7.08	23	0.837	5.90	0.22	4.53
	SW15-01a-190123	2019 01 23	-	21.3	-	6.97	43	< 4.0	-	-	53	< 10	-	7.87	< 100	11.0	9.4	9.4	< 1.0	< 1.0	< 1.0	< 0.10	-	2.8	64.2	7.13	28	0.849	5.27	0.22	4.55
	SW15-01A-190925	2019 09 25	6.69	35.6	-	6.27	90	< 2.0	1.33	-	136	< 10	-	25.3	< 100	11.3	4.6	4.6	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	25.9	12.9	30	0.848	7.90	0.22	8.06
	SW15-01-191205	2019 12 05	6.87	23.8	-	6.16	61	< 2.0	1.44	-	108	< 5.0	108	11.6	< 100	11.4	12.2	12.2	< 1.0	< 1.0	< 1.0	-	-	4.1	41.6	7.83	33	1.02	7.40	0.21	5.77
	SW15-01A-191205	2019 12 05	7.05	25.1	-	6.64	71	< 2.0	1.51	-	104	< 5.0	104	11.7	< 100	11.4	11.9	11.9	< 1.0	< 1.0	< 1.0	-	-	2.4	42.0	8.16	33	1.14	7.25	0.22	5.70
	SW15-01-200729	2020 07 29	6.68	26.4	-	6.33	77	-	1.33	-	108	< 1.0	-	17.1	22	9.27	6.7	6.7	< 1.0	< 1.0	< 1.0	0.058	-	-	26.0	9.38	23	0.731	7.34	0.179	6.34
	SW15-01-210112	2021 01 12	-	17.8	19.2	6.45	54	-	2.26	< 5.0	79.8	< 1.0	-	7.57	< 20	7.45	11.4	11.4	< 1.0	< 1.0	< 1.0	< 0.050	-	-	46.7	5.93	34	0.731	2.54	0.217	4.28
	SW-01A-08102021	2021 08 10	-	< 0.500	< 0.500	6.25	131	-	-	< 50	69	< 10	-	38.1	< 100	9.2	6.4	6.4	< 1.0	< 1.0	< 1.0	< 0.10	3.0	-	< 5.0	< 0.20	< 10	< 0.010	< 0.20	< 0.10	< 0.10
	SW15-01-220211	2022 02 11	-	17.3	17.3	6.14	51	-	-	-	34	< 10	-	9.06	< 100	7.9	7.3	7.3	< 1.0	< 1.0	< 1.0	< 0.10	-	2.7	51.8	5.78	40	0.688	5.48	0.16	4.37
	SW15-01A-220211	2022 02 11	-	17.3	17.3	6.14	50	-	-	-	34	< 10	-	9.30	< 100	8.2	7.2	7.2	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	59.0	5.79	41	0.695	5.31	0.17	4.43
	SW15-01A-221025	2022 10 25	-	21.1	21.1	6.84	38	-	-	< 50	116	< 10	-	13.0	< 100	8.5	5.5	5.5	< 1.0	< 1.0	< 1.0	< 0.10	2.0	-	34.2	7.38	83	0.636	6.97	0.21	5.45
	SW15-01A-230223	2023 02 23	-	19.2	19.2	7.28	37	-	-	< 50	64	< 10	-	8.63	< 100	7.3	7.6	7.6	< 1.0	< 1.0	< 1.0										

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																							
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L		
BC Guideline/Standard																										
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ⁱ	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k		
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ⁱ	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^j		
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ⁱ	400-1,100 ⁱ	n/a	2.5	100,000	2,500-15,000 ⁱ	200	5-150 ⁱ	30	10,000	n/a	850	n/a	750-16,500 ⁱ		
SW15-01	SW15-01-151117	2015 11 17	0.1	< 0.5	< 5	< 0.1	44	0.12	< 0.5	9.23	290	< 0.1	0.2	< 0.02	0.1	13.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	29		
	SW15-01A-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	55	0.17	0.6	14.6	461	< 0.1	0.1	< 0.02	< 0.1	19.1	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	43		
	SW15-01A-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	94	0.24	< 0.5	17.1	420	< 0.1	0.2	< 0.02	0.1	25.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	56		
	SW16-A-160906	Duplicate	0.1	< 0.5	< 5	< 0.1	94	0.23	< 0.5	17.7	438	< 0.1	0.2	< 0.02	0.1	26.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	58		
	QA/QC RPD%		*	*	*	*	0	4	*	3	4	*	*	*	*	4	*	*	*	*	-	*	*	4		
	SW15-01-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	56	0.15	< 0.5	14.6	361	< 0.1	0.2	< 0.02	0.1	18.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	41		
	SW15-01-161212	2016 12 12	0.1	< 0.5	< 5	< 0.1	25	0.10	0.5	7.74	318	< 0.1	0.1	< 0.02	< 0.1	9.8	< 0.5	0.08	< 0.02	< 5	-	< 0.02	< 1	24		
	SW15-01-170525	2017 05 25	< 0.10	< 0.50	< 5.0	< 0.10	34	0.079	0.53	7.36	305	< 0.10	0.11	< 0.02	< 0.10	9.49	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	22.9		
	SW15-01A-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	62.0	0.156	0.54	13.6	421	< 0.10	0.17	< 0.010	< 0.10	18.2	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	41.5		
	SW15-01A-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	72.9	0.150	< 0.50	12.3	397	< 0.20	0.17	< 0.010	< 0.10	17.0	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	38.6		
	SW15-01A-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	57.3	0.138	< 0.50	11.0	323	< 0.20	0.15	< 0.010	0.11	15.3	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	34.8		
	SW15-01-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	40.5	0.115	< 0.50	11.6	340	< 0.20	0.13	< 0.010	< 0.10	15.1	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	34.1		
	SW15-01-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	49.3	0.111	< 0.50	10.6	317	< 0.20	0.15	< 0.010	< 0.10	12.7	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	29.4		
	SW15-01-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	27.7	0.084	< 0.50	7.13	314	< 0.20	< 0.10	< 0.010	< 0.10	9.75	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	23.3		
	SW15-01-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	30.3	0.075	< 0.50	6.81	302	< 0.20	0.16	0.012	< 0.10	8.51	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	20.2		
	SW15-01A-180723	2018 07 23	< 0.20	< 0.50	< 5.0	< 0.10	54.9	0.175	< 0.50	13.5	401	< 0.20	0.20	< 0.010	< 0.10	18.1	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	41.9		
	SW15-01a-181029	2018 10 29	0.76	< 0.50	< 5.0	< 0.10	47.4	0.144	< 0.50	11.7	360	< 0.20	0.11	< 0.010	< 0.10	15.0	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	33.2		
	SW-B-181029	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	46.9	0.158	< 0.50	11.7	361	< 0.20	0.11	< 0.010	< 0.10	15.1	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	32.7		
	QA/QC RPD%		*	*	*	*	1	9	*	0	< 1	*	*	*	*	1	*	*	*	*	*	*	*	2		
	SW15-01-190123	2019 01 23	< 0.20	< 0.50	< 5.0	< 0.10	28.4	0.084	< 0.50	7.08	294	< 0.20	0.14	< 0.010	< 0.10	8.91	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	25.1		
	SW15-01a-190123	2019 01 23	< 0.20	< 0.50	< 5.0	< 0.10	28.6	0.073	< 0.50	6.64	283	< 0.20	0.13	< 0.010	< 0.10	8.54	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	24.4		
	SW15-01A-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	41.7	0.097	< 0.50	9.04	240	< 0.20	0.15	0.060	< 0.10	12.0	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	27.1		
	SW15-01-191205	2019 12 05	< 0.20	< 0.50	< 5.0	< 0.10	37.5	0.103	< 0.50	8.55	327	< 0.20	0.12	< 0.010	< 0.10	10.6	< 0.50	0.100	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	31.4		
	SW15-01A-191205	2019 12 05	< 0.20	< 0.50	< 5.0	< 0.10	39.0	0.097	< 0.50	8.74	332	< 0.20	0.12	< 0.010	< 0.10	11.0	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	31.0		
	SW15-01-200729	2020 07 29	< 0.10	< 0.10	0.44	< 0.020	40	0.0891	< 0.10	8.30	282	< 0.050	< 1.0	< 0.0050	0.071	10.5	0.133	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	23.6		
	SW15-01-210112	2021 01 12	< 0.10	< 0.10	0.56	< 0.020	20	0.0523	< 0.10	3.39	196	< 0.050	< 1.0	< 0.0050	0.132	5.14	0.239	0.014	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	13.1		
	SW-01A-08102021	2021 08 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	< 0.40 ^a	< 0.20	0.11	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW15-01-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.051	< 0.50	6.34	267	< 0.20	< 0.10	< 0.010	< 0.10	7.91	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	19.4		
	SW15-01A-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.078	< 0.50	6.24	259	< 0.20	< 0.10	< 0.010	< 0.10	7.89	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	19.3		
	SW15-01A-221025	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.091	< 0.50	7.68	213	< 0.20	0.12	< 0.010	< 0.10	9.84	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	22.6		
	SW15-01A-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.064	< 0.50	5.96	258	< 0.20	0.11	< 0.010	< 0.10	7.55	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	19.5		
	SW15-01A-230802	2023 08 02	< 0.050	< 0.050	0.44	< 0.010	28.1	0.0690	< 0.50	6.63	212	< 0.050	0.102	< 0.0050	0.053	8.38	0.14	0.012	< 0.0040	< 0.20	< 0.20	< 0.0010	< 1.00	20.7		
	SW-C-230802	Duplicate	< 0.050	< 0.050	0.43	< 0.010	28.0	0.0705	< 0.50	6.54	210	< 0.050	0.104	< 0.0050	0.051	8.65	0.12	0.010	< 0.0040	< 0.20	< 0.20	< 0.0010	< 1.00	19.3		
	QA/QC RPD%		*	*	*	*	0	2	*	1	1	*	*	*	4	3	*	*	*	*	*	*	*	7		
	SW15-01A-231113	2023 11 13	< 0.050	0.065	0.58	< 0.010	26.3	0.067																		

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.
Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Standard to protect freshwater aquatic life.
^d Guideline/standard varies with pH and temperature. Temperature assumed 15C

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-02	SW15-02-151117	2015 11 17	7.32	42.9	-	7.38	95	13	-	< 20	164	< 10	-	17.9	< 100	16.7	16	16	< 1	< 1	< 1	< 0.1	-	< 2	37	14.3	30	1.73	5.4	0.41	8.55	
	SW15-02-160629	2016 06 29	6.87	48.8	-	7.56	-	-	-	-	80	< 10	-	36.2	< 100	12.1	6	6	< 1	< 1	< 1	0.12	-	< 2	34	17.4	32	1.31	8.2	0.30	11.1	
	SW15-02-160906	2016 09 06	6.96	73.0	-	-	-	-	-	-	150	< 10	-	58.1	< 100	12.5	9	9	< 1	< 1	< 1	0.20	-	2	10	26.5	12	1.68	11.3	0.38	15.7	
	SW15-02-161017	2016 10 17	6.81	45.3	-	6.94	-	-	-	-	154	< 10	-	25.9	< 100	22.2	6	6	< 1	< 1	< 1	-	-	< 2	22	15.6	41	1.57	10.4	0.37	9.94	
	SW16-A-161017	Duplicate	6.88	45.2	-	6.94	-	-	-	-	152	< 10	-	25.8	< 100	22.1	6	6	< 1	< 1	< 1	-	-	< 2	21	15.6	40	1.50	9.9	0.35	9.66	
	QA/QC RPD%		1	< 1	-	0	-	-	-	-	1	*	-	< 1	*	< 1	0	0	*	*	*	-	-	*	*	0	*	5	5	*	3	
	SW15-02-161212	2016 12 12	7.07	24.0	-	6.94	-	-	-	-	140	< 5	140	6.71	< 100	7.9	17	17	< 1	< 1	< 1	-	-	< 2	39	7.7	36	1.19	4.4	0.31	5.30	
	SW15-02-170731	2017 07 31	6.80	45.3	-	6.56	-	-	-	-	95	< 10	-	36.6	120	11.5	6.2	6.2	< 1.0	< 1.0	< 1.0	-	-	2.1	27.5	15.8	23	1.39	6.44	0.30	10.9	
	SW15-02-170821	2017 08 21	6.82	42.4	-	6.97	-	-	-	-	82	< 10	-	33.4	< 100	10.7	6.9	6.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	22.2	15.1	22	1.16	5.85	-	9.54	
	SW15-02-170926	2017 09 26	6.55	37.7	-	7.66	-	-	-	-	122	< 10	-	31.6	< 100	11.2	7.3	7.3	< 1.0	< 1.0	< 1.0	-	-	5.4	27.9	13.4	39	1.03	6.50	0.22	9.05	
	SW15-02-171031	2017 10 31	6.21	31.0	-	6.55	-	-	-	-	169	< 10	-	18.9	< 100	16.9	6.5	6.5	< 1.0	< 1.0	< 1.0	-	-	4.0	33.9	10.9	37	0.939	5.70	0.21	6.96	
	SW15-02-171122	2017 11 22	6.62	32.5	-	6.88	-	-	-	-	118	< 10	-	19.9	140	11.8	9.3	9.3	< 1.0	< 1.0	< 1.0	-	-	43.0	34.3	11.1	27	1.15	6.73	0.20	6.77	
	SW15-02-171212	2017 12 12	6.54	23.1	-	7.19	-	-	-	-	114	< 10	-	6.55	< 100	10.9	12.2	12.2	< 1.0	< 1.0	< 1.0	-	-	3.4	37.2	7.65	25	0.973	3.62	0.24	4.97	
	SW15-02-180305	2018 03 05	6.62	20.0	-	7.01	-	-	-	-	72.4	< 5.0	72.4	7.04	< 100	10.5	10.4	10.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	42.8	6.54	27	0.884	3.46	0.24	4.70	
	SW15-02-180724	2018 07 24	6.40	53.2	-	7.17	-	-	-	-	68	< 10	-	39.9	110	9.9	3.7	3.7	< 1.0	< 1.0	< 1.0	-	-	2.0	27.3	19.2	29	1.29	7.90	0.31	11.3	
	SW-C-180724	Duplicate	6.31	51.8	-	7.17	-	-	-	-	67	< 10	-	45.6	< 100	9.9	2.7	2.7	< 1.0	< 1.0	< 1.0	-	-	< 2.0	32.0	18.6	22	1.27	7.58	0.30	11.1	
	QA/QC RPD%		1	3	-	0	-	-	-	-	1	*	-	13	*	0	*	*	*	*	*	-	-	*	*	16	3	*	2	4	*	2
	SW15-02-181029	2018 10 29	6.94	39.2	-	6.69	112	< 4.0	-	-	83	< 10	-	26.4	< 100	12.7	7.5	7.5	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	30.6	14.0	35	1.05	6.72	0.26	9.14	
	SW15-02-190123	2019 01 23	7.00	20.9	-	7	41	< 4.0	-	-	49	< 10	-	7.73	< 100	10.6	10.2	10.2	< 1.0	< 1.0	< 1.0	< 0.10	-	2.0	50.9	7.00	24	0.830	3.43	0.22	4.34	
	SW15-02-190925	2019 09 25	6.67	35.8	-	6.61	95	< 2.0	1.53	-	129	< 10	-	25.5	< 100	11.4	4.1	4.1	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	20.2	12.9	26	0.888	4.40	0.26	8.11	
	SW15-02-191205	2019 12 05	7.15	23.9	-	7.07	59	< 2.0	1.49	-	98.5	< 5.0	98.5	11.9	< 100	11.4	11.6	11.6	< 1.0	< 1.0	< 1.0	-	-	< 2.0	37.8	7.91	30	1.01	4.76	0.21	5.57	
	SW15-02-210812	2021 08 12	-	48.3	48.3	6.45	140	< 2.0	-	-	113	< 10	-	38.7	< 100	9.8	8.4	8.4	< 1.0	< 1.0	< 1.0	0.13	3.1	-	28.2	17.3	29	1.24	6.29	0.31	12.1	
SW15-02-220211	2022 02 11	-	17.4	17.4	6.29	39	-	-	-	30	< 10	-	9.09	< 100	8.3	7.2	7.2	< 1.0	< 1.0	< 1.0	< 0.10	2.6	-	47.0	5.78	39	0.708	3.69	0.16	4.40		
SW15-02-221025	2022 10 25	-	21.2	21.2	7.05	62	-	-	< 50	123	< 10	-	12.9	< 100	8.6	5.1	5.1	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	30.2	7.40	79	0.656	3.49	0.21	5.50		
SW15-02-230223	2023 02 23	-	18.9	18.9	7.32	47	-	-	< 50	62	< 10	-	8.68	< 100	7.4	7.3	7.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	52.4	6.33	35	0.753	4.01	0.20	4.31		
SW15-02-230802	2023 08 02	-	18.1	17.2	6.69	37	-	-	< 50	112	< 10	-	10.0	< 100	8.2	5.9	5.9	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	39.0	6.26	41.1	0.602	3.19	0.195	4.77		
SW15-02-231115	2023 11 15	-	21.7	21.9	6.42	42	-	-	< 50	115	< 10	-	10.9	< 100	10.4	5.8	5.8	< 1.0	< 1.0	< 1.0	< 0.1	3.7	-	50.1	7.37	42.9	0.803	4.03	0.206	5.38		
SW15-03	SW15-03-151117	2015 11 17	4.32	155	-	4.34	417	3	-	20	332	< 10	-	4.75	100	334	< 1	< 1	< 1	< 1	< 1	< 0.1	-	97	8,430	54.4	236	4.76	235	0.63	8.89	
	SW15-03-171123	2017 11 23	4.08	136	-	3.98	-	-	-	-	191	< 10	-	5.01	140	238	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	-	-	132	8,570	46.6	154	4.86	256	0.55	8.13	
	SW15-03-181101	2018 11 01	3.87	235	-	3.88	757	< 5.0	-	-	322	< 10	-	6.51	260	468	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 0.10	-	201	16,800	82.5	267	7.07	441	0.5		

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 60

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ^f	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^f	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ^f	100	400	200-900 ^f	400-1,100 ^f	n/a	2.5	100,000	2,500-15,000 ^f	200	5-150 ^f	30	10,000	n/a	850	n/a	750-16,500 ^f
SW15-02	SW15-02-151117	2015 11 17	< 0.1	< 0.5	< 5	< 0.1	42	0.12	< 0.5	8.97	244	< 0.1	0.2	< 0.02	0.1	13.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	30
	SW15-02-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	57	0.20	0.6	14.2	367	< 0.1	0.2	< 0.02	< 0.1	19.3	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	44
	SW15-02-160906	2016 09 06	0.1	< 0.5	< 5	< 0.1	98	0.24	< 0.5	17.2	401	< 0.1	0.2	< 0.02	< 0.1	24.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	58
	SW15-02-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	53	0.17	< 0.5	14.2	301	< 0.1	0.2	< 0.02	0.1	18.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	44
	SW16-A-161017	Duplicate	< 0.1	< 0.5	< 5	< 0.1	51	0.19	< 0.5	13.3	285	< 0.1	0.2	< 0.02	0.1	17.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	41
	QA/QC RPD%		*	*	*	*	4	11	*	7	5	*	*	*	*	5	*	*	*	*	-	*	*	7
	SW15-02-161212	2016 12 12	0.1	< 0.5	< 5	< 0.1	29	0.09	< 0.5	6.85	245	< 0.1	0.1	< 0.02	< 0.1	9.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	21
	SW15-02-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	62.2	0.168	< 0.50	13.2	345	< 0.10	0.18	< 0.010	< 0.10	18.5	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	44.4
	SW15-02-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	67.7	0.174	< 0.50	11.7	334	< 0.20	0.17	< 0.010	< 0.10	17.2	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	39.0
	SW15-02-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	56.6	0.151	< 0.50	10.6	292	< 0.20	0.15	< 0.010	< 0.10	15.0	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	35.3
	SW15-02-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	38.3	0.130	< 0.50	9.50	232	< 0.20	0.13	< 0.010	< 0.10	13.5	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	31.8
	SW15-02-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	46.6	0.113	< 0.50	9.98	247	< 0.20	0.16	< 0.010	< 0.10	12.4	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	30.0
	SW15-02-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	27.1	0.082	< 0.50	6.45	225	< 0.20	< 0.10	< 0.010	< 0.10	9.31	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	22.7
	SW15-02-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	30.4	0.081	0.84	5.94	195	< 0.20	0.16	0.019	< 0.10	7.98	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	19.6
	SW15-02-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	57.3	0.175	< 0.50	12.9	333	< 0.20	0.14	< 0.010	< 0.10	18.2	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	42.4
	SW-C-180724	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	47.0	0.161	< 0.50	13.0	320	< 0.20	0.13	< 0.010	< 0.10	18.1	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	41.9
	QA/QC RPD%		*	*	*	*	20	8	*	1	4	*	*	*	*	1	*	*	*	*	*	*	*	1
	SW15-02-181029	2018 10 29	< 0.20	< 0.50	< 5.0	< 0.10	45.4	0.126	< 0.50	11.0	288	< 0.20	0.11	< 0.010	< 0.10	15.3	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	35.2
	SW15-02-190123	2019 01 23	< 0.20	< 0.50	< 5.0	< 0.10	28.7	0.086	< 0.50	5.86	200	< 0.20	0.13	< 0.010	0.11	7.83	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	23.3
	SW15-02-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	43.8	0.116	< 0.50	7.79	182	< 0.20	0.17	0.052	< 0.10	11.5	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	29.2
	SW15-02-191205	2019 12 05	< 0.20	< 0.50	< 5.0	< 0.10	38.1	0.092	< 0.50	7.70	234	< 0.20	0.10	< 0.010	< 0.10	9.98	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	27.4
SW15-02-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	52.4	0.135	< 0.50	10.1	264	< 0.20	< 0.10	< 0.010	< 0.10	15.2	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	35.0	
SW15-02-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.063	< 0.50	5.55	202	< 0.20	< 0.10	< 0.010	< 0.10	7.36	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	18.1	
SW15-02-221025	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.091	< 0.50	6.42	164	< 0.20	0.12	< 0.010	< 0.10	8.98	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	21.7	
SW15-02-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.070	< 0.50	5.70	235	< 0.20	< 0.10	< 0.010	< 0.10	7.50	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	17.9	
SW15-02-230802	2023 08 02	< 0.050	< 0.050	0.43	< 0.010	28.1	0.0745	< 0.50	5.64	177	< 0.050	0.105	< 0.0050	0.057	7.78	0.14	0.011	< 0.0040	< 0.20	< 0.20	< 0.0010	< 1.00	19.4	
SW15-02-231115	2023 11 15	< 0.050	0.059	0.56	< 0.010	25.6	0.0789	< 0.50	6.35	200	< 0.050	0.123	< 0.0050	0.080	9.10	0.23	0.020	< 0.0040	< 0.20	< 0.20	0.0013	< 1.00	19.5	
SW15-03	SW15-03-151117	2015 11 17	< 0.1	< 0.5	< 5	0.2	18	1.13	< 0.5	152	29,700	0.1	3.2	< 0.02	< 0.1	233	0.5	< 0.05	< 0.02	< 5	-	0.02	< 1	268
	SW15-03-171123	2017 11 23	< 0.20	< 0.50	6.4	0.17	18.3	1.33	< 0.50	221	31,500	< 0.20	3.34	< 0.010	< 0.10	334	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	0.020	< 1.0	333
	SW15-03-181101	2018 11 01	< 0.20	< 0.50	5.1	0.34	22.2	2.29	0.60	374	61,000	0.25	5.84	< 0.010	< 0.10	519	0.80	0.132	< 0.020	< 5.0	< 1.0	0.035	< 1.0	571
SW15-04	SW15-04-151117	2015 11 17	< 0.1	< 0.5	< 5	0.3	16	1.45	< 0.5	185	39,200	0.4	3.6	< 0.02	< 0.1	264	0.9	< 0.05	< 0.02	< 5	-	0.02	< 1	358
	SW15-04-181101	2018 11 01	< 0.20	< 0.50	< 5.0	0.27	17.8	1.62	< 0.50	287	40,000	0.90	4.29	< 0.010	< 0.10	369	1.18	0.133	< 0.020	< 5.0	< 1.0	0.021	< 1.0	418
SW15-05	SW15-05-151117	2015 11 17	< 0.1	< 0.5	< 5	< 0.1	6	< 0.01	< 0.5	0.13	26.9	< 0.1</												

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B177

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-05	SW15-05-170822	2017 08 22	7.46	22.9	-	7.13	-	-	-	-	82	< 10	-	3.69	< 100	1.6	28.4	28.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	7.6	6.37	31	1.69	0.86	-	4.32	
	SW15-05-170927	2017 09 27	6.91	24.3	-	7.84	-	-	-	-	103	< 10	-	3.81	< 100	1.7	32.2	32.2	< 1.0	< 1.0	< 1.0	-	-	3.0	7.4	6.59	33	1.91	0.62	0.30	4.37	
	SW15-05-171101	2017 11 01	6.64	25.5	-	6.88	-	-	-	-	122	< 10	-	5.15	210	2.2	33.4	33.4	< 1.0	< 1.0	< 1.0	-	-	3.7	10.7	7.13	43	1.87	1.34	0.73	4.41	
	SW15-05-171123	2017 11 23	6.93	20.2	-	6.85	-	-	-	-	293	< 10	-	5.17	< 100	1.6	20.6	20.6	< 1.0	< 1.0	< 1.0	-	-	4.4	35.0	5.59	64	1.52	1.08	0.41	3.78	
	SW15-05-171213	2017 12 13	6.69	15.0	-	6.81	-	-	-	-	86	< 10	-	2.79	< 100	1.3	17.2	17.2	< 1.0	< 1.0	< 1.0	-	-	3.1	15.9	4.05	15	1.18	0.46	0.27	3.73	
	SW-A-171213	Duplicate	6.67	21.8	-	6.81	-	-	-	-	81	< 10	-	3.33	< 100	1.4	15.9	15.9	< 1.0	< 1.0	< 1.0	-	-	3.1	15.6	6.83	16	1.16	0.45	0.26	3.62	
	QA/QC RPD%		< 1	37	-	0	-	-	-	-	6	*	-	18	*	*	8	8	*	*	*	-	-	*	*	51	*	2	*	*	3	
SW15-06	SW15-05-180305	2018 03 05	6.85	14.3	-	7.08	-	-	-	-	108	< 5.0	108	3.46	< 100	1.6	16.3	16.3	< 1.0	< 1.0	< 1.0	-	-	< 2.0	15.8	3.86	11	1.14	0.36	0.28	3.64	
	SW15-06-151117	2015 11 17	7.06	17.7	-	7.28	45	34	-	42	333	< 10	-	6.31	130	1.8	53	53	< 1	< 1	< 1	< 0.1	-	< 2	92	4.9	77	1.35	2.4	0.57	4.03	
	SW15-06-160630	2016 06 30	7.54	26.6	-	8.31	-	-	-	-	150	< 10	-	5.10	< 100	2.9	29	29	< 1	< 1	< 1	< 0.10	-	< 2	8	7.4	< 10	1.98	0.4	0.34	5.06	
	SW15-06-160907	2016 09 07	7.52	29.5	-	-	-	-	-	-	177	< 10	-	4.56	< 100	2.8	32	32	< 1	< 1	< 1	< 0.10	-	< 2	7	8.2	12	2.20	0.2	0.33	5.35	
	SW15-06-161018	2016 10 18	7.41	29.8	-	6.54	-	-	-	-	824	< 10	-	6.27	< 100	3.1	27	27	< 1	< 1	< 1	-	-	< 2	20	8.3	25	2.23	0.9	0.56	4.84	
	SW15-06-161212	2016 12 12	6.47	14.7	-	6.87	-	-	-	-	194	< 5	194	3.57	< 100	1.4	13	13	< 1	< 1	< 1	-	-	< 2	34	3.9	19	1.23	0.5	0.32	3.76	
	SW15-06-170801	2017 08 01	7.51	23.1	-	6.24	-	-	-	-	129	< 10	-	5.35	< 100	1.7	26.7	26.7	< 1.0	< 1.0	< 1.0	-	-	< 2.0	8.6	6.31	13	1.78	0.64	0.28	4.75	
	SW15-06-170822	2017 08 22	7.47	24.0	-	6.57	-	-	-	-	79	< 10	-	3.72	< 100	1.7	30.4	30.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	6.8	6.70	12	1.76	0.72	-	4.38	
	SW15-06-170927	2017 09 27	6.92	25.4	-	7.51	-	-	-	-	129	< 10	-	3.94	< 100	2.0	32.4	32.4	< 1.0	< 1.0	< 1.0	-	-	3.0	6.4	6.85	13	2.02	0.83	0.31	4.49	
	SW15-06-171101	2017 11 01	6.67	26.4	-	6.85	-	-	-	-	180	< 10	-	4.60	130	2.0	33.7	33.7	< 1.0	< 1.0	< 1.0	-	-	3.8	8.6	7.39	16	1.92	0.89	0.40	4.33	
	SW15-06-171123	2017 11 23	6.94	19.4	-	6.89	-	-	-	-	315	< 10	-	5.20	< 100	1.7	21.4	21.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	37.3	5.29	30	1.49	0.91	0.38	3.62	
	SW-B-171123	Duplicate	6.88	21.0	-	6.89	-	-	-	-	313	< 10	-	8.16	270	2.4	20.3	20.3	< 1.0	< 1.0	< 1.0	-	-	11.1	40.3	5.75	33	1.61	1.02	0.46	3.99	
	QA/QC RPD%		1	8	-	0	-	-	-	-	1	*	-	44	*	*	5	5	*	*	*	-	-	*	*	8	8	*	8	*	*	10
SW15-07	SW15-06-171213	2017 12 13	6.70	14.6	-	7.48	-	-	-	-	89	< 10	-	3.03	< 100	1.3	16.9	16.9	< 1.0	< 1.0	< 1.0	-	-	3.0	16.8	3.90	11	1.17	0.33	0.27	3.66	
	SW15-06-180305	2018 03 05	6.87	15.5	-	7.07	-	-	-	-	118	< 5.0	118	3.49	< 100	1.7	16.6	16.6	< 1.0	< 1.0	< 1.0	-	-	< 2.0	16.2	4.25	< 10	1.18	0.30	0.28	3.69	
	SW15-07-151117	2015 11 17	7	44.4	-	6.69	92	< 2	-	34	153	< 10	-	18.9	< 100	17.9	16	16	< 1	< 1	< 1	< 0.1	-	4	42	14.8	35	1.84	7.5	0.38	8.61	
	SW15-07-160629	2016 06 29	6.91	47.4	-	6.78	-	-	-	-	82	< 10	-	35.5	< 100	11.9	7	7	< 1	< 1	< 1	0.12	-	< 2	42	17.1	32	1.12	12.0	0.26	10.5	
	SW15-07-160906	2016 09 06	6.95	75.7	-	-	-	-	-	-	116	< 10	-	61.1	< 100	12.3	8	8	< 1	< 1	< 1	0.20	-	2	10	27.8	< 10	1.55	14.6	0.36	16.4	
	SW15-07-161017	2016 10 17	6.85	45.6	-	6.72	-	-	-	-	156	< 10	-	25.2	< 100	21.6	7	7	< 1	< 1	< 1	-	-	< 2	25	15.8	43	1.53	11.8	0.36	9.60	
	SW15-07-161212	2016 12 12	7.16	25.1	-	6.96	-	-	-	-	138	< 5	138	6.79	< 100	9.5	17	17	< 1	< 1	< 1	-	-	< 2	44	8.0	40	1.25	6.2	0.30	5.28	
	SW15-07-170731	2017 07 31	6.83	46.1	-	5.99	-	-	-	-	59	< 10	-	35.0	< 100	11.4	6.4	6.4	< 1.0	< 1.0	< 1.0	-	-	2.2	57.4	16.4	38	1.24	10.5	0.27	11.2	
	SW15-07-170821	2017 08 21	6.80	40.7	-	6.98	-	-	-	-	81	< 10	-	32.7	< 100	10.7	6.7	6.7	< 1.0	< 1.0	< 1.0	-	-	< 2.0	27.2	14.6	25	0.992	8.67	-	9.30	
	SW15-07-170927	2017 09 27	6.37	37.4	-	6.73	-	-	-	-	118	< 10	-	30.2	< 100	10.8	6.9	6.9	< 1.0	< 1.0	< 1.0	-	-	4.0	29.9	13.4	39	0.954	8.49	0.21	9.05	
	SW15-07-171031	2017 10 31	6.22	30.9	-	6.26	-	-	-	-	172	< 10	-	18.8	< 100	16.8	7.3	7.3	< 1.0	< 1.0	< 1.0	-	-	4.2	38.5	10.8	38	0.928	9.43	0.20	7.01	
	SW15-07-171122	2017 11 22	6.57	31.2	-	6.62	-	-	-	-	149	< 10	-	19.5	100	13.9	10.8	10.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	33.0	10.6	24	1.14	8.90	0.16	6.25	
	SW15-07-171212	2017 12 12	6.50	23.2	-	7.07	-	-	-	-	104	< 10	-	6.34	< 100	11.5	13.3	13.3														

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B17

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																						
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L	
BC Guideline/Standard																									
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ^f	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k	
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^f	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ⁱ	400-1,100 ⁱ	n/a	2.5	100,000	2,500-15,000 ⁱ	200	5-150 ⁱ	30	10,000	n/a	850	n/a	750-16,500 ⁱ	
SW15-05	SW15-05-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	22.7	< 0.010	< 0.50	< 0.10	4.97	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW15-05-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	12.0	< 0.010	< 0.50	< 0.10	5.27	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW15-05-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	6.3	< 0.010	< 0.50	< 0.10	7.34	< 0.20	< 0.10	< 0.010	< 0.10	1.15	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	9.2	
	SW15-05-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	15.6	< 0.010	< 0.50	< 0.10	14.4	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW15-05-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	6.7	< 0.010	< 0.50	< 0.10	3.14	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW-A-171213	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	7.4	< 0.010	< 0.50	< 0.10	3.02	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	QA/QC RPD%		*	*	*	*	*	*	*	*	4	*	*	*	*	*	*	*	*	*	*	*	*	*	
SW15-06	SW15-05-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	7.5	< 0.010	< 0.50	< 0.10	2.84	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW15-06-151117	2015 11 17	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	< 0.5	0.18	31.2	< 0.1	< 0.1	< 0.02	< 0.1	0.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW15-06-160630	2016 06 30	< 0.1	< 0.5	< 5	< 0.1	13	< 0.01	< 0.5	0.06	16.2	< 0.1	< 0.1	< 0.02	0.1	0.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW15-06-160907	2016 09 07	< 0.1	< 0.5	< 5	< 0.1	8	< 0.01	< 0.5	< 0.05	19.1	< 0.1	< 0.1	< 0.02	0.1	0.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW15-06-161018	2016 10 18	< 0.1	< 0.5	< 5	< 0.1	12	< 0.01	0.7	0.10	41.1	< 0.1	< 0.1	< 0.02	0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	5	
	SW15-06-161212	2016 12 12	< 0.1	< 0.5	< 5	< 0.1	7	0.02	0.5	0.07	10.2	< 0.1	< 0.1	< 0.02	< 0.1	0.2	< 0.5	0.10	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW15-06-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	11.9	< 0.010	0.71	< 0.10	7.12	< 0.10	< 0.10	< 0.010	< 0.10	< 0.20	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW15-06-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	17.5	< 0.010	< 0.50	< 0.10	6.91	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW15-06-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	9.5	< 0.010	< 0.50	< 0.10	7.16	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW15-06-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	5.7	< 0.010	< 0.50	< 0.10	10.5	< 0.20	< 0.10	< 0.010	0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW15-06-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	11.7	< 0.010	< 0.50	< 0.10	16.9	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW-B-171123	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	5.1	< 0.010	< 0.50	< 0.10	18.4	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
QA/QC RPD%		*	*	*	*	*	*	*	*	8	*	*	*	*	*	*	*	*	*	*	*	*	*		
SW15-07	SW15-06-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	5.7	< 0.010	< 0.50	< 0.10	9.13	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	8.2	
	SW15-06-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	7.2	< 0.010	< 0.50	< 0.10	4.69	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW15-07-151117	2015 11 17	< 0.1	< 0.5	< 5	< 0.1	37	0.11	< 0.5	9.41	418	< 0.1	0.3	< 0.02	< 0.1	14.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	32	
	SW15-07-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	54	0.17	< 0.5	14.5	449	< 0.1	0.1	< 0.02	< 0.1	18.8	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	43	
	SW15-07-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	99	0.24	< 0.5	18.0	466	< 0.1	0.2	< 0.02	< 0.1	24.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	59	
	SW15-07-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	51	0.14	< 0.5	14.0	355	< 0.1	0.2	< 0.02	0.1	18.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	40	
	SW15-07-161212	2016 12 12	0.1	< 0.5	< 5	< 0.1	27	0.11	1.0	7.63	316	< 0.1	0.1	< 0.02	< 0.1	10.5	< 0.5	0.11	< 0.02	< 5	-	< 0.02	< 1	23	
	SW15-07-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	58.7	0.147	0.52	14.1	438	< 0.10	0.16	< 0.010	< 0.10	18.7	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	42.6	
	SW15-07-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	56.7	0.142	< 0.50	11.9	377	< 0.20	0.15	< 0.010	< 0.10	16.0	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	34.9	
	SW15-07-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	51.7	0.128	< 0.50	10.8	317	< 0.20	0.13	< 0.010	< 0.10	14.8	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	34.0	
	SW15-07-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	36.0	0.118	0.60	10.7	308	< 0.20	0.12	< 0.010	< 0.10	14.3	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	33.3	
	SW15-07-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	36.1	0.118	< 0.50	10.4	350	< 0.20	0.18	< 0.010	< 0.10	13.2	< 0.50	<							

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-07	SW15-07-190926	2019 09 26	6.70	34.0	-	6.43	92	< 2.0	1.44	-	133	< 10	-	23.9	< 100	12.3	5.2	5.2	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	28.0	12.2	29	0.873	7.55	0.24	7.65	
	SW15-07-191205	2019 12 05	7.12	24.3	-	6.95	73	< 2.0	1.47	-	108	< 5.0	108	11.5	< 100	11.6	12.7	12.7	< 1.0	< 1.0	< 1.0	-	< 2.0	43.4	8.03	32	1.03	7.03	0.21	5.52		
	SW15-07-210812	2021 08 12	-	47.9	47.9	6.86	142	7.6	1.05	-	104	< 10	-	38.5	< 100	9.3	7.6	7.6	< 1.0	< 1.0	< 1.0	0.13	3.2	-	36.3	17.4	33	1.09	9.04	0.26	11.9	
	SW15-07-220211	2022 02 11	-	17.5	17.5	6.44	54	-	-	-	46	< 10	-	9.10	< 100	8.3	7.7	7.7	< 1.0	< 1.0	< 1.0	< 0.10	2.3	-	51.2	5.85	39	0.692	4.96	0.17	4.29	
	SW15-07-221025	2022 10 25	-	21.5	21.5	6.92	89	-	-	< 50	124	< 10	-	13.2	< 100	8.7	5.1	5.1	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	37.2	7.51	84	0.658	6.69	0.20	5.54	
	SW15-07-230223	2023 02 23	-	19.2	19.2	7.32	25	-	-	< 50	58	< 10	-	8.60	< 100	7.4	7.4	7.4	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	51.0	6.39	37	0.778	4.44	0.21	4.38	
	SW15-07-230802	2023 08 02	-	18.3	16.5	6.44	41	-	-	< 50	126	< 10	-	10.1	< 100	8.3	5.5	5.5	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	46.4	6.32	46.6	0.599	5.04	0.195	4.86	
SW15-08	SW15-07-231115	2023 11 15	-	22.4	22.5	6.42	55	-	-	< 50	124	33	-	11.0	< 100	10.9	6.0	6.0	< 1.0	< 1.0	< 1.0	< 0.1	5.3	-	57.1	7.62	47.7	0.818	4.46	0.204	5.36	
	SW15-08-151118	2015 11 18	6.96	12.7	-	7.06	56	8	3.1	< 20	47	< 10	-	5.18	< 100	3.1	4	4	< 1	< 1	< 1	< 0.1	-	< 2	112	4.1	44	0.63	3.2	0.23	3.1	
	SW15-08-160629	2016 06 29	6.93	28.6	-	7.39	65	< 2	2.4	-	58	< 10	-	21.3	< 100	7.0	6	6	< 1	< 1	< 1	< 0.10	-	< 2	49	9.9	90	0.92	4.2	0.26	6.51	
	SW15-08-160906	2016 09 06	6.97	25.0	-	-	57	< 2	1.9	-	124	< 10	-	14.1	< 100	3.9	7	7	< 1	< 1	< 1	< 0.10	-	< 2	46	8.6	540	0.88	3.5	0.33	5.70	
	SW15-08-161017	2016 10 17	6.40	7.04	-	6.72	32	< 2	4.1	-	64	< 10	-	3.43	< 100	1.5	2	2	< 1	< 1	< 1	-	-	2	161	2.0	112	0.50	4.4	0.28	1.87	
	SW15-08-161212	2016 12 12	7.22	23.6	-	6.98	39	< 2	1.3	-	139	< 5	139	7.17	< 100	8.1	18	18	< 1	< 1	< 1	-	-	3	42	7.5	37	1.19	4.3	0.30	5.29	
	SW15-08-170731	2017 07 31	6.78	19.5	-	6.31	62	< 2.0	1.97	-	73	< 10	-	12.8	< 100	4.2	4.9	4.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	50.2	6.52	91	0.784	2.86	0.25	4.89	
	SW15-08-170821	2017 08 21	6.81	21.0	-	6.95	32	< 2.0	2.62	-	< 10	< 10	-	13.1	< 100	4.4	5.9	5.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	41.9	7.17	118	0.741	2.98	-	4.81	
	SW15-08-170926	2017 09 26	6.38	12.2	-	7.24	27	< 2.0	2.90	-	86.1	< 5.0	86.1	8.31	< 100	3.1	6.7	6.7	< 1.0	< 1.0	< 1.0	-	-	3.2	36.5	3.87	195	0.619	1.99	0.26	3.05	
	SW15-08-171031	2017 10 31	6.22	6.16	-	6.02	24	< 2.0	3.56	-	83	< 10	-	5.84	290	1.1	4.4	4.4	< 1.0	< 1.0	< 1.0	-	-	3.2	87.7	1.65	57	0.494	0.75	0.18	1.80	
	SW15-08-171122	2017 11 22	6.17	3.78	-	6.32	< 15	< 3.3	3.82	-	24	< 10	-	2.04	< 100	< 1.0	1.9	1.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	135	1.02	45	0.298	2.44	< 0.10	1.16	
	SW15-08-171212	2017 12 12	6.51	18.2	-	7.4	44	< 2.0	2.29	-	69	< 10	-	4.80	< 100	8.2	9.8	9.8	< 1.0	< 1.0	< 1.0	-	-	3.3	53.2	5.88	26	0.840	2.57	0.21	4.19	
	SW15-08-180305	2018 03 05	6.48	7.58	-	7.3	30	< 5.0	2.33	-	36.4	< 5.0	36.4	3.35	< 100	2.8	4.3	4.3	< 1.0	< 1.0	< 1.0	-	-	< 2.0	67.9	2.16	16	0.531	0.83	0.14	2.45	
	SW15-08-180725	2018 07 25	6.12	9.54	-	6.61	32	< 4.0	2.78	-	82	< 10	-	5.40	< 100	1.5	2.9	2.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	29.2	2.99	120	0.499	1.86	0.19	2.33	
	SW15-08-181030	2018 10 30	-	6.57	-	6.01	32	< 4.0	5.32	-	34	< 10	-	4.33	< 100	1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	160	1.62	45	0.610	0.82	0.12	2.62	
	SW-C-181030	Duplicate	-	6.27	-	6.01	22	< 4.0	5.01	-	37	< 10	-	4.31	< 100	1.0	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	2.3	157	1.54	45	0.586	0.86	0.11	2.55	
	QA/QC RPD%		-	5	-	0	*	*	6	-	*	*	-	< 1	*	*	*	*	*	*	*	*	-	*	2	5	*	4	*	*	3	
	SW15-08-190123	2019 01 23	-	4.63	-	6.12	< 15	< 4.0	4.20	-	< 10	< 10	-	2.82	< 100	< 1.0	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	2.6	154	1.20	32	0.398	0.76	< 0.10	1.73	
	SW15-08-190926	2019 09 26	6.09	7.18	-	6.34	25	2.2	8.15	-	48	< 10	-	3.06	< 100	< 1.0	1.6	1.6	< 1.0	< 1.0	< 1.0	< 0.10	-	2.3	190	1.93	173	0.570	5.22	0.16	2.04	
	SW15-08-191203	2019 12 03	7.08	20.3	-	5.78	54	< 2.0	2.14	-	40	< 10	-	9.35	< 100	8.1	10.0	10.0	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	61.8	6.62	34	0.923	3.39	0.23	4.71	
	SW15-08-210811	2021 08 11	-	8.13	8.13	7.67	< 15	< 2.0	2.08	-	143	< 10	-	4.50	< 100	1.5	7.8	7.8	< 1.0	< 1.0	< 1.0	< 0.10	2.5	-	57.4	2.22	474	0.626	1.63	0.29	2.20	
	SW15-08-220211	2022 02 11	-	4.92	4.92	6.63	< 15	< 3.3	2.47	-	16	< 10	-	2.02	< 100	< 1.0	3.7	3.7	< 1.0	< 1.0	< 1.0	< 0.10	2.2	-	85.6	1.31	27	0.398	0.59	< 0.10	1.76	
	SW15-08-221025	2022 10 25	-	10.0	10.0	7.18	85	-	1.83	< 50	126	< 10	-	8.28	< 100	4.8	5.6	5.6	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	31.9	2.98	262	0.628	1.35	0.30	2.67	
	SW15-08-230223	2023 02 23	-	7.41	7.41	7.11	58	-	2.37	< 50	20	< 10	-	3.15	< 100	1.0	2.6	2.6	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	83.4	2.14	21	0.500	0.95	0.12	2.49	
	SW15-08-230802	2023 08 02	-	9.15	9.02	6.86	21	< 3.3	1.82	< 50	104	< 10	-	4.01	< 100	2.9	5.8	5.8	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	38.0	2.83	85.1	0.506	1.31	0.24	2.47	
	SW15-08-231115	2023 11 15	-	8.61	8.18	6.57	50	< 2.0	3.08	< 50	36																					

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-Term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^l	n/a	n/a	n/a	n/a	0.038-1.418 ^f	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^f	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ^f	100	400	200-900 ^f	400-1,100 ^f	n/a	2.5	100,000	2,500-15,000 ^f	200	5-150 ^f	30	10,000	n/a	850	n/a	750-16,500 ^f
SW15-07	SW15-07-190926	2019 09 26	< 0.20	< 0.50	< 5.0	< 0.10	38.1	0.106	< 0.50	8.60	242	< 0.20	0.19	0.050	< 0.10	11.7	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	25.5
	SW15-07-191205	2019 12 05	< 0.20	< 0.50	< 5.0	< 0.10	36.3	0.093	< 0.50	8.81	342	< 0.20	0.12	< 0.010	< 0.10	11.3	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	31.7
	SW15-07-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	50.1	0.128	< 0.50	10.6	299	< 0.20	< 0.10	< 0.010	< 0.10	14.4	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	32.9
	SW15-07-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.072	< 0.50	5.90	244	< 0.20	< 0.10	< 0.010	< 0.10	7.51	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	18.8
	SW15-07-221025	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.083	< 0.50	7.70	204	< 0.20	0.14	< 0.010	< 0.10	9.97	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	22.4
	SW15-07-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.071	< 0.50	5.85	247	< 0.20	0.10	< 0.010	< 0.10	7.57	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	19.4
	SW15-07-230802	2023 08 02	< 0.050	0.057	0.45	< 0.010	28.2	0.0697	< 0.50	6.37	212	< 0.050	0.103	< 0.0050	0.051	8.28	0.17	0.013	< 0.0040	2.87	< 0.20	0.0012	< 1.00	20.7
	SW15-07-231115	2023 11 15	< 0.050	0.055	0.47	< 0.010	25.0	0.0783	< 0.50	6.64	226	< 0.050	0.119	< 0.0050	0.075	9.62	0.25	0.022	< 0.0040	0.24	< 0.20	0.0015	< 1.00	20.2
SW15-08	SW15-08-151118	2015 11 18	< 0.1	< 0.5	< 5	< 0.1	12	0.04	< 0.5	2.14	53.3	< 0.1	0.2	< 0.02	< 0.1	3.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	8
	SW15-08-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	32	0.09	< 0.5	7.12	156	< 0.1	0.1	< 0.02	< 0.1	9.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	22
	SW15-08-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	28	0.04	< 0.5	4.44	79.8	< 0.1	0.1	< 0.02	< 0.1	6.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	15
	SW15-08-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	7	0.01	< 0.5	0.63	15.2	< 0.1	0.1	< 0.02	< 0.1	0.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-08-161212	2016 12 12	< 0.1	< 0.5	< 5	< 0.1	27	0.09	< 0.5	6.95	246	< 0.1	0.1	< 0.02	< 0.1	9.3	< 0.5	0.45	< 0.02	< 5	-	< 0.02	< 1	21
	SW15-08-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	27.0	0.058	< 0.50	4.49	102	< 0.10	0.13	< 0.010	< 0.10	6.53	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	16.0
	SW15-08-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	33.6	0.069	< 0.50	4.75	116	< 0.20	0.15	< 0.010	< 0.10	6.99	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	16.7
	SW15-08-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	16.0	0.026	< 0.50	1.79	40.5	< 0.20	0.11	< 0.010	< 0.10	2.76	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	6.8
	SW15-08-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.69	< 0.10	1.95	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-08-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	8.2	< 0.010	< 0.50	0.33	15.1	< 0.20	0.11	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-08-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	19.4	0.064	< 0.50	4.37	148	< 0.20	< 0.10	< 0.010	< 0.10	6.32	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	16.0
	SW15-08-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	9.1	0.017	< 0.50	1.08	35.4	< 0.20	0.10	< 0.010	< 0.10	1.59	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.5
	SW15-08-180725	2018 07 25	< 0.20	< 0.50	< 5.0	< 0.10	15.6	0.019	< 0.50	1.12	25.6	< 0.20	0.14	< 0.010	< 0.10	1.74	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.4
	SW15-08-181030	2018 10 30	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.15	4.86	< 0.20	< 0.10	< 0.010	< 0.10	0.44	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW-C-181030	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.15	4.82	< 0.20	< 0.10	< 0.010	< 0.10	0.46	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	*
	SW15-08-190123	2019 01 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.10	4.19	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.3
	SW15-08-190926	2019 09 26	< 0.20	< 0.50	< 5.0	< 0.10	7.0	< 0.010	< 0.50	0.13	3.57	< 0.20	0.15	0.093	< 0.10	0.58	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-08-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	23.5	0.066	< 0.50	5.57	169	< 0.20	0.11	< 0.010	< 0.10	7.59	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	17.2
	SW15-08-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	3.06	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
SW15-08-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.014	< 0.50	0.48	17.8	< 0.20	< 0.10	< 0.010	< 0.10	0.80	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
SW15-08-221025	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.010	< 0.50	0.68	18.9	< 0.20	< 0.10	< 0.010	< 0.10	1.06	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
SW15-08-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	0.015	< 0.50	1.07	42.4	< 0.20	< 0.10	< 0.010	< 0.10	1.59	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	4.0	
SW15-08-230802	2023 08 02	< 0.050	0.069	1.97	< 0.010	8.5	0.0143	< 0.50	1.18	37.1	< 0.050	0.097	< 0.0050	0.020	1.77	< 0.10	< 0.010	< 0.0040	0.25	< 0.20	0.0011	< 1.00	4.9	
SW15-08-231115	2023 11 15	< 0.050	< 0.050	1.76	< 0.010	6.1	0.0159	< 0.50	1.16	37.3	< 0.050	0.082	< 0.0050	0.022	1.84	< 0.10	< 0.010	< 0.0040	0.38	< 0.20	0.0010	< 1.00	4.6	

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Standard to protect freshwater aquatic life.
^d Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^e Guideline/standard varies with Chloride.
^f Guideline/standard varies with Hardness.
^g Guideline varies with pH.
^h Guideline is temperature, pH, DOC and hardness dependent.
ⁱ Laboratory detection limit exceeds regulatory standard.
^j Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^k Guideline is hardness and DOC dependent.
^l Guideline is pH, DOC and hardness dependent.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a	
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
SW15-09	SW15-09-151118	2015 11 18	7.09	3.69	-	6.74	21	6	3.7	< 20	26	< 10	-	2.74	140	< 1	2	2	< 1	< 1	< 1	< 0.1	-	< 2	140	1	50	0.3	2.6	0.19	1.28	
	SW15-09-160629	2016 06 29	6.74	6.24	-	7.41	16	< 2	3.3	-	81	< 10	-	2.49	< 100	< 1.0	5	5	< 1	< 1	< 1	< 0.10	-	< 2	68	1.7	194	0.48	1.7	0.22	1.86	
	SW15-09-160906	2016 09 06	6.95	7.72	-	-	24	< 2	2.9	-	163	< 10	-	2.29	< 100	1.0	7	7	< 1	< 1	< 1	< 0.10	-	< 2	53	2.2	735	0.55	1.5	0.30	2.02	
	SW15-09-161017	2016 10 17	6.33	5.64	-	6.79	34	< 2	4.3	-	58	< 10	-	2.61	< 100	< 1.0	2	2	< 1	< 1	< 1	-	-	2	169	1.5	114	0.46	4.1	0.28	1.57	
	SW15-09-161212	2016 12 12	6.37	5.09	-	7.82	21	< 2	2.9	-	44	< 5	44	2.97	< 100	< 1.0	2	2	< 1	< 1	< 1	-	-	< 2	104	1.2	27	0.49	0.5	0.10	2.07	
	SW15-09-170731	2017 07 31	6.70	6.03	-	6.76	30	< 2.0	2.57	-	73	< 10	-	1.42	< 100	< 1.0	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	62.7	1.66	127	0.453	1.25	0.22	1.69	
	SW15-09-170821	2017 08 21	6.80	6.37	-	7.38	< 10	< 2.0	3.54	-	93	< 10	-	1.44	< 100	< 1.0	5.4	5.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	59.7	1.83	213	0.434	1.71	-	1.51	
	SW15-09-170926	2017 09 26	6.35	6.66	-	7.19	16	< 2.0	2.27	-	77.1	< 5.0	77.1	1.92	< 100	< 1.0	6.5	6.5	< 1.0	< 1.0	< 1.0	-	-	4.1	41.6	1.81	252	0.518	1.58	0.27	1.75	
	SW15-09-171031	2017 10 31	6.07	6.25	-	6.24	32	< 2.0	3.64	-	79	< 10	-	7.15	460	1.0	4.1	4.1	< 1.0	< 1.0	< 1.0	-	-	5.4	90.1	1.68	57	0.500	0.71	0.18	1.82	
	SW15-09-171122	2017 11 22	6.06	3.28	-	6.49	< 15	< 3.3	3.93	-	23	< 10	-	2.08	< 100	< 1.0	1.6	1.6	< 1.0	< 1.0	< 1.0	-	-	< 2.0	137	0.84	39	0.286	2.11	0.10	1.10	
	SW15-09-171212	2017 12 12	6.27	5.67	-	6.83	21	< 2.0	3.14	-	34	< 10	-	2.14	< 100	< 1.0	3.5	3.5	< 1.0	< 1.0	< 1.0	-	-	3.1	87.7	1.49	27	0.474	0.60	0.14	1.82	
	SW15-09-180305	2018 03 05	6.36	5.12	-	7.25	24	< 5.0	2.93	-	29.4	< 5.0	29.4	2.54	< 100	1.1	3.2	3.2	< 1.0	< 1.0	< 1.0	-	-	< 2.0	74.3	1.31	19	0.451	0.37	0.12	1.86	
	SW15-09-180725	2018 07 25	6.09	5.87	-	7.03	31	< 4.0	3.29	-	69	< 10	-	1.83	< 100	< 1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	52.5	1.66	166	0.418	1.42	0.18	1.53	
	SW-B-180725	Duplicate	6.07	9.28	-	7.03	33	< 4.0	-	-	75	< 10	-	5.28	< 100	1.5	2.8	2.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	43.7	2.91	126	0.486	1.88	0.18	2.26	
	QA/QC RPD%		< 1	45	-	0	*	*	*	-	8	*	-	97	*	*	*	*	*	*	*	*	-	-	*	18	55	27	15	28	*	39
	SW15-09-181030	2018 10 30	-	6.23	-	5.56	43	< 4.0	5.17	-	40	< 10	-	4.14	< 100	< 1.0	2.1	2.1	< 1.0	< 1.0	< 1.0	< 0.10	-	2.1	161	1.50	45	0.600	0.84	0.12	2.50	
	SW15-09-190123	2019 01 23	-	4.31	-	6.04	17	< 4.0	4.20	-	< 10	< 10	-	2.76	< 100	< 1.0	1.8	1.8	< 1.0	< 1.0	< 1.0	< 0.10	-	2.8	130	1.10	31	0.377	0.63	< 0.10	1.60	
	SW15-09-190926	2019 09 26	6.08	6.93	-	6.42	24	2.8	8.25	-	49	< 10	-	2.97	< 100	< 1.0	1.8	1.8	< 1.0	< 1.0	< 1.0	< 0.10	-	2.3	189	1.84	171	0.562	5.43	0.16	2.02	
	SW15-09-191204	2019 12 04	6.56	4.49	-	6.32	40	< 2.0	6.33	-	< 10	< 10	-	2.24	< 100	1.1	8.7	8.7	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	203	1.12	67	0.412	0.93	< 0.10	1.71	
	SW15-09-200729	2020 07 29	6.81	7.09	-	6.43	22	-	3.01	-	92.2	< 1.0	-	2.35	< 20	0.78	6.1	6.1	< 1.0	< 1.0	< 1.0	< 0.050	-	-	68.4	1.94	215	0.543	1.70	0.204	2.20	
	SW15-A-200729	Duplicate	6.78	6.85	-	6.43	23	-	2.97	-	91.1	< 1.0	-	2.34	< 20	0.77	6.1	6.1	< 1.0	< 1.0	< 1.0	< 0.050	-	-	65.9	1.85	206	0.539	1.52	0.195	2.07	
	QA/QC RPD%		< 1	3	-	0	*	-	1	-	1	*	-	< 1	*	*	0	0	*	*	*	*	-	-	4	5	4	1	11	*	6	
	SW15-09-210112	2021 01 12	-	3.22	3.60	5.08	24	-	4.06	< 5.0	28.9	< 1.0	-	1.88	< 20	0.60	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.050	-	-	130	0.826	37	0.282	1.52	0.127	1.28	
	SW15-09-210811	2021 08 11	-	7.91	7.91	7.13	< 15	< 2.0	2.32	-	142	< 10	-	2.02	< 100	< 1.0	8.0	8.0	< 1.0	< 1.0	< 1.0	< 0.10	2.2	-	55.0	2.12	445	0.634	1.34	0.33	2.15	
	SW-A-210811	Duplicate	-	7.84	7.84	7.13	< 15	< 2.0	2.06	-	147	< 10	-	1.96	< 100	< 1.0	7.8	7.8	< 1.0	< 1.0	< 1.0	< 0.10	2.3	-	48.9	2.15	453	0.598	1.50	0.30	2.13	
QA/QC RPD%		-	1	1	0	*	*	*	-	3	*	-	3	*	*	3	3	*	*	*	*	-	-	12	1	2	6	11	*	1		
SW15-09-220211	2022 02 11	-	3.85	3.85	6.62	< 15	< 3.3	2.51	-	14	< 10	-	1.80	< 100	< 1.0	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.10	2.2	-	112	0.95	19	0.361	0.21	< 0.10	1.52		
SW15-09-221025	2022 10 25	-	8.99	8.99	7.17	19	-	2.02	< 50	142	< 10	-	3.37	< 100	1.0	6.5	6.5	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	32.5	2.56	280	0.630	1.21	0.32	2.35		
SW15-09-230223	2023 02 23	-	4.56	4.56	6.52	< 15	-	2.57	< 50	20	< 10	-	3.01	< 100	< 1.0	3.4	3.4	< 1.0	< 1.0	< 1.0												

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ^l	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^l	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ^l	100	400	200-900 ^l	400-1,100 ^l	n/a	2.5	100,000	2,500-15,000 ^l	200	5-150 ^l	30	10,000	n/a	850	n/a	750-16,500 ^l
SW15-09	SW15-09-151118	2015 11 18	< 0.1	< 0.5	< 5	< 0.1	< 4	< 0.01	< 0.5	0.09	1.9	< 0.1	0.1	< 0.02	< 0.1	0.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-09-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	< 0.5	< 0.05	1.7	< 0.1	< 0.1	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-09-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	11	< 0.01	< 0.5	0.05	1.6	< 0.1	0.2	< 0.02	< 0.1	0.3	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-09-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	0.8	0.09	1.4	< 0.1	0.2	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-09-161212	2016 12 12	< 0.1	< 0.5	< 5	< 0.1	6	0.02	< 0.5	0.06	2.6	< 0.1	< 0.1	< 0.02	< 0.1	0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-09-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	8.4	< 0.010	< 0.50	< 0.10	1.57	< 0.10	0.13	< 0.010	< 0.10	0.39	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-09-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	12.5	< 0.010	< 0.50	< 0.10	1.58	< 0.20	0.13	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-09-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	6.3	< 0.010	< 0.50	< 0.10	1.31	< 0.20	0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-09-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	1.00	< 0.10	1.88	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	5.8	< 0.010	< 0.50	< 0.10	0.70	< 0.20	0.12	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.61	< 0.10	1.76	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.52	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-180725	2018 07 25	< 0.20	< 0.50	< 5.0	< 0.10	6.0	< 0.010	< 0.50	< 0.10	1.69	< 0.20	0.13	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW-B-180725	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	13.7	0.019	< 0.50	1.12	25.8	< 0.20	0.17	< 0.010	< 0.10	1.75	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.2
	QA/QC RPD%		*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
	SW15-09-181030	2018 10 30	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	2.60	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-190123	2019 01 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	2.50	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	5.0
	SW15-09-190926	2019 09 26	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.10	3.10	< 0.20	0.15	0.129	< 0.10	0.49	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-191204	2019 12 04	< 0.20	< 0.50	< 5.0	< 0.10	19.3	< 0.010	< 0.50	< 0.10	2.16	< 0.20	0.11	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-09-200729	2020 07 29	< 0.10	0.12	1.97	< 0.020	< 10	< 0.0050	0.16	< 0.10	2.50	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	11.2
	SW15-A-200729	Duplicate	< 0.10	0.12	1.97	< 0.020	< 10	< 0.0050	0.13	< 0.10	2.43	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	7.0
	QA/QC RPD%		*	*	0	*	*	*	*	*	3	*	*	*	*	*	*	*	*	*	*	*	*	*
	SW15-09-210112	2021 01 12	< 0.10	< 0.10	1.80	< 0.020	< 10	< 0.0050	0.17	< 0.10	1.21	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	0.054	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.8
SW15-09-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	1.73	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.7	
SW-A-210811	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	1.75	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	5.5	
QA/QC RPD%		*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	
SW15-09-220211	2022 02 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	1.93	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
SW15-09-221025	2022 10 25	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	1.72	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
SW15-09-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	0.57	< 0.10	1.63	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0	
SW15-09-230802	2023 08 02	< 0.050	0.098	2.33	< 0.010	3.1	0.0024	< 0.50	0.0284	1.72	< 0.050	0.095	< 0.0050	0.029	0.211	< 0.10	< 0.010	< 0.0040	0.29	< 0.20	0.0013	< 1.00	1.4	
SW15-09-231115	2023 11 15	< 0.050	< 0.050	1.88	< 0.010	2.2	0.0025	< 0.50	0.0383	1.94	< 0.050	0.												

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781,

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-10	SW15-10-170822	2017 08 22	6.83	6.48	-	7.04	19	< 2.0	3.93	-	100	< 10	-	2.14	< 100	1.1	5.4	5.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	57.9	1.85	204	0.452	1.84	-	1.72	
	SW15-10-170926	2017 09 26	6.31	7.55	-	6.88	28	< 2.0	1.81	-	79.9	< 5.0	79.9	2.75	< 100	1.2	5.9	5.9	< 1.0	< 1.0	< 1.0	-	-	3.4	43.1	2.13	206	0.540	1.62	0.25	2.03	
	SW15-10-171031	2017 10 31	6.06	7.67	-	5.68	18	< 2.0	3.58	-	92	< 10	-	6.61	330	1.6	4.3	4.3	< 1.0	< 1.0	< 1.0	-	-	3.8	88.4	2.14	51	0.561	1.00	0.19	2.22	
	SW15-10-171123	2017 11 23	6.07	3.85	-	6.07	< 15	< 3.3	3.94	-	37	< 10	-	4.53	150	1.1	1.8	1.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	132	1.05	40	0.298	2.15	0.10	1.14	
	SW15-10-171213	2017 12 13	6.24	6.58	-	7.24	28	< 2.0	2.74	-	44	< 10	-	2.05	< 100	2.1	3.7	3.7	< 1.0	< 1.0	< 1.0	-	-	3.2	78.7	1.80	24	0.506	1.16	0.14	1.91	
	SW15-10-180305	2018 03 05	6.33	5.61	-	6.92	33	< 5.0	3.19	-	48.5	< 5.0	48.5	2.62	< 100	4.8	3.2	3.2	< 1.0	< 1.0	< 1.0	-	-	< 2.0	73.8	1.48	15	0.463	0.42	0.12	1.96	
	SW15-10-180724	2018 07 24	6.28	6.90	-	7.16	< 15	< 4.0	2.72	-	59	< 10	-	6.30	220	1.8	2.0	2.0	< 1.0	< 1.0	< 1.0	-	-	< 2.0	62.1	2.02	117	0.450	1.29	0.26	1.84	
	SW15-10-181031	2018 10 31	6.27	5.79	-	5.62	34	< 4.0	3.91	-	28	< 10	-	4.18	< 100	1.3	2.2	2.2	< 1.0	< 1.0	< 1.0	< 0.10	-	2.6	130	1.43	36	0.535	0.81	< 0.10	2.26	
	SW15-10-190124	2019 01 24	6.40	5.30	-	6.51	25	< 4.0	3.22	-	< 10	< 10	-	2.92	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	118	1.41	26	0.433	0.64	< 0.10	1.94	
	SW15-10-190925	2019 09 25	6.31	7.14	-	6.69	23	< 2.0	6.31	-	45	< 10	-	3.03	< 100	< 1.0	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	137	1.94	191	0.556	2.38	0.19	2.14	
	SW15-10-191203	2019 12 03	6.68	6.80	-	6.2	32	< 2.0	3.88	-	19	< 10	-	3.03	< 100	1.6	4.6	4.6	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	112	1.84	38	0.533	0.99	0.13	2.12	
	SW15-10-191204	2019 12 04	6.55	4.52	-	6.32	50	< 2.0	6.48	-	12	< 10	-	2.33	< 100	1.2	8.7	8.7	< 1.0	< 1.0	< 1.0	< 0.10	-	2.6	193	1.15	65	0.398	1.05	< 0.10	1.67	
	SW15-10-200730	2020 07 30	7.00	7.37	-	6.68	26	-	3.00	-	91.2	< 1.0	-	2.89	< 20	1.12	5.5	5.5	< 1.0	< 1.0	< 1.0	< 0.050	-	-	66.0	2.11	216	0.509	1.95	0.205	2.01	
	SW15-10-210113	2021 01 13	-	3.35	3.49	5.47	15	-	3.57	< 5.0	27.9	< 1.0	-	1.74	< 20	0.68	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.050	-	-	129	0.884	38	0.278	1.99	0.135	1.25	
	SW15-C-210113	Duplicate	-	3.38	3.53	5.47	24	-	3.71	< 5.0	28.3	< 1.0	-	1.74	< 20	0.68	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.050	-	-	129	0.892	38	0.280	1.84	0.140	1.24	
	QA/QC RPD%		-	1	1	0	*	-	4	*	1	*	-	*	*	*	*	*	*	*	*	*	-	-	0	1	*	*	8	*	1	
	SW15-10-210811		2021 08 11	-	9.26	9.26	7.3	15	< 2.0	2.08	-	147	< 10	-	3.14	< 100	1.2	7.3	7.3	< 1.0	< 1.0	< 1.0	< 0.10	2.3	-	55.1	2.68	444	0.624	1.62	0.30	2.49
	SW15-10-220210		2022 02 10	-	4.20	4.20	6.6	< 15	< 2.0	3.01	-	< 10.0	< 5.0	9.9	1.99	< 100	1.2	3.6	3.6	< 1.0	< 1.0	< 1.0	< 0.10	2.1	-	86.1	1.13	17	0.331	0.33	< 0.10	1.48
	SW-A-220210		Duplicate	-	4.07	4.07	6.6	< 15	< 2.0	2.60	-	< 10.0	< 5.0	8.0	1.99	< 100	1.2	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	84.4	1.10	16	0.319	0.26	< 0.10	1.41
	QA/QC RPD%		-	-	3	3	0	*	*	15	-	*	*	*	*	*	*	*	*	*	*	*	-	-	2	3	*	*	*	*	*	5
	SW15-10-221027		2022 10 27	-	10.7	10.7	6.95	15	-	2.68	< 50	470	< 10	-	5.24	< 100	1.3	5.3	5.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	42.1	3.03	173	0.767	1.09	0.32	2.94
	SW15-10-230222		2023 02 22	-	4.76	4.76	6.71	18	-	2.82	< 50	16	< 10	-	3.17	< 100	1.1	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	104	1.18	23	0.439	0.46	0.11	2.12
	SW15-10-230801		2023 08 01	-	7.32	6.75	6.86	< 15	< 3.3	1.90	63	100	< 10	-	2.54	< 100	1.4	5.3	5.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	37.8	2.13	84.6	0.484	1.07	0.27	1.99
	SW-B-230801		Duplicate	-	7.23	6.93	6.86	< 15	< 3.3	2.71	< 50	90	< 10	-	2.54	< 100	1.4	5.8	5.8	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	39.2	2.08	85.8	0.49	1.09	0.277	2.05
	QA/QC RPD%		-	-	1	3	*	*	*	*	11	*	-	0	*	*	*	9	9	*	*	*	*	-	4	2	1	1	2	3	3	
	SW15-10-231114		2023 11 14	-	6.05	6.11	6.14	15	< 2.0	3.78	< 50	29	< 10	-	3.58	< 100	1.3	3.2	3.2	< 1.0	< 1.0	< 1.0	< 0.1	2.5	-	121	1.58	26.4	0.512	0.563	0.146	2.52
	SW-B-231114		Duplicate	-	6.12	6.06	6.14	15	< 2.0	3.68	< 50	28	< 10	-	3.59	< 100	1.3	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.1	2.7	-	122	1.6	26.9	0.515	0.586	0.148	2.54
	QA/QC RPD%		-	-	1	1	0	*	*	3	*	*	*	-	0	*	*	*	*	*	*	*	*	-	1	1	2	1	4	1	1	
SW15-11	SW15-11-151118	2015 11 18	6.53	3.89	-	6.63	14	6	3.6	< 20	29	< 10	-	2.76	110	< 1	2	2	< 1	< 1	< 1	< 0.1	-	< 2	139	1	46	0.31	2.5	0.19	1.3	
	SW15-11-160630	2016 06 30	6.77	7.68	-	6.99	28	< 2	3.2	-	76	< 10	-	3.49	< 100	1.3	6	6	< 1	< 1	< 1	< 0.10	-	< 2	69	2.2	185	0.51	1.8	0.22	2.13	
	SW15-11-160907	2016 09 07	6.92	9.30	-	-	28	< 2	2.0	-	154	< 10	-	4.16	< 100	1.4	6	6	< 1	< 1	< 1	< 0.10	-	< 2	49	2.8	649	0.59	1.7	0.31	2.34	
	SW15-11-161018	2016 10 18	6.36	7.62	-	6.05	31	< 2	3.8	-	66	< 10	-	3.10	< 100	1.2	3	3	< 1	< 1	< 1	-	-	2	145	2.1	93	0.59	3.2	0.29	1.70	

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213,

9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-Term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Standard to protect freshwater aquatic life.

^d Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^e Guideline/standard varies with Chloride.

^f Guideline/standard varies with Hardness.

^g Guideline varies with pH.

^h Guideline is temperature, pH, DOC and hardness dependent.

ⁱ Laboratory detection limit exceeds regulatory standard.

^j Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^k Guideline is hardness and DOC dependent.

^l Guideline is pH, DOC and hardness dependent.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ⁱ	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ⁱ	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ^j	400-1,100 ^j	n/a	2.5	100,000	2,500-15,000 ^j	200	5-150 ⁱ	30	10,000	n/a	850	n/a	750-16,500 ^j
SW15-10	SW15-10-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	9.6	< 0.010	< 0.50	0.31	10.7	< 0.20	0.12	< 0.010	< 0.10	0.75	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-10-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	6.7	0.013	< 0.50	0.57	27.8	< 0.20	0.10	< 0.010	< 0.10	1.12	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	5.3
	SW15-10-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.011	< 0.50	0.65	38.3	< 0.20	< 0.10	< 0.010	< 0.10	1.27	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	5.4	< 0.010	< 0.50	0.30	20.2	< 0.20	0.12	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.011	< 0.50	0.81	30.8	< 0.20	< 0.10	< 0.010	< 0.10	1.47	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.21	9.57	< 0.20	< 0.10	< 0.010	< 0.10	0.45	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	13.0	< 0.010	< 0.50	0.38	11.2	< 0.20	0.11	< 0.010	< 0.10	0.60	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.21	12.1	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.21	13.0	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.8
	SW15-10-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	7.7	< 0.010	< 0.50	0.15	5.01	< 0.20	0.13	0.065	< 0.10	0.54	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	9.8	< 0.010	< 0.50	0.40	16.1	< 0.20	< 0.10	< 0.010	< 0.10	0.78	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-191204	2019 12 04	< 0.20	< 0.50	< 5.0	< 0.10	13.8	< 0.010	< 0.50	0.18	9.25	< 0.20	0.11	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-10-200730	2020 07 30	< 0.10	0.12	1.88	< 0.020	< 10	< 0.0050	0.18	0.26	10.8	< 0.050	< 1.0	< 0.0050	< 0.050	0.57	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.3
	SW15-10-210113	2021 01 13	< 0.10	< 0.10	2.02	< 0.020	< 10	< 0.0050	0.19	0.11	9.05	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
	SW15-C-210113	Duplicate	< 0.10	< 0.10	2.18	< 0.020	< 10	< 0.0050	0.13	0.12	9.13	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	0.052	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
	QA/QC RPD%			*	*	8	*	*	*	*	1	*	*	*	*	*	*	*	*	*	*	*	*	*
	SW15-10-210811	2021 08 11	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.32	9.92	< 0.20	< 0.10	< 0.010	< 0.10	0.71	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	5.8
	SW15-10-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.18	8.95	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW-A-220210	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.18	8.50	< 0.20	< 0.10	< 0.010	< 0.10	0.43	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	QA/QC RPD%			*	*	*	*	*	*	*	5	*	*	*	*	*	*	*	*	*	*	*	*	*
	SW15-10-221027	2022 10 27	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.26	8.38	< 0.20	< 0.10	< 0.010	< 0.10	0.49	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW15-10-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	1.21	0.16	7.03	< 0.20	< 0.10	< 0.010	< 0.10	0.61	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	5.0
	SW15-10-230801	2023 08 01	< 0.050	0.076	2.22	< 0.010	4.5	0.0043	< 0.50	0.335	11.8	< 0.050	0.100	< 0.0050	0.022	0.636	< 0.10	< 0.010	< 0.0040	0.33	< 0.20	0.0011	< 1.00	3.5
	SW-B-230801	Duplicate	< 0.050	0.081	2.19	< 0.010	4.2	0.0061	< 0.50	0.323	11.9	< 0.050	0.096	< 0.0050	0.025	0.675	< 0.10	< 0.010	< 0.0040	0.32	< 0.20	< 0.0010	< 1.00	4.5
	QA/QC RPD%			*	*	1	*	*	*	*	4	*	*	*	*	*	6	*	*	*	*	*	*	*
	SW15-10-231114	2023 11 14	< 0.050	< 0.050	1.84	< 0.010	2.8	0.0057	< 0.50	0.256	10.4	< 0.050	0.079	< 0.0050	0.013	0.573	< 0.10	< 0.010	< 0.0040	0.47	< 0.20	< 0.0010	< 1.00	1.6
	SW-B-231114	Duplicate	< 0.050	< 0.050	1.90	< 0.010	2.9	0.0041	< 0.50	0.256	10.4	< 0.050	0.077	< 0.0050	0.014	0.536	< 0.10	< 0.010	< 0.0040	0.47	< 0.20	< 0.0010	< 1.00	1.5
	QA/QC RPD%			*	*	3	*	*	*	*	0	*	*	*	*	*	7	*	*	*	*	*	*	*
SW15-11	SW15-11-151118	2015 11 18	< 0.1	< 0.5	< 5	< 0.1	< 4	< 0.01	< 0.5	0.12	5.3	< 0.1	0.1	< 0.02	< 0.1	0.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-11-160630	2016 06 30	< 0.1	< 0.5	< 5	< 0.1	6	< 0.01	< 0.5	0.40	13.5	< 0.1	< 0.1	< 0.02	< 0.1	0.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-11-160907	2016 09 07	< 0.1	< 0.5	< 5	< 0.1	9	< 0.01	< 0.5	0.38	11.6	< 0.1	0.2	< 0.02	< 0.1	0.9	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-11-161018	2016 10 18	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	0.6	0.27	13.7	< 0.1	0.1	< 0.02	< 0.1	0.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	16

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213,

9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Standard to protect freshwater aquatic life.

^d Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^e Guideline/standard varies with Chloride.

^f Guideline/standard varies with Hardness.

^g Guideline varies with pH.

^h Guideline is temperature, pH, DOC and hardness dependent.

ⁱ Laboratory detection limit exceeds regulatory standard.

^j Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

^k Guideline is hardness and DOC dependent.

^l Guideline is pH, DOC and hardness dependent.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-11	SW15-11-161212	2016 12 12	6.67	8.52	-	6.81	14	3	2.7	-	129	< 5	129	3.61	< 100	2.1	8	8	< 1	< 1	< 1	-	-	2	82	2.2	24	0.73	0.7	0.18	2.67	
	SW15-11-170801	2017 08 01	6.74	7.15	-	6.35	29	< 2.0	2.43	-	74	< 10	-	2.33	< 100	1.3	5.2	5.2	< 1.0	< 1.0	< 1.0	-	-	< 2.0	64.7	2.02	126	0.511	1.44	0.23	2.02	
	SW15-11-170822	2017 08 22	6.88	7.10	-	6.67	26	< 2.0	3.38	-	97	< 10	-	2.27	< 100	1.1	6.4	6.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	55.5	2.04	197	0.484	1.86	-	1.80	
	SW15-11-170927	2017 09 27	6.34	8.06	-	7.36	23	< 2.0	2.67	-	81	< 10	-	2.78	< 100	1.1	6.2	6.2	< 1.0	< 1.0	< 1.0	-	-	3.2	44.2	2.31	250	0.556	1.88	0.26	2.01	
	SW15-11-171101	2017 11 01	6.06	7.31	-	6.05	28	< 2.0	3.67	-	82	< 10	-	4.03	130	1.7	4.4	4.4	< 1.0	< 1.0	< 1.0	-	-	3.7	86.6	2.03	56	0.541	1.08	0.19	2.06	
	SW15-11-171123	2017 11 23	6.32	7.53	-	6.12	< 15	< 3.3	3.70	-	69	< 10	-	2.83	< 100	1.7	4.0	4.0	< 1.0	< 1.0	< 1.0	-	-	8.5	106	2.12	33	0.542	1.75	0.16	1.60	
	SW15-11-171213	2017 12 13	6.37	8.93	-	7.12	19	< 2.0	2.33	-	60	< 10	-	2.62	< 100	1.4	6.8	6.8	< 1.0	< 1.0	< 1.0	-	-	3.4	61.4	2.41	20	0.707	0.73	0.18	2.50	
	SW15-11-180305	2018 03 05	6.49	7.24	-	7.29	31	< 5.0	2.56	-	41.6	< 5.0	41.6	2.75	< 100	1.5	5.3	5.3	< 1.0	< 1.0	< 1.0	-	-	< 2.0	66.4	1.90	13	0.608	0.50	0.16	2.31	
	SW15-11-180724	2018 07 24	6.39	6.99	-	7.2	15	< 4.0	2.56	-	59	< 10	-	3.06	< 100	1.2	2.4	2.4	< 1.0	< 1.0	< 1.0	-	-	< 2.0	46.3	2.04	116	0.456	1.31	0.25	1.81	
	SW15-11-181031	2018 10 31	6.31	5.67	-	5.7	46	< 4.0	4.91	-	25	< 10	-	4.07	< 100	< 1.0	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.10	-	2.3	138	1.38	38	0.535	0.79	< 0.10	2.23	
	SW15-11-190124	2019 01 24	6.40	5.22	-	6.1	< 15	< 4.0	-	-	< 10	< 10	-	2.81	< 100	1.0	2.8	2.8	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	98.8	1.37	20	0.437	0.56	< 0.10	1.94	
	SW15-11-190924	2019 09 24	6.21	8.05	-	6.52	19	< 2.0	6.33	-	40	< 10	-	2.87	< 100	< 1.0	1.9	1.9	< 1.0	< 1.0	< 1.0	< 0.10	-	2.2	139	2.30	219	0.560	3.07	0.28	3.43	
	SW15-11-191203	2019 12 03	6.72	6.92	-	6.85	24	< 2.0	3.50	-	18	< 10	-	3.00	< 100	1.5	5.0	5.0	< 1.0	< 1.0	< 1.0	< 0.10	-	2.1	112	1.88	37	0.540	0.94	0.14	2.11	
	SW15-11-200730	2020 07 30	7.01	7.58	-	6.57	25	-	3.00	-	88.6	< 1.0	-	2.91	< 20	1.20	6.0	6.0	< 1.0	< 1.0	< 1.0	< 0.050	-	-	67.2	2.18	214	0.520	1.96	0.208	2.02	
	SW15-11-210112	2021 01 12	-	3.46	3.76	5.73	24	-	3.46	< 5.0	30.3	< 1.0	-	1.88	< 20	0.62	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.050	-	-	129	0.903	36	0.294	1.34	0.127	1.30	
	SW15-11-210812	2021 08 12	-	9.46	9.46	6.92	< 15	< 2.0	2.05	-	149	< 10	-	3.09	< 100	1.2	8.3	8.3	< 1.0	< 1.0	< 1.0	< 0.10	2.4	-	60.3	2.70	500	0.660	1.90	0.31	2.51	
	SW15-11-220210	2022 02 10	-	5.16	5.16	6.54	< 15	< 2.0	2.63	-	52.6	< 5.0	52.6	2.02	< 100	1.2	4.3	4.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	83.5	1.43	18	0.387	0.35	0.10	1.59	
	SW15-11-221026	2022 10 26	-	10.7	10.7	7.24	28	-	2.55	< 50	365	< 10	-	4.66	< 100	1.5	5.8	5.8	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	38.9	3.06	228	0.736	1.26	0.33	2.78	
	SW-A-221026	Duplicate	-	10.7	10.7	7.24	44	-	2.55	< 50	375	< 10	-	4.64	< 100	1.5	5.6	5.6	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	38.7	3.07	230	0.738	1.36	0.34	2.77	
	QA/QC RPD%		-	0	0	*	*	-	0	*	3	*	-	0	*	*	4	4	*	*	*	*	*	-	1	0	1	0	8	*	0	
	SW15-11-230222	2023 02 22	-	5.22	5.22	5.98	< 15	-	2.92	61	22	< 10	-	3.27	< 100	1.1	3.8	3.8	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	110	1.32	25	0.469	0.95	0.12	2.19	
	SW15-11-230801	2023 08 01	-	7.69	6.92	7.59	< 15	< 3.3	2.54	104	90	< 10	-	2.52	< 100	1.5	5.8	5.8	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	40.9	2.22	90.3	0.519	1.20	0.255	2.02	
	SW15-11-231114	2023 11 14	-	6.19	6.26	6.40	< 10	< 2.0	4.40	< 50	33	< 10	-	3.65	< 100	1.3	3.8	3.8	< 1.0	< 1.0	< 1.0	< 0.1	2.7	-	120	1.61	28.1	0.527	0.613	0.148	2.53	
SW15-12	SW15-12-151118	2015 11 18	9.18	3.87	-	6.65	16	4	4.2	< 20	30	< 10	-	2.78	100	< 1	2	2	< 1	< 1	< 1	< 0.1	-	< 2	137	1	46	0.32	2.5	0.17	1.34	
	SW15-12-160630	2016 06 30	6.71	30.9	-	6.98	150	< 2	3.7	-	70	< 10	-	78.6	< 100	11.4	6	6	< 1	< 1	< 1	0.25	-	< 2	71	3.5	176	5.37	2.2	1.69	44.8	
	SW16-B-160630	Duplicate	6.93	29.5	-	6.98	145	< 2	3.2	-	66	< 10	-	78.1	< 100	11.6	6	6	< 1	< 1	< 1	0.27	-	< 2	73	3.4	177	5.09	2.3	1.62	42.6	
	QA/QC RPD%		3	5	-	0	3	*	14	-	6	*	-	1	*	2	0	0	*	*	*	*	-	*	3	3	1	5	4	4	4	5
	SW15-12-160907	2016 09 07	6.93	8.70	-	-	21	< 2	2.2	-	148	< 10	-	3.74	< 100	1.3	6	6	< 1	< 1	< 1	< 0.10	-	< 2	54	2.5	697	0.60	2.0	0.34	2.74	
	SW16-B-160907																															

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ^l	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^l	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ^l	100	400	200-900 ^l	400-1,100 ^l	n/a	2.5	100,000	2,500-15,000 ^l	200	5-150 ^l	30	10,000	n/a	850	n/a	750-16,500 ^l
SW15-11	SW15-11-161212	2016 12 12	< 0.1	< 0.5	< 5	< 0.1	6	0.03	< 0.5	0.48	25.7	< 0.1	< 0.1	< 0.02	< 0.1	1.0	< 0.5	0.06	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-11-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	7.7	< 0.010	0.63	0.38	13.7	< 0.10	0.13	< 0.010	< 0.10	0.79	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-11-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	10.5	< 0.010	< 0.50	0.34	13.2	< 0.20	0.13	< 0.010	< 0.10	0.72	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-11-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	6.3	< 0.010	< 0.50	0.35	12.7	< 0.20	0.11	< 0.040	< 0.10	0.75	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW15-11-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.43	15.3	< 0.20	< 0.10	< 0.010	< 0.10	0.95	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	5.4	< 0.010	< 0.50	0.42	21.6	< 0.20	0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.34	15.5	< 0.20	< 0.10	< 0.010	< 0.10	0.73	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.19	9.99	< 0.20	< 0.10	< 0.010	< 0.10	0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	9.5	< 0.010	< 0.50	0.40	13.1	< 0.20	< 0.10	< 0.010	< 0.10	0.46	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.11	5.08	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.16	8.86	< 0.20	< 0.10	< 0.010	< 0.10	0.55	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-190924	2019 09 24	< 0.20	< 0.50	< 5.0	< 0.10	20.2	< 0.010	< 0.50	0.13	5.29	< 0.20	0.22	< 0.010	< 0.10	0.48	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	8.8	< 0.010	< 0.50	0.33	13.2	< 0.20	< 0.10	< 0.010	< 0.10	0.70	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-200730	2020 07 30	< 0.10	0.12	1.84	< 0.020	< 10	0.0060	0.19	0.30	12.9	< 0.050	< 1.0	< 0.0050	< 0.050	0.64	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.2
	SW15-11-210112	2021 01 12	< 0.10	< 0.10	1.90	< 0.020	< 10	< 0.0050	0.17	< 0.10	3.30	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
	SW15-11-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.33	11.6	< 0.20	< 0.10	< 0.010	< 0.10	0.76	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.21	8.19	< 0.20	< 0.10	< 0.010	< 0.10	0.44	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW15-11-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.30	9.91	< 0.20	< 0.10	< 0.010	< 0.10	0.60	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW-A-221026	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.30	9.93	< 0.20	< 0.10	< 0.010	< 0.10	0.57	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	QA/QC RPD%			*	*	*	*	*	*	*	*	0	*	*	*	*	*	*	*	*	*	*	*	*
	SW15-11-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.15	6.61	< 0.20	< 0.10	< 0.010	< 0.10	0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW15-11-230801	2023 08 01	< 0.050	0.069	2.14	< 0.010	4.4	0.0041	< 0.50	0.350	13.8	< 0.050	0.094	< 0.0050	0.024	0.674	< 0.10	< 0.010	< 0.0040	0.26	< 0.20	0.0011	< 1.00	1.5
	SW15-11-231114	2023 11 14	< 0.050	< 0.050	1.87	< 0.010	2.7	0.0055	< 0.50	0.251	9.40	< 0.050	0.076	< 0.0050	0.020	0.604	< 0.10	< 0.010	< 0.0040	0.39	< 0.20	< 0.0010	< 1.00	1.6
SW15-12	SW15-12-151118	2015 11 18	< 0.1	< 0.5	< 5	< 0.1	< 4	< 0.01	< 0.5	0.1	4.3	< 0.1	0.1	< 0.02	< 0.1	0.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW15-12-160630	2016 06 30	< 0.1	< 0.5	< 5	< 0.1	23	< 0.01	0.6	0.37	13.9	< 0.1	0.8	< 0.02	< 0.1	0.5	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW16-B-160630	Duplicate	< 0.1	< 0.5	< 5	< 0.1	21	< 0.01	< 0.5	0.36	13.8	< 0.1	0.7	< 0.02	< 0.1	0.4	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	QA/QC RPD%		*	*	*	*	9	*	*	3	1	*	13	*	*	*	*	*	*	*	-	*	*	*
	SW15-12-160907	2016 09 07	< 0.1	< 0.5	< 5	< 0.1	9	< 0.01	< 0.5	0.27	9.3	< 0.1	0.2	< 0.02	< 0.1	0.7	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW16-B-160907	Duplicate	< 0.1	< 0.5	< 5	< 0.1	10	< 0.01	< 0.5	0.27	9.2	< 0.1	0.2	< 0.02	< 0.1	0.6	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	QA/QC RPD%		*	*	*	*	*	*	*	0	1	*	*	*	*	*	*	*	*	*	-	*	*	*
	SW15-12-161017	2016 10 17	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	0.6	0.12	2.8	< 0.1	0.1	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	SW16-B-161017	Duplicate	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	< 0.5	0.10	2.8	< 0.1	0.1	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4
	QA/QC RPD%		*	*	*	*	*	*	*	*	0	*	*	*	*	*	*	*	*	*	-	*	*	*
	SW15-12-161212	2016 12 12	< 0.1	< 0.5	< 5	< 0.1	43	< 0.01	< 0.5	0.12	4.9	< 0.1	2.5	< 0.02	0.1	0.7	< 0.5	0.17	< 0.02	< 5	-	0.03	< 1	5
	SW15-12-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	7.7	< 0.010																

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
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Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW15-12	SW15-12-170927	2017 09 27	6.35	7.71	-	7.1	< 15	< 2.0	2.76	-	78	< 10	-	3.05	< 100	1.3	6.2	6.2	< 1.0	< 1.0	< 1.0	-	-	3.2	42.5	2.15	241	0.568	1.87	0.28	2.18	
	SW15-12-171101	2017 11 01	6.05	130	-	6.47	383	< 2.0	3.61	-	93	< 10	-	209	< 100	30.3	5.8	5.8	< 1.0	< 1.0	< 1.0	-	-	4.6	85.4	10.6	48	25.2	2.00	7.80	206	
	SW15-12-171122	2017 11 22	6.06	3.43	-	6.23	< 15	< 3.3	3.90	-	32	< 10	-	4.79	150	< 1.0	1.4	1.4	< 1.0	< 1.0	< 1.0	-	-	4.0	133	0.90	38	0.284	2.44	0.11	1.09	
	SW15-12-171212	2017 12 12	6.33	6.84	-	6.86	15	< 2.0	2.54	-	48	< 10	-	3.74	< 100	2.1	5.2	5.2	< 1.0	< 1.0	< 1.0	-	-	3.4	80.3	1.73	28	0.610	1.09	0.21	3.17	
SW16-13	SW16-13-160629	2016 06 29	7.49	22.0	-	7.18	-	-	-	-	114	< 10	-	4.29	< 100	1.5	26	26	< 1	< 1	< 1	< 0.10	-	< 2	7	6.3	< 10	1.54	< 0.2	0.29	4.38	
	SW16-13-160906	2016 09 06	7.52	28.1	-	-	-	-	-	-	181	< 10	-	3.48	< 100	1.9	32	32	< 1	< 1	< 1	< 0.10	-	< 2	8	8.1	13	1.90	0.3	0.42	4.81	
	SW16-13-161018	2016 10 18	7.37	28.8	-	6.87	-	-	-	-	737	< 10	-	7.14	100	2.0	26	26	< 1	< 1	< 1	-	-	< 2	31	8.3	31	1.98	0.6	0.58	4.31	
	SW16-13-161212	2016 12 12	7.10	16.7	-	6.34	-	-	-	-	176	< 5	176	4.35	< 100	1.3	17	17	< 1	< 1	< 1	-	-	< 2	27	4.4	14	1.37	0.2	0.34	4.03	
	SW16-13-170801	2017 08 01	7.46	23.1	-	6.48	-	-	-	-	108	< 10	-	5.42	< 100	1.4	26.6	26.6	< 1.0	< 1.0	< 1.0	-	-	< 2.0	6.5	6.42	< 10	1.72	< 0.20	0.30	4.63	
	SW16-13-170821	2017 08 21	7.44	23.4	-	-	-	-	-	-	79	< 10	-	3.66	< 100	1.5	28.3	28.3	< 1.0	< 1.0	< 1.0	-	-	< 2.0	6.1	6.58	< 10	1.69	< 0.20	-	4.32	
	SW-A-170821	Duplicate	7.44	23.2	-	-	-	-	-	-	85	< 10	-	3.54	< 100	1.5	28.9	28.9	< 1.0	< 1.0	< 1.0	-	-	2.0	6.0	6.50	< 10	1.69	< 0.20	-	4.28	
	QA/QC RPD%		0	1	-	-	-	-	-	-	-	7	*	-	3	*	*	2	2	*	*	*	-	-	*	*	1	*	0	*	-	1
	SW16-13-170927	2017 09 27	6.79	25.2	-	7.32	-	-	-	-	-	127	< 10	-	3.72	< 100	1.6	32.7	32.7	< 1.0	< 1.0	< 1.0	-	-	3.5	6.1	6.94	< 10	1.90	0.40	0.34	4.35
	SW16-13-171101	2017 11 01	6.51	25.4	-	6.84	-	-	-	-	-	158	< 10	-	5.34	240	1.6	34.7	34.7	< 1.0	< 1.0	< 1.0	-	-	4.0	7.1	7.24	< 10	1.77	0.34	0.39	4.11
SW16-13-171122	2017 11 22	6.89	21.8	-	6.91	-	-	-	-	-	492	< 10	-	5.82	110	1.6	22.9	22.9	< 1.0	< 1.0	< 1.0	-	-	13.7	36.7	6.05	26	1.63	0.65	0.42	3.77	
SW16-13-171212	2017 12 12	6.63	15.5	-	7.33	-	-	-	-	-	75	< 10	-	2.67	< 100	1.2	17.1	17.1	< 1.0	< 1.0	< 1.0	-	-	2.9	15.8	4.23	< 10	1.19	0.25	0.27	3.82	
SW16-13-180306	2018 03 06	6.79	15.7	-	7.53	-	-	-	-	-	122	< 5.0	122	3.44	< 100	1.6	16.8	16.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	11.9	4.24	< 10	1.23	< 0.20	0.29	3.88	
SW17-14	SW17-14-170731	2017 07 31	6.63	5.94	-	6.26	18	< 2.0	2.49	-	165	< 10	-	1.49	< 100	< 1.0	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	66.2	1.64	139	0.448	1.30	0.22	1.66	
	SW17-14-170821	2017 08 21	6.86	6.32	-	7.24	22	< 2.0	3.92	-	94	< 10	-	1.43	< 100	< 1.0	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	< 2.0	61.8	1.80	224	0.443	1.85	-	1.56	
	SW17-14-170926	2017 09 26	6.43	6.48	-	7.31	< 15	< 2.0	2.20	-	75.2	< 5.0	75.2	1.93	< 100	< 1.0	6.6	6.6	< 1.0	< 1.0	< 1.0	-	-	3.0	42.2	1.77	255	0.497	1.57	0.25	1.69	
	SW17-14-171031	2017 10 31	6.17	6.44	-	6.26	21	< 2.0	3.89	-	83	< 10	-	4.14	250	1.1	4.6	4.6	< 1.0	< 1.0	< 1.0	-	-	3.0	92.8	1.71	59	0.525	0.76	0.19	1.87	
	SW17-14-171212	2017 12 12	6.31	5.62	-	6.86	20	< 2.0	2.63	-	29	< 10	-	1.86	< 100	< 1.0	3.8	3.8	< 1.0	< 1.0	< 1.0	-	-	3.1	88.7	1.46	26	0.480	0.56	0.14	1.85	
	SW17-14-180305	2018 03 05	6.33	5.02	-	7	15	< 4.9	2.70	-	33.6	< 5.0	33.6	2.66	< 100	1.2	3.0	3.0	< 1.0	< 1.0	< 1.0	-	-	< 2.0	70.0	1.27	15	0.450	0.33	0.13	1.90	

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.
Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD

SHADOW

SHADED

Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
Concentration greater than BCWQG Aquatic Life Long-Term Average (AW) guideline
Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Standard to protect freshwater aquatic life.
^d Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^e Guideline/standard varies with Chloride.
^f Guideline/standard varies with Hardness.
^g Guideline varies with pH.
^h Guideline is temperature, pH, DOC and hardness dependent.
ⁱ Laboratory detection limit exceeds regulatory standard.
^j Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^k Guideline is hardness and DOC dependent.
^l Guideline is pH, DOC and hardness dependent.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																						
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L	
BC Guideline/Standard																									
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ^f	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k	
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^f	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ^j	400-1,100 ^j	n/a	2.5	100,000	2,500-15,000 ^j	200	5-150 ^j	30	10,000	n/a	850	n/a	750-16,500 ^j	
SW15-12	SW15-12-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	6.0	< 0.010	< 0.50	0.34	13.1	< 0.20	0.15	0.052	< 0.10	0.73	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW15-12-171101	2017 11 01	< 0.20	< 0.50	6.3	< 0.10	33.6	< 0.010	0.77	0.43	15.2	< 0.20	3.33	< 0.010	0.22	0.99	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	0.064	< 1.0	< 4.0	
	SW15-12-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.75	< 0.20	0.13	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW15-12-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	1.03	0.37	16.6	< 0.20	< 0.10	< 0.010	< 0.10	0.81	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
SW16-13	SW16-13-160629	2016 06 29	< 0.1	< 0.5	< 5	< 0.1	5	< 0.01	0.6	< 0.05	< 0.2	< 0.1	< 0.1	< 0.02	0.4	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW16-13-160906	2016 09 06	< 0.1	< 0.5	< 5	< 0.1	< 4	< 0.01	< 0.5	< 0.05	1.0	< 0.1	0.1	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW16-13-161018	2016 10 18	< 0.1	< 0.5	< 5	< 0.1	6	< 0.01	< 0.5	< 0.05	1.6	< 0.1	< 0.1	< 0.02	< 0.1	< 0.2	< 0.5	< 0.05	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW16-13-161212	2016 12 12	< 0.1	< 0.5	< 5	< 0.1	8	0.01	< 0.5	< 0.05	0.9	< 0.1	< 0.1	< 0.02	< 0.1	< 0.2	< 0.5	0.07	< 0.02	< 5	-	< 0.02	< 1	< 4	
	SW16-13-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	7.1	< 0.010	< 0.50	< 0.10	0.50	< 0.10	< 0.10	< 0.010	< 0.10	< 0.20	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW16-13-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	10.9	< 0.010	< 0.50	< 0.10	0.48	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW-A-170821	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	8.7	< 0.010	< 0.50	< 0.10	< 0.40 ⁱ	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	QA/QC RPD%		*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*
	SW16-13-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	0.43	< 0.20	< 0.10	0.058	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW16-13-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	5.4	< 0.010	< 0.50	< 0.10	0.53	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
SW17-14	SW16-13-171122	2017 11 22	< 0.20	< 0.50	< 5.0	< 0.10	5.8	< 0.010	< 0.50	< 0.10	1.20	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW16-13-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	0.43	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW16-13-180306	2018 03 06	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.83	< 0.10	< 0.40 ⁱ	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW17-14-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	9.3	< 0.010	< 0.50	< 0.10	1.53	< 0.10	0.13	< 0.010	< 0.10	0.26	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW17-14-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	13.7	< 0.010	< 0.50	< 0.10	1.61	< 0.20	0.15	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW17-14-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.31	< 0.20	0.10	0.032	0.11	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0	
	SW17-14-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.86	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW17-14-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.69	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	
	SW17-14-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	< 0.10	1.46	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0	

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.
Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.
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< Denotes concentration less than indicated detection limit or RPD less than indicated value.
- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD
SHADOW
SHADED

Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
Concentration greater than BCWQG Aquatic Life Long-term Average (AW) guideline
Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").
^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").
^c Standard to protect freshwater aquatic life.
^d Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.
^e Guideline/standard varies with Chloride.
^f Guideline/standard varies with Hardness.
^g Guideline varies with pH.
^h Guideline is temperature, pH, DOC and hardness dependent.
ⁱ Laboratory detection limit exceeds regulatory standard.
^j Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023
^k Guideline is hardness and DOC dependent.
^l Guideline is pH, DOC and hardness dependent.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																					
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium
BC Guideline/Standard																															
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW17-15	SW17-15-170731	2017 07 31	6.76	8.75	-	6.53	20	< 2.0	2.43	-	72	< 10	-	5.54	< 100	2.1	5.0	5.0	< 1.0	< 1.0	< 1.0	-	-	< 2.0	66.3	2.63	128	0.527	1.88	0.24	2.39
	SW17-15-170821	2017 08 21	6.87	8.96	-	7.02	27	< 2.0	3.47	-	93	< 10	-	3.69	< 100	1.5	5.7	5.7	< 1.0	< 1.0	< 1.0	-	-	< 2.0	56.1	2.78	189	0.491	2.41	-	2.26
	SW17-15-170926	2017 09 26	6.32	9.70	-	7.17	39	< 2.0	1.85	-	75.5	< 5.0	75.5	4.89	< 100	< 1.0	6.2	6.2	< 1.0	< 1.0	< 1.0	-	-	3.0	41.7	2.96	227	0.557	2.19	0.26	2.49
	SW17-15-171031	2017 10 31	6.03	8.44	-	6.06	36	< 2.0	3.45	-	122	< 10	-	5.27	120	2.8	4.6	4.6	< 1.0	< 1.0	< 1.0	-	-	3.8	81.8	2.49	54	0.540	1.54	0.18	2.34
	SW-A-171031	Duplicate	6.02	7.98	-	6.06	33	< 2.0	3.49	-	91	< 10	-	5.19	170	2.9	4.5	4.5	< 1.0	< 1.0	< 1.0	-	-	3.5	78.2	2.35	53	0.512	1.42	0.17	2.23
	QA/QC RPD%		< 1	6	-	0	*	*	1	-	29	*	-	2	*	*	*	*	*	*	*	-	-	*	4	6	2	5	8	*	5
	SW17-15-171212	2017 12 12	6.33	7.42	-	7.39	24	< 2.0	2.37	-	36	< 10	-	2.37	< 100	2.1	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	2.8	83.9	2.07	29	0.544	1.05	0.14	2.24
SW17-16	SW17-15-180305	2018 03 05	6.29	5.07	-	7.22	22	< 5.0	3.41	-	29.6	< 5.0	29.6	2.73	< 100	1.8	3.1	3.1	< 1.0	< 1.0	< 1.0	-	-	< 2.0	67.4	1.34	14	0.418	0.36	0.10	1.76
	SW17-16-170801	2017 08 01	6.72	6.95	-	5.93	22	< 2.0	2.16	-	63	< 10	-	2.37	< 100	1.2	4.8	4.8	< 1.0	< 1.0	< 1.0	-	-	< 2.0	64.7	1.99	134	0.478	1.45	0.22	1.91
	SW17-16-170926	2017 09 26	6.33	7.45	-	6.9	17	< 2.0	1.75	-	73.7	< 5.0	73.7	2.73	< 100	1.3	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	3.3	41.2	2.12	235	0.523	1.60	0.26	1.92
	SW17-16-171031	2017 10 31	6.05	6.84	-	5.9	37	< 2.0	3.85	-	92	< 10	-	3.40	< 100	1.8	4.1	4.1	< 1.0	< 1.0	< 1.0	-	-	3.4	86.8	1.89	56	0.515	1.03	0.18	1.99
	SW17-16-171123	2017 11 23	6.08	3.81	-	5.38	16	< 2.0	3.81	-	26	< 10	-	3.95	170	1.0	1.7	1.7	< 1.0	< 1.0	< 1.0	-	-	< 2.0	139	1.01	42	0.309	2.09	0.15	1.24
	SW17-16-171213	2017 12 13	6.36	6.11	-	6.6	15	6.6	2.42	-	34	< 10	-	2.17	< 100	1.5	4.0	4.0	< 1.0	< 1.0	< 1.0	-	-	2.9	82.1	1.64	26	0.488	1.13	0.13	1.88
	SW17-16-180305	2018 03 05	6.33	6.85	-	6.86	33	< 5.0	3.06	-	34.8	< 5.0	34.8	2.66	< 100	7.4	3.1	3.1	< 1.0	< 1.0	< 1.0	-	-	< 2.0	61.1	1.88	12	0.519	0.75	0.14	1.97
SW17-17	SW17-17-170801	2017 08 01	6.76	7.22	-	6.35	29	< 2.0	3.39	-	63	< 10	-	2.37	< 100	1.3	4.9	4.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	65.4	2.11	127	0.470	1.47	0.23	1.91
	SW17-17-170822	2017 08 22	6.85	6.84	-	6.41	13	< 2.0	3.52	-	98	< 10	-	2.16	< 100	1.1	5.6	5.6	< 1.0	< 1.0	< 1.0	-	-	< 2.0	59.8	1.96	205	0.471	1.98	-	1.76
	SW17-17-170927	2017 09 27	6.38	7.59	-	7.29	17	< 2.0	2.06	-	78	< 10	-	2.76	< 100	1.1	7.2	7.2	< 1.0	< 1.0	< 1.0	-	-	3.3	41.4	2.16	240	0.531	1.76	0.25	4.95
	SW17-17-171101	2017 11 01	6.05	7.06	-	6.39	30	< 2.0	3.27	-	77	< 10	-	4.73	230	1.7	4.1	4.1	< 1.0	< 1.0	< 1.0	-	-	3.3	106	1.90	56	0.558	1.09	0.20	2.15
	SW17-17-171123	2017 11 23	6.10	4.02	-	5.55	< 15	< 2.0	3.70	-	27	< 10	-	3.45	100	< 1.0	1.8	1.8	< 1.0	< 1.0	< 1.0	-	-	6.1	137	1.08	39	0.322	1.93	0.14	1.25
	SW-A-171123	Duplicate	6.08	4.06	-	5.55	< 15	< 3.3	4.17	-	23	< 10	-	2.19	< 100	< 1.0	1.6	1.6	< 1.0	< 1.0	< 1.0	-	-	< 2.0	131	1.11	41	0.311	1.94	0.13	1.21
	QA/QC RPD%		< 1	1	-	0	*	*	12	-	*	*	-	45	*	*	*	*	*	*	*	-	-	*	4	3	*	3	1	*	3
	SW17-17-171213	2017 12 13	6.32	6.47	-	7.01	16	< 2.0	2.29	-	32	< 10	-	2.23	< 100	1.5	4.5	4.5	< 1.0	< 1.0	< 1.0	-	-	3.3	78.6	1.74	23	0.518	0.79	0.14	1.98
	SW17-17-180724	2018 07 24	6.33	6.95	-	7.24	18	< 4.0	2.17	-	63	< 10	-	3.70	140	1.3	2.3	2.3	< 1.0	< 1.0	< 1.0	-	-	< 2.0	60.7	2.03	116	0.453	1.50	0.23	1.78
	SW17-17-181101	2018 11 01	6.50	5.51	-	5.23	46	364	7.83	-	75	< 10	-	3.61	< 100	1.4	3.1	3.1	< 1.0	< 1.0	< 1.0	< 0.10	-	4.0	178	1.32	59	0.538	1.76	0.14	2.37
	SW17-17-190124	2019 01 24	6.47	6.50	-	5.97	< 15	< 4.0	2.97	-	17	< 10	-	2.95	< 100	1.3	3.2	3.2	< 1.0	< 1.0	< 1.0	< 0.10	-	2.1	99.8	1.76	27	0.509	0.74	0.11	2.19
	SW17-17-190924	2019 09 24	6.37	5.37	-	6.33	25	< 2.0	6.85	-	43	< 10	-	2.79	< 100	< 1.0	2.6	2.6	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	68.7	1.69	100	0.280	1.31	0.18	2.05
	SW17-17-191203	2019 12 03	6.77	7.16	-	6.81	34	< 2.0	3.43	-	19	< 10	-	3.06	< 100	1.6	5.8	5.8	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	106	1.95	36	0.554	0.94	0.14	2.14
	SW17-17-200730	2020 07 30	6.94	7.65	-	6.06	24	-	3.06	-	89.6	< 1.0	-	2.91	< 20	1.19	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	-	-	69.8	2.11	218	0.576	2.07	0.205	2.37
	SW17-17-200818	2020 08 18	6.99	8.24	-	7.27	29	-	3.17	-	134	< 1.0	-	3.10	< 20	1.25	6.3	6.3													

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																							
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L		
BC Guideline/Standard																										
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ⁱ	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k		
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^f	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h		
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ⁱ	400-1,100 ^f	n/a	2.5	100,000	2,500-15,000 ⁱ	200	5-150 ^f	30	10,000	n/a	850	n/a	750-16,500 ^f		
SW17-15	SW17-15-170731	2017 07 31	< 0.10	< 0.50	< 5.0	< 0.10	11.5	0.018	< 0.50	0.96	28.6	< 0.10	0.14	< 0.010	< 0.10	1.63	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	4.9		
	SW17-15-170821	2017 08 21	< 0.20	< 0.50	< 5.0	< 0.10	14.2	0.015	< 0.50	1.09	34.0	< 0.20	0.14	< 0.010	< 0.10	1.72	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	4.4		
	SW17-15-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	8.7	0.016	< 0.50	1.17	36.0	< 0.20	0.11	< 0.010	< 0.10	1.96	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	4.7		
	SW17-15-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	5.4	0.016	< 0.50	1.11	31.8	< 0.20	< 0.10	< 0.010	< 0.10	1.75	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.3		
	SW-A-171031	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	7.4	0.017	< 0.50	1.05	29.8	< 0.20	< 0.10	< 0.010	< 0.10	1.60	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.6		
	QA/QC RPD%		*	*	*	*	*	*	*	6	6	*	*	*	*	9	*	*	*	*	*	*	*	*	*	
	SW17-15-171212	2017 12 12	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.018	< 0.50	0.79	33.4	< 0.20	< 0.10	< 0.010	0.14	1.37	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.1		
SW17-16	SW17-15-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.13	4.71	< 0.20	0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-16-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	8.8	< 0.010	< 0.50	0.36	11.9	< 0.10	0.13	< 0.010	< 0.10	0.71	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0		
	SW17-16-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	5.4	< 0.010	< 0.50	0.35	11.2	< 0.20	0.11	< 0.010	< 0.10	0.69	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0		
	SW17-16-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.70	0.45	15.8	< 0.20	< 0.10	< 0.010	< 0.10	0.86	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-16-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.66	0.14	7.06	< 0.20	0.13	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-16-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	7.5	0.010	0.78	0.56	21.1	< 0.20	< 0.10	< 0.010	< 0.10	0.91	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-16-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	5.5	< 0.010	< 0.50	1.04	25.2	< 0.20	0.10	< 0.010	< 0.10	1.47	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
SW17-17	SW17-17-170801	2017 08 01	< 0.10	< 0.50	< 5.0	< 0.10	8.0	< 0.010	0.63	0.36	13.2	< 0.10	0.13	< 0.010	< 0.10	0.70	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0		
	SW17-17-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	9.7	< 0.010	< 0.50	0.32	13.1	< 0.20	0.14	< 0.010	< 0.10	0.69	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0		
	SW17-17-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	5.3	< 0.010	< 0.50	0.35	12.9	< 0.20	0.11	0.026	< 0.10	0.73	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0		
	SW17-17-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	0.71	0.42	15.3	< 0.20	< 0.10	< 0.010	< 0.10	0.95	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-17-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.12	5.55	< 0.20	0.13	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW-A-171123	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	5.3	< 0.010	< 0.50	0.16	10.0	< 0.20	0.15	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	QA/QC RPD%		*	*	*	*	*	*	*	*	57	*	*	*	*	*	*	*	*	*	*	*	*	*		
	SW17-17-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	6.3	< 0.010	< 0.50	0.38	16.3	< 0.20	< 0.10	< 0.010	< 0.10	0.78	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-17-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	8.2	0.011	< 0.50	0.43	14.0	0.30	< 0.10	< 0.010	< 0.10	0.67	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-17-181101	2018 11 01	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.016	< 0.50	0.16	8.38	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-17-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.010	1.87	0.25	13.4	< 0.20	< 0.10	< 0.010	< 0.10	1.04	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.9		
	SW17-17-190924	2019 09 24	< 0.20	< 0.50	< 5.0	< 0.10	8.9	< 0.010	< 0.50	< 0.10	2.55	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-17-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	8.0	< 0.010	< 0.50	0.33	13.2	< 0.20	< 0.10	< 0.010	< 0.10	0.70	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0		
	SW17-17-200730	2020 07 30	< 0.10	0.14	1.92	< 0.020	< 10	0.0077	0.16	0.34	14.1	0.052	< 1.0	< 0.0050	< 0.050	0.66	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	2.6		
	SW17-17-200818	2020 08 18	< 0.10	0.16	2.01	< 0.020	< 10	0.0055	0.13	0.35	13.7	0.053	< 1.0	< 0.0050	< 0.050	0.67	0.075	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	2.3		
	SW17-17-210112	2021 01 12	< 0.10	< 0.10	1.88	< 0.020	< 10	< 0.0050	0.17	< 0.10	4.50	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.4		
	SW17-17-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.33	10.9	< 0.20	< 0.10	< 0.010	< 0.10	0.69	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0					

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 90928

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW17-19	SW17-19-170822	2017 08 22	6.87	6.73	-	7.26	26	< 2.0	3.96	-	101	< 10	-	2.30	< 100	1.1	5.2	5.2	< 1.0	< 1.0	< 1.0	-	-	< 2.0	57.1	1.94	203	0.460	2.03	-	1.73	
	SW17-19-170927	2017 09 27	6.36	7.59	-	7.27	< 15	< 2.0	2.55	-	83	< 10	-	2.73	< 100	1.1	6.5	6.5	< 1.0	< 1.0	< 1.0	-	-	3.0	41.9	2.14	250	0.541	1.87	0.26	1.95	
	SW17-19-171101	2017 11 01	6.06	7.34	-	6.08	20	< 2.0	3.73	-	80	< 10	-	4.20	140	1.7	4.6	4.6	< 1.0	< 1.0	< 1.0	-	-	3.6	87.2	2.03	57	0.551	1.10	0.19	2.08	
	SW17-19-171123	2017 11 23	6.05	3.73	-	6.51	< 15	< 2.0	3.70	-	21	< 10	-	4.60	170	< 1.0	1.5	1.5	< 1.0	< 1.0	< 1.0	-	-	5.2	140	0.99	41	0.308	2.07	0.15	1.25	
	SW17-19-171213	2017 12 13	6.36	6.32	-	7.47	21	< 2.0	2.30	-	52	< 10	-	2.56	< 100	1.7	4.3	4.3	< 1.0	< 1.0	< 1.0	-	-	3.1	78.6	1.70	31	0.504	0.82	0.14	1.95	
	SW17-19-180305	2018 03 05	6.41	5.72	-	7.22	27	< 4.9	2.72	-	33.7	< 5.0	33.7	2.66	< 100	1.5	3.9	3.9	< 1.0	< 1.0	< 1.0	-	-	< 2.0	65.5	1.48	14	0.494	0.49	0.13	1.98	
	SW17-19-180724	2018 07 24	6.36	7.01	-	7.17	< 15	< 4.0	2.35	-	61	< 10	-	2.99	< 100	1.2	2.3	2.3	< 1.0	< 1.0	< 1.0	-	-	< 2.0	67.3	2.05	118	0.461	1.44	0.25	1.85	
	SW17-19-181031	2018 10 31	6.33	5.95	-	5.62	37	4.0	4.98	-	28	< 10	-	4.07	< 100	< 1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	-	2.6	143	1.48	39	0.546	0.90	< 0.10	2.32	
	SW17-19-190124	2019 01 24	6.41	5.28	-	6.36	< 15	< 4.0	-	-	< 10	< 10	-	2.89	< 100	1.1	3.0	3.0	< 1.0	< 1.0	< 1.0	< 0.10	-	2.0	101	1.40	20	0.431	0.49	< 0.10	1.87	
	SW-C-190124	Duplicate	6.35	5.20	-	6.36	< 15	< 4.0	3.08	-	< 10	< 10	-	2.90	< 100	1.1	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.10	-	2.7	118	1.38	24	0.425	0.76	< 0.10	1.92	
	QA/QC RPD%		1	2	-	0	*	*	*	-	*	*	-	0	*	*	*	*	*	*	*	*	-	*	16	1	*	1	*	*	3	
	SW17-19-190125	2019 01 25	-	-	-	-	-	-	2.63	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	SW17-19-190925	2019 09 25	6.28	7.25	-	7.5	35	< 2.0	6.98	-	44	< 10	-	3.02	< 100	< 1.0	2.1	2.1	< 1.0	< 1.0	< 1.0	< 0.10	-	2.1	139	1.98	194	0.561	2.52	0.19	2.17	
	SW17-19-191204	2019 12 04	6.54	4.52	-	6.24	36	< 2.0	5.41	-	24	< 10	-	2.30	< 100	1.1	9.4	9.4	< 1.0	< 1.0	< 1.0	< 0.10	-	2.4	189	1.15	123	0.397	1.65	< 0.10	1.66	
	SW17-19-210812	2021 08 12	-	9.32	9.32	6.03	< 15	< 2.0	2.10	-	156	< 10	-	3.04	< 100	1.2	7.7	7.7	< 1.0	< 1.0	< 1.0	< 0.10	2.4	-	58.8	2.70	504	0.624	2.00	0.30	2.48	
	SW17-19-220210	2022 02 10	-	4.57	4.57	6.72	< 15	< 2.0	2.60	-	24.7	< 5.0	24.7	2.00	< 100	1.3	3.8	3.8	< 1.0	< 1.0	< 1.0	< 0.10	2.0	-	90.2	1.23	18	0.360	0.33	0.10	1.53	
SW17-19-221026	2022 10 26	-	10.6	10.6	6.92	46	-	2.87	< 50	381	< 10	-	4.69	< 100	1.6	5.6	5.6	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	41.1	3.03	227	0.734	1.30	0.33	2.83		
SW17-19-230222	2023 02 22	-	4.89	4.89	6.01	16	-	2.78	< 50	17	< 10	-	3.12	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	106	1.23	20	0.441	0.41	0.11	2.13		
SW17-19-230801	2023 08 01	-	7.34	6.66	6.88	< 15	< 3.3	2.04	< 50	94	< 10	-	2.51	< 100	1.6	5.2	5.2	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	40.1	2.13	86.9	0.489	1.13	0.25	1.93		
SW17-19-231114	2023 11 14	-	6.05	6.03	6.37	15	< 2.0	4.20	< 50	29	< 10	-	3.62	< 100	1.3	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.1	2.7	-	124	1.57	27.8	0.516	0.602	0.153	2.52		
SW17-20	SW17-20-170822	2017 08 22	6.82	6.69	-	7.26	28	< 2.0	3.38	-	98	< 10	-	2.20	< 100	1.1	5.5	5.5	< 1.0	< 1.0	< 1.0	-	-	< 2.0	58.9	1.94	206	0.449	1.88	-	1.74	
	SW-C-170822	Duplicate	6.81	6.75	-	7.26	14	< 2.0	3.56	-	100	< 10	-	2.20	< 100	1.1	6.0	6.0	< 1.0	< 1.0	< 1.0	-	-	2.1	63.1	1.94	207	0.462	2.08	-	1.77	
	QA/QC RPD%		< 1	1	-	0	*	*	5	-	2	*	-	0	*	*	9	9	*	*	*	-	-	*	7	0	< 1	3	10	-	2	
SW17-21	SW17-21-170926	2017 09 26	6.34	7.54	-	7.03	27	< 2.0	1.93	-	73.0	< 5.0	73.0	2.72	< 100	1.2	6.2	6.2	< 1.0	< 1.0	< 1.0	-	-	3.0	40.7	2.13	241	0.536	1.64	0.27	1.98	
	SW17-21-171031	2017 10 31	6.04	7.45	-	5.89	22	< 2.0	3.72	-	84	< 10	-	7.26	400	1.6	4.3	4.3	< 1.0	< 1.0	< 1.0	-	-	3.6	94.3	2.07	60	0.550	1.08	0.19	2.17	
	SW17-21-171123	2017 11 23	6.02	3.90	-	7.33	< 15	< 2.0	4.15	-	26	< 10	-	4.61	150	< 1.0	1.6	1.6	< 1.0	< 1.0	< 1.0	-	-	4.5	148	1.04	46	0.318	2.22	0.16	1.28	
	SW17-21-171213	2017 12 13	6.31	6.13	-	7.17	19	< 2.0	2.41	-	33	< 10	-	2.31	< 100	1.5	4.3	4.3	< 1.0	< 1.0	< 1.0	-	-	3.3	81.9	1.65	25	0.489	0.78	0.13	1.92	
	SW17-21-180305	2018 03 05	6.34	5.30	-	6.72	32	< 5.0	3.11	-	29.4	< 5.0	29.4	2.64	< 100	1.4	3.0	3.0	< 1.0	< 1.0	< 1.0	-	-	< 2.0	68.2	1.41	14	0.431	0.42	0.11	1.81	
	SW17-21-180724	2018 07 24	6.33	6.79	-	6.68	15	< 4.0																								

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ⁱ	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^l	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ^l	400-1,100 ^l	n/a	2.5	100,000	2,500-15,000 ^l	200	5-150 ^l	30	10,000	n/a	850	n/a	750-16,500 ^l
SW17-19	SW17-19-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	6.8	< 0.010	< 0.50	0.34	12.0	< 0.20	0.14	< 0.010	< 0.10	0.73	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW17-19-170927	2017 09 27	< 0.20	< 0.50	< 5.0	< 0.10	5.2	0.019	< 0.50	0.37	12.0	< 0.20	0.10	0.019	< 0.10	0.74	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW17-19-171101	2017 11 01	< 0.20	< 0.50	< 5.0	< 0.10	5.9	< 0.010	0.70	0.48	16.4	< 0.20	< 0.10	< 0.010	< 0.10	0.96	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.11	4.43	< 0.20	0.12	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	5.5	0.032	0.68	0.36	14.2	< 0.20	< 0.10	< 0.010	< 0.10	0.78	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.23	11.1	< 0.20	< 0.10	< 0.010	< 0.10	0.45	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	6.1	< 0.010	0.54	0.41	13.1	< 0.20	< 0.10	< 0.010	< 0.10	0.69	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.11	5.57	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	5.2
	SW17-19-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.18	9.90	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW-C-190124	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.22	12.6	< 0.20	< 0.10	< 0.010	< 0.10	0.44	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.9
	QA/QC RPD%		*	*	*	*	*	*	*	*	24	*	*	*	*	*	*	*	*	*	*	*	*	*
	SW17-19-190125	2019 01 25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	SW17-19-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	5.3	0.010	< 0.50	0.17	6.09	< 0.20	0.14	0.199	< 0.10	0.52	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-191204	2019 12 04	< 0.20	< 0.50	< 5.0	< 0.10	11.7	< 0.010	3.58	0.13	5.43	< 0.20	0.11	< 0.010	0.95	0.41	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-19-210812	2021 08 12	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.33	11.1	< 0.20	< 0.10	< 0.010	< 0.10	0.94	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	7.1
	SW17-19-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.21	8.84	< 0.20	< 0.10	< 0.010	< 0.10	0.45	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	1.0	< 4.0
	SW17-19-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.33	10.3	< 0.20	< 0.10	< 0.010	< 0.10	0.61	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	5.3
	SW17-19-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.16	6.91	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW17-19-230801	2023 08 01	< 0.050	0.081	2.19	< 0.010	4.4	0.0051	< 0.50	0.354	13.2	< 0.050	0.091	< 0.0050	0.025	0.605	< 0.10	< 0.010	< 0.0040	0.25	< 0.20	< 0.0010	< 1.00	1.6
	SW17-19-231114	2023 11 14	< 0.050	< 0.050	1.85	< 0.010	2.7	0.0053	< 0.50	0.256	9.66	< 0.050	0.082	< 0.0050	0.018	0.648	< 0.10	< 0.010	< 0.0040	0.42	< 0.20	< 0.0010	< 1.00	1.9
SW17-20	SW17-20-170822	2017 08 22	< 0.20	< 0.50	< 5.0	< 0.10	11.2	< 0.010	< 0.50	0.33	10.4	< 0.20	0.13	< 0.010	< 0.10	0.67	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW-C-170822	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	6.9	< 0.010	< 0.50	0.30	10.5	< 0.20	0.13	< 0.010	< 0.10	0.64	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	10	1	*	*	*	*	*	*	*	*	*	-	*	*	*
SW17-21	SW17-21-170926	2017 09 26	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.33	8.84	< 0.20	< 0.10	0.038	< 0.10	0.65	< 0.50	< 0.050	< 0.020	< 5.0	-	< 0.020	< 1.0	< 4.0
	SW17-21-171031	2017 10 31	< 0.20	< 0.50	< 5.0	< 0.10	6.0	< 0.010	< 0.50	0.46	13.8	< 0.20	< 0.10	< 0.010	< 0.10	0.99	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-171123	2017 11 23	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.17	12.3	< 0.20	0.13	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-171213	2017 12 13	< 0.20	< 0.50	< 5.0	< 0.10	5.1	< 0.010	< 0.50	0.35	17.2	< 0.20	< 0.10	< 0.010	< 0.10	0.75	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-180305	2018 03 05	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.20	9.79	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-180724	2018 07 24	< 0.20	< 0.50	< 5.0	< 0.10	5.3	< 0.010	< 0.50	0.38	10.5	< 0.20	< 0.10	< 0.010	< 0.10	0.57	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.15	7.09	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.011	< 0.50	0.15	10.2	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.15	5.32	< 0.20	0.14	0.257	< 0.10	0.48	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW17-21-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10</																		

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Physical Parameters							Geochemical Indicators																						
			pH	Hardness	Total Hardness	pH (field)	Total Dissolved Solids	Total Suspended Solids	Dissolved Organic Carbon	Ammonia, total (as N)	Nitrate (as N)	Nitrite (as N)	Nitrate+Nitrite Nitrogen	Chloride	Fluoride	Sulfate	Total Alkalinity (as CaCO3)	Alkalinity, Bicarbonate (as CaCO3)	Alkalinity, Carbonate (as CaCO3)	Alkalinity, Hydroxide (as CaCO3)	Alkalinity, Phenolphthalein (as CaCO3)	Bromide	Total Acidity	Acidity (pH 8.3)	Dissolved Aluminum	Dissolved Calcium	Dissolved Iron	Dissolved Magnesium	Dissolved Manganese	Dissolved Potassium	Dissolved Sodium	
BC Guideline/Standard																																
BCWQG Aquatic Life Short-term Maximum (AW) ^a			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	19,700-24,500 ^d	32,800	60-600 ^e	n/a	600	400-1,678 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	350	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^b			6.5 - 9	n/a	n/a	6.5 - 9	n/a	n/a	n/a	1,770 ^d	3,000	20-200 ^e	n/a	150	n/a	128-429 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			n/a	n/a	n/a	n/a	n/a	n/a	n/a	184,000-185,500 ^d	4,000,000	2,000-20,000 ^e	4,000,000	15,000	20,000-30,000 ^f	12,800-42,900 ^f	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SW17-22	SW17-21-221026	2022 10 26	-	10.7	10.7	7.19	42	-	2.54	< 50	392	< 10	-	4.73	< 100	1.3	5.4	5.4	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	39.8	3.04	233	0.742	1.19	0.33	2.86	
	SW17-21-230222	2023 02 22	-	4.76	4.76	5.98	< 15	-	3.05	< 50	18	< 10	-	3.17	< 100	1.1	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	105	1.19	18	0.434	0.37	0.11	2.12	
	SW-A-230222	Duplicate	-	4.80	4.80	5.98	< 15	-	2.72	< 50	16	< 10	-	3.18	< 100	1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	105	1.19	18	0.443	0.39	0.11	2.12	
	QA/QC RPD%		-	1	1	*	*	-	11	*	*	*	-	0	*	*	*	*	*	*	*	*	*	-	0	0	*	2	*	*	0	
	SW17-21-230801	2023 08 01	-	7.43	6.55	6.57	< 15	< 3.3	2.28	< 50	90	< 10	-	2.50	< 100	1.5	6.4	6.4	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	38.5	2.17	79.6	0.489	1.03	0.25	1.97	
SW18-22	SW17-21-231114	2023 11 14	-	6.07	5.98	6.30	< 10	< 2.0	3.67	< 50	28	< 10	-	3.58	< 100	1.3	3.3	3.3	< 1.0	< 1.0	< 1.0	< 0.1	2.4	-	123	1.58	26.6	0.512	0.565	0.146	2.57	
	SW18-22-180726	2018 07 26	6.07	-	6.90	7.05	19	< 4.0	3.28	-	71	< 10	-	3.14	< 100	1.1	2.5	2.5	< 1.0	< 1.0	< 1.0	-	2.0	-	49.5	2.03	145	0.442	1.58	0.18	1.80	
	SW18-22-181031	2018 10 31	6.34	-	5.84	5.8	41	< 5.0	4.35	-	20	< 10	-	4.24	< 100	1.1	2.1	2.1	< 1.0	< 1.0	< 1.0	< 0.10	2.6	-	163	1.46	37	0.534	0.94	< 0.10	2.28	
	SW18-22-190124	2019 01 24	6.35	-	5.85	6.17	54	< 4.0	3.14	-	< 10	< 10	-	2.97	< 100	2.1	2.0	2.0	< 1.0	< 1.0	< 1.0	< 0.10	2.3	-	134	1.61	22	0.443	1.51	< 0.10	1.93	
	SW18-22-190925	2019 09 25	6.29	-	7.09	6.82	< 15	< 2.0	6.32	-	45	< 10	-	3.18	< 100	1.0	2.0	2.0	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	135	1.93	184	0.549	2.39	0.17	2.15	
	SW-C-190925	Duplicate	6.28	-	7.19	6.82	< 15	< 2.0	6.34	-	42	< 10	-	3.14	< 100	1.0	2.4	2.4	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	136	1.97	190	0.550	2.36	0.17	2.15	
	QA/QC RPD%		0	-	1	*	*	*	0	-	*	*	-	1	*	*	*	*	*	*	*	*	*	-	1	2	3	0	1	*	0	
	SW18-22-191203	2019 12 03	6.74	7.08	-	6.56	26	< 2.0	3.39	-	17	< 10	-	3.09	< 100	1.6	5.9	5.9	< 1.0	< 1.0	< 1.0	< 0.10	-	< 2.0	120	1.92	40	0.554	1.14	0.14	2.19	
	SW18-22-200729	2020 07 29	6.81	7.63	-	6.97	31	-	2.96	-	91.6	< 1.0	-	2.88	< 20	1.11	6.1	6.1	< 1.0	< 1.0	< 1.0	< 0.050	-	-	65.2	2.16	199	0.541	1.71	0.192	2.07	
	SW18-22-210113	2021 01 13	-	3.44	3.69	6	22	-	3.88	< 5.0	38.3	< 1.0	-	1.80	< 20	0.77	2.9	2.9	< 1.0	< 1.0	< 1.0	< 0.050	-	-	129	0.922	37	0.276	1.87	0.155	1.27	
	SW18-22-210811	2021 08 11	-	9.36	9.36	7.35	< 15	< 2.0	2.19	-	139	< 10	-	3.26	< 100	1.2	7.5	7.5	< 1.0	< 1.0	< 1.0	< 0.10	2.4	-	53.7	2.70	465	0.631	1.81	0.31	2.50	
	SW18-22-220210	2022 02 10	-	4.18	4.18	6.69	< 15	< 2.0	2.75	-	15.1	< 5.0	15.1	1.96	< 100	1.2	3.5	3.5	< 1.0	< 1.0	< 1.0	< 0.10	2.1	-	80.3	1.14	17	0.322	0.31	< 0.10	1.42	
	SW18-22-221026	2022 10 26	-	10.7	10.7	7.16	22	-	2.91	< 50	386	< 10	-	4.80	< 100	1.4	5.4	5.4	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	39.2	3.06	224	0.734	1.23	0.33	2.78	
	SW18-22-230223	2023 02 23	-	5.17	5.17	5.7	< 15	-	2.76	56	19	< 10	-	3.24	< 100	1.2	2.5	2.5	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	95.5	1.33	19	0.449	1.16	0.12	2.15	
	SW18-22-230801	2023 08 01	-	7.34	6.50	7.01	< 15	< 3.3	3.01	< 50	89	< 10	-	2.57	< 100	1.4	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	38.6	2.15	88.5	0.476	1.09	0.246	1.98	
	SW18-22-231115	2023 11 15	-	6.42	6.34	6.96	22	< 2.0	3.40	< 50	33	< 10	-	3.67	< 100	1.4	3.4	3.4	< 1.0	< 1.0	< 1.0	< 0.1	2.6	-	116	1.7	28.1	0.527	0.597	0.156	2.54	
SW20-24	SW20-24-200819	2020 08 19	6.84	7.16	-	7.09	26	-	2.74	-	134	< 1.0	-	2.23	< 20	0.81	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	-	-	66.5	1.84	363	0.520	2.50	0.243	1.82	
SW20-25	SW20-25-200819	2020 08 19	6.86	8.24	-	7.09	28	-	2.58	-	128	< 1.0	-	3.11	< 20	1.16	5.9	5.9	< 1.0	< 1.0	< 1.0	< 0.050	-	-	57.6	2.23	280	0.538	2.16	0.237	2.07	
SW20-26	SW20-26-200819	2020 08 19	6.87	7.22	-	7.16	27	-	2.68	-	122	< 1.0	-	2.14	< 20	0.75	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	-	-	67.0	1.79	378	0.506	2.69	0.242	1.85	
	SW20-A-200819	Duplicate	6.87	7.16	-	7.16	22	-	2.64	-	127	< 1.0	-	2.20	< 20	0.79	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	-	-	73.7	2.06	419	0.516	2.74	0.252	1.78	
QA/QC RPD%			0	1	-	0	*	-	2	-	4	*	-	*	*	*	0	0	*	*	*	*	-	-	10	14	10	2	2	*	4	
SW20-27	SW20-27-200908	2020 09 08	6.79	6.87	-	-	28	-	3.44	-	133	< 1.0	-	2.14	< 20	0.75	6.3	6.3	< 1.0	< 1.0	< 1.0	< 0.050	-	-	75.6	2.01	667	0.508	3.08	0.244	1.76	
SW20-28	SW20-28-200908	2020 09 08	6.84	6.66	-	-	31	-	3.41	-	126	< 1.0	-	2.06	< 20	0.68	6.2	6.2	< 1.0	< 1.0	< 1.0	< 0.050	-	-	71.4	1.98	632	0.497	2.75	0.253	1.79	
SW22-28	SW22-28-221027	2022 10 27	-	8.62	8.62	7.16	42	-	2.65	< 50	< 10	< 10	-	3.82	< 100	< 1.0	5.3	5.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	39.6	2.40	327	0.634	1.94	0.33	2.29	
	SW22-28-230223	2023 02 23	-	3.96	3.96	6.43	23	-	3.19	< 50	35	< 10	-	2.19	< 100	< 1.0	2.3	2.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	110	1.04	31	0.329	1.02	0.16	1.48	
	SW22-28-230803	2023 08 03	-	5.90	5.33	7.36	< 15	< 3.3	3.26	< 50	85	< 10	-	1.76	< 100	< 1.0	4.7	4.7	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	52.0	1.71	258	0.398	2.53	0.252	1.54	
	SW22-28-231116	2023 11 16	-	5.09	5.03	6.65	22	< 2.0	3.65	< 50	58	< 10	-	2.24	< 100	1.1	3.0	3.0	< 1.0	< 1.0	< 1.0	< 0.1	2.2	-	127	1.36	61.9	0.412	1.62	0.248	1.81	
SW23-29	SW23-29-230802	2023 08 02	-	8.00	7.89	6.55	21	< 3.3	2.49	< 50	101	< 10	-	4.05	< 100	1.8	5.3	5.3	< 1.0	< 1.0	< 1.0	< 0.10	< 2.0	-	41.9	2.23	97.8	0.591	1.10	0.278	2.87	

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-Term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

Table 6: Summary of Analytical Results for Dissolved Metals and Geochemical Indicators in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Dissolved Metals																					
			Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Boron µg/L	Cadmium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Lead µg/L	Lithium µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Selenium µg/L	Silver µg/L	Thallium µg/L	Titanium µg/L	Tungsten µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																								
BCWQG Aquatic Life Short-term Maximum (AW) ^a			250 ^j	n/a	n/a	n/a	n/a	0.038-1.418 ^f	n/a	n/a	0.2-9.5 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.6-57.2 ^k
BCWQG Aquatic Life Long-term Average (AW) ^b			n/a	n/a	n/a	n/a	n/a	0.0176-0.397 ^f	n/a	n/a	0.2-1.7 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	1.4-30.9 ^h
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^c			900	500	100,000	15	120,000	5-40 ⁱ	100	400	200-900 ⁱ	400-1,100 ⁱ	n/a	2.5	100,000	2,500-15,000 ⁱ	200	5-150 ⁱ	30	10,000	n/a	850	n/a	750-16,500 ⁱ
SW17-22	SW17-21-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.32	9.36	< 0.20	< 0.10	< 0.010	< 0.10	0.55	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW17-21-230222	2023 02 22	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.17	7.49	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW-A-230222	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.17	7.52	< 0.20	< 0.10	< 0.010	< 0.10	0.44	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	QA/QC RPD%		*	*	*	*	*	*	*	*	*	0	*	*	*	*	*	*	*	*	*	*	*	*
SW18-22	SW17-21-230801	2023 08 01	< 0.050	0.083	2.18	< 0.010	4.5	0.0062	< 0.50	0.347	11.7	< 0.050	0.100	< 0.0050	0.027	0.648	< 0.10	< 0.010	< 0.0040	0.25	< 0.20	< 0.0010	< 1.00	1.8
	SW17-21-231114	2023 11 14	< 0.050	< 0.050	1.94	< 0.010	2.8	0.0040	< 0.50	0.247	9.78	< 0.050	0.076	< 0.0050	0.016	0.561	< 0.10	< 0.010	< 0.0040	0.43	< 0.20	0.0011	< 1.00	1.9
	SW18-22-180726	2018 07 26	< 0.20	< 0.50	< 5.0	< 0.10	6.5	< 0.010	< 0.50	0.42	12.5	< 0.20	0.15	< 0.010	< 0.10	0.72	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	5.2
	SW18-22-181031	2018 10 31	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.20	16.7	< 0.20	< 0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW18-22-190124	2019 01 24	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.96	112	< 0.20	< 0.10	< 0.010	< 0.10	1.60	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	4.4
	SW18-22-190925	2019 09 25	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	< 0.010	< 0.50	0.19	7.61	< 0.20	0.13	0.072	< 0.10	0.56	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW-C-190925	Duplicate	< 0.20	< 0.50	< 5.0	< 0.10	< 5.0	0.010	< 0.50	0.19	7.61	< 0.20	0.13	0.039	< 0.10	0.51	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	10.7
	QA/QC RPD%		*	*	*	*	*	*	*	*	*	0	*	*	59	*	*	*	*	*	*	*	*	*
	SW18-22-191203	2019 12 03	< 0.20	< 0.50	< 5.0	< 0.10	7.5	< 0.010	< 0.50	0.41	22.0	< 0.20	< 0.10	< 0.010	< 0.10	0.79	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW18-22-200729	2020 07 29	< 0.10	0.12	1.81	< 0.020	< 10	< 0.0050	0.14	0.30	9.94	< 0.050	< 1.0	< 0.0050	< 0.050	0.55	< 0.050	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.9
	SW18-22-210113	2021 01 13	< 0.10	< 0.10	2.10	< 0.020	< 10	< 0.0050	0.13	0.13	13.4	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	2.1
	SW18-22-210811	2021 08 11	< 0.20	< 0.50	6.7	< 0.10	< 50.0	< 0.010	< 0.50	0.33	11.6	< 0.20	< 0.10	< 0.010	< 0.10	0.77	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	8.9
	SW18-22-220210	2022 02 10	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.17	8.14	< 0.20	< 0.10	< 0.010	< 0.10	0.47	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 1.0	< 4.0
	SW18-22-221026	2022 10 26	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.31	9.17	< 0.20	< 0.10	< 0.010	< 0.10	0.59	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW18-22-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	0.24	9.60	< 0.20	< 0.10	< 0.010	< 0.10	0.47	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW18-22-230801	2023 08 01	< 0.050	0.083	2.18	< 0.010	4.6	0.0037	< 0.50	0.335	11.4	< 0.050	0.099	< 0.0050	0.025	0.633	< 0.10	< 0.010	< 0.0040	0.26	< 0.20	< 0.0010	< 1.00	1.8
SW18-22-231115	2023 11 15	< 0.050	< 0.050	1.87	< 0.010	3.2	0.0053	< 0.50	0.300	10.3	< 0.050	0.079	< 0.0050	0.014	0.660	< 0.10	< 0.010	< 0.0040	0.40	< 0.20	0.0011	< 1.00	1.9	
SW20-24	SW20-24-200819	2020 08 19	< 0.10	0.20	2.03	< 0.020	< 10	< 0.0050	0.14	< 0.10	2.04	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
SW20-25	SW20-25-200819	2020 08 19	< 0.10	0.14	1.96	< 0.020	< 10	< 0.0050	0.12	0.43	12.4	< 0.050	< 1.0	< 0.0050	< 0.050	0.64	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	1.3
SW20-26	SW20-26-200819	2020 08 19	< 0.10	0.16	2.24	< 0.020	< 10	< 0.0050	0.15	< 0.10	1.73	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
	SW20-A-200819	Duplicate	< 0.10	0.18	2.13	< 0.020	< 10	< 0.0050	0.16	< 0.10	1.77	< 0.050	< 1.0	< 0.0050	< 0.050	< 0.50	0.058	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
QA/QC RPD%			*	*	5	*	*	*	*	*	2	*	*	*	*	*	*	*	*	*	*	*	*	
SW20-27	SW20-27-200908	2020 09 08	< 0.10	0.27	3.08	< 0.020	< 10	< 0.0050	0.18	< 0.10	0.98	0.054	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
SW20-28	SW20-28-200908	2020 09 08	< 0.10	0.29	3.29	< 0.020	< 10	< 0.0050	0.16	< 0.10	0.85	0.050	< 1.0	< 0.0050	< 0.050	< 0.50	< 0.0500	< 0.010	< 0.010	< 10.0	< 0.10	< 0.010	< 0.50	< 1.0
SW22-28	SW22-28-221027	2022 10 27	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	0.71	< 0.20	0.20	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW22-28-230223	2023 02 23	< 0.20	< 0.50	< 5.0	< 0.10	< 50.0	< 0.010	< 0.50	< 0.10	0.69	< 0.20	0.10	< 0.010	< 0.10	< 0.40	< 0.50	< 0.050	< 0.020	< 5.0	< 1.0	< 0.020	< 5.0	< 4.0
	SW22-28-230803	2023 08 03	< 0.050	0.148	3.61	< 0.010	2.9	< 0.0020	< 0.50	0.0425	0.66	< 0.050	0.170	< 0.0050	0.015	0.271	< 0.10	< 0.010	< 0.0040	0.39	< 0.20	0.0019	< 1.00	< 1.0
	SW22-28-231116	2023 11 16	< 0.050	0.098	3.39	< 0.010	2.0	0.0032	< 0.50	0.0477	0.75	< 0.050	0.149	< 0.0050	0.011	0.439	< 0.10	< 0.010	< 0.0040	0.61	< 0.20	0.0028	< 1.00	1.1
SW23-29	SW23-29-230802	2023 08 02	< 0.050	0.096	2.17	< 0.010	4.9	0.0053	< 0.50	0.328	13.7	< 0.050	0.121	< 0.0050	0.033	0.606	< 0.10	< 0.010	< 0.0040	0.22	< 0.20	0.0016	< 1.00	2.4

Associated ALS file(s): VA20B1740, VA20B1766, VA20B3206, VA20B3376, VA20B5029, VA21A0691, VA21A0716.

Associated Caro file(s): 5111409, 6070220, 6090497, 6101221, 6121082, 7052229, 7080265, 7082070, 7092754, 7110251, 7112152, 7121157, 8030690, 8072387, 8072648, 8102765, 8102781, 8110202, 8110213, 9011882, 9011884, 9011974, 9092679, 9092813, 9092820, 9120466, 9120554, 9120637, 21H1548, 22B1772, 22B2327, 21H1469, 22J3877, 23B2664, 23H1041, 23K2203, 23K2230.

All terms defined within the body of SNC-Lavalin's report.

< Denotes concentration less than indicated detection limit or RPD less than indicated value.

- Denotes analysis not conducted.

n/a Denotes no applicable standard/guideline.

RPD Denotes relative percent difference.

* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD	Concentration greater than BCWQG Aquatic Life Short-term Maximum (AW) guideline
SHADOW	Concentration greater than BCWQG Aquatic Life Long-term Average (AW) guideline
SHADED	Concentration greater than CSR Aquatic Life Upper Cap Concentration (AW UCC) standard

^a Guideline to protect freshwater aquatic life, short-term maximum (i.e. "acute").

^b Guideline to protect freshwater aquatic life, long-term average (i.e. "chronic").

^c Standard to protect freshwater aquatic life.

^d Guideline/standard varies with pH and temperature. Temperature assumed 15C BCWQG, 10C CSR.

^e Guideline/standard varies with Chloride.

^f Guideline/standard varies with Hardness.

^g Guideline varies with pH.

^h Guideline is temperature, pH, DOC and hardness dependent.

ⁱ Laboratory detection limit exceeds regulatory standard.

^j Guideline is for Antimony (III and V).Guideline came into affect November 17, 2023

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																					
					Aluminum μg/L	Antimony μg/L	Arsenic μg/L	Barium μg/L	Beryllium μg/L	Bismuth μg/L	Boron μg/L	Cadmium μg/L	Calcium μg/L	Chromium μg/L	Cobalt μg/L	Copper μg/L	Iron μg/L	Lead μg/L	Lithium μg/L	Magnesium μg/L	Manganese μg/L	Mercury μg/L	Molybdenum μg/L	Nickel μg/L	Phosphorous μg/L	Potassium μg/L	Selenium μg/L	Silicon μg/L	Silver μg/L	Sodium μg/L	Strontium μg/L	Sulphur μg/L	Tellurium μg/L	Thallium μg/L	Thorium μg/L	Tin μg/L	Titanium μg/L	Uranium μg/L	Vanadium μg/L	Zinc μg/L		
BC Guideline/Standard																																										
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ⁱ	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	n/a	10,000	850	n/a	750-16,500 ^e	
SW15-01	SW15-01-151117	2015 11 17	43.3	-	93	< 0.1	< 0.5	< 5	< 0.1	< 0.1	45	0.11	14,800	< 0.5	9.34	308	110	0.1	0.4	1,800	7.6	< 0.02	0.3	14	< 20	390	< 0.5	4,500	< 0.05	8,990	22	3,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	31		
	SW15-01A-160629	2016 06 29	47.1	-	66	< 0.1	< 0.5	< 5	< 0.1	< 0.1	70	0.18	19,800	0.7	15.2	484	60	< 0.1	0.2	1,300	12.1	< 0.02	< 0.1	20.0	< 20	290	< 0.5	3,900	< 0.05	10,900	25	1,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	45		
	SW15-01A-160906	2016 09 06	71.7	-	29	< 0.1	< 0.5	< 5	< 0.1	< 0.1	105	0.26	28,400	< 0.5	17.3	447	40	< 0.1	0.2	1,680	13.5	< 0.02	< 0.1	24.0	20	350	< 0.5	3,800	< 0.05	15,900	38	7,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	54		
	SW16-A-160906	Duplicate	73.8	-	29	< 0.1	< 0.5	< 5	< 0.1	< 0.1	96	0.24	28,400	< 0.5	17.2	445	40	< 0.1	0.2	1,650	13.6	< 0.02	0.1	23.7	20	350	< 0.5	3,700	< 0.05	15,900	37	7,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	54		
	QA/QC RPD%		3	-	0	*	*	*	*	*	9	8	0	*	1	< 1	*	*	*	2	1	*	*	1	*	0	*	3	*	0	3	0	*	*	*	*	*	*	*	0		
	SW15-01-161017	2016 10 17	46.3	-	51	0.1	< 0.5	< 5	< 0.1	< 0.1	59	0.19	15,000	< 0.5	15.8	395	80	< 0.1	0.2	1,610	14.1	< 0.02	0.1	20.7	< 20	410	< 0.5	< 500	< 0.05	10,200	26	8,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	47		
	SW15-01-161212	2016 12 12	23.5	-	71	< 0.1	< 0.5	< 5	< 0.1	< 0.1	35	0.10	8,500	< 0.5	7.68	330	60	< 0.1	0.2	1,190	6.8	< 0.02	< 0.1	10.0	< 20	340	< 0.5	4,900	< 0.05	5,290	16	3,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	25		
	SW15-01-170525	2017 05 25	16.5	-	88.0	< 0.10	< 0.50	< 5.0	< 0.10	< 0.10	34	0.080	5,770	< 0.50	7.80	357	53	< 0.10	0.12	693	6.12	< 0.02	< 0.10	9.69	< 50	150	< 0.50	3,500	< 0.050	4,480	8.7	4,000	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	24.2		
	SW15-01A-170731	2017 07 31	43.5	-	55.2	< 0.10	< 0.50	< 5.0	< 0.10	< 0.10	64.9	0.157	15,900	1.47	13.8	455	41	< 0.10	0.16	1,270	10.0	< 0.010	< 0.10	19.0	< 50	280	< 0.50	3,500	< 0.050	10,900	22.0	3,800	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	42.5		
	SW15-01A-170821	2017 08 21	42.5	-	76.6	< 0.20	< 0.50	< 5.0	< 0.10	-	89.9	0.179	16,100	< 0.50	12.8	424	45	0.35	0.17	1,130	9.37	< 0.010	< 0.10	17.8	-	-	< 0.50	-	< 0.050	10,000	22.6	-	-	< 0.020	-	-	< 5.0	< 0.020	< 1.0	54.1		
	SW15-01A-170926	2017 09 26	37.7	-	51.2	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	58.4	0.151	14,400	< 0.50	11.4	351	67	< 0.20	0.20	1,050	8.56	< 0.010	< 0.10	15.9	< 50	240	< 0.50	3,600	< 0.050	9,620	21.2	3,800	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	35.9		
	SW15-01-171031	2017 10 31	31.7	-	70.2	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	52.7	0.126	13,000	0.68	12.4	371	71	< 0.20	0.21	1,000	11.1	< 0.010	< 0.10	15.9	< 50	230	< 0.50	3,700	0.079	7,700	18.0	5,700	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	35.8		
	SW15-01-171122	2017 11 22	31.8	-	98.3	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	65.8	0.128	12,700	< 0.50	12.1	385	94	< 0.20	0.37	1,240	10.6	< 0.010	< 0.10	14.9	< 50	270	< 0.50	4,500	< 0.050	7,420	21.1	5,100	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	32.1		
	SW15-01-171212	2017 12 12	22.2	-	63.2	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	38.6	0.097	8,260	1.03	8.41	356	46	< 0.20	0.78	1,100	6.26	< 0.010	< 0.10	11.0	< 50	260	< 0.50	4,400	< 0.050	5,760	14.0	3,600	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	26.4		
	SW15-01-180305	2018 03 05	19.4	-	78.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	34.6	0.082	6,820	< 0.50	7.87	361	50	< 0.20	0.17	978	7.32	< 0.010	< 0.10	9.35	< 50	330	< 0.50	4,200	< 0.050	5,810	11.5	3,300	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	24.1		
	SW15-01A-180723	2018 07 23	55.4	-	59.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	80.3	0.180	20,400	< 0.50	14.6	406	36	< 0.20	0.37	1,280	11.7	< 0.010	< 0.10	19.6	< 50	290	< 0.50	3,600	< 0.050	11,800	35.2	5,100	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	43.0		
	SW15-01a-181029	2018 10 29	39.8	-	66.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	48.3	0.119	15,100	< 0.50	11.7	404	73	0.21	0.15	1,060	10.4	< 0.010	0.11	15.9	< 50	240	< 0.50	3,500	< 0.050	9,650	21.5	5,100	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	36.7		
	SW-B-181029	Duplicate	39.5	-	62.5	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	49.0	0.107	15,100	< 0.50	11.7	413	62	< 0.20	0.13	1,060	10.2	< 0.010	< 0.10	15.9	< 50	240	< 0.50	3,400	< 0.050	9,660	21.4	5,600	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	33.7		
	QA/QC RPD%		1	-	7																																					

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																				
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L	
BC Guideline/Standard																																									
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e	
SW15-02	SW15-02-151117	2015 11 17	42.9	-	467	< 0.1	< 0.5	< 5	< 0.1	< 0.1	47	0.13	15,100	0.9	11.9	507	790	0.9	0.4	1,970	24.9	< 0.02	0.1	15.9	< 20	430	< 0.5	5,300	< 0.05	8,960	23	4,000	< 0.2	< 0.02	< 0.1	< 0.2	12	< 0.02	2	34	
	SW15-02-160629	2016 06 29	48.8	-	50	< 0.1	< 0.5	< 5	< 0.1	< 0.1	63	0.17	18,500	0.5	13.9	372	40	< 0.1	0.2	1,350	8.1	< 0.02	0.1	19.0	< 20	300	< 0.5	3,700	< 0.05	10,600	24	1,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	43	
	SW15-02-160906	2016 09 06	73.0	-	38	< 0.1	< 0.5	< 5	< 0.1	< 0.1	110	0.24	27,900	< 0.5	17.3	437	60	< 0.1	0.3	1,760	11.3	< 0.02	0.1	25.0	< 20	380	< 0.5	4,100	< 0.05	16,100	39	5,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	57	
	SW15-02-161017	2016 10 17	45.3	-	80	0.1	< 0.5	< 5	< 0.1	< 0.1	62	0.17	15,700	0.6	15.2	372	140	0.1	0.2	1,650	12.8	< 0.02	0.1	20.2	< 20	370	< 0.5	< 500	< 0.05	10,300	27	9,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	47	
	SW16-A-161017	Duplicate	45.2	-	82	< 0.1	< 0.5	< 5	< 0.1	< 0.1	53	0.17	15,800	0.6	15.2	372	150	0.1	0.2	1,650	12.8	< 0.02	0.1	20.7	20	360	< 0.5	< 500	< 0.05	10,400	26	7,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	48	
	QA/QC RPD%			< 1	-	2	*	*	*	*	*	16	0	1	*	0	0	7	*	*	0	0	*	*	2	*	3	*	*	*	1	4	25	*	*	*	*	*	*	*	2
	SW15-02-161212	2016 12 12	24.0	-	59	< 0.1	< 0.5	< 5	< 0.1	< 0.1	33	0.09	8,700	< 0.5	6.89	265	60	< 0.1	0.2	1,180	4.7	< 0.02	< 0.1	9.3	< 20	320	< 0.5	5,100	< 0.05	5,400	16	4,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	21	
	SW15-02-170731	2017 07 31	45.3	-	47.1	< 0.10	< 0.50	< 5.0	< 0.10	< 0.10	63.2	0.174	16,500	< 0.50	13.3	366	37	< 0.10	0.18	1,460	6.62	< 0.010	< 0.10	18.9	< 50	340	< 0.50	3,700	< 0.050	11,300	22.7	4,200	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	44.4	
	SW15-02-170821	2017 08 21	42.4	-	56.1	< 0.20	< 0.50	< 5.0	< 0.10	-	80.6	0.182	16,000	< 0.50	12.2	368	53	< 0.20	0.20	1,210	6.59	< 0.010	< 0.10	17.6	-	-	< 0.50	-	< 0.050	9,800	22.3	-	-	< 0.020	-	-	< 5.0	< 0.020	< 1.0	41.7	
	SW15-02-170926	2017 09 26	37.7	-	50.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	58.2	0.160	14,000	< 0.50	11.0	321	66	< 0.20	0.19	1,070	7.04	< 0.010	< 0.10	15.8	< 50	250	< 0.50	3,600	< 0.050	9,550	21.3	3,700	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	36.3	
	SW15-02-171031	2017 10 31	31.0	-	54.9	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	46.3	0.131	11,700	0.71	9.92	253	58	< 0.20	0.20	992	5.85	< 0.010	< 0.10	13.9	< 50	220	< 0.50	3,500	0.061	7,360	17.5	5,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	32.1	
	SW15-02-171122	2017 11 22	32.5	-	202	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	54.0	0.148	12,400	< 0.50	11.4	440	274	0.29	0.38	1,230	13.2	< 0.010	0.10	14.0	< 50	240	< 0.50	4,400	< 0.050	7,150	19.3	3,500	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	33.4	
	SW15-02-171212	2017 12 12	23.1	-	54.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	36.0	0.083	8,260	0.75	7.27	263	44	< 0.20	0.69	1,050	3.84	< 0.010	< 0.10	10.2	< 50	240	< 0.50	4,400	< 0.050	5,670	13.5	3,300	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	23.6	
	SW15-02-180305	2018 03 05	20.0	-	65.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	32.8	0.072	6,820	< 0.50	6.45	231	49	< 0.20	0.20	929	3.85	< 0.010	< 0.10	8.09	< 50	320	< 0.50	4,000	< 0.050	5,520	10.8	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	20.1	
	SW15-02-180724	2018 07 24	53.2	-	72.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	65.7	0.168	21,000	< 0.50	13.7	348	34	< 0.20	0.17	1,270	8.11	< 0.010	0.12	19.5	< 50	300	< 0.50	3,500	< 0.050	11,400	28.9	4,400	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.1	41.7	
	SW-C-180724	Duplicate	51.8	-	54.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	61.2	0.188	20,700	< 0.50	13.5	351	34	< 0.20	0.15	1,280	8.10	< 0.010	< 0.10	19.4	< 50	300	< 0.50	3,500	< 0.050	11,400	28.9	4,100	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	42.2	
	QA/QC RPD%			3	-	28	*	*	*	*	*	7	11	1	*	1	1	*	*	*	1	< 1	*	*	1	*	*	*	*	*	0	0	*	*	*	*	*	*	*	*	1
	SW15-02-181029	2018 10 29	39.2	-	51.2	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	47.4	0.121	14,500	< 0.50	10.7	306	52	< 0.20	0.14	1,040	6.91	< 0.040 ^a	< 0.10	15.5	< 50	240	< 0.50	3,400	< 0.050	9,190	20.9	5,600	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	33.9	
	SW15-02-190123	2019 01 23	20.9	-	85.3	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	30.1	0.082	7,060	< 0.50	5.79	218	43	< 0.20	0.12	858	3.98	< 0.010	< 0.10	7.71	< 50	200	< 0.50	3,900	< 0.050	4,450	10.0	3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	20.4	
	SW15-02-190925	2019 09 25	35.8	-	44.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	57.7	0.126	13,800	< 0.50	7.61	206	51	< 0.20	0.21	905	4.60	< 0.010	< 0.10	11.9	< 50	220	< 0.50	3,200	< 0.050	8,330	19.8	3,800	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.5	27.2	
	SW15-02-191205	2019 12 05	23.9	-	63.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	37.0	0.092	8,700	0.66	7.87	267	75	< 0.20	0.21	1,020																					

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																			
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																																								
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e
SW15-03	SW15-03-151117	2015 11 17	155	-	8,560	< 0.1	< 0.5	5	0.2	< 0.1	18	1.09	54,600	0.5	152	29,700	470	0.3	3.2	4,770	233	< 0.02	< 0.1	236	< 20	630	0.5	20,700	< 0.05	8,990	64	83,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	0.02	< 1	263
	SW15-03-171123	2017 11 23	136	-	8,720	< 0.20	< 0.50	< 5.0	0.19	< 0.10	18.7	1.38	52,100	0.63	229	30,500	159	< 0.20	3.61	4,920	261	< 0.010	< 0.10	342	< 50	520	< 0.50	23,400	0.112	8,230	66.6	83,600	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	344
	SW15-03-181101	2018 11 01	235	-	22,100	< 0.20	< 0.50	5.4	0.34	< 0.10	29.1	2.40	88,400	0.87	382	62,300	642	0.38	6.69	7,500	445	< 0.010	< 0.10	562	< 50	660	0.83	36,400	0.249	13,700	92.4	162,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	0.036	< 1.0	554
SW15-04	SW15-04-151117	2015 11 17	153	-	10,300	< 0.1	< 0.5	< 5	0.2	< 0.1	17	1.38	57,500	0.6	191	40,400	180	0.4	3.8	4,380	301	< 0.02	< 0.1	274	< 20	590	1	35,000	< 0.05	11,400	70	94,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	0.02	< 1	357
	SW15-04-181101	2018 11 01	161	-	14,700	< 0.20	< 0.50	5.4	0.27	< 0.10	23.3	1.56	59,400	0.61	274	41,000	828	1.29	4.93	5,600	357	< 0.010	0.15	408	< 50	660	1.23	41,400	0.197	13,700	79.0	106,000	< 0.50	< 0.020	< 0.10	< 0.20	9.2	0.024	1.1	416
SW15-05	SW15-05-151117	2015 11 17	18.6	-	551	< 0.1	< 0.5	5	< 0.1	< 0.1	6	< 0.01	5,500	1.1	0.45	58.5	730	0.7	0.3	1,510	12.4	< 0.02	< 0.1	1.3	20	630	< 0.5	5,100	< 0.05	4,110	17	< 1,000	< 0.2	< 0.02	< 0.1	< 0.2	25	< 0.02	2	6
	SW15-05-160629	2016 06 29	22.5	-	61	< 0.1	< 0.5	< 5	< 0.1	< 0.1	14	< 0.01	7,600	< 0.5	0.11	7.0	140	< 0.1	< 0.1	1,760	4.4	< 0.02	0.1	0.2	< 20	300	< 0.5	7,000	< 0.05	4,510	20	< 1,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	8
	SW15-05-160907	2016 09 07	28.2	-	38	< 0.1	< 0.5	< 5	< 0.1	< 0.1	16	< 0.01	8,500	< 0.5	0.11	8.1	120	< 0.1	0.2	2,070	3.4	< 0.02	< 0.1	0.3	30	390	< 0.5	7,900	< 0.05	4,870	26	< 1,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	5
	SW15-05-161018	2016 10 18	26.1	-	229	< 0.1	< 0.5	< 5	< 0.1	< 0.1	18	< 0.01	7,300	1.1	0.22	28.9	350	0.2	0.2	2,090	5.7	< 0.02	< 0.1	0.6	20	590	< 0.5	< 500	< 0.05	4,820	26	< 1,000	< 0.2	< 0.02	< 0.1	< 0.2	10	< 0.02	1	7
	SW15-05-161212	2016 12 12	14.4	-	113	< 0.1	< 0.5	< 5	< 0.1	< 0.1	11	0.02	4,500	< 0.5	0.11	9.1	120	< 0.1	0.1	1,250	3.3	< 0.02	< 0.1	0.2	< 20	360	< 0.5	5,700	< 0.05	3,860	16	1,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	< 4
	SW15-05-170801	2017 08 01	22.4	-	46.9	< 0.10	< 0.50	< 5.0	< 0.10	< 0.10	14.5	< 0.010	6,490	0.56	< 0.10	7.86	132	< 0.10	< 0.10	1,840	3.03	< 0.010	< 0.10	0.27	< 50	280	< 0.50	7,300	< 0.050	4,960	19.2	< 3,000	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW15-05-170822	2017 08 22	22.9	-	73.6	< 0.20	< 0.50	< 5.0	< 0.10	-	30.3	< 0.010	6,720	< 0.50	0.13	10.6	232	< 0.20	0.10	1,740	5.89	< 0.010	< 0.10	< 0.40	-	-	< 0.50	-	< 0.050	4,550	20.4	-	-	< 0.020	-	-	< 5.0	< 0.020	< 1.0	< 4.0
	SW15-05-170927	2017 09 27	24.3	-	44.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	13.2	< 0.010	7,000	< 0.50	< 0.10	8.35	169	< 0.20	0.10	2,020	3.56	< 0.010	< 0.10	< 0.40	< 50	290	< 0.50	7,500	< 0.050	4,690	23.3	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW15-05-171101	2017 11 01	25.5	-	18.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	13.5	< 0.010	7,890	0.83	< 0.10	7.59	79	< 0.20	0.10	2,020	1.62	< 0.010	< 0.10	< 0.40	< 50	420	< 0.50	6,900	< 0.050	4,620	24.4	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW15-05-171123	2017 11 23	20.2	-	84.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	20.6	< 0.010	6,370	< 0.50	0.12	17.1	127	< 0.20	0.28	1,620	2.79	< 0.010	< 0.10	< 0.40	< 50	480	< 0.50	6,300	< 0.050	4,030	20.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW15-05-171213	2017 12 13	15.0	-	77.1	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	11.5	< 0.010	4,150	< 0.50	< 0.10	5.51	102	< 0.20	0.68	1,220	2.86	< 0.010	< 0.10	< 0.40	< 50	270	< 0.50	5,600	< 0.050	4,010	13.9	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.1	< 4.0
	SW-A-171213	Duplicate	21.8	-	86.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	5.6	< 0.010	4,100	< 0.50	< 0.10	6.17	119	< 0.20	0.21	1,190	3.26	< 0.010	< 0.10	< 0.40	< 50	250	< 0.50	5,300	< 0.050	3,850	13.9	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.0	< 4.0
	QA/QC RPD%			37	-	11	*	*	*	*	*	*	*	1	*	*	11	15	*	*	2	13	*	*	*	*	8	*	6	*	4	0	*	*	*	*	*	*	*	*
SW15-06	SW15-05-180305	2018 03 05	14.3	-	71.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	8.5	< 0.010	4,190	0.91	< 0.10	6.12	81	< 0.20																						

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																					
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L		
BC Guideline/Standard																																										
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e	
SW15-07	SW15-07-170927	2017 09 27	37.4	-	54.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	52.5	0.141	13,800	< 0.50	11.2	342	64	< 0.20	0.18	987	8.80	< 0.010	< 0.10	15.3	< 50	230	< 0.50	3,500	< 0.050	9,370	20.8	3,500	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	33.5		
	SW15-07-171031	2017 10 31	30.9	-	64.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	41.5	0.116	11,600	< 0.50	11.0	332	65	< 0.20	0.17	956	9.29	< 0.010	< 0.10	14.6	< 50	210	< 0.50	3,500	0.075	7,270	16.7	5,200	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	34.2		
	SW15-07-171122	2017 11 22	31.2	-	116	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	43.0	0.143	13,400	< 0.50	12.5	476	112	< 0.20	0.36	1,320	11.4	< 0.010	< 0.10	15.9	< 50	240	< 0.50	4,500	< 0.050	7,270	20.0	4,600	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	34.8		
	SW15-07-171212	2017 12 12	23.2	-	63.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	31.9	0.100	8,000	< 0.50	7.99	370	76	< 0.20	0.39	1,040	5.88	< 0.010	< 0.10	10.9	< 50	240	< 0.50	4,200	< 0.050	5,440	13.2	3,400	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	26.2		
	SW15-07-180305	2018 03 05	20.3	-	78.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	27.7	0.095	6,700	< 0.50	7.91	361	50	< 0.20	0.16	990	6.99	< 0.010	< 0.10	9.58	< 50	330	< 0.50	4,200	< 0.050	5,560	11.6	3,700	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	23.8		
	SW15-07-180725	2018 07 25	51.0	-	62.1	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	58.0	0.180	20,100	< 0.50	14.2	429	40	< 0.20	0.17	1,220	12.1	< 0.010	0.11	19.4	< 50	270	< 0.50	3,700	< 0.050	11,800	31.3	5,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	43.9		
	SW15-07-181030	2018 10 30	41.9	-	72.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	47.8	0.128	15,800	< 0.50	12.4	450	68	< 0.20	0.15	1,120	11.4	< 0.010	0.11	16.6	< 50	260	< 0.50	3,400	< 0.050	9,880	21.7	5,200	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	35.5		
	SW15-07-190123	2019 01 23	21.6	-	101	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	33.0	0.098	7,920	< 0.50	6.76	314	66	< 0.20	0.14	971	6.21	< 0.010	0.11	8.80	< 50	210	< 0.50	4,300	< 0.050	4,830	10.8	3,300	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	23.7		
	SW15-07-190926	2019 09 26	34.0	-	61.1	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	62.9	0.103	13,700	< 0.50	8.73	269	60	< 0.20	0.21	919	7.99	< 0.010	0.10	12.2	< 50	220	< 0.50	3,400	< 0.050	8,210	19.4	4,300	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.1	27.2		
	SW15-07-191205	2019 12 05	24.3	-	58.5	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	35.6	0.100	8,670	< 0.50	8.93	354	54	< 0.20	0.17	1,030	7.44	< 0.010	< 0.10	11.4	< 50	280	< 0.50	5,300	< 0.050	5,770	16.9	4,900	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	28.8		
	SW15-07-210812	2021 08 12	-	47.9	54.2	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	51.2	0.136	18,000	< 0.50	10.9	327	57	< 0.20	0.10	1,080	9.41	< 0.010	< 0.10	14.6	< 50	260	< 0.50	4,000	< 0.050	12,100	25.0	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	34.1		
	SW15-07-220211	2022 02 11	-	17.5	87.5	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	0.095	6,940	< 0.50	6.93	284	79	< 0.20	0.11	797	5.61	< 0.010	< 0.10	8.93	< 50	190	< 0.50	4,200	< 0.050	4,770	10.8	3,200	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	21.5		
	SW15-07-221025	2022 10 25	-	21.5	72.3	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	0.090	7,380	< 0.50	7.96	226	135	< 0.20	0.13	676	7.12	< 0.010	< 0.10	10.1	< 50	220	< 0.50	3,300	< 0.050	5,530	11.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	21.7		
	SW15-07-230223	2023 02 23	-	19.2	64.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	0.061	6,520	0.82	5.89	253	60	< 0.20	0.11	790	4.67	< 0.010	< 0.10	7.85	< 50	200	< 0.50	3,400	< 0.050	4,330	10.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	18.2		
	SW15-07-230802	2023 08 02	18.3	16.5	70.4	< 0.050	0.064	0.46	< 0.010	< 0.010	27.2	0.0753	5,640	< 0.50	6.44	224	79.6	< 0.050	0.092	596	5.19	< 0.0050	0.070	8.08	16	190	0.14	3,130	0.020	4,880	8.93	2,400	< 0.050	< 0.0040	< 0.010	< 0.050	0.38	0.0018	< 1.00	18.6		
	SW15-07-231115	2023 11 15	22.4	22.5	84.6	< 0.050	0.057	0.63	< 0.010	< 0.010	25.3	0.0756	7,640	< 0.50	6.60	236	79.9	< 0.050	0.119	821	4.73	< 0.0050	0.085	9.69	< 10	222	0.25	3,390	0.039	5,150	10.8	3,400	< 0.050	< 0.0040	< 0.010	< 0.050	0.67	0.0026	< 1.00	19.6		
SW15-08	SW15-08-151118	2015 11 18	12.7	-	400	< 0.1	< 0.5	< 5	< 0.1	< 0.1																																

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
					Aluminum μg/L	Antimony μg/L	Arsenic μg/L	Barium μg/L	Beryllium μg/L	Bismuth μg/L	Boron μg/L	Cadmium μg/L	Calcium μg/L	Chromium μg/L	Cobalt μg/L	Copper μg/L	Iron μg/L	Lead μg/L	Lithium μg/L	Magnesium μg/L	Manganese μg/L	Mercury μg/L	Molybdenum μg/L	Nickel μg/L	Phosphorous μg/L	Potassium μg/L	Selenium μg/L	Silicon μg/L	Silver μg/L	Sodium μg/L	Strontium μg/L	Sulphur μg/L	Tellurium μg/L	Thallium μg/L	Thorium μg/L	Tin μg/L	Titanium μg/L	Uranium μg/L	Vanadium μg/L	Zinc μg/L																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																				
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L	
BC Guideline/Standard																																									
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a	
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e	
SW15-10	SW15-10-171123	2017 11 23	3.85	-	231	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	7.1	< 0.010	1,200	0.68	0.42	23.2	141	< 0.20	0.34	334	6.65	< 0.010	< 0.10	< 0.40	< 50	120	< 0.50	1,200	< 0.050	1,260	5.5	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	5.1	< 0.020	< 1.0	< 4.0	
	SW15-10-171213	2017 12 13	6.58	-	130	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	7.9	0.012	2,060	1.35	1.08	38.8	114	0.25	0.29	561	4.70	< 0.010	< 0.10	2.15	< 50	150	< 0.50	2,000	< 0.050	2,190	6.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-180305	2018 03 05	5.61	-	75.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,510	< 0.50	0.23	10.5	22	< 0.20	< 0.10	478	0.84	< 0.010	< 0.10	< 0.40	< 50	170	< 0.50	1,700	< 0.050	2,130	4.7	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-180724	2018 07 24	6.90	-	64.9	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	27.2	< 0.010	2,220	0.56	0.43	13.2	158	< 0.20	0.12	461	2.01	< 0.010	< 0.10	0.85	< 50	260	< 0.50	1,600	< 0.050	1,930	8.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-181031	2018 10 31	5.79	-	171	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,860	1.17	0.28	19.3	68	< 0.20	0.13	587	1.97	< 0.010	< 0.10	1.04	< 50	130	< 0.50	2,500	< 0.050	2,470	6.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-190124	2019 01 24	5.30	-	118	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	10.1	< 0.010	1,600	< 0.50	0.20	13.1	47	< 0.20	0.12	423	1.68	< 0.010	< 0.10	0.41	< 50	< 100	< 0.50	1,600	< 0.050	1,780	4.5	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	4.4	
	SW15-10-190925	2019 09 25	7.14	-	147	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	12.5	< 0.010	1,940	< 0.50	0.17	5.28	221	< 0.20	0.15	562	4.09	< 0.010	< 0.10	0.59	< 50	160	< 0.50	1,800	< 0.050	2,200	8.7	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.2	< 4.0	
	SW15-10-191203	2019 12 03	6.80	-	113	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	22.9	< 0.010	2,000	< 0.50	0.42	16.9	45	< 0.20	0.12	525	1.43	< 0.010	< 0.10	0.83	< 50	130	< 0.50	2,200	< 0.050	2,070	6.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-191204	2019 12 04	4.52	-	230	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	16.2	< 0.010	1,300	0.51	0.22	10.5	91	< 0.20	0.14	447	2.48	< 0.010	< 0.10	< 0.40	< 50	110	< 0.50	2,000	< 0.050	1,880	5.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-200730	2020 07 30	7.37	-	78.0	< 0.10	0.16	1.94	< 0.020	< 0.050	< 10	0.0055	2,180	0.20	0.32	12.1	267	0.059	< 1.0	557	2.40	< 0.0050	< 0.050	0.64	< 50	190	0.062	1,980	< 0.010	2,280	9.06	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	< 3.0	
	SW15-10-210113	2021 01 13	3.35	3.49	177	< 0.10	< 0.10	2.41	< 0.020	< 0.050	< 10	< 0.0050	893	0.30	0.16	9.29	95	< 0.050	< 1.0	307	3.55	< 0.0050	< 0.050	< 0.50	< 50	142	< 0.050	1,250	< 0.010	1,210	5.55	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	< 3.0	
	SW15-C-210113	Duplicate	3.38	3.53	175	< 0.10	< 0.10	2.40	< 0.020	< 0.050	< 10	< 0.0050	908	0.26	0.17	9.45	99	< 0.050	< 1.0	307	3.72	< 0.0050	< 0.050	< 0.50	< 50	145	< 0.050	1,270	< 0.010	1,240	5.59	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	< 3.0	
	QA/QC RPD%			1	1	1	*	*	< 1	*	*	*	*	2	*	*	2	4	*	*	*	5	*	*	*	*	*	2	*	2	1	*	*	*	*	*	*	*	*	*	*
	SW15-10-210811	2021 08 11	-	9.26	63.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	2,740	< 0.50	0.41	11.3	560	< 0.20	0.10	647	2.40	< 0.010	< 0.10	0.70	< 50	290	< 0.50	2,200	< 0.050	2,500	9.9	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-10-220210	2022 02 10	-	4.20	112	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,320	< 0.50	0.22	10.5	38	< 0.20	< 0.10	380	0.86	< 0.010	< 0.10	0.49	< 50	110	< 0.50	2,200	< 0.050	1,650	4.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	5.6	
	SW-A-220210	Duplicate	-	4.07	96.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,250	< 0.50	0.20	9.78	21	< 0.20	< 0.10	372	0.38	< 0.010	< 0.10	0.48	< 50	120	< 0.50	2,000	< 0.050	1,620	4.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.2	< 4.0	
	QA/QC RPD%			-	3	15	*	*	*	*	*	*	*	5	*	*	7	*	*	*	*	*	*	*	*	*	*	10	*	2	*	*	*	*	*	*	*	*	*	*	*
	SW15-10-221027	2022 10 27	-	10.7	53.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	3,010	< 0.50	0.28	9.23	238	< 0.20	0.10	784	1.74	< 0.010	< 0.10	0.55	< 50	340	< 0.50	2,													

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Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																				
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L	
BC Guideline/Standard																																									
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e	
SW15-11	SW15-11-200730	2020 07 30	7.58	-	76.3	< 0.10	0.14	1.86	< 0.020	< 0.050	< 10	0.0058	2,250	0.20	0.33	14.0	254	0.070	< 1.0	561	2.33	< 0.0050	< 0.050	0.68	< 50	188	< 0.050	1,980	< 0.010	2,220	8.89	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	< 3.0	
	SW15-11-210112	2021 01 12	3.46	3.76	182	< 0.10	< 0.10	2.16	< 0.020	< 0.050	< 10	< 0.0050	957	0.30	0.12	3.75	94	< 0.050	< 1.0	333	3.17	< 0.0050	< 0.050	< 0.50	< 50	131	< 0.050	1,360	< 0.010	1,410	5.40	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	< 3.0	
	SW15-11-210812	2021 08 12	-	9.46	66.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	2,780	1.18	0.37	12.6	578	< 0.20	< 0.10	656	2.38	< 0.010	< 0.10	0.75	< 50	300	< 0.50	2,300	< 0.050	2,510	10.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-11-220210	2022 02 10	5.16		115	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,490	< 0.50	0.21	9.54	33	< 0.20	< 0.10	431	0.76	< 0.010	< 0.10	0.47	< 50	120	< 0.50	2,300	< 0.050	1,760	4.9	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0	
	SW15-11-221026	2022 10 26	-	10.7	47.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	2,950	< 0.50	0.32	11.0	283	< 0.20	0.10	748	1.85	< 0.010	< 0.10	0.63	< 50	340	< 0.50	2,200	< 0.050	2,790	13.8	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0	
	SW-A-221026	Duplicate	-	10.7	46.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	2,960	< 0.50	0.33	11.1	282	< 0.20	< 0.10	777	1.85	< 0.010	< 0.10	0.65	< 50	350	< 0.50	2,300	< 0.050	2,830	13.8	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0	
	QA/QC RPD%		-	0	3	*	*	*	*	*	*	*	0	*	*	1	0	*	*	4	0	*	*	*	*	*	*	*	1	0	*	*	*	*	*	*	*	*	*	*	
	SW15-11-230222	2023 02 22	-	5.22	106	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,320	< 0.50	0.15	6.41	22	< 0.20	< 0.10	468	0.40	< 0.010	< 0.10	0.45	< 50	110	< 0.50	1,900	< 0.050	2,110	4.9	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0	
	SW15-11-230801	2023 08 01	7.69	6.92	41.6	< 0.050	0.104	2.25	< 0.010	< 0.010	4.1	0.0061	1,960	< 0.50	0.339	13.7	104	< 0.050	0.098	492	1.32	< 0.0050	0.022	0.621	< 10	240	< 0.10	1,640	< 0.010	1,910	9.22	< 1,000	< 0.050	0.0040	< 0.010	< 0.050	0.42	0.0020	< 1.00	1.1	
	SW15-11-231114	2023 11 14	6.19	6.26	126	< 0.050	< 0.050	1.76	< 0.010	< 0.010	3.8	0.0044	1,650	< 0.50	0.256	9.85	36.5	< 0.050	0.070	520	1.01	< 0.0050	0.011	0.689	< 10	141	< 0.10	2,020	< 0.010	2,400	6.26	< 1,000	< 0.050	< 0.0040	< 0.010	< 0.050	0.67	0.0017	< 1.00	1.6	
	SW15-12	SW15-12-151118	2015 11 18	3.87	-	356	< 0.1	< 0.5	< 5	< 0.1	< 0.1	< 4	0.01	1,100	0.8	0.32	4.7	300	< 0.1	0.4	390	9.6	< 0.02	< 0.1	1	< 20	220	< 0.5	1,400	< 0.05	1,350	5	< 1,000	< 0.2	< 0.02	< 0.1	< 0.2	13	< 0.02	1	< 4
		SW15-12-160630	2016 06 30	30.9	-	75	< 0.1	< 0.5	< 5	< 0.1	< 0.1	25	< 0.01	3,900	0.6	0.37	14.5	210	< 0.1	0.7	5,670	2.5	< 0.02	< 0.1	0.7	< 20	1,810	< 0.5	2,100	< 0.05	43,800	34	2,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	< 4
		SW16-B-160630	Duplicate	29.5	-	81	< 0.1	< 0.5	< 5	< 0.1	< 0.1	26	< 0.01	4,100	< 0.5	0.39	15.2	220	< 0.1	0.8	5,930	2.6	< 0.02	< 0.1	0.8	< 20	1,850	< 0.5	2,200	< 0.05	45,100	35	2,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	< 4
		QA/QC RPD%		5	-	8	*	*	*	*	*	4	*	5	*	5	5	5	*	13	4	4	*	*	*	*	2	*	*	*	3	3	*	*	*	*	*	*	*	*	*
SW15-12-160907		2016 09 07	8.70	-	66	< 0.1	< 0.5	< 5	< 0.1	< 0.1	11	< 0.01	2,400	< 0.5	0.29	10.7	860	0.1	0.2	650	3.0	< 0.02	< 0.1	0.6	< 20	350	< 0.5	2,100	< 0.05	2,770	11	2,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	< 4	
SW16-B-160907		Duplicate	8.72	-	65	< 0.1	< 0.5	< 5	< 0.1	< 0.1	13	< 0.01	2,600	< 0.5	0.31	10.7	820	0.1	0.2	650	3.0	< 0.02	< 0.1	0.8	< 20	340	< 0.5	2,000	< 0.05	2,690	11	3,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	< 4	
QA/QC RPD%			< 1	-	2	*	*	*	*	*	*	*	8	*	7	0	5	*	*	0	0	*	*	*	*	3	*	*	*	*	3	0	*	*	*	*	*	*	*	*	
SW15-12-161017		2016 10 17	5.82	-	253	< 0.1	< 0.5	< 5	< 0.1	< 0.1	8	< 0.01	1,700	0.7	0.18	3.5	210	< 0.1	0.2	530	7.8	< 0.02	< 0.1	0.6	< 20	290	< 0.5	< 500	< 0.05	1,940	9	< 1,000	< 0.2	< 0.02	< 0.1	< 0.2	< 5	< 0.02	< 1	< 4	
SW16-B-161017		Duplicate	5.65	-	235	< 0.1	< 0.5	< 5	< 0.1	< 0.1	11	0.01	1,700	0.7	0.19	4.0	190	< 0.1	0.2	520	7.4	< 0.02	< 0.1	0.6	< 20	280	< 0.5	<													

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																			
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L
BC Guideline/Standard																																								
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e
SW17-14	SW17-14-170731	2017 07 31	5.94	-	79.1	0.27	< 0.50	< 5.0	< 0.10	< 0.10	5.2	< 0.010	1,710	< 0.50	< 0.10	2.04	238	< 0.10	0.13	470	2.29	0.049	< 0.10	0.42	< 50	280	< 0.50	1,600	< 0.050	1,730	7.1	< 3,000	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW17-14-170821	2017 08 21	6.32	-	94.0	< 0.20	< 0.50	< 5.0	< 0.10	-	9.3	< 0.010	1,820	< 0.50	< 0.10	2.29	322	< 0.20	0.13	473	3.05	< 0.010	< 0.10	3.55	-	-	< 0.50	-	< 0.050	1,620	8.2	-	-	< 0.020	-	-	< 5.0	< 0.020	< 1.0	< 4.0
	SW17-14-170926	2017 09 26	6.48	-	53.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,850	< 0.50	< 0.10	1.84	319	< 0.20	0.16	526	2.61	< 0.010	< 0.10	0.41	< 50	290	< 0.50	1,500	< 0.050	1,840	10.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW17-14-171031	2017 10 31	6.44	-	96.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	6.3	< 0.010	1,750	0.75	< 0.10	1.24	69	< 0.20	0.11	506	0.78	< 0.010	< 0.10	< 0.40	< 50	170	< 0.50	1,700	< 0.050	1,810	7.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW17-14-171212	2017 12 12	5.62	-	92.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,580	< 0.50	< 0.10	1.90	49	< 0.20	0.12	476	0.96	< 0.010	< 0.10	< 0.40	< 50	120	< 0.50	1,800	< 0.050	1,910	5.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
SW17-15	SW17-14-180305	2018 03 05	5.02	-	85.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,370	< 0.50	< 0.10	2.18	46	< 0.20	0.10	487	0.74	< 0.010	< 0.10	< 0.40	< 50	180	< 0.50	1,700	< 0.050	2,120	5.3	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	21.1
	SW17-15-170731	2017 07 31	8.75	-	74.9	< 0.10	< 0.50	< 5.0	< 0.10	< 0.10	9.9	0.020	3,200	< 0.50	1.54	45.7	159	< 0.10	0.13	557	2.75	< 0.010	< 0.10	2.31	< 50	250	< 0.50	1,900	< 0.050	2,760	8.5	< 3,000	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	6.7
	SW17-15-170821	2017 08 21	8.96	-	80.1	< 0.20	< 0.50	< 5.0	< 0.10	-	12.9	0.019	3,240	< 0.50	1.20	38.4	286	< 0.20	0.14	552	3.45	< 0.010	< 0.10	1.81	-	-	< 0.50	-	< 0.050	2,500	10.0	-	-	< 0.020	-	-	< 5.0	< 0.020	< 1.0	6.5
	SW17-15-170926	2017 09 26	9.70	-	53.9	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	9.0	0.019	3,100	< 0.50	1.22	36.4	293	< 0.20	0.16	575	3.36	< 0.010	< 0.10	1.88	< 50	260	< 0.50	1,700	< 0.050	2,600	11.3	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	4.8
	SW17-15-171031	2017 10 31	8.44	-	93.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	10.4	0.025	2,910	0.67	1.40	40.6	69	< 0.20	0.11	580	1.84	< 0.010	< 0.10	2.15	< 50	180	< 0.50	2,000	< 0.050	2,560	8.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	5.8
	SW-A-171031	Duplicate	7.98	-	91.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	9.4	0.015	2,800	< 0.50	1.35	39.8	68	< 0.20	0.10	560	1.81	< 0.010	< 0.10	2.15	< 50	170	< 0.50	1,900	< 0.050	2,480	8.4	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	5.5
	QA/QC RPD%		6	-	2	*	*	*	*	*	*	*	4	*	4	2	1	*	*	4	2	*	*	0	*	6	*	*	*	3	2	*	*	*	*	*	*	*	*	
	SW17-15-171212	2017 12 12	7.42	-	88.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	6.2	0.011	2,190	0.76	0.88	37.0	37	< 0.20	0.17	559	1.25	< 0.010	< 0.10	1.39	< 50	140	< 0.50	2,100	< 0.050	2,410	6.7	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW17-15-180305	2018 03 05	5.07	-	79.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,610	< 0.50	0.33	11.8	34	< 0.20	0.11	486	0.80	< 0.010	< 0.10	< 0.40	< 50	170	< 0.50	1,800	< 0.050	2,080	4.9	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
	SW17-16	SW17-16-170801	2017 08 01	6.95	-	80.4	0.14	< 0.50	< 5.0	< 0.10	< 0.10	6.0	< 0.010	2,060	0.60	0.38	12.5	175	< 0.10	0.13	512	2.07	< 0.010	< 0.10	0.85	< 50	250	< 0.50	1,800	< 0.050	2,020	7.5	< 3,000	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0
SW17-16-170926		2017 09 26	7.45	-	53.9	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	5.7	< 0.010	2,160	< 0.50	0.36	11.6	309	< 0.20	0.14	544	2.56	< 0.010	< 0.10	0.77	< 50	270	< 0.50	1,600	< 0.050	2,020	10.5	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
SW17-16-171031		2017 10 31	6.84	-	99.3	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	7.3	< 0.010	2,000	0.75	0.44	14.8	69	0.40	0.10	547	1.06	< 0.010	< 0.10	0.90	< 50	180	< 0.50	1,800	< 0.050	2,100	7.8	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0
SW17-16-171123		2017 11 23	3.81	-	221	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,100	0.90	0.21	8.33	135	< 0.20	0.32	336	5.49	< 0.010	< 0.10	< 0.40	< 50	130	< 0.50	1,200	< 0.050	1,250	5.5	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.	

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																						
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L			
BC Guideline/Standard																																											
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	8.5	n/a	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	n/a	30	n/a	n/a	10,000	850	n/a	750-16,500 ^e		
SW17-17	SW17-17-230801	2023 08 01	7.54	6.87	40.7	< 0.050	0.092	2.29	< 0.010	< 0.010	3.6	0.0079	1,950	< 0.50	0.342	13.7	105	< 0.050	0.075	487	1.32	< 0.0050	0.858	0.618	< 10	239	< 0.10	1,590	< 0.010	1,900	9.25	< 1,000	< 0.050	< 0.0040	< 0.010	< 0.050	0.39	0.0021	< 1.00	1.3			
	SW17-17-231114	2023 11 14	6.21	6.13	127	< 0.050	< 0.050	1.77	< 0.010	< 0.010	4.0	0.0047	1,600	< 0.50	0.245	10.4	35.8	< 0.050	0.071	519	0.946	< 0.0050	0.018	0.564	< 10	132	< 0.10	2,000	< 0.010	2,380	5.94	< 1,000	< 0.050	< 0.0040	< 0.010	< 0.050	0.88	0.0017	< 1.00	1.1			
SW17-18	SW17-18-170801	2017 08 01	7.68	-	93.9	< 0.10	< 0.50	< 5.0	< 0.10	< 0.10	12.4	< 0.010	1,700	< 0.50	0.33	16.3	189	< 0.10	1.13	993	3.49	< 0.010	< 0.10	0.70	< 50	810	< 0.50	1,900	< 0.050	12,400	8.5	< 3,000	< 0.20	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
SW17-19	SW17-19-170822	2017 08 22	6.73	-	82.6	< 0.20	< 0.50	< 5.0	< 0.10	-	7.9	< 0.010	2,160	< 0.50	0.37	13.3	265	< 0.20	0.13	486	2.85	< 0.010	< 0.10	0.78	-	-	< 0.50	-	< 0.050	1,800	8.5	-	-	< 0.020	-	-	< 5.0	< 0.020	< 1.0	4.4			
	SW17-19-170927	2017 09 27	7.59	-	52.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	5.4	< 0.010	2,240	< 0.50	0.41	13.4	317	< 0.20	0.15	563	2.65	< 0.010	< 0.10	0.95	< 50	270	< 0.50	1,600	< 0.050	2,050	10.7	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-171101	2017 11 01	7.34	-	88.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	6.7	< 0.010	2,020	< 0.50	0.44	14.3	64	< 0.20	< 0.10	505	1.08	< 0.010	< 0.10	0.87	< 50	160	< 0.50	1,700	< 0.050	1,890	7.3	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-171123	2017 11 23	3.73	-	221	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,150	0.53	0.17	5.16	125	< 0.20	0.36	345	5.35	< 0.010	< 0.10	< 0.40	< 50	130	< 0.50	1,200	< 0.050	1,320	5.7	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	4.4			
	SW17-19-171213	2017 12 13	6.32	-	96.4	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	2,020	0.82	0.42	17.1	65	< 0.20	0.22	543	1.63	< 0.010	< 0.10	0.77	< 50	130	< 0.50	2,100	< 0.050	2,180	6.4	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-180305	2018 03 05	5.72	-	71.0	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,640	< 0.50	0.24	11.5	21	< 0.20	< 0.10	509	0.81	< 0.010	< 0.10	< 0.40	< 50	170	< 0.50	1,900	< 0.050	1,990	5.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-180724	2018 07 24	7.01	-	72.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	17.0	< 0.010	2,310	0.51	0.45	15.2	152	< 0.20	0.11	471	2.02	< 0.010	< 0.10	0.91	< 50	240	< 0.50	1,700	< 0.050	1,910	8.8	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-181031	2018 10 31	5.95	-	192	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,810	< 0.50	0.17	7.47	85	0.33	0.14	606	2.36	< 0.010	< 0.10	0.46	< 50	130	< 0.50	2,500	< 0.050	2,480	6.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	8.5			
	SW17-19-190124	2019 01 24	5.28	-	105	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	5.9	< 0.010	1,440	< 0.50	0.17	9.99	24	< 0.20	< 0.10	432	0.80	< 0.010	< 0.10	0.54	< 50	< 100	< 0.50	1,500	< 0.050	1,810	3.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW-C-190124	Duplicate	5.20	-	138	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	10.6	< 0.010	1,560	0.52	0.22	14.6	50	0.43	0.11	455	1.49	< 0.010	< 0.10	0.86	< 50	< 100	< 0.50	1,600	0.056	1,900	3.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	0.043	< 1.0	6.6			
	QA/QC RPD%			2	-	27	*	*	*	*	*	*	8	*	*	37	*	*	*	5	*	*	*	*	*	*	*	*	*	5	*	*	*	*	*	*	*	*	*	*	*		
	SW17-19-190925	2019 09 25	7.25	-	154	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	9.2	< 0.010	1,980	< 0.50	0.19	6.35	229	< 0.20	0.15	572	4.31	< 0.010	< 0.10	0.52	< 50	170	< 0.50	1,900	< 0.050	2,220	9.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.2	< 4.0			
	SW17-19-191204	2019 12 04	4.52	-	232	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	14.9	< 0.010	1,320	< 0.50	0.15	6.16	95	< 0.20	0.14	454	2.43	< 0.010	< 0.10	< 0.40	< 50	110	< 0.50	2,100	< 0.050	1,910	5.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-210812	2021 08 12	-	9.32	61.8	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	0.010	2,770	< 0.50	0.36	12.3	581	< 0.20	0.14	656	2.36	< 0.010	< 0.10	0.72	< 50	310	< 0.50	2,600	< 0.050	2,520	10.0	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW17-19-220210	2022 02 10	-	4.57	108	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	&																															

Table 7: Summary of Analytical Results for Total Metals in Surface Water (Aquatic Life Comparison)

Sample Location	Sample ID	Sample Date (yyyy mm dd)	Hardness mg/L	Total Hardness mg/L	Total Metals																																						
					Aluminum µg/L	Antimony µg/L	Arsenic µg/L	Barium µg/L	Beryllium µg/L	Bismuth µg/L	Boron µg/L	Cadmium µg/L	Calcium µg/L	Chromium µg/L	Cobalt µg/L	Copper µg/L	Iron µg/L	Lead µg/L	Lithium µg/L	Magnesium µg/L	Manganese µg/L	Mercury µg/L	Molybdenum µg/L	Nickel µg/L	Phosphorous µg/L	Potassium µg/L	Selenium µg/L	Silicon µg/L	Silver µg/L	Sodium µg/L	Strontium µg/L	Sulphur µg/L	Tellurium µg/L	Thallium µg/L	Thorium µg/L	Tin µg/L	Titanium µg/L	Uranium µg/L	Vanadium µg/L	Zinc µg/L			
BC Guideline/Standard																																											
BCWQG Aquatic Life Short-term Maximum (AW) ^b			n/a	n/a	n/a	250 ^h	n/a	n/a	n/a	n/a	n/a	n/a	n/a	110	n/a	1,000	3-242.27 ^e	n/a	n/a	815-3,130 ^e	n/a	2,000	n/a	n/a	n/a	n/a	n/a	0.1-3 ^e	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
BCWQG Aquatic Life Long-term Average (AW) ^c			n/a	n/a	7.3-250 ^g	74 ^h	5	1,000	0.13	n/a	1,200	n/a	n/a	1 (Cr(+6))	4	n/a	n/a	3-12.76 ^e	n/a	n/a	767-1,639 ^e	0.02 ^f	1,000	25-150 ^e	n/a	n/a	2	n/a	0.05-1.5 ^e	n/a	n/a	n/a	n/a	0.8	n/a	n/a	n/a	n/a	n/a	8.5	n/a	n/a	n/a
CSR Aquatic Life Upper Cap Concentration (AW UCC) ^d			n/a	n/a	n/a	900	500	100,000	15	n/a	120,000	5-40 ^e	n/a	100	400	200-900 ^e	n/a	400-1,100 ^e	n/a	n/a	n/a	2.5	100,000	2,500-15,000 ^e	n/a	n/a	200	n/a	5-150 ^e	n/a	n/a	n/a	n/a	30	n/a	n/a	n/a	10,000	850	n/a	750-16,500 ^e		
SW17-21	SW17-21-221026	2022 10 26	-	10.7	47.1	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	3,010	< 0.50	0.32	10.1	280	< 0.20	< 0.10	741	1.76	< 0.010	< 0.10	0.62	< 50	340	< 0.50	2,200	< 0.050	2,760	13.5	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0			
	SW17-21-230222	2023 02 22	-	4.76	108	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,200	0.65	0.18	7.93	26	< 0.20	< 0.10	441	0.67	< 0.010	< 0.10	0.44	< 50	110	< 0.50	1,700	< 0.050	2,120	4.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0			
	SW-A-230222	Duplicate	-	4.80	107	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,210	1.64	0.18	7.64	29	< 0.20	< 0.10	441	0.73	< 0.010	< 0.10	0.97	< 50	100	< 0.50	1,700	< 0.050	2,160	4.4	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0			
	QA/QC RPD%		-	1	1	*	*	*	*	*	*	*	1	*	*	4	*	*	*	0	*	*	*	*	*	*	*	*	2	*	*	*	*	*	*	*	*	*	*	*	*		
	SW17-21-230801	2023 08 01	7.43	6.55	43.2	< 0.050	0.087	2.14	< 0.010	< 0.010	3.8	0.0047	1,850	< 0.50	0.331	11.9	108	< 0.050	0.081	468	1.31	< 0.0050	0.020	0.587	< 10	242	< 0.10	1,560	< 0.010	1,840	8.92	< 1,000	< 0.050	< 0.0040	< 0.010	< 0.050	0.43	0.0020	< 1.00	1.2			
SW17-21-231114	2023 11 14	6.07	5.98	124	< 0.050	0.050	1.83	< 0.010	< 0.010	3.7	0.0080	1,560	< 0.50	0.265	10.5	38.5	< 0.050	0.072	507	1.05	< 0.0050	0.029	0.546	< 10	125	< 0.10	2,000	< 0.010	2,440	5.97	< 1,000	< 0.050	< 0.0040	< 0.010	< 0.050	0.86	0.0017	< 1.00	1.2				
SW18-22	SW18-22-180726	2018 07 26	6.90	-	108	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	6.6	0.012	2,140	0.52	0.47	15.2	229	< 0.20	0.15	483	2.58	< 0.010	< 0.10	1.01	< 50	220	< 0.50	1,700	< 0.050	1,790	9.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.3	13.4			
	SW18-22-181031	2018 10 31	5.84	-	176	0.26	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	< 0.010	1,900	< 0.50	0.26	19.9	65	< 0.20	0.13	594	1.84	< 0.010	< 0.10	0.63	< 50	130	< 0.50	2,500	< 0.050	2,520	6.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW18-22-190124	2019 01 24	5.85	-	138	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 5.0	0.010	1,570	< 0.50	0.79	95.2	31	< 0.20	< 0.10	437	1.90	< 0.010	< 0.10	1.35	< 50	< 100	< 0.50	1,500	< 0.050	1,790	3.6	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW18-22-190925	2019 09 25	7.09	-	153	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	7.9	< 0.010	2,120	< 0.50	0.22	8.69	227	< 0.20	0.15	581	4.30	< 0.010	< 0.10	0.56	< 50	170	< 0.50	1,900	< 0.050	2,330	9.2	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	2.0	< 4.0			
	SW-C-190925	Duplicate	7.19	-	151	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	7.5	< 0.010	2,100	< 0.50	0.22	8.79	228	< 0.20	0.15	577	4.30	< 0.010	< 0.10	0.65	< 50	260	< 0.50	1,800	< 0.050	2,300	9.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	2.1	< 4.0			
	QA/QC RPD%		1	-	1	*	*	*	*	*	*	*	1	*	*	1	< 1	*	*	1	0	*	*	*	*	*	*	*	1	1	*	*	*	*	*	*	*	*	*	*			
	SW18-22-191203	2019 12 03	7.08	-	112	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	12.7	< 0.010	1,960	< 0.50	0.44	20.9	47	< 0.20	0.11	515	1.51	< 0.010	< 0.10	0.76	< 50	120	< 0.50	2,200	< 0.050	2,060	6.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW18-22-200729	2020 07 29	7.63	-	80.1	< 0.10	0.18	2.01	< 0.020	< 0.050	< 10	0.0065	2,110	0.21	0.31	10.9	246	0.051	< 1.0	534	2.41	< 0.0050	< 0.050	0.59	< 50	210	0.088	1,960	< 0.010	2,160	8.81	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	3.5			
	SW18-22-210113	2021 01 13	3.44	3.69	216	< 0.10	< 0.10	2.56	< 0.020	< 0.050	< 10	0.0050	944	0.57	0.21	15.2	163	0.178	< 1.0	323	4.57	< 0.0050	< 0.050	0.71	< 50	165	< 0.050	1,290	< 0.010	1,340	5.72	< 500	< 0.20	< 0.010	< 0.10	< 0.10	< 10.0	< 0.010	< 0.50	< 3.0			
	SW18-22-210811	2021 08 11	-	9.36	58.7	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	2,730	< 0.50	0.37	12.5	532	< 0.20	< 0.10	631	2.33	< 0.010	< 0.10	0.66	< 50	280	< 0.50	2,000	< 0.050	2,470	9.7	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 1.0	< 4.0			
	SW18-22-220210	2022 02 10	4.18	-	105	0.21	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,260	< 0.50	0.19	8.82	30	< 0.20	< 0.10	371	0.62	< 0.010	< 0.10	0.41	< 50	110	< 0.50	2,100	< 0.050	1,600	4.1	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	1.3	< 4.0			
	SW18-22-221026	2022 10 26	-	10.7	53.6	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	3,080	< 0.50	0.34	10.5	287	< 0.20	< 0.10	762	1.90	< 0.010	< 0.10	0.65	< 50	360	< 0.50	2,200	< 0.050	2,870	14.3	< 3,000	< 0.50	< 0.020	< 0.10	< 0.20	< 5.0	< 0.020	< 5.0	< 4.0			
SW18-22-230223	2023 02 23	-	5.17	102	< 0.20	< 0.50	< 5.0	< 0.10	< 0.10	< 50.0	< 0.010	1,340	1.60	0.26	10.2	54	< 0.20	< 0.10	466	0.81	< 0.010	< 0.10	1.03	< 50	120	< 0.50	1,800	< 0.050	2,200	5.2	< 3,000	< 0.50	<										

DRAWING

- 697296-001 – Monitoring Well and Water Sample Location Plan



Legend

- | | | |
|-----------------------------|---------------------------------|-------------------------------|
| Monitoring Well | Mine Access Road | AEC 5: 5100 Portal |
| Porewater Sample | Flow Direction | AEC 6: 1963 Mine Blowout |
| Surface Water Sample | Water Features | Inferred Extent of Ferricrete |
| Monitoring Well not Sampled | AEC 1: North Waste Rock Pile | Inferred Extent of Tailings |
| Porewater not Sampled | AEC 2: South Waste Rock Pile | Former Mine Buildings |
| Surface Water not Sampled | AEC 3: Concentrate Storage Area | Property Boundary |
| | AEC 4: Tailings Pipeline | Jordan River |

REFERENCES:

1. Service Layer Credits: Bing Maps Aerial: © 2024 Microsoft Corporation © 2024 Maxar ©CNES (2024) Distribution Airbus DS

NOTES:

1. Original in colour.
2. Numerical scale reflects full-size print. Print scaling will distort this scale, however scale bar will remain accurate.
3. Creeek locations are approximate.
4. SW15-01 and SW15-01A are the same location.

REVISIONS:

0 - AA - 2024-07-10 - FINAL - CP

CLIENT:
Teck Metals Ltd.

PROJECT LOCATION:
Sunro Mine, Jordan River

AtkinsRéalis

Monitoring Well and Water Sample Location Plan

BY: AA

SCALE: 1:1,000

DATE: 7/10/2024

REF No:

697296-001

REV: 0

CHK'D: CP

Proj Coord Sys: NAD 1983 UTM Zone 10N



ATTACHMENT 1

Analytical Laboratory Results

CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Jonathan Croston

PO NUMBER 669765

PROJECT 631848

PROJECT INFO Sunro Mine

WORK ORDER 21H1469

RECEIVED / TEMP 2021-08-12 08:45 / 10.3°C

REPORTED 2022-03-28 09:59

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

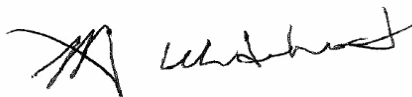
Work Order Comments:

Custody Seals Intact: yES

If you have any questions or concerns, please contact me at bwhitehead@caro.ca

Authorized By:

Brent Whitehead
Account Manager



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#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 21H1469
2022-03-28 09:59

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-01A-08102021_NP (21H1469-01) Matrix: Water Sampled: 2021-08-10 14:45					PRES
Anions					
Bromide	< 0.10	0.10	mg/L	2021-08-15	
Chloride	38.1	0.10	mg/L	2021-08-15	
Fluoride	< 0.10	0.10	mg/L	2021-08-15	
Nitrate (as N)	0.069	0.010	mg/L	2021-08-15	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2021-08-15	HT1
Sulfate	9.2	1.0	mg/L	2021-08-15	
Calculated Parameters					
Hardness, Total (as CaCO3)	< 0.500	0.500	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	< 0.0050	0.0050	mg/L	2021-08-15	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2021-08-15	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2021-08-15	
Barium, dissolved	< 0.0050	0.0050	mg/L	2021-08-15	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2021-08-15	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2021-08-15	
Boron, dissolved	< 0.0500	0.0500	mg/L	2021-08-15	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2021-08-15	
Calcium, dissolved	< 0.20	0.20	mg/L	2021-08-15	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2021-08-15	
Cobalt, dissolved	< 0.00010	0.00010	mg/L	2021-08-15	
Copper, dissolved	< 0.00040	0.00040	mg/L	2021-08-15	
Iron, dissolved	< 0.010	0.010	mg/L	2021-08-15	
Lead, dissolved	< 0.00020	0.00020	mg/L	2021-08-15	
Lithium, dissolved	0.00011	0.00010	mg/L	2021-08-15	
Magnesium, dissolved	< 0.010	0.010	mg/L	2021-08-15	
Manganese, dissolved	< 0.00020	0.00020	mg/L	2021-08-15	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2021-08-19	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2021-08-15	
Nickel, dissolved	< 0.00040	0.00040	mg/L	2021-08-15	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2021-08-15	
Potassium, dissolved	< 0.10	0.10	mg/L	2021-08-15	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2021-08-15	
Silicon, dissolved	< 1.0	1.0	mg/L	2021-08-15	
Silver, dissolved	< 0.000050	0.000050	mg/L	2021-08-15	
Sodium, dissolved	< 0.10	0.10	mg/L	2021-08-15	
Strontium, dissolved	< 0.0010	0.0010	mg/L	2021-08-15	
Sulfur, dissolved	< 3.0	3.0	mg/L	2021-08-15	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2021-08-15	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2021-08-15	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2021-08-15	
Tin, dissolved	< 0.00020	0.00020	mg/L	2021-08-15	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2021-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 21H1469
2022-03-28 09:59

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-01A-08102021_NP (21H1469-01) Matrix: Water Sampled: 2021-08-10 14:45, Continued					PRES
Dissolved Metals, Continued					
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2021-08-15	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2021-08-15	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2021-08-15	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2021-08-15	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2021-08-15	
General Parameters					
Acidity, Total (as CaCO ₃)	3.0	2.0	mg/L	2021-08-16	
Alkalinity, Total (as CaCO ₃)	6.4	1.0	mg/L	2021-08-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2021-08-17	
Alkalinity, Bicarbonate (as CaCO ₃)	6.4	1.0	mg/L	2021-08-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2021-08-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2021-08-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2021-08-16	
Solids, Total Dissolved	131	15	mg/L	2021-08-14	
Total Metals					
Aluminum, total	0.0614	0.0050	mg/L	2021-08-17	
Antimony, total	< 0.00020	0.00020	mg/L	2021-08-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2021-08-17	
Barium, total	< 0.0050	0.0050	mg/L	2021-08-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2021-08-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2021-08-17	
Boron, total	0.0545	0.0500	mg/L	2021-08-17	
Cadmium, total	0.000133	0.000010	mg/L	2021-08-17	
Calcium, total	18.0	0.20	mg/L	2021-08-17	
Chromium, total	< 0.00050	0.00050	mg/L	2021-08-17	
Cobalt, total	0.0108	0.00010	mg/L	2021-08-17	
Copper, total	0.327	0.00040	mg/L	2021-08-17	
Iron, total	0.054	0.010	mg/L	2021-08-17	
Lead, total	< 0.00020	0.00020	mg/L	2021-08-17	
Lithium, total	< 0.00010	0.00010	mg/L	2021-08-17	
Magnesium, total	1.15	0.010	mg/L	2021-08-17	
Manganese, total	0.00928	0.00020	mg/L	2021-08-17	
Mercury, total	< 0.000040	0.000040	mg/L	2021-08-17	CT5
Molybdenum, total	< 0.00010	0.00010	mg/L	2021-08-17	
Nickel, total	0.0149	0.00040	mg/L	2021-08-17	
Phosphorus, total	< 0.050	0.050	mg/L	2021-08-17	
Potassium, total	0.30	0.10	mg/L	2021-08-17	
Selenium, total	< 0.00050	0.00050	mg/L	2021-08-17	
Silicon, total	3.7	1.0	mg/L	2021-08-17	
Silver, total	< 0.000050	0.000050	mg/L	2021-08-17	
Sodium, total	12.5	0.10	mg/L	2021-08-17	
Strontium, total	0.0263	0.0010	mg/L	2021-08-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-01A-08102021_NP (21H1469-01) Matrix: Water Sampled: 2021-08-10 14:45, Continued					PRES

Total Metals, Continued

Sulfur, total	3.5	3.0	mg/L	2021-08-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2021-08-17	
Thallium, total	< 0.000020	0.000020	mg/L	2021-08-17	
Thorium, total	< 0.00010	0.00010	mg/L	2021-08-17	
Tin, total	< 0.00020	0.00020	mg/L	2021-08-17	
Titanium, total	< 0.0050	0.0050	mg/L	2021-08-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2021-08-17	
Uranium, total	< 0.000020	0.000020	mg/L	2021-08-17	
Vanadium, total	< 0.0010	0.0010	mg/L	2021-08-17	
Zinc, total	0.0340	0.0040	mg/L	2021-08-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2021-08-17	

Sample Qualifiers:

CT5 This sample has been incorrectly preserved for Hg analysis
HT1 The sample was prepared and/or analyzed past the recommended holding time.
PRES Sample has been preserved for Ammonia in the laboratory and the holding time has been extended.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2017)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2017)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2017)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2017)	Ion Chromatography	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2017)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	Solids in Water, Filtered / SM 2540 C* (2017)	Solids in Water, Filtered / Gravimetry (Dried at 103-105C)	✓	Kelowna
Total Metals in Water	EPA 200.2 / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association

General Comments:

The results in this report apply to the samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Samples will be disposed of 30 days after the test report has been issued or once samples expire, whichever comes first. Longer hold is possible if agreed to in writing.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: bwhitehead@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B1H1277									
Blank (B1H1277-BLK1)			Prepared: 2021-08-13, Analyzed: 2021-08-13						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B1H1277-BLK2)			Prepared: 2021-08-14, Analyzed: 2021-08-14						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B1H1277-BLK3)			Prepared: 2021-08-14, Analyzed: 2021-08-14						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B1H1277-BS1)			Prepared: 2021-08-13, Analyzed: 2021-08-13						
Bromide	4.07	0.10 mg/L	4.00		102	85-115			
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	3.98	0.10 mg/L	4.00		100	88-108			
Nitrate (as N)	4.07	0.010 mg/L	4.00		102	90-110			
Nitrite (as N)	1.98	0.010 mg/L	2.00		99	85-115			
Sulfate	16.1	1.0 mg/L	16.0		101	90-110			
LCS (B1H1277-BS2)			Prepared: 2021-08-14, Analyzed: 2021-08-14						
Bromide	4.12	0.10 mg/L	4.00		103	85-115			
Chloride	15.6	0.10 mg/L	16.0		98	90-110			
Fluoride	4.05	0.10 mg/L	4.00		101	88-108			
Nitrate (as N)	4.02	0.010 mg/L	4.00		101	90-110			
Nitrite (as N)	2.00	0.010 mg/L	2.00		100	85-115			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Anions, Batch B1H1277, Continued

LCS (B1H1277-BS2), Continued				Prepared: 2021-08-14, Analyzed: 2021-08-14					
Sulfate	16.1	1.0 mg/L	16.0		101	90-110			
LCS (B1H1277-BS3)				Prepared: 2021-08-14, Analyzed: 2021-08-14					
Bromide	4.11	0.10 mg/L	4.00		103	85-115			
Chloride	15.7	0.10 mg/L	16.0		98	90-110			
Fluoride	3.97	0.10 mg/L	4.00		99	88-108			
Nitrate (as N)	4.04	0.010 mg/L	4.00		101	90-110			
Nitrite (as N)	1.89	0.010 mg/L	2.00		95	85-115			
Sulfate	16.0	1.0 mg/L	16.0		100	90-110			

Dissolved Metals, Batch B1H1435

Blank (B1H1435-BLK1)				Prepared: 2021-08-15, Analyzed: 2021-08-15					
Aluminum, dissolved	< 0.0050	0.0050 mg/L							
Antimony, dissolved	< 0.00020	0.00020 mg/L							
Arsenic, dissolved	< 0.00050	0.00050 mg/L							
Barium, dissolved	< 0.0050	0.0050 mg/L							
Beryllium, dissolved	< 0.00010	0.00010 mg/L							
Bismuth, dissolved	< 0.00010	0.00010 mg/L							
Boron, dissolved	< 0.0500	0.0500 mg/L							
Cadmium, dissolved	< 0.000010	0.000010 mg/L							
Calcium, dissolved, dissolved	< 0.20	0.20 mg/L							
Chromium, dissolved	< 0.00050	0.00050 mg/L							
Cobalt, dissolved	< 0.00010	0.00010 mg/L							
Copper, dissolved	< 0.00040	0.00040 mg/L							
Iron, dissolved	< 0.010	0.010 mg/L							
Lead, dissolved	< 0.00020	0.00020 mg/L							
Lithium, dissolved	< 0.00010	0.00010 mg/L							
Magnesium, dissolved, dissolved	< 0.010	0.010 mg/L							
Manganese, dissolved	< 0.00020	0.00020 mg/L							
Molybdenum, dissolved	< 0.00010	0.00010 mg/L							
Nickel, dissolved	< 0.00040	0.00040 mg/L							
Phosphorus, dissolved	< 0.050	0.050 mg/L							
Potassium, dissolved	< 0.10	0.10 mg/L							
Selenium, dissolved	< 0.00050	0.00050 mg/L							
Silicon, dissolved	< 1.0	1.0 mg/L							
Silver, dissolved	< 0.000050	0.000050 mg/L							
Sodium, dissolved	< 0.10	0.10 mg/L							
Strontium, dissolved	< 0.0010	0.0010 mg/L							
Sulfur, dissolved	< 3.0	3.0 mg/L							
Tellurium, dissolved	< 0.00050	0.00050 mg/L							
Thallium, dissolved	< 0.000020	0.000020 mg/L							
Thorium, dissolved	< 0.00010	0.00010 mg/L							
Tin, dissolved	< 0.00020	0.00020 mg/L							
Titanium, dissolved	< 0.0050	0.0050 mg/L							
Tungsten, dissolved	< 0.0010	0.0010 mg/L							
Uranium, dissolved	< 0.000020	0.000020 mg/L							
Vanadium, dissolved	< 0.0010	0.0010 mg/L							
Zinc, dissolved	< 0.0040	0.0040 mg/L							
Zirconium, dissolved	< 0.00010	0.00010 mg/L							
LCS (B1H1435-BS1)				Prepared: 2021-08-15, Analyzed: 2021-08-15					
Aluminum, dissolved	0.0228	0.0050 mg/L	0.0199		115	80-120			
Antimony, dissolved	0.0194	0.00020 mg/L	0.0200		97	80-120			
Arsenic, dissolved	0.0185	0.00050 mg/L	0.0200		93	80-120			
Barium, dissolved	0.0193	0.0050 mg/L	0.0198		98	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 21H1469
2022-03-28 09:59

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B1H1435, Continued									
LCS (B1H1435-BS1), Continued					Prepared: 2021-08-15, Analyzed: 2021-08-15				
Beryllium, dissolved	0.0194	0.00010 mg/L	0.0198		98	80-120			
Bismuth, dissolved	0.0191	0.00010 mg/L	0.0200		96	80-120			
Boron, dissolved	< 0.0500	0.0500 mg/L	0.0200		111	80-120			
Cadmium, dissolved	0.0195	0.000010 mg/L	0.0199		98	80-120			
Calcium, dissolved, dissolved	1.88	0.20 mg/L	2.02		93	80-120			
Chromium, dissolved	0.0183	0.00050 mg/L	0.0198		92	80-120			
Cobalt, dissolved	0.0180	0.00010 mg/L	0.0199		90	80-120			
Copper, dissolved	0.0185	0.00040 mg/L	0.0200		93	80-120			
Iron, dissolved	1.90	0.010 mg/L	2.02		94	80-120			
Lead, dissolved	0.0211	0.00020 mg/L	0.0199		106	80-120			
Lithium, dissolved	0.0209	0.00010 mg/L	0.0200		105	80-120			
Magnesium, dissolved, dissolved	2.10	0.010 mg/L	2.02		104	80-120			
Manganese, dissolved	0.0171	0.00020 mg/L	0.0199		86	80-120			
Molybdenum, dissolved	0.0184	0.00010 mg/L	0.0200		92	80-120			
Nickel, dissolved	0.0184	0.00040 mg/L	0.0200		92	80-120			
Phosphorus, dissolved	2.03	0.050 mg/L	2.00		102	80-120			
Potassium, dissolved	2.07	0.10 mg/L	2.02		102	80-120			
Selenium, dissolved	0.0191	0.00050 mg/L	0.0200		95	80-120			
Silicon, dissolved	1.9	1.0 mg/L	2.00		95	80-120			
Silver, dissolved	0.0181	0.000050 mg/L	0.0200		91	80-120			
Sodium, dissolved	2.06	0.10 mg/L	2.02		102	80-120			
Strontium, dissolved	0.0177	0.0010 mg/L	0.0200		88	80-120			
Sulfur, dissolved	5.5	3.0 mg/L	5.00		110	80-120			
Tellurium, dissolved	0.0204	0.00050 mg/L	0.0200		102	80-120			
Thallium, dissolved	0.0186	0.000020 mg/L	0.0199		93	80-120			
Thorium, dissolved	0.0181	0.00010 mg/L	0.0200		91	80-120			
Tin, dissolved	0.0198	0.00020 mg/L	0.0200		99	80-120			
Titanium, dissolved	0.0188	0.0050 mg/L	0.0200		94	80-120			
Tungsten, dissolved	0.0208	0.0010 mg/L	0.0200		104	80-120			
Uranium, dissolved	0.0183	0.000020 mg/L	0.0200		92	80-120			
Vanadium, dissolved	0.0175	0.0010 mg/L	0.0200		88	80-120			
Zinc, dissolved	0.0224	0.0040 mg/L	0.0200		112	80-120			
Zirconium, dissolved	0.0194	0.00010 mg/L	0.0200		97	80-120			
Reference (B1H1435-SRM1)					Prepared: 2021-08-15, Analyzed: 2021-08-15				
Aluminum, dissolved	0.237	0.0050 mg/L	0.235		101	70-130			
Antimony, dissolved	0.0426	0.00020 mg/L	0.0431		99	70-130			
Arsenic, dissolved	0.414	0.00050 mg/L	0.423		98	70-130			
Barium, dissolved	2.89	0.0050 mg/L	3.30		88	70-130			
Beryllium, dissolved	0.202	0.00010 mg/L	0.209		96	70-130			
Boron, dissolved	1.56	0.0500 mg/L	1.65		95	70-130			
Cadmium, dissolved	0.211	0.000010 mg/L	0.221		95	70-130			
Calcium, dissolved, dissolved	6.95	0.20 mg/L	7.72		90	70-130			
Chromium, dissolved	0.390	0.00050 mg/L	0.434		90	70-130			
Cobalt, dissolved	0.113	0.00010 mg/L	0.124		91	70-130			
Copper, dissolved	0.733	0.00040 mg/L	0.815		90	70-130			
Iron, dissolved	1.20	0.010 mg/L	1.27		94	70-130			
Lead, dissolved	0.113	0.00020 mg/L	0.110		102	70-130			
Lithium, dissolved	0.103	0.00010 mg/L	0.100		103	70-130			
Magnesium, dissolved, dissolved	6.81	0.010 mg/L	6.59		103	70-130			
Manganese, dissolved	0.299	0.00020 mg/L	0.342		87	70-130			
Molybdenum, dissolved	0.380	0.00010 mg/L	0.404		94	70-130			
Nickel, dissolved	0.763	0.00040 mg/L	0.835		91	70-130			
Phosphorus, dissolved	0.491	0.050 mg/L	0.499		98	70-130			
Potassium, dissolved	3.03	0.10 mg/L	2.88		105	70-130			
Selenium, dissolved	0.0321	0.00050 mg/L	0.0324		99	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B1H1435, Continued									
Reference (B1H1435-SRM1), Continued				Prepared: 2021-08-15, Analyzed: 2021-08-15					
Sodium, dissolved	18.5	0.10 mg/L	18.0		103	70-130			
Strontium, dissolved	0.806	0.0010 mg/L	0.935		86	70-130			
Thallium, dissolved	0.0353	0.000020 mg/L	0.0385		92	70-130			
Uranium, dissolved	0.227	0.000020 mg/L	0.258		88	70-130			
Vanadium, dissolved	0.764	0.0010 mg/L	0.873		87	70-130			
Zinc, dissolved	0.815	0.0040 mg/L	0.848		96	70-130			

Dissolved Metals, Batch B1H1758

Blank (B1H1758-BLK1)				Prepared: 2021-08-18, Analyzed: 2021-08-19					
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Blank (B1H1758-BLK2)				Prepared: 2021-08-18, Analyzed: 2021-08-19					
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Reference (B1H1758-SRM1)				Prepared: 2021-08-18, Analyzed: 2021-08-19					
Mercury, dissolved	0.00613	0.000010 mg/L	0.00581		105	70-130			
Reference (B1H1758-SRM2)				Prepared: 2021-08-18, Analyzed: 2021-08-19					
Mercury, dissolved	0.00583	0.000010 mg/L	0.00581		100	70-130			

General Parameters, Batch B1H1383

Blank (B1H1383-BLK1)				Prepared: 2021-08-14, Analyzed: 2021-08-14					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B1H1383-BS1)				Prepared: 2021-08-14, Analyzed: 2021-08-14					
Solids, Total Dissolved	218	15 mg/L	240		91	85-115			

General Parameters, Batch B1H1478

Blank (B1H1478-BLK1)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B1H1478-BLK2)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B1H1478-BLK3)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B1H1478-BS1)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Ammonia, Total (as N)	0.972	0.050 mg/L	1.00		97	90-115			
LCS (B1H1478-BS2)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Ammonia, Total (as N)	0.969	0.050 mg/L	1.00		97	90-115			
LCS (B1H1478-BS3)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Ammonia, Total (as N)	0.987	0.050 mg/L	1.00		99	90-115			

General Parameters, Batch B1H1551

Blank (B1H1551-BLK1)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Acidity, Total (as CaCO3)	< 2.0	2.0 mg/L							
Blank (B1H1551-BLK2)				Prepared: 2021-08-16, Analyzed: 2021-08-16					
Acidity, Total (as CaCO3)	< 2.0	2.0 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 21H1469
2022-03-28 09:59

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B1H1551, Continued

LCS (B1H1551-BS1)			Prepared: 2021-08-16, Analyzed: 2021-08-16						
Acidity, Total (as CaCO ₃)	102	2.0 mg/L	100		102	80-120			
LCS (B1H1551-BS2)			Prepared: 2021-08-16, Analyzed: 2021-08-16						
Acidity, Total (as CaCO ₃)	101	2.0 mg/L	100		101	80-120			

General Parameters, Batch B1H1596

Blank (B1H1596-BLK1)			Prepared: 2021-08-17, Analyzed: 2021-08-17						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B1H1596-BLK2)			Prepared: 2021-08-17, Analyzed: 2021-08-17						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B1H1596-BS1)			Prepared: 2021-08-17, Analyzed: 2021-08-17						
Alkalinity, Total (as CaCO ₃)	107	1.0 mg/L	100		107	80-120			
LCS (B1H1596-BS2)			Prepared: 2021-08-17, Analyzed: 2021-08-17						
Alkalinity, Total (as CaCO ₃)	106	1.0 mg/L	100		106	80-120			

Total Metals, Batch B1H1706

Blank (B1H1706-BLK1)			Prepared: 2021-08-17, Analyzed: 2021-08-17						
Aluminum, total	< 0.0050	0.0050 mg/L							
Antimony, total	< 0.00020	0.00020 mg/L							
Arsenic, total	< 0.00050	0.00050 mg/L							
Barium, total	< 0.0050	0.0050 mg/L							
Beryllium, total	< 0.00010	0.00010 mg/L							
Bismuth, total	< 0.00010	0.00010 mg/L							
Boron, total	< 0.0500	0.0500 mg/L							
Cadmium, total	< 0.000010	0.000010 mg/L							
Calcium, total	< 0.20	0.20 mg/L							
Chromium, total	< 0.00050	0.00050 mg/L							
Cobalt, total	< 0.00010	0.00010 mg/L							
Copper, total	< 0.00040	0.00040 mg/L							
Iron, total	< 0.010	0.010 mg/L							
Lead, total	< 0.00020	0.00020 mg/L							
Lithium, total	< 0.00010	0.00010 mg/L							
Magnesium, total	< 0.010	0.010 mg/L							
Manganese, total	< 0.00020	0.00020 mg/L							
Mercury, total	< 0.000040	0.000040 mg/L							
Molybdenum, total	< 0.00010	0.00010 mg/L							
Nickel, total	< 0.00040	0.00040 mg/L							
Phosphorus, total	< 0.050	0.050 mg/L							
Potassium, total	< 0.10	0.10 mg/L							
Selenium, total	< 0.00050	0.00050 mg/L							
Silicon, total	< 1.0	1.0 mg/L							
Silver, total	< 0.000050	0.000050 mg/L							
Sodium, total	< 0.10	0.10 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 21H1469
2022-03-28 09:59

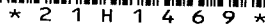
Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B1H1706, Continued									
Blank (B1H1706-BLK1), Continued				Prepared: 2021-08-17, Analyzed: 2021-08-17					
Strontium, total	< 0.0010	0.0010 mg/L							
Sulfur, total	< 3.0	3.0 mg/L							
Tellurium, total	< 0.00050	0.00050 mg/L							
Thallium, total	< 0.000020	0.000020 mg/L							
Thorium, total	< 0.00010	0.00010 mg/L							
Tin, total	< 0.00020	0.00020 mg/L							
Titanium, total	< 0.0050	0.0050 mg/L							
Tungsten, total	< 0.0010	0.0010 mg/L							
Uranium, total	< 0.000020	0.000020 mg/L							
Vanadium, total	< 0.0010	0.0010 mg/L							
Zinc, total	< 0.0040	0.0040 mg/L							
Zirconium, total	< 0.00010	0.00010 mg/L							
LCS (B1H1706-BS1)				Prepared: 2021-08-17, Analyzed: 2021-08-17					
Aluminum, total	0.0219	0.0050 mg/L	0.0199		110	80-120			
Antimony, total	0.0196	0.00020 mg/L	0.0200		98	80-120			
Arsenic, total	0.0199	0.00050 mg/L	0.0200		99	80-120			
Barium, total	0.0203	0.0050 mg/L	0.0198		102	80-120			
Beryllium, total	0.0208	0.00010 mg/L	0.0198		105	80-120			
Bismuth, total	0.0204	0.00010 mg/L	0.0200		102	80-120			
Boron, total	< 0.0500	0.0500 mg/L	0.0200		107	80-120			
Cadmium, total	0.0201	0.000010 mg/L	0.0199		101	80-120			
Calcium, total	2.05	0.20 mg/L	2.02		102	80-120			
Chromium, total	0.0205	0.00050 mg/L	0.0198		103	80-120			
Cobalt, total	0.0201	0.00010 mg/L	0.0199		101	80-120			
Copper, total	0.0197	0.00040 mg/L	0.0200		98	80-120			
Iron, total	2.01	0.010 mg/L	2.02		99	80-120			
Lead, total	0.0225	0.00020 mg/L	0.0199		113	80-120			
Lithium, total	0.0214	0.00010 mg/L	0.0200		107	80-120			
Magnesium, total	2.32	0.010 mg/L	2.02		115	80-120			
Manganese, total	0.0197	0.00020 mg/L	0.0199		99	80-120			
Mercury, total	0.000903	0.000040 mg/L	0.00100		90	80-120			
Molybdenum, total	0.0193	0.00010 mg/L	0.0200		97	80-120			
Nickel, total	0.0202	0.00040 mg/L	0.0200		101	80-120			
Phosphorus, total	2.06	0.050 mg/L	2.00		103	80-120			
Potassium, total	2.20	0.10 mg/L	2.02		109	80-120			
Selenium, total	0.0224	0.00050 mg/L	0.0200		112	80-120			
Silicon, total	2.4	1.0 mg/L	2.00		120	80-120			
Silver, total	0.0198	0.000050 mg/L	0.0200		99	80-120			
Sodium, total	2.31	0.10 mg/L	2.02		114	80-120			
Strontium, total	0.0187	0.0010 mg/L	0.0200		93	80-120			
Sulfur, total	5.8	3.0 mg/L	5.00		115	80-120			
Tellurium, total	0.0210	0.00050 mg/L	0.0200		105	80-120			
Thallium, total	0.0198	0.000020 mg/L	0.0199		100	80-120			
Thorium, total	0.0196	0.00010 mg/L	0.0200		98	80-120			
Tin, total	0.0201	0.00020 mg/L	0.0200		101	80-120			
Titanium, total	0.0209	0.0050 mg/L	0.0200		105	80-120			
Tungsten, total	0.0196	0.0010 mg/L	0.0200		98	80-120			
Uranium, total	0.0200	0.000020 mg/L	0.0200		100	80-120			
Vanadium, total	0.0202	0.0010 mg/L	0.0200		101	80-120			
Zinc, total	0.0216	0.0040 mg/L	0.0200		108	80-120			
Zirconium, total	0.0204	0.00010 mg/L	0.0200		102	80-120			
Reference (B1H1706-SRM1)				Prepared: 2021-08-17, Analyzed: 2021-08-17					
Aluminum, total	0.317	0.0050 mg/L	0.299		106	70-130			
Antimony, total	0.0495	0.00020 mg/L	0.0517		96	70-130			
Arsenic, total	0.120	0.00050 mg/L	0.119		101	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 21H1469
2022-03-28 09:59

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B1H1706, Continued									
Reference (B1H1706-SRM1), Continued				Prepared: 2021-08-17, Analyzed: 2021-08-17					
Barium, total	0.759	0.0050 mg/L	0.801		95	70-130			
Beryllium, total	0.0507	0.00010 mg/L	0.0501		101	70-130			
Boron, total	3.91	0.0500 mg/L	4.11		95	70-130			
Cadmium, total	0.0491	0.000010 mg/L	0.0503		98	70-130			
Calcium, total	9.82	0.20 mg/L	10.7		92	70-130			
Chromium, total	0.248	0.00050 mg/L	0.250		99	70-130			
Cobalt, total	0.0387	0.00010 mg/L	0.0384		101	70-130			
Copper, total	0.467	0.00040 mg/L	0.487		96	70-130			
Iron, total	0.503	0.010 mg/L	0.504		100	70-130			
Lead, total	0.294	0.00020 mg/L	0.278		106	70-130			
Lithium, total	0.419	0.00010 mg/L	0.398		105	70-130			
Magnesium, total	4.21	0.010 mg/L	3.59		117	70-130			
Manganese, total	0.106	0.00020 mg/L	0.111		96	70-130			
Mercury, total	0.00578	0.000040 mg/L	0.00581		100	70-130			
Molybdenum, total	0.193	0.00010 mg/L	0.196		98	70-130			
Nickel, total	0.247	0.00040 mg/L	0.248		99	70-130			
Phosphorus, total	0.239	0.050 mg/L	0.213		112	70-130			
Potassium, total	6.60	0.10 mg/L	5.89		112	70-130			
Selenium, total	0.123	0.00050 mg/L	0.120		103	70-130			
Sodium, total	10.0	0.10 mg/L	8.71		115	70-130			
Strontium, total	0.364	0.0010 mg/L	0.393		93	70-130			
Thallium, total	0.0782	0.000020 mg/L	0.0787		99	70-130			
Uranium, total	0.0340	0.000020 mg/L	0.0344		99	70-130			
Vanadium, total	0.381	0.0010 mg/L	0.391		98	70-130			
Zinc, total	2.46	0.0040 mg/L	2.50		98	70-130			



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WZB (Pmo)
8/12
8-25

Page 13 of 13

10,3%

CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Jonathan Croston

PO NUMBER 669765

PROJECT 631848

PROJECT INFO Sunro Mine

WORK ORDER 22B1771

RECEIVED / TEMP 2022-02-12 13:00 / 9.8°C

REPORTED 2022-02-22 16:21

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

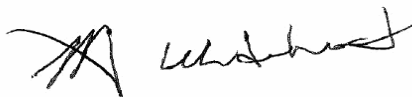
Work Order Comments:

Custody Seals Intact: YES

If you have any questions or concerns, please contact me at bwhitehead@caro.ca

Authorized By:

Brent Whitehead
Account Manager



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#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
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TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-17-02102022_NP (22B1771-01) Matrix: Water Sampled: 2022-02-10 10:25					PRESa
Anions					
Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.07	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0245	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO3)	5.08	0.500	mg/L	N/A	
Nitrate (as N)	0.0245	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0892	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.39	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00023	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00976	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.021	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.388	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00051	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00070	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.11	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.3	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.61	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0043	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-17-02102022_NP (22B1771-01) Matrix: Water Sampled: 2022-02-10 10:25, Continued					PRESa

Dissolved Metals, Continued

Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.9	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.9	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-17	

SU_PW-11-02102022_NP (22B1771-02) | Matrix: Water | Sampled: 2022-02-10 11:20

PRESa

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.06	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0242	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO ₃)	5.51	0.500	mg/L	N/A	
Nitrate (as N)	0.0242	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0981	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.54	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00020	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00824	0.00040	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-11-02102022_NP (22B1771-02) Matrix: Water Sampled: 2022-02-10 11:20, Continued					PRESa

Dissolved Metals, Continued

Iron, dissolved	0.025	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	0.00012	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.406	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00049	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00107	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.12	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.3	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.67	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0047	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	4.1	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	4.1	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	19	15	mg/L	2022-02-17	

SU_PW-19-02102022_NP (22B1771-03) | Matrix: Water | Sampled: 2022-02-10 11:50

PRESa

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.05	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-19-02102022_NP (22B1771-03) Matrix: Water Sampled: 2022-02-10 11:50, Continued					PRESa
Anions, Continued					
Nitrate+Nitrite (as N)	0.0231	0.0050	mg/L	2022-02-16	
Nitrite (as N)	0.0191	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.7	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO ₃)	4.73	0.500	mg/L	N/A	
Nitrate (as N)	< 0.0100	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.118	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.31	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00033	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.0247	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.055	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.355	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00156	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	0.00023	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00103	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.12	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.3	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.55	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0040	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-19-02102022_NP (22B1771-03) Matrix: Water Sampled: 2022-02-10 11:50, Continued					PRESa

Dissolved Metals, Continued

Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.7	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.7	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-17	

SU_PW-10-02102022_NP (22B1771-04) | Matrix: Water | Sampled: 2022-02-10 13:30

PRESa

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.99	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0221	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.3	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO ₃)	4.65	0.500	mg/L	N/A	
Nitrate (as N)	0.0221	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0896	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.27	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00024	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.0123	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.021	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-10-02102022_NP (22B1771-04) Matrix: Water Sampled: 2022-02-10 13:30, Continued					PRESa

Dissolved Metals, Continued

Magnesium, dissolved	0.356	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00057	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00076	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.11	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.2	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.56	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0042	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.5	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.5	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-17	

SU_PW-21-02102022_NP (22B1771-05) | Matrix: Water | Sampled: 2022-02-10 14:30

PRESa

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.00	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0213	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-21-02102022_NP (22B1771-05) Matrix: Water Sampled: 2022-02-10 14:30, Continued					PRESa
Calculated Parameters					
Hardness, Total (as CaCO ₃)	4.50	0.500	mg/L	N/A	
Nitrate (as N)	0.0213	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0903	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.22	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00015	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00669	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.018	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.353	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00037	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00049	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.1	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.54	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0039	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-21-02102022_NP (22B1771-05) Matrix: Water Sampled: 2022-02-10 14:30, Continued					PRESa
General Parameters, Continued					
Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.3	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.3	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	15	15	mg/L	2022-02-18	HT1

SU_PW-A-02102022_NP (22B1771-06) | Matrix: Water | Sampled: 2022-02-10 14:30 PRESa

Anions					
Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.98	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0202	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.1	1.0	mg/L	2022-02-15	

Calculated Parameters					
Hardness, Total (as CaCO ₃)	4.47	0.500	mg/L	N/A	
Nitrate (as N)	0.0202	0.0100	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	0.0891	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.20	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00017	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00674	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.018	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	0.00011	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.358	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00031	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1771
2022-02-22 16:21

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-A-02102022_NP (22B1771-06) Matrix: Water Sampled: 2022-02-10 14:30, Continued					PRESa

Dissolved Metals, Continued

Nickel, dissolved	0.00050	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.1	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.54	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0038	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.7	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.7	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	HT1

SU_PW-22-02102022_NP (22B1771-07) | Matrix: Water | Sampled: 2022-02-10 15:20

PRES

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.96	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0263	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	1.3	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO ₃)	4.69	0.500	mg/L	N/A	
Nitrate (as N)	0.0263	0.0100	mg/L	N/A	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-22-02102022_NP (22B1771-07) Matrix: Water Sampled: 2022-02-10 15:20, Continued					PRES
Dissolved Metals					
Aluminum, dissolved	0.0915	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.27	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00019	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.0208	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.025	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.370	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00073	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00063	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.12	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.1	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.58	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0042	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
General Parameters					
Acidity, Total (as CaCO3)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO3)	3.4	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2022-02-15	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-22-02102022_NP (22B1771-07) Matrix: Water Sampled: 2022-02-10 15:20, Continued					PRES
<i>General Parameters, Continued</i>					
Alkalinity, Bicarbonate (as CaCO ₃)	3.4	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-15	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	HT1

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.
 PRES Sample has been preserved for Hardness in the laboratory and the holding time has been extended.
 PRESa Sample has been preserved for Total metals in the laboratory and the holding time has been extended.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2017)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2017)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2017)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2017)	Ion Chromatography	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2017)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Nitrate+Nitrite in Water	SM 4500-NO ₃ - F (2017)	Automated Colorimetry (Cadmium Reduction)	✓	Kelowna
Nitrite in Water	SM 4500-NO ₂ B (2017)	Colorimetry	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association

General Comments:

The results in this report apply to the samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Samples will be disposed of 30 days after the test report has been issued or once samples expire, whichever comes first. Longer hold is possible if agreed to in writing.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: bwhitehead@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1486									
Blank (B2B1486-BLK1)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1486-BLK2)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	< 0.0050	0.0050 mg/L							
LCS (B2B1486-BS1)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	0.0489	0.0050 mg/L	0.0500		98	90-110			
LCS (B2B1486-BS2)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	0.0489	0.0050 mg/L	0.0500		98	90-110			
Duplicate (B2B1486-DUP1)			Source: 22B1771-01		Prepared: 2022-02-14, Analyzed: 2022-02-14				
Nitrite (as N)	< 0.0050	0.0050 mg/L		< 0.0050				10	
Duplicate (B2B1486-DUP2)			Source: 22B1771-02		Prepared: 2022-02-14, Analyzed: 2022-02-14				
Nitrite (as N)	< 0.0050	0.0050 mg/L		< 0.0050				10	
Matrix Spike (B2B1486-MS1)			Source: 22B1771-01		Prepared: 2022-02-14, Analyzed: 2022-02-14				
Nitrite (as N)	0.0460	0.0050 mg/L	0.0500	< 0.0050	91	80-120			
Matrix Spike (B2B1486-MS2)			Source: 22B1771-02		Prepared: 2022-02-14, Analyzed: 2022-02-14				
Nitrite (as N)	0.0459	0.0050 mg/L	0.0500	< 0.0050	90	80-120			

Anions, Batch B2B1570

Blank (B2B1570-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK3)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK4)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1570, Continued									
LCS (B2B1570-BS1)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Nitrate+Nitrite (as N)	0.527	0.0050 mg/L	0.500		105	91-108			
LCS (B2B1570-BS2)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Nitrate+Nitrite (as N)	0.520	0.0050 mg/L	0.500		104	91-108			
LCS (B2B1570-BS3)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Nitrate+Nitrite (as N)	0.521	0.0050 mg/L	0.500		104	91-108			
LCS (B2B1570-BS4)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Nitrate+Nitrite (as N)	0.525	0.0050 mg/L	0.500		105	91-108			
Anions, Batch B2B1586									
Blank (B2B1586-BLK1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B2B1586-BS1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Bromide	4.07	0.10 mg/L	4.00		102	85-115			
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	3.97	0.10 mg/L	4.00		99	88-108			
Sulfate	16.1	1.0 mg/L	16.0		100	90-110			
Duplicate (B2B1586-DUP1)				Source: 22B1771-01	Prepared: 2022-02-15, Analyzed: 2022-02-15				
Bromide	< 0.10	0.10 mg/L		< 0.10					10
Chloride	2.07	0.10 mg/L		2.07			< 1		10
Fluoride	< 0.10	0.10 mg/L		< 0.10					10
Sulfate	1.2	1.0 mg/L		1.2					10
Matrix Spike (B2B1586-MS1)				Source: 22B1771-01	Prepared: 2022-02-15, Analyzed: 2022-02-15				
Bromide	4.09	0.10 mg/L	4.00	< 0.10	102	80-120			
Chloride	18.1	0.10 mg/L	16.0	2.07	100	75-125			
Fluoride	4.02	0.10 mg/L	4.00	< 0.10	100	75-125			
Sulfate	17.3	1.0 mg/L	16.0	1.2	100	75-125			
Dissolved Metals, Batch B2B1631									
Blank (B2B1631-BLK1)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Aluminum, dissolved	< 0.0050	0.0050 mg/L							
Antimony, dissolved	< 0.00020	0.00020 mg/L							
Arsenic, dissolved	< 0.00050	0.00050 mg/L							
Barium, dissolved	< 0.0050	0.0050 mg/L							
Beryllium, dissolved	< 0.00010	0.00010 mg/L							
Bismuth, dissolved	< 0.00010	0.00010 mg/L							
Boron, dissolved	< 0.0500	0.0500 mg/L							
Cadmium, dissolved	< 0.000010	0.000010 mg/L							
Calcium, dissolved, dissolved	< 0.20	0.20 mg/L							
Chromium, dissolved	< 0.00050	0.00050 mg/L							
Cobalt, dissolved	< 0.00010	0.00010 mg/L							
Copper, dissolved	< 0.00040	0.00040 mg/L							
Iron, dissolved	< 0.010	0.010 mg/L							
Lead, dissolved	< 0.00020	0.00020 mg/L							
Lithium, dissolved	< 0.00010	0.00010 mg/L							
Magnesium, dissolved, dissolved	< 0.010	0.010 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B1631, Continued									
Blank (B2B1631-BLK1), Continued				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Manganese, dissolved	< 0.00020	0.00020 mg/L							
Molybdenum, dissolved	< 0.00010	0.00010 mg/L							
Nickel, dissolved	< 0.00040	0.00040 mg/L							
Phosphorus, dissolved	< 0.050	0.050 mg/L							
Potassium, dissolved	< 0.10	0.10 mg/L							
Selenium, dissolved	< 0.00050	0.00050 mg/L							
Silicon, dissolved	< 1.0	1.0 mg/L							
Silver, dissolved	< 0.000050	0.000050 mg/L							
Sodium, dissolved	< 0.10	0.10 mg/L							
Strontium, dissolved	< 0.0010	0.0010 mg/L							
Sulfur, dissolved	< 3.0	3.0 mg/L							
Tellurium, dissolved	< 0.00050	0.00050 mg/L							
Thallium, dissolved	< 0.000020	0.000020 mg/L							
Thorium, dissolved	< 0.00010	0.00010 mg/L							
Tin, dissolved	< 0.00020	0.00020 mg/L							
Titanium, dissolved	< 0.0050	0.0050 mg/L							
Tungsten, dissolved	< 0.0010	0.0010 mg/L							
Uranium, dissolved	< 0.000020	0.000020 mg/L							
Vanadium, dissolved	< 0.0010	0.0010 mg/L							
Zinc, dissolved	< 0.0040	0.0040 mg/L							
Zirconium, dissolved	< 0.00010	0.00010 mg/L							
LCS (B2B1631-BS1)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Aluminum, dissolved	0.0239	0.0050 mg/L	0.0200		120	80-120			
Antimony, dissolved	0.0206	0.00020 mg/L	0.0200		103	80-120			
Arsenic, dissolved	0.0194	0.00050 mg/L	0.0200		97	80-120			
Barium, dissolved	0.0192	0.0050 mg/L	0.0200		96	80-120			
Beryllium, dissolved	0.0194	0.00010 mg/L	0.0200		97	80-120			
Bismuth, dissolved	0.0174	0.00010 mg/L	0.0200		87	80-120			
Boron, dissolved	< 0.0500	0.0500 mg/L	0.0200		106	80-120			
Cadmium, dissolved	0.0186	0.000010 mg/L	0.0200		93	80-120			
Calcium, dissolved, dissolved	2.07	0.20 mg/L	2.00		103	80-120			
Chromium, dissolved	0.0194	0.00050 mg/L	0.0200		97	80-120			
Cobalt, dissolved	0.0197	0.00010 mg/L	0.0200		98	80-120			
Copper, dissolved	0.0189	0.00040 mg/L	0.0200		95	80-120			
Iron, dissolved	1.92	0.010 mg/L	2.00		96	80-120			
Lead, dissolved	0.0181	0.00020 mg/L	0.0200		91	80-120			
Lithium, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Magnesium, dissolved, dissolved	1.80	0.010 mg/L	2.00		90	80-120			
Manganese, dissolved	0.0182	0.00020 mg/L	0.0200		91	80-120			
Molybdenum, dissolved	0.0191	0.00010 mg/L	0.0200		95	80-120			
Nickel, dissolved	0.0195	0.00040 mg/L	0.0200		98	80-120			
Phosphorus, dissolved	2.17	0.050 mg/L	2.00		109	80-120			
Potassium, dissolved	1.75	0.10 mg/L	2.00		88	80-120			
Selenium, dissolved	0.0198	0.00050 mg/L	0.0200		99	80-120			
Silicon, dissolved	2.4	1.0 mg/L	2.00		119	80-120			
Silver, dissolved	0.0187	0.000050 mg/L	0.0200		94	80-120			
Sodium, dissolved	1.78	0.10 mg/L	2.00		89	80-120			
Strontium, dissolved	0.0206	0.0010 mg/L	0.0200		103	80-120			
Sulfur, dissolved	5.9	3.0 mg/L	5.00		118	80-120			
Tellurium, dissolved	0.0203	0.00050 mg/L	0.0200		101	80-120			
Thallium, dissolved	0.0174	0.000020 mg/L	0.0200		87	80-120			
Thorium, dissolved	0.0169	0.00010 mg/L	0.0200		84	80-120			
Tin, dissolved	0.0200	0.00020 mg/L	0.0200		100	80-120			
Titanium, dissolved	0.0219	0.0050 mg/L	0.0200		109	80-120			
Tungsten, dissolved	0.0194	0.0010 mg/L	0.0200		97	80-120			

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REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B2B1631, Continued

LCS (B2B1631-BS1), Continued			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Uranium, dissolved	0.0171	0.000020 mg/L	0.0200		85	80-120			
Vanadium, dissolved	0.0202	0.0010 mg/L	0.0200		101	80-120			
Zinc, dissolved	0.0198	0.0040 mg/L	0.0200		99	80-120			
Zirconium, dissolved	0.0200	0.00010 mg/L	0.0200		100	80-120			

Reference (B2B1631-SRM1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Aluminum, dissolved	0.217	0.0050 mg/L	0.198		109	70-130			
Antimony, dissolved	0.0250	0.00020 mg/L	0.0230		109	70-130			
Arsenic, dissolved	0.0212	0.00050 mg/L	0.0200		106	70-130			
Barium, dissolved	0.0157	0.0050 mg/L	0.0161		97	70-130			
Beryllium, dissolved	0.00424	0.00010 mg/L	0.00384		111	70-130			
Boron, dissolved	0.202	0.0500 mg/L	0.191		106	70-130			
Cadmium, dissolved	0.00397	0.000010 mg/L	0.00404		98	70-130			
Calcium, dissolved, dissolved	1.19	0.20 mg/L	0.938		127	70-130			
Chromium, dissolved	0.0261	0.00050 mg/L	0.0256		102	70-130			
Cobalt, dissolved	0.0223	0.00010 mg/L	0.0214		104	70-130			
Copper, dissolved	0.0313	0.00040 mg/L	0.0322		97	70-130			
Iron, dissolved	0.057	0.010 mg/L	0.0580		99	70-130			
Lead, dissolved	0.00805	0.00020 mg/L	0.00796		101	70-130			
Lithium, dissolved	0.0108	0.00010 mg/L	0.0102		106	70-130			
Magnesium, dissolved, dissolved	0.110	0.010 mg/L	0.112		98	70-130			
Manganese, dissolved	0.0119	0.00020 mg/L	0.0120		99	70-130			
Molybdenum, dissolved	0.0441	0.00010 mg/L	0.0438		101	70-130			
Nickel, dissolved	0.0398	0.00040 mg/L	0.0394		101	70-130			
Potassium, dissolved	0.78	0.10 mg/L	0.820		96	70-130			
Selenium, dissolved	0.116	0.00050 mg/L	0.117		99	70-130			
Sodium, dissolved	0.48	0.10 mg/L	0.490		98	70-130			
Strontium, dissolved	0.252	0.0010 mg/L	0.276		91	70-130			
Thallium, dissolved	0.0115	0.000020 mg/L	0.0118		97	70-130			
Uranium, dissolved	0.00907	0.000020 mg/L	0.00970		94	70-130			
Vanadium, dissolved	0.0291	0.0010 mg/L	0.0274		106	70-130			
Zinc, dissolved	0.0860	0.0040 mg/L	0.0884		97	70-130			

Dissolved Metals, Batch B2B1832

Blank (B2B1832-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-17						
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Blank (B2B1832-BLK2)			Prepared: 2022-02-16, Analyzed: 2022-02-17						
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Blank (B2B1832-BLK3)			Prepared: 2022-02-16, Analyzed: 2022-02-17						
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Reference (B2B1832-SRM1)			Prepared: 2022-02-16, Analyzed: 2022-02-18						
Mercury, dissolved	0.000506	0.000010 mg/L	0.000500		101	0-200			
Reference (B2B1832-SRM2)			Prepared: 2022-02-16, Analyzed: 2022-02-18						
Mercury, dissolved	0.000521	0.000010 mg/L	0.000500		104	0-200			
Reference (B2B1832-SRM3)			Prepared: 2022-02-16, Analyzed: 2022-02-17						
Mercury, dissolved	0.000501	0.000010 mg/L	0.000500		100	0-200			

General Parameters, Batch B2B1550

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1771
REPORTED 2022-02-22 16:21

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B1550, Continued									
Blank (B2B1550-BLK1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B1550-BLK2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B1550-BLK3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B2B1550-BS1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Ammonia, Total (as N)	0.992	0.050 mg/L	1.00		99	90-115			
LCS (B2B1550-BS2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Ammonia, Total (as N)	1.04	0.050 mg/L	1.00		104	90-115			
LCS (B2B1550-BS3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Ammonia, Total (as N)	1.06	0.050 mg/L	1.00		106	90-115			
General Parameters, Batch B2B1653									
Blank (B2B1653-BLK1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1653-BLK2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1653-BLK3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B2B1653-BS1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	104	1.0 mg/L	100		104	80-120			
LCS (B2B1653-BS2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	108	1.0 mg/L	100		108	80-120			
LCS (B2B1653-BS3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	106	1.0 mg/L	100		106	80-120			
General Parameters, Batch B2B1753									
Blank (B2B1753-BLK1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B1753-BS1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Solids, Total Dissolved	224	15 mg/L	240		93	85-115			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1771
REPORTED 2022-02-22 16:21

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B2010									
Blank (B2B2010-BLK1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B2010-BS1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Solids, Total Dissolved	225	15 mg/L	240		94	85-115			
General Parameters, Batch B2B2172									
Blank (B2B2172-BLK1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
Blank (B2B2172-BLK2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B2B2172-BS1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	107	2.0 mg/L	100		107	80-120			
LCS (B2B2172-BS2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	107	2.0 mg/L	100		107	80-120			
Duplicate (B2B2172-DUP1)				Source: 22B1771-03		Prepared: 2022-02-21, Analyzed: 2022-02-21			
Acidity, Total (as CaCO ₃)	2.2	2.0 mg/L		2.2				20	

Page 1 of 1

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CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Jonathan Croston

PO NUMBER 669765

PROJECT 631848

PROJECT INFO Sunro Mine

WORK ORDER 22B1772

RECEIVED / TEMP 2022-02-12 13:00 / 9.8°C

REPORTED 2022-02-22 16:19

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

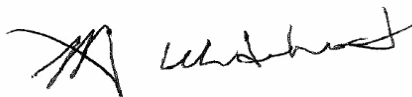
Work Order Comments:

Custody Seals Intact: YES

If you have any questions or concerns, please contact me at bwhitehead@caro.ca

Authorized By:

Brent Whitehead
Account Manager



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#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-17-02102022_NP (22B1772-01) Matrix: Water Sampled: 2022-02-11 10:20					
Anions					
Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.04	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0272	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	
Sulfate	1.4	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO3)	5.35	0.500	mg/L	N/A	
Nitrate (as N)	0.0272	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0861	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.50	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00022	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.0129	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.017	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.391	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00035	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00063	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.2	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.60	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0043	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-17-02102022_NP (22B1772-01) Matrix: Water Sampled: 2022-02-11 10:20, Continued					
Dissolved Metals, Continued					
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
General Parameters					
Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	4.2	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	4.2	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	2.53	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	
Total Metals					
Aluminum, total	0.105	0.0050	mg/L	2022-02-17	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.62	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00033	0.00010	mg/L	2022-02-17	
Copper, total	0.0171	0.00040	mg/L	2022-02-17	
Iron, total	0.033	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.444	0.010	mg/L	2022-02-17	
Manganese, total	0.00072	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	0.00080	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.12	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-17-02102022_NP (22B1772-01) Matrix: Water Sampled: 2022-02-11 10:20, Continued					
Total Metals, Continued					
Silicon, total	2.5	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, total	1.81	0.10	mg/L	2022-02-17	
Strontium, total	0.0048	0.0010	mg/L	2022-02-17	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-17	
Zinc, total	0.0056	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

SU_SW-11-02102022_NP (22B1772-02) | Matrix: Water | Sampled: 2022-02-11 11:10

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.02	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0526	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.024	0.010	mg/L	2022-02-15	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-15	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO3)	5.16	0.500	mg/L	N/A	
Nitrate (as N)	0.0526	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0835	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.43	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00021	0.00010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-11-02102022_NP (22B1772-02) | Matrix: Water | Sampled: 2022-02-11 11:10, Continued

Dissolved Metals, Continued

Copper, dissolved	0.00819	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.018	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.387	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00035	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00044	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.1	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.59	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0044	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	4.3	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	4.3	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	2.63	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	

Total Metals

Aluminum, total	0.115	0.0050	mg/L	2022-02-17	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-11-02102022_NP (22B1772-02) Matrix: Water Sampled: 2022-02-11 11:10, Continued					
<i>Total Metals, Continued</i>					
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.49	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00021	0.00010	mg/L	2022-02-17	
Copper, total	0.00954	0.00040	mg/L	2022-02-17	
Iron, total	0.033	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.431	0.010	mg/L	2022-02-17	
Manganese, total	0.00076	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	0.00047	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.12	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, total	2.3	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, total	1.76	0.10	mg/L	2022-02-17	
Strontium, total	0.0049	0.0010	mg/L	2022-02-17	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-17	
Zinc, total	< 0.0040	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

SU_SW-19-02102022_NP (22B1772-03) | Matrix: Water | Sampled: 2022-02-11 11:35

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.00	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-19-02102022_NP (22B1772-03) Matrix: Water Sampled: 2022-02-11 11:35, Continued					
Anions, Continued					
Nitrate+Nitrite (as N)	0.0247	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.024	0.010	mg/L	2022-02-15	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-15	HT1
Sulfate	1.3	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO3)	4.57	0.500	mg/L	N/A	
Nitrate (as N)	0.0247	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0902	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.23	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00021	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00884	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.018	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.360	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00033	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00045	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	2.1	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.53	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0038	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-19-02102022_NP (22B1772-03) Matrix: Water Sampled: 2022-02-11 11:35, Continued					
Dissolved Metals, Continued					
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
General Parameters					
Acidity, Total (as CaCO ₃)	2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.8	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.8	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	2.60	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	
Total Metals					
Aluminum, total	0.108	0.0050	mg/L	2022-02-17	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.33	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00024	0.00010	mg/L	2022-02-17	
Copper, total	0.0100	0.00040	mg/L	2022-02-17	
Iron, total	0.036	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.392	0.010	mg/L	2022-02-17	
Manganese, total	0.00074	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	0.00051	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.12	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, total	2.1	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-19-02102022_NP (22B1772-03) Matrix: Water Sampled: 2022-02-11 11:35, Continued					
Total Metals, Continued					
Sodium, total	1.64	0.10	mg/L	2022-02-17	
Strontium, total	0.0043	0.0010	mg/L	2022-02-17	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	0.0011	0.0010	mg/L	2022-02-17	
Zinc, total	< 0.0040	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

SU_SW-A-02102022_NP (22B1772-04) | Matrix: Water | Sampled: 2022-02-11 13:05

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.99	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0080	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.021	0.010	mg/L	2022-02-15	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-15	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO3)	4.07	0.500	mg/L	N/A	
Nitrate (as N)	< 0.0100	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0844	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.10	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00018	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00850	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.016	0.010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-A-02102022_NP (22B1772-04) | Matrix: Water | Sampled: 2022-02-11 13:05, Continued

Dissolved Metals, Continued

Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.319	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00026	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00043	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	1.7	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.41	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0035	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.5	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.5	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	2.60	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	

Total Metals

Aluminum, total	0.0966	0.0050	mg/L	2022-02-17	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-A-02102022_NP (22B1772-04) Matrix: Water Sampled: 2022-02-11 13:05, Continued					
<i>Total Metals, Continued</i>					
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.25	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00020	0.00010	mg/L	2022-02-17	
Copper, total	0.00978	0.00040	mg/L	2022-02-17	
Iron, total	0.021	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.372	0.010	mg/L	2022-02-17	
Manganese, total	0.00038	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	0.00048	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.12	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, total	2.0	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, total	1.62	0.10	mg/L	2022-02-17	
Strontium, total	0.0041	0.0010	mg/L	2022-02-17	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	0.0012	0.0010	mg/L	2022-02-17	
Zinc, total	< 0.0040	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

SU_SW-10-02102022_NP (22B1772-05) | Matrix: Water | Sampled: 2022-02-11 13:05

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.99	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0099	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.020	0.010	mg/L	2022-02-15	HT1

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-10-02102022_NP (22B1772-05) Matrix: Water Sampled: 2022-02-11 13:05, Continued					
Anions, Continued					
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-15	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO ₃)	4.20	0.500	mg/L	N/A	
Nitrate (as N)	< 0.0100	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0861	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.13	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00018	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00895	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.017	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.331	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00033	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	< 0.00040	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	1.9	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.48	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0036	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-10-02102022_NP (22B1772-05) Matrix: Water Sampled: 2022-02-11 13:05, Continued					
<i>Dissolved Metals, Continued</i>					
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	2.1	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.6	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.6	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	3.01	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	
<i>Total Metals</i>					
Aluminum, total	0.112	0.0050	mg/L	2022-02-17	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.32	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00022	0.00010	mg/L	2022-02-17	
Copper, total	0.0105	0.00040	mg/L	2022-02-17	
Iron, total	0.038	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.380	0.010	mg/L	2022-02-17	
Manganese, total	0.00086	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	0.00049	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.11	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, total	2.2	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, total	1.65	0.10	mg/L	2022-02-17	
Strontium, total	0.0042	0.0010	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-10-02102022_NP (22B1772-05) | Matrix: Water | Sampled: 2022-02-11 13:05, Continued

Total Metals, Continued

Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-17	
Zinc, total	0.0056	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

SU_SW-21-02102022_NP (22B1772-06) | Matrix: Water | Sampled: 2022-02-11 14:00

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.97	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0115	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.069	0.010	mg/L	2022-02-15	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-15	HT1
Sulfate	1.7	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO3)	4.27	0.500	mg/L	N/A	
Nitrate (as N)	0.0115	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0865	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.15	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00015	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00626	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.017	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-21-02102022_NP (22B1772-06) | Matrix: Water | Sampled: 2022-02-11 14:00, Continued

Dissolved Metals, Continued

Magnesium, dissolved	0.341	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00033	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	< 0.00040	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	1.8	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.48	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0036	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	2.3	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	2.6	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	2.6	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	2.64	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	

Total Metals

Aluminum, total	0.113	0.0050	mg/L	2022-02-17	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-21-02102022_NP (22B1772-06) Matrix: Water Sampled: 2022-02-11 14:00, Continued					
<i>Total Metals, Continued</i>					
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.32	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00019	0.00010	mg/L	2022-02-17	
Copper, total	0.00733	0.00040	mg/L	2022-02-17	
Iron, total	0.038	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.388	0.010	mg/L	2022-02-17	
Manganese, total	0.00102	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	< 0.00040	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.11	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, total	2.1	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, total	1.67	0.10	mg/L	2022-02-17	
Strontium, total	0.0042	0.0010	mg/L	2022-02-17	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	0.0010	0.0010	mg/L	2022-02-17	
Zinc, total	0.0050	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

SU_SW-22-02102022_NP (22B1772-07) | Matrix: Water | Sampled: 2022-02-11 15:20

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	1.96	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0151	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.019	0.010	mg/L	2022-02-15	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-15	HT1
Sulfate	1.2	1.0	mg/L	2022-02-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-22-02102022_NP (22B1772-07) Matrix: Water Sampled: 2022-02-11 15:20, Continued					
Calculated Parameters					
Hardness, Total (as CaCO ₃)	4.18	0.500	mg/L	N/A	
Nitrate (as N)	0.0151	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0803	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-16	
Calcium, dissolved	1.14	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.00017	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.00814	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.017	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	0.322	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00031	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00047	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	1.8	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	1.42	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0032	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-22-02102022_NP (22B1772-07) | Matrix: Water | Sampled: 2022-02-11 15:20, Continued

General Parameters, Continued

Acidity, Total (as CaCO ₃)	2.1	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.5	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	3.5	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Carbon, Dissolved Organic	2.75	0.50	mg/L	2022-02-16	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-18	
Solids, Total Suspended	< 2.0	2.0	mg/L	2022-02-15	
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-15	

Total Metals

Aluminum, total	0.105	0.0050	mg/L	2022-02-17	
Antimony, total	0.00021	0.00020	mg/L	2022-02-17	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, total	1.26	0.20	mg/L	2022-02-17	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, total	0.00019	0.00010	mg/L	2022-02-17	
Copper, total	0.00882	0.00040	mg/L	2022-02-17	
Iron, total	0.030	0.010	mg/L	2022-02-17	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Magnesium, total	0.371	0.010	mg/L	2022-02-17	
Manganese, total	0.00062	0.00020	mg/L	2022-02-17	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, total	0.00041	0.00040	mg/L	2022-02-17	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-17	
Potassium, total	0.11	0.10	mg/L	2022-02-17	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, total	2.1	1.0	mg/L	2022-02-17	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, total	1.60	0.10	mg/L	2022-02-17	
Strontium, total	0.0041	0.0010	mg/L	2022-02-17	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

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2022-02-22 16:19

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-22-02102022_NP (22B1772-07) Matrix: Water Sampled: 2022-02-11 15:20, Continued					
<i>Total Metals, Continued</i>					
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, total	0.0013	0.0010	mg/L	2022-02-17	
Zinc, total	< 0.0040	0.0040	mg/L	2022-02-17	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-17	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

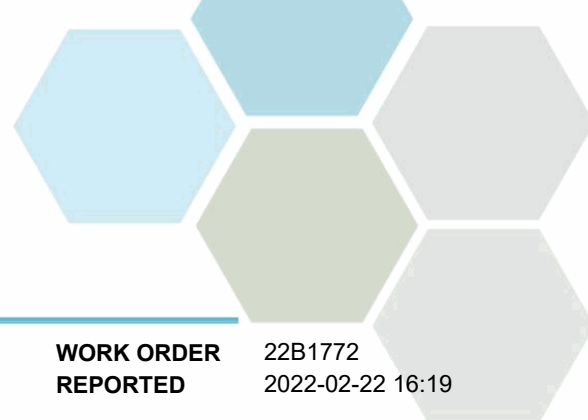
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2017)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2017)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2017)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2017)	Ion Chromatography	✓	Kelowna
Carbon, Dissolved Organic in Water	SM 5310 B (2017)	Combustion, Infrared CO ₂ Detection	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2017)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Mercury, total in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Nitrate+Nitrite in Water	SM 4500-NO ₃ - F (2017)	Automated Colorimetry (Cadmium Reduction)	✓	Kelowna
Nitrite in Water	SM 4500-NO ₂ B (2017)	Colorimetry	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna
Solids, Total Suspended in Water	SM 2540 D* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna
Sulfide, Total in Water	SM 4500-S ₂ D* (2017)	Colorimetry (Methylene Blue)	✓	Edmonton
Total Metals in Water	EPA 200.2 / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association



APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO SNC-Lavalin Inc. (Nelson)
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General Comments:

The results in this report apply to the samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Samples will be disposed of 30 days after the test report has been issued or once samples expire, whichever comes first. Longer hold is possible if agreed to in writing.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: bwhitehead@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1486									
Blank (B2B1486-BLK1)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1486-BLK2)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	< 0.0050	0.0050 mg/L							
LCS (B2B1486-BS1)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	0.0489	0.0050 mg/L	0.0500		98	90-110			
LCS (B2B1486-BS2)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	0.0489	0.0050 mg/L	0.0500		98	90-110			
Anions, Batch B2B1570									
Blank (B2B1570-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK3)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK4)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
LCS (B2B1570-BS1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.527	0.0050 mg/L	0.500		105	91-108			
LCS (B2B1570-BS2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.520	0.0050 mg/L	0.500		104	91-108			
LCS (B2B1570-BS3)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.521	0.0050 mg/L	0.500		104	91-108			
LCS (B2B1570-BS4)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.525	0.0050 mg/L	0.500		105	91-108			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Anions, Batch B2B1581

Blank (B2B1581-BLK1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							

LCS (B2B1581-BS1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Bromide	4.05	0.10 mg/L	4.00		101	85-115			
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	3.90	0.10 mg/L	4.00		98	88-108			
Nitrate (as N)	4.05	0.010 mg/L	4.00		101	90-110			
Nitrite (as N)	2.03	0.010 mg/L	2.00		101	85-115			
Sulfate	16.1	1.0 mg/L	16.0		100	90-110			

Dissolved Metals, Batch B2B1759

Blank (B2B1759-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Aluminum, dissolved	< 0.0050	0.0050 mg/L							
Antimony, dissolved	< 0.00020	0.00020 mg/L							
Arsenic, dissolved	< 0.00050	0.00050 mg/L							
Barium, dissolved	< 0.0050	0.0050 mg/L							
Beryllium, dissolved	< 0.00010	0.00010 mg/L							
Bismuth, dissolved	< 0.00010	0.00010 mg/L							
Boron, dissolved	< 0.0500	0.0500 mg/L							
Cadmium, dissolved	< 0.000010	0.000010 mg/L							
Calcium, dissolved, dissolved	< 0.20	0.20 mg/L							
Chromium, dissolved	< 0.00050	0.00050 mg/L							
Cobalt, dissolved	< 0.00010	0.00010 mg/L							
Copper, dissolved	< 0.00040	0.00040 mg/L							
Iron, dissolved	< 0.010	0.010 mg/L							
Lead, dissolved	< 0.00020	0.00020 mg/L							
Lithium, dissolved	< 0.00010	0.00010 mg/L							
Magnesium, dissolved, dissolved	< 0.010	0.010 mg/L							
Manganese, dissolved	< 0.00020	0.00020 mg/L							
Molybdenum, dissolved	< 0.00010	0.00010 mg/L							
Nickel, dissolved	< 0.00040	0.00040 mg/L							
Phosphorus, dissolved	< 0.050	0.050 mg/L							
Potassium, dissolved	< 0.10	0.10 mg/L							
Selenium, dissolved	< 0.00050	0.00050 mg/L							
Silicon, dissolved	< 1.0	1.0 mg/L							
Silver, dissolved	< 0.000050	0.000050 mg/L							
Sodium, dissolved	< 0.10	0.10 mg/L							
Strontium, dissolved	< 0.0010	0.0010 mg/L							
Sulfur, dissolved	< 3.0	3.0 mg/L							
Tellurium, dissolved	< 0.00050	0.00050 mg/L							
Thallium, dissolved	< 0.000020	0.000020 mg/L							
Thorium, dissolved	< 0.00010	0.00010 mg/L							
Tin, dissolved	< 0.00020	0.00020 mg/L							
Titanium, dissolved	< 0.0050	0.0050 mg/L							
Tungsten, dissolved	< 0.0010	0.0010 mg/L							
Uranium, dissolved	< 0.000020	0.000020 mg/L							
Vanadium, dissolved	< 0.0010	0.0010 mg/L							
Zinc, dissolved	< 0.0040	0.0040 mg/L							
Zirconium, dissolved	< 0.00010	0.00010 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 22B1772
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B1759, Continued									
LCS (B2B1759-BS1)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Aluminum, dissolved	0.0229	0.0050 mg/L	0.0200		115	80-120			
Antimony, dissolved	0.0205	0.00020 mg/L	0.0200		102	80-120			
Arsenic, dissolved	0.0193	0.00050 mg/L	0.0200		97	80-120			
Barium, dissolved	0.0189	0.0050 mg/L	0.0200		95	80-120			
Beryllium, dissolved	0.0192	0.00010 mg/L	0.0200		96	80-120			
Bismuth, dissolved	0.0178	0.00010 mg/L	0.0200		89	80-120			
Boron, dissolved	< 0.0500	0.0500 mg/L	0.0200		110	80-120			
Cadmium, dissolved	0.0189	0.000010 mg/L	0.0200		94	80-120			
Calcium, dissolved, dissolved	2.09	0.20 mg/L	2.00		104	80-120			
Chromium, dissolved	0.0192	0.00050 mg/L	0.0200		96	80-120			
Cobalt, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Copper, dissolved	0.0192	0.00040 mg/L	0.0200		96	80-120			
Iron, dissolved	1.89	0.010 mg/L	2.00		95	80-120			
Lead, dissolved	0.0186	0.00020 mg/L	0.0200		93	80-120			
Lithium, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Magnesium, dissolved, dissolved	1.88	0.010 mg/L	2.00		94	80-120			
Manganese, dissolved	0.0190	0.00020 mg/L	0.0200		95	80-120			
Molybdenum, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Nickel, dissolved	0.0201	0.00040 mg/L	0.0200		101	80-120			
Phosphorus, dissolved	2.01	0.050 mg/L	2.00		101	80-120			
Potassium, dissolved	1.77	0.10 mg/L	2.00		89	80-120			
Selenium, dissolved	0.0184	0.00050 mg/L	0.0200		92	80-120			
Silicon, dissolved	2.3	1.0 mg/L	2.00		115	80-120			
Silver, dissolved	0.0192	0.000050 mg/L	0.0200		96	80-120			
Sodium, dissolved	1.83	0.10 mg/L	2.00		92	80-120			
Strontium, dissolved	0.0205	0.0010 mg/L	0.0200		103	80-120			
Sulfur, dissolved	5.7	3.0 mg/L	5.00		115	80-120			
Tellurium, dissolved	0.0190	0.00050 mg/L	0.0200		95	80-120			
Thallium, dissolved	0.0179	0.000020 mg/L	0.0200		89	80-120			
Thorium, dissolved	0.0177	0.00010 mg/L	0.0200		89	80-120			
Tin, dissolved	0.0200	0.00020 mg/L	0.0200		100	80-120			
Titanium, dissolved	0.0224	0.0050 mg/L	0.0200		112	80-120			
Tungsten, dissolved	0.0196	0.0010 mg/L	0.0200		98	80-120			
Uranium, dissolved	0.0177	0.000020 mg/L	0.0200		88	80-120			
Vanadium, dissolved	0.0199	0.0010 mg/L	0.0200		100	80-120			
Zinc, dissolved	0.0190	0.0040 mg/L	0.0200		95	80-120			
Zirconium, dissolved	0.0204	0.00010 mg/L	0.0200		102	80-120			

Duplicate (B2B1759-DUP1)		Source: 22B1772-01		Prepared: 2022-02-16, Analyzed: 2022-02-16					
Aluminum, dissolved	0.0912	0.0050 mg/L	0.0861		6	20			
Antimony, dissolved	< 0.00020	0.00020 mg/L	< 0.00020			20			
Arsenic, dissolved	< 0.00050	0.00050 mg/L	< 0.00050			20			
Barium, dissolved	< 0.0050	0.0050 mg/L	< 0.0050			20			
Beryllium, dissolved	< 0.00010	0.00010 mg/L	< 0.00010			20			
Bismuth, dissolved	< 0.00010	0.00010 mg/L	< 0.00010			20			
Boron, dissolved	< 0.0500	0.0500 mg/L	< 0.0500			20			
Cadmium, dissolved	< 0.000010	0.000010 mg/L	< 0.000010			20			
Calcium, dissolved, dissolved	1.44	0.20 mg/L	1.50		4	20			
Chromium, dissolved	< 0.00050	0.00050 mg/L	< 0.00050			20			
Cobalt, dissolved	0.00030	0.00010 mg/L	0.00022			20			
Copper, dissolved	0.0141	0.00040 mg/L	0.0129		9	20			
Iron, dissolved	0.019	0.010 mg/L	0.017			20			
Lead, dissolved	< 0.00020	0.00020 mg/L	< 0.00020			20			
Lithium, dissolved	< 0.00010	0.00010 mg/L	< 0.00010			20			
Magnesium, dissolved, dissolved	0.402	0.010 mg/L	0.391		3	20			
Manganese, dissolved	0.00034	0.00020 mg/L	0.00035			20			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1772
REPORTED 2022-02-22 16:19

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B1759, Continued									
Duplicate (B2B1759-DUP1), Continued		Source: 22B1772-01		Prepared: 2022-02-16, Analyzed: 2022-02-16					
Molybdenum, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Nickel, dissolved	0.00089	0.00040 mg/L		0.00063				20	
Phosphorus, dissolved	< 0.050	0.050 mg/L		< 0.050				20	
Potassium, dissolved	0.10	0.10 mg/L		0.10				20	
Selenium, dissolved	< 0.00050	0.00050 mg/L		< 0.00050				20	
Silicon, dissolved	2.1	1.0 mg/L		2.2				20	
Silver, dissolved	< 0.000050	0.000050 mg/L		< 0.000050				20	
Sodium, dissolved	1.65	0.10 mg/L		1.60			3	20	
Strontium, dissolved	0.0048	0.0010 mg/L		0.0043				20	
Sulfur, dissolved	< 3.0	3.0 mg/L		< 3.0				20	
Tellurium, dissolved	< 0.00050	0.00050 mg/L		< 0.00050				20	
Thallium, dissolved	< 0.000020	0.000020 mg/L		< 0.000020				20	
Thorium, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Tin, dissolved	< 0.00020	0.00020 mg/L		< 0.00020				20	
Titanium, dissolved	< 0.0050	0.0050 mg/L		< 0.0050				20	
Tungsten, dissolved	< 0.0010	0.0010 mg/L		< 0.0010				20	
Uranium, dissolved	< 0.000020	0.000020 mg/L		< 0.000020				20	
Vanadium, dissolved	< 0.0010	0.0010 mg/L		< 0.0010				20	
Zinc, dissolved	< 0.0040	0.0040 mg/L		< 0.0040				20	
Zirconium, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	

Reference (B2B1759-SRM1)		Prepared: 2022-02-16, Analyzed: 2022-02-16							
Aluminum, dissolved	0.229	0.0050 mg/L	0.198		115	70-130			
Antimony, dissolved	0.0253	0.00020 mg/L	0.0230		110	70-130			
Arsenic, dissolved	0.0206	0.00050 mg/L	0.0200		103	70-130			
Barium, dissolved	0.0164	0.0050 mg/L	0.0161		102	70-130			
Beryllium, dissolved	0.00422	0.00010 mg/L	0.00384		110	70-130			
Boron, dissolved	0.180	0.0500 mg/L	0.191		94	70-130			
Cadmium, dissolved	0.00401	0.000010 mg/L	0.00404		99	70-130			
Calcium, dissolved, dissolved	1.19	0.20 mg/L	0.938		126	70-130			
Chromium, dissolved	0.0262	0.00050 mg/L	0.0256		102	70-130			
Cobalt, dissolved	0.0225	0.00010 mg/L	0.0214		105	70-130			
Copper, dissolved	0.0322	0.00040 mg/L	0.0322		100	70-130			
Iron, dissolved	0.058	0.010 mg/L	0.0580		101	70-130			
Lead, dissolved	0.00815	0.00020 mg/L	0.00796		102	70-130			
Lithium, dissolved	0.0108	0.00010 mg/L	0.0102		107	70-130			
Magnesium, dissolved, dissolved	0.111	0.010 mg/L	0.112		99	70-130			
Manganese, dissolved	0.0121	0.00020 mg/L	0.0120		101	70-130			
Molybdenum, dissolved	0.0459	0.00010 mg/L	0.0438		105	70-130			
Nickel, dissolved	0.0407	0.00040 mg/L	0.0394		103	70-130			
Potassium, dissolved	0.77	0.10 mg/L	0.820		94	70-130			
Selenium, dissolved	0.118	0.00050 mg/L	0.117		101	70-130			
Sodium, dissolved	0.49	0.10 mg/L	0.490		100	70-130			
Strontium, dissolved	0.257	0.0010 mg/L	0.276		93	70-130			
Thallium, dissolved	0.0117	0.000020 mg/L	0.0118		99	70-130			
Uranium, dissolved	0.00926	0.000020 mg/L	0.00970		95	70-130			
Vanadium, dissolved	0.0287	0.0010 mg/L	0.0274		105	70-130			
Zinc, dissolved	0.0868	0.0040 mg/L	0.0884		98	70-130			

Dissolved Metals, Batch B2B1832

Blank (B2B1832-BLK1)		Prepared: 2022-02-16, Analyzed: 2022-02-17							
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Blank (B2B1832-BLK2)		Prepared: 2022-02-16, Analyzed: 2022-02-17							
Mercury, dissolved	< 0.000010	0.000010 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B1832, Continued									
Blank (B2B1832-BLK3)			Prepared: 2022-02-16, Analyzed: 2022-02-17						
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Reference (B2B1832-SRM1)			Prepared: 2022-02-16, Analyzed: 2022-02-18						
Mercury, dissolved	0.000506	0.000010 mg/L	0.000500		101	0-200			
Reference (B2B1832-SRM2)			Prepared: 2022-02-16, Analyzed: 2022-02-18						
Mercury, dissolved	0.000521	0.000010 mg/L	0.000500		104	0-200			
Reference (B2B1832-SRM3)			Prepared: 2022-02-16, Analyzed: 2022-02-17						
Mercury, dissolved	0.000501	0.000010 mg/L	0.000500		100	0-200			
General Parameters, Batch B2B1390									
Blank (B2B1390-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B2B1390-BLK2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
LCS (B2B1390-BS1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Carbon, Dissolved Organic	9.66	0.50 mg/L	10.0		97	78-116			
LCS (B2B1390-BS2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Carbon, Dissolved Organic	9.31	0.50 mg/L	10.0		93	78-116			
General Parameters, Batch B2B1590									
Blank (B2B1590-BLK1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Sulfide, Total	< 0.020	0.020 mg/L							
LCS (B2B1590-BS1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Sulfide, Total	0.494	0.020 mg/L	0.498		99	80-120			
Matrix Spike (B2B1590-MS1)			Source: 22B1772-07		Prepared: 2022-02-15, Analyzed: 2022-02-15				
Sulfide, Total	0.511	0.020 mg/L	0.498	< 0.020	103	70-130			
General Parameters, Batch B2B1591									
Blank (B2B1591-BLK1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Solids, Total Suspended	< 2.0	2.0 mg/L							
LCS (B2B1591-BS1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Solids, Total Suspended	96.0	10.0 mg/L	100		96	85-115			
General Parameters, Batch B2B1653									
Blank (B2B1653-BLK1)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1653-BLK2)			Prepared: 2022-02-15, Analyzed: 2022-02-15						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B1653, Continued									
Blank (B2B1653-BLK2), Continued				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1653-BLK3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B2B1653-BS1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	104	1.0 mg/L	100		104	80-120			
LCS (B2B1653-BS2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	108	1.0 mg/L	100		108	80-120			
LCS (B2B1653-BS3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	106	1.0 mg/L	100		106	80-120			
General Parameters, Batch B2B1837									
Blank (B2B1837-BLK1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B1837-BLK2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B1837-BLK3)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B2B1837-BS1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	1.08	0.050 mg/L	1.00		108	90-115			
LCS (B2B1837-BS2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	1.06	0.050 mg/L	1.00		106	90-115			
LCS (B2B1837-BS3)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	1.04	0.050 mg/L	1.00		104	90-115			
Duplicate (B2B1837-DUP1)				Source: 22B1772-04		Prepared: 2022-02-17, Analyzed: 2022-02-17			
Ammonia, Total (as N)	< 0.050	0.050 mg/L		< 0.050				15	
Matrix Spike (B2B1837-MS1)				Source: 22B1772-04		Prepared: 2022-02-17, Analyzed: 2022-02-17			
Ammonia, Total (as N)	0.286	0.050 mg/L	0.250	< 0.050	107	75-125			
General Parameters, Batch B2B2010									
Blank (B2B2010-BLK1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B2010-BS1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Solids, Total Dissolved	225	15 mg/L	240		94	85-115			
General Parameters, Batch B2B2172									

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B2172, Continued									
Blank (B2B2172-BLK1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
Blank (B2B2172-BLK2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B2B2172-BS1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	107	2.0 mg/L	100		107	80-120			
LCS (B2B2172-BS2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	107	2.0 mg/L	100		107	80-120			
Total Metals, Batch B2B1721									
Blank (B2B1721-BLK1)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Aluminum, total	< 0.0050	0.0050 mg/L							
Antimony, total	< 0.00020	0.00020 mg/L							
Arsenic, total	< 0.00050	0.00050 mg/L							
Barium, total	< 0.0050	0.0050 mg/L							
Beryllium, total	< 0.00010	0.00010 mg/L							
Bismuth, total	< 0.00010	0.00010 mg/L							
Boron, total	< 0.0500	0.0500 mg/L							
Cadmium, total	< 0.000010	0.000010 mg/L							
Calcium, total	< 0.20	0.20 mg/L							
Chromium, total	< 0.00050	0.00050 mg/L							
Cobalt, total	< 0.00010	0.00010 mg/L							
Copper, total	< 0.00040	0.00040 mg/L							
Iron, total	< 0.010	0.010 mg/L							
Lead, total	< 0.00020	0.00020 mg/L							
Lithium, total	< 0.00010	0.00010 mg/L							
Magnesium, total	< 0.010	0.010 mg/L							
Manganese, total	< 0.00020	0.00020 mg/L							
Molybdenum, total	< 0.00010	0.00010 mg/L							
Nickel, total	< 0.00040	0.00040 mg/L							
Phosphorus, total	< 0.050	0.050 mg/L							
Potassium, total	< 0.10	0.10 mg/L							
Selenium, total	< 0.00050	0.00050 mg/L							
Silicon, total	< 1.0	1.0 mg/L							
Silver, total	< 0.000050	0.000050 mg/L							
Sodium, total	< 0.10	0.10 mg/L							
Strontium, total	< 0.0010	0.0010 mg/L							
Sulfur, total	< 3.0	3.0 mg/L							
Tellurium, total	< 0.00050	0.00050 mg/L							
Thallium, total	< 0.000020	0.000020 mg/L							
Thorium, total	< 0.00010	0.00010 mg/L							
Tin, total	< 0.00020	0.00020 mg/L							
Titanium, total	< 0.0050	0.0050 mg/L							
Tungsten, total	< 0.0010	0.0010 mg/L							
Uranium, total	< 0.000020	0.000020 mg/L							
Vanadium, total	< 0.0010	0.0010 mg/L							
Zinc, total	< 0.0040	0.0040 mg/L							
Zirconium, total	< 0.00010	0.00010 mg/L							
LCS (B2B1721-BS1)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Aluminum, total	0.0227	0.0050 mg/L	0.0200		113	80-120			
Antimony, total	0.0208	0.00020 mg/L	0.0200		104	80-120			
Arsenic, total	0.0189	0.00050 mg/L	0.0200		95	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B2B1721, Continued									
LCS (B2B1721-BS1), Continued					Prepared: 2022-02-16, Analyzed: 2022-02-17				
Barium, total	0.0202	0.0050 mg/L	0.0200		101	80-120			
Beryllium, total	0.0197	0.00010 mg/L	0.0200		99	80-120			
Bismuth, total	0.0187	0.00010 mg/L	0.0200		93	80-120			
Boron, total	< 0.0500	0.0500 mg/L	0.0200		103	80-120			
Cadmium, total	0.0185	0.000010 mg/L	0.0200		93	80-120			
Calcium, total	2.07	0.20 mg/L	2.00		103	80-120			
Chromium, total	0.0191	0.00050 mg/L	0.0200		95	80-120			
Cobalt, total	0.0191	0.00010 mg/L	0.0200		96	80-120			
Copper, total	0.0188	0.00040 mg/L	0.0200		94	80-120			
Iron, total	1.85	0.010 mg/L	2.00		92	80-120			
Lead, total	0.0195	0.00020 mg/L	0.0200		98	80-120			
Lithium, total	0.0203	0.00010 mg/L	0.0200		101	80-120			
Magnesium, total	1.83	0.010 mg/L	2.00		92	80-120			
Manganese, total	0.0184	0.00020 mg/L	0.0200		92	80-120			
Molybdenum, total	0.0189	0.00010 mg/L	0.0200		95	80-120			
Nickel, total	0.0189	0.00040 mg/L	0.0200		94	80-120			
Phosphorus, total	1.93	0.050 mg/L	2.00		96	80-120			
Potassium, total	1.74	0.10 mg/L	2.00		87	80-120			
Selenium, total	0.0186	0.00050 mg/L	0.0200		93	80-120			
Silicon, total	2.2	1.0 mg/L	2.00		112	80-120			
Silver, total	0.0189	0.000050 mg/L	0.0200		94	80-120			
Sodium, total	1.80	0.10 mg/L	2.00		90	80-120			
Strontium, total	0.0207	0.0010 mg/L	0.0200		103	80-120			
Sulfur, total	5.1	3.0 mg/L	5.00		102	80-120			
Tellurium, total	0.0193	0.00050 mg/L	0.0200		96	80-120			
Thallium, total	0.0188	0.000020 mg/L	0.0200		94	80-120			
Thorium, total	0.0182	0.00010 mg/L	0.0200		91	80-120			
Tin, total	0.0199	0.00020 mg/L	0.0200		100	80-120			
Titanium, total	0.0197	0.0050 mg/L	0.0200		98	80-120			
Tungsten, total	0.0192	0.0010 mg/L	0.0200		96	80-120			
Uranium, total	0.0183	0.000020 mg/L	0.0200		91	80-120			
Vanadium, total	0.0194	0.0010 mg/L	0.0200		97	80-120			
Zinc, total	0.0190	0.0040 mg/L	0.0200		95	80-120			
Zirconium, total	0.0198	0.00010 mg/L	0.0200		99	80-120			

Reference (B2B1721-SRM1)					Prepared: 2022-02-16, Analyzed: 2022-02-17				
Aluminum, total	0.202	0.0050 mg/L	0.198		102	70-130			
Antimony, total	0.0248	0.00020 mg/L	0.0230		108	70-130			
Arsenic, total	0.0202	0.00050 mg/L	0.0200		101	70-130			
Barium, total	0.0151	0.0050 mg/L	0.0161		94	70-130			
Beryllium, total	0.00413	0.00010 mg/L	0.00384		108	70-130			
Boron, total	0.176	0.0500 mg/L	0.191		92	70-130			
Cadmium, total	0.00379	0.000010 mg/L	0.00404		94	70-130			
Calcium, total	1.01	0.20 mg/L	0.938		107	70-130			
Chromium, total	0.0248	0.00050 mg/L	0.0256		97	70-130			
Cobalt, total	0.0213	0.00010 mg/L	0.0214		100	70-130			
Copper, total	0.0309	0.00040 mg/L	0.0322		96	70-130			
Iron, total	0.055	0.010 mg/L	0.0580		95	70-130			
Lead, total	0.00799	0.00020 mg/L	0.00796		100	70-130			
Lithium, total	0.0106	0.00010 mg/L	0.0102		104	70-130			
Magnesium, total	0.108	0.010 mg/L	0.112		97	70-130			
Manganese, total	0.0113	0.00020 mg/L	0.0120		94	70-130			
Molybdenum, total	0.0435	0.00010 mg/L	0.0438		99	70-130			
Nickel, total	0.0390	0.00040 mg/L	0.0394		99	70-130			
Potassium, total	0.73	0.10 mg/L	0.820		89	70-130			
Selenium, total	0.117	0.00050 mg/L	0.117		100	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1772
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B2B1721, Continued									
Reference (B2B1721-SRM1), Continued				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Sodium, total	0.45	0.10 mg/L	0.490		93	70-130			
Strontium, total	0.246	0.0010 mg/L	0.276		89	70-130			
Thallium, total	0.0116	0.000020 mg/L	0.0118		98	70-130			
Uranium, total	0.00913	0.000020 mg/L	0.00970		94	70-130			
Vanadium, total	0.0276	0.0010 mg/L	0.0274		101	70-130			
Zinc, total	0.0842	0.0040 mg/L	0.0884		95	70-130			

Total Metals, Batch B2B1833

Blank (B2B1833-BLK1)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, total	< 0.000010	0.000010 mg/L							
Blank (B2B1833-BLK2)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, total	< 0.000010	0.000010 mg/L							
Reference (B2B1833-SRM1)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, total	0.000467	0.000010 mg/L	0.000500		93	0-200			
Reference (B2B1833-SRM2)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, total	0.000421	0.000010 mg/L	0.000500		84	0-200			

Teck

COC ID:



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PROJECT/CLIENT	
Facility Name / Job#	Sunro Mine
Project Manager	Michelle Unger
Email	michelle.unger@teck.com
Address	601 Knighton Rd
City	Kimberly
Province	BC
Postal Code	V1A 1C7
Country	Canada
Phone Number	250-427-8422

RNAROUND TIME:		Regular		RUSH:	
LABORATORY			OTHER INFO		
Lab Name	Caro	Report Format / Distribution	Excel	PDF	EDD
Lab Contact	Brent Whitehead	Email 1:	michelle.unger@teck.com	X	X
Email	bwhitehead@caro.ca	Email 2:	craig.pinnell@snclavalin.com	X	X
Address	110 - 4011 Viking Way	Email 3:	jonathan.croston@snclavalin.com	X	X
		Email 4:	snclavalin@snclavalin.com	X	X
		Email 5:	ENV_West_BC_LabData@snclavalin.com	X	X
City	Richmond	Province	BC		
Postal Code	V6B2K9	Country	CA		
Phone Number	604-279-1499	PO number	649765		

SAMPLE DETAILS									ANALYSIS REQUESTED										Filtered - F: Field, L: Lab, FL: Field & Lab, N: None			
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	PHC	Yes													
								PRIORITY	Yes	Yes										Yes	Yes	Yes
								ANALYSIS	Total Metals + Hg in Water	Dissolved Metals + Hg in Water	TDS and TSS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	BTEX/VPH	LEPH/HEPH	Total and Speciated Alkalinity, Acidity	Sulfide	Ammonia	Dissolved Organic Carbon	HOLD		
✓ SU_SW-17-02102022_NP	SW17-17	SW	N	2022-02-10	10:20	G	9		X	X	X	X	X			X	X	X	X			
✓ SU_SW-11-02102022_NP	SW15-11	SW	N	2022-02-10	11:10	G	9		X	X	X	X	X			X	X	X	X			
✓ SU_SW-19-02102022_NP	SW17-19	SW	N	2022-02-10	11:35	G	9		X	X	X	X	X			X	X	X	X			
✓ SU_SW-A-02102022_NP		SW	N	2022-02-10	13:05	G	9		X	X	X	X	X			X	X	X	X			
✓ SU_SW-10-02102022_NP	SW15-10	SW	N	2022-02-10	13:05	G	9		X	X	X	X	X			X	X	X	X			
✓ SU_SW-21-02102022_NP	SW17-21	SW	N	2022-02-10	14:00	G	9		X	X	X	X	X			X	X	X	X			
✓ SU_SW-22-02102022_NP	SW18-22	SW	N	2022-02-10	15:20	G	9		X	X	X	X	X			X	X	X	X			

ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS		RELINQUISHED BY/AFFILIATION		DATE/TIME		ACCEPTED BY/AFFILIATION		DATE/TIME	
Please note custody seal condition. Dissolved metals and DOC field filtered.		CPakstas SNC-Lavalin Inc		22/02/11 1000					
Invoice to: SNC-Lavalin Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3									
Send to: payables@snclavalin.com Project number: 631848 Billing PO: 683703									
NB OF BOTTLES RETURNED/DESCRIPTION		Sampler's Name		Mobile #		Date/Time			
Regular (default) X		Josh Baran		250-418-1562					
Priority (2-3 business days) - 50% surcharge									
Emergency (1 Business Day) - 100% surcharge		Sampler's Signature							
For Emergency <1 Day, ASAP or Weekend								February 10, 2022	

* LL container 3/4 Full - limited volume

20 Pmt
2/12 13:00
9.00

CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Jonathan Croston

PO NUMBER 669765

PROJECT 631848

PROJECT INFO Sunro Mine

WORK ORDER 22B1773

RECEIVED / TEMP 2022-02-12 13:00 / 9.8°C

REPORTED 2022-02-25 15:54

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

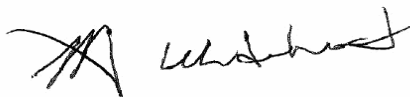
Work Order Comments:

Custody Seals Intact: YES

If you have any questions or concerns, please contact me at bwhitehead@caro.ca

Authorized By:

Brent Whitehead
Account Manager



1-888-311-8846 | www.caro.ca

#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-07-02092022_NP (22B1773-01) Matrix: Water Sampled: 2022-02-09 11:30					
Anions					
Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	9.89	0.10	mg/L	2022-02-15	
Fluoride	0.14	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.159	0.0050	mg/L	2022-02-16	
Nitrite (as N)	0.0077	0.0050	mg/L	2022-02-14	HT1
Sulfate	46.3	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO3)	55.0	0.500	mg/L	N/A	
Nitrate (as N)	0.151	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0212	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	0.000344	0.000010	mg/L	2022-02-16	
Calcium, dissolved	15.6	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.0463	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.590	0.00040	mg/L	2022-02-16	
Iron, dissolved	0.021	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	0.00029	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	3.90	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.00566	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.0789	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.36	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	9.0	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	7.87	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0439	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	16.8	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-07-02092022_NP (22B1773-01) | Matrix: Water | Sampled: 2022-02-09 11:30, Continued

Dissolved Metals, Continued

Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	0.0018	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	0.102	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO ₃)	4.3	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	25.1	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	25.1	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	0.095	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	132	15	mg/L	2022-02-17	HT1

SU_GW-05-02092022_NP (22B1773-02) | Matrix: Water | Sampled: 2022-02-09 12:00

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	5.06	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.152	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	40.1	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO ₃)	76.3	0.500	mg/L	N/A	
Nitrate (as N)	0.152	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0134	0.0050	mg/L	2022-02-16	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Barium, dissolved	0.0053	0.0050	mg/L	2022-02-16	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-16	
Cadmium, dissolved	0.000267	0.000010	mg/L	2022-02-16	
Calcium, dissolved	20.3	0.20	mg/L	2022-02-16	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Cobalt, dissolved	0.0205	0.00010	mg/L	2022-02-16	
Copper, dissolved	0.556	0.00040	mg/L	2022-02-16	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-05-02092022_NP (22B1773-02) | Matrix: Water | Sampled: 2022-02-09 12:00, Continued

Dissolved Metals, Continued

Iron, dissolved	< 0.010	0.010	mg/L	2022-02-16	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Lithium, dissolved	0.00065	0.00010	mg/L	2022-02-16	
Magnesium, dissolved	6.23	0.010	mg/L	2022-02-16	
Manganese, dissolved	0.527	0.00020	mg/L	2022-02-16	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Nickel, dissolved	0.00952	0.00040	mg/L	2022-02-16	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-16	
Potassium, dissolved	0.37	0.10	mg/L	2022-02-16	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Silicon, dissolved	13.5	1.0	mg/L	2022-02-16	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-16	
Sodium, dissolved	4.65	0.10	mg/L	2022-02-16	
Strontium, dissolved	0.0772	0.0010	mg/L	2022-02-16	
Sulfur, dissolved	13.2	3.0	mg/L	2022-02-16	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-16	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-16	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-16	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-16	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-16	
Zinc, dissolved	0.0510	0.0040	mg/L	2022-02-16	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-16	

General Parameters

Acidity, Total (as CaCO3)	9.0	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO3)	47.5	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO3)	47.5	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO3)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO3)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	139	15	mg/L	2022-02-17	HT1

SU_GW-09-02092022_NP (22B1773-03) | Matrix: Water | Sampled: 2022-02-09 12:40

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	12.0	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-09-02092022_NP (22B1773-03) Matrix: Water Sampled: 2022-02-09 12:40, Continued					
Anions, Continued					
Nitrate+Nitrite (as N)	0.0611	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	9.3	1.0	mg/L	2022-02-15	
Calculated Parameters					
Hardness, Total (as CaCO3)	44.7	0.500	mg/L	N/A	
Nitrate (as N)	0.0611	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0443	0.0050	mg/L	2022-02-17	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Calcium, dissolved	13.2	0.20	mg/L	2022-02-17	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Copper, dissolved	0.00529	0.00040	mg/L	2022-02-17	
Iron, dissolved	0.082	0.010	mg/L	2022-02-17	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, dissolved	0.00020	0.00010	mg/L	2022-02-17	
Magnesium, dissolved	2.82	0.010	mg/L	2022-02-17	
Manganese, dissolved	0.00317	0.00020	mg/L	2022-02-17	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	0.00020	0.00010	mg/L	2022-02-17	
Nickel, dissolved	0.00047	0.00040	mg/L	2022-02-17	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-17	
Potassium, dissolved	0.52	0.10	mg/L	2022-02-17	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, dissolved	8.9	1.0	mg/L	2022-02-17	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, dissolved	7.47	0.10	mg/L	2022-02-17	
Strontium, dissolved	0.0155	0.0010	mg/L	2022-02-17	
Sulfur, dissolved	4.1	3.0	mg/L	2022-02-17	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, dissolved	0.0030	0.0010	mg/L	2022-02-17	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-09-02092022_NP (22B1773-03) Matrix: Water Sampled: 2022-02-09 12:40, Continued					
<i>Dissolved Metals, Continued</i>					
Vanadium, dissolved	0.0017	0.0010	mg/L	2022-02-17	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-17	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	2.3	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	52.8	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	52.8	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	0.132	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	107	15	mg/L	2022-02-17	HT1

SU_GW-A-02092022_NP (22B1773-04) | Matrix: Water | Sampled: 2022-02-09 12:40

<i>Anions</i>					
Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	12.1	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0630	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	9.5	1.0	mg/L	2022-02-15	

<i>Calculated Parameters</i>					
Hardness, Total (as CaCO ₃)	44.6	0.500	mg/L	N/A	
Nitrate (as N)	0.0630	0.0100	mg/L	N/A	

<i>Dissolved Metals</i>					
Aluminum, dissolved	0.0451	0.0050	mg/L	2022-02-17	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, dissolved	0.000011	0.000010	mg/L	2022-02-17	
Calcium, dissolved	13.4	0.20	mg/L	2022-02-17	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Copper, dissolved	0.00524	0.00040	mg/L	2022-02-17	
Iron, dissolved	0.088	0.010	mg/L	2022-02-17	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, dissolved	0.00021	0.00010	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-A-02092022_NP (22B1773-04) | Matrix: Water | Sampled: 2022-02-09 12:40, Continued

Dissolved Metals, Continued

Magnesium, dissolved	2.71	0.010	mg/L	2022-02-17	
Manganese, dissolved	0.00331	0.00020	mg/L	2022-02-17	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	0.00023	0.00010	mg/L	2022-02-17	
Nickel, dissolved	0.00047	0.00040	mg/L	2022-02-17	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-17	
Potassium, dissolved	0.51	0.10	mg/L	2022-02-17	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, dissolved	8.6	1.0	mg/L	2022-02-17	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, dissolved	7.08	0.10	mg/L	2022-02-17	
Strontium, dissolved	0.0148	0.0010	mg/L	2022-02-17	
Sulfur, dissolved	3.3	3.0	mg/L	2022-02-17	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, dissolved	0.0029	0.0010	mg/L	2022-02-17	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, dissolved	0.0016	0.0010	mg/L	2022-02-17	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-17	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	

General Parameters

Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	51.7	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	51.7	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	0.166	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	105	15	mg/L	2022-02-17	HT1

SU_GW-03-02092022_NP (22B1773-05) | Matrix: Water | Sampled: 2022-02-09 14:00

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	5.24	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.283	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	5.4	1.0	mg/L	2022-02-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-03-02092022_NP (22B1773-05) Matrix: Water Sampled: 2022-02-09 14:00, Continued					
Calculated Parameters					
Hardness, Total (as CaCO ₃)	33.7	0.500	mg/L	N/A	
Nitrate (as N)	0.283	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0094	0.0050	mg/L	2022-02-17	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, dissolved	0.000013	0.000010	mg/L	2022-02-17	
Calcium, dissolved	9.41	0.20	mg/L	2022-02-17	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, dissolved	0.00032	0.00010	mg/L	2022-02-17	
Copper, dissolved	0.00799	0.00040	mg/L	2022-02-17	
Iron, dissolved	< 0.010	0.010	mg/L	2022-02-17	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, dissolved	0.00031	0.00010	mg/L	2022-02-17	
Magnesium, dissolved	2.47	0.010	mg/L	2022-02-17	
Manganese, dissolved	0.00057	0.00020	mg/L	2022-02-17	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	0.00033	0.00010	mg/L	2022-02-17	
Nickel, dissolved	0.00069	0.00040	mg/L	2022-02-17	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-17	
Potassium, dissolved	0.61	0.10	mg/L	2022-02-17	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, dissolved	6.5	1.0	mg/L	2022-02-17	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, dissolved	4.99	0.10	mg/L	2022-02-17	
Strontium, dissolved	0.0227	0.0010	mg/L	2022-02-17	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-17	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, dissolved	0.0016	0.0010	mg/L	2022-02-17	
Zinc, dissolved	0.0044	0.0040	mg/L	2022-02-17	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	

General Parameters

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-03-02092022_NP (22B1773-05) Matrix: Water Sampled: 2022-02-09 14:00, Continued					
<i>General Parameters, Continued</i>					
Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	41.7	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	41.7	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	77	15	mg/L	2022-02-17	HT1

SU_GW-04-02092022_NP (22B1773-06) | Matrix: Water | Sampled: 2022-02-09 14:45

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	2.87	0.10	mg/L	2022-02-16	
Fluoride	< 0.10	0.10	mg/L	2022-02-16	
Nitrate+Nitrite (as N)	0.258	0.0050	mg/L	2022-02-16	
Nitrate (as N)	0.300	0.010	mg/L	2022-02-16	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-16	HT1
Sulfate	22.3	1.0	mg/L	2022-02-16	

BCMOE Aggregate Hydrocarbons

VHw (6-10)	< 100	100	µg/L	2022-02-19	
VPHw	< 100	100	µg/L	N/A	

Calculated Parameters

Hardness, Total (as CaCO ₃)	46.9	0.500	mg/L	N/A	
Nitrate (as N)	0.258	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0144	0.0050	mg/L	2022-02-17	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, dissolved	0.0104	0.0050	mg/L	2022-02-17	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, dissolved	0.000118	0.000010	mg/L	2022-02-17	
Calcium, dissolved	13.5	0.20	mg/L	2022-02-17	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, dissolved	0.00754	0.00010	mg/L	2022-02-17	
Copper, dissolved	0.112	0.00040	mg/L	2022-02-17	
Iron, dissolved	0.031	0.010	mg/L	2022-02-17	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, dissolved	0.00028	0.00010	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-04-02092022_NP (22B1773-06) | Matrix: Water | Sampled: 2022-02-09 14:45, Continued

Dissolved Metals, Continued

Magnesium, dissolved	3.20	0.010	mg/L	2022-02-17	
Manganese, dissolved	0.00535	0.00020	mg/L	2022-02-17	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, dissolved	0.0129	0.00040	mg/L	2022-02-17	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-17	
Potassium, dissolved	0.34	0.10	mg/L	2022-02-17	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, dissolved	9.4	1.0	mg/L	2022-02-17	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, dissolved	3.40	0.10	mg/L	2022-02-17	
Strontium, dissolved	0.0314	0.0010	mg/L	2022-02-17	
Sulfur, dissolved	6.4	3.0	mg/L	2022-02-17	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-17	
Zinc, dissolved	0.0417	0.0040	mg/L	2022-02-17	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	

General Parameters

Acidity, Total (as CaCO ₃)	2.8	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	64.2	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	64.2	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	0.107	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	119	15	mg/L	2022-02-17	HT1

Volatile Organic Compounds (VOC)

Benzene	< 0.5	0.5	µg/L	2022-02-19	
Ethylbenzene	< 1.0	1.0	µg/L	2022-02-19	
Methyl tert-butyl ether	< 1.0	1.0	µg/L	2022-02-19	
Styrene	< 1.0	1.0	µg/L	2022-02-19	
Toluene	< 1.0	1.0	µg/L	2022-02-19	
Xylenes (total)	< 2.0	2.0	µg/L	2022-02-19	
Surrogate: Toluene-d8	119	70-130	%	2022-02-19	
Surrogate: 4-Bromofluorobenzene	94	70-130	%	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-B-02092022_NP (22B1773-07) | Matrix: Water | Sampled: 2022-02-09 14:45

BCMOE Aggregate Hydrocarbons

VHw (6-10)	< 100	100	µg/L	2022-02-19	
VPHw	< 100	100	µg/L	N/A	

Volatile Organic Compounds (VOC)

Benzene	< 0.5	0.5	µg/L	2022-02-19	
Ethylbenzene	< 1.0	1.0	µg/L	2022-02-19	
Methyl tert-butyl ether	< 1.0	1.0	µg/L	2022-02-19	
Styrene	< 1.0	1.0	µg/L	2022-02-19	
Toluene	< 1.0	1.0	µg/L	2022-02-19	
Xylenes (total)	< 2.0	2.0	µg/L	2022-02-19	
Surrogate: Toluene-d8	111	70-130	%	2022-02-19	
Surrogate: 4-Bromofluorobenzene	94	70-130	%	2022-02-19	

SU_GW-02-02102022_NP (22B1773-08) | Matrix: Water | Sampled: 2022-02-10 12:45

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	3.67	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.377	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	41.9	1.0	mg/L	2022-02-15	

Calculated Parameters

Hardness, Total (as CaCO3)	42.2	0.500	mg/L	N/A	
Nitrate (as N)	0.377	0.0100	mg/L	N/A	

Dissolved Metals

Aluminum, dissolved	0.0174	0.0050	mg/L	2022-02-17	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Barium, dissolved	0.0179	0.0050	mg/L	2022-02-17	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, dissolved	0.000387	0.000010	mg/L	2022-02-17	
Calcium, dissolved	12.8	0.20	mg/L	2022-02-17	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Cobalt, dissolved	0.0443	0.00010	mg/L	2022-02-17	
Copper, dissolved	1.13	0.00040	mg/L	2022-02-17	
Iron, dissolved	< 0.010	0.010	mg/L	2022-02-17	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Lithium, dissolved	0.00045	0.00010	mg/L	2022-02-17	
Magnesium, dissolved	2.50	0.010	mg/L	2022-02-17	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-02-02102022_NP (22B1773-08) | Matrix: Water | Sampled: 2022-02-10 12:45, Continued

Dissolved Metals, Continued

Manganese, dissolved	0.00222	0.00020	mg/L	2022-02-17	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Nickel, dissolved	0.0530	0.00040	mg/L	2022-02-17	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-17	
Potassium, dissolved	0.48	0.10	mg/L	2022-02-17	
Selenium, dissolved	0.00071	0.00050	mg/L	2022-02-17	
Silicon, dissolved	11.0	1.0	mg/L	2022-02-17	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, dissolved	6.74	0.10	mg/L	2022-02-17	
Strontium, dissolved	0.0365	0.0010	mg/L	2022-02-17	
Sulfur, dissolved	14.2	3.0	mg/L	2022-02-17	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-17	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-17	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-17	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-17	
Zinc, dissolved	0.106	0.0040	mg/L	2022-02-17	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	

General Parameters

Acidity, Total (as CaCO ₃)	6.8	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	15.7	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	15.7	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	0.154	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	105	15	mg/L	2022-02-17	

SU_GW-05-02102022_NP (22B1773-09) | Matrix: Water | Sampled: 2022-02-10 13:20

BCMOE Aggregate Hydrocarbons

VHw (6-10)	< 100	100	µg/L	2022-02-19	
VPHw	< 100	100	µg/L	N/A	
EPHw10-19	< 250	250	µg/L	2022-02-18	
EPHw19-32	< 250	250	µg/L	2022-02-18	
LEPHw	< 250	250	µg/L	N/A	
HEPHw	< 250	250	µg/L	N/A	
Surrogate: 2-Methylnonane (EPH/F2-4)	96	60-140	%	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-05-02102022_NP (22B1773-09) | Matrix: Water | Sampled: 2022-02-10 13:20, Continued

Polycyclic Aromatic Hydrocarbons (PAH)

Acenaphthene	< 0.050	0.050	µg/L	2022-02-18	
Acenaphthylene	< 0.200	0.200	µg/L	2022-02-18	
Acridine	< 0.050	0.050	µg/L	2022-02-18	
Anthracene	< 0.010	0.010	µg/L	2022-02-18	
Benz(a)anthracene	< 0.010	0.010	µg/L	2022-02-18	
Benzo(a)pyrene	< 0.010	0.010	µg/L	2022-02-18	
Benzo(b+j)fluoranthene	< 0.050	0.050	µg/L	2022-02-18	
Benzo(g,h,i)perylene	< 0.050	0.050	µg/L	2022-02-18	
Benzo(k)fluoranthene	< 0.050	0.050	µg/L	2022-02-18	
2-Chloronaphthalene	< 0.100	0.100	µg/L	2022-02-18	
Chrysene	< 0.050	0.050	µg/L	2022-02-18	
Dibenz(a,h)anthracene	< 0.010	0.010	µg/L	2022-02-18	
Fluoranthene	< 0.030	0.030	µg/L	2022-02-18	
Fluorene	< 0.050	0.050	µg/L	2022-02-18	
Indeno(1,2,3-cd)pyrene	< 0.050	0.050	µg/L	2022-02-18	
1-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-18	
2-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-18	
Naphthalene	< 0.200	0.200	µg/L	2022-02-18	
Phenanthrene	< 0.100	0.100	µg/L	2022-02-18	
Pyrene	< 0.020	0.020	µg/L	2022-02-18	
Quinoline	< 0.050	0.050	µg/L	2022-02-18	
Surrogate: Acridine-d9	85	50-140	%	2022-02-18	
Surrogate: Naphthalene-d8	88	50-140	%	2022-02-18	
Surrogate: Perylene-d12	166	50-140	%	2022-02-18	S02

Volatile Organic Compounds (VOC)

Benzene	< 0.5	0.5	µg/L	2022-02-19	
Ethylbenzene	< 1.0	1.0	µg/L	2022-02-19	
Methyl tert-butyl ether	< 1.0	1.0	µg/L	2022-02-19	
Styrene	< 1.0	1.0	µg/L	2022-02-19	
Toluene	< 1.0	1.0	µg/L	2022-02-19	
Xylenes (total)	< 2.0	2.0	µg/L	2022-02-19	
Surrogate: Toluene-d8	109	70-130	%	2022-02-19	
Surrogate: 4-Bromofluorobenzene	92	70-130	%	2022-02-19	

SU_GW-C-02102022_NP (22B1773-10) | Matrix: Water | Sampled: 2022-02-10 00:01

BCMOE Aggregate Hydrocarbons

EPHw10-19	< 250	250	µg/L	2022-02-24	
EPHw19-32	< 250	250	µg/L	2022-02-24	
LEPHw	< 250	250	µg/L	N/A	
HEPHw	< 250	250	µg/L	N/A	
Surrogate: 2-Methylnonane (EPH/F2-4)	92	60-140	%	2022-02-24	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-C-02102022_NP (22B1773-10) Matrix: Water Sampled: 2022-02-10 00:01, Continued					
Polycyclic Aromatic Hydrocarbons (PAH)					
Acenaphthene	< 0.050	0.050	µg/L	2022-02-25	
Acenaphthylene	< 0.200	0.200	µg/L	2022-02-25	
Acridine	< 0.050	0.050	µg/L	2022-02-25	
Anthracene	< 0.010	0.010	µg/L	2022-02-25	
Benz(a)anthracene	< 0.010	0.010	µg/L	2022-02-25	
Benzo(a)pyrene	< 0.010	0.010	µg/L	2022-02-25	
Benzo(b+j)fluoranthene	< 0.050	0.050	µg/L	2022-02-25	
Benzo(g,h,i)perylene	< 0.050	0.050	µg/L	2022-02-25	
Benzo(k)fluoranthene	< 0.050	0.050	µg/L	2022-02-25	
2-Chloronaphthalene	< 0.100	0.100	µg/L	2022-02-25	
Chrysene	< 0.050	0.050	µg/L	2022-02-25	
Dibenz(a,h)anthracene	< 0.010	0.010	µg/L	2022-02-25	
Fluoranthene	< 0.030	0.030	µg/L	2022-02-25	
Fluorene	< 0.050	0.050	µg/L	2022-02-25	
Indeno(1,2,3-cd)pyrene	< 0.050	0.050	µg/L	2022-02-25	
1-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-25	
2-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-25	
Naphthalene	< 0.200	0.200	µg/L	2022-02-25	
Phenanthrene	< 0.100	0.100	µg/L	2022-02-25	
Pyrene	< 0.020	0.020	µg/L	2022-02-25	
Quinoline	< 0.050	0.050	µg/L	2022-02-25	
Surrogate: Acridine-d9	76	50-140	%	2022-02-25	
Surrogate: Naphthalene-d8	79	50-140	%	2022-02-25	
Surrogate: Perylene-d12	156	50-140	%	2022-02-25	S02

SU_GW-06-02102022_NP (22B1773-11) | Matrix: Water | Sampled: 2022-02-10 13:50

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-15	
Chloride	5.78	0.10	mg/L	2022-02-15	
Fluoride	< 0.10	0.10	mg/L	2022-02-15	
Nitrate+Nitrite (as N)	0.0550	0.0050	mg/L	2022-02-16	
Nitrite (as N)	< 0.0050	0.0050	mg/L	2022-02-14	HT1
Sulfate	35.1	1.0	mg/L	2022-02-15	

BCMOE Aggregate Hydrocarbons

VHw (6-10)	< 100	100	µg/L	2022-02-19	
VPHw	< 100	100	µg/L	N/A	
EPHw10-19	< 250	250	µg/L	2022-02-18	
EPHw19-32	< 250	250	µg/L	2022-02-18	
LEPHw	< 250	250	µg/L	N/A	
HEPHw	< 250	250	µg/L	N/A	
Surrogate: 2-Methylnonane (EPH/F2-4)	97	60-140	%	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-06-02102022_NP (22B1773-11) Matrix: Water Sampled: 2022-02-10 13:50, Continued					
Calculated Parameters					
Hardness, Total (as CaCO ₃)	188	0.500	mg/L	N/A	
Nitrate (as N)	0.0550	0.0100	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.312	0.0050	mg/L	2022-02-17	
Antimony, dissolved	0.00044	0.00020	mg/L	2022-02-17	
Arsenic, dissolved	0.0288	0.00050	mg/L	2022-02-17	
Barium, dissolved	0.0252	0.0050	mg/L	2022-02-17	
Beryllium, dissolved	0.00024	0.00010	mg/L	2022-02-17	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-17	
Cadmium, dissolved	0.000263	0.000010	mg/L	2022-02-17	
Calcium, dissolved	58.9	0.20	mg/L	2022-02-17	
Chromium, dissolved	0.00971	0.00050	mg/L	2022-02-17	
Cobalt, dissolved	0.00567	0.00010	mg/L	2022-02-17	
Copper, dissolved	0.00374	0.00040	mg/L	2022-02-17	
Iron, dissolved	0.212	0.010	mg/L	2022-02-17	
Lead, dissolved	0.00024	0.00020	mg/L	2022-02-17	
Lithium, dissolved	0.00179	0.00010	mg/L	2022-02-17	
Magnesium, dissolved	10.0	0.010	mg/L	2022-02-17	
Manganese, dissolved	0.132	0.00020	mg/L	2022-02-17	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-17	
Molybdenum, dissolved	0.00166	0.00010	mg/L	2022-02-17	
Nickel, dissolved	0.00681	0.00040	mg/L	2022-02-17	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-17	
Potassium, dissolved	1.06	0.10	mg/L	2022-02-17	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Silicon, dissolved	12.4	1.0	mg/L	2022-02-17	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-17	
Sodium, dissolved	15.5	0.10	mg/L	2022-02-17	
Strontium, dissolved	0.0695	0.0010	mg/L	2022-02-17	
Sulfur, dissolved	11.6	3.0	mg/L	2022-02-17	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-17	
Thallium, dissolved	0.000035	0.000020	mg/L	2022-02-17	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	
Tin, dissolved	0.00138	0.00020	mg/L	2022-02-17	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-17	
Tungsten, dissolved	0.110	0.0010	mg/L	2022-02-17	
Uranium, dissolved	0.000490	0.000020	mg/L	2022-02-17	
Vanadium, dissolved	0.0014	0.0010	mg/L	2022-02-17	
Zinc, dissolved	0.0159	0.0040	mg/L	2022-02-17	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-17	

General Parameters

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_GW-06-02102022_NP (22B1773-11) | Matrix: Water | Sampled: 2022-02-10 13:50, Continued

General Parameters, Continued

Acidity, Total (as CaCO ₃)	4.7	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	187	1.0	mg/L	2022-02-15	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Bicarbonate (as CaCO ₃)	187	1.0	mg/L	2022-02-15	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-15	
Ammonia, Total (as N)	0.097	0.050	mg/L	2022-02-17	
Solids, Total Dissolved	518	15	mg/L	2022-02-17	

Polycyclic Aromatic Hydrocarbons (PAH)

Acenaphthene	< 0.050	0.050	µg/L	2022-02-18	
Acenaphthylene	< 0.200	0.200	µg/L	2022-02-18	
Acridine	< 0.050	0.050	µg/L	2022-02-18	
Anthracene	< 0.010	0.010	µg/L	2022-02-18	
Benz(a)anthracene	< 0.010	0.010	µg/L	2022-02-18	
Benzo(a)pyrene	< 0.010	0.010	µg/L	2022-02-18	
Benzo(b+j)fluoranthene	< 0.050	0.050	µg/L	2022-02-18	
Benzo(g,h,i)perylene	< 0.050	0.050	µg/L	2022-02-18	
Benzo(k)fluoranthene	< 0.050	0.050	µg/L	2022-02-18	
2-Chloronaphthalene	< 0.100	0.100	µg/L	2022-02-18	
Chrysene	< 0.050	0.050	µg/L	2022-02-18	
Dibenz(a,h)anthracene	< 0.010	0.010	µg/L	2022-02-18	
Fluoranthene	< 0.030	0.030	µg/L	2022-02-18	
Fluorene	< 0.050	0.050	µg/L	2022-02-18	
Indeno(1,2,3-cd)pyrene	< 0.050	0.050	µg/L	2022-02-18	
1-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-18	
2-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-18	
Naphthalene	< 0.200	0.200	µg/L	2022-02-18	
Phenanthrene	< 0.100	0.100	µg/L	2022-02-18	
Pyrene	< 0.020	0.020	µg/L	2022-02-18	
Quinoline	< 0.050	0.050	µg/L	2022-02-18	
Surrogate: Acridine-d9	81	50-140	%	2022-02-18	
Surrogate: Naphthalene-d8	88	50-140	%	2022-02-18	
Surrogate: Perylene-d12	160	50-140	%	2022-02-18	S02

Volatile Organic Compounds (VOC)

Benzene	< 0.5	0.5	µg/L	2022-02-19	
Ethylbenzene	< 1.0	1.0	µg/L	2022-02-19	
Methyl tert-butyl ether	< 1.0	1.0	µg/L	2022-02-19	
Styrene	< 1.0	1.0	µg/L	2022-02-19	
Toluene	< 1.0	1.0	µg/L	2022-02-19	
Xylenes (total)	< 2.0	2.0	µg/L	2022-02-19	
Surrogate: Toluene-d8	116	70-130	%	2022-02-19	
Surrogate: 4-Bromofluorobenzene	96	70-130	%	2022-02-19	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU_GW-04-02102022_NP (22B1773-12) Matrix: Water Sampled: 2022-02-10 14:15					
BCMOE Aggregate Hydrocarbons					
EPHw10-19	< 250	250	µg/L	2022-02-18	
EPHw19-32	< 250	250	µg/L	2022-02-18	
LEPHw	< 250	250	µg/L	N/A	
HEPHw	< 250	250	µg/L	N/A	
Surrogate: 2-Methylnonane (EPH/F2-4)	94	60-140	%	2022-02-18	
Polycyclic Aromatic Hydrocarbons (PAH)					
Acenaphthene	< 0.050	0.050	µg/L	2022-02-18	
Acenaphthylene	< 0.200	0.200	µg/L	2022-02-18	
Acridine	< 0.050	0.050	µg/L	2022-02-18	
Anthracene	< 0.010	0.010	µg/L	2022-02-18	
Benz(a)anthracene	< 0.010	0.010	µg/L	2022-02-18	
Benzo(a)pyrene	< 0.010	0.010	µg/L	2022-02-18	
Benzo(b+j)fluoranthene	< 0.050	0.050	µg/L	2022-02-18	
Benzo(g,h,i)perylene	< 0.050	0.050	µg/L	2022-02-18	
Benzo(k)fluoranthene	< 0.050	0.050	µg/L	2022-02-18	
2-Chloronaphthalene	< 0.100	0.100	µg/L	2022-02-18	
Chrysene	< 0.050	0.050	µg/L	2022-02-18	
Dibenz(a,h)anthracene	< 0.010	0.010	µg/L	2022-02-18	
Fluoranthene	< 0.030	0.030	µg/L	2022-02-18	
Fluorene	< 0.050	0.050	µg/L	2022-02-18	
Indeno(1,2,3-cd)pyrene	< 0.050	0.050	µg/L	2022-02-18	
1-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-18	
2-Methylnaphthalene	< 0.100	0.100	µg/L	2022-02-18	
Naphthalene	< 0.200	0.200	µg/L	2022-02-18	
Phenanthrene	< 0.100	0.100	µg/L	2022-02-18	
Pyrene	< 0.020	0.020	µg/L	2022-02-18	
Quinoline	< 0.050	0.050	µg/L	2022-02-18	
Surrogate: Acridine-d9	82	50-140	%	2022-02-18	
Surrogate: Naphthalene-d8	89	50-140	%	2022-02-18	
Surrogate: Perylene-d12	165	50-140	%	2022-02-18	S02

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.
S02 Surrogate recovery outside of control limits. Data accepted based on acceptable recovery of other surrogates.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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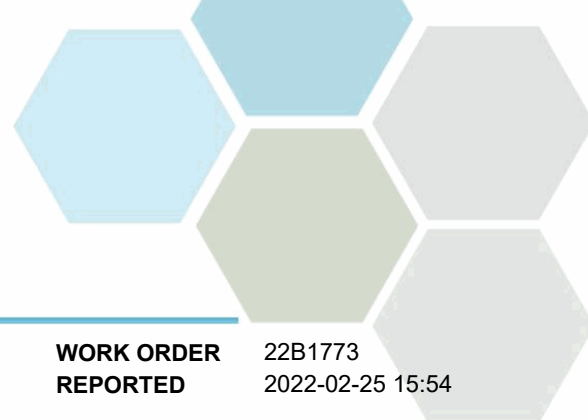
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2017)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2017)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2017)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2017)	Ion Chromatography	✓	Kelowna
BTEX in Water	EPA 5030B / EPA 8260D	Purge&Trap / GC-MSD (SIM)	✓	Richmond
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
EPH in Water	EPA 3511* / BCMOE EPHw	Hexane MicroExtraction (Base/Neutral) / Gas Chromatography (GC-FID)	✓	Richmond
Hardness in Water	SM 2340 B (2017)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
HEPHw in Water	BCMOE LEPH/HEPH	Calculation		N/A
LEPHw in Water	BCMOE LEPH/HEPH	Calculation		N/A
Mercury, dissolved in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Nitrate+Nitrite in Water	SM 4500-NO ₃ - F (2017)	Automated Colorimetry (Cadmium Reduction)	✓	Kelowna
Nitrite in Water	SM 4500-NO ₂ B (2017)	Colorimetry	✓	Richmond
Polycyclic Aromatic Hydrocarbons in Water	EPA 3511* / EPA 8270D	Hexane MicroExtraction (Base/Neutral) / GC-MSD (SIM)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna
VH in Water	EPA 5030B / BCMOE VHw	Purge&Trap / Gas Chromatography (GC-FID)	✓	Richmond
VPHw in Water	BCMOE VPH	Calculation: VH - (Benzene + Toluene + Ethylbenzene + Xylenes + Styrene)		N/A

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
BCMOE	British Columbia Environmental Laboratory Manual, British Columbia Ministry of Environment
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association



APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

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General Comments:

The results in this report apply to the samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Samples will be disposed of 30 days after the test report has been issued or once samples expire, whichever comes first. Longer hold is possible if agreed to in writing.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: bwhitehead@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1486									
Blank (B2B1486-BLK1)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1486-BLK2)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	< 0.0050	0.0050 mg/L							
LCS (B2B1486-BS1)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	0.0489	0.0050 mg/L	0.0500		98	90-110			
LCS (B2B1486-BS2)			Prepared: 2022-02-14, Analyzed: 2022-02-14						
Nitrite (as N)	0.0489	0.0050 mg/L	0.0500		98	90-110			
Anions, Batch B2B1570									
Blank (B2B1570-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK3)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
Blank (B2B1570-BLK4)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	< 0.0050	0.0050 mg/L							
LCS (B2B1570-BS1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.527	0.0050 mg/L	0.500		105	91-108			
LCS (B2B1570-BS2)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.520	0.0050 mg/L	0.500		104	91-108			
LCS (B2B1570-BS3)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.521	0.0050 mg/L	0.500		104	91-108			
LCS (B2B1570-BS4)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Nitrate+Nitrite (as N)	0.525	0.0050 mg/L	0.500		105	91-108			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1570, Continued									
Duplicate (B2B1570-DUP4)		Source: 22B1773-02		Prepared: 2022-02-16, Analyzed: 2022-02-16					
Nitrate+Nitrite (as N)	0.159	0.0050 mg/L		0.152			4	10	
Matrix Spike (B2B1570-MS4)		Source: 22B1773-02		Prepared: 2022-02-16, Analyzed: 2022-02-16					
Nitrate+Nitrite (as N)	0.300	0.0050 mg/L	0.125	0.152	119	80-120			
Anions, Batch B2B1585									
Blank (B2B1585-BLK1)		Prepared: 2022-02-15, Analyzed: 2022-02-15							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B2B1585-BLK2)		Prepared: 2022-02-16, Analyzed: 2022-02-16							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B2B1585-BS1)		Prepared: 2022-02-15, Analyzed: 2022-02-15							
Chloride	16.3	0.10 mg/L	16.0		102	90-110			
Fluoride	4.04	0.10 mg/L	4.00		101	88-108			
Nitrate (as N)	4.01	0.010 mg/L	4.00		100	90-110			
Nitrite (as N)	2.08	0.010 mg/L	2.00		104	85-115			
Sulfate	16.3	1.0 mg/L	16.0		102	90-110			
LCS (B2B1585-BS2)		Prepared: 2022-02-16, Analyzed: 2022-02-16							
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	4.03	0.10 mg/L	4.00		101	88-108			
Nitrate (as N)	4.11	0.010 mg/L	4.00		103	90-110			
Nitrite (as N)	2.14	0.010 mg/L	2.00		107	85-115			
Sulfate	16.3	1.0 mg/L	16.0		102	90-110			
Anions, Batch B2B1586									
Blank (B2B1586-BLK1)		Prepared: 2022-02-15, Analyzed: 2022-02-15							
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B2B1586-BS1)		Prepared: 2022-02-15, Analyzed: 2022-02-15							
Bromide	4.07	0.10 mg/L	4.00		102	85-115			
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	3.97	0.10 mg/L	4.00		99	88-108			
Sulfate	16.1	1.0 mg/L	16.0		100	90-110			
BCMOE Aggregate Hydrocarbons, Batch B2B1983									
Blank (B2B1983-BLK1)		Prepared: 2022-02-18, Analyzed: 2022-02-18							
EPHw10-19	< 250	250 µg/L							
EPHw19-32	< 250	250 µg/L							
Surrogate: 2-Methylnonane (EPH/F2-4)	2340	µg/L	2240		104	60-140			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1773
REPORTED 2022-02-25 15:54

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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BCMOE Aggregate Hydrocarbons, Batch B2B1983, Continued

LCS (B2B1983-BS2)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
EPHw10-19	16800	250 µg/L	15400		109	70-130			
EPHw19-32	23200	250 µg/L	22400		103	70-130			
Surrogate: 2-Methylnonane (EPH/F2-4)	1520	µg/L	2240		68	60-140			
LCS Dup (B2B1983-BSD2)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
EPHw10-19	16000	250 µg/L	15400		104	70-130	5	20	
EPHw19-32	21900	250 µg/L	22400		98	70-130	6	20	
Surrogate: 2-Methylnonane (EPH/F2-4)	1130	µg/L	2240		50	60-140			S09

BCMOE Aggregate Hydrocarbons, Batch B2B2049

Blank (B2B2049-BLK1)			Prepared: 2022-02-19, Analyzed: 2022-02-19						
VHw (6-10)	< 100	100 µg/L							
LCS (B2B2049-BS2)			Prepared: 2022-02-19, Analyzed: 2022-02-19						
VHw (6-10)	3060	100 µg/L	2650		116	70-130			

BCMOE Aggregate Hydrocarbons, Batch B2B2407

Blank (B2B2407-BLK1)			Prepared: 2022-02-24, Analyzed: 2022-02-24						
EPHw10-19	< 250	250 µg/L							
EPHw19-32	< 250	250 µg/L							
Surrogate: 2-Methylnonane (EPH/F2-4)	2150	µg/L	2240		96	60-140			
LCS (B2B2407-BS2)			Prepared: 2022-02-24, Analyzed: 2022-02-24						
EPHw10-19	15400	250 µg/L	15400		100	70-130			
EPHw19-32	23300	250 µg/L	22400		104	70-130			
Surrogate: 2-Methylnonane (EPH/F2-4)	1390	µg/L	2240		62	60-140			
LCS (B2B2407-BS5)			Prepared: 2022-02-24, Analyzed: 2022-02-24						
EPHw10-19	15400	250 µg/L	15400		100	70-130			
EPHw19-32	23200	250 µg/L	22400		103	70-130			
Surrogate: 2-Methylnonane (EPH/F2-4)	1380	µg/L	2240		61	60-140			
LCS (B2B2407-BS6)			Prepared: 2022-02-24, Analyzed: 2022-02-24						
EPHw10-19	15400	250 µg/L	15400		100	70-130			
EPHw19-32	23300	250 µg/L	22400		104	70-130			
Surrogate: 2-Methylnonane (EPH/F2-4)	1440	µg/L	2240		64	60-140			
LCS Dup (B2B2407-BSD2)			Prepared: 2022-02-24, Analyzed: 2022-02-24						
EPHw10-19	15400	250 µg/L	15400		100	70-130	< 1	20	
EPHw19-32	23300	250 µg/L	22400		104	70-130	< 1	20	
Surrogate: 2-Methylnonane (EPH/F2-4)	1400	µg/L	2240		62	60-140			

Dissolved Metals, Batch B2B1759

Blank (B2B1759-BLK1)			Prepared: 2022-02-16, Analyzed: 2022-02-16						
Aluminum, dissolved	< 0.0050	0.0050 mg/L							
Antimony, dissolved	< 0.00020	0.00020 mg/L							
Arsenic, dissolved	< 0.00050	0.00050 mg/L							
Barium, dissolved	< 0.0050	0.0050 mg/L							
Beryllium, dissolved	< 0.00010	0.00010 mg/L							
Bismuth, dissolved	< 0.00010	0.00010 mg/L							
Boron, dissolved	< 0.0500	0.0500 mg/L							
Cadmium, dissolved	< 0.000010	0.000010 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B2B1759, Continued

Blank (B2B1759-BLK1), Continued

Prepared: 2022-02-16, Analyzed: 2022-02-16

Calcium, dissolved, dissolved	< 0.20	0.20 mg/L							
Chromium, dissolved	< 0.00050	0.00050 mg/L							
Cobalt, dissolved	< 0.00010	0.00010 mg/L							
Copper, dissolved	< 0.00040	0.00040 mg/L							
Iron, dissolved	< 0.010	0.010 mg/L							
Lead, dissolved	< 0.00020	0.00020 mg/L							
Lithium, dissolved	< 0.00010	0.00010 mg/L							
Magnesium, dissolved, dissolved	< 0.010	0.010 mg/L							
Manganese, dissolved	< 0.00020	0.00020 mg/L							
Molybdenum, dissolved	< 0.00010	0.00010 mg/L							
Nickel, dissolved	< 0.00040	0.00040 mg/L							
Phosphorus, dissolved	< 0.050	0.050 mg/L							
Potassium, dissolved	< 0.10	0.10 mg/L							
Selenium, dissolved	< 0.00050	0.00050 mg/L							
Silicon, dissolved	< 1.0	1.0 mg/L							
Silver, dissolved	< 0.000050	0.000050 mg/L							
Sodium, dissolved	< 0.10	0.10 mg/L							
Strontium, dissolved	< 0.0010	0.0010 mg/L							
Sulfur, dissolved	< 3.0	3.0 mg/L							
Tellurium, dissolved	< 0.00050	0.00050 mg/L							
Thallium, dissolved	< 0.000020	0.000020 mg/L							
Thorium, dissolved	< 0.00010	0.00010 mg/L							
Tin, dissolved	< 0.00020	0.00020 mg/L							
Titanium, dissolved	< 0.0050	0.0050 mg/L							
Tungsten, dissolved	< 0.0010	0.0010 mg/L							
Uranium, dissolved	< 0.000020	0.000020 mg/L							
Vanadium, dissolved	< 0.0010	0.0010 mg/L							
Zinc, dissolved	< 0.0040	0.0040 mg/L							
Zirconium, dissolved	< 0.00010	0.00010 mg/L							

LCS (B2B1759-BS1)

Prepared: 2022-02-16, Analyzed: 2022-02-16

Aluminum, dissolved	0.0229	0.0050 mg/L	0.0200		115	80-120			
Antimony, dissolved	0.0205	0.00020 mg/L	0.0200		102	80-120			
Arsenic, dissolved	0.0193	0.00050 mg/L	0.0200		97	80-120			
Barium, dissolved	0.0189	0.0050 mg/L	0.0200		95	80-120			
Beryllium, dissolved	0.0192	0.00010 mg/L	0.0200		96	80-120			
Bismuth, dissolved	0.0178	0.00010 mg/L	0.0200		89	80-120			
Boron, dissolved	< 0.0500	0.0500 mg/L	0.0200		110	80-120			
Cadmium, dissolved	0.0189	0.000010 mg/L	0.0200		94	80-120			
Calcium, dissolved, dissolved	2.09	0.20 mg/L	2.00		104	80-120			
Chromium, dissolved	0.0192	0.00050 mg/L	0.0200		96	80-120			
Cobalt, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Copper, dissolved	0.0192	0.00040 mg/L	0.0200		96	80-120			
Iron, dissolved	1.89	0.010 mg/L	2.00		95	80-120			
Lead, dissolved	0.0186	0.00020 mg/L	0.0200		93	80-120			
Lithium, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Magnesium, dissolved, dissolved	1.88	0.010 mg/L	2.00		94	80-120			
Manganese, dissolved	0.0190	0.00020 mg/L	0.0200		95	80-120			
Molybdenum, dissolved	0.0196	0.00010 mg/L	0.0200		98	80-120			
Nickel, dissolved	0.0201	0.00040 mg/L	0.0200		101	80-120			
Phosphorus, dissolved	2.01	0.050 mg/L	2.00		101	80-120			
Potassium, dissolved	1.77	0.10 mg/L	2.00		89	80-120			
Selenium, dissolved	0.0184	0.00050 mg/L	0.0200		92	80-120			
Silicon, dissolved	2.3	1.0 mg/L	2.00		115	80-120			
Silver, dissolved	0.0192	0.000050 mg/L	0.0200		96	80-120			
Sodium, dissolved	1.83	0.10 mg/L	2.00		92	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B1759, Continued									
LCS (B2B1759-BS1), Continued				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Strontium, dissolved	0.0205	0.0010 mg/L	0.0200		103	80-120			
Sulfur, dissolved	5.7	3.0 mg/L	5.00		115	80-120			
Tellurium, dissolved	0.0190	0.00050 mg/L	0.0200		95	80-120			
Thallium, dissolved	0.0179	0.000020 mg/L	0.0200		89	80-120			
Thorium, dissolved	0.0177	0.00010 mg/L	0.0200		89	80-120			
Tin, dissolved	0.0200	0.00020 mg/L	0.0200		100	80-120			
Titanium, dissolved	0.0224	0.0050 mg/L	0.0200		112	80-120			
Tungsten, dissolved	0.0196	0.0010 mg/L	0.0200		98	80-120			
Uranium, dissolved	0.0177	0.000020 mg/L	0.0200		88	80-120			
Vanadium, dissolved	0.0199	0.0010 mg/L	0.0200		100	80-120			
Zinc, dissolved	0.0190	0.0040 mg/L	0.0200		95	80-120			
Zirconium, dissolved	0.0204	0.00010 mg/L	0.0200		102	80-120			
Reference (B2B1759-SRM1)				Prepared: 2022-02-16, Analyzed: 2022-02-16					
Aluminum, dissolved	0.229	0.0050 mg/L	0.198		115	70-130			
Antimony, dissolved	0.0253	0.00020 mg/L	0.0230		110	70-130			
Arsenic, dissolved	0.0206	0.00050 mg/L	0.0200		103	70-130			
Barium, dissolved	0.0164	0.0050 mg/L	0.0161		102	70-130			
Beryllium, dissolved	0.00422	0.00010 mg/L	0.00384		110	70-130			
Boron, dissolved	0.180	0.0500 mg/L	0.191		94	70-130			
Cadmium, dissolved	0.00401	0.000010 mg/L	0.00404		99	70-130			
Calcium, dissolved, dissolved	1.19	0.20 mg/L	0.938		126	70-130			
Chromium, dissolved	0.0262	0.00050 mg/L	0.0256		102	70-130			
Cobalt, dissolved	0.0225	0.00010 mg/L	0.0214		105	70-130			
Copper, dissolved	0.0322	0.00040 mg/L	0.0322		100	70-130			
Iron, dissolved	0.058	0.010 mg/L	0.0580		101	70-130			
Lead, dissolved	0.00815	0.00020 mg/L	0.00796		102	70-130			
Lithium, dissolved	0.0108	0.00010 mg/L	0.0102		107	70-130			
Magnesium, dissolved, dissolved	0.111	0.010 mg/L	0.112		99	70-130			
Manganese, dissolved	0.0121	0.00020 mg/L	0.0120		101	70-130			
Molybdenum, dissolved	0.0459	0.00010 mg/L	0.0438		105	70-130			
Nickel, dissolved	0.0407	0.00040 mg/L	0.0394		103	70-130			
Potassium, dissolved	0.77	0.10 mg/L	0.820		94	70-130			
Selenium, dissolved	0.118	0.00050 mg/L	0.117		101	70-130			
Sodium, dissolved	0.49	0.10 mg/L	0.490		100	70-130			
Strontium, dissolved	0.257	0.0010 mg/L	0.276		93	70-130			
Thallium, dissolved	0.0117	0.000020 mg/L	0.0118		99	70-130			
Uranium, dissolved	0.00926	0.000020 mg/L	0.00970		95	70-130			
Vanadium, dissolved	0.0287	0.0010 mg/L	0.0274		105	70-130			
Zinc, dissolved	0.0868	0.0040 mg/L	0.0884		98	70-130			
Dissolved Metals, Batch B2B1832									
Blank (B2B1832-BLK1)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Blank (B2B1832-BLK2)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Blank (B2B1832-BLK3)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Duplicate (B2B1832-DUP2)				Source: 22B1773-05		Prepared: 2022-02-16, Analyzed: 2022-02-17			
Mercury, dissolved	< 0.000010	0.000010 mg/L		< 0.000010			20		

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1773
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B1832, Continued									
Matrix Spike (B2B1832-MS2)		Source: 22B1773-06		Prepared: 2022-02-16, Analyzed: 2022-02-18					
Mercury, dissolved	0.000262	0.000010 mg/L	0.000250	< 0.000010	105	70-130			
Reference (B2B1832-SRM1)				Prepared: 2022-02-16, Analyzed: 2022-02-18					
Mercury, dissolved	0.000506	0.000010 mg/L	0.000500		101	0-200			
Reference (B2B1832-SRM2)				Prepared: 2022-02-16, Analyzed: 2022-02-18					
Mercury, dissolved	0.000521	0.000010 mg/L	0.000500		104	0-200			
Reference (B2B1832-SRM3)				Prepared: 2022-02-16, Analyzed: 2022-02-17					
Mercury, dissolved	0.000501	0.000010 mg/L	0.000500		100	0-200			
General Parameters, Batch B2B1653									
Blank (B2B1653-BLK1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1653-BLK2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1653-BLK3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B2B1653-BS1)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	104	1.0 mg/L	100		104	80-120			
LCS (B2B1653-BS2)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	108	1.0 mg/L	100		108	80-120			
LCS (B2B1653-BS3)				Prepared: 2022-02-15, Analyzed: 2022-02-15					
Alkalinity, Total (as CaCO ₃)	106	1.0 mg/L	100		106	80-120			
General Parameters, Batch B2B1753									
Blank (B2B1753-BLK1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B1753-BS1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Solids, Total Dissolved	224	15 mg/L	240		93	85-115			
General Parameters, Batch B2B1837									
Blank (B2B1837-BLK1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B1837, Continued									
Blank (B2B1837-BLK2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B1837-BLK3)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B2B1837-BS1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	1.08	0.050 mg/L	1.00		108	90-115			
LCS (B2B1837-BS2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	1.06	0.050 mg/L	1.00		106	90-115			
LCS (B2B1837-BS3)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Ammonia, Total (as N)	1.04	0.050 mg/L	1.00		104	90-115			
General Parameters, Batch B2B2172									
Blank (B2B2172-BLK1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO3)	< 2.0	2.0 mg/L							
Blank (B2B2172-BLK2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO3)	< 2.0	2.0 mg/L							
LCS (B2B2172-BS1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO3)	107	2.0 mg/L	100		107	80-120			
LCS (B2B2172-BS2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO3)	107	2.0 mg/L	100		107	80-120			
Duplicate (B2B2172-DUP2)				Source: 22B1773-06		Prepared: 2022-02-21, Analyzed: 2022-02-21			
Acidity, Total (as CaCO3)	2.9	2.0 mg/L		2.8				20	
Polycyclic Aromatic Hydrocarbons (PAH), Batch B2B1983									
Blank (B2B1983-BLK1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Acenaphthene	< 0.050	0.050 µg/L							
Acenaphthylene	< 0.200	0.200 µg/L							
Acridine	< 0.050	0.050 µg/L							
Anthracene	< 0.010	0.010 µg/L							
Benz(a)anthracene	< 0.010	0.010 µg/L							
Benzo(a)pyrene	< 0.010	0.010 µg/L							
Benzo(b+j)fluoranthene	< 0.050	0.050 µg/L							
Benzo(g,h,i)perylene	< 0.050	0.050 µg/L							
Benzo(k)fluoranthene	< 0.050	0.050 µg/L							
2-Chloronaphthalene	< 0.100	0.100 µg/L							
Chrysene	< 0.050	0.050 µg/L							
Dibenz(a,h)anthracene	< 0.010	0.010 µg/L							
Fluoranthene	< 0.030	0.030 µg/L							
Fluorene	< 0.050	0.050 µg/L							
Indeno(1,2,3-cd)pyrene	< 0.050	0.050 µg/L							
1-Methylnaphthalene	< 0.100	0.100 µg/L							
2-Methylnaphthalene	< 0.100	0.100 µg/L							
Naphthalene	< 0.200	0.200 µg/L							
Phenanthrene	< 0.100	0.100 µg/L							
Pyrene	< 0.020	0.020 µg/L							
Quinoline	< 0.050	0.050 µg/L							
Surrogate: Acridine-d9	19.2	µg/L	22.9		84	50-140			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1773
REPORTED 2022-02-25 15:54

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Polycyclic Aromatic Hydrocarbons (PAH), Batch B2B1983, Continued									
Blank (B2B1983-BLK1), Continued					Prepared: 2022-02-18, Analyzed: 2022-02-18				
Surrogate: Naphthalene-d8	19.9	µg/L	22.2		90	50-140			
Surrogate: Perylene-d12	36.5	µg/L	22.2		164	50-140			S02
LCS (B2B1983-BS1)					Prepared: 2022-02-18, Analyzed: 2022-02-18				
Acenaphthene	21.2	0.050 µg/L	22.2		96	50-140			
Acenaphthylene	20.2	0.200 µg/L	22.2		91	50-140			
Acridine	18.4	0.050 µg/L	21.9		84	50-140			
Anthracene	22.1	0.010 µg/L	22.2		99	50-140			
Benzo(a)anthracene	27.3	0.010 µg/L	22.2		123	50-140			
Benzo(a)pyrene	24.3	0.010 µg/L	22.2		109	50-140			
Benzo(b+j)fluoranthene	51.1	0.050 µg/L	44.0		116	50-140			
Benzo(g,h,i)perylene	23.0	0.050 µg/L	22.2		104	50-140			
Benzo(k)fluoranthene	25.8	0.050 µg/L	22.2		116	50-140			
2-Chloronaphthalene	19.4	0.100 µg/L	22.1		88	50-140			
Chrysene	28.0	0.050 µg/L	22.2		126	50-140			
Dibenz(a,h)anthracene	21.7	0.010 µg/L	22.2		98	50-140			
Fluoranthene	21.8	0.030 µg/L	22.2		98	50-140			
Fluorene	21.2	0.050 µg/L	22.2		95	50-140			
Indeno(1,2,3-cd)pyrene	23.1	0.050 µg/L	22.2		104	50-140			
1-Methylnaphthalene	21.0	0.100 µg/L	22.0		96	50-140			
2-Methylnaphthalene	21.7	0.100 µg/L	22.0		98	50-140			
Naphthalene	22.3	0.200 µg/L	22.2		100	50-140			
Phenanthrene	21.4	0.100 µg/L	22.2		96	50-140			
Pyrene	20.9	0.020 µg/L	22.2		94	50-140			
Quinoline	13.7	0.050 µg/L	22.1		62	50-140			
Surrogate: Acridine-d9	33.6	µg/L	45.8		73	50-140			
Surrogate: Naphthalene-d8	39.6	µg/L	44.4		89	50-140			
Surrogate: Perylene-d12	70.3	µg/L	44.4		158	50-140			S02
LCS Dup (B2B1983-BSD1)					Prepared: 2022-02-18, Analyzed: 2022-02-18				
Acenaphthene	22.3	0.050 µg/L	22.2		100	50-140	5	30	
Acenaphthylene	20.8	0.200 µg/L	22.2		94	50-140	3	30	
Acridine	22.6	0.050 µg/L	21.9		103	50-140	21	30	
Anthracene	22.5	0.010 µg/L	22.2		101	50-140	2	30	
Benzo(a)anthracene	27.5	0.010 µg/L	22.2		124	50-140	< 1	30	
Benzo(a)pyrene	24.8	0.010 µg/L	22.2		111	50-140	2	30	
Benzo(b+j)fluoranthene	52.4	0.050 µg/L	44.0		119	50-140	3	30	
Benzo(g,h,i)perylene	23.4	0.050 µg/L	22.2		105	50-140	2	30	
Benzo(k)fluoranthene	26.3	0.050 µg/L	22.2		118	50-140	2	30	
2-Chloronaphthalene	20.2	0.100 µg/L	22.1		91	50-140	4	30	
Chrysene	29.0	0.050 µg/L	22.2		131	50-140	3	30	
Dibenz(a,h)anthracene	22.2	0.010 µg/L	22.2		100	50-140	2	30	
Fluoranthene	22.2	0.030 µg/L	22.2		100	50-140	2	30	
Fluorene	21.7	0.050 µg/L	22.2		98	50-140	2	30	
Indeno(1,2,3-cd)pyrene	23.7	0.050 µg/L	22.2		107	50-140	2	30	
1-Methylnaphthalene	21.9	0.100 µg/L	22.0		99	50-140	4	30	
2-Methylnaphthalene	22.5	0.100 µg/L	22.0		102	50-140	4	30	
Naphthalene	23.5	0.200 µg/L	22.2		106	50-140	5	30	
Phenanthrene	22.3	0.100 µg/L	22.2		100	50-140	4	30	
Pyrene	21.3	0.020 µg/L	22.2		96	50-140	2	30	
Quinoline	26.5	0.050 µg/L	22.1		120	50-140	63	30	RPD
Surrogate: Acridine-d9	14.6	µg/L	22.9		64	50-140			
Surrogate: Naphthalene-d8	20.6	µg/L	22.2		93	50-140			
Surrogate: Perylene-d12	35.9	µg/L	22.2		161	50-140			S02

Polycyclic Aromatic Hydrocarbons (PAH), Batch B2B2407

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B1773
REPORTED 2022-02-25 15:54

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Polycyclic Aromatic Hydrocarbons (PAH), Batch B2B2407, Continued									
Blank (B2B2407-BLK1)					Prepared: 2022-02-24, Analyzed: 2022-02-25				
Acenaphthene	< 0.050	0.050 µg/L							
Acenaphthylene	< 0.200	0.200 µg/L							
Acridine	< 0.050	0.050 µg/L							
Anthracene	< 0.010	0.010 µg/L							
Benz(a)anthracene	< 0.010	0.010 µg/L							
Benzo(a)pyrene	< 0.010	0.010 µg/L							
Benzo(b+j)fluoranthene	< 0.050	0.050 µg/L							
Benzo(g,h,i)perylene	< 0.050	0.050 µg/L							
Benzo(k)fluoranthene	< 0.050	0.050 µg/L							
2-Chloronaphthalene	< 0.100	0.100 µg/L							
Chrysene	< 0.050	0.050 µg/L							
Dibenz(a,h)anthracene	< 0.010	0.010 µg/L							
Fluoranthene	< 0.030	0.030 µg/L							
Fluorene	< 0.050	0.050 µg/L							
Indeno(1,2,3-cd)pyrene	< 0.050	0.050 µg/L							
1-Methylnaphthalene	< 0.100	0.100 µg/L							
2-Methylnaphthalene	< 0.100	0.100 µg/L							
Naphthalene	< 0.200	0.200 µg/L							
Phenanthrene	< 0.100	0.100 µg/L							
Pyrene	< 0.020	0.020 µg/L							
Quinoline	< 0.050	0.050 µg/L							
Surrogate: Acridine-d9	21.6	µg/L	22.9		94	50-140			
Surrogate: Naphthalene-d8	22.2	µg/L	22.2		100	50-140			
Surrogate: Perylene-d12	46.2	µg/L	22.2		208	50-140			S02

LCS (B2B2407-BS1)					Prepared: 2022-02-24, Analyzed: 2022-02-25				
Acenaphthene	21.0	0.050 µg/L	22.2		95	50-140			
Acenaphthylene	20.1	0.200 µg/L	22.2		91	50-140			
Acridine	18.4	0.050 µg/L	22.2		83	50-140			
Anthracene	20.7	0.010 µg/L	22.2		93	50-140			
Benz(a)anthracene	24.7	0.010 µg/L	22.2		111	50-140			
Benzo(a)pyrene	21.8	0.010 µg/L	22.2		98	50-140			
Benzo(b+j)fluoranthene	48.8	0.050 µg/L	44.4		110	50-140			
Benzo(g,h,i)perylene	21.1	0.050 µg/L	22.2		95	50-140			
Benzo(k)fluoranthene	23.3	0.050 µg/L	22.2		105	50-140			
2-Chloronaphthalene	19.1	0.100 µg/L	22.4		85	50-140			
Chrysene	25.4	0.050 µg/L	22.2		114	50-140			
Dibenz(a,h)anthracene	19.5	0.010 µg/L	22.2		88	50-140			
Fluoranthene	19.6	0.030 µg/L	22.2		88	50-140			
Fluorene	20.5	0.050 µg/L	22.2		92	50-140			
Indeno(1,2,3-cd)pyrene	20.7	0.050 µg/L	22.2		93	50-140			
1-Methylnaphthalene	20.5	0.100 µg/L	22.1		93	50-140			
2-Methylnaphthalene	21.6	0.100 µg/L	22.2		97	50-140			
Naphthalene	22.9	0.200 µg/L	22.2		103	50-140			
Phenanthrene	20.7	0.100 µg/L	22.4		92	50-140			
Pyrene	18.8	0.020 µg/L	22.2		85	50-140			
Quinoline	25.0	0.050 µg/L	22.4		111	50-140			
Surrogate: Acridine-d9	14.6	µg/L	22.9		64	50-140			
Surrogate: Naphthalene-d8	21.0	µg/L	22.2		94	50-140			
Surrogate: Perylene-d12	29.7	µg/L	22.2		134	50-140			

Volatile Organic Compounds (VOC), Batch B2B2049

Blank (B2B2049-BLK1)					Prepared: 2022-02-19, Analyzed: 2022-02-19				
Benzene	< 0.5	0.5 µg/L							
Ethylbenzene	< 1.0	1.0 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B1773
2022-02-25 15:54

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Volatile Organic Compounds (VOC), Batch B2B2049, Continued									
Blank (B2B2049-BLK1), Continued					Prepared: 2022-02-19, Analyzed: 2022-02-19				
Methyl tert-butyl ether	< 1.0	1.0 µg/L							
Styrene	< 1.0	1.0 µg/L							
Toluene	< 1.0	1.0 µg/L							
Xylenes (total)	< 2.0	2.0 µg/L							
Surrogate: Toluene-d8	30.6	µg/L	25.0		122	70-130			
Surrogate: 4-Bromofluorobenzene	24.9	µg/L	25.0		99	70-130			
LCS (B2B2049-BS1)					Prepared: 2022-02-19, Analyzed: 2022-02-19				
Benzene	18.0	0.5 µg/L	20.0		90	70-130			
Ethylbenzene	12.7	1.0 µg/L	20.0		63	70-130			SPK
Methyl tert-butyl ether	20.8	1.0 µg/L	20.0		104	70-130			
Styrene	14.4	1.0 µg/L	20.0		72	70-130			
Toluene	18.3	1.0 µg/L	20.0		91	70-130			
Xylenes (total)	41.5	2.0 µg/L	60.0		69	70-130			SPK
Surrogate: Toluene-d8	28.2	µg/L	25.0		113	70-130			
Surrogate: 4-Bromofluorobenzene	35.3	µg/L	25.0		141	70-130			S02
LCS (B2B2049-BS2)					Prepared: 2022-02-19, Analyzed: 2022-02-19				

QC Qualifiers:

RPD	Relative percent difference (RPD) of duplicate analysis are outside of control limits for unknown reason(s).
S02	Surrogate recovery outside of control limits. Data accepted based on acceptable recovery of other surrogates.
S09	The surrogate recovery for this sample is outside of established control limits Likely due to syringe error, data not affected
SPK	The recovery of this analyte was outside of established control limits.

Teck

COC ID:

PROJECT/



Page 1 of 2

TURNAROUND TIME:

Regular

RUSH:

LABORATORY

OTHER INFO

Facility Name / Job# Sunro Mine		Project Manager Michelle Unger		Email michelle.unger@teck.com		Address 601 Knighton Rd		City Kimberley		Province BC	Postal Code V1A 1C7	Country Canada	Phone Number 250-427-8422	
Lab Name Caro		Lab Contact Brent Whitehead		Email bwhitehead@caro.ca		Address 110 - 4011 Viking Way		City Richmond		Province BC	Postal Code V6B2K9	Country CA	Phone Number 604-279-1499	
Report Format / Distribution		Excel		PDF		EDD		Email 1: michelle.unger@teck.com		Email 2: craig.pinnell@snc-lavalin.com		Email 3: jonathan.croston@snc-lavalin.com		
Email 4: josh.baran@snc-lavalin.com		Email 5: jhw.West_BC_LabData@snc-lavalin.com												
PO number		669765												
SAMPLE DETAILS														
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	ANALYSIS REQUESTED	Filtered - F: Field, L: Lab, FL: Field & Lab, N: None					
								Yes						
								Yes	Yes			Yes	Yes	
								Total Metals + Hg in Water	Dissolved Metals + Hg in Water	TDS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	BTEX/VPH	
								LEPH/HEPH	Total and Speciated Alkalinity, Acidity	Sulfide	Ammonia		HOLD	
✓ SU_GW-07-02092022_NP	MW18-07	GW	N	2022-02-09	11:30	G	4		x	x	x	x		
✓ SU_GW-05-02092022_NP	MW18-05	GW	N	2022-02-09	12:00	G	4		x	x	x	x		
✓ SU_GW-09-02092022_NP	MW18-09	GW	N	2022-02-09	12:40	G	4		x	x	x	x		
✓ SU_GW-A-02092022_NP	~	GW	N	2022-02-09	12:40	G	4		x	x	x	x		
✓ SU_GW-03-02092022_NP	MW18-03	GW	N	2022-02-09	14:00	G	4		x	x	x	x		
✓ SU_GW-04-02092022_NP	MW18-04	GW	N	2022-02-09	14:45	G	6		x	x	x	x		
✓ SU_GW-B-02092022_NP	~	GW	N	2022-02-09	14:45	G	2					x		
✓ SU_GW-02-02102022_NP	MW18-02	GW	N	2022-02-10	12:45	G	4		x	x	x	x		
✓ SU_GW-05-02102022_NP	MW18-05	GW	N	2022-02-10	13:20	G	4					x		
✓ SU_GW-C-02102022_NP	~	GW	N	2022-02-10	00:01	G	2					x		
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS				RELINQUISHED BY/AFFILIATION				DATE/TIME		ACCEPTED BY/AFFILIATION		DATE/TIME		
Please note custody seal condition. Dissolved metals field filtered.				SNC-Lavalin Inc										
Invoice to: SNC-Lavalin Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3														
Send to: payables@snc-lavalin.com Project number: 631848 Billing PO: 683703														
NB OF BOTTLES RETURNED/DESCRIPTION														
Regular (default) X				Sampler's Name				Josh Baran		Mobile #		250-418-1562		
Priority (2-3 business days) - 50% surcharge				Sampler's Signature						Date/Time		February 10, 2022		
Emergency (1 Business Day) - 100% surcharge														
For Emergency <1 Day, ASAP or Weekend														

KD RW
2/12/22 13:00
9.8°C

PROJECT/CLIENT INFO								LABORATORY						OTHER INFO										
COC ID:								TURNAROUND TIME:						RUSH:										
Facility Name / Job#								Lab Name						Report Format / Distribution										
Project Manager								Lab Contact						Email 1:										
Email								Email						Excel PDF EDD										
Address								Address						Email 2:										
														Email 3:										
City								City						Email 4:										
Postal Code								Postal Code						Email 5:										
Phone Number								Phone Number						PO number										
Sunro Mine								Caro						michelle.unger@teck.com										
Michelle Unger								Brent Whitehead						x x x										
michelle.unger@teck.com								bwhitehead@caro.ca						x x x										
601 Knighton Rd								110 - 4011 Viking Way						x x x										
														x x x										
Kimberly								Richmond						x x x										
VIA IC7								V6B2K9						x x x										
Province BC								Province BC						x x x										
Country Canada								Country CA						x x x										
250-427-8422								604-279-1499						669765										
SAMPLE DETAILS								ANALYSIS REQUESTED																
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	PRI.		Yes														
								PRESERV.	Yes	Yes					Yes	Yes		Yes	Yes					
								ANALYSIS	Total Metals + Hg in Water	Dissolved Metals + Hg in Water	TDS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	BTEX/VPH	LEPH/HEPH	Total and Speciated Alkalinity, Acidity	Sulfide	Ammonia						
SU_GW-06-02102022_NP	MW18-06	GW	N	2022-02-10	13:50	G	8		x	x	x	x	x	x	x	x	x							
SU_GW-04-02102022_NP	MW18-04	GW	N	2022-02-10	14:15	G	2								x									
SU_GW-02-02102022_NP	-	GW	N	2022-02-10	14:15	G	2		x	x	x	x	x	x	x	x	x							
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS								RELINQUISHED BY/AFFILIATION								DATE/TIME								
Please note custody seal condition. Dissolved metals field filtered.								Chakras SNC-Lavalin Inc								22/02/11/1000								
Invoice to: SNC-Lavalin Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3																								
Send to: payables@snc-lavalin.com Project number: 631848 Billing PO: 683703																								
NB OF BOTTLES RETURNED/DESCRIPTION																								
Regular (default) X								Sampler's Name								Josh Baran								
Priority (2-3 business days) - 50% surcharge								Sampler's Signature								Mobile #								
Emergency (1 Business Day) - 100% surcharge																250-418-1562								
For Emergency/ <1 Day, A.S. or Weekend																Date/Time								
																February 10, 2022								

KD Pur
2/12 13:00
9.8°C

CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Jonathan Croston

PO NUMBER 669765

PROJECT 631848

PROJECT INFO Sunro Mine

WORK ORDER 22B2326

RECEIVED / TEMP 2022-02-16 09:45 / 5.6°C

REPORTED 2022-02-24 16:27

COC NUMBER No Number

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

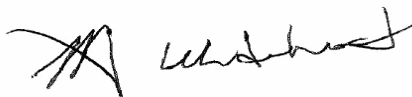
Work Order Comments:

Custody Seals Intact: YES

If you have any questions or concerns, please contact me at bwhitehead@caro.ca

Authorized By:

Brent Whitehead
Account Manager



1-888-311-8846 | www.caro.ca

#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2326
2022-02-24 16:27

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-08-02112022_NP (22B2326-01) Matrix: Water Sampled: 2022-02-11 10:20					
Anions					
Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	5.21	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.026	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	4.2	1.0	mg/L	2022-02-18	
Calculated Parameters					
Hardness, Total (as CaCO3)	10.6	0.500	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0915	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	0.000031	0.000010	mg/L	2022-02-18	
Calcium, dissolved	3.36	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	0.00250	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.0844	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.031	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.541	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00162	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	0.00373	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	0.13	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	2.4	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	2.83	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0070	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2326
2022-02-24 16:27

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-08-02112022_NP (22B2326-01) Matrix: Water Sampled: 2022-02-11 10:20, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	0.0092	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	5.0	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	5.0	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Solids, Total Dissolved	39	15	mg/L	2022-02-22	HT1

SU_PW-09-02112022_NP (22B2326-02) | Matrix: Water | Sampled: 2022-02-11 11:30

<i>Anions</i>					
Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	1.81	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.014	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	< 1.0	1.0	mg/L	2022-02-18	

<i>Calculated Parameters</i>					
Hardness, Total (as CaCO ₃)	3.57	0.500	mg/L	N/A	

<i>Dissolved Metals</i>					
Aluminum, dissolved	0.0898	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-18	
Calcium, dissolved	0.84	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.00185	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.018	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2326
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU_PW-09-02112022_NP (22B2326-02) Matrix: Water Sampled: 2022-02-11 11:30, Continued					
Dissolved Metals, Continued					
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.353	0.010	mg/L	2022-02-18	
Manganese, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	< 0.00040	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	1.7	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	1.50	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0038	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
General Parameters					
Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.4	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	3.4	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Solids, Total Dissolved	22	15	mg/L	2022-02-22	HT1

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2017)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2017)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2017)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2017)	Ion Chromatography	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2017)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association

General Comments:

The results in this report apply to the samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Samples will be disposed of 30 days after the test report has been issued or once samples expire, whichever comes first. Longer hold is possible if agreed to in writing.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: bwhitehead@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (BLK):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1803									
Blank (B2B1803-BLK1)			Prepared: 2022-02-17, Analyzed: 2022-02-17						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B2B1803-BLK2)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B2B1803-BS1)			Prepared: 2022-02-17, Analyzed: 2022-02-17						
Bromide	4.07	0.10 mg/L	4.00		102	85-115			
Chloride	16.3	0.10 mg/L	16.0		102	90-110			
Fluoride	4.03	0.10 mg/L	4.00		101	88-108			
Nitrate (as N)	3.95	0.010 mg/L	4.00		99	90-110			
Nitrite (as N)	2.06	0.010 mg/L	2.00		103	85-115			
Sulfate	16.0	1.0 mg/L	16.0		100	90-110			
LCS (B2B1803-BS2)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
Bromide	4.03	0.10 mg/L	4.00		101	85-115			
Chloride	16.2	0.10 mg/L	16.0		101	90-110			
Fluoride	3.96	0.10 mg/L	4.00		99	88-108			
Nitrate (as N)	3.95	0.010 mg/L	4.00		99	90-110			
Nitrite (as N)	1.99	0.010 mg/L	2.00		100	85-115			
Sulfate	16.1	1.0 mg/L	16.0		100	90-110			

Dissolved Metals, Batch B2B2057

Blank (B2B2057-BLK1)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
Aluminum, dissolved	< 0.0050	0.0050 mg/L							
Antimony, dissolved	< 0.00020	0.00020 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
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WORK ORDER 22B2326
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B2B2057, Continued

Blank (B2B2057-BLK1), Continued

Prepared: 2022-02-18, Analyzed: 2022-02-18

Arsenic, dissolved	< 0.00050	0.00050 mg/L							
Barium, dissolved	< 0.0050	0.0050 mg/L							
Beryllium, dissolved	< 0.00010	0.00010 mg/L							
Bismuth, dissolved	< 0.00010	0.00010 mg/L							
Boron, dissolved	< 0.0500	0.0500 mg/L							
Cadmium, dissolved	< 0.000010	0.000010 mg/L							
Calcium, dissolved, dissolved	< 0.20	0.20 mg/L							
Chromium, dissolved	< 0.00050	0.00050 mg/L							
Cobalt, dissolved	< 0.00010	0.00010 mg/L							
Copper, dissolved	< 0.00040	0.00040 mg/L							
Iron, dissolved	< 0.010	0.010 mg/L							
Lead, dissolved	< 0.00020	0.00020 mg/L							
Lithium, dissolved	< 0.00010	0.00010 mg/L							
Magnesium, dissolved, dissolved	< 0.010	0.010 mg/L							
Manganese, dissolved	< 0.00020	0.00020 mg/L							
Molybdenum, dissolved	< 0.00010	0.00010 mg/L							
Nickel, dissolved	< 0.00040	0.00040 mg/L							
Phosphorus, dissolved	< 0.050	0.050 mg/L							
Potassium, dissolved	< 0.10	0.10 mg/L							
Selenium, dissolved	< 0.00050	0.00050 mg/L							
Silicon, dissolved	< 1.0	1.0 mg/L							
Silver, dissolved	< 0.000050	0.000050 mg/L							
Sodium, dissolved	< 0.10	0.10 mg/L							
Strontium, dissolved	< 0.0010	0.0010 mg/L							
Sulfur, dissolved	< 3.0	3.0 mg/L							
Tellurium, dissolved	< 0.00050	0.00050 mg/L							
Thallium, dissolved	< 0.000020	0.000020 mg/L							
Thorium, dissolved	< 0.00010	0.00010 mg/L							
Tin, dissolved	< 0.00020	0.00020 mg/L							
Titanium, dissolved	< 0.0050	0.0050 mg/L							
Tungsten, dissolved	< 0.0010	0.0010 mg/L							
Uranium, dissolved	< 0.000020	0.000020 mg/L							
Vanadium, dissolved	< 0.0010	0.0010 mg/L							
Zinc, dissolved	< 0.0040	0.0040 mg/L							
Zirconium, dissolved	< 0.00010	0.00010 mg/L							

LCS (B2B2057-BS1)

Prepared: 2022-02-18, Analyzed: 2022-02-18

Aluminum, dissolved	0.0234	0.0050 mg/L	0.0200		117	80-120
Antimony, dissolved	0.0165	0.00020 mg/L	0.0200		82	80-120
Arsenic, dissolved	0.0172	0.00050 mg/L	0.0200		86	80-120
Barium, dissolved	0.0174	0.0050 mg/L	0.0200		87	80-120
Beryllium, dissolved	0.0184	0.00010 mg/L	0.0200		92	80-120
Bismuth, dissolved	0.0189	0.00010 mg/L	0.0200		95	80-120
Boron, dissolved	< 0.0500	0.0500 mg/L	0.0200		99	80-120
Cadmium, dissolved	0.0180	0.000010 mg/L	0.0200		90	80-120
Calcium, dissolved, dissolved	1.72	0.20 mg/L	2.00		86	80-120
Chromium, dissolved	0.0178	0.00050 mg/L	0.0200		89	80-120
Cobalt, dissolved	0.0185	0.00010 mg/L	0.0200		93	80-120
Copper, dissolved	0.0197	0.00040 mg/L	0.0200		99	80-120
Iron, dissolved	1.83	0.010 mg/L	2.00		92	80-120
Lead, dissolved	0.0194	0.00020 mg/L	0.0200		97	80-120
Lithium, dissolved	0.0187	0.00010 mg/L	0.0200		93	80-120
Magnesium, dissolved, dissolved	1.93	0.010 mg/L	2.00		97	80-120
Manganese, dissolved	0.0197	0.00020 mg/L	0.0200		99	80-120
Molybdenum, dissolved	0.0185	0.00010 mg/L	0.0200		93	80-120
Nickel, dissolved	0.0189	0.00040 mg/L	0.0200		94	80-120

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2326
2022-02-24 16:27

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B2057, Continued									
LCS (B2B2057-BS1), Continued				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Phosphorus, dissolved	1.97	0.050 mg/L	2.00		98	80-120			
Potassium, dissolved	1.80	0.10 mg/L	2.00		90	80-120			
Selenium, dissolved	0.0186	0.00050 mg/L	0.0200		93	80-120			
Silicon, dissolved	2.0	1.0 mg/L	2.00		98	80-120			
Silver, dissolved	0.0185	0.000050 mg/L	0.0200		93	80-120			
Sodium, dissolved	1.86	0.10 mg/L	2.00		93	80-120			
Strontium, dissolved	0.0170	0.0010 mg/L	0.0200		85	80-120			
Sulfur, dissolved	4.9	3.0 mg/L	5.00		98	80-120			
Tellurium, dissolved	0.0174	0.00050 mg/L	0.0200		87	80-120			
Thallium, dissolved	0.0189	0.000020 mg/L	0.0200		94	80-120			
Thorium, dissolved	0.0176	0.00010 mg/L	0.0200		88	80-120			
Tin, dissolved	0.0191	0.00020 mg/L	0.0200		95	80-120			
Titanium, dissolved	0.0190	0.0050 mg/L	0.0200		95	80-120			
Tungsten, dissolved	0.0182	0.0010 mg/L	0.0200		91	80-120			
Uranium, dissolved	0.0195	0.000020 mg/L	0.0200		97	80-120			
Vanadium, dissolved	0.0167	0.0010 mg/L	0.0200		84	80-120			
Zinc, dissolved	0.0195	0.0040 mg/L	0.0200		97	80-120			
Zirconium, dissolved	0.0186	0.00010 mg/L	0.0200		93	80-120			
Duplicate (B2B2057-DUP1)				Source: 22B2326-01	Prepared: 2022-02-18, Analyzed: 2022-02-18				
Aluminum, dissolved	0.0823	0.0050 mg/L		0.0915			11	20	
Antimony, dissolved	< 0.00020	0.00020 mg/L		< 0.00020				20	
Arsenic, dissolved	< 0.00050	0.00050 mg/L		< 0.00050				20	
Barium, dissolved	< 0.0050	0.0050 mg/L		< 0.0050				20	
Beryllium, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Bismuth, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Boron, dissolved	< 0.0500	0.0500 mg/L		< 0.0500				20	
Cadmium, dissolved	0.000029	0.000010 mg/L		0.000031				20	
Calcium, dissolved, dissolved	3.28	0.20 mg/L		3.36			2	20	
Chromium, dissolved	< 0.00050	0.00050 mg/L		< 0.00050				20	
Cobalt, dissolved	0.00239	0.00010 mg/L		0.00250			4	20	
Copper, dissolved	0.0812	0.00040 mg/L		0.0844			4	20	
Iron, dissolved	0.029	0.010 mg/L		0.031				20	
Lead, dissolved	< 0.00020	0.00020 mg/L		< 0.00020				20	
Lithium, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Magnesium, dissolved, dissolved	0.523	0.010 mg/L		0.541			3	20	
Manganese, dissolved	0.00173	0.00020 mg/L		0.00162			7	20	
Molybdenum, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Nickel, dissolved	0.00339	0.00040 mg/L		0.00373			9	20	
Phosphorus, dissolved	< 0.050	0.050 mg/L		< 0.050				20	
Potassium, dissolved	0.12	0.10 mg/L		0.13				20	
Selenium, dissolved	< 0.00050	0.00050 mg/L		< 0.00050				20	
Silicon, dissolved	2.4	1.0 mg/L		2.4				20	
Silver, dissolved	< 0.000050	0.000050 mg/L		< 0.000050				20	
Sodium, dissolved	2.75	0.10 mg/L		2.83			3	20	
Strontium, dissolved	0.0066	0.0010 mg/L		0.0070			6	20	
Sulfur, dissolved	< 3.0	3.0 mg/L		< 3.0				20	
Tellurium, dissolved	< 0.00050	0.00050 mg/L		< 0.00050				20	
Thallium, dissolved	< 0.000020	0.000020 mg/L		< 0.000020				20	
Thorium, dissolved	< 0.00010	0.00010 mg/L		< 0.00010				20	
Tin, dissolved	< 0.00020	0.00020 mg/L		< 0.00020				20	
Titanium, dissolved	< 0.0050	0.0050 mg/L		< 0.0050				20	
Tungsten, dissolved	< 0.0010	0.0010 mg/L		< 0.0010				20	
Uranium, dissolved	< 0.000020	0.000020 mg/L		< 0.000020				20	
Vanadium, dissolved	< 0.0010	0.0010 mg/L		< 0.0010				20	
Zinc, dissolved	0.0088	0.0040 mg/L		0.0092				20	

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2326
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B2057, Continued									
Duplicate (B2B2057-DUP1), Continued		Source: 22B2326-01		Prepared: 2022-02-18, Analyzed: 2022-02-18					
Zirconium, dissolved	< 0.00010	0.00010 mg/L		< 0.00010			20		
Reference (B2B2057-SRM1)		Prepared: 2022-02-18, Analyzed: 2022-02-18							
Aluminum, dissolved	0.235	0.0050 mg/L		0.235	100	70-130			
Antimony, dissolved	0.0470	0.00020 mg/L		0.0431	109	70-130			
Arsenic, dissolved	0.415	0.00050 mg/L		0.423	98	70-130			
Barium, dissolved	2.85	0.0050 mg/L		3.30	86	70-130			
Beryllium, dissolved	0.198	0.00010 mg/L		0.209	95	70-130			
Boron, dissolved	1.56	0.0500 mg/L		1.65	94	70-130			
Cadmium, dissolved	0.208	0.000010 mg/L		0.221	94	70-130			
Calcium, dissolved, dissolved	6.74	0.20 mg/L		7.72	87	70-130			
Chromium, dissolved	0.405	0.00050 mg/L		0.434	93	70-130			
Cobalt, dissolved	0.122	0.00010 mg/L		0.124	98	70-130			
Copper, dissolved	0.830	0.00040 mg/L		0.815	102	70-130			
Iron, dissolved	1.22	0.010 mg/L		1.27	96	70-130			
Lead, dissolved	0.109	0.00020 mg/L		0.110	99	70-130			
Lithium, dissolved	0.0978	0.00010 mg/L		0.100	98	70-130			
Magnesium, dissolved, dissolved	6.48	0.010 mg/L		6.59	98	70-130			
Manganese, dissolved	0.342	0.00020 mg/L		0.342	100	70-130			
Molybdenum, dissolved	0.392	0.00010 mg/L		0.404	97	70-130			
Nickel, dissolved	0.818	0.00040 mg/L		0.835	98	70-130			
Phosphorus, dissolved	0.440	0.050 mg/L		0.499	88	70-130			
Potassium, dissolved	2.75	0.10 mg/L		2.88	95	70-130			
Selenium, dissolved	0.0320	0.00050 mg/L		0.0324	99	70-130			
Sodium, dissolved	17.6	0.10 mg/L		18.0	98	70-130			
Strontium, dissolved	0.820	0.0010 mg/L		0.935	88	70-130			
Thallium, dissolved	0.0377	0.000020 mg/L		0.0385	98	70-130			
Uranium, dissolved	0.238	0.000020 mg/L		0.258	92	70-130			
Vanadium, dissolved	0.805	0.0010 mg/L		0.873	92	70-130			
Zinc, dissolved	0.831	0.0040 mg/L		0.848	98	70-130			

Dissolved Metals, Batch B2B2367

Blank (B2B2367-BLK1)		Prepared: 2022-02-23, Analyzed: 2022-02-23							
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Reference (B2B2367-SRM1)		Prepared: 2022-02-23, Analyzed: 2022-02-23							
Mercury, dissolved	0.000527	0.000010 mg/L		0.000500	105	0-200			
Reference (B2B2367-SRM2)		Prepared: 2022-02-23, Analyzed: 2022-02-23							
Mercury, dissolved	0.000491	0.000010 mg/L		0.000500	98	0-200			
Reference (B2B2367-SRM3)		Prepared: 2022-02-23, Analyzed: 2022-02-23							
Mercury, dissolved	0.000502	0.000010 mg/L		0.000500	100	0-200			

General Parameters, Batch B2B1847

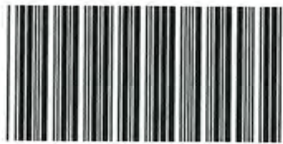
Blank (B2B1847-BLK1)		Prepared: 2022-02-17, Analyzed: 2022-02-17							
Alkalinity, Total (as CaCO3)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO3)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO3)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO3)	< 1.0	1.0 mg/L							
Blank (B2B1847-BLK2)		Prepared: 2022-02-17, Analyzed: 2022-02-17							
Alkalinity, Total (as CaCO3)	< 1.0	1.0 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B2326
REPORTED 2022-02-24 16:27

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B1847, Continued									
Blank (B2B1847-BLK2), Continued				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B2B1847-BS1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Total (as CaCO ₃)	106	1.0 mg/L	100		106	80-120			
LCS (B2B1847-BS2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Total (as CaCO ₃)	108	1.0 mg/L	100		108	80-120			
General Parameters, Batch B2B2173									
Blank (B2B2173-BLK1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
Blank (B2B2173-BLK2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B2B2173-BS1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	106	2.0 mg/L	100		106	80-120			
LCS (B2B2173-BS2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	107	2.0 mg/L	100		107	80-120			
Duplicate (B2B2173-DUP2)				Source: 22B2326-02		Prepared: 2022-02-21, Analyzed: 2022-02-21			
Acidity, Total (as CaCO ₃)	2.3	2.0 mg/L		2.2				20	
General Parameters, Batch B2B2193									
Blank (B2B2193-BLK1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B2193-BLK2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B2193-BLK3)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B2B2193-BS1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	90-115			
LCS (B2B2193-BS2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	1.06	0.050 mg/L	1.00		106	90-115			
LCS (B2B2193-BS3)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	1.09	0.050 mg/L	1.00		109	90-115			
General Parameters, Batch B2B2375									
Blank (B2B2375-BLK1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B2375-BS1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Solids, Total Dissolved	238	15 mg/L	240		99	85-115			



RUSH for nitrite/nitrate

COC ID:		TURNAROUND TIME:		Regular		RUSH:										
PROJECT/CLIENT INFO				LABORATORY				OTHER INFO								
Facility Name / Job#				Lab Name				Report Format / Distribution								
Project Manager				Lab Contact				Email 1:								
Email				Email				Email 2:								
Address				Address				Email 3:								
City				City				Email 4:								
Province				Province				Email 5:								
Postal Code				Postal Code				PO number								
Phone Number				Phone Number												
SAMPLE DETAILS								ANALYSIS REQUESTED								
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	Anal. Preserv.	Anal. Matrix	Anal. Matrix	Anal. Matrix	Anal. Matrix	Anal. Matrix	Anal. Matrix	Anal. Matrix	Anal. Matrix
SU_PW-08-02112022_NP	PW15-08	PW	N	2022-02-11	10:20	G	4	Yes	Yes	Total Metals + Hg in Water	Dissolved Metals + Hg in Water	TDS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	BTEX/VPH	LEPH/HEPH
SU_PW-09-02112022_NP	PW15-09	PW	N	2022-02-11	11:30	G	4	Yes	Yes							
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS								RELINQUISHED BY/AFFILIATION				DATE/TIME				
Please note custody seal condition. Dissolved metals field filtered.								Josh Baran				22/02/14 13:00				
Invoice to:								SNC-Lavalin Inc								
Jonathan Croston / Craig Pinnell (Nelson BC)																
455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3																
Send to: payables@snc-lavalin.com																
Project number: 631848																
Billing PO: 669765																
NB OF BOTTLES RETURNED/DESCRIPTION								SAMPLER'S NAME				DATE/TIME				
Regular (default) X								Josh Baran				250-418-1562				
Priority (2-3 business days) - 50% surcharge								Sampler's Signature				February 14, 2022				
Emergency (1 Business Day) - 100% surcharge																
For Emergency <1 Day, ASAP or Weekend																

LG 100
9.45
5.60C

Custody seal ✓

CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Jonathan Croston

PO NUMBER 669765

PROJECT 631848

PROJECT INFO Sunro Mine

WORK ORDER 22B2327

RECEIVED / TEMP 2022-02-16 09:45 / 5.6°C

REPORTED 2022-02-24 16:25

COC NUMBER No Number

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

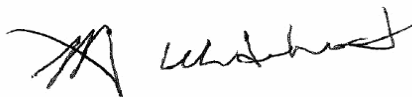
Work Order Comments:

Custody Seals Intact: YES

If you have any questions or concerns, please contact me at bwhitehead@caro.ca

Authorized By:

Brent Whitehead
Account Manager



1-888-311-8846 | www.caro.ca

#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-02-02112022_NP (22B2327-01) Matrix: Water Sampled: 2022-02-11 10:00					
Anions					
Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	9.09	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.030	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	8.3	1.0	mg/L	2022-02-18	
Calculated Parameters					
Hardness, Total (as CaCO3)	17.4	0.500	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0470	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	0.000063	0.000010	mg/L	2022-02-18	
Calcium, dissolved	5.78	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	0.00555	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.202	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.039	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.708	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00369	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	0.00736	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	0.16	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	3.3	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	4.40	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0095	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-02-02112022_NP (22B2327-01) | Matrix: Water | Sampled: 2022-02-11 10:00, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	0.0181	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	

General Parameters

Acidity, Total (as CaCO ₃)	2.6	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	7.2	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	7.2	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Solids, Total Dissolved	39	15	mg/L	2022-02-23	HT1

Total Metals

Aluminum, total	0.169	0.0050	mg/L	2022-02-19	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-19	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	0.000069	0.000010	mg/L	2022-02-19	
Calcium, total	6.93	0.20	mg/L	2022-02-19	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Cobalt, total	0.00627	0.00010	mg/L	2022-02-19	
Copper, total	0.266	0.00040	mg/L	2022-02-19	
Iron, total	0.179	0.010	mg/L	2022-02-19	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-19	
Lithium, total	0.00017	0.00010	mg/L	2022-02-19	
Magnesium, total	0.799	0.010	mg/L	2022-02-19	
Manganese, total	0.00631	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-22	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-19	
Nickel, total	0.00831	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	0.19	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	3.8	1.0	mg/L	2022-02-19	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-19	
Sodium, total	4.57	0.10	mg/L	2022-02-19	
Strontium, total	0.0103	0.0010	mg/L	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-02-02112022_NP (22B2327-01) Matrix: Water Sampled: 2022-02-11 10:00, Continued					
Total Metals, Continued					
Sulfur, total	5.0	3.0	mg/L	2022-02-19	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	0.0233	0.0040	mg/L	2022-02-19	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-19	

SU_SW-B-02112022_NP (22B2327-02) | Matrix: Water | Sampled: 2022-02-11 10:00

Calculated Parameters					
Hardness, Total (as CaCO3)	19.3	0.500	mg/L	N/A	
Total Metals					
Aluminum, total	0.160	0.0050	mg/L	2022-02-19	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-19	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	0.000077	0.000010	mg/L	2022-02-19	
Calcium, total	6.41	0.20	mg/L	2022-02-19	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Cobalt, total	0.00633	0.00010	mg/L	2022-02-19	
Copper, total	0.267	0.00040	mg/L	2022-02-19	
Iron, total	0.182	0.010	mg/L	2022-02-19	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-19	
Lithium, total	0.00015	0.00010	mg/L	2022-02-19	
Magnesium, total	0.789	0.010	mg/L	2022-02-19	
Manganese, total	0.00590	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000040	0.000040	mg/L	2022-02-19	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-19	
Nickel, total	0.00868	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	0.18	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	3.8	1.0	mg/L	2022-02-19	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-B-02112022_NP (22B2327-02) Matrix: Water Sampled: 2022-02-11 10:00, Continued					
<i>Total Metals, Continued</i>					
Sodium, total	4.56	0.10	mg/L	2022-02-19	
Strontium, total	0.0106	0.0010	mg/L	2022-02-19	
Sulfur, total	4.4	3.0	mg/L	2022-02-19	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	0.0219	0.0040	mg/L	2022-02-19	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-19	

SU_SW-08-02112022_NP (22B2327-03) | Matrix: Water | Sampled: 2022-02-11 10:10

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	2.02	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.016	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	< 1.0	1.0	mg/L	2022-02-18	

Calculated Parameters

Hardness, Total (as CaCO ₃)	4.92	0.500	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	0.0856	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	0.000014	0.000010	mg/L	2022-02-18	
Calcium, dissolved	1.31	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	0.00048	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.0178	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.027	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-08-02112022_NP (22B2327-03) | Matrix: Water | Sampled: 2022-02-11 10:10, Continued

Dissolved Metals, Continued

Magnesium, dissolved	0.398	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00059	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	0.00080	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	1.9	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	1.76	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0043	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	0.00024	0.00010	mg/L	2022-02-18	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	3.7	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	3.7	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Carbon, Dissolved Organic	2.47	0.50	mg/L	2022-02-18	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-22	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2022-02-22	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-18	

Total Metals

Aluminum, total	0.111	0.0050	mg/L	2022-02-19	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-08-02112022_NP (22B2327-03) Matrix: Water Sampled: 2022-02-11 10:10, Continued					
<i>Total Metals, Continued</i>					
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-19	
Calcium, total	1.54	0.20	mg/L	2022-02-19	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Cobalt, total	0.00052	0.00010	mg/L	2022-02-19	
Copper, total	0.0236	0.00040	mg/L	2022-02-19	
Iron, total	0.049	0.010	mg/L	2022-02-19	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-19	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Magnesium, total	0.428	0.010	mg/L	2022-02-19	
Manganese, total	0.00096	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-22	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-19	
Nickel, total	0.00092	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	< 0.10	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	2.1	1.0	mg/L	2022-02-19	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-19	
Sodium, total	1.87	0.10	mg/L	2022-02-19	
Strontium, total	0.0047	0.0010	mg/L	2022-02-19	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-19	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	0.0058	0.0040	mg/L	2022-02-19	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-19	

SU_SW-07-02112022_NP (22B2327-04) | Matrix: Water | Sampled: 2022-02-11 10:30

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	9.10	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.046	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	8.3	1.0	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-07-02112022_NP (22B2327-04) Matrix: Water Sampled: 2022-02-11 10:30, Continued					
Calculated Parameters					
Hardness, Total (as CaCO ₃)	17.5	0.500	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	0.0512	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	0.000072	0.000010	mg/L	2022-02-18	
Calcium, dissolved	5.85	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	0.00590	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.244	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.039	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.692	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00496	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	0.00751	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	0.17	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	3.2	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	4.29	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0093	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	0.0188	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	

General Parameters

Acidity, Total (as CaCO₃) 2.3 2.0 mg/L 2022-02-21

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-07-02112022_NP (22B2327-04) | Matrix: Water | Sampled: 2022-02-11 10:30, Continued

General Parameters, Continued

Alkalinity, Total (as CaCO ₃)	7.7	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	7.7	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Solids, Total Dissolved	54	15	mg/L	2022-02-22	HT1

Total Metals

Aluminum, total	0.0875	0.0050	mg/L	2022-02-19	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-19	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	0.000095	0.000010	mg/L	2022-02-19	
Calcium, total	6.94	0.20	mg/L	2022-02-19	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Cobalt, total	0.00693	0.00010	mg/L	2022-02-19	
Copper, total	0.284	0.00040	mg/L	2022-02-19	
Iron, total	0.079	0.010	mg/L	2022-02-19	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-19	
Lithium, total	0.00011	0.00010	mg/L	2022-02-19	
Magnesium, total	0.797	0.010	mg/L	2022-02-19	
Manganese, total	0.00561	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-22	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-19	
Nickel, total	0.00893	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	0.19	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	4.2	1.0	mg/L	2022-02-19	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-19	
Sodium, total	4.77	0.10	mg/L	2022-02-19	
Strontium, total	0.0108	0.0010	mg/L	2022-02-19	
Sulfur, total	3.2	3.0	mg/L	2022-02-19	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-07-02112022_NP (22B2327-04) | Matrix: Water | Sampled: 2022-02-11 10:30, Continued

Total Metals, Continued

Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	0.0215	0.0040	mg/L	2022-02-19	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-19	

SU_SW-09-02112022_NP (22B2327-05) | Matrix: Water | Sampled: 2022-02-11 10:45

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	1.80	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.014	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	< 1.0	1.0	mg/L	2022-02-18	

Calculated Parameters

Hardness, Total (as CaCO3)	3.85	0.500	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	0.112	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	< 0.000010	0.000010	mg/L	2022-02-18	
Calcium, dissolved	0.95	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.00193	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.019	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.361	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00021	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	< 0.00040	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	< 0.10	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	1.7	1.0	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-09-02112022_NP (22B2327-05) | Matrix: Water | Sampled: 2022-02-11 10:45, Continued

Dissolved Metals, Continued

Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	1.52	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0041	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	< 0.0040	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	

General Parameters

Acidity, Total (as CaCO3)	2.2	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO3)	3.5	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO3)	3.5	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO3)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO3)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	0.066	0.050	mg/L	2022-02-22	
Carbon, Dissolved Organic	2.51	0.50	mg/L	2022-02-18	
Solids, Total Dissolved	< 15	15	mg/L	2022-02-22	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2022-02-22	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2022-02-18	

Total Metals

Aluminum, total	0.102	0.0050	mg/L	2022-02-19	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-19	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	< 0.000010	0.000010	mg/L	2022-02-19	
Calcium, total	1.12	0.20	mg/L	2022-02-19	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Cobalt, total	< 0.00010	0.00010	mg/L	2022-02-19	
Copper, total	0.00186	0.00040	mg/L	2022-02-19	
Iron, total	0.026	0.010	mg/L	2022-02-19	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-19	
Lithium, total	< 0.00010	0.00010	mg/L	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-09-02112022_NP (22B2327-05) Matrix: Water Sampled: 2022-02-11 10:45, Continued					
Total Metals, Continued					
Magnesium, total	0.396	0.010	mg/L	2022-02-19	
Manganese, total	0.00041	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-22	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-19	
Nickel, total	< 0.00040	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	< 0.10	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	2.0	1.0	mg/L	2022-02-19	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-19	
Sodium, total	1.64	0.10	mg/L	2022-02-19	
Strontium, total	0.0042	0.0010	mg/L	2022-02-19	
Sulfur, total	< 3.0	3.0	mg/L	2022-02-19	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	< 0.0040	0.0040	mg/L	2022-02-19	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-19	

SU_SW-01-02112022_NP (22B2327-06) | Matrix: Water | Sampled: 2022-02-11 11:40

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-18	
Chloride	9.06	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.034	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	7.9	1.0	mg/L	2022-02-18	

Calculated Parameters

Hardness, Total (as CaCO ₃)	17.3	0.500	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	0.0518	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-01-02112022_NP (22B2327-06) | Matrix: Water | Sampled: 2022-02-11 11:40, Continued

Dissolved Metals, Continued

Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	0.000051	0.000010	mg/L	2022-02-18	
Calcium, dissolved	5.78	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	0.00634	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.267	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.040	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.688	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00548	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	0.00791	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	0.16	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	3.2	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	4.37	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0091	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	0.0194	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	

General Parameters

Acidity, Total (as CaCO ₃)	2.7	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	7.3	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	7.3	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Solids, Total Dissolved	51	15	mg/L	2022-02-22	HT1

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-01-02112022_NP (22B2327-06) Matrix: Water Sampled: 2022-02-11 11:40, Continued					
Total Metals					
Aluminum, total	0.488	0.0050	mg/L	2022-02-19	
Antimony, total	0.00030	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	0.0060	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	0.00032	0.00010	mg/L	2022-02-19	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	0.000109	0.000010	mg/L	2022-02-19	
Calcium, total	8.88	0.20	mg/L	2022-02-19	
Chromium, total	0.00758	0.00050	mg/L	2022-02-19	
Cobalt, total	0.00688	0.00010	mg/L	2022-02-19	
Copper, total	0.324	0.00040	mg/L	2022-02-19	
Iron, total	0.989	0.010	mg/L	2022-02-19	
Lead, total	0.00299	0.00020	mg/L	2022-02-19	
Lithium, total	0.00037	0.00010	mg/L	2022-02-19	
Magnesium, total	1.00	0.010	mg/L	2022-02-19	
Manganese, total	0.0165	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-22	
Molybdenum, total	0.00025	0.00010	mg/L	2022-02-19	
Nickel, total	0.0383	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	0.68	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	3.5	1.0	mg/L	2022-02-19	
Silver, total	0.000254	0.000050	mg/L	2022-02-19	
Sodium, total	5.25	0.10	mg/L	2022-02-19	
Strontium, total	0.0144	0.0010	mg/L	2022-02-19	
Sulfur, total	4.3	3.0	mg/L	2022-02-19	
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	0.000075	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	0.0693	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	< 0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	0.272	0.0040	mg/L	2022-02-19	
Zirconium, total	0.00118	0.00010	mg/L	2022-02-19	

SU_SW-01A-02112022_NP (22B2327-07) | Matrix: Water | Sampled: 2022-02-11 11:50

Anions

Bromide	< 0.10	0.10	mg/L	2022-02-18	
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TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU_SW-01A-02112022_NP (22B2327-07) | Matrix: Water | Sampled: 2022-02-11 11:50, Continued

Anions, Continued

Chloride	9.30	0.10	mg/L	2022-02-18	
Fluoride	< 0.10	0.10	mg/L	2022-02-18	
Nitrate (as N)	0.034	0.010	mg/L	2022-02-18	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2022-02-18	HT1
Sulfate	8.2	1.0	mg/L	2022-02-18	

Calculated Parameters

Hardness, Total (as CaCO3)	17.3	0.500	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	0.0590	0.0050	mg/L	2022-02-18	
Antimony, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Arsenic, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Barium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Beryllium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Bismuth, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Boron, dissolved	< 0.0500	0.0500	mg/L	2022-02-18	
Cadmium, dissolved	0.000078	0.000010	mg/L	2022-02-18	
Calcium, dissolved	5.79	0.20	mg/L	2022-02-18	
Chromium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Cobalt, dissolved	0.00624	0.00010	mg/L	2022-02-18	
Copper, dissolved	0.259	0.00040	mg/L	2022-02-18	
Iron, dissolved	0.041	0.010	mg/L	2022-02-18	
Lead, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Lithium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Magnesium, dissolved	0.695	0.010	mg/L	2022-02-18	
Manganese, dissolved	0.00531	0.00020	mg/L	2022-02-18	
Mercury, dissolved	< 0.000010	0.000010	mg/L	2022-02-23	
Molybdenum, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Nickel, dissolved	0.00789	0.00040	mg/L	2022-02-18	
Phosphorus, dissolved	< 0.050	0.050	mg/L	2022-02-18	
Potassium, dissolved	0.17	0.10	mg/L	2022-02-18	
Selenium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Silicon, dissolved	3.1	1.0	mg/L	2022-02-18	
Silver, dissolved	< 0.000050	0.000050	mg/L	2022-02-18	
Sodium, dissolved	4.43	0.10	mg/L	2022-02-18	
Strontium, dissolved	0.0094	0.0010	mg/L	2022-02-18	
Sulfur, dissolved	< 3.0	3.0	mg/L	2022-02-18	
Tellurium, dissolved	< 0.00050	0.00050	mg/L	2022-02-18	
Thallium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Thorium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
Tin, dissolved	< 0.00020	0.00020	mg/L	2022-02-18	
Titanium, dissolved	< 0.0050	0.0050	mg/L	2022-02-18	
Tungsten, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-01A-02112022_NP (22B2327-07) Matrix: Water Sampled: 2022-02-11 11:50, Continued					
Dissolved Metals, Continued					
Uranium, dissolved	< 0.000020	0.000020	mg/L	2022-02-18	
Vanadium, dissolved	< 0.0010	0.0010	mg/L	2022-02-18	
Zinc, dissolved	0.0193	0.0040	mg/L	2022-02-18	
Zirconium, dissolved	< 0.00010	0.00010	mg/L	2022-02-18	
General Parameters					
Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2022-02-21	
Alkalinity, Total (as CaCO ₃)	7.2	1.0	mg/L	2022-02-17	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Bicarbonate (as CaCO ₃)	7.2	1.0	mg/L	2022-02-17	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2022-02-17	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2022-02-22	
Solids, Total Dissolved	50	15	mg/L	2022-02-22	HT1
Total Metals					
Aluminum, total	0.0787	0.0050	mg/L	2022-02-19	
Antimony, total	< 0.00020	0.00020	mg/L	2022-02-19	
Arsenic, total	< 0.00050	0.00050	mg/L	2022-02-19	
Barium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Beryllium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Bismuth, total	< 0.00010	0.00010	mg/L	2022-02-19	
Boron, total	< 0.0500	0.0500	mg/L	2022-02-19	
Cadmium, total	0.000095	0.000010	mg/L	2022-02-19	
Calcium, total	7.25	0.20	mg/L	2022-02-19	
Chromium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Cobalt, total	0.00649	0.00010	mg/L	2022-02-19	
Copper, total	0.272	0.00040	mg/L	2022-02-19	
Iron, total	0.076	0.010	mg/L	2022-02-19	
Lead, total	< 0.00020	0.00020	mg/L	2022-02-19	
Lithium, total	0.00013	0.00010	mg/L	2022-02-19	
Magnesium, total	0.742	0.010	mg/L	2022-02-19	
Manganese, total	0.00571	0.00020	mg/L	2022-02-19	
Mercury, total	< 0.000010	0.000010	mg/L	2022-02-22	
Molybdenum, total	< 0.00010	0.00010	mg/L	2022-02-19	
Nickel, total	0.00821	0.00040	mg/L	2022-02-19	
Phosphorus, total	< 0.050	0.050	mg/L	2022-02-19	
Potassium, total	0.18	0.10	mg/L	2022-02-19	
Selenium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Silicon, total	3.6	1.0	mg/L	2022-02-19	
Silver, total	< 0.000050	0.000050	mg/L	2022-02-19	
Sodium, total	4.45	0.10	mg/L	2022-02-19	
Strontium, total	0.0102	0.0010	mg/L	2022-02-19	
Sulfur, total	3.2	3.0	mg/L	2022-02-19	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL	Units	Analyzed	Qualifier
SU_SW-01A-02112022_NP (22B2327-07) Matrix: Water Sampled: 2022-02-11 11:50, Continued					
<i>Total Metals, Continued</i>					
Tellurium, total	< 0.00050	0.00050	mg/L	2022-02-19	
Thallium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Thorium, total	< 0.00010	0.00010	mg/L	2022-02-19	
Tin, total	< 0.00020	0.00020	mg/L	2022-02-19	
Titanium, total	< 0.0050	0.0050	mg/L	2022-02-19	
Tungsten, total	< 0.0010	0.0010	mg/L	2022-02-19	
Uranium, total	< 0.000020	0.000020	mg/L	2022-02-19	
Vanadium, total	0.0010	0.0010	mg/L	2022-02-19	
Zinc, total	0.0216	0.0040	mg/L	2022-02-19	
Zirconium, total	< 0.00010	0.00010	mg/L	2022-02-19	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

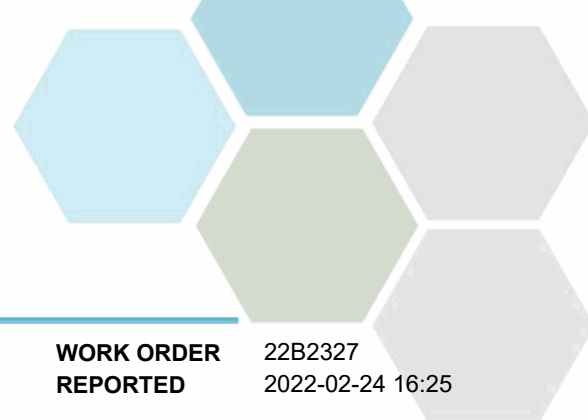
WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2017)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2017)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2017)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2017)	Ion Chromatography	✓	Kelowna
Carbon, Dissolved Organic in Water	SM 5310 B (2017)	Combustion, Infrared CO ₂ Detection	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2017)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Mercury, total in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna
Solids, Total Suspended in Water	SM 2540 D* (2017)	Gravimetry (Dried at 103-105C)	✓	Kelowna
Sulfide, Total in Water	SM 4500-S ₂ D* (2017)	Colorimetry (Methylene Blue)	✓	Edmonton
Total Metals in Water	EPA 200.2 / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association



APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B2327
REPORTED 2022-02-24 16:25

General Comments:

The results in this report apply to the samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Samples will be disposed of 30 days after the test report has been issued or once samples expire, whichever comes first. Longer hold is possible if agreed to in writing.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: bwhitehead@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (BLK):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B2B1803									
Blank (B2B1803-BLK1)			Prepared: 2022-02-17, Analyzed: 2022-02-17						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B2B1803-BLK2)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B2B1803-BS1)			Prepared: 2022-02-17, Analyzed: 2022-02-17						
Bromide	4.07	0.10 mg/L	4.00		102	85-115			
Chloride	16.3	0.10 mg/L	16.0		102	90-110			
Fluoride	4.03	0.10 mg/L	4.00		101	88-108			
Nitrate (as N)	3.95	0.010 mg/L	4.00		99	90-110			
Nitrite (as N)	2.06	0.010 mg/L	2.00		103	85-115			
Sulfate	16.0	1.0 mg/L	16.0		100	90-110			
LCS (B2B1803-BS2)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
Bromide	4.03	0.10 mg/L	4.00		101	85-115			
Chloride	16.2	0.10 mg/L	16.0		101	90-110			
Fluoride	3.96	0.10 mg/L	4.00		99	88-108			
Nitrate (as N)	3.95	0.010 mg/L	4.00		99	90-110			
Nitrite (as N)	1.99	0.010 mg/L	2.00		100	85-115			
Sulfate	16.1	1.0 mg/L	16.0		100	90-110			

Dissolved Metals, Batch B2B2057

Blank (B2B2057-BLK1)			Prepared: 2022-02-18, Analyzed: 2022-02-18						
Aluminum, dissolved	< 0.0050	0.0050 mg/L							
Antimony, dissolved	< 0.00020	0.00020 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B2B2057, Continued

Blank (B2B2057-BLK1), Continued

Prepared: 2022-02-18, Analyzed: 2022-02-18

Arsenic, dissolved	< 0.00050	0.00050 mg/L							
Barium, dissolved	< 0.0050	0.0050 mg/L							
Beryllium, dissolved	< 0.00010	0.00010 mg/L							
Bismuth, dissolved	< 0.00010	0.00010 mg/L							
Boron, dissolved	< 0.0500	0.0500 mg/L							
Cadmium, dissolved	< 0.000010	0.000010 mg/L							
Calcium, dissolved, dissolved	< 0.20	0.20 mg/L							
Chromium, dissolved	< 0.00050	0.00050 mg/L							
Cobalt, dissolved	< 0.00010	0.00010 mg/L							
Copper, dissolved	< 0.00040	0.00040 mg/L							
Iron, dissolved	< 0.010	0.010 mg/L							
Lead, dissolved	< 0.00020	0.00020 mg/L							
Lithium, dissolved	< 0.00010	0.00010 mg/L							
Magnesium, dissolved, dissolved	< 0.010	0.010 mg/L							
Manganese, dissolved	< 0.00020	0.00020 mg/L							
Molybdenum, dissolved	< 0.00010	0.00010 mg/L							
Nickel, dissolved	< 0.00040	0.00040 mg/L							
Phosphorus, dissolved	< 0.050	0.050 mg/L							
Potassium, dissolved	< 0.10	0.10 mg/L							
Selenium, dissolved	< 0.00050	0.00050 mg/L							
Silicon, dissolved	< 1.0	1.0 mg/L							
Silver, dissolved	< 0.000050	0.000050 mg/L							
Sodium, dissolved	< 0.10	0.10 mg/L							
Strontium, dissolved	< 0.0010	0.0010 mg/L							
Sulfur, dissolved	< 3.0	3.0 mg/L							
Tellurium, dissolved	< 0.00050	0.00050 mg/L							
Thallium, dissolved	< 0.000020	0.000020 mg/L							
Thorium, dissolved	< 0.00010	0.00010 mg/L							
Tin, dissolved	< 0.00020	0.00020 mg/L							
Titanium, dissolved	< 0.0050	0.0050 mg/L							
Tungsten, dissolved	< 0.0010	0.0010 mg/L							
Uranium, dissolved	< 0.000020	0.000020 mg/L							
Vanadium, dissolved	< 0.0010	0.0010 mg/L							
Zinc, dissolved	< 0.0040	0.0040 mg/L							
Zirconium, dissolved	< 0.00010	0.00010 mg/L							

LCS (B2B2057-BS1)

Prepared: 2022-02-18, Analyzed: 2022-02-18

Aluminum, dissolved	0.0234	0.0050 mg/L	0.0200		117	80-120
Antimony, dissolved	0.0165	0.00020 mg/L	0.0200		82	80-120
Arsenic, dissolved	0.0172	0.00050 mg/L	0.0200		86	80-120
Barium, dissolved	0.0174	0.0050 mg/L	0.0200		87	80-120
Beryllium, dissolved	0.0184	0.00010 mg/L	0.0200		92	80-120
Bismuth, dissolved	0.0189	0.00010 mg/L	0.0200		95	80-120
Boron, dissolved	< 0.0500	0.0500 mg/L	0.0200		99	80-120
Cadmium, dissolved	0.0180	0.000010 mg/L	0.0200		90	80-120
Calcium, dissolved, dissolved	1.72	0.20 mg/L	2.00		86	80-120
Chromium, dissolved	0.0178	0.00050 mg/L	0.0200		89	80-120
Cobalt, dissolved	0.0185	0.00010 mg/L	0.0200		93	80-120
Copper, dissolved	0.0197	0.00040 mg/L	0.0200		99	80-120
Iron, dissolved	1.83	0.010 mg/L	2.00		92	80-120
Lead, dissolved	0.0194	0.00020 mg/L	0.0200		97	80-120
Lithium, dissolved	0.0187	0.00010 mg/L	0.0200		93	80-120
Magnesium, dissolved, dissolved	1.93	0.010 mg/L	2.00		97	80-120
Manganese, dissolved	0.0197	0.00020 mg/L	0.0200		99	80-120
Molybdenum, dissolved	0.0185	0.00010 mg/L	0.0200		93	80-120
Nickel, dissolved	0.0189	0.00040 mg/L	0.0200		94	80-120

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B2B2057, Continued

LCS (B2B2057-BS1), Continued				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Phosphorus, dissolved	1.97	0.050 mg/L	2.00		98	80-120			
Potassium, dissolved	1.80	0.10 mg/L	2.00		90	80-120			
Selenium, dissolved	0.0186	0.00050 mg/L	0.0200		93	80-120			
Silicon, dissolved	2.0	1.0 mg/L	2.00		98	80-120			
Silver, dissolved	0.0185	0.000050 mg/L	0.0200		93	80-120			
Sodium, dissolved	1.86	0.10 mg/L	2.00		93	80-120			
Strontium, dissolved	0.0170	0.0010 mg/L	0.0200		85	80-120			
Sulfur, dissolved	4.9	3.0 mg/L	5.00		98	80-120			
Tellurium, dissolved	0.0174	0.00050 mg/L	0.0200		87	80-120			
Thallium, dissolved	0.0189	0.000020 mg/L	0.0200		94	80-120			
Thorium, dissolved	0.0176	0.00010 mg/L	0.0200		88	80-120			
Tin, dissolved	0.0191	0.00020 mg/L	0.0200		95	80-120			
Titanium, dissolved	0.0190	0.0050 mg/L	0.0200		95	80-120			
Tungsten, dissolved	0.0182	0.0010 mg/L	0.0200		91	80-120			
Uranium, dissolved	0.0195	0.000020 mg/L	0.0200		97	80-120			
Vanadium, dissolved	0.0167	0.0010 mg/L	0.0200		84	80-120			
Zinc, dissolved	0.0195	0.0040 mg/L	0.0200		97	80-120			
Zirconium, dissolved	0.0186	0.00010 mg/L	0.0200		93	80-120			

Reference (B2B2057-SRM1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Aluminum, dissolved	0.235	0.0050 mg/L	0.235		100	70-130			
Antimony, dissolved	0.0470	0.00020 mg/L	0.0431		109	70-130			
Arsenic, dissolved	0.415	0.00050 mg/L	0.423		98	70-130			
Barium, dissolved	2.85	0.0050 mg/L	3.30		86	70-130			
Beryllium, dissolved	0.198	0.00010 mg/L	0.209		95	70-130			
Boron, dissolved	1.56	0.0500 mg/L	1.65		94	70-130			
Cadmium, dissolved	0.208	0.000010 mg/L	0.221		94	70-130			
Calcium, dissolved, dissolved	6.74	0.20 mg/L	7.72		87	70-130			
Chromium, dissolved	0.405	0.00050 mg/L	0.434		93	70-130			
Cobalt, dissolved	0.122	0.00010 mg/L	0.124		98	70-130			
Copper, dissolved	0.830	0.00040 mg/L	0.815		102	70-130			
Iron, dissolved	1.22	0.010 mg/L	1.27		96	70-130			
Lead, dissolved	0.109	0.00020 mg/L	0.110		99	70-130			
Lithium, dissolved	0.0978	0.00010 mg/L	0.100		98	70-130			
Magnesium, dissolved, dissolved	6.48	0.010 mg/L	6.59		98	70-130			
Manganese, dissolved	0.342	0.00020 mg/L	0.342		100	70-130			
Molybdenum, dissolved	0.392	0.00010 mg/L	0.404		97	70-130			
Nickel, dissolved	0.818	0.00040 mg/L	0.835		98	70-130			
Phosphorus, dissolved	0.440	0.050 mg/L	0.499		88	70-130			
Potassium, dissolved	2.75	0.10 mg/L	2.88		95	70-130			
Selenium, dissolved	0.0320	0.00050 mg/L	0.0324		99	70-130			
Sodium, dissolved	17.6	0.10 mg/L	18.0		98	70-130			
Strontium, dissolved	0.820	0.0010 mg/L	0.935		88	70-130			
Thallium, dissolved	0.0377	0.000020 mg/L	0.0385		98	70-130			
Uranium, dissolved	0.238	0.000020 mg/L	0.258		92	70-130			
Vanadium, dissolved	0.805	0.0010 mg/L	0.873		92	70-130			
Zinc, dissolved	0.831	0.0040 mg/L	0.848		98	70-130			

Dissolved Metals, Batch B2B2367

Blank (B2B2367-BLK1)				Prepared: 2022-02-23, Analyzed: 2022-02-23					
Mercury, dissolved	< 0.000010	0.000010 mg/L							
Duplicate (B2B2367-DUP2)				Source: 22B2327-06 Prepared: 2022-02-23, Analyzed: 2022-02-23					
Mercury, dissolved	< 0.000010	0.000010 mg/L		< 0.000010				20	

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B2B2367, Continued									
Matrix Spike (B2B2367-MS2)		Source: 22B2327-07		Prepared: 2022-02-23, Analyzed: 2022-02-23					
Mercury, dissolved	0.000263	0.000010 mg/L	0.000250	< 0.000010	105	70-130			
Reference (B2B2367-SRM1)				Prepared: 2022-02-23, Analyzed: 2022-02-23					
Mercury, dissolved	0.000527	0.000010 mg/L	0.000500		105	0-200			
Reference (B2B2367-SRM2)				Prepared: 2022-02-23, Analyzed: 2022-02-23					
Mercury, dissolved	0.000491	0.000010 mg/L	0.000500		98	0-200			
Reference (B2B2367-SRM3)				Prepared: 2022-02-23, Analyzed: 2022-02-23					
Mercury, dissolved	0.000502	0.000010 mg/L	0.000500		100	0-200			
General Parameters, Batch B2B1780									
Blank (B2B1780-BLK1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B2B1780-BLK2)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
LCS (B2B1780-BS1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Carbon, Dissolved Organic	10.7	0.50 mg/L	10.0		107	78-116			
LCS (B2B1780-BS2)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Carbon, Dissolved Organic	10.2	0.50 mg/L	10.0		102	78-116			
General Parameters, Batch B2B1847									
Blank (B2B1847-BLK1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B2B1847-BLK2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B2B1847-BS1)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Total (as CaCO ₃)	106	1.0 mg/L	100		106	80-120			
LCS (B2B1847-BS2)				Prepared: 2022-02-17, Analyzed: 2022-02-17					
Alkalinity, Total (as CaCO ₃)	108	1.0 mg/L	100		108	80-120			
General Parameters, Batch B2B2006									
Blank (B2B2006-BLK1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Sulfide, Total	< 0.020	0.020 mg/L							
LCS (B2B2006-BS1)				Prepared: 2022-02-18, Analyzed: 2022-02-18					
Sulfide, Total	0.495	0.020 mg/L	0.498		99	80-120			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B2B2173									
Blank (B2B2173-BLK1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
Blank (B2B2173-BLK2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B2B2173-BS1)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	106	2.0 mg/L	100		106	80-120			
LCS (B2B2173-BS2)				Prepared: 2022-02-21, Analyzed: 2022-02-21					
Acidity, Total (as CaCO ₃)	107	2.0 mg/L	100		107	80-120			
General Parameters, Batch B2B2193									
Blank (B2B2193-BLK1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B2193-BLK2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B2B2193-BLK3)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B2B2193-BS1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	90-115			
LCS (B2B2193-BS2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	1.06	0.050 mg/L	1.00		106	90-115			
LCS (B2B2193-BS3)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Ammonia, Total (as N)	1.09	0.050 mg/L	1.00		109	90-115			
General Parameters, Batch B2B2246									
Blank (B2B2246-BLK1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Solids, Total Suspended	< 2.0	2.0 mg/L							
Blank (B2B2246-BLK2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Solids, Total Suspended	< 2.0	2.0 mg/L							
LCS (B2B2246-BS1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Solids, Total Suspended	85.0	10.0 mg/L	100		85	85-115			
LCS (B2B2246-BS2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Solids, Total Suspended	91.0	10.0 mg/L	100		91	85-115			
General Parameters, Batch B2B2266									
Blank (B2B2266-BLK1)				Prepared: 2022-02-23, Analyzed: 2022-02-23					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B2266-BS1)				Prepared: 2022-02-23, Analyzed: 2022-02-23					
Solids, Total Dissolved	230	15 mg/L	240		96	85-115			

General Parameters, Batch B2B2375

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B2B2375, Continued

Blank (B2B2375-BLK1)			Prepared: 2022-02-22, Analyzed: 2022-02-22						
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B2B2375-BS1)			Prepared: 2022-02-22, Analyzed: 2022-02-22						
Solids, Total Dissolved	238	15 mg/L	240		99	85-115			

Total Metals, Batch B2B2098

Blank (B2B2098-BLK1)			Prepared: 2022-02-19, Analyzed: 2022-02-19						
Aluminum, total	< 0.0050	0.0050 mg/L							
Antimony, total	< 0.00020	0.00020 mg/L							
Arsenic, total	< 0.00050	0.00050 mg/L							
Barium, total	< 0.0050	0.0050 mg/L							
Beryllium, total	< 0.00010	0.00010 mg/L							
Bismuth, total	< 0.00010	0.00010 mg/L							
Boron, total	< 0.0500	0.0500 mg/L							
Cadmium, total	< 0.000010	0.000010 mg/L							
Calcium, total	< 0.20	0.20 mg/L							
Chromium, total	< 0.00050	0.00050 mg/L							
Cobalt, total	< 0.00010	0.00010 mg/L							
Copper, total	< 0.00040	0.00040 mg/L							
Iron, total	< 0.010	0.010 mg/L							
Lead, total	< 0.00020	0.00020 mg/L							
Lithium, total	< 0.00010	0.00010 mg/L							
Magnesium, total	< 0.010	0.010 mg/L							
Manganese, total	< 0.00020	0.00020 mg/L							
Mercury, total	< 0.000040	0.000040 mg/L							
Molybdenum, total	< 0.00010	0.00010 mg/L							
Nickel, total	< 0.00040	0.00040 mg/L							
Phosphorus, total	< 0.050	0.050 mg/L							
Potassium, total	< 0.10	0.10 mg/L							
Selenium, total	< 0.00050	0.00050 mg/L							
Silicon, total	< 1.0	1.0 mg/L							
Silver, total	< 0.000050	0.000050 mg/L							
Sodium, total	< 0.10	0.10 mg/L							
Strontium, total	< 0.0010	0.0010 mg/L							
Sulfur, total	< 3.0	3.0 mg/L							
Tellurium, total	< 0.00050	0.00050 mg/L							
Thallium, total	< 0.000020	0.000020 mg/L							
Thorium, total	< 0.00010	0.00010 mg/L							
Tin, total	< 0.00020	0.00020 mg/L							
Titanium, total	< 0.0050	0.0050 mg/L							
Tungsten, total	< 0.0010	0.0010 mg/L							
Uranium, total	< 0.000020	0.000020 mg/L							
Vanadium, total	< 0.0010	0.0010 mg/L							
Zinc, total	< 0.0040	0.0040 mg/L							
Zirconium, total	< 0.00010	0.00010 mg/L							

LCS (B2B2098-BS1)			Prepared: 2022-02-19, Analyzed: 2022-02-19						
Aluminum, total	0.0223	0.0050 mg/L	0.0200		112	80-120			
Antimony, total	0.0191	0.00020 mg/L	0.0200		96	80-120			
Arsenic, total	0.0189	0.00050 mg/L	0.0200		95	80-120			
Barium, total	0.0185	0.0050 mg/L	0.0200		93	80-120			
Beryllium, total	0.0189	0.00010 mg/L	0.0200		94	80-120			
Bismuth, total	0.0175	0.00010 mg/L	0.0200		87	80-120			
Boron, total	< 0.0500	0.0500 mg/L	0.0200		104	80-120			
Cadmium, total	0.0186	0.000010 mg/L	0.0200		93	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
631848

WORK ORDER REPORTED 22B2327
2022-02-24 16:25

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B2B2098, Continued									
LCS (B2B2098-BS1), Continued					Prepared: 2022-02-19, Analyzed: 2022-02-19				
Calcium, total	1.98	0.20 mg/L	2.00		99	80-120			
Chromium, total	0.0194	0.00050 mg/L	0.0200		97	80-120			
Cobalt, total	0.0190	0.00010 mg/L	0.0200		95	80-120			
Copper, total	0.0196	0.00040 mg/L	0.0200		98	80-120			
Iron, total	1.91	0.010 mg/L	2.00		95	80-120			
Lead, total	0.0189	0.00020 mg/L	0.0200		95	80-120			
Lithium, total	0.0191	0.00010 mg/L	0.0200		95	80-120			
Magnesium, total	1.95	0.010 mg/L	2.00		98	80-120			
Manganese, total	0.0190	0.00020 mg/L	0.0200		95	80-120			
Mercury, total	0.000959	0.000040 mg/L	0.00101		95	80-120			
Molybdenum, total	0.0185	0.00010 mg/L	0.0200		93	80-120			
Nickel, total	0.0190	0.00040 mg/L	0.0200		95	80-120			
Phosphorus, total	1.92	0.050 mg/L	2.00		96	80-120			
Potassium, total	1.77	0.10 mg/L	2.00		89	80-120			
Selenium, total	0.0192	0.00050 mg/L	0.0200		96	80-120			
Silicon, total	2.1	1.0 mg/L	2.00		107	80-120			
Silver, total	0.0183	0.000050 mg/L	0.0200		91	80-120			
Sodium, total	1.90	0.10 mg/L	2.00		95	80-120			
Strontium, total	0.0176	0.0010 mg/L	0.0200		88	80-120			
Sulfur, total	4.1	3.0 mg/L	5.00		83	80-120			
Tellurium, total	0.0188	0.00050 mg/L	0.0200		94	80-120			
Thallium, total	0.0170	0.000020 mg/L	0.0200		85	80-120			
Thorium, total	0.0178	0.00010 mg/L	0.0200		89	80-120			
Tin, total	0.0190	0.00020 mg/L	0.0200		95	80-120			
Titanium, total	0.0192	0.0050 mg/L	0.0200		96	80-120			
Tungsten, total	0.0187	0.0010 mg/L	0.0200		94	80-120			
Uranium, total	0.0178	0.000020 mg/L	0.0200		89	80-120			
Vanadium, total	0.0195	0.0010 mg/L	0.0200		98	80-120			
Zinc, total	0.0205	0.0040 mg/L	0.0200		102	80-120			
Zirconium, total	0.0195	0.00010 mg/L	0.0200		97	80-120			
Reference (B2B2098-SRM1)					Prepared: 2022-02-19, Analyzed: 2022-02-19				
Aluminum, total	0.209	0.0050 mg/L	0.198		105	70-130			
Antimony, total	0.0230	0.00020 mg/L	0.0230		100	70-130			
Arsenic, total	0.0205	0.00050 mg/L	0.0200		103	70-130			
Barium, total	0.0154	0.0050 mg/L	0.0161		96	70-130			
Beryllium, total	0.00381	0.00010 mg/L	0.00384		99	70-130			
Boron, total	0.192	0.0500 mg/L	0.191		101	70-130			
Cadmium, total	0.00389	0.000010 mg/L	0.00404		96	70-130			
Calcium, total	1.00	0.20 mg/L	0.938		107	70-130			
Chromium, total	0.0258	0.00050 mg/L	0.0256		101	70-130			
Cobalt, total	0.0215	0.00010 mg/L	0.0214		100	70-130			
Copper, total	0.0322	0.00040 mg/L	0.0322		100	70-130			
Iron, total	0.057	0.010 mg/L	0.0580		99	70-130			
Lead, total	0.00786	0.00020 mg/L	0.00796		99	70-130			
Lithium, total	0.00946	0.00010 mg/L	0.0102		93	70-130			
Magnesium, total	0.113	0.010 mg/L	0.112		101	70-130			
Manganese, total	0.0117	0.00020 mg/L	0.0120		97	70-130			
Molybdenum, total	0.0428	0.00010 mg/L	0.0438		98	70-130			
Nickel, total	0.0394	0.00040 mg/L	0.0394		100	70-130			
Potassium, total	0.72	0.10 mg/L	0.820		87	70-130			
Selenium, total	0.115	0.00050 mg/L	0.117		98	70-130			
Sodium, total	0.49	0.10 mg/L	0.490		99	70-130			
Strontium, total	0.254	0.0010 mg/L	0.276		92	70-130			
Thallium, total	0.0106	0.000020 mg/L	0.0118		90	70-130			
Uranium, total	0.00882	0.000020 mg/L	0.00970		91	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 631848

WORK ORDER 22B2327
REPORTED 2022-02-24 16:25

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B2B2098, Continued									
Reference (B2B2098-SRM1), Continued				Prepared: 2022-02-19, Analyzed: 2022-02-19					
Vanadium, total	0.0280	0.0010 mg/L	0.0274		102	70-130			
Zinc, total	0.0880	0.0040 mg/L	0.0884		100	70-130			
Total Metals, Batch B2B2213									
Blank (B2B2213-BLK1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Mercury, total	< 0.000010	0.000010 mg/L							
Duplicate (B2B2213-DUP2)				Source: 22B2327-05	Prepared: 2022-02-22, Analyzed: 2022-02-22				
Mercury, total	< 0.000010	0.000010 mg/L		< 0.000010			20		
Matrix Spike (B2B2213-MS2)				Source: 22B2327-06	Prepared: 2022-02-22, Analyzed: 2022-02-22				
Mercury, total	0.000231	0.000010 mg/L	0.000250	< 0.000010	92	70-130			
Reference (B2B2213-SRM1)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Mercury, total	0.000484	0.000010 mg/L	0.000500		97	0-200			
Reference (B2B2213-SRM2)				Prepared: 2022-02-22, Analyzed: 2022-02-22					
Mercury, total	0.000485	0.000010 mg/L	0.000500		97	0-200			



RUSH for nitrate/nitrite

COC ID:		TURNAROUND TIME:		Regular		RUSH:					
PROJECT/CLIENT INFO				LABORATORY				OTHER INFO			
Facility Name / Job# Sunro Mine				Lab Name Caro		Report Format / Distribution		Excel	PDF	EDI	
Project Manager Michelle Unger				Lab Contact Brent Whitehead		Email 1: michelle.unger@teck.com		X	X	X	
Email michele.unger@teck.com				Email bwhitehead@caro.ca		Email 2: craig.pinnell@snc-lavalin.com		X	X	X	
Address 601 Knighton Rd				Address 101 - 3677 Highway 97N		Email 3: jonathan.croston@snc-lavalin.com		X	X	X	
						Email 4: josh.baran@snc-lavalin.com		X	X	X	
City Kimberly Province BC				City Kelowna Province BC		Email 5: ENV_West_BC_LabData@snc-lavalin.com		X	X	X	
Postal Code V1A 1C7 Country Canada				Postal Code V1X5C3 Country CA							
Phone Number 250-427-8422				Phone Number 250-765-9646		PO number		669765			
SAMPLE DETAILS								ANALYSIS REQUESTED			
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	ANALYSIS	Yes	Yes	Yes
SU_SW-02-02112022_NP	SW15-02	SW	N	2022-02-11	10:00	G	6	Total Metals + Hg in Water	X	X	X
SU_SW-B-02112022_NP	-	SW	N	2022-02-11	10:00	G	2	Dissolved Metals + Hg in Water	X	X	X
SU_SW-08-02112022_NP	SW15-08	SW	N	2022-02-11	10:10	G	9	TDS	X	X	X
SU_SW-07-02112022_NP	SW15-07	SW	N	2022-02-11	10:30	G	6	Hardness	X	X	X
SU_SW-09-02112022_NP	SW15-09	SW	N	2022-02-11	10:45	G	9	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	X	X	X
SU_SW-01-02112022_NP	SW15-01	SW	N	2022-02-11	11:40	G	6	BTEX/VPH	X	X	X
SU_SW-01A-02112022_NP	SW15-01A	SW	N	2022-02-11	11:50	G	6	TSS	X	X	X
								Total and Speciated Alkalinity, Acidity	X	X	X
								Sulfide	X	X	X
								Ammonia	X	X	X
								Dissolved Organic Carbon	X	X	X
								HOLD			
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS				RELINQUISHED BY/AFFILIATION		DATE/TIME		ACCEPTED BY/AFFILIATION		DATE/TIME	
Please note custody seal condition. Dissolved metals and DOC field filtered.				Josh Baran SNC-Lavalin Inc		22/02/14 15:00					
Invoice to: SNC-Lavalin Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3											
Send to: payables@snc-lavalin.com Project number: 631848 Billing PO: 682701										LG / Purv 9.45 Feb 16 5.60C	
NB OF BOTTLES RETURNED/DESCRIPTION				Sampler's Name		Mobile #		Date/Time			
Regular (default) X				Josh Baran		250-418-1562					
Priority (2-3 business days) - 50% surcharge											
Emergency (1 Business Day) - 100% surcharge											
For Emergency <1 Day, ASAP or Weekend											

custody seal ✓

CERTIFICATE OF ANALYSIS

REPORTED TO SNC-Lavalin Inc. (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Craig Pinnell

PO NUMBER 697296

PROJECT 697296

PROJECT INFO Teck Sunro Mine

WORK ORDER 23H1041

RECEIVED / TEMP 2023-08-08 11:45 / 20.0°C

REPORTED 2023-08-15 17:02

COC NUMBER 697296-20230803-01

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

Work Order Comments:

Custody Seals Intact: YES

By engaging our services, you are agreeing to CARO Analytical Service's Standard Terms and Conditions outlined here:
<https://www.caro.ca/terms-conditions>

If you have any questions or concerns, please contact me at rpschyk@caro.ca

Authorized By:

Regan Pshyk
Account Manager



1-888-311-8846 | www.caro.ca

#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-02-08012023-NP (23H1041-01) Matrix: Ground Water Sampled: 2023-08-01 10:25					
Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	4.71	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.604	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	25.3	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	42.3	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	5.2	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Barium, dissolved	11.8	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	21.5	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.191	0.0020	µg/L	2023-08-10	
Calcium, dissolved	13100	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	27.2	0.0050	µg/L	2023-08-10	
Copper, dissolved	373	0.10	µg/L	2023-08-10	
Iron, dissolved	< 2.0	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.234	0.050	µg/L	2023-08-10	
Magnesium, dissolved	2310	5.0	µg/L	2023-08-10	
Manganese, dissolved	0.570	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.060	0.010	µg/L	2023-08-11	
Nickel, dissolved	31.2	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	581	20	µg/L	2023-08-10	
Selenium, dissolved	0.59	0.10	µg/L	2023-08-10	
Silicon, dissolved	9060	100	µg/L	2023-08-10	
Silver, dissolved	0.011	0.010	µg/L	2023-08-10	
Sodium, dissolved	6570	20	µg/L	2023-08-10	
Strontium, dissolved	34.9	0.10	µg/L	2023-08-10	
Sulfur, dissolved	8760	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-02-08012023-NP (23H1041-01) Matrix: Ground Water Sampled: 2023-08-01 10:25, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0028	0.0010	µg/L	2023-08-11	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	61.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	11.8	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	22.7	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	22.7	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.060	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	66	15	mg/L	2023-08-10	HT1

SU-GW-03-07312023-NP (23H1041-02) | Matrix: Ground Water | Sampled: 2023-07-31 17:10

<i>Anions</i>					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	4.96	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.278	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	3.7	1.0	mg/L	2023-08-09	

<i>Calculated Parameters</i>					
Hardness, Dissolved (as CaCO ₃)	39.3	0.125	mg/L	N/A	

<i>Dissolved Metals</i>					
Aluminum, dissolved	3.9	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.057	0.050	µg/L	2023-08-10	
Barium, dissolved	2.89	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	25.1	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0101	0.0020	µg/L	2023-08-10	
Calcium, dissolved	11000	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.190	0.0050	µg/L	2023-08-10	
Copper, dissolved	4.55	0.10	µg/L	2023-08-10	
Iron, dissolved	2.8	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-03-07312023-NP (23H1041-02) | Matrix: Ground Water | Sampled: 2023-07-31 17:10, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.106	0.050	µg/L	2023-08-10	
Magnesium, dissolved	2870	5.0	µg/L	2023-08-10	
Manganese, dissolved	0.199	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.319	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.560	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	651	20	µg/L	2023-08-10	
Selenium, dissolved	0.14	0.10	µg/L	2023-08-10	
Silicon, dissolved	6750	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	5850	20	µg/L	2023-08-10	
Strontium, dissolved	25.0	0.10	µg/L	2023-08-10	
Sulfur, dissolved	1150	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	0.0050	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0028	0.0010	µg/L	2023-08-11	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	3.0	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	6.9	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	44.9	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	44.9	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	54	15	mg/L	2023-08-10	HT1

SU-GW-04-07312023-NP (23H1041-03) | Matrix: Ground Water | Sampled: 2023-07-31 13:25

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	3.65	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.518	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-04-07312023-NP (23H1041-03) Matrix: Ground Water Sampled: 2023-07-31 13:25, Continued					
Anions, Continued					
Sulfate	21.5	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO ₃)	67.6	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	8.3	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.192	0.050	µg/L	2023-08-10	
Barium, dissolved	9.10	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	15.9	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0511	0.0020	µg/L	2023-08-10	
Calcium, dissolved	19300	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	1.48	0.0050	µg/L	2023-08-10	
Copper, dissolved	56.4	0.10	µg/L	2023-08-10	
Iron, dissolved	5.5	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.392	0.050	µg/L	2023-08-10	
Magnesium, dissolved	4700	5.0	µg/L	2023-08-10	
Manganese, dissolved	0.672	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.153	0.010	µg/L	2023-08-10	
Nickel, dissolved	5.13	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	467	20	µg/L	2023-08-10	
Selenium, dissolved	0.21	0.10	µg/L	2023-08-10	
Silicon, dissolved	9800	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	4820	20	µg/L	2023-08-10	
Strontium, dissolved	36.2	0.10	µg/L	2023-08-10	
Sulfur, dissolved	7420	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	0.0050	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	3.75	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0297	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	10.8	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-04-07312023-NP (23H1041-03) Matrix: Ground Water Sampled: 2023-07-31 13:25, Continued					
General Parameters					
Acidity, Total (as CaCO ₃)	3.2	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	61.3	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	61.3	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.060	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	113	15	mg/L	2023-08-10	HT1

SU-GW-06-08012023-NP (23H1041-04) | Matrix: Ground Water | Sampled: 2023-08-01 11:25

Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	7.06	0.10	mg/L	2023-08-09	
Fluoride	0.17	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.019	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	37.0	1.0	mg/L	2023-08-09	

Calculated Parameters					
Hardness, Dissolved (as CaCO ₃)	225	0.125	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	95.0	1.0	µg/L	2023-08-10	
Antimony, dissolved	0.142	0.050	µg/L	2023-08-10	
Arsenic, dissolved	1.39	0.050	µg/L	2023-08-10	
Barium, dissolved	62.0	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	27.9	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0298	0.0020	µg/L	2023-08-10	
Calcium, dissolved	72000	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	6.65	0.0050	µg/L	2023-08-10	
Copper, dissolved	2.18	0.10	µg/L	2023-08-10	
Iron, dissolved	16.4	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	8.07	0.050	µg/L	2023-08-10	
Magnesium, dissolved	11000	5.0	µg/L	2023-08-10	
Manganese, dissolved	429	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	2.76	0.010	µg/L	2023-08-10	
Nickel, dissolved	1.40	0.040	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-06-08012023-NP (23H1041-04) | Matrix: Ground Water | Sampled: 2023-08-01 11:25, Continued

Dissolved Metals, Continued

Phosphorus, dissolved	12	10	µg/L	2023-08-10	
Potassium, dissolved	2570	20	µg/L	2023-08-10	
Selenium, dissolved	0.11	0.10	µg/L	2023-08-10	
Silicon, dissolved	10400	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	54600	20	µg/L	2023-08-10	
Strontium, dissolved	121	0.10	µg/L	2023-08-10	
Sulfur, dissolved	13400	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	0.0222	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	0.987	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	36.4	0.20	µg/L	2023-08-10	
Uranium, dissolved	1.73	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	5.8	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.032	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	7.8	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	294	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	294	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.078	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	346	15	mg/L	2023-08-10	HT1

SU-GW-07-07312023-NP (23H1041-05) | Matrix: Ground Water | Sampled: 2023-07-31 15:35

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	20.7	0.10	mg/L	2023-08-09	
Fluoride	0.15	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.653	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	79.9	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	109	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	98.5	1.0	µg/L	2023-08-10	
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TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-07-07312023-NP (23H1041-05) | Matrix: Ground Water | Sampled: 2023-07-31 15:35, Continued

Dissolved Metals, Continued

Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.076	0.050	µg/L	2023-08-10	
Barium, dissolved	7.38	0.10	µg/L	2023-08-10	
Beryllium, dissolved	0.070	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	77.8	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.755	0.0020	µg/L	2023-08-10	
Calcium, dissolved	31600	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	91.0	0.0050	µg/L	2023-08-10	
Copper, dissolved	1620	0.10	µg/L	2023-08-10	
Iron, dissolved	35.3	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.344	0.050	µg/L	2023-08-10	
Magnesium, dissolved	7370	5.0	µg/L	2023-08-10	
Manganese, dissolved	12.2	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.037	0.010	µg/L	2023-08-11	
Nickel, dissolved	153	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	11	10	µg/L	2023-08-10	
Potassium, dissolved	541	20	µg/L	2023-08-10	
Selenium, dissolved	0.20	0.10	µg/L	2023-08-10	
Silicon, dissolved	9680	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	12000	20	µg/L	2023-08-10	
Strontium, dissolved	87.8	0.10	µg/L	2023-08-10	
Sulfur, dissolved	28700	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.29	0.20	µg/L	2023-08-10	
Tungsten, dissolved	2.23	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0120	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	207	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.021	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	12.7	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	15.3	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	15.3	1.0	mg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-07-07312023-NP (23H1041-05) Matrix: Ground Water Sampled: 2023-07-31 15:35, Continued					
<i>General Parameters, Continued</i>					
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	172	15	mg/L	2023-08-10	HT1

SU-GW-09-08012023-NP (23H1041-06) | Matrix: Ground Water | Sampled: 2023-08-01 10:20

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	1.91	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.109	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.1	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.82	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	40.7	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.073	0.050	µg/L	2023-08-10	
Barium, dissolved	2.29	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	3.2	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0023	0.0020	µg/L	2023-08-10	
Calcium, dissolved	1930	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.0375	0.0050	µg/L	2023-08-10	
Copper, dissolved	1.68	0.10	µg/L	2023-08-10	
Iron, dissolved	85.5	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.098	0.050	µg/L	2023-08-10	
Magnesium, dissolved	484	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.21	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.033	0.010	µg/L	2023-08-11	
Nickel, dissolved	0.369	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	255	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1720	100	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-09-08012023-NP (23H1041-06) | Matrix: Ground Water | Sampled: 2023-08-01 10:20, Continued

Dissolved Metals, Continued

Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1830	20	µg/L	2023-08-10	
Strontium, dissolved	9.42	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.26	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.8	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	9.0	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	9.0	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

SU-GW-B-07312023-NP (23H1041-07) | Matrix: Ground Water | Sampled: 2023-07-31 12:00

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	3.64	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.518	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	21.6	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	67.4	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	5.5	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.204	0.050	µg/L	2023-08-10	
Barium, dissolved	9.50	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-B-07312023-NP (23H1041-07) | Matrix: Ground Water | Sampled: 2023-07-31 12:00, Continued

Dissolved Metals, Continued

Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	15.4	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0516	0.0020	µg/L	2023-08-10	
Calcium, dissolved	19200	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	1.46	0.0050	µg/L	2023-08-10	
Copper, dissolved	56.3	0.10	µg/L	2023-08-10	
Iron, dissolved	4.0	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.396	0.050	µg/L	2023-08-10	
Magnesium, dissolved	4710	5.0	µg/L	2023-08-10	
Manganese, dissolved	0.772	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.175	0.010	µg/L	2023-08-10	
Nickel, dissolved	5.14	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	464	20	µg/L	2023-08-10	
Selenium, dissolved	0.22	0.10	µg/L	2023-08-10	
Silicon, dissolved	9630	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	4880	20	µg/L	2023-08-10	
Strontium, dissolved	36.3	0.10	µg/L	2023-08-10	
Sulfur, dissolved	7360	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	0.0048	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	4.35	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0307	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	11.1	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	60.4	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	60.4	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	92	15	mg/L	2023-08-10	HT1

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-EQUIPMENT-08012023-NP (23H1041-08) Matrix: Water Sampled: 2023-08-01 13:20					
Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	< 0.10	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	< 1.0	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	< 0.125	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	< 1.0	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Barium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	< 2.0	2.0	µg/L	2023-08-10	
Cadmium, dissolved	< 0.0020	0.0020	µg/L	2023-08-10	
Calcium, dissolved	< 50	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	< 0.0050	0.0050	µg/L	2023-08-10	
Copper, dissolved	< 0.10	0.10	µg/L	2023-08-11	
Iron, dissolved	< 2.0	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Magnesium, dissolved	< 5.0	5.0	µg/L	2023-08-10	
Manganese, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Nickel, dissolved	< 0.040	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	< 20	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	< 100	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	< 20	20	µg/L	2023-08-10	
Strontium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-EQUIPMENT-08012023-NP (23H1041-08) | Matrix: Water | Sampled: 2023-08-01 13:20, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	< 1.0	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	2.0	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	2.0	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	< 0.50	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	< 2.0	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	< 0.050	0.050	µg/L	2023-08-15	
Barium, total	< 0.10	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	< 2.0	2.0	µg/L	2023-08-15	
Cadmium, total	< 0.0020	0.0020	µg/L	2023-08-15	
Calcium, total	< 50	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	< 0.0050	0.0050	µg/L	2023-08-15	
Copper, total	< 0.20	0.20	µg/L	2023-08-15	
Iron, total	< 2.0	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	< 0.050	0.050	µg/L	2023-08-15	
Magnesium, total	< 5.0	5.0	µg/L	2023-08-15	
Manganese, total	< 0.050	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	< 0.010	0.010	µg/L	2023-08-15	
Nickel, total	< 0.040	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	< 20	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	< 100	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-EQUIPMENT-08012023-NP (23H1041-08) Matrix: Water Sampled: 2023-08-01 13:20, Continued					
Total Metals, Continued					
Sodium, total	29	20	µg/L	2023-08-15	RE2
Strontium, total	< 0.10	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	< 0.20	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	< 0.0010	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	22.1	1.0	µg/L	2023-08-15	RE2
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-FIELD-08012023-NP (23H1041-09) | Matrix: Water | Sampled: 2023-08-01 15:00

Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	< 0.10	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	< 1.0	1.0	mg/L	2023-08-09	

Calculated Parameters					
Hardness, Total (as CaCO ₃)	< 0.125	0.125	mg/L	N/A	

General Parameters					
Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

Total Metals					
Aluminum, total	< 2.0	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	< 0.050	0.050	µg/L	2023-08-15	
Barium, total	< 0.10	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-FIELD-08012023-NP (23H1041-09) Matrix: Water Sampled: 2023-08-01 15:00, Continued					
Total Metals, Continued					
Boron, total	< 2.0	2.0	µg/L	2023-08-15	
Cadmium, total	< 0.0020	0.0020	µg/L	2023-08-15	
Calcium, total	< 50	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	< 0.0050	0.0050	µg/L	2023-08-15	
Copper, total	< 0.20	0.20	µg/L	2023-08-15	
Iron, total	< 2.0	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	< 0.050	0.050	µg/L	2023-08-15	
Magnesium, total	< 5.0	5.0	µg/L	2023-08-15	
Manganese, total	< 0.050	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.045	0.010	µg/L	2023-08-15	RE2
Nickel, total	< 0.040	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	< 20	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	< 100	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	< 20	20	µg/L	2023-08-15	
Strontium, total	< 0.10	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	< 0.20	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	< 0.0010	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	< 1.0	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-TRIP-07272023-NP (23H1041-10) | Matrix: Water | Sampled: 2023-07-27 14:25

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	< 0.10	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	< 1.0	1.0	mg/L	2023-08-09	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-TRIP-07272023-NP (23H1041-10) | Matrix: Water | Sampled: 2023-07-27 14:25, Continued

Calculated Parameters

Hardness, Total (as CaCO3)	< 0.125	0.125	mg/L	N/A	
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General Parameters

Acidity, Total (as CaCO3)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Total Organic	< 0.50	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	< 2.0	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	< 0.050	0.050	µg/L	2023-08-15	
Barium, total	< 0.10	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	< 2.0	2.0	µg/L	2023-08-15	
Cadmium, total	< 0.0020	0.0020	µg/L	2023-08-15	
Calcium, total	< 50	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	< 0.0050	0.0050	µg/L	2023-08-15	
Copper, total	< 0.20	0.20	µg/L	2023-08-15	
Iron, total	< 2.0	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	< 0.050	0.050	µg/L	2023-08-15	
Magnesium, total	< 5.0	5.0	µg/L	2023-08-15	
Manganese, total	< 0.050	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	< 0.010	0.010	µg/L	2023-08-15	
Nickel, total	< 0.040	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	< 20	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	< 100	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	< 20	20	µg/L	2023-08-15	
Strontium, total	< 0.10	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-TRIP-07272023-NP (23H1041-10) Matrix: Water Sampled: 2023-07-27 14:25, Continued					
<i>Total Metals, Continued</i>					
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	< 0.20	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	< 0.0010	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	< 1.0	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-PW-10-08012023-NP (23H1041-11) | Matrix: Potable Water | Sampled: 2023-08-01 15:15

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.31	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.111	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.4	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	7.31	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	34.4	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.073	0.050	µg/L	2023-08-10	
Barium, dissolved	2.01	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	5.5	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0078	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2110	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.303	0.0050	µg/L	2023-08-10	
Copper, dissolved	13.8	0.10	µg/L	2023-08-10	
Iron, dissolved	74.4	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.097	0.050	µg/L	2023-08-10	
Magnesium, dissolved	494	5.0	µg/L	2023-08-10	
Manganese, dissolved	0.958	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.031	0.010	µg/L	2023-08-11	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-10-08012023-NP (23H1041-11) | Matrix: Potable Water | Sampled: 2023-08-01 15:15, Continued

Dissolved Metals, Continued

Nickel, dissolved	0.671	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	248	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1810	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2020	20	µg/L	2023-08-10	
Strontium, dissolved	8.83	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.22	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.6	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.026	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.2	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.2	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

SU-PW-11-08012023-NP (23H1041-12) | Matrix: Potable Water | Sampled: 2023-08-01 13:30

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.34	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.123	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.6	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	8.03	0.125	mg/L	N/A	
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Dissolved Metals

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-11-08012023-NP (23H1041-12) | Matrix: Potable Water | Sampled: 2023-08-01 13:30, Continued

Dissolved Metals, Continued

Aluminum, dissolved	53.0	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.067	0.050	µg/L	2023-08-10	
Barium, dissolved	2.22	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	5.2	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0125	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2290	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.436	0.0050	µg/L	2023-08-10	
Copper, dissolved	15.7	0.10	µg/L	2023-08-10	
Iron, dissolved	92.1	2.0	µg/L	2023-08-10	
Lead, dissolved	0.075	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.109	0.050	µg/L	2023-08-10	
Magnesium, dissolved	558	5.0	µg/L	2023-08-10	
Manganese, dissolved	3.20	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.041	0.010	µg/L	2023-08-11	
Nickel, dissolved	0.872	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	272	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	2040	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2090	20	µg/L	2023-08-10	
Strontium, dissolved	9.18	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	1.12	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	3.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.023	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	6.1	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-11-08012023-NP (23H1041-12) Matrix: Potable Water Sampled: 2023-08-01 13:30, Continued					
<i>General Parameters, Continued</i>					
Alkalinity, Bicarbonate (as CaCO ₃)	6.1	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.092	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

SU-PW-17-08012023-NP (23H1041-13) | Matrix: Potable Water | Sampled: 2023-08-01 13:10

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.33	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.105	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.4	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	7.55	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	24.3	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.059	0.050	µg/L	2023-08-10	
Barium, dissolved	0.43	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	5.3	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0292	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2200	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.576	0.0050	µg/L	2023-08-10	
Copper, dissolved	43.6	0.10	µg/L	2023-08-10	
Iron, dissolved	52.5	2.0	µg/L	2023-08-10	
Lead, dissolved	0.063	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.081	0.050	µg/L	2023-08-10	
Magnesium, dissolved	497	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.36	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.053	0.010	µg/L	2023-08-11	
Nickel, dissolved	1.77	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	241	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-17-08012023-NP (23H1041-13) | Matrix: Potable Water | Sampled: 2023-08-01 13:10, Continued

Dissolved Metals, Continued

Silicon, dissolved	1980	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2070	20	µg/L	2023-08-10	
Strontium, dissolved	6.27	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.26	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	3.9	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	6.0	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	6.0	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	15	15	mg/L	2023-08-10	HT1

SU-PW-19-08012023-NP (23H1041-14) | Matrix: Potable Water | Sampled: 2023-08-01 14:45

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.35	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.118	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.5	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	7.48	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	59.4	1.0	µg/L	2023-08-10	
Antimony, dissolved	0.053	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.068	0.050	µg/L	2023-08-10	
Barium, dissolved	1.21	0.10	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-19-08012023-NP (23H1041-14) | Matrix: Potable Water | Sampled: 2023-08-01 14:45, Continued

Dissolved Metals, Continued

Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	5.3	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0119	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2100	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.280	0.0050	µg/L	2023-08-10	
Copper, dissolved	16.2	0.10	µg/L	2023-08-10	
Iron, dissolved	94.5	2.0	µg/L	2023-08-10	
Lead, dissolved	0.173	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.086	0.050	µg/L	2023-08-10	
Magnesium, dissolved	542	5.0	µg/L	2023-08-10	
Manganese, dissolved	2.64	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.050	0.010	µg/L	2023-08-11	
Nickel, dissolved	0.847	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	251	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1960	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2030	20	µg/L	2023-08-10	
Strontium, dissolved	8.43	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	1.43	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0014	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	2.6	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.022	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.5	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.5	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-19-08012023-NP (23H1041-14) | Matrix: Potable Water | Sampled: 2023-08-01 14:45, Continued

General Parameters, Continued

Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
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SU-PW-08-0802023-NP (23H1041-15) | Matrix: Potable Water | Sampled: 2023-08-02 12:05

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.91	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.103	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.9	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	8.26	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	38.7	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.085	0.050	µg/L	2023-08-10	
Barium, dissolved	2.28	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	7.0	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0098	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2480	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.754	0.0050	µg/L	2023-08-10	
Copper, dissolved	23.4	0.10	µg/L	2023-08-10	
Iron, dissolved	77.8	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.102	0.050	µg/L	2023-08-10	
Magnesium, dissolved	500	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.33	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.034	0.010	µg/L	2023-08-11	
Nickel, dissolved	1.24	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	242	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1990	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2250	20	µg/L	2023-08-10	
Strontium, dissolved	8.95	0.10	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-08-0802023-NP (23H1041-15) | Matrix: Potable Water | Sampled: 2023-08-02 12:05, Continued

Dissolved Metals, Continued

Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.34	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	3.0	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	4.9	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	4.9	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	24	15	mg/L	2023-08-10	HT1

SU-PW-09-0802023-NP (23H1041-16) | Matrix: Potable Water | Sampled: 2023-08-02 12:10

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	8.99	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.487	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	33.5	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	84.5	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	13.4	1.0	µg/L	2023-08-10	
Antimony, dissolved	0.167	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.256	0.050	µg/L	2023-08-10	
Barium, dissolved	7.20	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	31.2	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0152	0.0020	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-09-08022023-NP (23H1041-16) | Matrix: Potable Water | Sampled: 2023-08-02 12:10, Continued

Dissolved Metals, Continued

Calcium, dissolved	25800	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.193	0.0050	µg/L	2023-08-10	
Copper, dissolved	2.91	0.10	µg/L	2023-08-10	
Iron, dissolved	8.7	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.429	0.050	µg/L	2023-08-10	
Magnesium, dissolved	4860	5.0	µg/L	2023-08-10	
Manganese, dissolved	12.5	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.961	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.628	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	931	20	µg/L	2023-08-10	
Selenium, dissolved	1.37	0.10	µg/L	2023-08-10	
Silicon, dissolved	7910	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	9540	20	µg/L	2023-08-10	
Strontium, dissolved	24.5	0.10	µg/L	2023-08-10	
Sulfur, dissolved	11000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	0.0041	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	0.193	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	9.13	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0300	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	1.16	1.00	µg/L	2023-08-10	
Zinc, dissolved	4.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	68.9	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	68.9	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	139	15	mg/L	2023-08-10	HT1

SU-PW-21-08022023-NP (23H1041-17) | Matrix: Potable Water | Sampled: 2023-08-02 10:40

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-21-08022023-NP (23H1041-17) Matrix: Potable Water Sampled: 2023-08-02 10:40, Continued					
Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.34	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.102	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.3	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	7.44	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	40.8	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.087	0.050	µg/L	2023-08-10	
Barium, dissolved	2.28	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	5.4	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0053	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2170	50	µg/L	2023-08-10	
Chromium, dissolved	0.91	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.346	0.0050	µg/L	2023-08-10	
Copper, dissolved	11.5	0.10	µg/L	2023-08-10	
Iron, dissolved	92.6	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.105	0.050	µg/L	2023-08-10	
Magnesium, dissolved	489	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.28	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.069	0.010	µg/L	2023-08-11	
Nickel, dissolved	0.981	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	250	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1800	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1970	20	µg/L	2023-08-10	
Strontium, dissolved	9.22	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.29	0.20	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-21-08022023-NP (23H1041-17) Matrix: Potable Water Sampled: 2023-08-02 10:40, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	4.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.3	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.3	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

SU-PW-22-08022023-NP (23H1041-18) | Matrix: Potable Water | Sampled: 2023-08-02 10:35

<i>Anions</i>					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.31	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.114	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.4	1.0	mg/L	2023-08-09	
<i>Calculated Parameters</i>					
Hardness, Dissolved (as CaCO ₃)	7.53	0.125	mg/L	N/A	
<i>Dissolved Metals</i>					
Aluminum, dissolved	46.8	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.078	0.050	µg/L	2023-08-10	
Barium, dissolved	2.36	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.9	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0051	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2150	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.209	0.0050	µg/L	2023-08-10	
Copper, dissolved	9.11	0.10	µg/L	2023-08-10	
Iron, dissolved	85.5	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-22-08022023-NP (23H1041-18) | Matrix: Potable Water | Sampled: 2023-08-02 10:35, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.092	0.050	µg/L	2023-08-10	
Magnesium, dissolved	524	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.96	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.034	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.600	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	259	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1820	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1980	20	µg/L	2023-08-10	
Strontium, dissolved	9.30	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	1.01	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	< 1.0	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.035	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.4	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.4	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

SU-PW-29-08022023-NP (23H1041-19) | Matrix: Potable Water | Sampled: 2023-08-02 14:30

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	7.93	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.132	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-29-08022023-NP (23H1041-19) | Matrix: Potable Water | Sampled: 2023-08-02 14:30, Continued

Anions, Continued

Sulfate	3.4	1.0	mg/L	2023-08-09	
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Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	4.50	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	25.0	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.065	0.050	µg/L	2023-08-10	
Barium, dissolved	0.26	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	9.1	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0156	0.0020	µg/L	2023-08-10	
Calcium, dissolved	1070	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.385	0.0050	µg/L	2023-08-10	
Copper, dissolved	56.7	0.10	µg/L	2023-08-10	
Iron, dissolved	42.0	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.256	0.050	µg/L	2023-08-10	
Magnesium, dissolved	443	5.0	µg/L	2023-08-10	
Manganese, dissolved	0.949	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.096	0.010	µg/L	2023-08-10	
Nickel, dissolved	1.08	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	508	20	µg/L	2023-08-10	
Selenium, dissolved	0.13	0.10	µg/L	2023-08-10	
Silicon, dissolved	2680	100	µg/L	2023-08-10	
Silver, dissolved	0.012	0.010	µg/L	2023-08-10	
Sodium, dissolved	9590	20	µg/L	2023-08-10	
Strontium, dissolved	3.97	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.27	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0027	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	2.2	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-29-08022023-NP (23H1041-19) Matrix: Potable Water Sampled: 2023-08-02 14:30, Continued					
General Parameters					
Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	9.8	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	9.8	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	31	15	mg/L	2023-08-10	HT1

SU-PW-B-08012023-NP (23H1041-20) | Matrix: Potable Water | Sampled: 2023-08-01 12:00

Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.60	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.118	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.6	1.0	mg/L	2023-08-09	

Calculated Parameters					
Hardness, Dissolved (as CaCO ₃)	7.95	0.125	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	52.1	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.078	0.050	µg/L	2023-08-10	
Barium, dissolved	2.19	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.6	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0103	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2290	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.438	0.0050	µg/L	2023-08-10	
Copper, dissolved	15.7	0.10	µg/L	2023-08-10	
Iron, dissolved	91.9	2.0	µg/L	2023-08-10	
Lead, dissolved	0.077	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.101	0.050	µg/L	2023-08-10	
Magnesium, dissolved	540	5.0	µg/L	2023-08-10	
Manganese, dissolved	3.18	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.042	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.850	0.040	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-B-08012023-NP (23H1041-20) | Matrix: Potable Water | Sampled: 2023-08-01 12:00, Continued

Dissolved Metals, Continued

Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	274	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	2010	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2030	20	µg/L	2023-08-10	
Strontium, dissolved	8.95	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	1.48	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.9	1.0	µg/L	2023-08-10	
Zirconium, dissolved	0.024	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.9	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.9	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1

SU-SW-17-08012023-NP (23H1041-21) | Matrix: Surface Water | Sampled: 2023-08-01 12:35

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.57	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.093	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.5	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	7.54	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	39.3	1.0	µg/L	2023-08-10	
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TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-17-08012023-NP (23H1041-21) | Matrix: Surface Water | Sampled: 2023-08-01 12:35, Continued

Dissolved Metals, Continued

Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.071	0.050	µg/L	2023-08-10	
Barium, dissolved	2.30	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.5	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0044	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2190	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.340	0.0050	µg/L	2023-08-10	
Copper, dissolved	13.4	0.10	µg/L	2023-08-10	
Iron, dissolved	81.0	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.092	0.050	µg/L	2023-08-10	
Magnesium, dissolved	501	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.06	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.023	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.623	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	253	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1870	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2000	20	µg/L	2023-08-10	
Strontium, dissolved	8.98	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.23	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	2.8	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO3)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO3)	5.7	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO3)	5.7	1.0	mg/L	2023-08-10	

TEST RESULTS

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SU-SW-17-08012023-NP (23H1041-21) | Matrix: Surface Water | Sampled: 2023-08-01 12:35, Continued

General Parameters, Continued

Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.07	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	40.7	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.092	0.050	µg/L	2023-08-15	
Barium, total	2.29	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	3.6	2.0	µg/L	2023-08-15	
Cadmium, total	0.0079	0.0020	µg/L	2023-08-15	
Calcium, total	1950	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.342	0.0050	µg/L	2023-08-15	
Copper, total	13.7	0.20	µg/L	2023-08-15	
Iron, total	105	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.075	0.050	µg/L	2023-08-15	
Magnesium, total	487	5.0	µg/L	2023-08-15	
Manganese, total	1.32	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.858	0.010	µg/L	2023-08-15	
Nickel, total	0.618	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	239	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1590	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1900	20	µg/L	2023-08-15	
Strontium, total	9.25	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.39	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	

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SU-SW-17-08012023-NP (23H1041-21) | Matrix: Surface Water | Sampled: 2023-08-01 12:35, Continued

Total Metals, Continued

Uranium, total	0.0021	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.3	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-19-08012023-NP (23H1041-22) | Matrix: Surface Water | Sampled: 2023-08-01 14:00

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.51	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.094	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.6	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	7.34	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	40.1	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.081	0.050	µg/L	2023-08-10	
Barium, dissolved	2.19	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.4	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0051	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2130	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.354	0.0050	µg/L	2023-08-10	
Copper, dissolved	13.2	0.10	µg/L	2023-08-10	
Iron, dissolved	86.9	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.091	0.050	µg/L	2023-08-10	
Magnesium, dissolved	489	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.13	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.025	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.605	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	250	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1780	100	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-19-08012023-NP (23H1041-22) | Matrix: Surface Water | Sampled: 2023-08-01 14:00, Continued

Dissolved Metals, Continued

Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1930	20	µg/L	2023-08-10	
Strontium, dissolved	8.95	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.25	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.6	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.2	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.2	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.04	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	42.3	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.093	0.050	µg/L	2023-08-15	
Barium, total	2.11	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	3.7	2.0	µg/L	2023-08-15	
Cadmium, total	0.0071	0.0020	µg/L	2023-08-15	
Calcium, total	1870	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.364	0.0050	µg/L	2023-08-15	
Copper, total	13.8	0.20	µg/L	2023-08-15	
Iron, total	109	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.094	0.050	µg/L	2023-08-15	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-19-08012023-NP (23H1041-22) | Matrix: Surface Water | Sampled: 2023-08-01 14:00, Continued

Total Metals, Continued

Magnesium, total	483	5.0	µg/L	2023-08-15	
Manganese, total	1.33	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.024	0.010	µg/L	2023-08-15	
Nickel, total	0.620	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	251	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1550	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1890	20	µg/L	2023-08-15	
Strontium, total	9.24	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.38	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0018	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	3.0	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-21-08012023-NP (23H1041-23) | Matrix: Surface Water | Sampled: 2023-08-01 15:25

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.50	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.090	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.5	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	7.43	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	38.5	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.083	0.050	µg/L	2023-08-10	
Barium, dissolved	2.18	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-21-08012023-NP (23H1041-23) | Matrix: Surface Water | Sampled: 2023-08-01 15:25, Continued

Dissolved Metals, Continued

Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.5	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0062	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2170	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.347	0.0050	µg/L	2023-08-10	
Copper, dissolved	11.7	0.10	µg/L	2023-08-10	
Iron, dissolved	79.6	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.100	0.050	µg/L	2023-08-10	
Magnesium, dissolved	489	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.03	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.027	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.648	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	250	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1760	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1970	20	µg/L	2023-08-10	
Strontium, dissolved	9.14	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.25	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.8	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	6.4	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	6.4	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.28	0.50	mg/L	2023-08-10	

TEST RESULTS

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SU-SW-21-08012023-NP (23H1041-23) | Matrix: Surface Water | Sampled: 2023-08-01 15:25, Continued

General Parameters, Continued

Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	43.2	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.087	0.050	µg/L	2023-08-15	
Barium, total	2.14	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	3.8	2.0	µg/L	2023-08-15	
Cadmium, total	0.0047	0.0020	µg/L	2023-08-15	
Calcium, total	1850	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.331	0.0050	µg/L	2023-08-15	
Copper, total	11.9	0.20	µg/L	2023-08-15	
Iron, total	108	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.081	0.050	µg/L	2023-08-15	
Magnesium, total	468	5.0	µg/L	2023-08-15	
Manganese, total	1.31	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.020	0.010	µg/L	2023-08-15	
Nickel, total	0.587	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	242	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1560	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1840	20	µg/L	2023-08-15	
Strontium, total	8.92	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.43	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0020	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.2	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-22-08012023-NP (23H1041-24) Matrix: Surface Water Sampled: 2023-08-01 16:00					
Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.57	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.089	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.4	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	7.34	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	38.6	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.083	0.050	µg/L	2023-08-10	
Barium, dissolved	2.18	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.6	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0037	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2150	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.335	0.0050	µg/L	2023-08-10	
Copper, dissolved	11.4	0.10	µg/L	2023-08-10	
Iron, dissolved	88.5	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.099	0.050	µg/L	2023-08-10	
Magnesium, dissolved	476	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.09	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.025	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.633	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	246	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1770	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1980	20	µg/L	2023-08-10	
Strontium, dissolved	8.90	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.26	0.20	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-22-08012023-NP (23H1041-24) Matrix: Surface Water Sampled: 2023-08-01 16:00, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.8	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	6.2	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	6.2	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	3.01	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1
<i>Total Metals</i>					
Aluminum, total	40.7	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.096	0.050	µg/L	2023-08-15	
Barium, total	2.27	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	3.7	2.0	µg/L	2023-08-15	
Cadmium, total	0.0043	0.0020	µg/L	2023-08-15	
Calcium, total	1830	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.345	0.0050	µg/L	2023-08-15	
Copper, total	11.9	0.20	µg/L	2023-08-15	
Iron, total	110	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.086	0.050	µg/L	2023-08-15	
Magnesium, total	470	5.0	µg/L	2023-08-15	
Manganese, total	1.37	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.026	0.010	µg/L	2023-08-15	
Nickel, total	0.617	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	245	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1540	100	µg/L	2023-08-15	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-22-08012023-NP (23H1041-24) Matrix: Surface Water Sampled: 2023-08-01 16:00, Continued					
<i>Total Metals, Continued</i>					
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1900	20	µg/L	2023-08-15	
Strontium, total	9.10	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.43	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0019	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.4	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-11-08012023-NP (23H1041-25) | Matrix: Surface Water | Sampled: 2023-08-01 12:47

<i>Anions</i>					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.52	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.090	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.5	1.0	mg/L	2023-08-09	

<i>Calculated Parameters</i>					
Hardness, Dissolved (as CaCO3)	7.69	0.125	mg/L	N/A	

<i>Dissolved Metals</i>					
Aluminum, dissolved	40.9	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.069	0.050	µg/L	2023-08-10	
Barium, dissolved	2.14	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.4	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0041	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2220	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.350	0.0050	µg/L	2023-08-10	
Copper, dissolved	13.8	0.10	µg/L	2023-08-10	
Iron, dissolved	90.3	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-11-08012023-NP (23H1041-25) | Matrix: Surface Water | Sampled: 2023-08-01 12:47, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.094	0.050	µg/L	2023-08-10	
Magnesium, dissolved	519	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.20	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.024	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.674	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	255	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1890	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2020	20	µg/L	2023-08-10	
Strontium, dissolved	9.31	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.26	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.5	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.8	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.8	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.104	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.54	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	41.6	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.104	0.050	µg/L	2023-08-15	
Barium, total	2.25	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-11-08012023-NP (23H1041-25) | Matrix: Surface Water | Sampled: 2023-08-01 12:47, Continued

Total Metals, Continued

Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	4.1	2.0	µg/L	2023-08-15	
Cadmium, total	0.0061	0.0020	µg/L	2023-08-15	
Calcium, total	1960	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.339	0.0050	µg/L	2023-08-15	
Copper, total	13.7	0.20	µg/L	2023-08-15	
Iron, total	104	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.098	0.050	µg/L	2023-08-15	
Magnesium, total	492	5.0	µg/L	2023-08-15	
Manganese, total	1.32	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.022	0.010	µg/L	2023-08-15	
Nickel, total	0.621	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	240	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1640	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1910	20	µg/L	2023-08-15	
Strontium, total	9.22	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.42	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0020	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.1	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-10-08012023-NP (23H1041-26) | Matrix: Surface Water | Sampled: 2023-08-01 14:30

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.54	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.100	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.4	1.0	mg/L	2023-08-09	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-10-08012023-NP (23H1041-26) | Matrix: Surface Water | Sampled: 2023-08-01 14:30, Continued

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	7.32	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	37.8	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.076	0.050	µg/L	2023-08-10	
Barium, dissolved	2.22	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.5	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0043	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2130	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.335	0.0050	µg/L	2023-08-10	
Copper, dissolved	11.8	0.10	µg/L	2023-08-10	
Iron, dissolved	84.6	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.100	0.050	µg/L	2023-08-10	
Magnesium, dissolved	484	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.07	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.022	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.636	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	270	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1790	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1990	20	µg/L	2023-08-10	
Strontium, dissolved	9.17	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.33	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	3.5	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
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TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-10-08012023-NP (23H1041-26) | Matrix: Surface Water | Sampled: 2023-08-01 14:30, Continued

General Parameters, Continued

Alkalinity, Total (as CaCO ₃)	5.3	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.3	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.063	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	1.90	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	41.2	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.090	0.050	µg/L	2023-08-15	
Barium, total	2.23	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	4.0	2.0	µg/L	2023-08-15	
Cadmium, total	0.0066	0.0020	µg/L	2023-08-15	
Calcium, total	1920	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.325	0.0050	µg/L	2023-08-15	
Copper, total	12.1	0.20	µg/L	2023-08-15	
Iron, total	108	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.099	0.050	µg/L	2023-08-15	
Magnesium, total	476	5.0	µg/L	2023-08-15	
Manganese, total	1.27	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.024	0.010	µg/L	2023-08-15	
Nickel, total	0.591	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	241	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1560	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1860	20	µg/L	2023-08-15	
Strontium, total	8.92	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	0.0060	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-10-08012023-NP (23H1041-26) | Matrix: Surface Water | Sampled: 2023-08-01 14:30, Continued

Total Metals, Continued

Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.43	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0022	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.6	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-B-08012023-NP (23H1041-27) | Matrix: Surface Water | Sampled: 2023-08-01 12:00

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.54	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.090	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.4	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	7.23	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	39.2	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.081	0.050	µg/L	2023-08-10	
Barium, dissolved	2.19	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.2	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0061	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2080	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.323	0.0050	µg/L	2023-08-10	
Copper, dissolved	11.9	0.10	µg/L	2023-08-10	
Iron, dissolved	85.8	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.096	0.050	µg/L	2023-08-10	
Magnesium, dissolved	490	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.09	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.025	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.675	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-B-08012023-NP (23H1041-27) | Matrix: Surface Water | Sampled: 2023-08-01 12:00, Continued

Dissolved Metals, Continued

Potassium, dissolved	277	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1780	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2050	20	µg/L	2023-08-10	
Strontium, dissolved	8.96	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.32	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	4.5	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO3)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO3)	5.8	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO3)	5.8	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.71	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	HT1
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	43.5	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.099	0.050	µg/L	2023-08-15	
Barium, total	2.23	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	4.1	2.0	µg/L	2023-08-15	
Cadmium, total	0.0053	0.0020	µg/L	2023-08-15	
Calcium, total	1970	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.338	0.0050	µg/L	2023-08-15	
Copper, total	12.3	0.20	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-B-08012023-NP (23H1041-27) | Matrix: Surface Water | Sampled: 2023-08-01 12:00, Continued

Total Metals, Continued

Iron, total	109	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.102	0.050	µg/L	2023-08-15	
Magnesium, total	487	5.0	µg/L	2023-08-15	
Manganese, total	1.34	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.019	0.010	µg/L	2023-08-15	
Nickel, total	0.616	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	249	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1580	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1930	20	µg/L	2023-08-15	
Strontium, total	9.48	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.42	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0020	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.9	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-07-08022023-NP (23H1041-28) | Matrix: Surface Water | Sampled: 2023-08-02 11:13

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	10.1	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.126	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	8.3	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	18.3	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	46.4	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-07-08022023-NP (23H1041-28) | Matrix: Surface Water | Sampled: 2023-08-02 11:13, Continued

Dissolved Metals, Continued

Arsenic, dissolved	0.057	0.050	µg/L	2023-08-10	
Barium, dissolved	0.45	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	28.2	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0697	0.0020	µg/L	2023-08-10	
Calcium, dissolved	6320	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	6.37	0.0050	µg/L	2023-08-10	
Copper, dissolved	212	0.10	µg/L	2023-08-10	
Iron, dissolved	46.6	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.103	0.050	µg/L	2023-08-10	
Magnesium, dissolved	599	5.0	µg/L	2023-08-10	
Manganese, dissolved	5.04	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.051	0.010	µg/L	2023-08-10	
Nickel, dissolved	8.28	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	195	20	µg/L	2023-08-10	
Selenium, dissolved	0.17	0.10	µg/L	2023-08-10	
Silicon, dissolved	3290	100	µg/L	2023-08-10	
Silver, dissolved	0.013	0.010	µg/L	2023-08-10	
Sodium, dissolved	4860	20	µg/L	2023-08-10	
Strontium, dissolved	8.88	0.10	µg/L	2023-08-10	
Sulfur, dissolved	2540	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	2.87	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0012	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	20.7	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.5	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.5	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-07-08022023-NP (23H1041-28) | Matrix: Surface Water | Sampled: 2023-08-02 11:13, Continued

General Parameters, Continued

Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	41	15	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	70.4	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.064	0.050	µg/L	2023-08-15	
Barium, total	0.46	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	27.2	2.0	µg/L	2023-08-15	
Cadmium, total	0.0753	0.0020	µg/L	2023-08-15	
Calcium, total	5640	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	6.44	0.0050	µg/L	2023-08-15	
Copper, total	224	0.20	µg/L	2023-08-15	
Iron, total	79.6	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.092	0.050	µg/L	2023-08-15	
Magnesium, total	596	5.0	µg/L	2023-08-15	
Manganese, total	5.19	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.070	0.010	µg/L	2023-08-15	
Nickel, total	8.08	0.040	µg/L	2023-08-15	
Phosphorus, total	16	10	µg/L	2023-08-15	
Potassium, total	190	20	µg/L	2023-08-15	
Selenium, total	0.14	0.10	µg/L	2023-08-15	
Silicon, total	3130	100	µg/L	2023-08-15	
Silver, total	0.020	0.010	µg/L	2023-08-15	
Sodium, total	4880	20	µg/L	2023-08-15	
Strontium, total	8.93	0.10	µg/L	2023-08-15	
Sulfur, total	2400	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.38	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0018	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	18.6	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-08-08022023-NP (23H1041-29) Matrix: Surface Water Sampled: 2023-08-02 11:40					
Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	4.01	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.104	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	2.9	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	9.15	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	38.0	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.069	0.050	µg/L	2023-08-10	
Barium, dissolved	1.97	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	8.5	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0143	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2830	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	1.18	0.0050	µg/L	2023-08-10	
Copper, dissolved	37.1	0.10	µg/L	2023-08-10	
Iron, dissolved	85.1	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.097	0.050	µg/L	2023-08-10	
Magnesium, dissolved	506	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.31	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.020	0.010	µg/L	2023-08-10	
Nickel, dissolved	1.77	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	240	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	2030	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2470	20	µg/L	2023-08-10	
Strontium, dissolved	9.04	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.25	0.20	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-08-08022023-NP (23H1041-29) Matrix: Surface Water Sampled: 2023-08-02 11:40, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	4.9	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.8	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.8	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	1.82	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	21	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1
<i>Total Metals</i>					
Aluminum, total	44.5	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.091	0.050	µg/L	2023-08-15	
Barium, total	2.09	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	8.0	2.0	µg/L	2023-08-15	
Cadmium, total	0.0222	0.0020	µg/L	2023-08-15	
Calcium, total	2760	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	1.27	0.0050	µg/L	2023-08-15	
Copper, total	41.8	0.20	µg/L	2023-08-15	
Iron, total	109	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.106	0.050	µg/L	2023-08-15	
Magnesium, total	517	5.0	µg/L	2023-08-15	
Manganese, total	1.66	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.024	0.010	µg/L	2023-08-15	
Nickel, total	1.77	0.040	µg/L	2023-08-15	
Phosphorus, total	< 20	10	µg/L	2023-08-15	RA3
Potassium, total	247	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1840	100	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-08-08022023-NP (23H1041-29) Matrix: Surface Water Sampled: 2023-08-02 11:40, Continued					
Total Metals, Continued					
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	2440	20	µg/L	2023-08-15	
Strontium, total	9.53	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.40	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0020	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	4.9	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-02-08022023-NP (23H1041-30) | Matrix: Surface Water | Sampled: 2023-08-02 12:25

Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	10.0	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.112	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	8.2	1.0	mg/L	2023-08-09	

Calculated Parameters					
Hardness, Dissolved (as CaCO3)	18.1	0.125	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	39.0	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Barium, dissolved	0.43	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	28.1	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0745	0.0020	µg/L	2023-08-10	
Calcium, dissolved	6260	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	5.64	0.0050	µg/L	2023-08-10	
Copper, dissolved	177	0.10	µg/L	2023-08-10	
Iron, dissolved	41.1	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-02-08022023-NP (23H1041-30) | Matrix: Surface Water | Sampled: 2023-08-02 12:25, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.105	0.050	µg/L	2023-08-10	
Magnesium, dissolved	602	5.0	µg/L	2023-08-10	
Manganese, dissolved	3.19	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.057	0.010	µg/L	2023-08-10	
Nickel, dissolved	7.78	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	195	20	µg/L	2023-08-10	
Selenium, dissolved	0.14	0.10	µg/L	2023-08-10	
Silicon, dissolved	3310	100	µg/L	2023-08-10	
Silver, dissolved	0.011	0.010	µg/L	2023-08-10	
Sodium, dissolved	4770	20	µg/L	2023-08-10	
Strontium, dissolved	8.76	0.10	µg/L	2023-08-10	
Sulfur, dissolved	2540	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	19.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.9	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.9	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	37	15	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	58.1	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.057	0.050	µg/L	2023-08-15	
Barium, total	0.41	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	27.0	2.0	µg/L	2023-08-15	
Cadmium, total	0.0633	0.0020	µg/L	2023-08-15	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-02-08022023-NP (23H1041-30) | Matrix: Surface Water | Sampled: 2023-08-02 12:25, Continued

Total Metals, Continued

Calcium, total	5890	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	5.64	0.0050	µg/L	2023-08-15	
Copper, total	190	0.20	µg/L	2023-08-15	
Iron, total	70.5	2.0	µg/L	2023-08-15	
Lead, total	0.057	0.050	µg/L	2023-08-15	
Lithium, total	0.108	0.050	µg/L	2023-08-15	
Magnesium, total	605	5.0	µg/L	2023-08-15	
Manganese, total	3.35	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.060	0.010	µg/L	2023-08-15	
Nickel, total	7.72	0.040	µg/L	2023-08-15	
Phosphorus, total	< 20	10	µg/L	2023-08-15	RA3
Potassium, total	189	20	µg/L	2023-08-15	
Selenium, total	0.15	0.10	µg/L	2023-08-15	
Silicon, total	3110	100	µg/L	2023-08-15	
Silver, total	0.020	0.010	µg/L	2023-08-15	
Sodium, total	4890	20	µg/L	2023-08-15	
Strontium, total	8.85	0.10	µg/L	2023-08-15	
Sulfur, total	2500	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.35	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0018	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	17.4	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-29-08022023-NP (23H1041-31) | Matrix: Surface Water | Sampled: 2023-08-02 13:50

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	4.05	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.101	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.8	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	8.00	0.125	mg/L	N/A	
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TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-29-08022023-NP (23H1041-31) Matrix: Surface Water Sampled: 2023-08-02 13:50, Continued					
Dissolved Metals					
Aluminum, dissolved	41.9	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.096	0.050	µg/L	2023-08-10	
Barium, dissolved	2.17	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	4.9	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0053	0.0020	µg/L	2023-08-10	
Calcium, dissolved	2230	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.328	0.0050	µg/L	2023-08-10	
Copper, dissolved	13.7	0.10	µg/L	2023-08-10	
Iron, dissolved	97.8	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-15	
Lithium, dissolved	0.121	0.050	µg/L	2023-08-10	
Magnesium, dissolved	591	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.10	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.033	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.606	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	278	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	
Silicon, dissolved	1830	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	2870	20	µg/L	2023-08-10	
Strontium, dissolved	9.73	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.22	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0016	0.0010	µg/L	2023-08-15	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	2.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	
General Parameters					
Acidity, Total (as CaCO3)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO3)	5.3	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2023-08-10	

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REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-29-08022023-NP (23H1041-31) | Matrix: Surface Water | Sampled: 2023-08-02 13:50, Continued

General Parameters, Continued

Alkalinity, Bicarbonate (as CaCO ₃)	5.3	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.49	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	21	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	43.0	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.104	0.050	µg/L	2023-08-15	
Barium, total	2.14	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	4.8	2.0	µg/L	2023-08-15	
Cadmium, total	0.0062	0.0020	µg/L	2023-08-15	
Calcium, total	2150	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.325	0.0050	µg/L	2023-08-15	
Copper, total	14.3	0.20	µg/L	2023-08-15	
Iron, total	109	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.126	0.050	µg/L	2023-08-15	
Magnesium, total	608	5.0	µg/L	2023-08-15	
Manganese, total	1.28	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.033	0.010	µg/L	2023-08-15	
Nickel, total	0.624	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	279	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1640	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	2880	20	µg/L	2023-08-15	
Strontium, total	9.84	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.40	0.20	µg/L	2023-08-15	

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-29-08022023-NP (23H1041-31) | Matrix: Surface Water | Sampled: 2023-08-02 13:50, Continued

Total Metals, Continued

Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0020	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.9	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-09-08022023-NP (23H1041-32) | Matrix: Surface Water | Sampled: 2023-08-02 11:30

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	2.08	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.106	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	1.0	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	6.58	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	91.6	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	0.098	0.050	µg/L	2023-08-10	
Barium, dissolved	2.33	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	3.1	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0024	0.0020	µg/L	2023-08-10	
Calcium, dissolved	1860	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	0.0284	0.0050	µg/L	2023-08-10	
Copper, dissolved	1.72	0.10	µg/L	2023-08-10	
Iron, dissolved	98.9	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.095	0.050	µg/L	2023-08-10	
Magnesium, dissolved	467	5.0	µg/L	2023-08-10	
Manganese, dissolved	1.05	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.029	0.010	µg/L	2023-08-10	
Nickel, dissolved	0.211	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	248	20	µg/L	2023-08-10	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-09-08022023-NP (23H1041-32) | Matrix: Surface Water | Sampled: 2023-08-02 11:30, Continued

Dissolved Metals, Continued

Silicon, dissolved	1670	100	µg/L	2023-08-10	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	1780	20	µg/L	2023-08-10	
Strontium, dissolved	8.87	0.10	µg/L	2023-08-10	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	0.29	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	0.0013	0.0010	µg/L	2023-08-11	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	1.4	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	6.5	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	6.5	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	2.30	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	HT1
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	41.8	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.098	0.050	µg/L	2023-08-15	
Barium, total	2.48	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	2.7	2.0	µg/L	2023-08-15	
Cadmium, total	0.0041	0.0020	µg/L	2023-08-15	
Calcium, total	1860	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.0329	0.0050	µg/L	2023-08-15	
Copper, total	1.65	0.20	µg/L	2023-08-15	
Iron, total	119	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	

TEST RESULTS

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SU-SW-09-08022023-NP (23H1041-32) | Matrix: Surface Water | Sampled: 2023-08-02 11:30, Continued

Total Metals, Continued

Lithium, total	0.104	0.050	µg/L	2023-08-15	
Magnesium, total	482	5.0	µg/L	2023-08-15	
Manganese, total	1.27	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.023	0.010	µg/L	2023-08-15	
Nickel, total	0.171	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	250	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1500	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1750	20	µg/L	2023-08-15	
Strontium, total	9.29	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.33	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0019	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	< 1.0	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-01A-08022023-NP (23H1041-33) | Matrix: Surface Water | Sampled: 2023-08-02 15:30

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	10.3	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.126	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	8.3	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	18.2	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	35.9	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Barium, dissolved	0.44	0.10	µg/L	2023-08-10	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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SU-SW-01A-08022023-NP (23H1041-33) | Matrix: Surface Water | Sampled: 2023-08-02 15:30, Continued

Dissolved Metals, Continued

Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	28.1	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0690	0.0020	µg/L	2023-08-10	
Calcium, dissolved	6300	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	6.63	0.0050	µg/L	2023-08-10	
Copper, dissolved	212	0.10	µg/L	2023-08-10	
Iron, dissolved	37.6	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.102	0.050	µg/L	2023-08-10	
Magnesium, dissolved	597	5.0	µg/L	2023-08-10	
Manganese, dissolved	5.33	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.053	0.010	µg/L	2023-08-10	
Nickel, dissolved	8.38	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	193	20	µg/L	2023-08-10	
Selenium, dissolved	0.14	0.10	µg/L	2023-08-10	
Silicon, dissolved	3330	100	µg/L	2023-08-10	
Silver, dissolved	0.012	0.010	µg/L	2023-08-10	
Sodium, dissolved	4820	20	µg/L	2023-08-10	
Strontium, dissolved	9.04	0.10	µg/L	2023-08-10	
Sulfur, dissolved	2560	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-10	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	20.7	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.7	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.7	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	0.347	0.050	mg/L	2023-08-09	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-01A-08022023-NP (23H1041-33) | Matrix: Surface Water | Sampled: 2023-08-02 15:30, Continued

General Parameters, Continued

Solids, Total Dissolved	35	15	mg/L	2023-08-10	HT1
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Total Metals

Aluminum, total	67.9	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.055	0.050	µg/L	2023-08-15	
Barium, total	0.42	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	27.6	2.0	µg/L	2023-08-15	
Cadmium, total	0.0767	0.0020	µg/L	2023-08-15	
Calcium, total	6140	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	6.54	0.0050	µg/L	2023-08-15	
Copper, total	228	0.20	µg/L	2023-08-15	
Iron, total	80.1	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.113	0.050	µg/L	2023-08-15	
Magnesium, total	609	5.0	µg/L	2023-08-15	
Manganese, total	5.44	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.060	0.010	µg/L	2023-08-15	
Nickel, total	8.24	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	189	20	µg/L	2023-08-15	
Selenium, total	0.17	0.10	µg/L	2023-08-15	
Silicon, total	3160	100	µg/L	2023-08-15	
Silver, total	0.019	0.010	µg/L	2023-08-15	
Sodium, total	4910	20	µg/L	2023-08-15	
Strontium, total	9.20	0.10	µg/L	2023-08-15	
Sulfur, total	2500	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.43	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0016	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	18.5	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-C-08022023-NP (23H1041-34) | Matrix: Surface Water | Sampled: 2023-08-02 12:00

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-C-08022023-NP (23H1041-34) Matrix: Surface Water Sampled: 2023-08-02 12:00, Continued					
Anions					
Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	10.2	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.119	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	8.5	1.0	mg/L	2023-08-09	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	18.4	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	34.4	1.0	µg/L	2023-08-10	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Barium, dissolved	0.43	0.10	µg/L	2023-08-10	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Boron, dissolved	28.0	2.0	µg/L	2023-08-10	
Cadmium, dissolved	0.0705	0.0020	µg/L	2023-08-10	
Calcium, dissolved	6370	50	µg/L	2023-08-10	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-10	
Cobalt, dissolved	6.54	0.0050	µg/L	2023-08-10	
Copper, dissolved	210	0.10	µg/L	2023-08-10	
Iron, dissolved	35.5	2.0	µg/L	2023-08-10	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Lithium, dissolved	0.104	0.050	µg/L	2023-08-10	
Magnesium, dissolved	594	5.0	µg/L	2023-08-10	
Manganese, dissolved	5.32	0.050	µg/L	2023-08-10	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.051	0.010	µg/L	2023-08-10	
Nickel, dissolved	8.65	0.040	µg/L	2023-08-10	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-10	
Potassium, dissolved	188	20	µg/L	2023-08-10	
Selenium, dissolved	0.12	0.10	µg/L	2023-08-10	
Silicon, dissolved	3350	100	µg/L	2023-08-10	
Silver, dissolved	0.010	0.010	µg/L	2023-08-10	
Sodium, dissolved	4680	20	µg/L	2023-08-10	
Strontium, dissolved	8.90	0.10	µg/L	2023-08-10	
Sulfur, dissolved	2480	1000	µg/L	2023-08-10	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-10	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-10	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-10	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-08-10	

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-C-08022023-NP (23H1041-34) | Matrix: Surface Water | Sampled: 2023-08-02 12:00, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-10	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-08-11	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-10	
Zinc, dissolved	19.3	1.0	µg/L	2023-08-10	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-08-10	

General Parameters

Acidity, Total (as CaCO ₃)	2.1	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	5.9	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	5.9	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Solids, Total Dissolved	38	15	mg/L	2023-08-10	HT1

Total Metals

Aluminum, total	67.7	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.064	0.050	µg/L	2023-08-15	
Barium, total	0.41	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	27.9	2.0	µg/L	2023-08-15	
Cadmium, total	0.0690	0.0020	µg/L	2023-08-15	
Calcium, total	6100	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	6.43	0.0050	µg/L	2023-08-15	
Copper, total	225	0.20	µg/L	2023-08-15	
Iron, total	78.8	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.110	0.050	µg/L	2023-08-15	
Magnesium, total	600	5.0	µg/L	2023-08-15	
Manganese, total	5.43	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.058	0.010	µg/L	2023-08-15	
Nickel, total	8.26	0.040	µg/L	2023-08-15	
Phosphorus, total	< 20	10	µg/L	2023-08-15	RA3
Potassium, total	196	20	µg/L	2023-08-15	
Selenium, total	0.16	0.10	µg/L	2023-08-15	
Silicon, total	3180	100	µg/L	2023-08-15	
Silver, total	0.022	0.010	µg/L	2023-08-15	
Sodium, total	4910	20	µg/L	2023-08-15	
Strontium, total	8.95	0.10	µg/L	2023-08-15	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-C-08022023-NP (23H1041-34) | Matrix: Surface Water | Sampled: 2023-08-02 12:00, Continued

Total Metals, Continued

Sulfur, total	2500	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.40	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0016	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	19.5	1.0	µg/L	2023-08-15	
Zirconium, total	< 0.020	0.020	µg/L	2023-08-15	

SU-SW-28-08022023-NP (23H1041-35) | Matrix: Surface Water | Sampled: 2023-08-03 11:15

Anions

Bromide	< 0.10	0.10	mg/L	2023-08-09	
Chloride	1.76	0.10	mg/L	2023-08-09	
Fluoride	< 0.10	0.10	mg/L	2023-08-09	
Nitrate (as N)	0.085	0.010	mg/L	2023-08-09	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-08-09	HT1
Sulfate	< 1.0	1.0	mg/L	2023-08-09	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	5.90	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	52.0	1.0	µg/L	2023-08-11	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-08-11	
Arsenic, dissolved	0.148	0.050	µg/L	2023-08-11	
Barium, dissolved	3.61	0.10	µg/L	2023-08-11	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-08-11	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-08-11	
Boron, dissolved	2.9	2.0	µg/L	2023-08-11	
Cadmium, dissolved	< 0.0020	0.0020	µg/L	2023-08-11	
Calcium, dissolved	1710	50	µg/L	2023-08-11	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-08-11	
Cobalt, dissolved	0.0425	0.0050	µg/L	2023-08-11	
Copper, dissolved	0.66	0.10	µg/L	2023-08-11	
Iron, dissolved	258	2.0	µg/L	2023-08-11	
Lead, dissolved	< 0.050	0.050	µg/L	2023-08-11	
Lithium, dissolved	0.170	0.050	µg/L	2023-08-11	
Magnesium, dissolved	398	5.0	µg/L	2023-08-11	
Manganese, dissolved	2.53	0.050	µg/L	2023-08-11	

TEST RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-28-08022023-NP (23H1041-35) | Matrix: Surface Water | Sampled: 2023-08-03 11:15, Continued

Dissolved Metals, Continued

Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, dissolved	0.015	0.010	µg/L	2023-08-11	
Nickel, dissolved	0.271	0.040	µg/L	2023-08-11	
Phosphorus, dissolved	< 10	10	µg/L	2023-08-11	
Potassium, dissolved	252	20	µg/L	2023-08-11	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-08-11	
Silicon, dissolved	1650	100	µg/L	2023-08-11	
Silver, dissolved	< 0.010	0.010	µg/L	2023-08-11	
Sodium, dissolved	1540	20	µg/L	2023-08-11	
Strontium, dissolved	9.16	0.10	µg/L	2023-08-11	
Sulfur, dissolved	< 1000	1000	µg/L	2023-08-11	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-08-11	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-08-11	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-08-11	
Tin, dissolved	< 0.050	0.050	µg/L	2023-08-11	
Titanium, dissolved	0.39	0.20	µg/L	2023-08-11	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-08-11	
Uranium, dissolved	0.0019	0.0010	µg/L	2023-08-11	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-08-11	
Zinc, dissolved	< 1.0	1.0	µg/L	2023-08-11	
Zirconium, dissolved	0.023	0.020	µg/L	2023-08-11	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-08-10	
Alkalinity, Total (as CaCO ₃)	4.7	1.0	mg/L	2023-08-10	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Bicarbonate (as CaCO ₃)	4.7	1.0	mg/L	2023-08-10	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-08-10	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-08-09	
Carbon, Dissolved Organic	3.26	0.50	mg/L	2023-08-10	
Solids, Total Dissolved	< 15	15	mg/L	2023-08-10	
Solids, Total Suspended	< 3.3	2.0	mg/L	2023-08-09	
Sulfide, Total	< 0.020	0.020	mg/L	2023-08-10	

Total Metals

Aluminum, total	64.7	2.0	µg/L	2023-08-15	
Antimony, total	< 0.050	0.050	µg/L	2023-08-15	
Arsenic, total	0.184	0.050	µg/L	2023-08-15	
Barium, total	3.60	0.10	µg/L	2023-08-15	
Beryllium, total	< 0.010	0.010	µg/L	2023-08-15	
Bismuth, total	< 0.010	0.010	µg/L	2023-08-15	
Boron, total	2.2	2.0	µg/L	2023-08-15	
Cadmium, total	0.0027	0.0020	µg/L	2023-08-15	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-28-08022023-NP (23H1041-35) Matrix: Surface Water Sampled: 2023-08-03 11:15, Continued					
<i>Total Metals, Continued</i>					
Calcium, total	1480	50	µg/L	2023-08-15	
Chromium, total	< 0.50	0.50	µg/L	2023-08-15	
Cobalt, total	0.0749	0.0050	µg/L	2023-08-15	
Copper, total	0.80	0.20	µg/L	2023-08-15	
Iron, total	422	2.0	µg/L	2023-08-15	
Lead, total	< 0.050	0.050	µg/L	2023-08-15	
Lithium, total	0.160	0.050	µg/L	2023-08-15	
Magnesium, total	397	5.0	µg/L	2023-08-15	
Manganese, total	5.16	0.050	µg/L	2023-08-15	
Mercury, total	< 0.0050	0.0050	µg/L	2023-08-12	
Molybdenum, total	0.016	0.010	µg/L	2023-08-15	
Nickel, total	0.277	0.040	µg/L	2023-08-15	
Phosphorus, total	< 10	10	µg/L	2023-08-15	
Potassium, total	254	20	µg/L	2023-08-15	
Selenium, total	< 0.10	0.10	µg/L	2023-08-15	
Silicon, total	1420	100	µg/L	2023-08-15	
Silver, total	< 0.010	0.010	µg/L	2023-08-15	
Sodium, total	1460	20	µg/L	2023-08-15	
Strontium, total	9.19	0.10	µg/L	2023-08-15	
Sulfur, total	< 1000	1000	µg/L	2023-08-15	
Tellurium, total	< 0.050	0.050	µg/L	2023-08-15	
Thallium, total	< 0.0040	0.0040	µg/L	2023-08-15	
Thorium, total	< 0.010	0.010	µg/L	2023-08-15	
Tin, total	< 0.050	0.050	µg/L	2023-08-15	
Titanium, total	0.80	0.20	µg/L	2023-08-15	
Tungsten, total	< 0.20	0.20	µg/L	2023-08-15	
Uranium, total	0.0038	0.0010	µg/L	2023-08-15	
Vanadium, total	< 1.00	1.00	µg/L	2023-08-15	
Zinc, total	1.4	1.0	µg/L	2023-08-15	
Zirconium, total	0.022	0.020	µg/L	2023-08-15	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.
RA3 The Reporting Limit has been raised due to comparable level detected in the blank(s).
RE2 Result was confirmed by re-analysis prior to reporting.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2020)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2021)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2021)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2020)	Ion Chromatography	✓	Kelowna
Carbon, Dissolved Organic in Water	SM 5310 B (2022)	Combustion, Infrared CO ₂ Detection	✓	Kelowna
Carbon, Total Organic in Water	SM 5310 B (2022)	Combustion, Infrared CO ₂ Detection	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2021)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Mercury, total, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	Solids in Water, Filtered / SM 2540 C* (2020)	Solids in Water, Filtered / Gravimetry (Dried at 103-105C)	✓	Kelowna
Solids, Total Suspended in Water	Solids in Water, Filtered / SM 2540 D* (2020)	Solids in Water, Filtered / Gravimetry (Dried at 103-105C)	✓	Kelowna
Sulfide, Total in Water	SM 4500-S ₂ D* (2021)	Colorimetry (Methylene Blue)	✓	Edmonton
Total Metals in Water	EPA 200.2* / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association



APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 697296

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General Comments:

The results in this report apply to the received samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Caro will dispose of all samples within 30 days of sample receipt, unless otherwise agreed.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: rpshyk@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 697296

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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B3H0711									
Blank (B3H0711-BLK1)			Prepared: 2023-08-09, Analyzed: 2023-08-09						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B3H0711-BLK2)			Prepared: 2023-08-09, Analyzed: 2023-08-09						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B3H0711-BLK3)			Prepared: 2023-08-09, Analyzed: 2023-08-09						
Bromide	< 0.10	0.10 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B3H0711-BS1)			Prepared: 2023-08-09, Analyzed: 2023-08-09						
Bromide	4.05	0.10 mg/L	4.00		101	85-115			
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	4.03	0.10 mg/L	4.00		101	88-108			
Nitrate (as N)	4.00	0.010 mg/L	4.00		100	90-110			
Nitrite (as N)	2.01	0.010 mg/L	2.00		100	85-115			
Sulfate	16.0	1.0 mg/L	16.0		100	90-110			
LCS (B3H0711-BS2)			Prepared: 2023-08-09, Analyzed: 2023-08-09						
Bromide	3.85	0.10 mg/L	4.00		96	85-115			
Chloride	16.0	0.10 mg/L	16.0		100	90-110			
Fluoride	3.96	0.10 mg/L	4.00		99	88-108			
Nitrate (as N)	3.90	0.010 mg/L	4.00		97	90-110			
Nitrite (as N)	1.95	0.010 mg/L	2.00		97	85-115			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B3H0711, Continued									
LCS (B3H0711-BS2), Continued				Prepared: 2023-08-09, Analyzed: 2023-08-09					
Sulfate	15.7	1.0 mg/L	16.0		98	90-110			
LCS (B3H0711-BS3)				Prepared: 2023-08-09, Analyzed: 2023-08-09					
Bromide	3.82	0.10 mg/L	4.00		96	85-115			
Chloride	15.9	0.10 mg/L	16.0		99	90-110			
Fluoride	4.01	0.10 mg/L	4.00		100	88-108			
Nitrate (as N)	3.87	0.010 mg/L	4.00		97	90-110			
Nitrite (as N)	2.00	0.010 mg/L	2.00		100	85-115			
Sulfate	16.0	1.0 mg/L	16.0		100	90-110			
Duplicate (B3H0711-DUP2)				Source: 23H1041-33		Prepared: 2023-08-09, Analyzed: 2023-08-09			
Bromide	< 0.10	0.10 mg/L		< 0.10				10	
Chloride	10.2	0.10 mg/L		10.3			1	10	
Fluoride	< 0.10	0.10 mg/L		< 0.10				10	
Nitrate (as N)	0.122	0.010 mg/L		0.126			2	10	
Nitrite (as N)	< 0.010	0.010 mg/L		< 0.010				15	
Sulfate	8.2	1.0 mg/L		8.3			< 1	10	
Matrix Spike (B3H0711-MS2)				Source: 23H1041-33		Prepared: 2023-08-09, Analyzed: 2023-08-09			
Bromide	4.19	0.10 mg/L	4.00	< 0.10	105	80-120			
Chloride	25.9	0.10 mg/L	16.0	10.3	97	75-125			
Fluoride	3.99	0.10 mg/L	4.00	< 0.10	100	75-125			
Nitrate (as N)	3.94	0.010 mg/L	4.00	0.126	95	75-125			
Nitrite (as N)	2.02	0.010 mg/L	2.00	< 0.010	101	80-120			
Sulfate	24.3	1.0 mg/L	16.0	8.3	100	75-125			

Dissolved Metals, Batch B3H0989

Blank (B3H0989-BLK1)			Prepared: 2023-08-10, Analyzed: 2023-08-10						
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							
Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 697296

WORK ORDER 23H1041
REPORTED 2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3H0989, Continued									
Blank (B3H0989-BLK1), Continued					Prepared: 2023-08-10, Analyzed: 2023-08-10				
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							
LCS (B3H0989-BS1)					Prepared: 2023-08-10, Analyzed: 2023-08-10				
Aluminum, dissolved	4040	1.0 µg/L	4000		101	80-120			
Antimony, dissolved	41.3	0.050 µg/L	40.0		103	80-120			
Arsenic, dissolved	408	0.050 µg/L	400		102	80-120			
Barium, dissolved	40.8	0.10 µg/L	40.0		102	80-120			
Beryllium, dissolved	40.9	0.010 µg/L	40.0		102	80-120			
Bismuth, dissolved	40.0	0.010 µg/L	40.0		100	80-120			
Boron, dissolved	412	2.0 µg/L	400		103	80-120			
Cadmium, dissolved	40.7	0.0020 µg/L	40.0		102	80-120			
Calcium, dissolved	4110	50 µg/L	4000		103	80-120			
Chromium, dissolved	40.9	0.50 µg/L	40.0		102	80-120			
Cobalt, dissolved	40.6	0.0050 µg/L	40.0		101	80-120			
Copper, dissolved	40.1	0.10 µg/L	40.0		100	80-120			
Iron, dissolved	4110	2.0 µg/L	4000		103	80-120			
Lead, dissolved	39.7	0.050 µg/L	40.0		99	80-120			
Lithium, dissolved	40.6	0.050 µg/L	40.0		101	80-120			
Magnesium, dissolved	4080	5.0 µg/L	4000		102	80-120			
Manganese, dissolved	40.7	0.050 µg/L	40.0		102	80-120			
Molybdenum, dissolved	39.5	0.010 µg/L	40.0		99	80-120			
Nickel, dissolved	40.0	0.040 µg/L	40.0		100	80-120			
Phosphorus, dissolved	4060	10 µg/L	4000		101	80-120			
Potassium, dissolved	3980	20 µg/L	4000		99	80-120			
Selenium, dissolved	412	0.10 µg/L	400		103	80-120			
Silicon, dissolved	4230	100 µg/L	4000		106	80-120			
Silver, dissolved	35.2	0.010 µg/L	40.0		88	80-120			
Sodium, dissolved	4010	20 µg/L	4000		100	80-120			
Strontium, dissolved	39.9	0.10 µg/L	40.0		100	80-120			
Sulfur, dissolved	41400	1000 µg/L	40000		103	80-120			
Tellurium, dissolved	40.8	0.050 µg/L	40.0		102	80-120			
Thallium, dissolved	38.9	0.0040 µg/L	40.0		97	80-120			
Thorium, dissolved	38.1	0.010 µg/L	40.0		95	80-120			
Tin, dissolved	40.6	0.050 µg/L	40.0		101	80-120			
Titanium, dissolved	40.2	0.20 µg/L	40.0		101	80-120			
Tungsten, dissolved	40.5	0.20 µg/L	40.0		101	80-120			
Uranium, dissolved	40.7	0.0010 µg/L	40.0		102	80-120			
Vanadium, dissolved	40.7	1.00 µg/L	40.0		102	80-120			
Zinc, dissolved	407	1.0 µg/L	400		102	80-120			
Zirconium, dissolved	41.6	0.020 µg/L	40.0		104	80-120			

Dissolved Metals, Batch B3H0991

Blank (B3H0991-BLK1)					Prepared: 2023-08-10, Analyzed: 2023-08-10				
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							
Arsenic, dissolved	< 0.050	0.050 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3H0991, Continued

Blank (B3H0991-BLK1), Continued

Prepared: 2023-08-10, Analyzed: 2023-08-10

Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							

LCS (B3H0991-BS1)

Prepared: 2023-08-10, Analyzed: 2023-08-10

Aluminum, dissolved	4030	1.0 µg/L	4000		101	80-120			
Antimony, dissolved	40.7	0.050 µg/L	40.0		102	80-120			
Arsenic, dissolved	405	0.050 µg/L	400		101	80-120			
Barium, dissolved	39.2	0.10 µg/L	40.0		98	80-120			
Beryllium, dissolved	41.4	0.010 µg/L	40.0		103	80-120			
Bismuth, dissolved	40.0	0.010 µg/L	40.0		100	80-120			
Boron, dissolved	410	2.0 µg/L	400		102	80-120			
Cadmium, dissolved	40.3	0.0020 µg/L	40.0		101	80-120			
Calcium, dissolved	4110	50 µg/L	4000		103	80-120			
Chromium, dissolved	40.6	0.50 µg/L	40.0		102	80-120			
Cobalt, dissolved	40.2	0.0050 µg/L	40.0		100	80-120			
Copper, dissolved	39.9	0.10 µg/L	40.0		100	80-120			
Iron, dissolved	4060	2.0 µg/L	4000		101	80-120			
Lead, dissolved	39.8	0.050 µg/L	40.0		100	80-120			
Lithium, dissolved	40.3	0.050 µg/L	40.0		101	80-120			
Magnesium, dissolved	4120	5.0 µg/L	4000		103	80-120			
Manganese, dissolved	40.1	0.050 µg/L	40.0		100	80-120			
Molybdenum, dissolved	38.6	0.010 µg/L	40.0		97	80-120			
Nickel, dissolved	40.2	0.040 µg/L	40.0		101	80-120			
Phosphorus, dissolved	4040	10 µg/L	4000		101	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3H0991, Continued									
LCS (B3H0991-BS1), Continued				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Potassium, dissolved	3960	20 µg/L	4000		99	80-120			
Selenium, dissolved	410	0.10 µg/L	400		102	80-120			
Silicon, dissolved	4310	100 µg/L	4000		108	80-120			
Silver, dissolved	34.6	0.010 µg/L	40.0		87	80-120			
Sodium, dissolved	3920	20 µg/L	4000		98	80-120			
Strontium, dissolved	40.3	0.10 µg/L	40.0		101	80-120			
Sulfur, dissolved	41800	1000 µg/L	40000		104	80-120			
Tellurium, dissolved	40.8	0.050 µg/L	40.0		102	80-120			
Thallium, dissolved	40.0	0.0040 µg/L	40.0		100	80-120			
Thorium, dissolved	37.3	0.010 µg/L	40.0		93	80-120			
Tin, dissolved	40.4	0.050 µg/L	40.0		101	80-120			
Titanium, dissolved	40.1	0.20 µg/L	40.0		100	80-120			
Tungsten, dissolved	40.1	0.20 µg/L	40.0		100	80-120			
Uranium, dissolved	39.9	0.0010 µg/L	40.0		100	80-120			
Vanadium, dissolved	41.1	1.00 µg/L	40.0		103	80-120			
Zinc, dissolved	401	1.0 µg/L	400		100	80-120			
Zirconium, dissolved	41.2	0.020 µg/L	40.0		103	80-120			
Duplicate (B3H0991-DUP1)				Source: 23H1041-01	Prepared: 2023-08-10, Analyzed: 2023-08-10				
Aluminum, dissolved	3.3	1.0 µg/L		5.2			46	20	
Antimony, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Arsenic, dissolved	0.060	0.050 µg/L		< 0.050				20	
Barium, dissolved	12.0	0.10 µg/L		11.8			1	20	
Beryllium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Bismuth, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Boron, dissolved	21.5	2.0 µg/L		21.5			< 1	20	
Cadmium, dissolved	0.195	0.0020 µg/L		0.191			2	20	
Calcium, dissolved	13200	50 µg/L		13100			< 1	20	
Chromium, dissolved	< 0.50	0.50 µg/L		< 0.50				20	
Cobalt, dissolved	27.3	0.0050 µg/L		27.2			< 1	20	
Copper, dissolved	378	0.10 µg/L		373			1	20	
Iron, dissolved	< 2.0	2.0 µg/L		< 2.0				20	
Lead, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Lithium, dissolved	0.235	0.050 µg/L		0.234				20	
Magnesium, dissolved	2310	5.0 µg/L		2310			< 1	20	
Manganese, dissolved	0.570	0.050 µg/L		0.570			< 1	20	
Molybdenum, dissolved	0.063	0.010 µg/L		0.060			5	20	
Nickel, dissolved	31.6	0.040 µg/L		31.2			1	20	
Phosphorus, dissolved	< 10	10 µg/L		< 10				20	
Potassium, dissolved	580	20 µg/L		581			< 1	20	
Selenium, dissolved	0.71	0.10 µg/L		0.59			19	20	
Silicon, dissolved	9150	100 µg/L		9060			1	20	
Silver, dissolved	0.010	0.010 µg/L		0.011				20	
Sodium, dissolved	6460	20 µg/L		6570			2	20	
Strontium, dissolved	34.9	0.10 µg/L		34.9			< 1	20	
Sulfur, dissolved	8820	1000 µg/L		8760			< 1	20	
Tellurium, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Thallium, dissolved	0.0043	0.0040 µg/L		< 0.0040				20	
Thorium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Tin, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Titanium, dissolved	< 0.20	0.20 µg/L		< 0.20				20	
Tungsten, dissolved	< 0.20	0.20 µg/L		0.20				20	
Uranium, dissolved	0.0023	0.0010 µg/L		0.0028				20	
Vanadium, dissolved	< 1.00	1.00 µg/L		< 1.00				20	
Zinc, dissolved	62.4	1.0 µg/L		61.4			2	20	
Zirconium, dissolved	< 0.020	0.020 µg/L		< 0.020				20	

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3H0991, Continued									
Matrix Spike (B3H0991-MS1)		Source: 23H1041-02		Prepared: 2023-08-10, Analyzed: 2023-08-10					
Aluminum, dissolved	4290	1.0 µg/L	4000	3.9	107	70-130			
Antimony, dissolved	37.4	0.050 µg/L	40.0	< 0.050	93	70-130			
Arsenic, dissolved	431	0.050 µg/L	400	0.057	108	70-130			
Barium, dissolved	45.0	0.10 µg/L	40.0	2.89	105	70-130			
Beryllium, dissolved	42.7	0.010 µg/L	40.0	< 0.010	107	70-130			
Bismuth, dissolved	26.7	0.010 µg/L	40.0	< 0.010	67	70-130			MES
Boron, dissolved	450	2.0 µg/L	400	25.1	106	70-130			
Cadmium, dissolved	43.0	0.0020 µg/L	40.0	0.0101	108	70-130			
Calcium, dissolved	14700	50 µg/L	4000	11000	92	70-130			
Chromium, dissolved	44.5	0.50 µg/L	40.0	< 0.50	111	70-130			
Cobalt, dissolved	42.1	0.0050 µg/L	40.0	0.190	105	70-130			
Copper, dissolved	46.0	0.10 µg/L	40.0	4.55	104	70-130			
Iron, dissolved	4140	2.0 µg/L	4000	2.8	103	70-130			
Lead, dissolved	41.3	0.050 µg/L	40.0	< 0.050	103	70-130			
Lithium, dissolved	41.9	0.050 µg/L	40.0	0.106	105	70-130			
Magnesium, dissolved	7240	5.0 µg/L	4000	2870	109	70-130			
Manganese, dissolved	42.2	0.050 µg/L	40.0	0.199	105	70-130			
Molybdenum, dissolved	38.1	0.010 µg/L	40.0	0.319	94	70-130			
Nickel, dissolved	42.9	0.040 µg/L	40.0	0.560	106	70-130			
Phosphorus, dissolved	4400	10 µg/L	4000	< 10	110	70-130			
Potassium, dissolved	4970	20 µg/L	4000	651	108	70-130			
Selenium, dissolved	434	0.10 µg/L	400	0.14	109	70-130			
Silicon, dissolved	10900	100 µg/L	4000	6750	103	70-130			
Silver, dissolved	26.8	0.010 µg/L	40.0	< 0.010	67	70-130			MES
Sodium, dissolved	9890	20 µg/L	4000	5850	101	70-130			
Strontium, dissolved	66.4	0.10 µg/L	40.0	25.0	104	70-130			
Sulfur, dissolved	43700	1000 µg/L	40000	1150	106	70-130			
Tellurium, dissolved	38.7	0.050 µg/L	40.0	< 0.050	97	70-130			
Thallium, dissolved	41.2	0.0040 µg/L	40.0	0.0050	103	70-130			
Thorium, dissolved	37.6	0.010 µg/L	40.0	< 0.010	94	70-130			
Tin, dissolved	40.9	0.050 µg/L	40.0	< 0.050	102	70-130			
Titanium, dissolved	39.7	0.20 µg/L	40.0	< 0.20	99	70-130			
Tungsten, dissolved	40.0	0.20 µg/L	40.0	< 0.20	100	70-130			
Uranium, dissolved	39.9	0.0010 µg/L	40.0	0.0028	100	70-130			
Vanadium, dissolved	43.6	1.00 µg/L	40.0	< 1.00	107	70-130			
Zinc, dissolved	438	1.0 µg/L	400	3.0	109	70-130			
Zirconium, dissolved	42.9	0.020 µg/L	40.0	< 0.020	107	70-130			

Dissolved Metals, Batch B3H0993

Blank (B3H0993-BLK1)		Prepared: 2023-08-10, Analyzed: 2023-08-10							
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							
Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3H0993, Continued

Blank (B3H0993-BLK1), Continued

Prepared: 2023-08-10, Analyzed: 2023-08-10

Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							

LCS (B3H0993-BS1)

Prepared: 2023-08-10, Analyzed: 2023-08-10

Aluminum, dissolved	3990	1.0 µg/L	4000		100	80-120			
Antimony, dissolved	41.0	0.050 µg/L	40.0		102	80-120			
Arsenic, dissolved	405	0.050 µg/L	400		101	80-120			
Barium, dissolved	39.2	0.10 µg/L	40.0		98	80-120			
Beryllium, dissolved	39.8	0.010 µg/L	40.0		99	80-120			
Bismuth, dissolved	39.4	0.010 µg/L	40.0		98	80-120			
Boron, dissolved	397	2.0 µg/L	400		99	80-120			
Cadmium, dissolved	40.5	0.0020 µg/L	40.0		101	80-120			
Calcium, dissolved	4050	50 µg/L	4000		101	80-120			
Chromium, dissolved	40.4	0.50 µg/L	40.0		101	80-120			
Cobalt, dissolved	40.5	0.0050 µg/L	40.0		101	80-120			
Copper, dissolved	40.1	0.10 µg/L	40.0		100	80-120			
Iron, dissolved	4060	2.0 µg/L	4000		101	80-120			
Lead, dissolved	39.7	0.050 µg/L	40.0		99	80-120			
Lithium, dissolved	39.2	0.050 µg/L	40.0		98	80-120			
Magnesium, dissolved	4000	5.0 µg/L	4000		100	80-120			
Manganese, dissolved	40.2	0.050 µg/L	40.0		101	80-120			
Molybdenum, dissolved	38.8	0.010 µg/L	40.0		97	80-120			
Nickel, dissolved	40.2	0.040 µg/L	40.0		100	80-120			
Phosphorus, dissolved	3990	10 µg/L	4000		100	80-120			
Potassium, dissolved	3910	20 µg/L	4000		98	80-120			
Selenium, dissolved	408	0.10 µg/L	400		102	80-120			
Silicon, dissolved	4090	100 µg/L	4000		102	80-120			
Silver, dissolved	34.6	0.010 µg/L	40.0		87	80-120			
Sodium, dissolved	3990	20 µg/L	4000		100	80-120			
Strontium, dissolved	39.2	0.10 µg/L	40.0		98	80-120			
Sulfur, dissolved	39600	1000 µg/L	40000		99	80-120			
Tellurium, dissolved	40.9	0.050 µg/L	40.0		102	80-120			
Thallium, dissolved	39.1	0.0040 µg/L	40.0		98	80-120			
Thorium, dissolved	38.5	0.010 µg/L	40.0		96	80-120			
Tin, dissolved	40.2	0.050 µg/L	40.0		101	80-120			
Titanium, dissolved	40.9	0.20 µg/L	40.0		102	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3H0993, Continued									
LCS (B3H0993-BS1), Continued				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Tungsten, dissolved	40.4	0.20 µg/L	40.0		101	80-120			
Uranium, dissolved	40.2	0.0010 µg/L	40.0		100	80-120			
Vanadium, dissolved	40.8	1.00 µg/L	40.0		102	80-120			
Zinc, dissolved	404	1.0 µg/L	400		101	80-120			
Zirconium, dissolved	41.1	0.020 µg/L	40.0		103	80-120			
Duplicate (B3H0993-DUP1)				Source: 23H1041-18	Prepared: 2023-08-10, Analyzed: 2023-08-10				
Aluminum, dissolved	49.0	1.0 µg/L		46.8			4	20	
Antimony, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Arsenic, dissolved	0.076	0.050 µg/L		0.078				20	
Barium, dissolved	2.27	0.10 µg/L		2.36			4	20	
Beryllium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Bismuth, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Boron, dissolved	4.7	2.0 µg/L		4.9				20	
Cadmium, dissolved	0.0039	0.0020 µg/L		0.0051				20	
Calcium, dissolved	2130	50 µg/L		2150			< 1	20	
Chromium, dissolved	< 0.50	0.50 µg/L		< 0.50				20	
Cobalt, dissolved	0.209	0.0050 µg/L		0.209			< 1	20	
Copper, dissolved	8.99	0.10 µg/L		9.11			1	20	
Iron, dissolved	80.5	2.0 µg/L		85.5			6	20	
Lead, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Lithium, dissolved	0.093	0.050 µg/L		0.092				20	
Magnesium, dissolved	513	5.0 µg/L		524			2	20	
Manganese, dissolved	1.94	0.050 µg/L		1.96			1	20	
Molybdenum, dissolved	0.042	0.010 µg/L		0.034				20	
Nickel, dissolved	0.634	0.040 µg/L		0.600			5	20	
Phosphorus, dissolved	< 10	10 µg/L		< 10				20	
Potassium, dissolved	258	20 µg/L		259			< 1	20	
Selenium, dissolved	< 0.10	0.10 µg/L		< 0.10				20	
Silicon, dissolved	1820	100 µg/L		1820			< 1	20	
Silver, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Sodium, dissolved	1960	20 µg/L		1980			< 1	20	
Strontium, dissolved	9.23	0.10 µg/L		9.30			< 1	20	
Sulfur, dissolved	< 1000	1000 µg/L		< 1000				20	
Tellurium, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Thallium, dissolved	< 0.0040	0.0040 µg/L		< 0.0040				20	
Thorium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Tin, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Titanium, dissolved	0.98	0.20 µg/L		1.01			4	20	
Tungsten, dissolved	< 0.20	0.20 µg/L		< 0.20				20	
Uranium, dissolved	< 0.0010	0.0010 µg/L		0.0011				20	
Vanadium, dissolved	< 1.00	1.00 µg/L		< 1.00				20	
Zinc, dissolved	1.1	1.0 µg/L		< 1.0				20	
Zirconium, dissolved	0.043	0.020 µg/L		0.035				20	
Matrix Spike (B3H0993-MS1)				Source: 23H1041-19	Prepared: 2023-08-11, Analyzed: 2023-08-11				
Aluminum, dissolved	4260	1.0 µg/L	4000	25.0	106	70-130			
Antimony, dissolved	35.6	0.050 µg/L	40.0	< 0.050	89	70-130			
Arsenic, dissolved	419	0.050 µg/L	400	0.065	105	70-130			
Barium, dissolved	41.2	0.10 µg/L	40.0	0.26	102	70-130			
Beryllium, dissolved	43.5	0.010 µg/L	40.0	< 0.010	109	70-130			
Bismuth, dissolved	29.7	0.010 µg/L	40.0	< 0.010	74	70-130			
Boron, dissolved	408	2.0 µg/L	400	9.1	100	70-130			
Cadmium, dissolved	42.5	0.0020 µg/L	40.0	0.0156	106	70-130			
Calcium, dissolved	5260	50 µg/L	4000	1070	105	70-130			
Chromium, dissolved	43.3	0.50 µg/L	40.0	< 0.50	108	70-130			
Cobalt, dissolved	41.5	0.0050 µg/L	40.0	0.385	103	70-130			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3H0993, Continued									
Matrix Spike (B3H0993-MS1), Continued		Source: 23H1041-19		Prepared: 2023-08-11, Analyzed: 2023-08-11					
Copper, dissolved	94.6	0.10 µg/L	40.0	56.7	95	70-130			MS2
Iron, dissolved	4000	2.0 µg/L	4000	42.0	99	70-130			
Lead, dissolved	40.1	0.050 µg/L	40.0	< 0.050	100	70-130			
Lithium, dissolved	42.3	0.050 µg/L	40.0	0.256	105	70-130			
Magnesium, dissolved	4690	5.0 µg/L	4000	443	106	70-130			
Manganese, dissolved	42.3	0.050 µg/L	40.0	0.949	103	70-130			
Molybdenum, dissolved	38.2	0.010 µg/L	40.0	0.096	95	70-130			
Nickel, dissolved	42.8	0.040 µg/L	40.0	1.08	104	70-130			
Phosphorus, dissolved	4240	10 µg/L	4000	< 10	106	70-130			
Potassium, dissolved	4740	20 µg/L	4000	508	106	70-130			
Selenium, dissolved	420	0.10 µg/L	400	0.13	105	70-130			
Silicon, dissolved	6940	100 µg/L	4000	2680	106	70-130			
Silver, dissolved	31.7	0.010 µg/L	40.0	0.012	79	70-130			
Sodium, dissolved	13300	20 µg/L	4000	9590	93	70-130			MS2
Strontium, dissolved	44.9	0.10 µg/L	40.0	3.97	102	70-130			
Sulfur, dissolved	43200	1000 µg/L	40000	< 1000	106	70-130			
Tellurium, dissolved	38.4	0.050 µg/L	40.0	< 0.050	96	70-130			
Thallium, dissolved	39.8	0.0040 µg/L	40.0	< 0.0040	99	70-130			
Thorium, dissolved	38.3	0.010 µg/L	40.0	< 0.010	96	70-130			
Tin, dissolved	41.2	0.050 µg/L	40.0	< 0.050	103	70-130			
Titanium, dissolved	40.4	0.20 µg/L	40.0	0.27	100	70-130			
Tungsten, dissolved	39.1	0.20 µg/L	40.0	< 0.20	98	70-130			
Uranium, dissolved	39.4	0.0010 µg/L	40.0	0.0027	98	70-130			
Vanadium, dissolved	42.1	1.00 µg/L	40.0	< 1.00	105	70-130			
Zinc, dissolved	431	1.0 µg/L	400	2.2	107	70-130			
Zirconium, dissolved	43.9	0.020 µg/L	40.0	< 0.020	110	70-130			

Dissolved Metals, Batch B3H1057

Blank (B3H1057-BLK1)		Prepared: 2023-08-10, Analyzed: 2023-08-12							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3H1057-BLK2)		Prepared: 2023-08-10, Analyzed: 2023-08-12							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3H1057-BLK3)		Prepared: 2023-08-10, Analyzed: 2023-08-12							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3H1057-BLK4)		Prepared: 2023-08-10, Analyzed: 2023-08-13							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
LCS (B3H1057-BS1)		Prepared: 2023-08-10, Analyzed: 2023-08-12							
Mercury, dissolved	0.282	0.0050 µg/L	0.250		113	80-120			
LCS (B3H1057-BS2)		Prepared: 2023-08-10, Analyzed: 2023-08-12							
Mercury, dissolved	0.295	0.0050 µg/L	0.250		118	80-120			
LCS (B3H1057-BS3)		Prepared: 2023-08-10, Analyzed: 2023-08-13							
Mercury, dissolved	0.236	0.0050 µg/L	0.250		95	80-120			
LCS (B3H1057-BS4)		Prepared: 2023-08-10, Analyzed: 2023-08-13							
Mercury, dissolved	0.243	0.0050 µg/L	0.250		97	80-120			
Duplicate (B3H1057-DUP2)		Source: 23H1041-15		Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, dissolved	< 0.0050	0.0050 µg/L		< 0.0050				20	

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3H1057, Continued									
Matrix Spike (B3H1057-MS2)		Source: 23H1041-16		Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, dissolved	0.278	0.0050 µg/L	0.250	< 0.0050	111	70-130			
General Parameters, Batch B3H0811									
Blank (B3H0811-BLK1)		Prepared: 2023-08-09, Analyzed: 2023-08-09							
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3H0811-BLK2)		Prepared: 2023-08-09, Analyzed: 2023-08-09							
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B3H0811-BS1)		Prepared: 2023-08-09, Analyzed: 2023-08-09							
Ammonia, Total (as N)	1.05	0.050 mg/L	1.00		105	85-115			
LCS (B3H0811-BS2)		Prepared: 2023-08-09, Analyzed: 2023-08-09							
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
Duplicate (B3H0811-DUP1)		Source: 23H1041-06		Prepared: 2023-08-09, Analyzed: 2023-08-09					
Ammonia, Total (as N)	< 0.050	0.050 mg/L		< 0.050				15	
Duplicate (B3H0811-DUP2)		Source: 23H1041-21		Prepared: 2023-08-09, Analyzed: 2023-08-09					
Ammonia, Total (as N)	< 0.050	0.050 mg/L		< 0.050				15	
Matrix Spike (B3H0811-MS1)		Source: 23H1041-06		Prepared: 2023-08-09, Analyzed: 2023-08-09					
Ammonia, Total (as N)	0.231	0.050 mg/L	0.204	< 0.050	103	75-125			
Matrix Spike (B3H0811-MS2)		Source: 23H1041-21		Prepared: 2023-08-09, Analyzed: 2023-08-09					
Ammonia, Total (as N)	0.212	0.050 mg/L	0.204	< 0.050	104	75-125			
General Parameters, Batch B3H0834									
Blank (B3H0834-BLK1)		Prepared: 2023-08-09, Analyzed: 2023-08-09							
Solids, Total Suspended	< 2.0	2.0 mg/L							
LCS (B3H0834-BS1)		Prepared: 2023-08-09, Analyzed: 2023-08-09							
Solids, Total Suspended	86.0	10.0 mg/L	100		86	85-115			
General Parameters, Batch B3H0855									
Blank (B3H0855-BLK1)		Prepared: 2023-08-10, Analyzed: 2023-08-10							
Acidity, Total (as CaCO3)	< 2.0	2.0 mg/L							
Blank (B3H0855-BLK2)		Prepared: 2023-08-10, Analyzed: 2023-08-10							
Acidity, Total (as CaCO3)	< 2.0	2.0 mg/L							
LCS (B3H0855-BS1)		Prepared: 2023-08-10, Analyzed: 2023-08-10							
Acidity, Total (as CaCO3)	106	2.0 mg/L	100		106	80-120			
LCS (B3H0855-BS2)		Prepared: 2023-08-10, Analyzed: 2023-08-10							
Acidity, Total (as CaCO3)	97.4	2.0 mg/L	100		97	80-120			
Duplicate (B3H0855-DUP1)		Source: 23H1041-01		Prepared: 2023-08-10, Analyzed: 2023-08-10					
Acidity, Total (as CaCO3)	12.0	2.0 mg/L		11.8			2	20	
Duplicate (B3H0855-DUP2)		Source: 23H1041-20		Prepared: 2023-08-10, Analyzed: 2023-08-10					
Acidity, Total (as CaCO3)	2.1	2.0 mg/L		< 2.0				20	

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B3H0857									
Blank (B3H0857-BLK1)				Prepared: 2023-08-09, Analyzed: 2023-08-10					
Carbon, Total Organic	< 0.50	0.50 mg/L							
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3H0857-BLK2)				Prepared: 2023-08-09, Analyzed: 2023-08-10					
Carbon, Total Organic	< 0.50	0.50 mg/L							
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3H0857-BLK3)				Prepared: 2023-08-09, Analyzed: 2023-08-10					
Carbon, Total Organic	< 0.50	0.50 mg/L							
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3H0857-BLK4)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Carbon, Total Organic	< 0.50	0.50 mg/L							
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
LCS (B3H0857-BS1)				Prepared: 2023-08-09, Analyzed: 2023-08-10					
Carbon, Total Organic	9.21	0.50 mg/L	10.0		92	78-116			
Carbon, Dissolved Organic	9.80	0.50 mg/L	10.0		98	78-116			
LCS (B3H0857-BS2)				Prepared: 2023-08-09, Analyzed: 2023-08-10					
Carbon, Total Organic	9.92	0.50 mg/L	10.0		99	78-116			
Carbon, Dissolved Organic	9.77	0.50 mg/L	10.0		98	78-116			
LCS (B3H0857-BS3)				Prepared: 2023-08-09, Analyzed: 2023-08-10					
Carbon, Total Organic	9.06	0.50 mg/L	10.0		91	78-116			
Carbon, Dissolved Organic	9.18	0.50 mg/L	10.0		92	78-116			
LCS (B3H0857-BS4)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Carbon, Total Organic	9.27	0.50 mg/L	10.0		93	78-116			
Carbon, Dissolved Organic	9.97	0.50 mg/L	10.0		100	78-116			
Duplicate (B3H0857-DUP2)				Source: 23H1041-23		Prepared: 2023-08-09, Analyzed: 2023-08-10			
Carbon, Dissolved Organic	2.40	0.50 mg/L		2.28				15	
Matrix Spike (B3H0857-MS2)				Source: 23H1041-23		Prepared: 2023-08-09, Analyzed: 2023-08-10			
Carbon, Dissolved Organic	11.3	0.50 mg/L	10.0	2.28	90	70-130			
General Parameters, Batch B3H0875									
Blank (B3H0875-BLK1)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Solids, Total Dissolved	< 15	15 mg/L							
Blank (B3H0875-BLK2)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Solids, Total Dissolved	< 15	15 mg/L							
LCS (B3H0875-BS1)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Solids, Total Dissolved	239	15 mg/L	240		100	85-115			
LCS (B3H0875-BS2)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Solids, Total Dissolved	240	15 mg/L	240		100	85-115			
Duplicate (B3H0875-DUP1)				Source: 23H1041-03		Prepared: 2023-08-10, Analyzed: 2023-08-10			
Solids, Total Dissolved	120	15 mg/L		113			6	15	
Duplicate (B3H0875-DUP2)				Source: 23H1041-15		Prepared: 2023-08-10, Analyzed: 2023-08-10			
Solids, Total Dissolved	24	15 mg/L		24				15	

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B3H0968									
Blank (B3H0968-BLK1)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3H0968-BLK2)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3H0968-BLK3)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B3H0968-BS1)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Alkalinity, Total (as CaCO ₃)	104	1.0 mg/L	100		104	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	41.0	1.0 mg/L	50.0		82	0-200			
LCS (B3H0968-BS2)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Alkalinity, Total (as CaCO ₃)	109	1.0 mg/L	100		109	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	34.1	1.0 mg/L	50.0		68	0-200			
LCS (B3H0968-BS3)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Alkalinity, Total (as CaCO ₃)	110	1.0 mg/L	100		110	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	32.4	1.0 mg/L	50.0		65	0-200			
General Parameters, Batch B3H1028									
Blank (B3H1028-BLK1)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Sulfide, Total	< 0.020	0.020 mg/L							
LCS (B3H1028-BS1)				Prepared: 2023-08-10, Analyzed: 2023-08-10					
Sulfide, Total	0.408	0.020 mg/L	0.470		87	80-120			
Total Metals, Batch B3H1056									
Blank (B3H1056-BLK1)				Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, total	< 0.0050	0.0050 µg/L							
Blank (B3H1056-BLK2)				Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, total	< 0.0050	0.0050 µg/L							
Blank (B3H1056-BLK3)				Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, total	< 0.0050	0.0050 µg/L							
Blank (B3H1056-BLK4)				Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, total	< 0.0050	0.0050 µg/L							
LCS (B3H1056-BS1)				Prepared: 2023-08-10, Analyzed: 2023-08-12					
Mercury, total	0.270	0.0050 µg/L	0.250		108	80-120			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3H1056, Continued									
LCS (B3H1056-BS2)			Prepared: 2023-08-10, Analyzed: 2023-08-12						
Mercury, total	0.243	0.0050 µg/L	0.250		97	80-120			
LCS (B3H1056-BS3)			Prepared: 2023-08-10, Analyzed: 2023-08-12						
Mercury, total	0.273	0.0050 µg/L	0.250		109	80-120			
LCS (B3H1056-BS4)			Prepared: 2023-08-10, Analyzed: 2023-08-12						
Mercury, total	0.275	0.0050 µg/L	0.250		110	80-120			
Duplicate (B3H1056-DUP2)			Source: 23H1041-29		Prepared: 2023-08-10, Analyzed: 2023-08-12				
Mercury, total	< 0.0050	0.0050 µg/L		< 0.0050				20	
Matrix Spike (B3H1056-MS2)			Source: 23H1041-30		Prepared: 2023-08-10, Analyzed: 2023-08-12				
Mercury, total	0.239	0.0050 µg/L	0.250	< 0.0050	96	70-130			
Total Metals, Batch B3H1453									
Blank (B3H1453-BLK1)			Prepared: 2023-08-15, Analyzed: 2023-08-15						
Aluminum, total	< 2.0	2.0 µg/L							
Antimony, total	< 0.050	0.050 µg/L							
Arsenic, total	< 0.050	0.050 µg/L							
Barium, total	< 0.10	0.10 µg/L							
Beryllium, total	< 0.010	0.010 µg/L							
Bismuth, total	< 0.010	0.010 µg/L							
Boron, total	< 2.0	2.0 µg/L							
Cadmium, total	< 0.0020	0.0020 µg/L							
Calcium, total	< 50	50 µg/L							
Chromium, total	< 0.50	0.50 µg/L							
Cobalt, total	< 0.0050	0.0050 µg/L							
Copper, total	< 0.20	0.20 µg/L							
Iron, total	< 2.0	2.0 µg/L							
Lead, total	< 0.050	0.050 µg/L							
Lithium, total	< 0.050	0.050 µg/L							
Magnesium, total	< 5.0	5.0 µg/L							
Manganese, total	< 0.050	0.050 µg/L							
Molybdenum, total	< 0.010	0.010 µg/L							
Nickel, total	< 0.040	0.040 µg/L							
Phosphorus, total	13	10 µg/L							BLK
Potassium, total	< 20	20 µg/L							
Selenium, total	< 0.10	0.10 µg/L							
Silicon, total	< 100	100 µg/L							
Silver, total	< 0.010	0.010 µg/L							
Sodium, total	< 20	20 µg/L							
Strontium, total	< 0.10	0.10 µg/L							
Sulfur, total	< 1000	1000 µg/L							
Tellurium, total	< 0.050	0.050 µg/L							
Thallium, total	< 0.0040	0.0040 µg/L							
Thorium, total	< 0.010	0.010 µg/L							
Tin, total	< 0.050	0.050 µg/L							
Titanium, total	< 0.20	0.20 µg/L							
Tungsten, total	< 0.20	0.20 µg/L							
Uranium, total	< 0.0010	0.0010 µg/L							
Vanadium, total	< 1.00	1.00 µg/L							
Zinc, total	< 1.0	1.0 µg/L							
Zirconium, total	< 0.020	0.020 µg/L							

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REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3H1453, Continued									
LCS (B3H1453-BS1)				Prepared: 2023-08-15, Analyzed: 2023-08-15					
Aluminum, total	3910	2.0 µg/L	4000		98	80-120			
Antimony, total	40.0	0.050 µg/L	40.0		100	80-120			
Arsenic, total	394	0.050 µg/L	400		99	80-120			
Barium, total	39.8	0.10 µg/L	40.0		100	80-120			
Beryllium, total	38.5	0.010 µg/L	40.0		96	80-120			
Bismuth, total	39.3	0.010 µg/L	40.0		98	80-120			
Boron, total	372	2.0 µg/L	400		93	80-120			
Cadmium, total	39.0	0.0020 µg/L	40.0		97	80-120			
Calcium, total	3870	50 µg/L	4000		97	80-120			
Chromium, total	40.3	0.50 µg/L	40.0		101	80-120			
Cobalt, total	39.9	0.0050 µg/L	40.0		100	80-120			
Copper, total	39.5	0.20 µg/L	40.0		99	80-120			
Iron, total	4050	2.0 µg/L	4000		101	80-120			
Lead, total	39.6	0.050 µg/L	40.0		99	80-120			
Lithium, total	36.0	0.050 µg/L	40.0		90	80-120			
Magnesium, total	3860	5.0 µg/L	4000		96	80-120			
Manganese, total	39.8	0.050 µg/L	40.0		99	80-120			
Molybdenum, total	38.2	0.010 µg/L	40.0		96	80-120			
Nickel, total	39.1	0.040 µg/L	40.0		98	80-120			
Phosphorus, total	3910	10 µg/L	4000		98	80-120			
Potassium, total	3920	20 µg/L	4000		98	80-120			
Selenium, total	392	0.10 µg/L	400		98	80-120			
Silicon, total	3940	100 µg/L	4000		99	80-120			
Silver, total	34.1	0.010 µg/L	40.0		85	80-120			
Sodium, total	3950	20 µg/L	4000		99	80-120			
Strontium, total	40.4	0.10 µg/L	40.0		101	80-120			
Sulfur, total	38500	1000 µg/L	40000		96	80-120			
Tellurium, total	39.2	0.050 µg/L	40.0		98	80-120			
Thallium, total	39.7	0.0040 µg/L	40.0		99	80-120			
Thorium, total	39.8	0.010 µg/L	40.0		100	80-120			
Tin, total	39.9	0.050 µg/L	40.0		100	80-120			
Titanium, total	38.4	0.20 µg/L	40.0		96	80-120			
Tungsten, total	41.3	0.20 µg/L	40.0		103	80-120			
Uranium, total	40.0	0.0010 µg/L	40.0		100	80-120			
Vanadium, total	40.2	1.00 µg/L	40.0		100	80-120			
Zinc, total	389	1.0 µg/L	400		97	80-120			
Zirconium, total	40.5	0.020 µg/L	40.0		101	80-120			

Duplicate (B3H1453-DUP1)		Source: 23H1041-21		Prepared: 2023-08-15, Analyzed: 2023-08-15					
Aluminum, total	41.4	2.0 µg/L	40.7				2	20	
Antimony, total	< 0.050	0.050 µg/L	< 0.050					20	
Arsenic, total	0.103	0.050 µg/L	0.092					20	
Barium, total	2.31	0.10 µg/L	2.29				1	20	
Beryllium, total	< 0.010	0.010 µg/L	< 0.010					20	
Bismuth, total	< 0.010	0.010 µg/L	< 0.010					20	
Boron, total	3.7	2.0 µg/L	3.6					20	
Cadmium, total	0.0058	0.0020 µg/L	0.0079					20	
Calcium, total	1940	50 µg/L	1950				< 1	20	
Chromium, total	< 0.50	0.50 µg/L	< 0.50					20	
Cobalt, total	0.344	0.0050 µg/L	0.342				< 1	20	
Copper, total	13.9	0.20 µg/L	13.7				2	20	
Iron, total	106	2.0 µg/L	105				< 1	20	
Lead, total	< 0.050	0.050 µg/L	< 0.050					20	
Lithium, total	0.080	0.050 µg/L	0.075					20	
Magnesium, total	493	5.0 µg/L	487				1	20	
Manganese, total	1.35	0.050 µg/L	1.32				2	20	

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT SNC-Lavalin Inc. (Nelson)
697296

WORK ORDER REPORTED 23H1041
2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3H1453, Continued									
Duplicate (B3H1453-DUP1), Continued		Source: 23H1041-21		Prepared: 2023-08-15, Analyzed: 2023-08-15					
Molybdenum, total	0.852	0.010 µg/L		0.858			< 1	20	
Nickel, total	0.602	0.040 µg/L		0.618			3	20	
Phosphorus, total	< 10	10 µg/L		< 10				20	
Potassium, total	239	20 µg/L		239			< 1	20	
Selenium, total	< 0.10	0.10 µg/L		< 0.10				20	
Silicon, total	1610	100 µg/L		1590			2	20	
Silver, total	< 0.010	0.010 µg/L		< 0.010				20	
Sodium, total	1920	20 µg/L		1900			1	20	
Strontium, total	9.34	0.10 µg/L		9.25			< 1	20	
Sulfur, total	< 1000	1000 µg/L		< 1000				20	
Tellurium, total	< 0.050	0.050 µg/L		< 0.050				20	
Thallium, total	< 0.0040	0.0040 µg/L		< 0.0040				20	
Thorium, total	< 0.010	0.010 µg/L		< 0.010				20	
Tin, total	0.053	0.050 µg/L		< 0.050				20	
Titanium, total	0.32	0.20 µg/L		0.39				20	
Tungsten, total	< 0.20	0.20 µg/L		< 0.20				20	
Uranium, total	0.0021	0.0010 µg/L		0.0021				20	
Vanadium, total	< 1.00	1.00 µg/L		< 1.00				20	
Zinc, total	1.4	1.0 µg/L		1.3				20	
Zirconium, total	< 0.020	0.020 µg/L		< 0.020				20	
Matrix Spike (B3H1453-MS1)		Source: 23H1041-22		Prepared: 2023-08-15, Analyzed: 2023-08-15					
Aluminum, total	4540	2.0 µg/L	4000	42.3	112	70-130			
Antimony, total	36.4	0.050 µg/L	40.0	< 0.050	91	70-130			
Arsenic, total	411	0.050 µg/L	400	0.093	103	70-130			
Barium, total	43.1	0.10 µg/L	40.0	2.11	103	70-130			
Beryllium, total	45.5	0.010 µg/L	40.0	< 0.010	114	70-130			
Bismuth, total	40.7	0.010 µg/L	40.0	< 0.010	102	70-130			
Boron, total	459	2.0 µg/L	400	3.7	114	70-130			
Cadmium, total	41.3	0.0020 µg/L	40.0	0.0071	103	70-130			
Calcium, total	6240	50 µg/L	4000	1870	109	70-130			
Chromium, total	42.9	0.50 µg/L	40.0	< 0.50	107	70-130			
Cobalt, total	42.4	0.0050 µg/L	40.0	0.364	105	70-130			
Copper, total	54.3	0.20 µg/L	40.0	13.8	101	70-130			
Iron, total	4400	2.0 µg/L	4000	109	107	70-130			
Lead, total	41.0	0.050 µg/L	40.0	< 0.050	103	70-130			
Lithium, total	49.8	0.050 µg/L	40.0	0.094	124	70-130			
Magnesium, total	5140	5.0 µg/L	4000	483	116	70-130			
Manganese, total	43.7	0.050 µg/L	40.0	1.33	106	70-130			
Molybdenum, total	39.1	0.010 µg/L	40.0	0.024	98	70-130			
Nickel, total	42.4	0.040 µg/L	40.0	0.620	105	70-130			
Phosphorus, total	4360	10 µg/L	4000	< 10	109	70-130			
Potassium, total	4750	20 µg/L	4000	251	112	70-130			
Selenium, total	414	0.10 µg/L	400	< 0.10	104	70-130			
Silicon, total	6610	100 µg/L	4000	1550	127	70-130			
Silver, total	28.8	0.010 µg/L	40.0	< 0.010	72	70-130			
Sodium, total	6620	20 µg/L	4000	1890	118	70-130			
Strontium, total	49.9	0.10 µg/L	40.0	9.24	102	70-130			
Sulfur, total	46800	1000 µg/L	40000	< 1000	116	70-130			
Tellurium, total	40.6	0.050 µg/L	40.0	< 0.050	101	70-130			
Thallium, total	41.1	0.0040 µg/L	40.0	< 0.0040	103	70-130			
Thorium, total	41.5	0.010 µg/L	40.0	< 0.010	104	70-130			
Tin, total	41.0	0.050 µg/L	40.0	< 0.050	103	70-130			
Titanium, total	44.9	0.20 µg/L	40.0	0.38	111	70-130			
Tungsten, total	42.0	0.20 µg/L	40.0	< 0.20	105	70-130			
Uranium, total	42.6	0.0010 µg/L	40.0	0.0018	107	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO SNC-Lavalin Inc. (Nelson)
PROJECT 697296

WORK ORDER 23H1041
REPORTED 2023-08-15 17:02

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3H1453, Continued									
Matrix Spike (B3H1453-MS1), Continued		Source: 23H1041-22		Prepared: 2023-08-15, Analyzed: 2023-08-15					
Vanadium, total	42.6	1.00 µg/L	40.0	< 1.00	106	70-130			
Zinc, total	418	1.0 µg/L	400	3.0	104	70-130			
Zirconium, total	41.5	0.020 µg/L	40.0	< 0.020	104	70-130			

QC Qualifiers:

BLK Analyte concentration in the Method Blank is above the Reporting Limit (RL).
 MES A number up to 10% (rounded down) of the analytes in a Multi-Element Scan may exceed control limits by up to 10% (absolute).
 MS2 The native sample concentration is greater than the spike concentration hence the matrix spike limits do not apply.

Teck

Teck  * 2 3 H 1 0 4 1 *																																																											
COC ID:		697296-20230803-01																																																									
PROJECT/CLIENT INFO																																																											
Facility Name / Job:		Sunro Mine																																																									
Project Manager:		Michelle Unger																																																									
Email:		m.ung@teck.com																																																									
Address:		601 Knighton Rd				Address:		110-4011 Viking Way																																																			
City:		Kimberly		Province:		BC		City:		Richmond		Province:		BC																																													
Postal Code:		V1A 1C7		Country:		Canada		Postal Code:		V6B 2K9		Country:		CA																																													
Phone Number:		250-427-8422				Phone Number:		604-279-1499				PO number:		697296																																													
SAMPLE DETAILS										ANALYSIS REQUESTED																																																	
Sample ID		Sample Location		Field Matrix		Hazardous Material (Yes/No)		Date		Time (24hr)		G=Grab C=Comp		# Of Cont.		DOC		TOXIC		ANALYSIS		OTHER		FIELD		LAB		TOTAL																															
																Yes		Yes		Yes		Yes		Yes		Yes		Yes		Yes		Yes		Yes		Yes		Yes																					
																Total Metals + Hg in Water		Dissolved Metals + Hg in Water		TDS		Hardness		Major Anions (sulfate, chloride, bromide, nitrate/nitrite)		Dissolved Organic Carbon		Ammonia		Total and Speciated Alkalinity, Acidity		Sulfide		HOLD																									
SU_GW-02-08012023_NP		MW18-02		GW		N		2023-08-01		10:25		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_GW-03-07312023_NP		MW18-03		GW		N		2023-07-31		17:10		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_GW-04-07312023_NP		MW18-04		GW		N		2023-07-31		13:25		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_GW-06-08012023_NP		MW18-06		GW		N		2023-08-01		11:25		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_GW-07-07312023_NP		MW18-07		GW		N		2023-07-31		15:35		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_GW-09-08012023_NP		MW18-09		GW		N		2023-08-01		10:20		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_GW-8-07312023_NP		MW18-B		GW		N		2023-07-31		12:00		G		4		x		x		x		x		x		x		x		x		x		x																									
SU_EQUIPMENT-08012023_NP				Water		N		2023-08-01		13:20		G		9		x		x		x		x		x		x		x		x		x		x																									
SU_FIELD-08012023_NP				Water		N		2023-08-01		15:00		G		5		x		x		x		x		x		x		x		x		x		x																									
SU_TRIP-07272023_NP				Water		N		2023-07-27		14:25		G		7		x		x		x		x		x		x		x		x		x		x																									
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS										RELINQUISHED BY/AFFILIATION										DATE/TIME										ACCEPTED BY/AFFILIATION										DATE/TIME																			
Please note custody seal condition. Dissolved metals and DOC field filtered. Trip Blank was removed from bag to label bottles with correct ID.										Bonn Johnston SNC-Lavalin Inc										2023/08/03																																							
Invoice to: SNC-Lavalin Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3																																																											
Send to: payables@snc-lavalin.com Project number: 631848																																																											
NB OF BOTTLES RETURNED/DESCRIPTION										SAMPLER'S NAME										SAMPLER'S SIGNATURE																																							

COC ID: 697296-20230803-02		TURNAROUND TIME:		Regular		RUSH:												
PROJECT/CLIENT INFO				LABORATORY		OTHER INFO												
Facility Name / Job# Sunro Mine				Lab Name CARO		Report Format / Distribution												
Project Manager Michelle Unger				Lab Contact Regan Pslyk		Email 1: michelle.unger@teck.com X X X												
Email michelle.unger@teck.com				Email regan.pslyk@teck.com		Email 2: craig.pinnell@unclavalin.com X X X												
Address 601 Knighion Rd				Address 110-4011 Viking Way		Email 3: jonathan.croston@unclavalin.com X X X												
City Kimberly Province BC				City Richmond Province BC		Email 4: jonathan.croston@unclavalin.com X X X												
Postal Code V1A 1C7				Postal Code V6B 2K9		Email 5: tony_vincent@unclavalin.com X X X												
Country Canada				Country CA														
Phone Number 250-427-8422				Phone Number 604-279-1499		PO number 697296												
SAMPLE DETAILS				ANALYSIS REQUESTED														
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	Filtered - P: Field, L: Lab, FL: Field & Lab, N: None										
								DOC	Yes			Yes						
								PURSURY	Yes	Yes			Yes	Yes		Yes		
								ANALYSIS	Total Metals + Hg in Water	Dissolved Metals + Hg in Water	TDS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	Dissolved Organic Carbon	Ammonia	Total and Speciated Alkalinity, Acidity	Sulfide	HOLD
11 SU_PW-10-08012023_NP	PW15-10	PW	N	2023-08-01	15:15	G	4		X	X	X	X	X	X	X			
12 SU_PW-11-08012023_NP	PW15-11	PW	N	2023-08-01	13:30	G	4		X	X	X	X	X	X	X			
13 SU_PW-17-08012023_NP	PW17-17	PW	N	2023-08-01	13:10	G	4		X	X	X	X	X	X	X			
14 SU_PW-19-08012023_NP	PW17-19	PW	N	2023-08-01	14:45	G	4		X	X	X	X	X	X	X			
15 SU_PW-05-08022023_NP	PW15-08	PW	N	2023-08-02	12:05	G	4		X	X	X	X	X	X	X			
16 SU_PW-09-08022023_NP	PW15-09	PW	N	2023-08-02	12:10	G	4		X	X	X	X	X	X	X			
17 SU_PW-21-08022023_NP	PW17-21	PW	N	2023-08-02	10:40	G	4		X	X	X	X	X	X	X			
18 SU_PW-22-08022023_NP	PW18-22	PW	N	2023-08-02	10:35	G	4		X	X	X	X	X	X	X			
19 SU_PW-29-08022023_NP	PW23-29	PW	N	2023-08-02	14:30	G	4		X	X	X	X	X	X	X			
20 SU_PW-8-08012023_NP	PW-B	PW	N	2023-08-01	12:00	G	4		X	X	X	X	X	X	X			
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS				RELINQUISHED BY/AFFILIATION		DATE/TIME		ACCEPTED BY/AFFILIATION		DATE/TIME								
Please note custody seal condition. Dissolved metals and DOC field filtered.				Bonn Johnston SNC-Lavalin Inc		2023/08/03												
Invoice to: SNC-Lavalin Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3																		
Send to: payables@unclavalin.com Project number: 631848 Billing PO: 697296																		
NB OF BOTTLES RETURNED/DESCRIPTION				Sampler's Name		Bonn Johnston and Chris Walker		Mobile #		236-464-2830, 250-505-6625								
Regular (default) X				Sampler's Signature				Date/Time										
Priority (2-3 business days) - 50% surcharge																		
Emergency (1 Business Day) - 100% surcharge																		
For Emergency <1 Day, ASAP or Weekend																		

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[illegible]

CERTIFICATE OF ANALYSIS

REPORTED TO AtkinsRéalis (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Craig Pinnell

PO NUMBER 697296

PROJECT 697296

PROJECT INFO Teck Sunro Mine- 20231116-4

WORK ORDER 23K2203

RECEIVED / TEMP 2023-11-20 09:00 / 9.4°C

REPORTED 2023-11-27 12:55

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

Work Order Comments:

Custody Seals Intact: YES

By engaging our services, you are agreeing to CARO Analytical Service's Standard Terms and Conditions outlined here:
<https://www.caro.ca/terms-conditions>

If you have any questions or concerns, please contact me at rpshyk@caro.ca

Authorized By:

Regan Pshyk
Account Manager

1-888-311-8846 | www.caro.ca

#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-17-11142023-NP (23K2203-01) Matrix: Water Sampled: 2023-11-14 11:40					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.66	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.035	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	6.21	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	123	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.84	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	3.1	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0053	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1630	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.247	0.0050	µg/L	2023-11-22	
Copper, dissolved	9.93	0.10	µg/L	2023-11-22	
Iron, dissolved	28.0	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.081	0.050	µg/L	2023-11-22	
Magnesium, dissolved	520	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.603	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.017	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.555	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	147	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2150	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2490	20	µg/L	2023-11-22	
Strontium, dissolved	6.43	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.45	0.20	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-17-11142023-NP (23K2203-01) Matrix: Water Sampled: 2023-11-14 11:40, Continued					
Dissolved Metals, Continued					
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.6	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.050	0.020	µg/L	2023-11-22	
General Parameters					
Acidity, Total (as CaCO ₃)	3.2	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	5.8	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	5.8	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.77	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	15	10	mg/L	2023-11-21	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	
Total Metals					
Aluminum, total	127	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	1.77	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	4.0	2.0	µg/L	2023-11-21	
Cadmium, total	0.0047	0.0020	µg/L	2023-11-20	
Calcium, total	1600	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.245	0.0050	µg/L	2023-11-20	
Copper, total	10.4	0.20	µg/L	2023-11-20	
Iron, total	35.8	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.071	0.050	µg/L	2023-11-20	
Magnesium, total	519	5.0	µg/L	2023-11-20	
Manganese, total	0.946	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.018	0.010	µg/L	2023-11-20	
Nickel, total	0.564	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	132	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-21	
Silicon, total	2000	100	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-17-11142023-NP (23K2203-01) Matrix: Water Sampled: 2023-11-14 11:40, Continued					
Total Metals, Continued					
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2380	20	µg/L	2023-11-20	
Strontium, total	5.94	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.88	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0017	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	1.1	1.0	µg/L	2023-11-20	
Zirconium, total	0.045	0.020	µg/L	2023-11-20	

SU-SW-11-11142023-NP (23K2203-02) | Matrix: Water | Sampled: 2023-11-14 12:15

Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.65	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.033	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	

Calculated Parameters					
Hardness, Dissolved (as CaCO3)	6.19	0.125	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	120	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.87	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.7	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0055	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1610	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.251	0.0050	µg/L	2023-11-22	
Copper, dissolved	9.40	0.10	µg/L	2023-11-22	
Iron, dissolved	28.1	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-11-11142023-NP (23K2203-02) | Matrix: Water | Sampled: 2023-11-14 12:15, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.076	0.050	µg/L	2023-11-22	
Magnesium, dissolved	527	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.613	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.020	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.604	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	148	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2140	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2530	20	µg/L	2023-11-22	
Strontium, dissolved	6.39	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.39	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.6	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.058	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.7	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.8	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.8	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	4.40	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	< 10	10	mg/L	2023-11-21	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	126	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	1.76	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-11-11142023-NP (23K2203-02) | Matrix: Water | Sampled: 2023-11-14 12:15, Continued

Total Metals, Continued

Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	3.8	2.0	µg/L	2023-11-21	
Cadmium, total	0.0044	0.0020	µg/L	2023-11-20	
Calcium, total	1650	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.256	0.0050	µg/L	2023-11-20	
Copper, total	9.85	0.20	µg/L	2023-11-20	
Iron, total	36.5	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.070	0.050	µg/L	2023-11-20	
Magnesium, total	520	5.0	µg/L	2023-11-20	
Manganese, total	1.01	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.011	0.010	µg/L	2023-11-20	
Nickel, total	0.689	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	141	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-21	
Silicon, total	2020	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2400	20	µg/L	2023-11-20	
Strontium, total	6.26	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.67	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0017	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	1.6	1.0	µg/L	2023-11-20	
Zirconium, total	0.071	0.020	µg/L	2023-11-20	

SU-SW-B-11142023-NP (23K2203-03) | Matrix: Water | Sampled: 2023-11-14 15:20

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.59	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.028	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-B-11142023-NP (23K2203-03) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.12	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	122	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.90	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.9	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0041	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1600	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.256	0.0050	µg/L	2023-11-22	
Copper, dissolved	10.4	0.10	µg/L	2023-11-22	
Iron, dissolved	26.9	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.077	0.050	µg/L	2023-11-22	
Magnesium, dissolved	515	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.586	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.014	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.536	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	148	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2140	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2540	20	µg/L	2023-11-22	
Strontium, dissolved	6.30	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.47	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.5	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.053	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.7	2.0	mg/L	2023-11-23	
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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-B-11142023-NP (23K2203-03) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

General Parameters, Continued

Alkalinity, Total (as CaCO ₃)	3.4	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.4	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.68	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	15	10	mg/L	2023-11-21	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	125	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.051	0.050	µg/L	2023-11-20	
Barium, total	1.80	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	3.7	2.0	µg/L	2023-11-21	
Cadmium, total	0.0035	0.0020	µg/L	2023-11-20	
Calcium, total	1580	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.260	0.0050	µg/L	2023-11-20	
Copper, total	10.9	0.20	µg/L	2023-11-20	
Iron, total	35.9	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.069	0.050	µg/L	2023-11-20	
Magnesium, total	512	5.0	µg/L	2023-11-20	
Manganese, total	1.01	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.016	0.010	µg/L	2023-11-20	
Nickel, total	0.602	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	136	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-21	
Silicon, total	2020	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2380	20	µg/L	2023-11-20	
Strontium, total	5.87	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-B-11142023-NP (23K2203-03) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

Total Metals, Continued

Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.86	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0020	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	1.8	1.0	µg/L	2023-11-20	
Zirconium, total	0.067	0.020	µg/L	2023-11-20	

SU-SW-19-11142023-NP (23K2203-04) | Matrix: Water | Sampled: 2023-11-14 13:00

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.62	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.029	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	6.05	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	124	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.85	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.7	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0053	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1570	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.256	0.0050	µg/L	2023-11-22	
Copper, dissolved	9.66	0.10	µg/L	2023-11-22	
Iron, dissolved	27.8	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.082	0.050	µg/L	2023-11-22	
Magnesium, dissolved	516	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.602	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.018	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.648	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-19-11142023-NP (23K2203-04) | Matrix: Water | Sampled: 2023-11-14 13:00, Continued

Dissolved Metals, Continued

Potassium, dissolved	153	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2140	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2520	20	µg/L	2023-11-22	
Strontium, dissolved	6.46	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.42	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.9	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.062	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO3)	2.7	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO3)	3.4	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO3)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO3)	3.4	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO3)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO3)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	4.20	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	15	10	mg/L	2023-11-21	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	125	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	1.96	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	3.8	2.0	µg/L	2023-11-21	
Cadmium, total	0.0052	0.0020	µg/L	2023-11-20	
Calcium, total	1580	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.265	0.0050	µg/L	2023-11-20	
Copper, total	10.1	0.20	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-19-11142023-NP (23K2203-04) | Matrix: Water | Sampled: 2023-11-14 13:00, Continued

Total Metals, Continued

Iron, total	35.2	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.071	0.050	µg/L	2023-11-20	
Magnesium, total	503	5.0	µg/L	2023-11-20	
Manganese, total	1.42	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.014	0.010	µg/L	2023-11-20	
Nickel, total	0.598	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	125	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-21	
Silicon, total	2010	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2420	20	µg/L	2023-11-20	
Strontium, total	5.54	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.77	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0015	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	1.8	1.0	µg/L	2023-11-20	
Zirconium, total	0.062	0.020	µg/L	2023-11-20	

SU-SW-10-11142023-NP (23K2203-05) | Matrix: Water | Sampled: 2023-11-14 15:20

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.58	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.029	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.05	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	121	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-10-11142023-NP (23K2203-05) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

Dissolved Metals, Continued

Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.84	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.8	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0057	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1580	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.256	0.0050	µg/L	2023-11-22	
Copper, dissolved	10.4	0.10	µg/L	2023-11-22	
Iron, dissolved	26.4	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.079	0.050	µg/L	2023-11-22	
Magnesium, dissolved	512	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.563	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.013	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.573	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	146	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2150	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2520	20	µg/L	2023-11-22	
Strontium, dissolved	6.15	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.47	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.6	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.052	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.5	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.2	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.2	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-10-11142023-NP (23K2203-05) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

General Parameters, Continued

Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.78	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	15	10	mg/L	2023-11-21	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	128	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.050	0.050	µg/L	2023-11-20	
Barium, total	1.80	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	3.8	2.0	µg/L	2023-11-21	
Cadmium, total	0.0042	0.0020	µg/L	2023-11-20	
Calcium, total	1600	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.268	0.0050	µg/L	2023-11-20	
Copper, total	10.9	0.20	µg/L	2023-11-20	
Iron, total	36.1	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.072	0.050	µg/L	2023-11-20	
Magnesium, total	512	5.0	µg/L	2023-11-20	
Manganese, total	1.01	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	< 0.010	0.010	µg/L	2023-11-20	
Nickel, total	0.538	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	135	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	2010	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2420	20	µg/L	2023-11-20	
Strontium, total	5.95	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.83	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0016	0.0010	µg/L	2023-11-20	

TEST RESULTS

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SU-SW-10-11142023-NP (23K2203-05) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

Total Metals, Continued

Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	1.4	1.0	µg/L	2023-11-20	
Zirconium, total	0.049	0.020	µg/L	2023-11-20	

SU-SW-21-11142023-NP (23K2203-06) | Matrix: Water | Sampled: 2023-11-14 15:20

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.58	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.028	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.07	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	123	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.94	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.8	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0040	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1580	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.247	0.0050	µg/L	2023-11-22	
Copper, dissolved	9.78	0.10	µg/L	2023-11-22	
Iron, dissolved	26.6	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.076	0.050	µg/L	2023-11-22	
Magnesium, dissolved	512	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.565	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.016	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.561	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	146	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2160	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-21-11142023-NP (23K2203-06) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

Dissolved Metals, Continued

Sodium, dissolved	2570	20	µg/L	2023-11-22	
Strontium, dissolved	6.47	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.43	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.9	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.053	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.3	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.3	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.67	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	< 10	10	mg/L	2023-11-21	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	124	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	1.83	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	3.7	2.0	µg/L	2023-11-21	
Cadmium, total	0.0080	0.0020	µg/L	2023-11-20	
Calcium, total	1560	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.265	0.0050	µg/L	2023-11-20	
Copper, total	10.5	0.20	µg/L	2023-11-20	
Iron, total	38.5	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.072	0.050	µg/L	2023-11-20	
Magnesium, total	507	5.0	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-21-11142023-NP (23K2203-06) | Matrix: Water | Sampled: 2023-11-14 15:20, Continued

Total Metals, Continued

Manganese, total	1.05	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.029	0.010	µg/L	2023-11-20	
Nickel, total	0.546	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	125	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	2000	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2440	20	µg/L	2023-11-20	
Strontium, total	5.97	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.86	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0017	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	1.2	1.0	µg/L	2023-11-20	
Zirconium, total	0.049	0.020	µg/L	2023-11-20	

SU-SW-22-11152023-NP (23K2203-07) | Matrix: Water | Sampled: 2023-11-15 10:00

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.67	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.033	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.4	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.42	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	116	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.87	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-22-11152023-NP (23K2203-07) | Matrix: Water | Sampled: 2023-11-15 10:00, Continued

Dissolved Metals, Continued

Boron, dissolved	3.2	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0053	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1700	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.300	0.0050	µg/L	2023-11-22	
Copper, dissolved	10.3	0.10	µg/L	2023-11-22	
Iron, dissolved	28.1	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.079	0.050	µg/L	2023-11-22	
Magnesium, dissolved	527	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.597	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.014	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.660	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	156	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2180	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2540	20	µg/L	2023-11-22	
Strontium, dissolved	6.36	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.40	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0011	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.9	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.061	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.6	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.4	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.4	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.40	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	22	10	mg/L	2023-11-23	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-22-11152023-NP (23K2203-07) | Matrix: Water | Sampled: 2023-11-15 10:00, Continued

General Parameters, Continued

Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-22	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	117	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.054	0.050	µg/L	2023-11-20	
Barium, total	1.72	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	3.7	2.0	µg/L	2023-11-21	
Cadmium, total	0.0048	0.0020	µg/L	2023-11-20	
Calcium, total	1670	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.298	0.0050	µg/L	2023-11-20	
Copper, total	11.0	0.20	µg/L	2023-11-20	
Iron, total	37.0	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.073	0.050	µg/L	2023-11-20	
Magnesium, total	524	5.0	µg/L	2023-11-20	
Manganese, total	1.03	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.014	0.010	µg/L	2023-11-20	
Nickel, total	0.848	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	138	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	2090	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2550	20	µg/L	2023-11-20	
Strontium, total	5.97	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.64	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0018	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	50.0	1.0	µg/L	2023-11-20	
Zirconium, total	0.045	0.020	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-09-11152023-NP (23K2203-08) Matrix: Water Sampled: 2023-11-15 12:50					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.32	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.035	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.0	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	5.62	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	117	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.88	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.2	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0025	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1410	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.0383	0.0050	µg/L	2023-11-22	
Copper, dissolved	1.94	0.10	µg/L	2023-11-22	
Iron, dissolved	27.2	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.078	0.050	µg/L	2023-11-22	
Magnesium, dissolved	511	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.444	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.016	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.283	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	141	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2130	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2420	20	µg/L	2023-11-22	
Strontium, dissolved	6.26	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.36	0.20	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-09-11152023-NP (23K2203-08) | Matrix: Water | Sampled: 2023-11-15 12:50, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	< 1.0	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.048	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.4	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.3	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.3	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.65	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	52	10	mg/L	2023-11-23	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-22	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	119	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	1.78	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	< 2.0	2.0	µg/L	2023-11-20	
Cadmium, total	0.0021	0.0020	µg/L	2023-11-20	
Calcium, total	1400	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.0443	0.0050	µg/L	2023-11-20	
Copper, total	1.92	0.20	µg/L	2023-11-20	
Iron, total	32.8	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.068	0.050	µg/L	2023-11-20	
Magnesium, total	503	5.0	µg/L	2023-11-20	
Manganese, total	0.862	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.013	0.010	µg/L	2023-11-20	
Nickel, total	0.274	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	122	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	2080	100	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-09-11152023-NP (23K2203-08) Matrix: Water Sampled: 2023-11-15 12:50, Continued					
Total Metals, Continued					
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2290	20	µg/L	2023-11-20	
Strontium, total	5.71	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.66	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0021	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	< 1.0	1.0	µg/L	2023-11-20	
Zirconium, total	0.041	0.020	µg/L	2023-11-20	

SU-SW-08-11152023-NP (23K2203-09) | Matrix: Water | Sampled: 2023-11-15 14:00

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	4.70	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.036	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	2.7	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	8.61	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	103	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	1.76	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	6.1	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0159	0.0020	µg/L	2023-11-22	
Calcium, dissolved	2510	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	1.16	0.0050	µg/L	2023-11-22	
Copper, dissolved	37.3	0.10	µg/L	2023-11-22	
Iron, dissolved	29.6	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-08-11152023-NP (23K2203-09) | Matrix: Water | Sampled: 2023-11-15 14:00, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.082	0.050	µg/L	2023-11-22	
Magnesium, dissolved	565	5.0	µg/L	2023-11-22	
Manganese, dissolved	1.06	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.022	0.010	µg/L	2023-11-22	
Nickel, dissolved	1.84	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	158	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	2390	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	2950	20	µg/L	2023-11-22	
Strontium, dissolved	7.21	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.38	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0010	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	4.6	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.045	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.8	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.8	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.8	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.08	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	50	10	mg/L	2023-11-23	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-22	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	112	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.055	0.050	µg/L	2023-11-20	
Barium, total	1.66	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-08-11152023-NP (23K2203-09) | Matrix: Water | Sampled: 2023-11-15 14:00, Continued

Total Metals, Continued

Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	5.4	2.0	µg/L	2023-11-20	
Cadmium, total	0.0140	0.0020	µg/L	2023-11-20	
Calcium, total	2370	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	1.00	0.0050	µg/L	2023-11-20	
Copper, total	35.8	0.20	µg/L	2023-11-20	
Iron, total	43.2	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.075	0.050	µg/L	2023-11-20	
Magnesium, total	550	5.0	µg/L	2023-11-20	
Manganese, total	1.43	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.022	0.010	µg/L	2023-11-20	
Nickel, total	1.62	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	141	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	2330	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	2770	20	µg/L	2023-11-20	
Strontium, total	6.51	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.80	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0019	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	3.8	1.0	µg/L	2023-11-20	
Zirconium, total	0.036	0.020	µg/L	2023-11-20	

SU-SW-28-11162023-NP (23K2203-10) | Matrix: Water | Sampled: 2023-11-16 10:35

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	2.24	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.058	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.1	1.0	mg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-28-11162023-NP (23K2203-10) | Matrix: Water | Sampled: 2023-11-16 10:35, Continued

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	5.09	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	127	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	0.098	0.050	µg/L	2023-11-22	
Barium, dissolved	3.39	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	2.0	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0032	0.0020	µg/L	2023-11-22	
Calcium, dissolved	1360	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.0477	0.0050	µg/L	2023-11-22	
Copper, dissolved	0.75	0.10	µg/L	2023-11-22	
Iron, dissolved	61.9	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.149	0.050	µg/L	2023-11-22	
Magnesium, dissolved	412	5.0	µg/L	2023-11-22	
Manganese, dissolved	1.62	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.011	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.439	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	248	20	µg/L	2023-11-22	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-22	
Silicon, dissolved	1790	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	1810	20	µg/L	2023-11-22	
Strontium, dissolved	7.86	0.10	µg/L	2023-11-22	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	0.61	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0028	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	1.1	1.0	µg/L	2023-11-22	
Zirconium, dissolved	0.031	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	2.2	2.0	mg/L	2023-11-23	
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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-28-11162023-NP (23K2203-10) | Matrix: Water | Sampled: 2023-11-16 10:35, Continued

General Parameters, Continued

Alkalinity, Total (as CaCO ₃)	3.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	3.65	0.50	mg/L	2023-11-21	
Solids, Total Dissolved	22	10	mg/L	2023-11-24	
Solids, Total Suspended	< 2.0	2.0	mg/L	2023-11-23	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	139	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.109	0.050	µg/L	2023-11-20	
Barium, total	3.11	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	< 2.0	2.0	µg/L	2023-11-20	
Cadmium, total	0.0023	0.0020	µg/L	2023-11-20	
Calcium, total	1330	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	0.0746	0.0050	µg/L	2023-11-20	
Copper, total	0.77	0.20	µg/L	2023-11-20	
Iron, total	91.2	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.146	0.050	µg/L	2023-11-20	
Magnesium, total	415	5.0	µg/L	2023-11-20	
Manganese, total	3.40	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.016	0.010	µg/L	2023-11-20	
Nickel, total	0.424	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	243	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	1760	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	1700	20	µg/L	2023-11-20	
Strontium, total	7.28	0.10	µg/L	2023-11-20	
Sulfur, total	1200	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	



TEST RESULTS

REPORTED TO PROJECT	AtkinsRéalis (Nelson) 697296	WORK ORDER REPORTED	23K2203 2023-11-27 12:55
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-28-11162023-NP (23K2203-10) Matrix: Water Sampled: 2023-11-16 10:35, Continued					
Total Metals, Continued					
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	1.39	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0037	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	14.5	1.0	µg/L	2023-11-20	
Zirconium, total	0.036	0.020	µg/L	2023-11-20	

Sample Qualifiers:	
HT1	The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2020)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2021)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2021)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2020)	Ion Chromatography	✓	Richmond
Carbon, Dissolved Organic in Water	SM 5310 B (2022)	Combustion, Infrared CO ₂ Detection	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2021)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Mercury, total, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2020)	Gravimetry (Dried at 103-105C)		Richmond
Solids, Total Suspended in Water	SM 2540 D* (2020)	Gravimetry (Dried at 103-105C)	✓	Richmond
Sulfide, Total in Water	SM 4500-S ₂ D* (2021)	Colorimetry (Methylene Blue)	✓	Edmonton
Total Metals in Water	EPA 200.2* / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association



APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO AtkinsRéalis (Nelson)
PROJECT 697296

WORK ORDER 23K2203
REPORTED 2023-11-27 12:55

General Comments:

The results in this report apply to the received samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Caro will dispose of all samples within 30 days of sample receipt, unless otherwise agreed.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: rpshyk@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (BLK):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B3K1988									
Blank (B3K1988-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B3K1988-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B3K1988-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		98	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.276	0.010 mg/L	0.300		92	85-115			
Nitrite (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		105	85-115			
LCS (B3K1988-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		97	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Nitrite (as N)	0.273	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		106	85-115			
Duplicate (B3K1988-DUP1)			Source: 23K2203-01		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Bromide	< 0.1	0.1 mg/L		< 0.1					20
Chloride	3.60	0.10 mg/L		3.66			2		20
Fluoride	< 0.10	0.10 mg/L		< 0.10					20
Nitrate (as N)	0.035	0.010 mg/L		0.035					20
Nitrite (as N)	< 0.010	0.010 mg/L		< 0.010					20

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Anions, Batch B3K1988, Continued

Duplicate (B3K1988-DUP1), Continued		Source: 23K2203-01		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Sulfate	1.3	1.0 mg/L		1.3				20	
Duplicate (B3K1988-DUP2)		Source: 23K2203-02		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Bromide	< 0.1	0.1 mg/L		< 0.1				20	
Chloride	3.59	0.10 mg/L		3.65			2	20	
Fluoride	< 0.10	0.10 mg/L		< 0.10				20	
Nitrate (as N)	0.033	0.010 mg/L		0.033				20	
Nitrite (as N)	< 0.010	0.010 mg/L		< 0.010				20	
Sulfate	1.3	1.0 mg/L		1.3				20	

Dissolved Metals, Batch B3K1993

Blank (B3K1993-BLK1)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							
Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.10	0.10 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							
LCS (B3K1993-BS1)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Aluminum, dissolved	4090	1.0 µg/L		4000		102		80-120	
Antimony, dissolved	40.6	0.050 µg/L		40.0		101		80-120	
Arsenic, dissolved	412	0.050 µg/L		400		103		80-120	
Barium, dissolved	40.3	0.10 µg/L		40.0		101		80-120	

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1993, Continued									
LCS (B3K1993-BS1), Continued				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Beryllium, dissolved	40.4	0.010 µg/L	40.0		101	80-120			
Bismuth, dissolved	40.6	0.010 µg/L	40.0		102	80-120			
Boron, dissolved	400	2.0 µg/L	400		100	80-120			
Cadmium, dissolved	40.6	0.0020 µg/L	40.0		102	80-120			
Calcium, dissolved	4050	50 µg/L	4000		101	80-120			
Chromium, dissolved	41.3	0.10 µg/L	40.0		103	80-120			
Cobalt, dissolved	40.8	0.0050 µg/L	40.0		102	80-120			
Copper, dissolved	40.9	0.10 µg/L	40.0		102	80-120			
Iron, dissolved	4070	2.0 µg/L	4000		102	80-120			
Lead, dissolved	41.0	0.050 µg/L	40.0		102	80-120			
Lithium, dissolved	40.7	0.050 µg/L	40.0		102	80-120			
Magnesium, dissolved	4160	5.0 µg/L	4000		104	80-120			
Manganese, dissolved	41.2	0.050 µg/L	40.0		103	80-120			
Molybdenum, dissolved	39.8	0.010 µg/L	40.0		100	80-120			
Nickel, dissolved	40.2	0.040 µg/L	40.0		101	80-120			
Phosphorus, dissolved	4210	10 µg/L	4000		105	80-120			
Potassium, dissolved	4280	20 µg/L	4000		107	80-120			
Selenium, dissolved	401	0.10 µg/L	400		100	80-120			
Silicon, dissolved	3980	100 µg/L	4000		99	80-120			
Silver, dissolved	40.4	0.010 µg/L	40.0		101	80-120			
Sodium, dissolved	4110	20 µg/L	4000		103	80-120			
Strontium, dissolved	41.1	0.10 µg/L	40.0		103	80-120			
Sulfur, dissolved	40200	1000 µg/L	40000		100	80-120			
Tellurium, dissolved	40.2	0.050 µg/L	40.0		100	80-120			
Thallium, dissolved	40.5	0.0040 µg/L	40.0		101	80-120			
Thorium, dissolved	37.9	0.010 µg/L	40.0		95	80-120			
Tin, dissolved	40.1	0.050 µg/L	40.0		100	80-120			
Titanium, dissolved	40.2	0.20 µg/L	40.0		101	80-120			
Tungsten, dissolved	40.0	0.20 µg/L	40.0		100	80-120			
Uranium, dissolved	40.8	0.0010 µg/L	40.0		102	80-120			
Vanadium, dissolved	41.1	1.00 µg/L	40.0		103	80-120			
Zinc, dissolved	407	1.0 µg/L	400		102	80-120			
Zirconium, dissolved	38.9	0.020 µg/L	40.0		97	80-120			
Duplicate (B3K1993-DUP1)				Source: 23K2203-01		Prepared: 2023-11-22, Analyzed: 2023-11-22			
Aluminum, dissolved	124	1.0 µg/L		123			< 1	20	
Antimony, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Arsenic, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Barium, dissolved	1.89	0.10 µg/L		1.84			3	20	
Beryllium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Bismuth, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Boron, dissolved	3.3	2.0 µg/L		3.1				20	
Cadmium, dissolved	0.0053	0.0020 µg/L		0.0053				20	
Calcium, dissolved	1650	50 µg/L		1630			1	20	
Chromium, dissolved	0.29	0.50 µg/L		< 0.50				20	
Cobalt, dissolved	0.247	0.0050 µg/L		0.247			< 1	20	
Copper, dissolved	9.86	0.10 µg/L		9.93			< 1	20	
Iron, dissolved	27.8	2.0 µg/L		28.0			1	20	
Lead, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Lithium, dissolved	0.083	0.050 µg/L		0.081				20	
Magnesium, dissolved	522	5.0 µg/L		520			< 1	11	
Manganese, dissolved	0.616	0.050 µg/L		0.603			2	15	
Molybdenum, dissolved	0.018	0.010 µg/L		0.017				17	
Nickel, dissolved	0.592	0.040 µg/L		0.555			6	20	
Phosphorus, dissolved	< 10	10 µg/L		< 10				15	
Potassium, dissolved	148	20 µg/L		147			< 1	20	

APPENDIX 2: QUALITY CONTROL RESULTS

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WORK ORDER REPORTED 23K2203
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1993, Continued									
Duplicate (B3K1993-DUP1), Continued		Source: 23K2203-01		Prepared: 2023-11-22, Analyzed: 2023-11-22					
Selenium, dissolved	< 0.10	0.10 µg/L		< 0.10				20	
Silicon, dissolved	2170	100 µg/L		2150			< 1	11	
Silver, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Sodium, dissolved	2500	20 µg/L		2490			< 1	10	
Strontium, dissolved	6.19	0.10 µg/L		6.43			4	12	
Sulfur, dissolved	< 1000	1000 µg/L		< 1000				20	
Tellurium, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Thallium, dissolved	< 0.0040	0.0040 µg/L		< 0.0040				20	
Thorium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Tin, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Titanium, dissolved	0.49	0.20 µg/L		0.45				15	
Tungsten, dissolved	< 0.20	0.20 µg/L		< 0.20				20	
Uranium, dissolved	< 0.0010	0.0010 µg/L		0.0010				16	
Vanadium, dissolved	< 1.00	1.00 µg/L		< 1.00				20	
Zinc, dissolved	1.7	1.0 µg/L		1.6				15	
Zirconium, dissolved	0.053	0.020 µg/L		0.050				20	
Matrix Spike (B3K1993-MS1)		Source: 23K2203-02		Prepared: 2023-11-22, Analyzed: 2023-11-22					
Aluminum, dissolved	4490	1.0 µg/L	4000	120	109	70-130			
Antimony, dissolved	36.0	0.050 µg/L	40.0	< 0.050	90	70-130			
Arsenic, dissolved	430	0.050 µg/L	400	< 0.050	107	70-130			
Barium, dissolved	46.7	0.10 µg/L	40.0	1.87	112	70-130			
Beryllium, dissolved	43.5	0.010 µg/L	40.0	< 0.010	109	70-130			
Bismuth, dissolved	26.4	0.010 µg/L	40.0	< 0.010	66	70-130			MES
Boron, dissolved	393	2.0 µg/L	400	2.7	98	70-130			
Cadmium, dissolved	43.8	0.0020 µg/L	40.0	0.0055	110	70-130			
Calcium, dissolved	5800	50 µg/L	4000	1610	105	70-130			
Chromium, dissolved	42.2	0.10 µg/L	40.0	< 0.50	105	85-117			
Cobalt, dissolved	42.7	0.0050 µg/L	40.0	0.251	106	70-130			
Copper, dissolved	51.7	0.10 µg/L	40.0	9.40	106	88-113			
Iron, dissolved	4080	2.0 µg/L	4000	28.1	101	80-115			
Lead, dissolved	42.9	0.050 µg/L	40.0	< 0.050	107	84-121			
Lithium, dissolved	44.7	0.050 µg/L	40.0	0.076	111	70-130			
Magnesium, dissolved	4860	5.0 µg/L	4000	527	108	70-130			
Manganese, dissolved	44.1	0.050 µg/L	40.0	0.613	109	70-130			
Molybdenum, dissolved	34.5	0.010 µg/L	40.0	0.020	86	70-130			
Nickel, dissolved	43.2	0.040 µg/L	40.0	0.604	106	83-121			
Phosphorus, dissolved	4320	10 µg/L	4000	< 10	108	70-130			
Potassium, dissolved	4650	20 µg/L	4000	148	112	70-130			
Selenium, dissolved	426	0.10 µg/L	400	< 0.10	107	91-122			
Silicon, dissolved	6750	100 µg/L	4000	2140	115	70-130			
Silver, dissolved	30.1	0.010 µg/L	40.0	< 0.010	75	74-120			
Sodium, dissolved	6550	20 µg/L	4000	2530	100	70-130			
Strontium, dissolved	49.4	0.10 µg/L	40.0	6.39	108	70-130			
Sulfur, dissolved	42900	1000 µg/L	40000	< 1000	107	70-130			
Tellurium, dissolved	36.8	0.050 µg/L	40.0	< 0.050	92	70-130			
Thallium, dissolved	43.5	0.0040 µg/L	40.0	< 0.0040	109	79-119			
Thorium, dissolved	40.1	0.010 µg/L	40.0	< 0.010	100	70-130			
Tin, dissolved	40.8	0.050 µg/L	40.0	< 0.050	102	70-130			
Titanium, dissolved	40.6	0.20 µg/L	40.0	0.39	100	70-130			
Tungsten, dissolved	37.4	0.20 µg/L	40.0	< 0.20	93	70-130			
Uranium, dissolved	41.8	0.0010 µg/L	40.0	< 0.0010	104	70-130			
Vanadium, dissolved	43.2	1.00 µg/L	40.0	< 1.00	108	80-115			
Zinc, dissolved	441	1.0 µg/L	400	1.6	110	89-123			
Zirconium, dissolved	41.6	0.020 µg/L	40.0	0.058	104	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K2139									
Duplicate (B3K2139-DUP2)		Source: 23K2203-06		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Mercury, dissolved	< 0.0050	0.0050 µg/L		< 0.0050				20	
Dissolved Metals, Batch B3K2259									
Blank (B3K2259-BLK1)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK2)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK3)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK4)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK5)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK6)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK7)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK8)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
LCS (B3K2259-BS1)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.249	0.0050 µg/L		0.250	100	80-120			
LCS (B3K2259-BS2)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.223	0.0050 µg/L		0.250	89	80-120			
LCS (B3K2259-BS3)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.236	0.0050 µg/L		0.250	94	80-120			
LCS (B3K2259-BS4)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.242	0.0050 µg/L		0.250	97	80-120			
LCS (B3K2259-BS5)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.261	0.0050 µg/L		0.250	104	80-120			
LCS (B3K2259-BS6)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.246	0.0050 µg/L		0.250	98	80-120			
LCS (B3K2259-BS7)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.254	0.0050 µg/L		0.250	102	80-120			
LCS (B3K2259-BS8)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Mercury, dissolved	0.252	0.0050 µg/L		0.250	101	80-120			
General Parameters, Batch B3K1963									
Blank (B3K1963-BLK1)		Prepared: 2023-11-20, Analyzed: 2023-11-21							
Solids, Total Dissolved	< 10	10 mg/L							

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B3K1963, Continued									
Blank (B3K1963-BLK2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K1963-BS1)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	45	10 mg/L	50.0		90	85-115			
LCS (B3K1963-BS2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	198	10 mg/L	208		95	85-115			
General Parameters, Batch B3K2028									
Blank (B3K2028-BLK1)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3K2028-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3K2028-BLK3)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
LCS (B3K2028-BS1)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	10.1	0.50 mg/L	10.0		101	78-116			
LCS (B3K2028-BS2)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	9.58	0.50 mg/L	10.0		96	78-116			
LCS (B3K2028-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	9.79	0.50 mg/L	10.0		98	78-116			
Duplicate (B3K2028-DUP3)		Source: 23K2203-10		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Carbon, Dissolved Organic	4.06	0.50 mg/L		3.65			11	15	
General Parameters, Batch B3K2063									
Blank (B3K2063-BLK1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Solids, Total Suspended	< 2.0	2.0 mg/L							
Blank (B3K2063-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Solids, Total Suspended	< 2.0	2.0 mg/L							
LCS (B3K2063-BS1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Solids, Total Suspended	87.0	5.0 mg/L	100		87	83-107			
LCS (B3K2063-BS2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Solids, Total Suspended	93.0	5.0 mg/L	100		93	83-107			
General Parameters, Batch B3K2066									
Blank (B3K2066-BLK1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK3)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B3K2066, Continued									
Blank (B3K2066-BLK4)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B3K2066-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	0.980	0.050 mg/L	1.00		98	85-115			
LCS (B3K2066-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
LCS (B3K2066-BS3)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.02	0.050 mg/L	1.00		102	85-115			
LCS (B3K2066-BS4)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
Duplicate (B3K2066-DUP2)			Source: 23K2203-01		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Ammonia, Total (as N)	< 0.050	0.050 mg/L		< 0.050				15	
Matrix Spike (B3K2066-MS2)			Source: 23K2203-01		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Ammonia, Total (as N)	0.251	0.050 mg/L	0.204	< 0.050	117	75-125			
General Parameters, Batch B3K2079									
Blank (B3K2079-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Sulfide, Total	< 0.020	0.020 mg/L							
LCS (B3K2079-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Sulfide, Total	0.472	0.020 mg/L	0.480		98	80-120			
Matrix Spike (B3K2079-MS1)			Source: 23K2203-04		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Sulfide, Total	0.493	0.020 mg/L	0.480	< 0.020	103	70-130			
General Parameters, Batch B3K2147									
Blank (B3K2147-BLK1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK2)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK3)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B3K2147-BS1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	100	1.0 mg/L	100		100	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	34.1	1.0 mg/L	50.0		68	0-200			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B3K2147, Continued

LCS (B3K2147-BS2)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	114	1.0 mg/L	100		114	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.8	1.0 mg/L	50.0		40	0-200			
LCS (B3K2147-BS3)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	112	1.0 mg/L	100		112	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.4	1.0 mg/L	50.0		39	0-200			

General Parameters, Batch B3K2221

Blank (B3K2221-BLK1)			Prepared: 2023-11-22, Analyzed: 2023-11-23						
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K2221-BS1)			Prepared: 2023-11-22, Analyzed: 2023-11-23						
Solids, Total Dissolved	55	10 mg/L	50.0		110	85-115			
LCS (B3K2221-BS2)			Prepared: 2023-11-22, Analyzed: 2023-11-23						
Solids, Total Dissolved	212	10 mg/L	208		102	85-115			

General Parameters, Batch B3K2238

Blank (B3K2238-BLK1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Solids, Total Suspended	< 2.0	2.0 mg/L							
Blank (B3K2238-BLK2)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Solids, Total Suspended	< 2.0	2.0 mg/L							
LCS (B3K2238-BS1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Solids, Total Suspended	101	5.0 mg/L	100		101	83-107			
LCS (B3K2238-BS2)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Solids, Total Suspended	100	5.0 mg/L	100		100	83-107			

General Parameters, Batch B3K2318

Blank (B3K2318-BLK1)			Prepared: 2023-11-23, Analyzed: 2023-11-23						
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B3K2318-BS1)			Prepared: 2023-11-23, Analyzed: 2023-11-23						
Acidity, Total (as CaCO ₃)	94.0	2.0 mg/L	100		94	80-120			
Duplicate (B3K2318-DUP1)			Source: 23K2203-03		Prepared: 2023-11-23, Analyzed: 2023-11-23				
Acidity, Total (as CaCO ₃)	2.6	2.0 mg/L		2.7				20	

General Parameters, Batch B3K2339

Blank (B3K2339-BLK1)			Prepared: 2023-11-23, Analyzed: 2023-11-24						
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K2339-BS1)			Prepared: 2023-11-23, Analyzed: 2023-11-24						
Solids, Total Dissolved	42	10 mg/L	50.0		85	85-115			
LCS (B3K2339-BS2)			Prepared: 2023-11-23, Analyzed: 2023-11-24						
Solids, Total Dissolved	210	10 mg/L	208		101	85-115			

General Parameters, Batch B3K2341

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B3K2341, Continued

Blank (B3K2341-BLK1)

Prepared: 2023-11-23, Analyzed: 2023-11-23

Solids, Total Suspended	< 2.0	2.0 mg/L
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LCS (B3K2341-BS1)

Prepared: 2023-11-23, Analyzed: 2023-11-23

Solids, Total Suspended	103	5.0 mg/L	100	103	83-107
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Total Metals, Batch B3K2005

Blank (B3K2005-BLK1)

Prepared: 2023-11-20, Analyzed: 2023-11-20

Aluminum, total	< 2.0	2.0 µg/L
Antimony, total	< 0.050	0.050 µg/L
Arsenic, total	< 0.050	0.050 µg/L
Barium, total	< 0.10	0.10 µg/L
Beryllium, total	< 0.010	0.010 µg/L
Bismuth, total	< 0.010	0.010 µg/L
Boron, total	< 2.0	2.0 µg/L
Cadmium, total	< 0.0020	0.0020 µg/L
Calcium, total	< 50	50 µg/L
Chromium, total	< 0.50	0.50 µg/L
Cobalt, total	< 0.0050	0.0050 µg/L
Copper, total	< 0.20	0.20 µg/L
Iron, total	< 2.0	2.0 µg/L
Lead, total	< 0.050	0.050 µg/L
Lithium, total	< 0.050	0.050 µg/L
Magnesium, total	< 5.0	5.0 µg/L
Manganese, total	< 0.050	0.050 µg/L
Molybdenum, total	< 0.010	0.010 µg/L
Nickel, total	< 0.040	0.040 µg/L
Phosphorus, total	< 10	10 µg/L
Potassium, total	< 20	20 µg/L
Selenium, total	< 0.10	0.10 µg/L
Silicon, total	< 100	100 µg/L
Silver, total	< 0.010	0.010 µg/L
Sodium, total	< 20	20 µg/L
Strontium, total	< 0.10	0.10 µg/L
Sulfur, total	< 1000	1000 µg/L
Tellurium, total	< 0.050	0.050 µg/L
Thallium, total	< 0.0040	0.0040 µg/L
Thorium, total	< 0.010	0.010 µg/L
Tin, total	< 0.050	0.050 µg/L
Titanium, total	< 0.20	0.20 µg/L
Tungsten, total	< 0.20	0.20 µg/L
Uranium, total	< 0.0010	0.0010 µg/L
Vanadium, total	< 1.00	1.00 µg/L
Zinc, total	< 1.0	1.0 µg/L
Zirconium, total	< 0.020	0.020 µg/L

LCS (B3K2005-BS1)

Prepared: 2023-11-20, Analyzed: 2023-11-20

Aluminum, total	4070	2.0 µg/L	4000	102	80-120
Antimony, total	38.7	0.050 µg/L	40.0	97	80-120
Arsenic, total	405	0.050 µg/L	400	101	80-120
Barium, total	39.8	0.10 µg/L	40.0	99	80-120
Beryllium, total	39.8	0.010 µg/L	40.0	99	80-120
Bismuth, total	38.2	0.010 µg/L	40.0	95	80-120
Boron, total	409	2.0 µg/L	400	102	80-120
Cadmium, total	38.8	0.0020 µg/L	40.0	97	80-120
Calcium, total	4010	50 µg/L	4000	100	80-120

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2203
2023-11-27 12:55

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3K2005, Continued									
LCS (B3K2005-BS1), Continued				Prepared: 2023-11-20, Analyzed: 2023-11-20					
Chromium, total	41.2	0.50 µg/L	40.0		103	80-120			
Cobalt, total	41.3	0.0050 µg/L	40.0		103	80-120			
Copper, total	41.0	0.20 µg/L	40.0		102	80-120			
Iron, total	4100	2.0 µg/L	4000		102	80-120			
Lead, total	38.8	0.050 µg/L	40.0		97	80-120			
Lithium, total	38.5	0.050 µg/L	40.0		96	80-120			
Magnesium, total	4180	5.0 µg/L	4000		104	80-120			
Manganese, total	41.1	0.050 µg/L	40.0		103	80-120			
Molybdenum, total	39.5	0.010 µg/L	40.0		99	80-120			
Nickel, total	41.1	0.040 µg/L	40.0		103	80-120			
Phosphorus, total	4000	10 µg/L	4000		100	80-120			
Potassium, total	4030	20 µg/L	4000		101	80-120			
Selenium, total	399	0.10 µg/L	400		100	80-120			
Silicon, total	3910	100 µg/L	4000		98	80-120			
Silver, total	39.8	0.010 µg/L	40.0		99	80-120			
Sodium, total	4180	20 µg/L	4000		105	80-120			
Strontium, total	40.0	0.10 µg/L	40.0		100	80-120			
Sulfur, total	38600	1000 µg/L	40000		97	80-120			
Tellurium, total	38.7	0.050 µg/L	40.0		97	80-120			
Thallium, total	38.8	0.0040 µg/L	40.0		97	80-120			
Thorium, total	39.5	0.010 µg/L	40.0		99	80-120			
Tin, total	40.6	0.050 µg/L	40.0		102	80-120			
Titanium, total	39.8	0.20 µg/L	40.0		99	80-120			
Tungsten, total	39.7	0.20 µg/L	40.0		99	80-120			
Uranium, total	40.1	0.0010 µg/L	40.0		100	80-120			
Vanadium, total	41.3	1.00 µg/L	40.0		103	80-120			
Zinc, total	405	1.0 µg/L	400		101	80-120			
Zirconium, total	38.5	0.020 µg/L	40.0		96	80-120			
Matrix Spike (B3K2005-MS1)				Source: 23K2203-01 Prepared: 2023-11-20, Analyzed: 2023-11-20					
Aluminum, total	4570	2.0 µg/L	4000	127	111	70-130			
Antimony, total	31.9	0.050 µg/L	40.0	< 0.050	80	70-130			
Arsenic, total	443	0.050 µg/L	400	< 0.050	111	70-130			
Barium, total	45.8	0.10 µg/L	40.0	1.77	110	70-130			
Beryllium, total	44.0	0.010 µg/L	40.0	< 0.010	110	70-130			
Bismuth, total	42.7	0.010 µg/L	40.0	< 0.010	107	70-130			
Boron, total	449	2.0 µg/L	400	4.0	111	70-130			
Cadmium, total	42.7	0.0020 µg/L	40.0	0.0047	107	70-130			
Calcium, total	5960	50 µg/L	4000	1600	109	70-130			
Chromium, total	44.8	0.50 µg/L	40.0	< 0.50	111	70-130			
Cobalt, total	44.9	0.0050 µg/L	40.0	0.245	112	70-130			
Copper, total	55.1	0.20 µg/L	40.0	10.4	112	70-130			
Iron, total	4420	2.0 µg/L	4000	35.8	110	70-130			
Lead, total	42.8	0.050 µg/L	40.0	< 0.050	107	70-130			
Lithium, total	42.6	0.050 µg/L	40.0	0.071	106	70-130			
Magnesium, total	5100	5.0 µg/L	4000	519	114	70-130			
Manganese, total	46.5	0.050 µg/L	40.0	0.946	114	70-130			
Molybdenum, total	40.7	0.010 µg/L	40.0	0.018	102	70-130			
Nickel, total	45.6	0.040 µg/L	40.0	0.564	113	70-130			
Phosphorus, total	4340	10 µg/L	4000	< 10	109	70-130			
Potassium, total	4580	20 µg/L	4000	132	111	70-130			
Selenium, total	437	0.10 µg/L	400	< 0.10	109	70-130			
Silicon, total	6530	100 µg/L	4000	2000	113	70-130			
Silver, total	34.8	0.010 µg/L	40.0	< 0.010	87	70-130			
Sodium, total	6620	20 µg/L	4000	2380	106	70-130			
Strontium, total	50.8	0.10 µg/L	40.0	5.94	112	70-130			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2203
2023-11-27 12:55

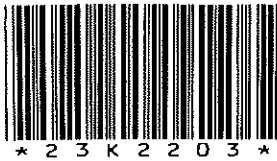
Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3K2005, Continued									
Matrix Spike (B3K2005-MS1), Continued		Source: 23K2203-01		Prepared: 2023-11-20, Analyzed: 2023-11-20					
Sulfur, total	43200	1000 µg/L	40000	< 1000	108	70-130			
Tellurium, total	42.8	0.050 µg/L	40.0	< 0.050	107	70-130			
Thallium, total	42.3	0.0040 µg/L	40.0	< 0.0040	106	70-130			
Thorium, total	40.7	0.010 µg/L	40.0	< 0.010	102	70-130			
Tin, total	43.8	0.050 µg/L	40.0	< 0.050	110	70-130			
Titanium, total	43.9	0.20 µg/L	40.0	0.88	108	70-130			
Tungsten, total	42.8	0.20 µg/L	40.0	< 0.20	107	70-130			
Uranium, total	43.5	0.0010 µg/L	40.0	0.0017	109	70-130			
Vanadium, total	44.6	1.00 µg/L	40.0	< 1.00	111	70-130			
Zinc, total	448	1.0 µg/L	400	1.1	112	70-130			
Zirconium, total	42.2	0.020 µg/L	40.0	0.045	105	70-130			

Total Metals, Batch B3K2139

Blank (B3K2139-BLK2)		Prepared: 2023-11-21, Analyzed: 2023-11-21							
Mercury, total	< 0.0050	0.0050 µg/L							
LCS (B3K2139-BS2)		Prepared: 2023-11-21, Analyzed: 2023-11-21							
Mercury, total	0.256	0.0050 µg/L	0.250		102	80-120			
Duplicate (B3K2139-DUP2)		Source: 23K2203-06		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Mercury, total	< 0.0050	0.0050 µg/L		< 0.0050			20		

QC Qualifiers:

MES A number up to 10% (rounded down) of the analytes in a Multi-Element Scan may exceed control limits by up to 10% (absolute).



SSD Puvg 11-20 9:045 9.4°C

COC ID: 697296-20231116-4		TURNAROUND TIME: Regular		RUSH:													
PROJECT/CLIENT INFO				LABORATORY				OTHER INFO									
/ Job# Sunro Mine				Lab Name Caro				Report Format / Distribution									
Inager Michelle Unger				Lab Contact Regan Pshyk				Email 1: michelle.unger@teck.com									
Email michelle.unger@teck.com				Email rpschyk@caro.ca				Email 2: craig.pinnell@atkinsrealis.com									
Address 601 Knighton Rd				Address 110 - 4011 Viking Way				Email 3: jonathan.croston@atkinsrealis.com									
City Kimberly				City Richmond				Email 4: bonn.johnston@atkinsrealis.com									
Province BC				Province BC				Email 5: thiv.West_BC_LabData@atkinsrealis.com									
Postal Code V1A 1C7				Postal Code V6B2K9				Country Canada									
Phone Number 250-427-8422				Phone Number 604-279-1499				PO number 697296									
SAMPLE DETAILS									ANALYSIS REQUESTED								
Filtered - F: Field, L: Lab, FL: Field & Lab, X: None																	
Sample ID									Sample Location								
Field Matrix									Hazardous Material (Yes/No)								
Date									Time (24hr)								
G=Grab C=Comp									# Of Cont.								
BIB									PRESERV.								
ANALYSIS									Total Metals + Hg								
									Dissolved Metals + Hg								
									TDS & TSS								
									Hardness								
									Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)								
									Total and Speciated Alkalinity, Acidity								
									Sulphide								
									Ammonia								
									Dissolved Organic Carbon								
									HOLD								
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS									RELINQUISHED BY/AFFILIATION								
Ultra low detection for metals required, Detection limits suitable for CSR and BCWQG									Atkins Realis Inc								
Invoice to: Atkins Realis Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3									DATE/TIME								
Send to: payables@atkinsrealis.com Billing PO: 697296									ACCEPTED BY/AFFILIATION								
DATE/TIME									DATE/TIME								
NO OF BOTTLES RETURNED/DESCRIPTION																	
Regular (default) X									Sampler's Name								
Priority (2-3 business days) - 50% surcharge									Bonn Johnston and Chris Walker								
Emergency (1 Business Day) - 100% surcharge									Mobile #								
For Emergency <1 Day, ASAP or Weekend									236-464-2830								
Sampler's Signature									Date/Time								
									2023-11-16 17:00:00 PM								

CERTIFICATE OF ANALYSIS

REPORTED TO AtkinsRéalis (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Craig Pinnell

PO NUMBER 697296

PROJECT 697296

PROJECT INFO Teck Sunro Mine-20231116-01

WORK ORDER 23K2213

RECEIVED / TEMP 2023-11-20 09:00 / 9.4°C

REPORTED 2023-11-27 11:40

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

Work Order Comments:

Custody Seals Intact: YES

By engaging our services, you are agreeing to CARO Analytical Service's Standard Terms and Conditions outlined here:
<https://www.caro.ca/terms-conditions>

If you have any questions or concerns, please contact me at rpsyhk@caro.ca

Authorized By:

Regan Pshyk
Account Manager

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#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-02-11132023-NP (23K2213-01) Matrix: Water Sampled: 2023-11-13 15:10					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	5.61	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.961	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	45.3	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	51.1	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	13.9	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Barium, dissolved	21.9	0.10	µg/L	2023-11-22	
Beryllium, dissolved	0.018	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	23.0	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.471	0.0020	µg/L	2023-11-22	
Calcium, dissolved	15700	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	50.2	0.0050	µg/L	2023-11-22	
Copper, dissolved	1620	0.10	µg/L	2023-11-22	
Iron, dissolved	4.8	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Lithium, dissolved	0.437	0.050	µg/L	2023-11-22	
Magnesium, dissolved	2870	5.0	µg/L	2023-11-22	
Manganese, dissolved	7.71	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.058	0.010	µg/L	2023-11-22	
Nickel, dissolved	61.6	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	563	20	µg/L	2023-11-22	
Selenium, dissolved	0.80	0.10	µg/L	2023-11-22	
Silicon, dissolved	11200	100	µg/L	2023-11-22	
Silver, dissolved	0.017	0.010	µg/L	2023-11-22	
Sodium, dissolved	8110	20	µg/L	2023-11-22	
Strontium, dissolved	45.9	0.10	µg/L	2023-11-22	
Sulfur, dissolved	14900	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	0.0047	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-02-11132023-NP (23K2213-01) | Matrix: Water | Sampled: 2023-11-13 15:10, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0018	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	122	1.0	µg/L	2023-11-22	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	11.2	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	12.1	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	12.1	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	92	10	mg/L	2023-11-21	

SU-GW-03-11132023-NP (23K2213-02) | Matrix: Water | Sampled: 2023-11-13 15:15

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	5.60	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.650	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	5.4	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	43.2	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	9.4	1.0	µg/L	2023-11-22	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Arsenic, dissolved	0.062	0.050	µg/L	2023-11-22	
Barium, dissolved	3.48	0.10	µg/L	2023-11-22	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Boron, dissolved	23.1	2.0	µg/L	2023-11-22	
Cadmium, dissolved	0.0121	0.0020	µg/L	2023-11-22	
Calcium, dissolved	12000	50	µg/L	2023-11-22	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-22	
Cobalt, dissolved	0.271	0.0050	µg/L	2023-11-22	
Copper, dissolved	6.66	0.10	µg/L	2023-11-22	
Iron, dissolved	8.9	2.0	µg/L	2023-11-22	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-03-11132023-NP (23K2213-02) | Matrix: Water | Sampled: 2023-11-13 15:15, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.116	0.050	µg/L	2023-11-22	
Magnesium, dissolved	3200	5.0	µg/L	2023-11-22	
Manganese, dissolved	0.470	0.050	µg/L	2023-11-22	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.271	0.010	µg/L	2023-11-22	
Nickel, dissolved	0.698	0.040	µg/L	2023-11-22	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-22	
Potassium, dissolved	710	20	µg/L	2023-11-22	
Selenium, dissolved	0.15	0.10	µg/L	2023-11-22	
Silicon, dissolved	7040	100	µg/L	2023-11-22	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Sodium, dissolved	6460	20	µg/L	2023-11-22	
Strontium, dissolved	29.5	0.10	µg/L	2023-11-22	
Sulfur, dissolved	1370	1000	µg/L	2023-11-22	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-22	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-22	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-22	
Titanium, dissolved	2.36	0.20	µg/L	2023-11-22	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-22	
Uranium, dissolved	0.0038	0.0010	µg/L	2023-11-22	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-22	
Zinc, dissolved	3.4	1.0	µg/L	2023-11-22	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-22	

General Parameters

Acidity, Total (as CaCO ₃)	5.9	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	43.3	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	43.3	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	68	10	mg/L	2023-11-21	

SU-GW-A-11132023-NP (23K2213-03) | Matrix: Water | Sampled: 2023-11-13 11:35

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	10.9	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.250	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-A-11132023-NP (23K2213-03) Matrix: Water Sampled: 2023-11-13 11:35, Continued					
Anions, Continued					
Sulfate	38.4	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO ₃)	83.4	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	8.4	1.0	µg/L	2023-11-21	
Antimony, dissolved	0.093	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.153	0.050	µg/L	2023-11-21	
Barium, dissolved	7.51	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	31.8	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0126	0.0020	µg/L	2023-11-21	
Calcium, dissolved	24900	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.0829	0.0050	µg/L	2023-11-21	
Copper, dissolved	3.77	0.10	µg/L	2023-11-21	
Iron, dissolved	11.4	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.240	0.050	µg/L	2023-11-21	
Magnesium, dissolved	5160	5.0	µg/L	2023-11-21	
Manganese, dissolved	3.50	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.256	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.468	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	851	20	µg/L	2023-11-21	
Selenium, dissolved	1.29	0.10	µg/L	2023-11-21	
Silicon, dissolved	8300	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	11100	20	µg/L	2023-11-21	
Strontium, dissolved	29.5	0.10	µg/L	2023-11-21	
Sulfur, dissolved	12900	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	0.066	0.050	µg/L	2023-11-21	
Titanium, dissolved	0.22	0.20	µg/L	2023-11-21	
Tungsten, dissolved	2.80	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0144	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	2.2	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-A-11132023-NP (23K2213-03) Matrix: Water Sampled: 2023-11-13 11:35, Continued					
General Parameters					
Acidity, Total (as CaCO ₃)	4.2	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	51.7	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	51.7	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	158	10	mg/L	2023-11-21	

SU-GW-04-11132023-NP (23K2213-04) | Matrix: Water | Sampled: 2023-11-13 11:30

Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	4.61	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	1.20	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	25.6	1.0	mg/L	2023-11-21	

Calculated Parameters					
Hardness, Dissolved (as CaCO ₃)	61.8	0.125	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	11.2	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.126	0.050	µg/L	2023-11-21	
Barium, dissolved	18.7	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	11.2	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.210	0.0020	µg/L	2023-11-21	
Calcium, dissolved	17300	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	17.3	0.0050	µg/L	2023-11-21	
Copper, dissolved	349	0.10	µg/L	2023-11-21	
Iron, dissolved	13.9	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.414	0.050	µg/L	2023-11-21	
Magnesium, dissolved	4480	5.0	µg/L	2023-11-21	
Manganese, dissolved	9.74	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.028	0.010	µg/L	2023-11-21	
Nickel, dissolved	24.6	0.040	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-04-11132023-NP (23K2213-04) | Matrix: Water | Sampled: 2023-11-13 11:30, Continued

Dissolved Metals, Continued

Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	519	20	µg/L	2023-11-21	
Selenium, dissolved	0.14	0.10	µg/L	2023-11-21	
Silicon, dissolved	10500	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	5170	20	µg/L	2023-11-21	
Strontium, dissolved	49.1	0.10	µg/L	2023-11-21	
Sulfur, dissolved	7780	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	0.0103	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Tungsten, dissolved	0.43	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0115	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	74.6	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.026	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	6.7	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	43.3	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	43.3	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	135	10	mg/L	2023-11-21	

SU-GW-05-11142023-NP (23K2213-05) | Matrix: Water | Sampled: 2023-11-14 10:05

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	6.47	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.534	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	99.7	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	269	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	3.7	1.0	µg/L	2023-11-21	
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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-05-11142023-NP (23K2213-05) | Matrix: Water | Sampled: 2023-11-14 10:05, Continued

Dissolved Metals, Continued

Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.198	0.050	µg/L	2023-11-21	
Barium, dissolved	21.6	0.10	µg/L	2023-11-21	
Beryllium, dissolved	0.013	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	8.7	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.241	0.0020	µg/L	2023-11-21	
Calcium, dissolved	83300	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	10.6	0.0050	µg/L	2023-11-21	
Copper, dissolved	561	0.10	µg/L	2023-11-21	
Iron, dissolved	< 2.0	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	1.23	0.050	µg/L	2023-11-21	
Magnesium, dissolved	14700	5.0	µg/L	2023-11-21	
Manganese, dissolved	327	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-22	
Molybdenum, dissolved	0.119	0.010	µg/L	2023-11-21	
Nickel, dissolved	7.93	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	886	20	µg/L	2023-11-21	
Selenium, dissolved	0.12	0.10	µg/L	2023-11-21	
Silicon, dissolved	13900	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	8030	20	µg/L	2023-11-21	
Strontium, dissolved	185	0.10	µg/L	2023-11-21	
Sulfur, dissolved	34500	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	0.0190	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Tungsten, dissolved	2.03	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.225	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	48.0	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	12.8	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	145	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	145	1.0	mg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-05-11142023-NP (23K2213-05) Matrix: Water Sampled: 2023-11-14 10:05, Continued					
General Parameters, Continued					
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	358	10	mg/L	2023-11-21	

SU-GW-06-11132023-NP (23K2213-06) | Matrix: Water | Sampled: 2023-11-13 10:35

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	9.35	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	47.3	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	261	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	6.0	1.0	µg/L	2023-11-21	
Antimony, dissolved	0.054	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.418	0.050	µg/L	2023-11-21	
Barium, dissolved	22.9	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	16.1	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0109	0.0020	µg/L	2023-11-21	
Calcium, dissolved	79200	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	1.40	0.0050	µg/L	2023-11-21	
Copper, dissolved	1.19	0.10	µg/L	2023-11-21	
Iron, dissolved	5.6	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	1.04	0.050	µg/L	2023-11-21	
Magnesium, dissolved	15200	5.0	µg/L	2023-11-21	
Manganese, dissolved	41.2	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.558	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.407	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	909	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	12300	100	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2213
2023-11-27 11:40

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-06-11132023-NP (23K2213-06) | Matrix: Water | Sampled: 2023-11-13 10:35, Continued

Dissolved Metals, Continued

Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	15400	20	µg/L	2023-11-21	
Strontium, dissolved	81.9	0.10	µg/L	2023-11-21	
Sulfur, dissolved	14200	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	0.0065	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	0.130	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Tungsten, dissolved	60.4	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.309	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	< 1.0	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	8.6	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	232	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	232	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	305	10	mg/L	2023-11-21	

SU-GW-07-11132023-NP (23K2213-07) | Matrix: Water | Sampled: 2023-11-13 13:15

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	14.4	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	1.57	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	138	1.0	mg/L	2023-11-24	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	149	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	25.5	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Barium, dissolved	2.83	0.10	µg/L	2023-11-21	
Beryllium, dissolved	0.045	0.010	µg/L	2023-11-21	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-GW-07-11132023-NP (23K2213-07) | Matrix: Water | Sampled: 2023-11-13 13:15, Continued

Dissolved Metals, Continued

Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	56.6	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.814	0.0020	µg/L	2023-11-21	
Calcium, dissolved	40800	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	97.8	0.0050	µg/L	2023-11-21	
Copper, dissolved	1420	0.10	µg/L	2023-11-21	
Iron, dissolved	5.9	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.254	0.050	µg/L	2023-11-21	
Magnesium, dissolved	11500	5.0	µg/L	2023-11-21	
Manganese, dissolved	15.1	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.048	0.010	µg/L	2023-11-21	
Nickel, dissolved	172	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	618	20	µg/L	2023-11-21	
Selenium, dissolved	0.40	0.10	µg/L	2023-11-21	
Silicon, dissolved	8700	100	µg/L	2023-11-21	
Silver, dissolved	0.015	0.010	µg/L	2023-11-21	
Sodium, dissolved	16500	20	µg/L	2023-11-21	
Strontium, dissolved	126	0.10	µg/L	2023-11-21	
Sulfur, dissolved	46200	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	0.0045	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Tungsten, dissolved	2.23	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0061	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	214	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	8.3	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	18.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	18.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	242	10	mg/L	2023-11-21	

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-09-11132023-NP (23K2213-08) Matrix: Water Sampled: 2023-11-13 11:35					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	10.9	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.275	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	40.5	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	83.5	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	7.4	1.0	µg/L	2023-11-21	
Antimony, dissolved	0.112	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.155	0.050	µg/L	2023-11-21	
Barium, dissolved	7.37	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	32.1	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0132	0.0020	µg/L	2023-11-21	
Calcium, dissolved	24500	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.0802	0.0050	µg/L	2023-11-21	
Copper, dissolved	3.88	0.10	µg/L	2023-11-21	
Iron, dissolved	15.6	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.245	0.050	µg/L	2023-11-21	
Magnesium, dissolved	5390	5.0	µg/L	2023-11-21	
Manganese, dissolved	3.40	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.260	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.499	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	863	20	µg/L	2023-11-21	
Selenium, dissolved	1.35	0.10	µg/L	2023-11-21	
Silicon, dissolved	8070	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	11200	20	µg/L	2023-11-21	
Strontium, dissolved	29.4	0.10	µg/L	2023-11-21	
Sulfur, dissolved	12200	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	

TEST RESULTS

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-GW-09-11132023-NP (23K2213-08) Matrix: Water Sampled: 2023-11-13 11:35, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	3.37	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0156	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	1.03	1.00	µg/L	2023-11-21	
Zinc, dissolved	2.2	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	3.5	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	52.1	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	52.1	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	152	10	mg/L	2023-11-21	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2020)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2021)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2021)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2020)	Ion Chromatography	✓	Richmond
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2021)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2020)	Gravimetry (Dried at 103-105C)		Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association

General Comments:

The results in this report apply to the received samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Caro will dispose of all samples within 30 days of sample receipt, unless otherwise agreed.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: rpshyk@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Anions, Batch B3K1988

Blank (B3K1988-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							

Blank (B3K1988-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							

LCS (B3K1988-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		98	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.276	0.010 mg/L	0.300		92	85-115			
Nitrite (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		105	85-115			

LCS (B3K1988-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		97	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Nitrite (as N)	0.273	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		106	85-115			

Dissolved Metals, Batch B3K1990

Blank (B3K1990-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3K1990, Continued

Blank (B3K1990-BLK1), Continued

Prepared: 2023-11-21, Analyzed: 2023-11-21

Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							

LCS (B3K1990-BS1)

Prepared: 2023-11-21, Analyzed: 2023-11-21

Aluminum, dissolved	3900	1.0 µg/L	4000	97	80-120
Antimony, dissolved	39.4	0.050 µg/L	40.0	99	80-120
Arsenic, dissolved	404	0.050 µg/L	400	101	80-120
Barium, dissolved	38.6	0.10 µg/L	40.0	96	80-120
Beryllium, dissolved	39.1	0.010 µg/L	40.0	98	80-120
Bismuth, dissolved	39.1	0.010 µg/L	40.0	98	80-120
Boron, dissolved	393	2.0 µg/L	400	98	80-120
Cadmium, dissolved	40.3	0.0020 µg/L	40.0	101	80-120
Calcium, dissolved	4060	50 µg/L	4000	102	80-120
Chromium, dissolved	40.1	0.50 µg/L	40.0	100	80-120
Cobalt, dissolved	40.1	0.0050 µg/L	40.0	100	80-120
Copper, dissolved	40.2	0.10 µg/L	40.0	100	80-120
Iron, dissolved	4000	2.0 µg/L	4000	100	80-120
Lead, dissolved	39.6	0.050 µg/L	40.0	99	80-120
Lithium, dissolved	38.6	0.050 µg/L	40.0	96	80-120
Magnesium, dissolved	3930	5.0 µg/L	4000	98	80-120
Manganese, dissolved	40.0	0.050 µg/L	40.0	100	80-120
Molybdenum, dissolved	40.3	0.010 µg/L	40.0	101	80-120
Nickel, dissolved	40.9	0.040 µg/L	40.0	102	80-120

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1990, Continued									
LCS (B3K1990-BS1), Continued				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Phosphorus, dissolved	4000	10 µg/L	4000		100	80-120			
Potassium, dissolved	3980	20 µg/L	4000		99	80-120			
Selenium, dissolved	411	0.10 µg/L	400		103	80-120			
Silicon, dissolved	3940	100 µg/L	4000		98	80-120			
Silver, dissolved	40.7	0.010 µg/L	40.0		102	80-120			
Sodium, dissolved	4010	20 µg/L	4000		100	80-120			
Strontium, dissolved	39.5	0.10 µg/L	40.0		99	80-120			
Sulfur, dissolved	38700	1000 µg/L	40000		97	80-120			
Tellurium, dissolved	38.3	0.050 µg/L	40.0		96	80-120			
Thallium, dissolved	39.5	0.0040 µg/L	40.0		99	80-120			
Thorium, dissolved	40.0	0.010 µg/L	40.0		100	80-120			
Tin, dissolved	40.0	0.050 µg/L	40.0		100	80-120			
Titanium, dissolved	39.2	0.20 µg/L	40.0		98	80-120			
Tungsten, dissolved	39.3	0.20 µg/L	40.0		98	80-120			
Uranium, dissolved	40.0	0.0010 µg/L	40.0		100	80-120			
Vanadium, dissolved	40.5	1.00 µg/L	40.0		101	80-120			
Zinc, dissolved	399	1.0 µg/L	400		100	80-120			
Zirconium, dissolved	39.5	0.020 µg/L	40.0		99	80-120			
Duplicate (B3K1990-DUP1)				Source: 23K2213-03	Prepared: 2023-11-21, Analyzed: 2023-11-21				
Aluminum, dissolved	9.5	1.0 µg/L		8.4			12	20	
Antimony, dissolved	0.107	0.050 µg/L		0.093				20	
Arsenic, dissolved	0.146	0.050 µg/L		0.153				20	
Barium, dissolved	7.21	0.10 µg/L		7.51			4	20	
Beryllium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Bismuth, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Boron, dissolved	32.2	2.0 µg/L		31.8			1	20	
Cadmium, dissolved	0.0136	0.0020 µg/L		0.0126			8	20	
Calcium, dissolved	24600	50 µg/L		24900			1	20	
Chromium, dissolved	< 0.50	0.50 µg/L		< 0.50				20	
Cobalt, dissolved	0.0845	0.0050 µg/L		0.0829			2	20	
Copper, dissolved	3.84	0.10 µg/L		3.77			2	20	
Iron, dissolved	11.5	2.0 µg/L		11.4			< 1	20	
Lead, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Lithium, dissolved	0.239	0.050 µg/L		0.240				20	
Magnesium, dissolved	5270	5.0 µg/L		5160			2	20	
Manganese, dissolved	3.55	0.050 µg/L		3.50			1	20	
Molybdenum, dissolved	0.256	0.010 µg/L		0.256			< 1	20	
Nickel, dissolved	0.500	0.040 µg/L		0.468			7	20	
Phosphorus, dissolved	< 10	10 µg/L		< 10				20	
Potassium, dissolved	851	20 µg/L		851			< 1	20	
Selenium, dissolved	1.38	0.10 µg/L		1.29			7	20	
Silicon, dissolved	8200	100 µg/L		8300			1	20	
Silver, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Sodium, dissolved	11200	20 µg/L		11100			2	20	
Strontium, dissolved	30.0	0.10 µg/L		29.5			1	20	
Sulfur, dissolved	12700	1000 µg/L		12900			2	20	
Tellurium, dissolved	< 0.050	0.050 µg/L		< 0.050				20	
Thallium, dissolved	0.0057	0.0040 µg/L		< 0.0040				20	
Thorium, dissolved	< 0.010	0.010 µg/L		< 0.010				20	
Tin, dissolved	0.061	0.050 µg/L		0.066				20	
Titanium, dissolved	< 0.20	0.20 µg/L		0.22				20	
Tungsten, dissolved	2.69	0.20 µg/L		2.80			4	20	
Uranium, dissolved	0.0165	0.0010 µg/L		0.0144			14	20	
Vanadium, dissolved	1.01	1.00 µg/L		1.00				20	
Zinc, dissolved	2.3	1.0 µg/L		2.2				20	

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1990, Continued									
Duplicate (B3K1990-DUP1), Continued		Source: 23K2213-03		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Zirconium, dissolved	< 0.020	0.020 µg/L		< 0.020				20	
Matrix Spike (B3K1990-MS1)		Source: 23K2213-04		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Aluminum, dissolved	4520	1.0 µg/L	4000	11.2	113	70-130			
Antimony, dissolved	42.8	0.050 µg/L	40.0	< 0.050	107	70-130			
Arsenic, dissolved	451	0.050 µg/L	400	0.126	113	70-130			
Barium, dissolved	63.6	0.10 µg/L	40.0	18.7	112	70-130			
Beryllium, dissolved	45.0	0.010 µg/L	40.0	< 0.010	113	70-130			
Bismuth, dissolved	38.4	0.010 µg/L	40.0	< 0.010	96	70-130			
Boron, dissolved	416	2.0 µg/L	400	11.2	101	70-130			
Cadmium, dissolved	45.0	0.0020 µg/L	40.0	0.210	112	70-130			
Calcium, dissolved	20600	50 µg/L	4000	17300	81	70-130			
Chromium, dissolved	44.8	0.50 µg/L	40.0	< 0.50	112	70-130			
Cobalt, dissolved	60.5	0.0050 µg/L	40.0	17.3	108	70-130			
Copper, dissolved	378	0.10 µg/L	40.0	349	71	70-130			
Iron, dissolved	4400	2.0 µg/L	4000	13.9	110	70-130			
Lead, dissolved	44.7	0.050 µg/L	40.0	< 0.050	112	70-130			
Lithium, dissolved	44.8	0.050 µg/L	40.0	0.414	111	70-130			
Magnesium, dissolved	9070	5.0 µg/L	4000	4480	115	70-130			
Manganese, dissolved	53.9	0.050 µg/L	40.0	9.74	110	70-130			
Molybdenum, dissolved	41.7	0.010 µg/L	40.0	0.028	104	70-130			
Nickel, dissolved	68.5	0.040 µg/L	40.0	24.6	110	70-130			
Phosphorus, dissolved	4570	10 µg/L	4000	< 10	114	70-130			
Potassium, dissolved	4960	20 µg/L	4000	519	111	70-130			
Selenium, dissolved	453	0.10 µg/L	400	0.14	113	70-130			
Silicon, dissolved	14500	100 µg/L	4000	10500	100	70-130			
Silver, dissolved	31.5	0.010 µg/L	40.0	< 0.010	79	70-130			
Sodium, dissolved	9680	20 µg/L	4000	5170	113	70-130			
Strontium, dissolved	93.0	0.10 µg/L	40.0	49.1	110	70-130			
Sulfur, dissolved	47700	1000 µg/L	40000	7780	100	70-130			
Tellurium, dissolved	45.6	0.050 µg/L	40.0	< 0.050	114	70-130			
Thallium, dissolved	44.7	0.0040 µg/L	40.0	0.0103	112	70-130			
Thorium, dissolved	45.8	0.010 µg/L	40.0	< 0.010	114	70-130			
Tin, dissolved	44.2	0.050 µg/L	40.0	< 0.050	110	70-130			
Titanium, dissolved	42.7	0.20 µg/L	40.0	< 0.20	106	70-130			
Tungsten, dissolved	43.9	0.20 µg/L	40.0	0.43	109	70-130			
Uranium, dissolved	45.8	0.0010 µg/L	40.0	0.0115	114	70-130			
Vanadium, dissolved	44.9	1.00 µg/L	40.0	< 1.00	112	70-130			
Zinc, dissolved	520	1.0 µg/L	400	74.6	111	70-130			
Zirconium, dissolved	44.6	0.020 µg/L	40.0	0.026	112	70-130			

Dissolved Metals, Batch B3K1993

Blank (B3K1993-BLK1)		Prepared: 2023-11-22, Analyzed: 2023-11-22							
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							
Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.10	0.10 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1993, Continued									
Blank (B3K1993-BLK1), Continued					Prepared: 2023-11-22, Analyzed: 2023-11-22				
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							
LCS (B3K1993-BS1)					Prepared: 2023-11-22, Analyzed: 2023-11-22				
Aluminum, dissolved	4090	1.0 µg/L	4000		102	80-120			
Antimony, dissolved	40.6	0.050 µg/L	40.0		101	80-120			
Arsenic, dissolved	412	0.050 µg/L	400		103	80-120			
Barium, dissolved	40.3	0.10 µg/L	40.0		101	80-120			
Beryllium, dissolved	40.4	0.010 µg/L	40.0		101	80-120			
Bismuth, dissolved	40.6	0.010 µg/L	40.0		102	80-120			
Boron, dissolved	400	2.0 µg/L	400		100	80-120			
Cadmium, dissolved	40.6	0.0020 µg/L	40.0		102	80-120			
Calcium, dissolved	4050	50 µg/L	4000		101	80-120			
Chromium, dissolved	41.3	0.10 µg/L	40.0		103	80-120			
Cobalt, dissolved	40.8	0.0050 µg/L	40.0		102	80-120			
Copper, dissolved	40.9	0.10 µg/L	40.0		102	80-120			
Iron, dissolved	4070	2.0 µg/L	4000		102	80-120			
Lead, dissolved	41.0	0.050 µg/L	40.0		102	80-120			
Lithium, dissolved	40.7	0.050 µg/L	40.0		102	80-120			
Magnesium, dissolved	4160	5.0 µg/L	4000		104	80-120			
Manganese, dissolved	41.2	0.050 µg/L	40.0		103	80-120			
Molybdenum, dissolved	39.8	0.010 µg/L	40.0		100	80-120			
Nickel, dissolved	40.2	0.040 µg/L	40.0		101	80-120			
Phosphorus, dissolved	4210	10 µg/L	4000		105	80-120			
Potassium, dissolved	4280	20 µg/L	4000		107	80-120			
Selenium, dissolved	401	0.10 µg/L	400		100	80-120			
Silicon, dissolved	3980	100 µg/L	4000		99	80-120			
Silver, dissolved	40.4	0.010 µg/L	40.0		101	80-120			
Sodium, dissolved	4110	20 µg/L	4000		103	80-120			
Strontium, dissolved	41.1	0.10 µg/L	40.0		103	80-120			
Sulfur, dissolved	40200	1000 µg/L	40000		100	80-120			
Tellurium, dissolved	40.2	0.050 µg/L	40.0		100	80-120			
Thallium, dissolved	40.5	0.0040 µg/L	40.0		101	80-120			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1993, Continued									
LCS (B3K1993-BS1), Continued				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Thorium, dissolved	37.9	0.010 µg/L	40.0		95	80-120			
Tin, dissolved	40.1	0.050 µg/L	40.0		100	80-120			
Titanium, dissolved	40.2	0.20 µg/L	40.0		101	80-120			
Tungsten, dissolved	40.0	0.20 µg/L	40.0		100	80-120			
Uranium, dissolved	40.8	0.0010 µg/L	40.0		102	80-120			
Vanadium, dissolved	41.1	1.00 µg/L	40.0		103	80-120			
Zinc, dissolved	407	1.0 µg/L	400		102	80-120			
Zirconium, dissolved	38.9	0.020 µg/L	40.0		97	80-120			

Dissolved Metals, Batch B3K2259

Blank (B3K2259-BLK1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK4)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK5)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK6)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK7)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2259-BLK8)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	< 0.0050	0.0050 µg/L							
LCS (B3K2259-BS1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.249	0.0050 µg/L	0.250		100	80-120			
LCS (B3K2259-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.223	0.0050 µg/L	0.250		89	80-120			
LCS (B3K2259-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.236	0.0050 µg/L	0.250		94	80-120			
LCS (B3K2259-BS4)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.242	0.0050 µg/L	0.250		97	80-120			
LCS (B3K2259-BS5)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.261	0.0050 µg/L	0.250		104	80-120			
LCS (B3K2259-BS6)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.246	0.0050 µg/L	0.250		98	80-120			
LCS (B3K2259-BS7)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.254	0.0050 µg/L	0.250		102	80-120			
LCS (B3K2259-BS8)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.252	0.0050 µg/L	0.250		101	80-120			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3K2259, Continued

Matrix Spike (B3K2259-MS8)		Source: 23K2213-05		Prepared: 2023-11-22, Analyzed: 2023-11-22					
Mercury, dissolved	0.218	0.0050 µg/L	0.250	< 0.0050	87	70-130			

Dissolved Metals, Batch B3K2521

Blank (B3K2521-BLK1)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK2)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK3)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK4)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK5)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK6)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK7)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	< 0.0050	0.0050 µg/L							
LCS (B3K2521-BS1)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.251	0.0050 µg/L	0.250		101	80-120			
LCS (B3K2521-BS2)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.256	0.0050 µg/L	0.250		102	80-120			
LCS (B3K2521-BS3)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.263	0.0050 µg/L	0.250		105	80-120			
LCS (B3K2521-BS4)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.290	0.0050 µg/L	0.250		116	80-120			
LCS (B3K2521-BS5)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.282	0.0050 µg/L	0.250		113	80-120			
LCS (B3K2521-BS6)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.245	0.0050 µg/L	0.250		98	80-120			
LCS (B3K2521-BS7)		Prepared: 2023-11-24, Analyzed: 2023-11-24							
Mercury, dissolved	0.240	0.0050 µg/L	0.250		96	80-120			
Duplicate (B3K2521-DUP5)		Source: 23K2213-07		Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	< 0.0050	0.0050 µg/L		< 0.0050				20	

General Parameters, Batch B3K1963

Blank (B3K1963-BLK1)		Prepared: 2023-11-20, Analyzed: 2023-11-21							
Solids, Total Dissolved	< 10	10 mg/L							
Blank (B3K1963-BLK2)		Prepared: 2023-11-20, Analyzed: 2023-11-21							
Solids, Total Dissolved	< 10	10 mg/L							

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B3K1963, Continued									
LCS (B3K1963-BS1)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	45	10 mg/L	50.0		90	85-115			
LCS (B3K1963-BS2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	198	10 mg/L	208		95	85-115			
Duplicate (B3K1963-DUP1)				Source: 23K2213-02		Prepared: 2023-11-20, Analyzed: 2023-11-21			
Solids, Total Dissolved	78	10 mg/L		68			14	20	

General Parameters, Batch B3K2066

Blank (B3K2066-BLK1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK3)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK4)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B3K2066-BS1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	0.980	0.050 mg/L	1.00		98	85-115			
LCS (B3K2066-BS2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
LCS (B3K2066-BS3)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	1.02	0.050 mg/L	1.00		102	85-115			
LCS (B3K2066-BS4)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
Duplicate (B3K2066-DUP3)				Source: 23K2213-06		Prepared: 2023-11-21, Analyzed: 2023-11-21			
Ammonia, Total (as N)	< 0.050	0.050 mg/L		< 0.050				15	
Matrix Spike (B3K2066-MS3)				Source: 23K2213-06		Prepared: 2023-11-21, Analyzed: 2023-11-21			
Ammonia, Total (as N)	0.254	0.050 mg/L	0.204	< 0.050	124	75-125			

General Parameters, Batch B3K2147

Blank (B3K2147-BLK1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							

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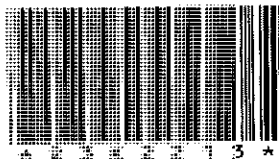
Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B3K2147, Continued

Blank (B3K2147-BLK3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B3K2147-BS1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	100	1.0 mg/L	100		100	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	34.1	1.0 mg/L	50.0		68	0-200			
LCS (B3K2147-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	114	1.0 mg/L	100		114	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.8	1.0 mg/L	50.0		40	0-200			
LCS (B3K2147-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	112	1.0 mg/L	100		112	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.4	1.0 mg/L	50.0		39	0-200			

General Parameters, Batch B3K2318

Blank (B3K2318-BLK1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B3K2318-BS1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	94.0	2.0 mg/L	100		94	80-120			



JSD Puro 11-20 9:05 9.4.c

COC ID: 697296-20231116-01		TURNAROUND TIME: Regular		RUSH:													
PROJECT/CLIENT INFO				LABORATORY				OTHER INFO									
/ Job# Sunro Mine				Lab Name Caro				Report Format / Distribution									
Manager Michelle Unger				Lab Contact Regan Pshyk				Email 1: michelle.unger@teck.com									
Email michelle.unger@teck.com				Email rpschyk@caro.ca				Email 2: craig.pinnell@atkinsrealis.com									
Address 601 Knighton Rd				Address 110 - 4011 Viking Way				Email 3: jonathan.croston@atkinsrealis.com									
								Email 4: beva.johnston@atkinsrealis.com									
								Email 5: envr_west_bc_labdata@atkinsrealis.com									
City Kimberly		Province BC		City Richmond		Province BC											
Postal Code V1A 1C7		Country Canada		Postal Code V6B2K9		Country Canada											
Phone Number 250-427-8422				Phone Number 604-279-1499				PO number		697296							
SAMPLE DETAILS				ANALYSIS REQUESTED				Filtered - 1: Field, 1: Lab, 1: Field & Lab, X: None									
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	Total Metals + Hg	Dissolved Metals + Hg	TDS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	Total and Speciated Alkalinity, Acidity	Sulphide	Ammonia	Dissolved Organic Carbon	HOLD
1 SU_GW-02-11132023_NP	MW18-02	Water	N	2023-11-13	15:10	G	4	x	x	x	x	x	x	x	x		
2 SU_GW-03-11132023_NP	MW18-03	Water	N	2023-11-13	15:15	G	4	x	x	x	x	x	x	x	x		
3 SU_GW-A-11132023_NP	MW-A	Water	N	2023-11-13	11:35	G	4	x	x	x	x	x	x	x	x		
4 SU_GW-04-11132023_NP	MW18-04	Water	N	2023-11-13	11:30	G	4	x	x	x	x	x	x	x	x		
5 SU_GW-05-11142023_NP	MW18-05	Water	N	2023-11-14	10:05	G	4	x	x	x	x	x	x	x	x		
6 SU_GW-06-11132023_NP	MW18-06	Water	N	2023-11-13	10:35	G	4	x	x	x	x	x	x	x	x		
7 SU_GW-07-11132023_NP	MW18-07	Water	N	2023-11-13	13:15	G	4	x	x	x	x	x	x	x	x		
8 SU_GW-09-11132023_NP	MW18-09	Water	N	2023-11-13	11:35	G	4	x	x	x	x	x	x	x	x		
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS				RELINQUISHED BY/AFFILIATION				DATE/TIME		ACCEPTED BY/AFFILIATION		DATE/TIME					
Ultra low detection for metals required. Detection limits suitable for CSR and BCWQG				Atkins Realis Inc													
Invoice to: Atkins Realis Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3																	
Send to: payables@atkinsrealis.com Billing PO: 697296																	
NB OF BOTTLES RETURNED/DESCRIPTION																	
Regular (default) X				Sampler's Name				Dann Johnston and Chris Walker		Mobile #		236-464-2830					
Priority (2-3 business days) - 50% surcharge				Sampler's Signature						Date/Time		2023-11-16 17:00:00 PM					
Emergency (1 Business Day) - 100% surcharge																	
For Emergency <1 Day, ASAP or Weekend																	

CERTIFICATE OF ANALYSIS

REPORTED TO AtkinsRéalis (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Craig Pinnell

PO NUMBER 697296

PROJECT 697296

PROJECT INFO Teck Sunro Mine-20231116-02

WORK ORDER 23K2229

RECEIVED / TEMP 2023-11-20 09:00 / 9.4°C

REPORTED 2023-11-27 08:00

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

Work Order Comments:

Custody Seals Intact: YES

By engaging our services, you are agreeing to CARO Analytical Service's Standard Terms and Conditions outlined here:
<https://www.caro.ca/terms-conditions>

If you have any questions or concerns, please contact me at rpsyhk@caro.ca

Authorized By:

Regan Pshyk
Account Manager

1-888-311-8846 | www.caro.ca

#110 4011 Viking Way Richmond, BC V6V 2K9 | #102 3677 Highway 97N Kelowna, BC V1X 5C3 | 17225 109 Avenue Edmonton, AB T5S 1H7 |
#108 4475 Wayburne Drive Burnaby, BC V5G 4X4

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-10-11142023-NP (23K2229-01) Matrix: Water Sampled: 2023-11-14 15:50					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.59	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.030	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	5.98	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	118	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.054	0.050	µg/L	2023-11-21	
Barium, dissolved	1.91	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	3.3	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0050	0.0020	µg/L	2023-11-21	
Calcium, dissolved	1550	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.281	0.0050	µg/L	2023-11-21	
Copper, dissolved	12.0	0.10	µg/L	2023-11-21	
Iron, dissolved	30.6	2.0	µg/L	2023-11-21	
Lead, dissolved	0.059	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.074	0.050	µg/L	2023-11-21	
Magnesium, dissolved	511	5.0	µg/L	2023-11-21	
Manganese, dissolved	0.929	0.050	µg/L	2023-11-21	
Mercury, dissolved	0.0058	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.035	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.710	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	112	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2070	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	2410	20	µg/L	2023-11-21	
Strontium, dissolved	6.34	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	0.63	0.20	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-10-11142023-NP (23K2229-01) | Matrix: Water | Sampled: 2023-11-14 15:50, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	2.0	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.060	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	2.1	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.3	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.3	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	32	10	mg/L	2023-11-21	

SU-PW-11-11142023-NP (23K2229-02) | Matrix: Water | Sampled: 2023-11-14 12:40

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.14	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.032	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.9	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	8.94	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	141	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.054	0.050	µg/L	2023-11-21	
Barium, dissolved	1.24	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	3.4	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0537	0.0020	µg/L	2023-11-21	
Calcium, dissolved	2560	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	1.13	0.0050	µg/L	2023-11-21	
Copper, dissolved	118	0.10	µg/L	2023-11-21	
Iron, dissolved	103	2.0	µg/L	2023-11-21	
Lead, dissolved	0.174	0.050	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-11-11142023-NP (23K2229-02) | Matrix: Water | Sampled: 2023-11-14 12:40, Continued

Dissolved Metals, Continued

Lithium, dissolved	0.087	0.050	µg/L	2023-11-21	
Magnesium, dissolved	614	5.0	µg/L	2023-11-21	
Manganese, dissolved	5.35	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.082	0.010	µg/L	2023-11-21	
Nickel, dissolved	2.77	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	228	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2360	100	µg/L	2023-11-21	
Silver, dissolved	0.017	0.010	µg/L	2023-11-21	
Sodium, dissolved	2270	20	µg/L	2023-11-21	
Strontium, dissolved	7.12	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	3.93	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0022	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	6.3	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.433	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	3.5	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	6.5	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	6.5	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	52	10	mg/L	2023-11-21	

SU-PW-A-11142023-NP (23K2229-03) | Matrix: Water | Sampled: 2023-11-14 15:50

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.59	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.032	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-A-11142023-NP (23K2229-03) Matrix: Water Sampled: 2023-11-14 15:50, Continued					
<i>Anions, Continued</i>					
Sulfate	1.4	1.0	mg/L	2023-11-21	
<i>Calculated Parameters</i>					
Hardness, Dissolved (as CaCO ₃)	6.06	0.125	mg/L	N/A	
<i>Dissolved Metals</i>					
Aluminum, dissolved	126	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.059	0.050	µg/L	2023-11-21	
Barium, dissolved	1.91	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	3.5	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0072	0.0020	µg/L	2023-11-21	
Calcium, dissolved	1580	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.338	0.0050	µg/L	2023-11-21	
Copper, dissolved	22.0	0.10	µg/L	2023-11-21	
Iron, dissolved	51.7	2.0	µg/L	2023-11-21	
Lead, dissolved	0.099	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.077	0.050	µg/L	2023-11-21	
Magnesium, dissolved	509	5.0	µg/L	2023-11-21	
Manganese, dissolved	2.16	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.033	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.723	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	122	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2130	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	2370	20	µg/L	2023-11-21	
Strontium, dissolved	6.55	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	1.44	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0017	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	1.5	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.065	0.020	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-A-11142023-NP (23K2229-03) Matrix: Water Sampled: 2023-11-14 15:50, Continued					
General Parameters					
Acidity, Total (as CaCO ₃)	5.8	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	4.1	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	4.1	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	40	10	mg/L	2023-11-21	

SU-PW-17-11142023-NP (23K2229-04) | Matrix: Water | Sampled: 2023-11-14 12:15

Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.93	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.297	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	11.6	1.0	mg/L	2023-11-21	

Calculated Parameters					
Hardness, Dissolved (as CaCO ₃)	27.9	0.125	mg/L	N/A	

Dissolved Metals					
Aluminum, dissolved	21.2	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Barium, dissolved	1.35	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	6.7	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.159	0.0020	µg/L	2023-11-21	
Calcium, dissolved	8260	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	3.94	0.0050	µg/L	2023-11-21	
Copper, dissolved	159	0.10	µg/L	2023-11-21	
Iron, dissolved	16.7	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.131	0.050	µg/L	2023-11-21	
Magnesium, dissolved	1750	5.0	µg/L	2023-11-21	
Manganese, dissolved	0.712	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.073	0.010	µg/L	2023-11-21	
Nickel, dissolved	9.35	0.040	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-17-11142023-NP (23K2229-04) | Matrix: Water | Sampled: 2023-11-14 12:15, Continued

Dissolved Metals, Continued

Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	344	20	µg/L	2023-11-21	
Selenium, dissolved	0.82	0.10	µg/L	2023-11-21	
Silicon, dissolved	5220	100	µg/L	2023-11-21	
Silver, dissolved	0.011	0.010	µg/L	2023-11-21	
Sodium, dissolved	4110	20	µg/L	2023-11-21	
Strontium, dissolved	20.4	0.10	µg/L	2023-11-21	
Sulfur, dissolved	3610	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	0.22	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0012	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	18.1	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.030	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	8.1	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	17.3	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	17.3	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	12	10	mg/L	2023-11-21	

SU-PW-19-11142023-NP (23K2229-05) | Matrix: Water | Sampled: 2023-11-14 13:40

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.48	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.030	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.3	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.67	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	150	1.0	µg/L	2023-11-21	
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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-19-11142023-NP (23K2229-05) | Matrix: Water | Sampled: 2023-11-14 13:40, Continued

Dissolved Metals, Continued

Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.073	0.050	µg/L	2023-11-21	
Barium, dissolved	1.24	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	3.5	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0242	0.0020	µg/L	2023-11-21	
Calcium, dissolved	1850	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.509	0.0050	µg/L	2023-11-21	
Copper, dissolved	40.4	0.10	µg/L	2023-11-21	
Iron, dissolved	120	2.0	µg/L	2023-11-21	
Lead, dissolved	24.7	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.069	0.050	µg/L	2023-11-21	
Magnesium, dissolved	494	5.0	µg/L	2023-11-21	
Manganese, dissolved	3.41	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.041	0.010	µg/L	2023-11-21	
Nickel, dissolved	1.22	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	162	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2120	100	µg/L	2023-11-21	
Silver, dissolved	0.014	0.010	µg/L	2023-11-21	
Sodium, dissolved	2260	20	µg/L	2023-11-21	
Strontium, dissolved	6.74	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	3.45	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0026	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	4.4	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.076	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	3.6	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	4.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	4.0	1.0	mg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-19-11142023-NP (23K2229-05) | Matrix: Water | Sampled: 2023-11-14 13:40, Continued

General Parameters, Continued

Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	55	10	mg/L	2023-11-21	

SU-PW-21-11142023-NP (23K2229-06) | Matrix: Water | Sampled: 2023-11-14 15:50

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.59	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.030	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.4	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	6.20	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	128	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.052	0.050	µg/L	2023-11-21	
Barium, dissolved	1.76	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	3.4	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0085	0.0020	µg/L	2023-11-21	
Calcium, dissolved	1620	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.346	0.0050	µg/L	2023-11-21	
Copper, dissolved	22.0	0.10	µg/L	2023-11-21	
Iron, dissolved	51.0	2.0	µg/L	2023-11-21	
Lead, dissolved	0.111	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.075	0.050	µg/L	2023-11-21	
Magnesium, dissolved	522	5.0	µg/L	2023-11-21	
Manganese, dissolved	2.15	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.029	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.776	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	127	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2070	100	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-21-11142023-NP (23K2229-06) | Matrix: Water | Sampled: 2023-11-14 15:50, Continued

Dissolved Metals, Continued

Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	2420	20	µg/L	2023-11-21	
Strontium, dissolved	6.56	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	1.44	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0017	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	1.7	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.088	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	3.2	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.5	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.5	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	60	10	mg/L	2023-11-21	

SU-PW-09-11152023-NP (23K2229-07) | Matrix: Water | Sampled: 2023-11-15 13:20

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.34	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.030	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.0	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	5.50	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	113	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.054	0.050	µg/L	2023-11-21	
Barium, dissolved	1.89	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-PW-09-11152023-NP (23K2229-07) | Matrix: Water | Sampled: 2023-11-15 13:20, Continued

Dissolved Metals, Continued

Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	2.8	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0039	0.0020	µg/L	2023-11-21	
Calcium, dissolved	1360	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.0504	0.0050	µg/L	2023-11-21	
Copper, dissolved	2.45	0.10	µg/L	2023-11-21	
Iron, dissolved	31.1	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.078	0.050	µg/L	2023-11-21	
Magnesium, dissolved	511	5.0	µg/L	2023-11-21	
Manganese, dissolved	0.944	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.020	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.388	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	123	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2080	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	2340	20	µg/L	2023-11-21	
Strontium, dissolved	6.36	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	0.71	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0016	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	1.8	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.044	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	3.1	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.5	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.5	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	38	10	mg/L	2023-11-23	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-22-11152023-NP (23K2229-08) Matrix: Water Sampled: 2023-11-15 10:40					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	3.64	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.031	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	1.5	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	6.34	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	110	1.0	µg/L	2023-11-21	
Antimony, dissolved	0.064	0.050	µg/L	2023-11-21	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Barium, dissolved	1.81	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	3.5	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0059	0.0020	µg/L	2023-11-21	
Calcium, dissolved	1660	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	0.249	0.0050	µg/L	2023-11-21	
Copper, dissolved	18.6	0.10	µg/L	2023-11-21	
Iron, dissolved	36.1	2.0	µg/L	2023-11-21	
Lead, dissolved	0.613	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.070	0.050	µg/L	2023-11-21	
Magnesium, dissolved	532	5.0	µg/L	2023-11-21	
Manganese, dissolved	0.851	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.022	0.010	µg/L	2023-11-21	
Nickel, dissolved	0.686	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	125	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	2110	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	2450	20	µg/L	2023-11-21	
Strontium, dissolved	6.99	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	0.94	0.20	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-PW-22-11152023-NP (23K2229-08) Matrix: Water Sampled: 2023-11-15 10:40, Continued					
<i>Dissolved Metals, Continued</i>					
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0014	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	1.9	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.056	0.020	µg/L	2023-11-21	
<i>General Parameters</i>					
Acidity, Total (as CaCO ₃)	3.6	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	3.5	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	3.5	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	28	10	mg/L	2023-11-23	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2020)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2021)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2021)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2020)	Ion Chromatography	✓	Richmond
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2021)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2020)	Gravimetry (Dried at 103-105C)		Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association

General Comments:

The results in this report apply to the received samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Caro will dispose of all samples within 30 days of sample receipt, unless otherwise agreed.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: rpshyk@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (Blk):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Anions, Batch B3K1988

Blank (B3K1988-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							

Blank (B3K1988-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							

LCS (B3K1988-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		98	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.276	0.010 mg/L	0.300		92	85-115			
Nitrite (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		105	85-115			

LCS (B3K1988-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		97	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Nitrite (as N)	0.273	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		106	85-115			

Dissolved Metals, Batch B3K1990

Blank (B3K1990-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3K1990, Continued

Blank (B3K1990-BLK1), Continued

Prepared: 2023-11-21, Analyzed: 2023-11-21

Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							

LCS (B3K1990-BS1)

Prepared: 2023-11-21, Analyzed: 2023-11-21

Aluminum, dissolved	3900	1.0 µg/L	4000	97	80-120
Antimony, dissolved	39.4	0.050 µg/L	40.0	99	80-120
Arsenic, dissolved	404	0.050 µg/L	400	101	80-120
Barium, dissolved	38.6	0.10 µg/L	40.0	96	80-120
Beryllium, dissolved	39.1	0.010 µg/L	40.0	98	80-120
Bismuth, dissolved	39.1	0.010 µg/L	40.0	98	80-120
Boron, dissolved	393	2.0 µg/L	400	98	80-120
Cadmium, dissolved	40.3	0.0020 µg/L	40.0	101	80-120
Calcium, dissolved	4060	50 µg/L	4000	102	80-120
Chromium, dissolved	40.1	0.50 µg/L	40.0	100	80-120
Cobalt, dissolved	40.1	0.0050 µg/L	40.0	100	80-120
Copper, dissolved	40.2	0.10 µg/L	40.0	100	80-120
Iron, dissolved	4000	2.0 µg/L	4000	100	80-120
Lead, dissolved	39.6	0.050 µg/L	40.0	99	80-120
Lithium, dissolved	38.6	0.050 µg/L	40.0	96	80-120
Magnesium, dissolved	3930	5.0 µg/L	4000	98	80-120
Manganese, dissolved	40.0	0.050 µg/L	40.0	100	80-120
Molybdenum, dissolved	40.3	0.010 µg/L	40.0	101	80-120
Nickel, dissolved	40.9	0.040 µg/L	40.0	102	80-120

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1990, Continued									
LCS (B3K1990-BS1), Continued					Prepared: 2023-11-21, Analyzed: 2023-11-21				
Phosphorus, dissolved	4000	10 µg/L	4000		100	80-120			
Potassium, dissolved	3980	20 µg/L	4000		99	80-120			
Selenium, dissolved	411	0.10 µg/L	400		103	80-120			
Silicon, dissolved	3940	100 µg/L	4000		98	80-120			
Silver, dissolved	40.7	0.010 µg/L	40.0		102	80-120			
Sodium, dissolved	4010	20 µg/L	4000		100	80-120			
Strontium, dissolved	39.5	0.10 µg/L	40.0		99	80-120			
Sulfur, dissolved	38700	1000 µg/L	40000		97	80-120			
Tellurium, dissolved	38.3	0.050 µg/L	40.0		96	80-120			
Thallium, dissolved	39.5	0.0040 µg/L	40.0		99	80-120			
Thorium, dissolved	40.0	0.010 µg/L	40.0		100	80-120			
Tin, dissolved	40.0	0.050 µg/L	40.0		100	80-120			
Titanium, dissolved	39.2	0.20 µg/L	40.0		98	80-120			
Tungsten, dissolved	39.3	0.20 µg/L	40.0		98	80-120			
Uranium, dissolved	40.0	0.0010 µg/L	40.0		100	80-120			
Vanadium, dissolved	40.5	1.00 µg/L	40.0		101	80-120			
Zinc, dissolved	399	1.0 µg/L	400		100	80-120			
Zirconium, dissolved	39.5	0.020 µg/L	40.0		99	80-120			

Dissolved Metals, Batch B3K2521

Blank (B3K2521-BLK1)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK2)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK3)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK4)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK5)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK6)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK7)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
LCS (B3K2521-BS1)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.251	0.0050 µg/L	0.250		101	80-120			
LCS (B3K2521-BS2)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.256	0.0050 µg/L	0.250		102	80-120			
LCS (B3K2521-BS3)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.263	0.0050 µg/L	0.250		105	80-120			
LCS (B3K2521-BS4)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.290	0.0050 µg/L	0.250		116	80-120			
LCS (B3K2521-BS5)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.282	0.0050 µg/L	0.250		113	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K2521, Continued									
LCS (B3K2521-BS6)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.245	0.0050 µg/L	0.250		98	80-120			
LCS (B3K2521-BS7)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.240	0.0050 µg/L	0.250		96	80-120			
Matrix Spike (B3K2521-MS4)				Source: 23K2229-07		Prepared: 2023-11-24, Analyzed: 2023-11-24			
Mercury, dissolved	0.298	0.0050 µg/L	0.250	< 0.0050	119	70-130			

General Parameters, Batch B3K1963

Blank (B3K1963-BLK1)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	< 10	10 mg/L							
Blank (B3K1963-BLK2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K1963-BS1)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	45	10 mg/L	50.0		90	85-115			
LCS (B3K1963-BS2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	198	10 mg/L	208		95	85-115			

General Parameters, Batch B3K2066

Blank (B3K2066-BLK1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK3)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK4)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B3K2066-BS1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	0.980	0.050 mg/L	1.00		98	85-115			
LCS (B3K2066-BS2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
LCS (B3K2066-BS3)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	1.02	0.050 mg/L	1.00		102	85-115			
LCS (B3K2066-BS4)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			

General Parameters, Batch B3K2147

Blank (B3K2147-BLK1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2229
2023-11-27 08:00

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B3K2147, Continued

Blank (B3K2147-BLK2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B3K2147-BS1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	100	1.0 mg/L	100		100	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	34.1	1.0 mg/L	50.0		68	0-200			
LCS (B3K2147-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	114	1.0 mg/L	100		114	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.8	1.0 mg/L	50.0		40	0-200			
LCS (B3K2147-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	112	1.0 mg/L	100		112	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.4	1.0 mg/L	50.0		39	0-200			

General Parameters, Batch B3K2221

Blank (B3K2221-BLK1)				Prepared: 2023-11-22, Analyzed: 2023-11-23					
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K2221-BS1)				Prepared: 2023-11-22, Analyzed: 2023-11-23					
Solids, Total Dissolved	55	10 mg/L	50.0		110	85-115			
LCS (B3K2221-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-23					
Solids, Total Dissolved	212	10 mg/L	208		102	85-115			

General Parameters, Batch B3K2318

Blank (B3K2318-BLK1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B3K2318-BS1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	94.0	2.0 mg/L	100		94	80-120			

General Parameters, Batch B3K2319

Blank (B3K2319-BLK1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B3K2319-BS1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	106	2.0 mg/L	100		106	80-120			



COC ID:		697296-20231116-02		TURNAROUND TIME:		Regular		RUSH:							
PROJECT/CLIENT INFO				LABORATORY				OTHER INFO							
/ Job#	Sunro Mine			Lab Name	Caro		Report Format / Distribution		Excel	PDF	EDD				
anager	Michelle Unger			Lab Contact	Regan Pshyk		Email 1:	michelle.unger@teck.com	X	X	X				
Email	michelle.unger@teck.com			Email	rpshyk@caro.ca		Email 2:	craig.pinnell@atkinsrealis.com	X	X	X				
Address	601 Knighton Rd			Address	110 - 4011 Viking Way		Email 3:	johncroston@atkinsrealis.com	X	X	X				
							Email 4:	benn.johnston@atkinsrealis.com	X	X	X				
City	Kimberly	Province	BC	City	Richmond	Province	BC	Email 5:	ttw_west_bc_labinfo@atkinsrealis.com	X	X				
Postal Code	VIA 1C7	Country	Canada	Postal Code	V6B2K9	Country	Canada								
Phone Number	250-427-8422			Phone Number	604-279-1499			PO number	697296						
SAMPLE DETAILS								ANALYSIS REQUESTED							
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	FIL	PRESERV.	ANALYSES	Filtered - F: Field; L: Lab; FL: Field & Lab; N: None				
SU_PW-10-11142023_NP	PW15-10	Water	N	2023-11-14	15:50	G	4			Total Metals + Hg	No	No	No	No	No
SU_PW-11-11142023_NP	PW15-11	Water	N	2023-11-14	12:40	G	4			Dissolved Metals + Hg	No	No	No	No	No
SU_PW-A-11142023_NP	PW-A	Water	N	2023-11-14	15:50	G	4			TDS	No	No	No	No	No
SU_PW-17-11142023_NP	PW17-17	Water	N	2023-11-14	12:15	G	4			Hardness	No	No	No	No	No
SU_PW-19-11142023_NP	PW17-19	Water	N	2023-11-14	13:40	G	4			Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	No	No	No	No	No
SU_PW-21-11142023_NP	PW17-21	Water	N	2023-11-14	15:50	G	4			Total and Speciated Alkalinity, Acidity	No	No	No	No	No
SU_PW-09-11152023_NP	PW15-09	Water	N	2023-11-15	13:20	G	4			Sulphide	No	No	No	No	No
SU_PW-22-11152023_NP	PW18-22	Water	N	2023-11-15	10:40	G	4			Ammonia	No	No	No	No	No
										Dissolved Organic Carbon	No	No	No	No	No
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS Ultra low detection for metals required. Detection limits suitable for CSR and BCWQG Invoice to: Atkins Realis Inc, Jonathan Croston / Craig Pinnel (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3 Send to: payables@atkinsrealis.com Billing PO: 697296								RELINQUISHED BY/AFFILIATION		DATE/TIME		ACCEPTED BY/AFFILIATION		DATE/TIME	
								Atkins Realis Inc							
NB OF BOTTLES RETURNED / DESCRIPTION								SAMPLER'S SIGNATURE AND DATE							
Regular (default) X								Sampler's Name		Bonn Johnston and Chris Walker		Mobile #		236-464-2830	
Priority (2-3 business days) - 50% surcharge								Sampler's Signature				Date/Time		2023-11-16 17:00:00 PM	
Emergency (1 Business Day) - 100% surcharge For Emergency <1 Day, ASAP or Weekend															

BTA Puro

2011
9:00

9.4

CERTIFICATE OF ANALYSIS

REPORTED TO AtkinsRéalis (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Craig Pinnell

PO NUMBER 697296

PROJECT 697296

PROJECT INFO Teck Sunro Mine-20231116-03

WORK ORDER 23K2230

RECEIVED / TEMP 2023-11-20 09:00 / 8.3°C

REPORTED 2023-11-27 07:58

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

Work Order Comments:

Custody Seals Intact: YES

By engaging our services, you are agreeing to CARO Analytical Service's Standard Terms and Conditions outlined here:
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If you have any questions or concerns, please contact me at rpsyhk@caro.ca

Authorized By:

Regan Pshyk
Account Manager



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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-01A-11132023-NP (23K2230-01) Matrix: Water Sampled: 2023-11-13 12:20					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	9.63	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.136	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	9.8	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	19.7	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	45.8	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.065	0.050	µg/L	2023-11-21	
Barium, dissolved	0.58	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	26.3	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0670	0.0020	µg/L	2023-11-21	
Calcium, dissolved	6640	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	6.28	0.0050	µg/L	2023-11-21	
Copper, dissolved	216	0.10	µg/L	2023-11-21	
Iron, dissolved	40.6	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.118	0.050	µg/L	2023-11-21	
Magnesium, dissolved	761	5.0	µg/L	2023-11-21	
Manganese, dissolved	4.65	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.064	0.010	µg/L	2023-11-21	
Nickel, dissolved	8.90	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	209	20	µg/L	2023-11-21	
Selenium, dissolved	0.19	0.10	µg/L	2023-11-21	
Silicon, dissolved	3350	100	µg/L	2023-11-21	
Silver, dissolved	0.012	0.010	µg/L	2023-11-21	
Sodium, dissolved	5150	20	µg/L	2023-11-21	
Strontium, dissolved	10.8	0.10	µg/L	2023-11-21	
Sulfur, dissolved	2580	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-01A-11132023-NP (23K2230-01) Matrix: Water Sampled: 2023-11-13 12:20, Continued					
Dissolved Metals, Continued					
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0017	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	20.3	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.023	0.020	µg/L	2023-11-21	
General Parameters					
Acidity, Total (as CaCO ₃)	5.0	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	6.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	6.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	38	10	mg/L	2023-11-21	
Total Metals					
Aluminum, total	137	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.105	0.050	µg/L	2023-11-20	
Barium, total	0.71	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	26.4	2.0	µg/L	2023-11-20	
Cadmium, total	0.0734	0.0020	µg/L	2023-11-20	
Calcium, total	6900	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	6.32	0.0050	µg/L	2023-11-20	
Copper, total	246	0.20	µg/L	2023-11-20	
Iron, total	158	2.0	µg/L	2023-11-20	
Lead, total	0.089	0.050	µg/L	2023-11-20	
Lithium, total	0.131	0.050	µg/L	2023-11-20	
Magnesium, total	748	5.0	µg/L	2023-11-20	
Manganese, total	6.31	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.067	0.010	µg/L	2023-11-20	
Nickel, total	8.75	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	219	20	µg/L	2023-11-20	
Selenium, total	0.24	0.10	µg/L	2023-11-20	
Silicon, total	3410	100	µg/L	2023-11-20	
Silver, total	0.027	0.010	µg/L	2023-11-20	
Sodium, total	4850	20	µg/L	2023-11-20	
Strontium, total	10.2	0.10	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-01A-11132023-NP (23K2230-01) | Matrix: Water | Sampled: 2023-11-13 12:20, Continued

Total Metals, Continued

Sulfur, total	3000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	2.61	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0025	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	19.5	1.0	µg/L	2023-11-20	
Zirconium, total	0.051	0.020	µg/L	2023-11-20	

SU-SW-02-11152023-NP (23K2230-02) | Matrix: Water | Sampled: 2023-11-15 13:25

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	10.9	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.115	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	10.4	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	21.7	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	50.1	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.059	0.050	µg/L	2023-11-21	
Barium, dissolved	0.56	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	25.6	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0789	0.0020	µg/L	2023-11-21	
Calcium, dissolved	7370	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	6.35	0.0050	µg/L	2023-11-21	
Copper, dissolved	200	0.10	µg/L	2023-11-21	
Iron, dissolved	42.9	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.123	0.050	µg/L	2023-11-21	
Magnesium, dissolved	803	5.0	µg/L	2023-11-21	
Manganese, dissolved	4.03	0.050	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-02-11152023-NP (23K2230-02) | Matrix: Water | Sampled: 2023-11-15 13:25, Continued

Dissolved Metals, Continued

Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.080	0.010	µg/L	2023-11-21	
Nickel, dissolved	9.10	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	206	20	µg/L	2023-11-21	
Selenium, dissolved	0.23	0.10	µg/L	2023-11-21	
Silicon, dissolved	3340	100	µg/L	2023-11-21	
Silver, dissolved	0.020	0.010	µg/L	2023-11-21	
Sodium, dissolved	5380	20	µg/L	2023-11-21	
Strontium, dissolved	11.3	0.10	µg/L	2023-11-21	
Sulfur, dissolved	3170	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0013	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	19.5	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	3.7	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	5.8	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	5.8	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	42	10	mg/L	2023-11-23	

Total Metals

Aluminum, total	95.4	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.056	0.050	µg/L	2023-11-20	
Barium, total	0.58	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	25.5	2.0	µg/L	2023-11-20	
Cadmium, total	0.0762	0.0020	µg/L	2023-11-20	
Calcium, total	7480	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	6.34	0.0050	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-02-11152023-NP (23K2230-02) | Matrix: Water | Sampled: 2023-11-15 13:25, Continued

Total Metals, Continued

Copper, total	224	0.20	µg/L	2023-11-20	
Iron, total	97.2	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.116	0.050	µg/L	2023-11-20	
Magnesium, total	785	5.0	µg/L	2023-11-20	
Manganese, total	4.76	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.084	0.010	µg/L	2023-11-20	
Nickel, total	8.79	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	221	20	µg/L	2023-11-20	
Selenium, total	0.24	0.10	µg/L	2023-11-20	
Silicon, total	3310	100	µg/L	2023-11-20	
Silver, total	0.032	0.010	µg/L	2023-11-20	
Sodium, total	5140	20	µg/L	2023-11-20	
Strontium, total	10.9	0.10	µg/L	2023-11-20	
Sulfur, total	3200	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	1.16	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0026	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	19.4	1.0	µg/L	2023-11-20	
Zirconium, total	0.024	0.020	µg/L	2023-11-20	

SU-SW-A-11132023-NP (23K2230-03) | Matrix: Water | Sampled: 2023-11-13 12:20

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	9.62	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.136	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	9.9	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO ₃)	19.6	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	43.7	1.0	µg/L	2023-11-21	
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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-A-11132023-NP (23K2230-03) | Matrix: Water | Sampled: 2023-11-13 12:20, Continued

Dissolved Metals, Continued

Antimony, dissolved	0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.060	0.050	µg/L	2023-11-21	
Barium, dissolved	0.46	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	25.8	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0735	0.0020	µg/L	2023-11-21	
Calcium, dissolved	6620	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	6.17	0.0050	µg/L	2023-11-21	
Copper, dissolved	215	0.10	µg/L	2023-11-21	
Iron, dissolved	40.5	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.113	0.050	µg/L	2023-11-21	
Magnesium, dissolved	735	5.0	µg/L	2023-11-21	
Manganese, dissolved	4.58	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.065	0.010	µg/L	2023-11-21	
Nickel, dissolved	8.76	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	199	20	µg/L	2023-11-21	
Selenium, dissolved	0.20	0.10	µg/L	2023-11-21	
Silicon, dissolved	3380	100	µg/L	2023-11-21	
Silver, dissolved	0.014	0.010	µg/L	2023-11-21	
Sodium, dissolved	5060	20	µg/L	2023-11-21	
Strontium, dissolved	10.6	0.10	µg/L	2023-11-21	
Sulfur, dissolved	2600	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0015	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	19.8	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.021	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	5.0	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	5.9	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	5.9	1.0	mg/L	2023-11-22	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-A-11132023-NP (23K2230-03) | Matrix: Water | Sampled: 2023-11-13 12:20, Continued

General Parameters, Continued

Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	50	10	mg/L	2023-11-21	

Total Metals

Aluminum, total	136	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.077	0.050	µg/L	2023-11-20	
Barium, total	0.71	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	26.3	2.0	µg/L	2023-11-20	
Cadmium, total	0.0812	0.0020	µg/L	2023-11-20	
Calcium, total	6850	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	6.27	0.0050	µg/L	2023-11-20	
Copper, total	244	0.20	µg/L	2023-11-20	
Iron, total	159	2.0	µg/L	2023-11-20	
Lead, total	0.192	0.050	µg/L	2023-11-20	
Lithium, total	0.130	0.050	µg/L	2023-11-20	
Magnesium, total	745	5.0	µg/L	2023-11-20	
Manganese, total	6.16	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.068	0.010	µg/L	2023-11-20	
Nickel, total	8.72	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	211	20	µg/L	2023-11-20	
Selenium, total	0.20	0.10	µg/L	2023-11-20	
Silicon, total	3390	100	µg/L	2023-11-20	
Silver, total	0.027	0.010	µg/L	2023-11-20	
Sodium, total	4820	20	µg/L	2023-11-20	
Strontium, total	10.3	0.10	µg/L	2023-11-20	
Sulfur, total	3000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	2.58	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0027	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	19.2	1.0	µg/L	2023-11-20	

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SU-SW-A-11132023-NP (23K2230-03) | Matrix: Water | Sampled: 2023-11-13 12:20, Continued

Total Metals, Continued

Zirconium, total	0.034	0.020	µg/L	2023-11-20	
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SU-SW-07-11152023-NP (23K2230-04) | Matrix: Water | Sampled: 2023-11-15 14:15

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	11.0	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	0.124	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	0.033	0.010	mg/L	2023-11-21	HT1
Sulfate	10.9	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Dissolved (as CaCO3)	22.4	0.125	mg/L	N/A	
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Dissolved Metals

Aluminum, dissolved	57.1	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	0.055	0.050	µg/L	2023-11-21	
Barium, dissolved	0.47	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	25.0	2.0	µg/L	2023-11-21	
Cadmium, dissolved	0.0783	0.0020	µg/L	2023-11-21	
Calcium, dissolved	7620	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	6.64	0.0050	µg/L	2023-11-21	
Copper, dissolved	226	0.10	µg/L	2023-11-21	
Iron, dissolved	47.7	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	0.119	0.050	µg/L	2023-11-21	
Magnesium, dissolved	818	5.0	µg/L	2023-11-21	
Manganese, dissolved	4.46	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	0.075	0.010	µg/L	2023-11-21	
Nickel, dissolved	9.62	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	204	20	µg/L	2023-11-21	
Selenium, dissolved	0.25	0.10	µg/L	2023-11-21	
Silicon, dissolved	3360	100	µg/L	2023-11-21	
Silver, dissolved	0.022	0.010	µg/L	2023-11-21	
Sodium, dissolved	5360	20	µg/L	2023-11-21	
Strontium, dissolved	11.3	0.10	µg/L	2023-11-21	

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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-SW-07-11152023-NP (23K2230-04) | Matrix: Water | Sampled: 2023-11-15 14:15, Continued

Dissolved Metals, Continued

Sulfur, dissolved	3040	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	0.24	0.20	µg/L	2023-11-21	
Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	0.0015	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	20.2	1.0	µg/L	2023-11-21	
Zirconium, dissolved	0.031	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	5.3	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	6.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	6.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	55	10	mg/L	2023-11-23	

Total Metals

Aluminum, total	84.6	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	0.057	0.050	µg/L	2023-11-20	
Barium, total	0.63	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	25.3	2.0	µg/L	2023-11-20	
Cadmium, total	0.0756	0.0020	µg/L	2023-11-20	
Calcium, total	7640	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	6.60	0.0050	µg/L	2023-11-20	
Copper, total	236	0.20	µg/L	2023-11-20	
Iron, total	79.9	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	0.119	0.050	µg/L	2023-11-20	
Magnesium, total	821	5.0	µg/L	2023-11-20	
Manganese, total	4.73	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	0.085	0.010	µg/L	2023-11-20	
Nickel, total	9.69	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-20	

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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-SW-07-11152023-NP (23K2230-04) Matrix: Water Sampled: 2023-11-15 14:15, Continued					
<i>Total Metals, Continued</i>					
Potassium, total	222	20	µg/L	2023-11-20	
Selenium, total	0.25	0.10	µg/L	2023-11-20	
Silicon, total	3390	100	µg/L	2023-11-20	
Silver, total	0.039	0.010	µg/L	2023-11-20	
Sodium, total	5150	20	µg/L	2023-11-20	
Strontium, total	10.8	0.10	µg/L	2023-11-20	
Sulfur, total	3400	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	0.67	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	0.0026	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	19.6	1.0	µg/L	2023-11-20	
Zirconium, total	0.021	0.020	µg/L	2023-11-20	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2020)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2021)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2021)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2020)	Ion Chromatography	✓	Richmond
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectrometry (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2021)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Mercury, total, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2020)	Gravimetry (Dried at 103-105C)		Richmond
Total Metals in Water	EPA 200.2* / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectrometry (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association

General Comments:

The results in this report apply to the received samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Caro will dispose of all samples within 30 days of sample receipt, unless otherwise agreed.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: rpschyk@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (BLK):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B3K1988									
Blank (B3K1988-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B3K1988-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B3K1988-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		98	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.276	0.010 mg/L	0.300		92	85-115			
Nitrite (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		105	85-115			
LCS (B3K1988-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		97	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Nitrite (as N)	0.273	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		106	85-115			
Dissolved Metals, Batch B3K1990									
Blank (B3K1990-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3K1990, Continued

Blank (B3K1990-BLK1), Continued

Prepared: 2023-11-21, Analyzed: 2023-11-21

Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							

LCS (B3K1990-BS1)

Prepared: 2023-11-21, Analyzed: 2023-11-21

Aluminum, dissolved	3900	1.0 µg/L	4000	97	80-120
Antimony, dissolved	39.4	0.050 µg/L	40.0	99	80-120
Arsenic, dissolved	404	0.050 µg/L	400	101	80-120
Barium, dissolved	38.6	0.10 µg/L	40.0	96	80-120
Beryllium, dissolved	39.1	0.010 µg/L	40.0	98	80-120
Bismuth, dissolved	39.1	0.010 µg/L	40.0	98	80-120
Boron, dissolved	393	2.0 µg/L	400	98	80-120
Cadmium, dissolved	40.3	0.0020 µg/L	40.0	101	80-120
Calcium, dissolved	4060	50 µg/L	4000	102	80-120
Chromium, dissolved	40.1	0.50 µg/L	40.0	100	80-120
Cobalt, dissolved	40.1	0.0050 µg/L	40.0	100	80-120
Copper, dissolved	40.2	0.10 µg/L	40.0	100	80-120
Iron, dissolved	4000	2.0 µg/L	4000	100	80-120
Lead, dissolved	39.6	0.050 µg/L	40.0	99	80-120
Lithium, dissolved	38.6	0.050 µg/L	40.0	96	80-120
Magnesium, dissolved	3930	5.0 µg/L	4000	98	80-120
Manganese, dissolved	40.0	0.050 µg/L	40.0	100	80-120
Molybdenum, dissolved	40.3	0.010 µg/L	40.0	101	80-120
Nickel, dissolved	40.9	0.040 µg/L	40.0	102	80-120

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1990, Continued									
LCS (B3K1990-BS1), Continued					Prepared: 2023-11-21, Analyzed: 2023-11-21				
Phosphorus, dissolved	4000	10 µg/L	4000		100	80-120			
Potassium, dissolved	3980	20 µg/L	4000		99	80-120			
Selenium, dissolved	411	0.10 µg/L	400		103	80-120			
Silicon, dissolved	3940	100 µg/L	4000		98	80-120			
Silver, dissolved	40.7	0.010 µg/L	40.0		102	80-120			
Sodium, dissolved	4010	20 µg/L	4000		100	80-120			
Strontium, dissolved	39.5	0.10 µg/L	40.0		99	80-120			
Sulfur, dissolved	38700	1000 µg/L	40000		97	80-120			
Tellurium, dissolved	38.3	0.050 µg/L	40.0		96	80-120			
Thallium, dissolved	39.5	0.0040 µg/L	40.0		99	80-120			
Thorium, dissolved	40.0	0.010 µg/L	40.0		100	80-120			
Tin, dissolved	40.0	0.050 µg/L	40.0		100	80-120			
Titanium, dissolved	39.2	0.20 µg/L	40.0		98	80-120			
Tungsten, dissolved	39.3	0.20 µg/L	40.0		98	80-120			
Uranium, dissolved	40.0	0.0010 µg/L	40.0		100	80-120			
Vanadium, dissolved	40.5	1.00 µg/L	40.0		101	80-120			
Zinc, dissolved	399	1.0 µg/L	400		100	80-120			
Zirconium, dissolved	39.5	0.020 µg/L	40.0		99	80-120			

Dissolved Metals, Batch B3K2521

Blank (B3K2521-BLK1)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK2)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK3)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK4)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK5)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK6)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK7)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	< 0.0050	0.0050 µg/L							
LCS (B3K2521-BS1)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.251	0.0050 µg/L	0.250		101	80-120			
LCS (B3K2521-BS2)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.256	0.0050 µg/L	0.250		102	80-120			
LCS (B3K2521-BS3)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.263	0.0050 µg/L	0.250		105	80-120			
LCS (B3K2521-BS4)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.290	0.0050 µg/L	0.250		116	80-120			
LCS (B3K2521-BS5)					Prepared: 2023-11-24, Analyzed: 2023-11-24				
Mercury, dissolved	0.282	0.0050 µg/L	0.250		113	80-120			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Dissolved Metals, Batch B3K2521, Continued

LCS (B3K2521-BS6)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	0.245	0.0050 µg/L	0.250		98	80-120			
LCS (B3K2521-BS7)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	0.240	0.0050 µg/L	0.250		96	80-120			

General Parameters, Batch B3K1963

Blank (B3K1963-BLK1)			Prepared: 2023-11-20, Analyzed: 2023-11-21						
Solids, Total Dissolved	< 10	10 mg/L							
Blank (B3K1963-BLK2)			Prepared: 2023-11-20, Analyzed: 2023-11-21						
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K1963-BS1)			Prepared: 2023-11-20, Analyzed: 2023-11-21						
Solids, Total Dissolved	45	10 mg/L	50.0		90	85-115			
LCS (B3K1963-BS2)			Prepared: 2023-11-20, Analyzed: 2023-11-21						
Solids, Total Dissolved	198	10 mg/L	208		95	85-115			

General Parameters, Batch B3K2066

Blank (B3K2066-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK3)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK4)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B3K2066-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	0.980	0.050 mg/L	1.00		98	85-115			
LCS (B3K2066-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
LCS (B3K2066-BS3)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.02	0.050 mg/L	1.00		102	85-115			
LCS (B3K2066-BS4)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
Duplicate (B3K2066-DUP4)			Source: 23K2230-01		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Ammonia, Total (as N)	< 0.050	0.050 mg/L		< 0.050				15	
Matrix Spike (B3K2066-MS4)			Source: 23K2230-01		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Ammonia, Total (as N)	0.265	0.050 mg/L	0.204	< 0.050	122	75-125			

General Parameters, Batch B3K2147

Blank (B3K2147-BLK1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B3K2147, Continued

Blank (B3K2147-BLK1), Continued				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B3K2147-BS1)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	100	1.0 mg/L	100		100	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	34.1	1.0 mg/L	50.0		68	0-200			
LCS (B3K2147-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	114	1.0 mg/L	100		114	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.8	1.0 mg/L	50.0		40	0-200			
LCS (B3K2147-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	112	1.0 mg/L	100		112	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.4	1.0 mg/L	50.0		39	0-200			

General Parameters, Batch B3K2221

Blank (B3K2221-BLK1)				Prepared: 2023-11-22, Analyzed: 2023-11-23					
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K2221-BS1)				Prepared: 2023-11-22, Analyzed: 2023-11-23					
Solids, Total Dissolved	55	10 mg/L	50.0		110	85-115			
LCS (B3K2221-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-23					
Solids, Total Dissolved	212	10 mg/L	208		102	85-115			

General Parameters, Batch B3K2319

Blank (B3K2319-BLK1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B3K2319-BS1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	106	2.0 mg/L	100		106	80-120			

Total Metals, Batch B3K2005

Blank (B3K2005-BLK1)				Prepared: 2023-11-20, Analyzed: 2023-11-20					
Aluminum, total	< 2.0	2.0 µg/L							
Antimony, total	< 0.050	0.050 µg/L							
Arsenic, total	< 0.050	0.050 µg/L							
Barium, total	< 0.10	0.10 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Total Metals, Batch B3K2005, Continued

Blank (B3K2005-BLK1), Continued

Prepared: 2023-11-20, Analyzed: 2023-11-20

Beryllium, total	< 0.010	0.010 µg/L							
Bismuth, total	< 0.010	0.010 µg/L							
Boron, total	< 2.0	2.0 µg/L							
Cadmium, total	< 0.0020	0.0020 µg/L							
Calcium, total	< 50	50 µg/L							
Chromium, total	< 0.50	0.50 µg/L							
Cobalt, total	< 0.0050	0.0050 µg/L							
Copper, total	< 0.20	0.20 µg/L							
Iron, total	< 2.0	2.0 µg/L							
Lead, total	< 0.050	0.050 µg/L							
Lithium, total	< 0.050	0.050 µg/L							
Magnesium, total	< 5.0	5.0 µg/L							
Manganese, total	< 0.050	0.050 µg/L							
Molybdenum, total	< 0.010	0.010 µg/L							
Nickel, total	< 0.040	0.040 µg/L							
Phosphorus, total	< 10	10 µg/L							
Potassium, total	< 20	20 µg/L							
Selenium, total	< 0.10	0.10 µg/L							
Silicon, total	< 100	100 µg/L							
Silver, total	< 0.010	0.010 µg/L							
Sodium, total	< 20	20 µg/L							
Strontium, total	< 0.10	0.10 µg/L							
Sulfur, total	< 1000	1000 µg/L							
Tellurium, total	< 0.050	0.050 µg/L							
Thallium, total	< 0.0040	0.0040 µg/L							
Thorium, total	< 0.010	0.010 µg/L							
Tin, total	< 0.050	0.050 µg/L							
Titanium, total	< 0.20	0.20 µg/L							
Tungsten, total	< 0.20	0.20 µg/L							
Uranium, total	< 0.0010	0.0010 µg/L							
Vanadium, total	< 1.00	1.00 µg/L							
Zinc, total	< 1.0	1.0 µg/L							
Zirconium, total	< 0.020	0.020 µg/L							

LCS (B3K2005-BS1)

Prepared: 2023-11-20, Analyzed: 2023-11-20

Aluminum, total	4070	2.0 µg/L	4000	102	80-120
Antimony, total	38.7	0.050 µg/L	40.0	97	80-120
Arsenic, total	405	0.050 µg/L	400	101	80-120
Barium, total	39.8	0.10 µg/L	40.0	99	80-120
Beryllium, total	39.8	0.010 µg/L	40.0	99	80-120
Bismuth, total	38.2	0.010 µg/L	40.0	95	80-120
Boron, total	409	2.0 µg/L	400	102	80-120
Cadmium, total	38.8	0.0020 µg/L	40.0	97	80-120
Calcium, total	4010	50 µg/L	4000	100	80-120
Chromium, total	41.2	0.50 µg/L	40.0	103	80-120
Cobalt, total	41.3	0.0050 µg/L	40.0	103	80-120
Copper, total	41.0	0.20 µg/L	40.0	102	80-120
Iron, total	4100	2.0 µg/L	4000	102	80-120
Lead, total	38.8	0.050 µg/L	40.0	97	80-120
Lithium, total	38.5	0.050 µg/L	40.0	96	80-120
Magnesium, total	4180	5.0 µg/L	4000	104	80-120
Manganese, total	41.1	0.050 µg/L	40.0	103	80-120
Molybdenum, total	39.5	0.010 µg/L	40.0	99	80-120
Nickel, total	41.1	0.040 µg/L	40.0	103	80-120
Phosphorus, total	4000	10 µg/L	4000	100	80-120
Potassium, total	4030	20 µg/L	4000	101	80-120

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2230
2023-11-27 07:58

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3K2005, Continued									
LCS (B3K2005-BS1), Continued					Prepared: 2023-11-20, Analyzed: 2023-11-20				
Selenium, total	399	0.10 µg/L	400		100	80-120			
Silicon, total	3910	100 µg/L	4000		98	80-120			
Silver, total	39.8	0.010 µg/L	40.0		99	80-120			
Sodium, total	4180	20 µg/L	4000		105	80-120			
Strontium, total	40.0	0.10 µg/L	40.0		100	80-120			
Sulfur, total	38600	1000 µg/L	40000		97	80-120			
Tellurium, total	38.7	0.050 µg/L	40.0		97	80-120			
Thallium, total	38.8	0.0040 µg/L	40.0		97	80-120			
Thorium, total	39.5	0.010 µg/L	40.0		99	80-120			
Tin, total	40.6	0.050 µg/L	40.0		102	80-120			
Titanium, total	39.8	0.20 µg/L	40.0		99	80-120			
Tungsten, total	39.7	0.20 µg/L	40.0		99	80-120			
Uranium, total	40.1	0.0010 µg/L	40.0		100	80-120			
Vanadium, total	41.3	1.00 µg/L	40.0		103	80-120			
Zinc, total	405	1.0 µg/L	400		101	80-120			
Zirconium, total	38.5	0.020 µg/L	40.0		96	80-120			

Total Metals, Batch B3K2139

Blank (B3K2139-BLK2)					Prepared: 2023-11-21, Analyzed: 2023-11-21				
Mercury, total	< 0.0050	0.0050 µg/L							
LCS (B3K2139-BS2)					Prepared: 2023-11-21, Analyzed: 2023-11-21				
Mercury, total	0.256	0.0050 µg/L	0.250		102	80-120			



BDA puro 2011 9:00 8.0

CERTIFICATE OF ANALYSIS

REPORTED TO AtkinsRéalis (Nelson)
#3-520 Lake Street
Nelson, BC V1L 4C6

ATTENTION Craig Pinnell

PO NUMBER 697296

PROJECT 697296

PROJECT INFO Teck Sunro Mine-20231116-05

WORK ORDER 23K2233

RECEIVED / TEMP 2023-11-20 09:00 / 8.3°C

REPORTED 2023-11-27 12:53

COC NUMBER No #

Introduction:

CARO Analytical Services is a testing laboratory full of smart, engaged scientists driven to make the world a safer and healthier place. Through our clients' projects we become an essential element for a better world. We employ methods conducted in accordance with recognized professional standards using accepted testing methodologies and quality control efforts. CARO is accredited by the Canadian Association for Laboratories Accreditation (CALA) to ISO/IEC 17025:2017 for specific tests listed in the scope of accreditation approved by CALA.

Big Picture Sidekicks



You know that the sample you collected after snowshoeing to site, digging 5 meters, and racing to get it on a plane so you can submit it to the lab for time sensitive results needed to make important and expensive decisions (whew) is VERY important. We know that too.

We've Got Chemistry



It's simple. We figure the more you enjoy working with our fun and engaged team members; the more likely you are to give us continued opportunities to support you.

Ahead of the Curve



Through research, regulation knowledge, and instrumentation, we are your analytical centre for the technical knowledge you need, BEFORE you need it, so you can stay up to date and in the know.

Work Order Comments:

Custody Seals Intact: YES

By engaging our services, you are agreeing to CARO Analytical Service's Standard Terms and Conditions outlined here:
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If you have any questions or concerns, please contact me at rpsyhk@caro.ca

Authorized By:

Regan Pshyk
Account Manager

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TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2233
2023-11-27 12:53

Analyte	Result	RL	Units	Analyzed	Qualifier
SU-EQUIPMENT-11142023-NP (23K2233-01) Matrix: Water Sampled: 2023-11-14 14:00					
Anions					
Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	< 0.10	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	< 1.0	1.0	mg/L	2023-11-21	
Calculated Parameters					
Hardness, Dissolved (as CaCO3)	< 0.125	0.125	mg/L	N/A	
Dissolved Metals					
Aluminum, dissolved	< 1.0	1.0	µg/L	2023-11-21	
Antimony, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Arsenic, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Barium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Beryllium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Bismuth, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Boron, dissolved	< 2.0	2.0	µg/L	2023-11-21	
Cadmium, dissolved	< 0.0020	0.0020	µg/L	2023-11-21	
Calcium, dissolved	< 50	50	µg/L	2023-11-21	
Chromium, dissolved	< 0.50	0.50	µg/L	2023-11-21	
Cobalt, dissolved	< 0.0050	0.0050	µg/L	2023-11-21	
Copper, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Iron, dissolved	< 2.0	2.0	µg/L	2023-11-21	
Lead, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Lithium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Magnesium, dissolved	< 5.0	5.0	µg/L	2023-11-21	
Manganese, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Mercury, dissolved	< 0.0050	0.0050	µg/L	2023-11-24	
Molybdenum, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Nickel, dissolved	< 0.040	0.040	µg/L	2023-11-21	
Phosphorus, dissolved	< 10	10	µg/L	2023-11-21	
Potassium, dissolved	< 20	20	µg/L	2023-11-21	
Selenium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Silicon, dissolved	< 100	100	µg/L	2023-11-21	
Silver, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Sodium, dissolved	< 20	20	µg/L	2023-11-21	
Strontium, dissolved	< 0.10	0.10	µg/L	2023-11-21	
Sulfur, dissolved	< 1000	1000	µg/L	2023-11-21	
Tellurium, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Thallium, dissolved	< 0.0040	0.0040	µg/L	2023-11-21	
Thorium, dissolved	< 0.010	0.010	µg/L	2023-11-21	
Tin, dissolved	< 0.050	0.050	µg/L	2023-11-21	
Titanium, dissolved	< 0.20	0.20	µg/L	2023-11-21	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2233
2023-11-27 12:53

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-EQUIPMENT-11142023-NP (23K2233-01) | Matrix: Water | Sampled: 2023-11-14 14:00, Continued

Dissolved Metals, Continued

Tungsten, dissolved	< 0.20	0.20	µg/L	2023-11-21	
Uranium, dissolved	< 0.0010	0.0010	µg/L	2023-11-21	
Vanadium, dissolved	< 1.00	1.00	µg/L	2023-11-21	
Zinc, dissolved	< 1.0	1.0	µg/L	2023-11-21	
Zirconium, dissolved	< 0.020	0.020	µg/L	2023-11-21	

General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Solids, Total Dissolved	< 10	10	mg/L	2023-11-21	

Total Metals

Aluminum, total	< 2.0	2.0	µg/L	2023-11-21	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	< 0.10	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	< 2.0	2.0	µg/L	2023-11-20	
Cadmium, total	< 0.0020	0.0020	µg/L	2023-11-20	
Calcium, total	< 50	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	< 0.0050	0.0050	µg/L	2023-11-20	
Copper, total	< 0.20	0.20	µg/L	2023-11-20	
Iron, total	< 2.0	2.0	µg/L	2023-11-21	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	< 0.050	0.050	µg/L	2023-11-20	
Magnesium, total	< 5.0	5.0	µg/L	2023-11-20	
Manganese, total	< 0.050	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	< 0.010	0.010	µg/L	2023-11-20	
Nickel, total	< 0.040	0.040	µg/L	2023-11-21	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	< 20	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	< 100	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	< 20	20	µg/L	2023-11-20	
Strontium, total	< 0.10	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-EQUIPMENT-11142023-NP (23K2233-01) | Matrix: Water | Sampled: 2023-11-14 14:00, Continued

Total Metals, Continued

Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	< 0.20	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	< 0.0010	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	< 1.0	1.0	µg/L	2023-11-21	
Zirconium, total	< 0.020	0.020	µg/L	2023-11-20	

SU-TRIP-11162023-NP (23K2233-02) | Matrix: Water | Sampled: 2023-11-16 15:30

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	< 0.10	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	< 1.0	1.0	mg/L	2023-11-21	

Calculated Parameters

Hardness, Total (as CaCO ₃)	< 0.125	0.125	mg/L	N/A	
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General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Carbon, Dissolved Organic	< 0.50	0.50	mg/L	2023-11-23	
Solids, Total Dissolved	< 10	10	mg/L	2023-11-24	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	< 2.0	2.0	µg/L	2023-11-20	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	< 0.10	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	< 2.0	2.0	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-TRIP-11162023-NP (23K2233-02) | Matrix: Water | Sampled: 2023-11-16 15:30, Continued

Total Metals, Continued

Cadmium, total	< 0.0020	0.0020	µg/L	2023-11-20	
Calcium, total	< 50	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	< 0.0050	0.0050	µg/L	2023-11-20	
Copper, total	< 0.20	0.20	µg/L	2023-11-20	
Iron, total	< 2.0	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	< 0.050	0.050	µg/L	2023-11-20	
Magnesium, total	< 5.0	5.0	µg/L	2023-11-20	
Manganese, total	< 0.050	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	< 0.010	0.010	µg/L	2023-11-20	
Nickel, total	< 0.040	0.040	µg/L	2023-11-20	
Phosphorus, total	< 10	10	µg/L	2023-11-21	
Potassium, total	< 20	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	< 100	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	< 20	20	µg/L	2023-11-20	
Strontium, total	< 0.10	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	< 0.20	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	< 0.0010	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	< 1.0	1.0	µg/L	2023-11-20	
Zirconium, total	< 0.020	0.020	µg/L	2023-11-20	

SU-FIELD-11142023-NP (23K2233-03) | Matrix: Water | Sampled: 2023-11-14 11:45

Anions

Bromide	< 0.1	0.1	mg/L	2023-11-21	
Chloride	< 0.10	0.10	mg/L	2023-11-21	
Fluoride	< 0.10	0.10	mg/L	2023-11-21	
Nitrate (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Nitrite (as N)	< 0.010	0.010	mg/L	2023-11-21	HT1
Sulfate	< 1.0	1.0	mg/L	2023-11-21	

Calculated Parameters

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2233
2023-11-27 12:53

Analyte	Result	RL	Units	Analyzed	Qualifier
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SU-FIELD-11142023-NP (23K2233-03) | Matrix: Water | Sampled: 2023-11-14 11:45, Continued

Calculated Parameters, Continued

Hardness, Total (as CaCO ₃)	< 0.125	0.125	mg/L	N/A	
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General Parameters

Acidity, Total (as CaCO ₃)	< 2.0	2.0	mg/L	2023-11-23	
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0	mg/L	2023-11-22	
Ammonia, Total (as N)	< 0.050	0.050	mg/L	2023-11-21	
Solids, Total Dissolved	< 10	10	mg/L	2023-11-21	
Sulfide, Total	< 0.020	0.020	mg/L	2023-11-21	

Total Metals

Aluminum, total	< 2.0	2.0	µg/L	2023-11-21	
Antimony, total	< 0.050	0.050	µg/L	2023-11-20	
Arsenic, total	< 0.050	0.050	µg/L	2023-11-20	
Barium, total	< 0.10	0.10	µg/L	2023-11-20	
Beryllium, total	< 0.010	0.010	µg/L	2023-11-20	
Bismuth, total	< 0.010	0.010	µg/L	2023-11-20	
Boron, total	< 2.0	2.0	µg/L	2023-11-20	
Cadmium, total	< 0.0020	0.0020	µg/L	2023-11-20	
Calcium, total	< 50	50	µg/L	2023-11-20	
Chromium, total	< 0.50	0.50	µg/L	2023-11-20	
Cobalt, total	< 0.0050	0.0050	µg/L	2023-11-20	
Copper, total	< 0.20	0.20	µg/L	2023-11-20	
Iron, total	< 2.0	2.0	µg/L	2023-11-20	
Lead, total	< 0.050	0.050	µg/L	2023-11-20	
Lithium, total	< 0.050	0.050	µg/L	2023-11-20	
Magnesium, total	< 5.0	5.0	µg/L	2023-11-20	
Manganese, total	< 0.050	0.050	µg/L	2023-11-20	
Mercury, total	< 0.0050	0.0050	µg/L	2023-11-21	
Molybdenum, total	< 0.010	0.010	µg/L	2023-11-20	
Nickel, total	< 0.040	0.040	µg/L	2023-11-21	
Phosphorus, total	< 10	10	µg/L	2023-11-20	
Potassium, total	< 20	20	µg/L	2023-11-20	
Selenium, total	< 0.10	0.10	µg/L	2023-11-20	
Silicon, total	< 100	100	µg/L	2023-11-20	
Silver, total	< 0.010	0.010	µg/L	2023-11-20	
Sodium, total	< 20	20	µg/L	2023-11-20	
Strontium, total	< 0.10	0.10	µg/L	2023-11-20	
Sulfur, total	< 1000	1000	µg/L	2023-11-20	
Tellurium, total	< 0.050	0.050	µg/L	2023-11-20	
Thallium, total	< 0.0040	0.0040	µg/L	2023-11-20	

TEST RESULTS

REPORTED TO PROJECT AtkinsRéalys (Nelson)
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Analyte	Result	RL	Units	Analyzed	Qualifier
SU-FIELD-11142023-NP (23K2233-03) Matrix: Water Sampled: 2023-11-14 11:45, Continued					
<i>Total Metals, Continued</i>					
Thorium, total	< 0.010	0.010	µg/L	2023-11-20	
Tin, total	< 0.050	0.050	µg/L	2023-11-20	
Titanium, total	< 0.20	0.20	µg/L	2023-11-20	
Tungsten, total	< 0.20	0.20	µg/L	2023-11-20	
Uranium, total	< 0.0010	0.0010	µg/L	2023-11-20	
Vanadium, total	< 1.00	1.00	µg/L	2023-11-20	
Zinc, total	< 1.0	1.0	µg/L	2023-11-20	
Zirconium, total	< 0.020	0.020	µg/L	2023-11-20	

Sample Qualifiers:

HT1 The sample was prepared and/or analyzed past the recommended holding time.

APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2233
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Analysis Description	Method Ref.	Technique	Accredited	Location
Acidity in Water	SM 2310 B (2020)	Titration with NaOH to pH 8.3	✓	Kelowna
Alkalinity in Water	SM 2320 B* (2021)	Titration with H ₂ SO ₄	✓	Kelowna
Ammonia, Total in Water	SM 4500-NH ₃ G* (2021)	Automated Colorimetry (Phenate)	✓	Kelowna
Anions in Water	SM 4110 B (2020)	Ion Chromatography	✓	Richmond
Carbon, Dissolved Organic in Water	SM 5310 B (2022)	Combustion, Infrared CO ₂ Detection	✓	Kelowna
Dissolved Metals in Water	EPA 200.8 / EPA 6020B	0.45 µm Filtration / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond
Hardness in Water	SM 2340 B (2021)	Calculation: 2.497 [diss Ca] + 4.118 [diss Mg]	✓	N/A
Mercury, dissolved, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Mercury, total, Low level in Water	EPA 245.7*	BrCl ₂ Oxidation / Cold Vapor Atomic Fluorescence Spectrometry (CVAFS)	✓	Richmond
Solids, Total Dissolved in Water	SM 2540 C* (2020)	Gravimetry (Dried at 103-105C)		Richmond
Sulfide, Total in Water	SM 4500-S ₂ D* (2021)	Colorimetry (Methylene Blue)	✓	Edmonton
Total Metals in Water	EPA 200.2* / EPA 6020B	HNO ₃ +HCl Hot Block Digestion / Inductively Coupled Plasma-Mass Spectroscopy (ICP-MS)	✓	Richmond

Note: An asterisk in the Method Reference indicates that the CARO method has been modified from the reference method

Glossary of Terms:

RL	Reporting Limit (default)
<	Less than the specified Reporting Limit (RL) - the actual RL may be higher than the default RL due to various factors
mg/L	Milligrams per litre
µg/L	Micrograms per litre
EPA	United States Environmental Protection Agency Test Methods
SM	Standard Methods for the Examination of Water and Wastewater, American Public Health Association



APPENDIX 1: SUPPORTING INFORMATION

REPORTED TO AtkinsRéalis (Nelson)
PROJECT 697296

WORK ORDER 23K2233
REPORTED 2023-11-27 12:53

General Comments:

The results in this report apply to the received samples analyzed in accordance with the Chain of Custody document. This analytical report must be reproduced in its entirety. CARO is not responsible for any loss or damage resulting directly or indirectly from error or omission in the conduct of testing. Liability is limited to the cost of analysis. Caro will dispose of all samples within 30 days of sample receipt, unless otherwise agreed.

Results in **Bold** indicate values that are above CARO's method reporting limits. Any results that are above regulatory limits are highlighted **red**. Please note that results will only be highlighted red if the regulatory limits are included on the CARO report. Any Bold and/or highlighted results do not take into account method uncertainty. If you would like method uncertainty or regulatory limits to be included on your report, please contact your Account Manager: rpshyk@caro.ca

Please note any regulatory guidelines applied to this report are added as a convenience to the client, at their request, to help provide some initial context to analytical results obtained. Although CARO makes every effort to ensure accuracy of the associated regulatory guideline(s) applied, the guidelines applied cannot be assumed to be correct due to a variety of factors and as such CARO Analytical Services assumes no liability or responsibility for the use of those guidelines to make any decisions. The original source of the regulation should be verified and a review of the guideline(s) should be validated as correct in order to make any decisions arising from the comparison of the analytical data obtained to the relevant regulatory guideline for one's particular circumstances. Further, CARO Analytical Services assumes no liability or responsibility for any loss attributed from the use of these guidelines in any way.

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2233
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The following section displays the quality control (QC) data that is associated with your sample data. Groups of samples are prepared in "batches" and analyzed in conjunction with QC samples that ensure your data is of the highest quality. Common QC types include:

- **Method Blank (BLK):** A blank sample that undergoes sample processing identical to that carried out for the test samples. Method blank results are used to assess contamination from the laboratory environment and reagents.
- **Duplicate (Dup):** An additional or second portion of a randomly selected sample in the analytical run carried through the entire analytical process. Duplicates provide a measure of the analytical method's precision (reproducibility).
- **Blank Spike (BS):** A sample of known concentration which undergoes processing identical to that carried out for test samples, also referred to as a laboratory control sample (LCS). Blank spikes provide a measure of the analytical method's accuracy.
- **Matrix Spike (MS):** A second aliquot of sample is fortified with a known concentration of target analytes and carried through the entire analytical process. Matrix spikes evaluate potential matrix effects that may affect the analyte recovery.
- **Reference Material (SRM):** A homogenous material of similar matrix to the samples, certified for the parameter(s) listed. Reference Materials ensure that the analytical process is adequate to achieve acceptable recoveries of the parameter(s) tested.

Each QC type is analyzed at a 5-10% frequency, i.e. one blank/duplicate/spike for every 10-20 samples. For all types of QC, the specified recovery (% Rec) and relative percent difference (RPD) limits are derived from long-term method performance averages and/or prescribed by the reference method.

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Anions, Batch B3K1988									
Blank (B3K1988-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
Blank (B3K1988-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	< 0.1	0.1 mg/L							
Chloride	< 0.10	0.10 mg/L							
Fluoride	< 0.10	0.10 mg/L							
Nitrate (as N)	< 0.010	0.010 mg/L							
Nitrite (as N)	< 0.010	0.010 mg/L							
Sulfate	< 1.0	1.0 mg/L							
LCS (B3K1988-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		98	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.276	0.010 mg/L	0.300		92	85-115			
Nitrite (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		105	85-115			
LCS (B3K1988-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Bromide	0.9	0.1 mg/L	1.00		87	85-115			
Chloride	0.29	0.10 mg/L	0.300		97	85-115			
Fluoride	0.20	0.10 mg/L	0.200		102	85-115			
Nitrate (as N)	0.274	0.010 mg/L	0.300		91	85-115			
Nitrite (as N)	0.273	0.010 mg/L	0.300		91	85-115			
Sulfate	1.6	1.0 mg/L	1.50		106	85-115			
Matrix Spike (B3K1988-MS1)			Source: 23K2233-02		Prepared: 2023-11-21, Analyzed: 2023-11-21				
Bromide	9.6	0.1 mg/L	10.0	< 0.1	96	75-125			
Chloride	2.88	0.10 mg/L	3.00	< 0.10	95	75-125			
Fluoride	2.07	0.10 mg/L	2.00	< 0.10	103	75-125			
Nitrate (as N)	2.90	0.010 mg/L	3.00	< 0.010	97	75-125			
Nitrite (as N)	2.85	0.010 mg/L	3.00	< 0.010	95	75-125			

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
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WORK ORDER REPORTED 23K2233
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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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Anions, Batch B3K1988, Continued

Matrix Spike (B3K1988-MS1), Continued		Source: 23K2233-02		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Sulfate	15.4	1.0 mg/L	15.0	< 1.0	103	75-125			
Matrix Spike (B3K1988-MS2)		Source: 23K2233-03		Prepared: 2023-11-21, Analyzed: 2023-11-21					
Bromide	9.5	0.1 mg/L	10.0	< 0.1	95	75-125			
Chloride	2.85	0.10 mg/L	3.00	< 0.10	95	75-125			
Fluoride	2.08	0.10 mg/L	2.00	< 0.10	104	75-125			
Nitrate (as N)	2.90	0.010 mg/L	3.00	< 0.010	96	75-125			
Nitrite (as N)	2.83	0.010 mg/L	3.00	< 0.010	94	75-125			
Sulfate	15.4	1.0 mg/L	15.0	< 1.0	103	75-125			

Dissolved Metals, Batch B3K1990

Blank (B3K1990-BLK1)		Prepared: 2023-11-21, Analyzed: 2023-11-21							
Aluminum, dissolved	< 1.0	1.0 µg/L							
Antimony, dissolved	< 0.050	0.050 µg/L							
Arsenic, dissolved	< 0.050	0.050 µg/L							
Barium, dissolved	< 0.10	0.10 µg/L							
Beryllium, dissolved	< 0.010	0.010 µg/L							
Bismuth, dissolved	< 0.010	0.010 µg/L							
Boron, dissolved	< 2.0	2.0 µg/L							
Cadmium, dissolved	< 0.0020	0.0020 µg/L							
Calcium, dissolved	< 50	50 µg/L							
Chromium, dissolved	< 0.50	0.50 µg/L							
Cobalt, dissolved	< 0.0050	0.0050 µg/L							
Copper, dissolved	< 0.10	0.10 µg/L							
Iron, dissolved	< 2.0	2.0 µg/L							
Lead, dissolved	< 0.050	0.050 µg/L							
Lithium, dissolved	< 0.050	0.050 µg/L							
Magnesium, dissolved	< 5.0	5.0 µg/L							
Manganese, dissolved	< 0.050	0.050 µg/L							
Molybdenum, dissolved	< 0.010	0.010 µg/L							
Nickel, dissolved	< 0.040	0.040 µg/L							
Phosphorus, dissolved	< 10	10 µg/L							
Potassium, dissolved	< 20	20 µg/L							
Selenium, dissolved	< 0.10	0.10 µg/L							
Silicon, dissolved	< 100	100 µg/L							
Silver, dissolved	< 0.010	0.010 µg/L							
Sodium, dissolved	< 20	20 µg/L							
Strontium, dissolved	< 0.10	0.10 µg/L							
Sulfur, dissolved	< 1000	1000 µg/L							
Tellurium, dissolved	< 0.050	0.050 µg/L							
Thallium, dissolved	< 0.0040	0.0040 µg/L							
Thorium, dissolved	< 0.010	0.010 µg/L							
Tin, dissolved	< 0.050	0.050 µg/L							
Titanium, dissolved	< 0.20	0.20 µg/L							
Tungsten, dissolved	< 0.20	0.20 µg/L							
Uranium, dissolved	< 0.0010	0.0010 µg/L							
Vanadium, dissolved	< 1.00	1.00 µg/L							
Zinc, dissolved	< 1.0	1.0 µg/L							
Zirconium, dissolved	< 0.020	0.020 µg/L							

LCS (B3K1990-BS1)		Prepared: 2023-11-21, Analyzed: 2023-11-21							
Aluminum, dissolved	3900	1.0 µg/L	4000		97	80-120			
Antimony, dissolved	39.4	0.050 µg/L	40.0		99	80-120			
Arsenic, dissolved	404	0.050 µg/L	400		101	80-120			
Barium, dissolved	38.6	0.10 µg/L	40.0		96	80-120			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K1990, Continued									
LCS (B3K1990-BS1), Continued					Prepared: 2023-11-21, Analyzed: 2023-11-21				
Beryllium, dissolved	39.1	0.010 µg/L	40.0		98	80-120			
Bismuth, dissolved	39.1	0.010 µg/L	40.0		98	80-120			
Boron, dissolved	393	2.0 µg/L	400		98	80-120			
Cadmium, dissolved	40.3	0.0020 µg/L	40.0		101	80-120			
Calcium, dissolved	4060	50 µg/L	4000		102	80-120			
Chromium, dissolved	40.1	0.50 µg/L	40.0		100	80-120			
Cobalt, dissolved	40.1	0.0050 µg/L	40.0		100	80-120			
Copper, dissolved	40.2	0.10 µg/L	40.0		100	80-120			
Iron, dissolved	4000	2.0 µg/L	4000		100	80-120			
Lead, dissolved	39.6	0.050 µg/L	40.0		99	80-120			
Lithium, dissolved	38.6	0.050 µg/L	40.0		96	80-120			
Magnesium, dissolved	3930	5.0 µg/L	4000		98	80-120			
Manganese, dissolved	40.0	0.050 µg/L	40.0		100	80-120			
Molybdenum, dissolved	40.3	0.010 µg/L	40.0		101	80-120			
Nickel, dissolved	40.9	0.040 µg/L	40.0		102	80-120			
Phosphorus, dissolved	4000	10 µg/L	4000		100	80-120			
Potassium, dissolved	3980	20 µg/L	4000		99	80-120			
Selenium, dissolved	411	0.10 µg/L	400		103	80-120			
Silicon, dissolved	3940	100 µg/L	4000		98	80-120			
Silver, dissolved	40.7	0.010 µg/L	40.0		102	80-120			
Sodium, dissolved	4010	20 µg/L	4000		100	80-120			
Strontium, dissolved	39.5	0.10 µg/L	40.0		99	80-120			
Sulfur, dissolved	38700	1000 µg/L	40000		97	80-120			
Tellurium, dissolved	38.3	0.050 µg/L	40.0		96	80-120			
Thallium, dissolved	39.5	0.0040 µg/L	40.0		99	80-120			
Thorium, dissolved	40.0	0.010 µg/L	40.0		100	80-120			
Tin, dissolved	40.0	0.050 µg/L	40.0		100	80-120			
Titanium, dissolved	39.2	0.20 µg/L	40.0		98	80-120			
Tungsten, dissolved	39.3	0.20 µg/L	40.0		98	80-120			
Uranium, dissolved	40.0	0.0010 µg/L	40.0		100	80-120			
Vanadium, dissolved	40.5	1.00 µg/L	40.0		101	80-120			
Zinc, dissolved	399	1.0 µg/L	400		100	80-120			
Zirconium, dissolved	39.5	0.020 µg/L	40.0		99	80-120			

Dissolved Metals, Batch B3K2521

Blank (B3K2521-BLK1)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK2)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK3)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK4)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK5)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK6)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							
Blank (B3K2521-BLK7)			Prepared: 2023-11-24, Analyzed: 2023-11-24						
Mercury, dissolved	< 0.0050	0.0050 µg/L							

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Dissolved Metals, Batch B3K2521, Continued									
LCS (B3K2521-BS1)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.251	0.0050 µg/L	0.250		101	80-120			
LCS (B3K2521-BS2)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.256	0.0050 µg/L	0.250		102	80-120			
LCS (B3K2521-BS3)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.263	0.0050 µg/L	0.250		105	80-120			
LCS (B3K2521-BS4)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.290	0.0050 µg/L	0.250		116	80-120			
LCS (B3K2521-BS5)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.282	0.0050 µg/L	0.250		113	80-120			
LCS (B3K2521-BS6)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.245	0.0050 µg/L	0.250		98	80-120			
LCS (B3K2521-BS7)				Prepared: 2023-11-24, Analyzed: 2023-11-24					
Mercury, dissolved	0.240	0.0050 µg/L	0.250		96	80-120			
General Parameters, Batch B3K1963									
Blank (B3K1963-BLK1)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	< 10	10 mg/L							
Blank (B3K1963-BLK2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K1963-BS1)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	45	10 mg/L	50.0		90	85-115			
LCS (B3K1963-BS2)				Prepared: 2023-11-20, Analyzed: 2023-11-21					
Solids, Total Dissolved	198	10 mg/L	208		95	85-115			
General Parameters, Batch B3K2028									
Blank (B3K2028-BLK1)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3K2028-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
Blank (B3K2028-BLK3)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	< 0.50	0.50 mg/L							
LCS (B3K2028-BS1)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	10.1	0.50 mg/L	10.0		101	78-116			
LCS (B3K2028-BS2)				Prepared: 2023-11-21, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	9.58	0.50 mg/L	10.0		96	78-116			
LCS (B3K2028-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Carbon, Dissolved Organic	9.79	0.50 mg/L	10.0		98	78-116			

General Parameters, Batch B3K2066

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
General Parameters, Batch B3K2066, Continued									
Blank (B3K2066-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK3)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
Blank (B3K2066-BLK4)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	< 0.050	0.050 mg/L							
LCS (B3K2066-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	0.980	0.050 mg/L	1.00		98	85-115			
LCS (B3K2066-BS2)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
LCS (B3K2066-BS3)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.02	0.050 mg/L	1.00		102	85-115			
LCS (B3K2066-BS4)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Ammonia, Total (as N)	1.01	0.050 mg/L	1.00		101	85-115			
General Parameters, Batch B3K2079									
Blank (B3K2079-BLK1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Sulfide, Total	< 0.020	0.020 mg/L							
LCS (B3K2079-BS1)			Prepared: 2023-11-21, Analyzed: 2023-11-21						
Sulfide, Total	0.472	0.020 mg/L	0.480		98	80-120			
General Parameters, Batch B3K2147									
Blank (B3K2147-BLK1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK2)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
Blank (B3K2147-BLK3)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Phenolphthalein (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Bicarbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Carbonate (as CaCO ₃)	< 1.0	1.0 mg/L							
Alkalinity, Hydroxide (as CaCO ₃)	< 1.0	1.0 mg/L							
LCS (B3K2147-BS1)			Prepared: 2023-11-22, Analyzed: 2023-11-22						
Alkalinity, Total (as CaCO ₃)	100	1.0 mg/L	100		100	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	34.1	1.0 mg/L	50.0		68	0-200			

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
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General Parameters, Batch B3K2147, Continued

LCS (B3K2147-BS2)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	114	1.0 mg/L	100		114	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.8	1.0 mg/L	50.0		40	0-200			
LCS (B3K2147-BS3)				Prepared: 2023-11-22, Analyzed: 2023-11-22					
Alkalinity, Total (as CaCO ₃)	112	1.0 mg/L	100		112	80-120			
Alkalinity, Phenolphthalein (as CaCO ₃)	19.4	1.0 mg/L	50.0		39	0-200			

General Parameters, Batch B3K2319

Blank (B3K2319-BLK1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	< 2.0	2.0 mg/L							
LCS (B3K2319-BS1)				Prepared: 2023-11-23, Analyzed: 2023-11-23					
Acidity, Total (as CaCO ₃)	106	2.0 mg/L	100		106	80-120			

General Parameters, Batch B3K2339

Blank (B3K2339-BLK1)				Prepared: 2023-11-23, Analyzed: 2023-11-24					
Solids, Total Dissolved	< 10	10 mg/L							
LCS (B3K2339-BS1)				Prepared: 2023-11-23, Analyzed: 2023-11-24					
Solids, Total Dissolved	42	10 mg/L	50.0		85	85-115			
LCS (B3K2339-BS2)				Prepared: 2023-11-23, Analyzed: 2023-11-24					
Solids, Total Dissolved	210	10 mg/L	208		101	85-115			

Total Metals, Batch B3K2005

Blank (B3K2005-BLK1)				Prepared: 2023-11-20, Analyzed: 2023-11-20					
Aluminum, total	< 2.0	2.0 µg/L							
Antimony, total	< 0.050	0.050 µg/L							
Arsenic, total	< 0.050	0.050 µg/L							
Barium, total	< 0.10	0.10 µg/L							
Beryllium, total	< 0.010	0.010 µg/L							
Bismuth, total	< 0.010	0.010 µg/L							
Boron, total	< 2.0	2.0 µg/L							
Cadmium, total	< 0.0020	0.0020 µg/L							
Calcium, total	< 50	50 µg/L							
Chromium, total	< 0.50	0.50 µg/L							
Cobalt, total	< 0.0050	0.0050 µg/L							
Copper, total	< 0.20	0.20 µg/L							
Iron, total	< 2.0	2.0 µg/L							
Lead, total	< 0.050	0.050 µg/L							
Lithium, total	< 0.050	0.050 µg/L							
Magnesium, total	< 5.0	5.0 µg/L							
Manganese, total	< 0.050	0.050 µg/L							
Molybdenum, total	< 0.010	0.010 µg/L							
Nickel, total	< 0.040	0.040 µg/L							
Phosphorus, total	< 10	10 µg/L							
Potassium, total	< 20	20 µg/L							
Selenium, total	< 0.10	0.10 µg/L							
Silicon, total	< 100	100 µg/L							
Silver, total	< 0.010	0.010 µg/L							
Sodium, total	< 20	20 µg/L							
Strontium, total	< 0.10	0.10 µg/L							

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Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3K2005, Continued									
Blank (B3K2005-BLK1), Continued					Prepared: 2023-11-20, Analyzed: 2023-11-20				
Sulfur, total	< 1000	1000 µg/L							
Tellurium, total	< 0.050	0.050 µg/L							
Thallium, total	< 0.0040	0.0040 µg/L							
Thorium, total	< 0.010	0.010 µg/L							
Tin, total	< 0.050	0.050 µg/L							
Titanium, total	< 0.20	0.20 µg/L							
Tungsten, total	< 0.20	0.20 µg/L							
Uranium, total	< 0.0010	0.0010 µg/L							
Vanadium, total	< 1.00	1.00 µg/L							
Zinc, total	< 1.0	1.0 µg/L							
Zirconium, total	< 0.020	0.020 µg/L							
LCS (B3K2005-BS1)					Prepared: 2023-11-20, Analyzed: 2023-11-20				
Aluminum, total	4070	2.0 µg/L	4000		102	80-120			
Antimony, total	38.7	0.050 µg/L	40.0		97	80-120			
Arsenic, total	405	0.050 µg/L	400		101	80-120			
Barium, total	39.8	0.10 µg/L	40.0		99	80-120			
Beryllium, total	39.8	0.010 µg/L	40.0		99	80-120			
Bismuth, total	38.2	0.010 µg/L	40.0		95	80-120			
Boron, total	409	2.0 µg/L	400		102	80-120			
Cadmium, total	38.8	0.0020 µg/L	40.0		97	80-120			
Calcium, total	4010	50 µg/L	4000		100	80-120			
Chromium, total	41.2	0.50 µg/L	40.0		103	80-120			
Cobalt, total	41.3	0.0050 µg/L	40.0		103	80-120			
Copper, total	41.0	0.20 µg/L	40.0		102	80-120			
Iron, total	4100	2.0 µg/L	4000		102	80-120			
Lead, total	38.8	0.050 µg/L	40.0		97	80-120			
Lithium, total	38.5	0.050 µg/L	40.0		96	80-120			
Magnesium, total	4180	5.0 µg/L	4000		104	80-120			
Manganese, total	41.1	0.050 µg/L	40.0		103	80-120			
Molybdenum, total	39.5	0.010 µg/L	40.0		99	80-120			
Nickel, total	41.1	0.040 µg/L	40.0		103	80-120			
Phosphorus, total	4000	10 µg/L	4000		100	80-120			
Potassium, total	4030	20 µg/L	4000		101	80-120			
Selenium, total	399	0.10 µg/L	400		100	80-120			
Silicon, total	3910	100 µg/L	4000		98	80-120			
Silver, total	39.8	0.010 µg/L	40.0		99	80-120			
Sodium, total	4180	20 µg/L	4000		105	80-120			
Strontium, total	40.0	0.10 µg/L	40.0		100	80-120			
Sulfur, total	38600	1000 µg/L	40000		97	80-120			
Tellurium, total	38.7	0.050 µg/L	40.0		97	80-120			
Thallium, total	38.8	0.0040 µg/L	40.0		97	80-120			
Thorium, total	39.5	0.010 µg/L	40.0		99	80-120			
Tin, total	40.6	0.050 µg/L	40.0		102	80-120			
Titanium, total	39.8	0.20 µg/L	40.0		99	80-120			
Tungsten, total	39.7	0.20 µg/L	40.0		99	80-120			
Uranium, total	40.1	0.0010 µg/L	40.0		100	80-120			
Vanadium, total	41.3	1.00 µg/L	40.0		103	80-120			
Zinc, total	405	1.0 µg/L	400		101	80-120			
Zirconium, total	38.5	0.020 µg/L	40.0		96	80-120			

Total Metals, Batch B3K2127

Blank (B3K2127-BLK1)					Prepared: 2023-11-21, Analyzed: 2023-11-21				
Mercury, total	< 0.0050	0.0050 µg/L							

APPENDIX 2: QUALITY CONTROL RESULTS

REPORTED TO PROJECT AtkinsRéalis (Nelson)
697296

WORK ORDER REPORTED 23K2233
2023-11-27 12:53

Analyte	Result	RL Units	Spike Level	Source Result	% REC	REC Limit	% RPD	RPD Limit	Qualifier
Total Metals, Batch B3K2127, Continued									
LCS (B3K2127-BS1)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Mercury, total	0.257	0.0050 µg/L	0.250		103	80-120			
Duplicate (B3K2127-DUP1)				Source: 23K2233-02		Prepared: 2023-11-21, Analyzed: 2023-11-21			
Mercury, total	< 0.0050	0.0050 µg/L		< 0.0050				20	
Total Metals, Batch B3K2139									
Blank (B3K2139-BLK2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Mercury, total	< 0.0050	0.0050 µg/L							
LCS (B3K2139-BS2)				Prepared: 2023-11-21, Analyzed: 2023-11-21					
Mercury, total	0.256	0.0050 µg/L	0.250		102	80-120			



 2 3 K 2 2 3 3 *		Page 1 of 1																						
		COC ID: 697296-20231116-005				TURNAROUND TIME:				Regular				RUSH:										
		PROJECT/CLIENT INFO								LABORATORY				OTHER INFO										
		Job# Sunro Mine								Lab Name Caro				Report Format / Distribution				Excel	PDF	EDD				
		ger Michelle Unger								Lab Contact Regan Pshyk				Email 1: michelle.unger@teck.com				X	X	X				
		email michelle.unger@teck.com								Email rpshyk@caro.ca				Email 2: craig.pinnell@atkinsrealis.com				X	X	X				
Address 601 Knighton Rd								Address 110 - 4011 Viding Way				Email 3: Jonathan.croston@atkinsrealis.com				X	X	X						
												Email 4: bonn.johnston@atkinsrealis.com				X	X	X						
City Kimberly Province BC								City Richmond Province BC				Email 5: tlv_west_bc_atkdat@atkinsrealis.com				X	X	X						
Postal Code V1A 1C7 Country Canada								Postal Code V6B2K9 Country Canada																
Phone Number 250-427-8422								Phone Number 604-279-1499				PO number				697296								
SAMPLE DETAILS								ANALYSIS REQUESTED								Filtered - F: Field, L: Lab, FL: Field & Lab, N: None								
Sample ID	Sample Location	Field Matrix	Hazardous Material (Yes/No)	Date	Time (24hr)	G=Grab C=Comp	# Of Cont.	FILE	PRESERV.	ANALYSIS	Total Metals + Hg	Dissolved Metals + Hg	TDS	Hardness	Major Anions (sulphate, chloride, bromide, fluoride, nitrate/nitrite)	Total and Speciated Alkalinity, Acidity	Sulphide	Ammonia	Dissolved Organic Carbon				HOLD	
SU_EQUIPMENT-11142023_NP	SW15-10	Water	N	2023-11-14	14:00	G	5				x	x	x	x	x	x	x							
SU_TRIP-11162023_NP	Trip Blank	Water	N	2023-11-16	15:30	G	7				x		x	x	x	x	x	x	x					
SU_FIELD-11142023_NP	SW17-17	Water	N	2023-11-14	11:45	G	5				x		x	x	x	x	x	x						
ADDITIONAL COMMENTS/SPECIAL INSTRUCTIONS			RELINQUISHED BY/AFFILIATION					DATE/TIME		ACCEPTED BY/AFFILIATION				DATE/TIME										
Ultra low detection for metals required. Detection limits suitable for CSR and BCWQG Invoice to: Atkins Realis Inc. Jonathan Croston / Craig Pinnell (Nelson BC) 455 Rene-Levesque Blvd. West, Montreal, QC, H2Z 1Z3 Send to: payables@atkinsrealis.com Billing PO: 697296			Atkins Realis Inc																					
NB OF BOTTLES RETURNED/DESCRIPTION			Sampler's Name					Bonn Johnston and Chris Walker					Mobile #											
Regular (default) X													236-464-2830											
Priority (2-3 business days) - 50% surcharge																								
Emergency (1 Business Day) - 100% surcharge																								
For Emergency <1 Day, ASAP or Weekend			Sampler's Signature										Date/Time											
													2023-11-16 17:00:00 PM											